
notes

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CONTENTS:

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1.1 cppcon

- Opening Keynote Meeting C++ 2019 - Howard Hinnant - Design Rationale for the chrono Library

https://www.youtube.com/watch?v=adSAN282YIw&ab_channel=MeetingCpp

1.2 Meeting C++

- <https://meetingcpp.com/mcpp/slides/>

1.3 getline

See https://en.cppreference.com/w/cpp/string/basic_string/getline.

Listing 1: ./code/getline/main.cc

```
1  #include <cassert>
2  #include <iostream>
3  #include <sstream>
4  #include <string>
5
6  int main() {
7      std::string s = "1,2,3";
8      std::stringstream ss(s);
9      std::getline(ss, s, ',');
10     assert(s == "1");
11
12     std::getline(ss, s, ',');
13     assert(s == "2");
14 }
```

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```
15     std::getline(ss, s, ',');
16     assert(s == "3");
17     return 0;
18 }
```

1.4 ratio

- <https://en.cppreference.com/w/cpp/numeric/ratio/ratio>

Listing 2: ./code/ratio/main.cc

```
1  #include <assert.h>
2  #include <ratio>
3
4  int main() {
5      {
6          std::ratio<1, 3> r;
7          assert(r.num == 1);
8          assert(r.den == 3);
9      }
10
11     // some predefined constants
12     {
13         std::pico r;
14         assert(r.num == 1);
15         assert(r.den == 1e12);
16     }
17
18     {
19         std::nano r;
20         assert(r.num == 1);
21         assert(r.den == 1e9);
22     }
23
24     {
25         std::micro r;
26         assert(r.num == 1);
27         assert(r.den == 1e6);
28     }
29
30     {
31         std::milli r;
```

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```
32     assert(r.num == 1);
33     assert(r.den == 1e3);
34 }
35
36 {
37     std::kilo r;
38     assert(r.num == 1e3);
39     assert(r.den == 1);
40 }
41
42 {
43     std::mega r;
44     assert(r.num == 1e6);
45     assert(r.den == 1);
46 }
47
48 {
49     std::giga r;
50     assert(r.num == 1e9);
51     assert(r.den == 1);
52 }
53 return 0;
54 }
```

1.5 chrono

- https://www.youtube.com/watch?v=P32hvk8b13M&ab_channel=CppCon

<https://github.com/CppCon/CppCon2016/tree/master/Tutorials/A%20chrono%20Tutorial>

Everything is inside the namespace `std::chrono`.

Differences between:

- `std::time_t`
- `std::tm`
- `std::time()`

`std::time()` to get the number of elapsed seconds since the epoch time 1970.1.1 00:00:00. The returned type is `time_t`. Note its precision is in seconds.

`std::localtime` converts `time_t` to a struct `tm` in local time.

`std::mktime()` converts a struct `tm` into `time_t` in local time.

`std::asctime()` converts a `struct tm` into a `char*` in local time.

`std::ctime()` converts a `time_t` into a `char*` in local time.

`std::strftime` is more powerful.

`std::this_thread::sleep_until` takes a timepoint as argument, while `std::this_thread::sleep_for` takes a duration as argument.

After using namespace `std::chrono_literals`; or using `std::chrono::operator""s`, we can use `auto s = 1s`.

Listing 3: `./code/chrono/main.cc`

```
1 // Requires c++14
2 // g++ -std=c++14 main.cc
3 #include <assert.h>
4 #include <chrono>
5 #include <ctime>
6 #include <iostream>
7 #include <string>
8 #include <thread>
9 #include <type_traits>
10 #include <vector>
11
12 void test_duration() {
13     std::chrono::seconds s(1);
14     static_assert(
15         std::is_same<decltype(s),
16             std::chrono::duration<long long, std::ratio<1>>>::value,
17         "");
18     static_assert(sizeof(s) == sizeof(long long), "");
19
20     assert(s.count() == 1);
21
22     std::chrono::milliseconds ms(1);
23     static_assert(
24         std::is_same<decltype(ms), std::chrono::duration<
25             long long, std::ratio<1, 1000>>>::value,
26         "");
27
28     static_assert(sizeof(ms) == sizeof(long long), "");
29
30     assert(ms.count() == 1);
31     ms = s; // implicit cast
32     assert(ms.count() == 1000);
33 }
```

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```

34 s = std::chrono::duration_cast<std::chrono::seconds>(ms);
35 assert(s.count() == 1);
36
37 using namespace std::chrono_literals;
38 auto s2 = 10s;
39 static_assert(std::is_same<decltype(s), decltype(s2)>::value, "");
40 assert(s2.count() == 10);
41
42 ms = s2;
43 assert(ms.count() == 1e4);
44 ms = ms + s;
45 assert(ms.count() == 11000);
46
47 // s = 3; // error
48 // std::cout << s << "\n"; // error
49
50 s = 1s;
51 ms = 1ms;
52 ms = s + ms;
53 assert(ms.count() == 1001);
54 s = std::chrono::duration_cast<std::chrono::seconds>(ms);
55 assert(s.count() == 1);
56
57 ms = 1100ms;
58 s = std::chrono::duration_cast<std::chrono::seconds>(ms); // truncated
59 assert(s.count() == 1);
60
61 std::chrono::minutes m = 1min;
62 s = m;
63 assert(s.count() == 60);
64
65 std::chrono::hours h = 1h;
66 s = h;
67 assert(s.count() == 60 * 60);
68
69 std::chrono::seconds s3; // s3.count() is uninitialized
70 // assert(s3.count() == 0); // does not hold
71 }
72
73 void test_time_1() {
74     // the resolution of std::time() is in seconds
75     std::time_t t = std::time(nullptr);
76     t = 1669039051;

```

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```
77 // 16690369051
78 // 16690369052
79 // 16690369053
80 // t is an integer, representing the elapsed seconds since 1970.1.1 00:00 UTC
81
82 // the returned value points to a static allocated memory. Not thread safe.
83 // it is overwritten in the next call
84 std::tm *tm = std::localtime(&t);
85
86 // 2022-11-21 21:57:31
87 printf("%d-%d-%d %d:%d:%d\n", tm->tm_year + 1900, tm->tm_mon + 1, tm->tm_mday,
88        tm->tm_hour, tm->tm_min, tm->tm_sec);
89 // tm_year, years since 1900
90 // tm_mon: months since january (0-11)
91 std::vector<std::string> days = {"sunday", "saturday", "monday", "tuesday",
92                                "wednesday", "thursday", "friday"};
93 // tm_wday: days since sunday (0-6)
94 printf("weekday: %s, %d\n", days[tm->tm_wday].c_str(), tm->tm_wday);
95 // weekday: saturday, 1
96
97 // tm_yday: days since january 1 (0-365)
98 printf("day of the year: %d\n", tm->tm_yday);
99 // day of the year: 324
100
101 // std::asctime() is not threadsafe
102 //
103 // Mon Nov 21 21:57:31 2022
104 printf("%s\n", std::asctime(tm));
105
106 // Mon Nov 21 13:57:31 2022
107 std::tm *gm = std::gmtime(&t); // UTC
108 printf("%s\n", std::asctime(gm));
109
110 // Mon Nov 21 21:57:31 2022
111 printf("%s\n", std::ctime(&t));
112 // std::ctime(&t) == std::asctime(std::localtime(&t))
113
114 // tm is in localtime
115 //
116 // https://en.cppreference.com/w/cpp/chrono/c/strftime
117 {
118     char s[100];
119     std::strftime(s, sizeof(s), "%Y", tm);
```

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```
120     assert(s == std::string("2022"));
121 }
122
123 {
124     char s[100];
125     std::strftime(s, sizeof(s), "%y", tm);
126     assert(s == std::string("22")); // the last 2 digits of the year
127 }
128
129 {
130     char s[100];
131     std::strftime(s, sizeof(s), "%C", tm);
132     assert(s == std::string("20")); // the first 2 digits of the year
133 }
134
135 {
136     char s[100];
137     std::strftime(s, sizeof(s), "%b", tm);
138     assert(s == std::string("Nov"));
139 }
140
141 {
142     char s[100];
143     std::strftime(s, sizeof(s), "%B", tm);
144     assert(s == std::string("November"));
145 }
146
147 {
148     char s[100];
149     std::strftime(s, sizeof(s), "%m", tm);
150     assert(s == std::string("11")); // month
151 }
152
153 {
154     char s[100];
155     std::strftime(s, sizeof(s), "%U", tm);
156     // week of the year. Sunday is the first day of the week
157     assert(s == std::string("47"));
158 }
159
160 {
161     char s[100];
162     std::strftime(s, sizeof(s), "%W", tm);
```

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```
163     // week of the year. Monday is the first day of the week
164     assert(s == std::string("47"));
165 }
166
167 {
168     char s[100];
169     std::strftime(s, sizeof(s), "%j", tm);
170     // day of the year: 1-366
171     assert(s == std::string("325"));
172 }
173
174 {
175     char s[100];
176     std::strftime(s, sizeof(s), "%d", tm);
177     // day of the month: 1-31
178     assert(s == std::string("21"));
179 }
180 }
181
182 void test_time_2() {
183     struct std::tm tm {};
184     tm.tm_year = 1970 - 1900;
185     tm.tm_mon = 0;
186     tm.tm_mday = 1;
187     tm.tm_sec = 1;
188     tm.tm_isdst = 0;
189     // Cannot use std::mktime() here since it treats tm as localtime
190     // std::time_t t = std::mktime(&tm);
191
192     std::time_t t = timegm(&tm);
193     assert(t == 1); // epoch time is 1970, january 1, 00:00:00
194 }
195
196 void test_timepoint() {
197     {
198         // conversion between timepoint and time_t
199         std::time_t t = std::time(nullptr);
200
201         std::chrono::system_clock::time_point tp =
202             std::chrono::system_clock::from_time_t(t);
203
204         auto duration = tp.time_since_epoch();
205         auto s = std::chrono::duration_cast<std::chrono::seconds>(duration);
```

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```

206     assert(t == s.count());
207 }
208
209 {
210     std::chrono::system_clock::time_point now =
211         std::chrono::system_clock::now();
212     std::time_t t = std::chrono::system_clock::to_time_t(now);
213     std::time_t t2 = std::time(nullptr);
214     assert(t == t2);
215 }
216 }
217
218 void test_sleep() {
219     std::time_t t = std::time(nullptr);
220     // note: std::ctime contains "\n"
221     std::cout << "start to sleep at " << std::ctime(&t);
222     auto now = std::chrono::steady_clock::now();
223     using std::chrono::operator"s;
224     std::this_thread::sleep_until(now + 1s);
225     t = std::time(nullptr);
226     std::cout << "finish sleeping at " << std::ctime(&t);
227 }
228
229 void test_sleep_2() {
230     auto start = std::chrono::steady_clock::now();
231     using std::chrono::operator"ms;
232     // use sleep_until with a timepoint
233     std::this_thread::sleep_until(start + 500ms);
234     auto end = std::chrono::steady_clock::now();
235     std::cout << "finish sleeping at "
236         << std::chrono::duration<double, std::milli>(end - start).count()
237         << " ms\n";
238 }
239
240 void test_sleep_3() {
241     auto start = std::chrono::steady_clock::now();
242     using std::chrono::operator"ms;
243     // use sleep_for with a duration
244     std::this_thread::sleep_for(10ms);
245     auto end = std::chrono::steady_clock::now();
246     std::cout << "finish sleeping at "
247         << std::chrono::duration<double, std::milli>(end - start).count()
248         << " ms\n";

```

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```
249 }
250
251 int main() {
252     test_duration();
253     test_time_1();
254     test_time_2();
255     test_timepoint();
256     test_sleep();
257     test_sleep_2();
258     test_sleep_3();
259     return 0;
260 }
```

1.6 iomanip

1.6.1 quoted

See <https://en.cppreference.com/w/cpp/io/manip/quoted>.

Listing 4: <https://en.cppreference.com/w/cpp/io/manip/quoted>

```
1 void TestQuoted() {
2     // -std=c++14
3     std::string s = "abc";
4     std::ostringstream os;
5
6     // "abc"
7     os << std::quoted(s) << "\n";
8
9     std::cout << os.str() << "\n";
10 }
```

1.7 Shared libs

- [How to write shared libraries](#)
- <https://gcc.gnu.org/wiki/Visibility>
- Position Independent Code (PIC) in shared libraries

<https://eli.thegreenplace.net/2011/11/03/position-independent-code-pic-in-shared-libraries/>

- Position Independent Code (PIC) in shared libraries

<https://eli.thegreenplace.net/2011/11/03/position-independent-code-pic-in-shared-libraries>

- OpenFST does not recognize FST type "vector" #1

```
gcc commandline: -fvisibility=hidden
```

1.7.1 --whole-archive

See

- https://ftp.gnu.org/old-gnu/Manuals/ld-2.9.1/html_node/ld_3.html
- <https://stackoverflow.com/questions/805555/ld-linker-question-the-whole-archive-option>
- <https://jonathanhamberg.com/post/gcc-archive-linker-oddity/>

```
g++ main.cc -Wl,--whole-archive -lhttp -Wl,--no-whole-archive -lhandlers
```

```
# https://jonathanhamberg.com/post/gcc-archive-linker-oddity/
target_link_options(hal PUBLIC
    -Wl,--whole-archive libhal.a -Wl,--no-whole-archive
)
```

1.8 regex

Listing 5: ./code/regex/main.cc

```
1 // g++ -std=c++11 ./main.cc
2 #include <algorithm>
3 #include <cctype>
4 #include <codecvt>
5 #include <iostream>
6 #include <locale>
7 #include <regex>
8 #include <string>
9
10 std::string ToLowerCase(const std::string &s) {
11     std::string ans(s.size(), 0);
12     std::transform(s.begin(), s.end(), ans.begin(),
13                   [](unsigned char c) { return std::tolower(c); });
14     return ans;
15 }
```

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```

16
17 std::wstring ToWideString(const std::string &s) {
18     // see
19     // https://stackoverflow.com/questions/2573834/c-convert-string-or-char-to-wstring-
    ↳ or-wchar-t
20     std::wstring_convert<std::codecvt_utf8_utf16<wchar_t>> converter;
21     return converter.from_bytes(s);
22 }
23
24 std::string ToString(const std::wstring &s) {
25     // see
26     // https://stackoverflow.com/questions/2573834/c-convert-string-or-char-to-wstring-
    ↳ or-wchar-t
27     std::wstring_convert<std::codecvt_utf8_utf16<wchar_t>> converter;
28     return converter.to_bytes(s);
29 }
30
31 int32_t main() {
32     std::string s = "你 (好) 吗, HoW    are You doIng? 包含\"中文\"和'英文'。are "
33                     "you ok? 谢谢! 好的; 说: 一句话…也 “可以” ";
34     std::cout << s << "\n";
35
36     std::vector<std::pair<std::string, std::string>> replace_str_pairs = {
37         {", ", " , " ,"}, {"; ", " ;"}, {"。 ", " ."}, {"! ", " !"}, {"\\s+", " "},
38     };
39     for (const auto &p : replace_str_pairs) {
40         std::regex re(p.first);
41         s = std::regex_replace(s, re, p.second);
42     }
43     std::cout << s << "\n";
44
45     s = ToLowerCase(s);
46
47     std::cout << s << "\n";
48
49     auto ws = ToWideString(s);
50
51     s = ToString(ws);
52     std::cout << s << "\n";
53
54     // https://en.cppreference.com/w/cpp/regex
55     // https://stackoverflow.com/questions/37989081/how-to-use-unicode-range-in-c-regex
56     std::string expr =

```

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```

57     "([;:, .?!'\"...\\(\\) '0' ])|([\\u4e00-\\u9fff])+|([\\u0000-\\u007f])+";
58
59     std::wstring wexpr = ToWideString(expr);
60     std::wregex we(wexpr);
61
62     auto begin = std::wsregex_iterator(ws.begin(), ws.end(), we);
63     auto end = std::wsregex_iterator();
64     for (std::wsregex_iterator i = begin; i != end; ++i) {
65         std::wsmatch match = *i;
66         std::wstring match_str = match.str();
67         auto ms = ToString(match_str);
68         uint8_t c = reinterpret_cast<const uint8_t*>(ms.data())[0];
69         if (c < 0x80) {
70             std::cout << "Non-Chinese: " << ms << "\n";
71         } else {
72             std::cout << "Chinese: " << ms << "\n";
73         }
74     }
75
76     return 0;
77 }
78 /*
79 你好吗, HoW are You doIng? 包含中文和英文。are you ok? 谢谢
80 你好吗, HoW are You doIng? 包含中文和英文。are you ok? 谢谢
81 你好吗, how are you doing? 包含中文和英文。are you ok? 谢谢
82 你好吗, how are you doing? 包含中文和英文。are you ok? 谢谢
83 Chinese: 你好吗
84 Non-Chinese: ,how are you doing?
85 Chinese: 包含中文和英文
86 Non-Chinese: .are you ok?
87 Chinese: 谢谢
88 */

```

1.9 float32

Listing 6: ./code/float32/main.cc

```

1 #include <assert.h>
2 #include <cmath>
3 #include <iostream>
4 #include <limits>

```

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```
5
6 // 1 sign
7 // 8 exp
8 // 23 significand, or mantissa
9 //
10 // bias is 127
11
12 bool IsInfinity(float x) {
13     // exp: all 1
14     // significant: all 0
15     // when sign is 0 -> positive infinity
16     // when sign is 1 -> negative infinity
17     union {
18         float f;
19         int32_t i;
20     } w;
21     w.f = x;
22     int32_t exp = (w.i >> 23) & 0xff;
23     int32_t significand = w.i & 0x7fffff;
24     return exp == 0xff && significand == 0;
25 }
26
27 bool IsNaN(float x) {
28     // exp: all 1
29     // significant: non-zero
30     //
31     // quiet nan: the leftmost significand is 1
32     // signaling nan: the leftmost significand is 0
33     union {
34         float f;
35         int32_t i;
36     } w;
37     w.f = x;
38     int32_t exp = (w.i >> 23) & 0xff;
39     int32_t significand = w.i & 0x7fffff;
40     return exp == 0xff && significand != 0;
41 }
42
43 bool IsDenormalized(float x) {
44     // exp, all 0
45     // significand is not 0
46     union {
47         float f;
```

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```

48     int32_t i;
49 } w;
50 w.f = x;
51 int32_t exp = (w.i >> 23) & 0xff;
52 int32_t significand = w.i & 0x7fffff;
53 return exp == 0 && significand != 0;
54 }
55
56 int32_t main() {
57     float pos_inf = std::numeric_limits<float>::infinity(); // 0x7f80_0000
58     float neg_inf = -std::numeric_limits<float>::infinity(); // 0xff80_0000
59     assert(IsInfinity(pos_inf));
60     assert(IsInfinity(neg_inf));
61     assert(std::fpclassify(pos_inf) == FP_INFINITE);
62     assert(std::fpclassify(neg_inf) == FP_INFINITE);
63
64     float quiet_nan = std::numeric_limits<float>::quiet_NaN(); // 0x7fc0_0000
65     float signaling_nan =
66         std::numeric_limits<float>::signaling_NaN(); // 0x7fa0_0000
67
68     assert(IsNaN(quiet_nan));
69     assert(IsNaN(signaling_nan));
70     assert(std::fpclassify(quiet_nan) == FP_NAN);
71     assert(std::fpclassify(signaling_nan) == FP_NAN);
72
73     // the smallest positive normal value
74     //
75     // Note the bias is 127
76     float d1 = std::numeric_limits<float>::min(); // 0x80_0000, 1/(1<<126)
77
78     float d2 = std::numeric_limits<float>::min() / 2; // 0x40_0000
79
80     assert(std::fpclassify(d1) == FP_NORMAL);
81     assert(std::fpclassify(d2) == FP_SUBNORMAL);
82     assert(IsDenormalized(d1) == false);
83     assert(IsDenormalized(d2) == true);
84 }

```

See also

- <https://www.h-schmidt.net/FloatConverter/IEEE754.html>
- <https://github.com/AnthonyBongers/CppFloatsInDepth?tab=readme-ov-file>

This page describes how this website is setup.

2.1 Setup

1. Install the dependencies in `./docs/requirements.txt`.

```
recommonmark
sphinx
docutils==0.17
sphinx_rtd_theme
sphinxcontrib-bibtex
sphinx-tabs
sphinx-copybutton
```

2. Use `sphinx-quickstart` to generate the skeleton. When it prompts:

```
Separate source and build directories (y/n)
```

Answer yes.

3. Edit `docs/source/conf.py` and add the following lines to it:

```
import sphinx_rtd_theme
extensions = [
    'sphinx.ext.autodoc',
    'sphinx.ext.autosummary',
    'sphinx.ext.githubpages',
    'sphinx.ext.mathjax',
    'sphinx.ext.napoleon',
    'sphinx.ext.todo',
    'sphinx.ext.viewcode',
    'sphinxcontrib.bibtex',
```

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```
]

html_theme = 'sphinx_rtd_theme'

master_doc = 'index'
pygments_style = 'sphinx'
html_theme_path = [sphinx_rtd_theme.get_html_theme_path()]
smartquotes = False
html_show_sourcelink = True

html_context = {
    'display_github': True,
    'github_user': 'csu-fangjun',
    'github_repo': 'notes',
    'github_version': 'master',
    'conf_py_path': '/docs/source/',
}

html_theme_options = {
    'logo_only': False,
    'display_version': True,
    'prev_next_buttons_location': 'bottom',
    'style_external_links': True,
}

latex_engine = 'xelatex'
```

4. To generate the notes in pdf format, use `make latex`, which generates lots of `tex` files in `./build/latex`. Switch to `build/latex` and run `make`. Assume that you have installed the software to compile `tex` files. It will generate `notes.pdf`.

2.2 How to include code from a file

See <https://www.sphinx-doc.org/en/master/usage/restructuredtext/directives.html#directive-literalinclude>.

1. Show line number: `:linenos:`. By default, line number counts from 0. To add an offset, e.g., 10, to the line number, use `:lineno-start: 10`. Note: It still includes all the contents of the file.
2. To emphasize a line, specified lines, or specified line ranges, use: `:emphasize-lines: 10`, `:emphasize-lines: 10,12,14`, and `:emphasize-lines: 12,15-18` Note: `emphasize` means to change the background color.
3. Set the language, e.g., `:language: python`.
4. Set the caption, e.g., `:caption: hello world`.

5. To include a function from the python file, use `:pyobject: my_func`
6. To include specified lines, use `:lines:1,3,5-10,15-`. Note that if using this option, line number counts from 0. Use `:lineno-start: xx` to change the offset for display.

2.3 Link

See <https://sublime-and-sphinx-guide.readthedocs.io/en/latest/references.html> and <https://www.sphinx-doc.org/en/master/usage/restructuredtext/basics.html#hyperlinks>

2.3.1 hello

Here is a link to *hello*.

```
.. _Link to hello:

hello
-----

Here is a link to :ref:`Link to hello`.
```


This page describes commonly used git commands.

3.1 Branch

After renaming `main` to `master`, do the following locally:

```
git branch -m main master
git fetch origin
git branch -u origin/master master
git remote set-head origin -a

git ls-remote --tags origin # view the tags of the remote origin
git push origin --delete tag master # delete the master tag from the remote origin
```

3.2 Commands

3.2.1 rev-parse

It is quite common to get the root directory of the repository with the command:

```
git rev-parse --show-toplevel
```

For instance, the above command executed in this repository prints something like as follows:

```
/xxx/notes
```

The following shows its usage in a Python script:

```
#!/usr/bin/env python3

import subprocess

d = (
    subprocess.check_output(["git", "rev-parse", "--show-toplevel"])
    .decode("ascii")
    .strip() # remove the trailing \n
)
print(d) # /path/to/notes
```

It can also be used in bash script:

```
root_dir=$(git rev-parse --show-toplevel)
echo "root_dir ${root_dir}"
```

`help git-rev-parse` outputs helpful information for `git rev-parse`. In particular, it explains the differences among `HEAD~`, `HEAD~n`, `HEAD^`, and `HEAD^n`. The following shows the help information about it:

```
<rev>^[<n>], e.g. HEAD^, v1.5.1^0
  A suffix ^ to a revision parameter means the first parent of that commit object.
→ ^<n> means the <n>th parent
  (i.e. <rev>^ is equivalent to <rev>^1). As a special rule, <rev>^0 means the
→ commit itself and is used when
  <rev> is the object name of a tag object that refers to a commit object.

<rev>~[<n>], e.g. HEAD~, master~3
  A suffix ~ to a revision parameter means the first parent of that commit object.
→ A suffix ~<n> to a revision
  parameter means the commit object that is the <n>th generation ancestor of the
→ named commit object, following
  only the first parents. I.e. <rev>~3 is equivalent to <rev>^^^ which is
→ equivalent to <rev>^1^1^1. See below
  for an illustration of the usage of this form.
```

```

G   H   I   J
 \ /     \ /
  D   E   F
   \ | / \
    \ | /  |
     \|/   |
      B    C
       \  /
```

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```

      \ /
      A

A =      = A^0
B = A^    = A^1    = A~1
C =      = A^2
D = A^^   = A^1^1   = A~2
E = B^2   = A^^2
F = B^3   = A^^3
G = A^^^  = A^1^1^1 = A~3
H = D^2   = B^^2    = A^^^2 = A~2^2
I = F^    = B^3^    = A^^3^
J = F^2   = B^3^2   = A^^3^2

```

3.3 tag

```

# list remote tags
git ls-remote --tags dan

# remove the xxx tag
git push --delete dan refs/tags/xxx

git push origin :refs/tags/xxx

```

3.4 auto complete

3.4.1 macos

```
brew install git bash-completion
```

3.4.2 linux

```

sudo apt-get install bash-completion
source /etc/bash_completion

```


4.1 Installation

4.1.1 macos

Refer to <https://docs.docker.com/desktop/mac/install/>.

4.1.2 Change directory to save images

```
docker info -f '{{ .DockerRootDir }}'
mkdir -p /tmp/new-docker-root
sudo vi /etc/docker/daemon.json

cat /etc/docker/daemon.json
{
  "data-root": "/tmp/new-docker-root"
}

sudo systemctl daemon-reload
sudo systemctl restart docker
docker info -f '{{ .DockerRootDir }}'
```

4.1.3 Run an image

```
sudo docker run -it pytorch/manylinux-builder:cuda10.2 /bin/bash
```

Map a host directory to docker container:

```
sudo docker run --rm -it --name my_container_name -v /host/path:/container/path_
↳pytorch/manylinux-builder:cuda10.2 /bin/bash
```

4.1.4 copy data between container and host

```
sudo docker cp container:src_path dst_path  
sudo docker cp src_path container:dst_path
```

Use `docker ps` to view the container ID.

4.1.5 pull an image

```
sudo docker pull nvcr.io/nvidia/tritonserver:24.11-py3
```

4.1.6 map ports

```
# Start the container  
-p 8080:80
```

Accessing 8080 on the host is mapped to the container's port 80.

5.1 TikZ

5.1.1 Basics

This page describes commonly used git commands.

6.1 Decoding

```
CompactLattice compact_lat;  
decoder.GetLattice(true, &compact_lat);  
  
CompactLattice compact_best_path;  
CompactLatticeShortestPath(compact_lat, &compact_best_path);  
  
Lattice best_path;  
ConvertLattice(compact_best_path, best_path);  
  
std::vector<int32_t> tokens;  
std::vector<int32_t> words;  
LatticeWeight weight;  
GetLinearSymbolSequence(best_path, &tokens, &words, &weight);
```

- `decoder/simple-decoder.{h,cc}`

6.2 mfa

```
mfa model download acoustic english_mfa  
mfa model download dictionary english_us_mfa
```

We can also use:

```
wget https://github.com/MontrealCorpusTools/mfa-models/releases/download/acoustic-  
↪english_mfa-v3.1.0/english_mfa.zip
```

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```
wget https://github.com/MontrealCorpusTools/mfa-models/releases/download/dictionary-  
↪english_us_mfa-v3.1.0/english_us_mfa.dict
```

See also https://mfa-models.readthedocs.io/en/latest/acoustic/English/English%20MFA%20acoustic%20model%20v3_1_0.html https://mfa-models.readthedocs.io/en/latest/dictionary/English/English%20%28US%29%20MFA%20dictionary%20v3_1_0.html

```
mfa align_one -t ./temp -j 1 --verbose ./wav2/LJ001-0001.wav ./wav2/LJ001-0001.txt  
↪english_us_mfa ./t/english_mfa ./ljs_aligned.txt
```

- `command_line/align_one.py: def align_one_cli()`
- `online/alignment.py: def align_utterance_online().`
- `kalpy/fstext/lexicon.py`

7.1 sort

Sort files in the folder `t`. The filename has the pattern `xxx.n.txt`, where `n` is some numerical value. Also, exclude `xxx.100.txt`.

```
find ./t -name "xxx*.txt" ! -name "xxx.100.txt" -print0 | sort -z -t. -k2 -n | xargs -  
↪r0  
  
# -z line delimiter is NUL, not new line  
# -t file separator  
# -k sort via a key  
# -n numeric sort
```

Convert 32-bit floating pcm to wave:

```
# sample rate: 22050  
# c: channel  
sox -t raw -r 22050 -b 32 -e floating-point -c 1 ./t.pcm ./t.wav
```

7.2 echo

Generate a binary file:

```
echo -n -e '\x30\x31\x32' > a.bin  
hexdump a.bin
```

7.3 ffmpeg

```
ffprobe xxx.opus
```

7.3.1 Convert format

- opus to wav

```
ffmpeg -i input.opus output.wav

ffmpeg -i input.opus -acodec pcm_s16le -ac 1 -ar 16000 output.wav

# compress mov
# a larger value for crf -> a smaller file
ffmpeg -i ./kokoro-1.0-tts-engine-apk-demo.mov -c:v libx264 -c:a copy -crf 25 \
->newmovie.mov
```

- Extract part of a file

```
# extract 30 seconds starting at offset 1 minute
ffmpeg -i input.opus -ss 60 -t 30 output.wav
# or use HH:MM:SS format
ffmpeg -i input.opus -ss 0:01:00 -t 0:00:30 output.wav

ffmpeg -ss 00:00:16 -to 00:00:30 -i ./abc.mov -c copy part.mp4
```

7.3.2 References

See <https://gist.github.com/whizkydee/804d7e290f46c73f55a84db8a8936d74>

7.4 wget

7.4.1 cookies

1. Install the chrome extension: editthiscookie
2. Copy the cookie of the selected website to clipboard
3. Save it to a text file, e.g., a.txt
4. Load the cookie with `wget --load-cookie a.txt`

7.5 Hello

See <http://mywiki.woledge.org/BashGuide>

7.6 pkg-config

```
# Show the pre-defined search paths
pkg-config --variable pc_path pkg-config

echo $PKG_CONFIG_PATH
```

7.7 sed

```
find . -name *.yaml -type f -exec sed -i.bak 's/1\.10\.17/1\.10\.18/g' {} \;
```

7.8 rsync

7.8.1 Copy a folder

```
rsync -av -P /path/to/src/folder /path/to/dst
```

It will create a folder inside /path/to/dst

7.9 htop

If there are toooo many CPUs, refer to <https://superuser.com/questions/806614/how-to-compress-or-hide-the-processors-at-top-of-htop-on-large-machines> to configure htop.

8.1 Installation

8.1.1 CUDA 10.1.243

```
./cuda_10.1.243_418.87.00_linux.run --silent --toolkit --installpath=/ceph-data4/  
↳fangjun/software/cuda-10.1.243 --no-opengl-libs --no-drm --no-man-page  
  
# Install cuDNN  
cd /ceph-data4/fangjun/software/cuda-10.1.243  
tar xvf /ceph-sh0/fangjun/cudnn/cudnn-10.1-linux-x64-v8.0.4.30.tgz --strip-  
↳components=1
```

```
#!/usr/bin/env bash  
  
export CUDA_HOME=/ceph-data4/fangjun/software/cuda-10.1.243  
export PATH=$CUDA_HOME/bin:$PATH  
export LD_LIBRARY_PATH=$CUDA_HOME/lib64:$LD_LIBRARY_PATH  
  
# See /ceph-fj/fangjun/py38/lib/python3.8/site-packages/torch/share/cmake/Caffe2/  
↳Modules_CUDA_fix/upstream/FindCUDA.cmake  
export CUDA_TOOLKIT_ROOT_DIR=$CUDA_HOME  
export CUDA_TOOLKIT_ROOT=$CUDA_HOME  
export CUDA_BIN_PATH=$CUDA_HOME  
export CUDA_PATH=$CUDA_HOME  
export CUDA_INC_PATH=$CUDA_HOME/targets/x86_64-linux
```

8.1.2 CUDA 10.2.89

```
./cuda_10.2.89_440.33.01_linux.run --silent --toolkit --installpath=/ceph-data4/  
↪fangjun/software/cuda-10.2.89 --no-opengl-libs --no-drm --no-man-page  
# Install cuDNN  
wget https://huggingface.co/csukuangfj/cudnn/resolve/main/cudnn-10.2-linux-x64-v8.0.2.  
↪39.tgz  
tar xvf cudnn-10.2-linux-x64-v8.0.2.39.tgz --strip-components=1 -C /ceph-data4/  
↪fangjun/software/cuda-10.2.89
```

```
#!/usr/bin/env bash  
  
export CUDA_HOME=/ceph-sh1/fangjun/software/cuda-10.2.89  
export PATH=$CUDA_HOME/bin:$PATH  
export LD_LIBRARY_PATH=$CUDA_HOME/lib64:$LD_LIBRARY_PATH  
export LD_LIBRARY_PATH=$CUDA_HOME/lib:$LD_LIBRARY_PATH  
  
# See /ceph-fj/fangjun/py38/lib/python3.8/site-packages/torch/share/cmake/Caffe2/  
↪Modules_CUDA_fix/upstream/FindCUDA.cmake  
export CUDA_TOOLKIT_ROOT_DIR=$CUDA_HOME  
export CUDA_TOOLKIT_ROOT=$CUDA_HOME  
export CUDA_BIN_PATH=$CUDA_HOME  
export CUDA_PATH=$CUDA_HOME  
export CUDA_INC_PATH=$CUDA_HOME/targets/x86_64-linux  
export CFLAGS=-I$CUDA_HOME/targets/x86_64-linux/include
```

8.1.3 CUDA 11.0.3

```
./cuda_11.0.3_450.51.06_linux.run --silent --toolkit --installpath=/ceph-data4/  
↪fangjun/software/cuda-11.0.3 --no-opengl-libs --no-drm --no-man-page  
  
# Install cuDNN  
cd /ceph-data4/fangjun/software/cuda-11.0.3  
tar xvf /ceph-sh0/fangjun/cudnn/cudnn-11.0-linux-x64-v8.0.4.30.tgz --strip-  
↪components=1
```

8.1.4 CUDA 11.3.1

```
./cuda_11.3.1_465.19.01_linux.run --silent --toolkit --installpath=/ceph-data4/  
↪fangjun/software/cuda-11.3.1 --no-opengl-libs --no-drm --no-man-page  
cd /ceph-data4/fangjun/software/cuda-11.3.1  
tar xvf /ceph-sh0/fangjun/cudnn/cudnn-11.3-linux-x64-v8.2.1.32.tgz --strip-  
↪components=1
```

8.1.5 CUDA 11.5.2

```
./cuda_11.5.2_495.29.05_linux.run --silent --toolkit --installpath=/ceph-data4/  
↪fangjun/software/cuda-11.5.2 --no-opengl-libs --no-drm --no-man-page  
cd /ceph-data4/fangjun/software/cuda-11.5.2  
tar xvf /ceph-sh0/fangjun/cudnn/cudnn-linux-x86_64-8.3.2.44_cuda11.5-archive.tar.xz --  
↪strip-components=1
```

8.1.6 CUDA 11.6.1

```
./cuda_11.6.1_510.47.03_linux.run --silent --toolkit --installpath=/ceph-data4/  
↪fangjun/software/cuda-11.6.1 --no-opengl-libs --no-drm --no-man-page  
cd /ceph-data4/fangjun/software/cuda-11.6.1  
tar xvf /ceph-sh0/fangjun/cudnn/cudnn-11.3-linux-x64-v8.2.1.32.tgz --strip-  
↪components=1
```

8.1.7 CUDA 11.7.1

```
wget https://developer.download.nvidia.com/compute/cuda/11.7.1/local_installers/cuda_  
↪11.7.1_515.65.01_linux.run  
chmod +x cuda_11.7.1_515.65.01_linux.run  
./cuda_11.7.1_515.65.01_linux.run --silent --toolkit --installpath=/ceph-data4/  
↪fangjun/software/cuda-11.7.1 --no-opengl-libs --no-drm --no-man-page  
cd /ceph-data4/fangjun/software/cuda-11.7.1
```

8.2 blogs

- Best books on GPGPU programming with CUDA?

8.3 Projects using triton

- <https://github.com/unslothai/unsloth/tree/main/unsloth/kernels>

It has many useful kernels. GitHub stars: 41k

8.4 References

- CUDA C++ Programming Guide

8.5 0-Basics

```
KernelA<<<nBlk, nTid>>>(args);
```

A grid of blocks

- gridDim, number of blocks in each dim
- blockIdx, the index of the block in the grid
- blockDim, number of threads in a block

A block of threads

- threadIdx, the index of the thread in the block

```
// 1-d
int global_thread_id = blockDim.x * blockIdx.x + threadIdx.x;

// 2-d
int global_x = blockDim.x * blockIdx.x + threadIdx.x;
int global_y = blockDim.y * blockIdx.y + threadIdx.y;

// 3-d
int global_x = blockDim.x * blockIdx.x + threadIdx.x;
int global_y = blockDim.y * blockIdx.y + threadIdx.y;
int global_z = blockDim.z * blockIdx.z + threadIdx.z;
```

Note: There is no `gridIdx`.

- `__host__`: default, calls from CPU, runs on CPU
- `__device__`: calls from GPU, runs on GPU
- `__global__`: calls from CPU, runs on GPU

```
dim3 dimGrid(32, 1, 1);  
dim3 dimBlock(128, 1, 1);  
vecAddKernel<<<dimGrid, dimBlock>>>(xxxx);
```


9.1 torch.load and torch.save

Listing 1: ./code/load-and-save.py

```
1  #!/usr/bin/env python3
2
3  import torch
4  import tempfile
5
6
7  def main():
8      a = torch.arange(3)
9      with tempfile.NamedTemporaryFile() as f:
10         torch.save(a, f)
11         f.seek(0)
12         b = torch.load(f)
13         assert torch.all(torch.eq(a, b)), (a, b)
14
15
16 if __name__ == "__main__":
17     main()
```

9.2 torch.gather

Listing 2: ./code/gather.py

```
1  #!/usr/bin/env python3
2
3  import torch
4
```

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```

5
6 def main():
7     left_context = 0
8     N = 1
9     T = 1
10    H = 5 # time1
11    W = 2 * H - 1 + left_context # 2time1 - 1 + left_context
12    a = torch.randn(N, T, H, W)
13    a = torch.arange(N * T * H * W).reshape(N, T, H, W).contiguous()
14
15    if True:
16        rows = torch.arange(start=H - 1, end=-1, step=-1).unsqueeze(-1)
17        cols = torch.arange(H + left_context)
18        indexes = rows + cols
19
20        indexes = torch.tile(indexes, (N * T, 1))
21    else:
22        rows = torch.arange(start=H - 1, end=-1, step=-1)
23        cols = torch.arange(H + left_context)
24        rows = torch.cat([rows] * (N * T)).unsqueeze(-1)
25        indexes = rows + cols
26
27    print(indexes.shape)
28
29    ta = a.reshape(-1, W)
30
31    b = torch.gather(ta, dim=1, index=indexes)
32    b = b.reshape(N, T, H, -1)
33
34    c = a.as_strided(
35        (N, T, H, H + left_context),
36        (T * H * W, H * W, W - 1, 1),
37        storage_offset=H - 1,
38    )
39    assert torch.equal(b, c), (b, c)
40
41
42 if __name__ == "__main__":
43     torch.manual_seed(20220727)
44     main()

```


9.3 DDP

9.3.1 Initialization

9.4 TorchScript

9.4.1 doxygen doc

See

9.4.2 Hello

See https://pytorch.org/tutorials/beginner/Intro_to_TorchScript_tutorial.html.

torch.jit.script as a decorator

Listing 3: ./code/1-ex.py

```

1  @torch.jit.script
2  def adder(x: int):
3      return x + 1
4
5
6  def test_adder():
7      assert isinstance(adder, torch.jit.ScriptFunction)
8      print(adder.graph)
9      print("-" * 10)
10     print(adder.code)
11     adder.save("adder.pt")
12
13     my_adder = torch.jit.load("adder.pt")
14
15     assert isinstance(my_adder, torch.jit._script.RecursiveScriptModule)
16     assert isinstance(my_adder, torch.jit.ScriptModule)
17     assert not isinstance(my_adder, torch.jit.ScriptFunction)
18     print(my_adder(torch.tensor([3])))
19
20
21     """
22     graph(%x.1 : int):
23         %2 : int = prim::Constant[value=1]() # ./1-ex.py:8:15

```

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```

24     %3 : int = aten::add(%x.1, %2) # ./1-ex.py:8:11
25     return (%3)
26
27     -----
28     def adder(x: int) -> int:
29         return torch.add(x, 1)
30
31     4
32     """

```

`torch.Value` has the following attributes (output of `dir()`):

```

['_class__', '__delattr__', '__dir__', '__doc__', '__eq__', '__format__',
'__ge__', '__getattr__', '__gt__', '__hash__', '__init__',
'__init_subclass__', '__le__', '__lt__', '__module__', '__ne__', '__new__',
'__reduce__', '__reduce_ex__', '__repr__', '__setattr__', '__sizeof__',
'__str__', '__subclasshook__', 'copyMetadata', 'debugName', 'inferTypeFrom',
'isCompleteTensor', 'node', 'offset', 'replaceAllUsesAfterNodeWith',
'replaceAllUsesWith', 'requiresGrad', 'requires_grad', 'setDebugName',
'setType', 'setTypeAs', 'toIValue', 'type', 'unique', 'uses']

```

Listing 4: `./code/1-ex.py`

```

1  def print_graph():
2      assert isinstance(adder.graph, torch._C.Graph)
3      assert isinstance(adder.graph, torch.Graph)
4
5      # It should have only 1 input
6      assert len(list(adder.graph.inputs())) == 1
7
8      x = next(adder.graph.inputs())
9      assert isinstance(x, torch.Value)
10     assert isinstance(x.debugName(), str)
11     assert x.debugName() == "x.1"
12     print(type(x.uses()[0]))
13     print(dir(x.uses()[0]))
14     print(x.uses()[0].user)
15     assert isinstance(x.uses()[0].user, torch.Node)
16
17     x.setDebugName("x.2")
18     assert next(adder.graph.inputs()).debugName() == "x.2"
19     assert isinstance(x.type(), torch.IntType)
20

```

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```

21 print(x.node())
22 assert isinstance(x.node(), torch.Node)
23 print(dir(x.node()))
24 n = x.node()
25 assert isinstance(n.kind(), str)
26 assert n.kind() == "prim::Param", n.kind()
27 print(n.kind())
28 # a node as input and output
29 assert list(n.inputs()) == []
30
31 # n has only one output, i.e., x
32 assert len(list(n.outputs())) == 1
33 x2 = next(n.outputs()) # its type is torch.Value
34 assert x2 is x
35 assert len(list(n.blocks())) == 0

```

`torch.Node` has the following attributes (output from `dir()`):

```

['__class__', '__delattr__', '__dir__', '__doc__', '__eq__', '__format__',
'__ge__', '__getattr__', '__getitem__', '__gt__', '__hash__', '__init__',
'__init_subclass__', '__le__', '__lt__', '__module__', '__ne__', '__new__',
'__reduce__', '__reduce_ex__', '__repr__', '__setattr__', '__sizeof__',
'__str__', '__subclasshook__', 'addBlock', 'addInput', 'addOutput',
'attributeNames', 'blocks', 'c', 'c_', 'conv', 'copyAttributes', 'copyMetadata',
'destroy', 'eraseOutput', 'f', 'f_', 'findAllNodes', 'findNode', 'fs',
'fs_', 'g', 'g_', 'getModuleHierarchy', 'gs', 'gs_', 'hasAttribute',
'hasAttributes', 'hasMultipleOutputs', 'hasUses', 'i', 'i_', 'input',
'inputs', 'inputsAt', 'inputsSize', 'insertAfter', 'insertBefore', 'is',
'isAfter', 'isBefore', 'isNondeterministic', 'is_', 'ival', 'ival_', 'kind',
'kindOf', 'matches', 'moveAfter', 'moveBefore', 'mustBeNone', 'namedInput',
'output', 'outputs', 'outputsAt', 'outputsSize', 'owningBlock', 'prev',
'pyname', 'pyobj', 'removeAllInputs', 'removeAttribute', 'removeInput',
'replaceAllUsesWith', 'replaceInput', 'replaceInputWith', 's', 's_', 'scalar_args',
'schema', 'scopeName', 'sourceRange', 'ss', 'ss_', 't', 't_', 'ts', 'ts_',
'ty_', 'tys_', 'z', 'z_', 'zs', 'zs_']

```

torch.jit.script as a function

Listing 5: ./code/2-ex.py

```
1 def adder(x: int):
2     return x + 2
3
4
5 def test_adder():
6     adder_func = torch.jit.script(adder)
7     assert isinstance(adder_func, torch.jit.ScriptFunction)
8     print(adder_func.graph)
9     print(adder_func(3))
10
11
12 """
13 graph(%x.1 : int):
14     %2 : int = prim::Constant[value=2]() # ./2-ex.py:6:15
15     %3 : int = aten::add(%x.1, %2) # ./2-ex.py:6:11
16     return (%3)
17
18 5
19 """
```

torchscript a module

Listing 6: ./code/3-ex.py

```
1 class MyModel(torch.nn.Module):
2     def __init__(self):
3         super().__init__()
4         self.p = torch.nn.Parameter(torch.tensor([2.0]))
5
6     def forward(self, x: torch.Tensor):
7         return self.p * x
8
9
10 def test_my_model():
11     model = MyModel()
12     scripted_model = torch.jit.script(model)
13     print(scripted_model.graph)
14     print("-" * 10)
15     print(scripted_model.code)
```

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```

16     print(scripted_model(torch.tensor([10])))
17
18
19 """
20 graph(%self : __torch__.MyModel,
21       %x.1 : Tensor):
22     %p : Tensor = prim::GetAttr[name="p"](%self)
23     %4 : Tensor = aten::mul(%p, %x.1) # ./3-ex.py:12:15
24     return (%4)
25
26 -----
27 def forward(self,
28             x: Tensor) -> Tensor:
29     p = self.p
30     return torch.mul(p, x)
31 """

```

trace a module

Listing 7: ./code/trace/ex0.py

```

1  #!/usr/bin/env python3
2
3  import torch
4
5  import torch.nn as nn
6  from typing import List
7
8
9  class Foo(nn.Module):
10     def __init__(self):
11         super().__init__()
12         self.relu = nn.ReLU()
13
14     def forward(self, x):
15         return self.relu(x)
16
17
18 def test_foo():
19     f = Foo()
20     m = torch.jit.trace(f, torch.rand(2, 3))
21

```

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```

22     print(m(torch.rand(2)))
23     print(m(torch.rand(2, 3, 4)))
24     # Note: The input shape is dynamic, not fixed.
25
26
27 def simple(x: List[torch.Tensor], y: torch.Tensor):
28     x = x[0].item()
29     if x > 2:
30         return y + x + 1
31     elif x < 1:
32         return y
33     else:
34         return y + x
35
36
37 def test_simple():
38     f0 = torch.jit.trace(simple, ([torch.tensor([0])], torch.rand(2, 3)))
39     # print(dir(f0))
40     """
41     ['__call__', '__class__', '__delattr__', '__dict__', '__dir__', '__doc__',
42     '__eq__', '__format__', '__ge__', '__getattribute__', '__gt__', '__hash__',
43     '__init__', '__init_subclass__', '__le__', '__lt__', '__module__', '__ne__',
44     '__new__', '__reduce__', '__reduce_ex__', '__repr__', '__setattr__',
45     '__sizeof__', '__str__', '__subclasshook__', '_debug_flush_compilation_cache',
46     'code', 'get_debug_state', 'graph', 'graph_for', 'inlined_graph', 'name',
47     'qualified_name', 'save', 'save_to_buffer', 'schema']
48     """
49     # print(f0.schema) # simple(Tensor[] x, Tensor y) -> (Tensor)
50     # print(f0.code)
51     """
52     def simple(x: List[Tensor],
53               y: Tensor) -> Tensor:
54         return y
55     """
56     # print(f0.graph)
57     """
58     graph(%x : Tensor[],
59           %y : Float(2, 3, strides=[3, 1], requires_grad=0, device=cpu)):
60         return (%y)
61     """
62     # print(f0.inlined_graph) # same as the above one
63     # print(f0.name) # simple
64     print(f0.qualified_name) # __torch__.simple

```

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```

65
66
67 def main():
68     # test_foo()
69     test_simple()
70
71
72 if __name__ == "__main__":
73     main()

```

Export and ignore methods

1. Use `@torch.jit.export` decorator to export a method.
2. Use `torch.jit.export` function call to export a method.
3. Use `@torch.jit.ignore` decorator to ignore a method.
4. Use `torch.jit.ignore` function call to ignore a method.
5. Use `@torch.jit.unused` or `torch.jit.unused` to ignore a method.

See *Load in C++* to load the saved file.

Listing 8: `./code/4-ex.py`

```

1 class MyModel(torch.nn.Module):
2     def __init__(self):
3         super().__init__()
4         self.p = torch.nn.Parameter(torch.tensor([2.0]))
5
6     def foobar(self, x: torch.Tensor):
7         return x + 3
8
9     def foo(self, x: torch.Tensor):
10        return self.foobar(x)
11
12    def bar(self, x: torch.Tensor):
13        return self.p - x
14
15    @torch.jit.export
16    def baz(self, x: torch.Tensor):
17        return self.p + x + 2
18
19    def forward(self, x: torch.Tensor):

```

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```

20         return self.p * x
21
22
23 def test_my_model():
24     MyModel.foo = torch.jit.export(MyModel.foo)  # manually export
25
26     # Note: forward is exported by default. We ignore it here manually
27     MyModel.forward = torch.jit.ignore(MyModel.forward)
28
29     model = MyModel()
30     scripted_model = torch.jit.script(model)
31     assert hasattr(scripted_model, "foo")
32     assert hasattr(scripted_model, "baz")
33     assert hasattr(scripted_model, "foobar")  # because it is called by `foo`
34     assert not hasattr(scripted_model, "bar")
35
36     scripted_model.save("foo.pt")
37
38     m = torch.jit.load("foo.pt")
39     print(m.foo(torch.tensor([1])))
40     print(m.baz(torch.tensor([1])))
41
42
43 """
44 graph(%self : __torch__.MyModel,
45       %x.1 : Tensor):
46     %p : Tensor = prim::GetAttr[name="p"](%self)
47     %4 : Tensor = aten::mul(%p, %x.1) # ./3-ex.py:12:15
48     return (%4)
49
50 -----
51 def forward(self,
52             x: Tensor) -> Tensor:
53     p = self.p
54     return torch.mul(p, x)
55 """

```


9.4.3 Load in C++

See https://pytorch.org/tutorials/advanced/cpp_export.html.

Load the saved `foo.pt` in C++ from *Export and ignore methods*.

Listing 9: `./code/load-in-cpp/Makefile`

```

1  USE_CXX11_ABI := $(shell python3 -c 'import torch; print(int(torch.compiled_with_
   ↪ cxx11_abi()))')
2  TORCH_INSTALL_DIR := $(shell python3 -c 'import os; import torch; print(os.path.
   ↪ dirname(torch.__file__))')
3
4  $(info USE_CXX11_ABI $(USE_CXX11_ABI))
5  $(info TORCH_INSTALL_DIR $(TORCH_INSTALL_DIR))
6
7  CXXFLAGS := -I$(TORCH_INSTALL_DIR)/include
8  CXXFLAGS += -I$(TORCH_INSTALL_DIR)/include/torch/csrc/api/include
9  CXXFLAGS += -I$(TORCH_INSTALL_DIR)/include/TH
10 CXXFLAGS += -I$(TORCH_INSTALL_DIR)/include/THC
11 CXXFLAGS += -std=c++14
12 CXXFLAGS += -D_GLIBCXX_USE_CXX11_ABI=$(USE_CXX11_ABI)
13
14 CXXFLAGS += -Wno-unknown-pragmas # disable omp warnings
15
16 LDFLAGS := -L$(TORCH_INSTALL_DIR)/lib
17 LDFLAGS += -lc10 -ltorch -ltorch_cpu
18 # LDFLAGS += -lc10 -ltorch
19 LDFLAGS += -Wl,-rpath,$(TORCH_INSTALL_DIR)/lib
20
21 HAS_CUDA := $(shell python3 -c 'import torch; print("yes" if torch.cuda.is_
   ↪ available() else "no")')
22 HAS_CUDA := yes
23 $(info has cuda $(HAS_CUDA))
24
25 ifeq ($(HAS_CUDA),yes)
26 CUDA_HOME := $(shell which nvcc | xargs dirname | xargs dirname)
27 CXXFLAGS += -I$(CUDA_HOME)/include
28 LDFLAGS += -L$(CUDA_HOME)/lib64
29 LDFLAGS += -lcudart -lc10_cuda -ltorch_cuda
30 LDFLAGS += -Wl,-rpath,$(CUDA_HOME)/lib64
31 endif
32
33 .PHONY: clean
34

```

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```

35 main: main.o
36     $(CXX) -o $@ $< $(LDFLAGS)
37
38 main.o: main.cc
39     $(CXX) $(CXXFLAGS) -c -o $@ $<
40
41 clean:
42     $(RM) main.o main

```

Note: `torch::jit::script::Module` is deprecated, use `torch::jit::Module` instead.

Listing 10: `./code/load-in-cpp/main.cc`

```

1  #include "torch/script.h"
2
3  int main() {
4      // see torch/csrc/jit/module.h
5      torch::jit::Module m = torch::jit::load("../foo.pt");
6      std::cout << "is training: " << m.is_training() << "\n";
7      m.eval();
8      std::cout << "after m.eval(): is training: " << m.is_training() << "\n";
9      torch::Tensor x = torch::tensor({1, 2, 3}, torch::kFloat);
10     torch::Tensor y = m.run_method("baz", x).toTensor();
11     std::cout << y << "\n";
12
13     return 0;
14 }

```

The output of make is:

```

USE_CXX11_ABI 0
TORCH_INSTALL_DIR /ceph-fj/fangjun/software/py38/lib/python3.8/site-packages/torch
has cuda yes
g++ -I/ceph-fj/fangjun/software/py38/lib/python3.8/site-packages/torch/include \
    -I/ceph-fj/fangjun/software/py38/lib/python3.8/site-packages/torch/include/torch/
↪csrc/api/include \
    -I/ceph-fj/fangjun/software/py38/lib/python3.8/site-packages/torch/include/TH \
    -I/ceph-fj/fangjun/software/py38/lib/python3.8/site-packages/torch/include/THC \
    -std=c++14 \
    -D_GLIBCXX_USE_CXX11_ABI=0 \
    -Wno-unknown-pragmas \
    -I/ceph-sh1/fangjun/software/cuda-10.2.89/include \

```

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```

-c -o main.o main.cc
g++ -o main main.o \
-L/ceph-fj/fangjun/software/py38/lib/python3.8/site-packages/torch/lib \
-lc10 -ltorch -ltorch_cpu \
-Wl,-rpath,/ceph-fj/fangjun/software/py38/lib/python3.8/site-packages/torch/lib \
-L/ceph-sh1/fangjun/software/cuda-10.2.89/lib64 \
-lcudart -lc10_cuda -ltorch_cuda \
-Wl,-rpath,/ceph-sh1/fangjun/software/cuda-10.2.89/lib64

```

The output of `./main` is:

```

is training: 1
after m.eval(): is training: 0
5
6
7
[ CPUFloatType{3} ]

```

9.4.4 ArrayRef

See `c10/Utils/ArrayRef.h`.

Caution: `IntArrayRef` is an alias to `ArrayRef<int64_t>`.

`ArrayRef<T>` contains only two members: A const data pointer and a size. It is trivially copyable and assignable.

It has similar methods like `std::vector`. It also has two methods to get the front and back: `front()` and `back()`; both return a const reference.

Its method `vec()` converts itself to a `std::vector` by **copying** the underlying data.

Constructors

Data members

Listing 11: `./code/array_ref/main.cc` (Check size)

```

1 struct Foo {
2     const int32_t *p;
3     size_t len;
4 };

```

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```
5
6 static void TestSize() {
7     // Note: The data pointer in ArrayRef is const!
8     static_assert(sizeof(torch::ArrayRef<int32_t>) == sizeof(Foo), "");
9 }
```

Default constructed

Listing 12: ./code/array_ref/main.cc (Default constructor)

```
1 static void TestDefaultConstructor() {
2     torch::ArrayRef<int32_t> a;
3     TORCH_CHECK(a.data() == nullptr);
4     TORCH_CHECK(a.size() == 0);
5     TORCH_CHECK(a.empty() == true);
6
7     TORCH_CHECK(a.begin() == nullptr);
8     TORCH_CHECK(a.end() == nullptr);
9 }
```

From a single element

Listing 13: ./code/array_ref/main.cc (From a single element)

```
1 static void TestFromSingleElement() {
2     int32_t a = 10;
3     torch::ArrayRef<int32_t> b(a);
4     TORCH_CHECK(b[0] == a);
5     TORCH_CHECK(b.data() == &a);
6     TORCH_CHECK(b.size() == 1);
7 }
```

From an initializer list

Listing 14: ./code/array_ref/main.cc (From an initializer list)

```
1 static void TestFromInitializerList() {
2     torch::ArrayRef<int32_t> a = {1, 2, 3};
3     TORCH_CHECK(a.size() == 3);
```

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```

4  TORCH_CHECK(a[0] == 1);
5  TORCH_CHECK(a[1] == 2);
6  TORCH_CHECK(a[2] == 3);
7  }

```

Other types of constructors

- From two pointers: begin and end
- From a pointer and a length
- From a `std::vector`
- From a container that has `data()` and `size()` methods
- From a C array
- From a `std::array`

9.4.5 ScalarType

See `c10/core/ScalarType.h`. and <https://github.com/pytorch/pytorch/blob/master/torch/csrc/api/include/torch/types.h>.

`ScalarType` is an enum class, i.e., `enum class ScalarType : int8_t { ... }`.

Members

It has the following members:

Listing 15: `./code/scalar-type/members.cc`

```

1  #define AT_FORALL_SCALAR_TYPES_WITH_COMPLEX_EXCEPT_COMPLEX_HALF(_) \
2      _(uint8_t, Byte) \
3      _(int8_t, Char) \
4      _(int16_t, Short) \
5      _(int, Int) \
6      _(int64_t, Long) \
7      _(at::Half, Half) \
8      _(float, Float) \
9      _(double, Double) \
10     _(c10::complex<float>, ComplexFloat) \
11     _(c10::complex<double>, ComplexDouble) \
12     _(bool, Bool) \
13     _(at::BFloat16, BFloat16)

```

Some aliases

Listing 16: ./code/scalar-type/main.cc (alias)

```
1 static void TestAlias() {
2     static_assert(c10::ScalarType::Int == c10::kInt, "");
3     static_assert(c10::ScalarType::Byte == c10::kByte, "");
4 }
```

Listing 17: ./code/scalar-type/alias.cc

```
1 // See torch/csrc/api/include/torch/types.h
2 using Dtype = at::ScalarType;
3
4 /// Fixed width dtypes.
5 constexpr auto kUInt8 = at::kByte;
6 constexpr auto kInt8 = at::kChar;
7 constexpr auto kInt16 = at::kShort;
8 constexpr auto kInt32 = at::kInt;
9 constexpr auto kInt64 = at::kLong;
10 constexpr auto kFloat16 = at::kHalf;
11 constexpr auto kFloat32 = at::kFloat;
12 constexpr auto kFloat64 = at::kDouble;
13
14 /// Rust-style short dtypes.
15 constexpr auto kU8 = kUInt8;
16 constexpr auto kI8 = kInt8;
17 constexpr auto kI16 = kInt16;
18 constexpr auto kI32 = kInt32;
19 constexpr auto kI64 = kInt64;
20 constexpr auto kF16 = kFloat16;
21 constexpr auto kF32 = kFloat32;
22 constexpr auto kF64 = kFloat64;
```

ScalarType to CPP type

Listing 18: ./code/scalar-type/main.cc

```
1 static void TestScalarTypeToCppType() {
2     static_assert(
3         std::is_same<
4             int32_t, //
5             c10::impl::ScalarTypeToCppType<c10::ScalarType::Int>::type::value,
```

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```

6     "");
7 }

```

CPP type to ScalarType

Listing 19: ./code/scalar-type/main.cc

```

1 static void TestCppTypeToScalarType() {
2     static_assert(
3         c10::CppTypeToScalarType<float>::value == c10::ScalarType::Float, "");
4 }

```

Note: It is `c10::impl::ScalarTypeToCPPTType`, but it is `c10::CppTypeToScalarType`.

9.4.6 TypeMeta

See

- <https://github.com/pytorch/pytorch/blob/master/c10/util/typeid.h>
- <https://github.com/pytorch/pytorch/blob/master/c10/core/ScalarTypeToTypeMeta.h>

`struct TypeMeta` contains only a single `int16_t` data member:

Listing 20: ./code/type-meta/main.cc (Check size)

```

1 static void TestSize() {
2     static_assert(sizeof(caffe2::TypeMeta) == sizeof(int16_t), "");
3 }

```

Constructors

Listing 21: ./code/type-meta/main.cc (Make)

```

1 static void TestConstructor() {
2     caffe2::TypeMeta t = caffe2::TypeMeta::Make<int32_t>();
3     TORCH_CHECK(t.Match<int32_t>());
4
5     TORCH_CHECK(t.isScalarType());
6 }

```

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```
7 TORCH_CHECK(t.isScalarType(torch::kInt));
8 TORCH_CHECK(t.isScalarType(torch::kFloat) == false);
9
10 TORCH_CHECK(t.name() == "int");
11 }
```

Operations with ScalarType

Listing 22: ./code/type-meta/main.cc (Operations with ScalarType)

```
1 static void TestFromScalarType() {
2     caffe2::TypeMeta t = caffe2::TypeMeta::fromScalarType(torch::kDouble);
3
4     TORCH_CHECK(t.isScalarType(torch::kDouble));
5     TORCH_CHECK(t.name() == "double");
6
7     TORCH_CHECK(t.toScalarType() == torch::kDouble);
8     TORCH_CHECK(t == torch::kDouble);
9     TORCH_CHECK(t != torch::kFloat);
10    TORCH_CHECK(torch::kInt != t);
11 }
```

9.4.7 torch::Device

See

- <https://github.com/pytorch/pytorch/blob/master/c10/core/DeviceType.h>
- <https://github.com/pytorch/pytorch/blob/master/c10/core/Device.h>

DeviceType

`torch::DeviceType` is defined as `enum class Device: int8_t {...}`. The most commonly used types are `torch::DeviceType::CPU` and `torch::DeviceType::CUDA`, which are aliased to `torch::kCPU` and `torch::kCUDA`.

Listing 23: ./code/device/main.cc

```
1 void TestDeviceType() {
2     torch::DeviceType d = torch::kCPU;
3     std::ostream os;
4     os << d;
5     TORCH_CHECK(os.str() == "cpu");
6 }
```

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```

6
7 TORCH_CHECK(DeviceTypeName(d /*,lower_case=false*/ ) == "CPU");
8 TORCH_CHECK(DeviceTypeName(d, /*lower_case*/ true) == "cpu");

```

Device

A `torch::Device` class has two members: a `torch::DeviceType` and an `int8_t` index.

Listing 24: `./code/device/main.cc` (Constructors)

```

1 void TestDeviceConstructorCPU() {
2     torch::Device d(torch::kCPU);
3     TORCH_CHECK(d.is_cpu() == true);
4     TORCH_CHECK(d.is_cuda() == false);
5     TORCH_CHECK(d.type() == torch::kCPU);
6     TORCH_CHECK(d.has_index() == false);
7     TORCH_CHECK(d.index() == -1);
8     TORCH_CHECK(d.str() == "cpu");
9 }
10
11 void TestDeviceConstructorCUDA() {
12     torch::Device d(torch::kCUDA, 3);
13     TORCH_CHECK(d.is_cpu() == false);
14     TORCH_CHECK(d.is_cuda() == true);
15     TORCH_CHECK(d.type() == torch::kCUDA);
16     TORCH_CHECK(d.has_index() == true);
17     TORCH_CHECK(d.index() == 3);
18     TORCH_CHECK(d.str() == "cuda:3");
19
20     d.set_index(2);
21     TORCH_CHECK(d.index() == 2);
22     TORCH_CHECK(d.str() == "cuda:2");
23
24     d = torch::Device("cpu");
25     TORCH_CHECK(d.is_cpu() == true);
26
27     d = torch::Device("CPU");
28     TORCH_CHECK(d.is_cpu() == true);
29
30     d = torch::Device("cuda:1");
31     TORCH_CHECK(d.is_cuda() == true);
32     TORCH_CHECK(d.index() == 1);
33

```

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```

34 d = torch::Device("CUDA:1");
35 TORCH_CHECK(d.is_cuda() == true);
36 TORCH_CHECK(d.index() == 1);
37 }

```

9.4.8 TensorOptions

See <https://github.com/pytorch/pytorch/blob/master/c10/core/TensorOptions.h>

Constructors (not recommended)

Listing 25: ./code/tensor-options/main.cc (Not recommended constructors)

```

1 void TestConstructor() {
2     // not recommended
3     torch::TensorOptions opt1(torch::kCPU);
4     torch::TensorOptions opt2(torch::Device(torch::kCPU));
5     torch::TensorOptions opt3(torch::Device({torch::kCUDA, 1}));
6     torch::TensorOptions opt4("cpu");
7     // torch::TensorOptions opt5("CPU") // error;
8     torch::TensorOptions opt6("cuda:1");
9     // torch::TensorOptions opt7("CUDA:1"); // error
10
11     // not recommended, from a scalar type (implicit)
12     torch::TensorOptions opt8(torch::kInt32);
13 }

```

Constructors (Recommended)

Listing 26: ./code/tensor-options/main.cc (Recommended constructors)

```

1 void TestConstructor2() {
2     // recommended
3     torch::TensorOptions opt1 = torch::dtype(torch::kFloat);
4     torch::TensorOptions opt2 = torch::dtype(caffe2::TypeMeta::Make<float>());
5     torch::TensorOptions opt3 = torch::device(torch::kCPU);
6     torch::TensorOptions opt4 = torch::device({torch::kCUDA, 1});
7     // Note: torch::device() returns a TensorOptions
8     // while torch::Device() is the constructor of a class

```

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```
9
10 torch::TensorOptions opt5 = torch::requires_grad(true);
11 std::cout << opt5 << "\n";
12 // TensorOptions(dtype=float (default), device=cpu (default), layout=Strided
13 // (default), requires_grad=true, pinned_memory=false (default),
14 // memory_format=(nullopt))
15
16 torch::TensorOptions opt6 = torch::dtype<float>();
17 std::cout << torch::toString(opt6) << "\n";
18 // TensorOptions(dtype=float, device=cpu (default), layout=Strided (default),
19 // requires_grad=false (default), pinned_memory=false (default),
20 // memory_format=(nullopt))
21
22 std::cout << "default:" << torch::TensorOptions() << "\n";
23 // default:TensorOptions(dtype=float (default), device=cpu (default),
24 // layout=Strided (default), requires_grad=false (default),
25 // pinned_memory=false (default), memory_format=(nullopt))
26 }
```

Methods

Listing 27: ./code/tensor-options/main.cc (Methods)

```
1 void TestMethods() {
2     torch::TensorOptions opts = torch::dtype<float>();
3     TORCH_CHECK(opts.device() == torch::Device(torch::kCPU));
4     // It has not device_type()!
5     TORCH_CHECK(opts.device() == torch::kCPU);
6     TORCH_CHECK(opts.device().type() == torch::kCPU);
7     TORCH_CHECK(opts.requires_grad() == false);
8
9     torch::TensorOptions opts2 =
10         opts.device("cuda:2").dtype(torch::kInt).requires_grad(false);
11
12     TORCH_CHECK(opts2.dtype() == caffe2::TypeMeta::Make<int32_t>());
13     TORCH_CHECK(opts2.dtype() == torch::kInt32);
14     TORCH_CHECK(opts2.requires_grad() == false);
15 }
```

9.4.9 Tensor Creation

See

TensorDataContainer

Note: data is **copied** to the returned tensor!

See

- <https://github.com/pytorch/pytorch/blob/master/torch/csrc/api/include/torch/detail/TensorDataContainer.h>
- https://github.com/pytorch/pytorch/blob/master/tools/autograd/templates/variable_factories.h
- <https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/Utils.cpp>

Support the following data types:

- From a `std::vector<T>`
- From a scalar
- From an initializer list
- From an `ArrayRef<T>`.

From std::vector

Listing 28: ./code/tensor-creation/main.cc

```
1 static void FromStdVecotr() {
2     torch::Tensor t1 = torch::tensor(std::vector<int32_t>{1, 2, 3});
3     TORCH_CHECK(t1.scalar_type() == torch::kLong);
4     t1 = t1.to(torch::kInt);
5     const int32_t *p1 = t1.data_ptr<int32_t>();
6     TORCH_CHECK(p1[0] == 1);
7     TORCH_CHECK(p1[1] == 2);
8     TORCH_CHECK(p1[2] == 3);
9
10    torch::Tensor t2 = torch::tensor(std::vector<float>{1, 2, 3});
11    TORCH_CHECK(t2.scalar_type() == torch::kFloat);
12
13    torch::Tensor t3 =
14        torch::tensor(std::vector<double>{1, 2, 3}, torch::kDouble);
15    TORCH_CHECK(t3.scalar_type() == torch::kDouble);
16
17    torch::Tensor t4 =
18        torch::tensor(std::vector<double>{1, 2, 3},
19                      torch::dtype(torch::kDouble).device("cuda:0"));
20    TORCH_CHECK(t4.is_cuda());
21 }
```

From scalar

Listing 29: ./code/tensor-creation/main.cc

```
1 static void FromScalar() {
2     torch::Tensor t = torch::tensor(3);
3     TORCH_CHECK(t.item<int64_t>() == 3);
4
5     torch::Tensor t2 = torch::tensor(0.5);
6     TORCH_CHECK(t2.scalar_type() == torch::kFloat);
7 }
```

From initializer list

Listing 30: ./code/tensor-creation/main.cc

```
1 static void FromInitializerList() {
2     torch::Tensor t1 = torch::tensor({1, 2, 3});
3     torch::Tensor t2 = torch::tensor(std::vector<int32_t>{1, 2, 3});
4     TORCH_CHECK(torch::allclose(t1, t2));
5
6     torch::Tensor t3 = torch::tensor({{1, 2, 3}, {4, 5, 6}});
7     TORCH_CHECK(t3.dim() == 2);
8
9     torch::Tensor t4 = torch::tensor({1, 2, 3});
10    torch::Tensor t5 = torch::tensor({4, 5, 6});
11    TORCH_CHECK(torch::allclose(t3[0], t4));
12    TORCH_CHECK(torch::allclose(t3[1], t5));
13 }
```

From ArrayRef

Listing 31: ./code/tensor-creation/main.cc

```
1 static void FromArrayRef() {
2     int32_t i[] = {1, 2, 3};
3     torch::ArrayRef<int32_t> a(i);
4     torch::Tensor t = torch::tensor(a);
5     // Data is copied to t
6
7     TORCH_CHECK(t[0].item<int64_t>(), 1);
8     TORCH_CHECK(t[1].item<int64_t>(), 2);
9     TORCH_CHECK(t[2].item<int64_t>(), 3);
10 }
```

9.4.10 Tensor

See

- <https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/core/TensorBase.h>
- <https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/templates/TensorBody.h>
- <https://github.com/pytorch/pytorch/blob/master/c10/core/TensorImpl.h>

Common methods

Listing 32: ./code/tensor/main.cc (Not recommended constructors)

```

1  static void TestCommonMethods() {
2      torch::Tensor t = torch::rand({2, 3, 4});
3
4      TORCH_CHECK(t.dim() == 3);           // 3-d tensor
5      TORCH_CHECK(t.ndimension() == t.dim()); // same
6      TORCH_CHECK(t.numel() == 2 * 3 * 4);
7      TORCH_CHECK(t.is_contiguous() == true);
8      TORCH_CHECK(t.contiguous().is_contiguous() == true);
9
10     t.fill_(10); // fill all entries to 0
11     t.zero_();   // zero out all entries
12
13     t = t.to(torch::kInt);
14     TORCH_CHECK(t.is_floating_point() == false);
15     TORCH_CHECK(t.is_signed() == true);
16
17     TORCH_CHECK(t.size(0) == 2);
18     TORCH_CHECK(t.size(1) == 3);
19     TORCH_CHECK(t.size(2) == 4);
20     TORCH_CHECK(t.sizes() == torch::ArrayRef<int64_t>({2, 3, 4}));
21
22     t = t.contiguous();
23     TORCH_CHECK(t.stride(0) == 3 * 4);
24     TORCH_CHECK(t.stride(1) == 4);
25     TORCH_CHECK(t.stride(2) == 1);
26     TORCH_CHECK(t.strides() == torch::ArrayRef<int64_t>({12, 4, 1}));
27
28     TORCH_CHECK(t.defined() == true);
29     {
30         torch::Tensor a;
31         TORCH_CHECK(a.defined() == false);

```

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```

32     a = t;
33     TORCH_CHECK(a.defined() == true);
34     a.reset();
35     TORCH_CHECK(a.defined() == false);
36 }
37
38 t = t.to(torch::kShort);
39 TORCH_CHECK(t.itemsize() == sizeof(int16_t));
40 TORCH_CHECK(t.nbytes() == t.numel() * t.itemsize());
41 TORCH_CHECK(t.itemsize() == t.element_size()); // same
42
43 TORCH_CHECK(t.scalar_type() == torch::kShort);
44 TORCH_CHECK(t.dtype() == caffe2::TypeMeta::Make<int16_t>());
45 TORCH_CHECK(t.dtype().toScalarType() == torch::kShort);
46
47 TORCH_CHECK(t.device() == torch::Device("cpu"));
48 TORCH_CHECK(t.device() == torch::Device(torch::kCPU));
49
50 // Note: t.device() return an instance of torch::Device
51 // t.get_device() returns the device index.
52 TORCH_CHECK(t.get_device() == t.device().index());
53
54 TORCH_CHECK(t.is_cpu() == true);
55 TORCH_CHECK(t.is_cuda() == false);
56
57 t = t.to(torch::kInt);
58 int32_t *p = t.data_ptr<int32_t>();
59 p[0] = 100;
60
61 torch::TensorAccessor<int32_t, 3> acc = t.accessor<int32_t, 3>();
62 TORCH_CHECK(acc[0][0][0] == p[0]);
63 p[12] = -2;
64 TORCH_CHECK(acc[1][0][0] == -2);
65
66 acc[1][1][2] = 3;
67 TORCH_CHECK(*(p + 12 + 4 + 2) == 3);
68
69 t = t.to(torch::kFloat);
70 t.set_requires_grad(true);
71 TORCH_CHECK(t.requires_grad() == true);
72
73 t.set_requires_grad(false);
74 TORCH_CHECK(t.requires_grad() == false);

```

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```

75
76 t = t.cuda();
77 TORCH_CHECK(t.device().type() == torch::kCUDA);
78 t = t.cpu();
79
80 torch::TensorOptions opts = t.options();
81 TORCH_CHECK(opts.device() == t.device());

```

slice

Listing 33: torch::slice

```

1 static void TestSlice() {
2     auto t = torch::tensor({1, 2, 3, 4, 5}, torch::kInt);
3     torch::TensorAccessor<int32_t, 1> acc = t.accessor<int32_t, 1>();
4
5     // t2 = t[1:3]
6     torch::Tensor t2 = t.slice(/*dim*/ 0, /*start*/ 1,
7                               /*end, exclusive*/ 3); // memory is shared
8     torch::TensorAccessor<int32_t, 1> acc2 = t2.accessor<int32_t, 1>();
9     TORCH_CHECK(acc2[0] == 2);
10    TORCH_CHECK(acc2[1] == 3);
11
12    acc2[0] = 10; // also changes t since the memory is shared
13    TORCH_CHECK(acc[1] == 10);

```

Listing 34: torch::slice

```

1 void TestSlice2() {
2     auto t = torch::full({2, 3}, -1);
3     std::cout << t << "\n";
4
5     // set the last column to 0
6     t.index({torch::indexing::Slice(), -1}) = 0;
7     std::cout << t << "\n";
8 }

```

Listing 35: torch::slice

```

1 void TestSlice3() {
2     auto a = torch::tensor({1, 3, 10}, torch::kFloat);
3     auto count = torch::zeros({10}, torch::kFloat);

```

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```

4
5 count.slice(0, 1, 4).add_(a);
6 // count: 0 1 3 10 0 0 0 0 0
7 std::cout << "count: " << count << "\n";
8 count.slice(0, 2, 5).add_(a);
9 std::cout << "count: " << count << "\n";
10 // count: 0 1 4 13 10 0 0 0 0
11 }

```

topk

Listing 36: torch::topk

```

1 // https://pytorch.org/docs/stable/generated/torch.topk.html
2 static void TestTopK() {
3     auto t = torch::tensor({1, 0, 3, -1}, torch::kInt).to(torch::kFloat);
4     torch::Tensor values, indexes;
5     std::tie(values, indexes) =
6         t.topk(/*k*/ 2, /*dim*/ 0, /*largest*/ true, /*sorted*/ true);
7     auto values_acc = values.accessor<float, 1>();
8     auto indexes_acc = indexes.accessor<int64_t, 1>(); // Note: it is int64_t
9
10    TORCH_CHECK(values.numel() == 2); // k in topk is 2
11    TORCH_CHECK(values_acc[0] == 3); // the largest value is 3, at t[2]
12    TORCH_CHECK(values_acc[1] == 1); // the second largest value is 1, at t[0]
13    //
14    TORCH_CHECK(indexes_acc[0] == 2); // the largest value is t[2]
15    TORCH_CHECK(indexes_acc[1] == 0); // the second largest value is t[0]

```

floor_divide

Listing 37: torch::floor_divide

```

1 static void TestFloorDivide() {
2     auto t = torch::tensor({1, 0, 3, 5, 9}, torch::kInt);
3     auto p = torch::floor_divide(t, 2);
4     auto acc = p.accessor<int32_t, 1>();
5     TORCH_CHECK(acc[0] == 1 / 2);
6     TORCH_CHECK(acc[1] == 0 / 2);
7     TORCH_CHECK(acc[2] == 3 / 2);
8     TORCH_CHECK(acc[3] == 5 / 2);

```

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```
9 TORCH_CHECK(acc[4] == 9 / 2);
```

div

Listing 38: torch::div

```
1 // https://pytorch.org/docs/stable/generated/torch.div.html
2 static void TestDiv() {
3     auto t = torch::tensor({1, 0, 3, 5, 9}, torch::kInt);
4     // the rounding mode is supported in torch >= 1.8.0
5     auto p = torch::div(t, 2, /*rounding_mode*/ "trunc");
6     auto acc = p.accessor<int32_t, 1>();
7     TORCH_CHECK(acc[0] == 1 / 2);
8     TORCH_CHECK(acc[1] == 0 / 2);
9     TORCH_CHECK(acc[2] == 3 / 2);
10    TORCH_CHECK(acc[3] == 5 / 2);
11    TORCH_CHECK(acc[4] == 9 / 2);
```

remainder

Listing 39: torch::remainder

```
1 // https://pytorch.org/docs/1.6.0/generated/torch.remainder.html
2 static void TestRemainder() {
3     auto t = torch::tensor({1, 3, 8}, torch::kInt);
4     auto p = torch::remainder(t, 3);
5     auto acc = p.accessor<int32_t, 1>();
6     TORCH_CHECK(acc[0] == 1);
7     TORCH_CHECK(acc[1] == 0);
8     TORCH_CHECK(acc[2] == 2);
```

empty

Listing 40: torch::empty

```
1 static void TestEmpty() {
2     auto t = torch::empty({3}, torch::kInt);
3     TORCH_CHECK(t.scalar_type() == torch::kInt);
4     TORCH_CHECK(t.numel() == 3);
```

stack

Listing 41: torch::stack

```
1 static void TestStack() {
2     auto t = torch::empty({6, 5}, torch::kInt);
3     auto a = torch::stack({t, t}, /*dim*/ 1);
4     TORCH_CHECK(a.sizes() == torch::ArrayRef<int64_t>({6, 2, 5}));
5
6     a = torch::stack({t, t}, /*dim*/ 0);
7     TORCH_CHECK(a.sizes() == torch::ArrayRef<int64_t>({2, 6, 5}));
8
9     a = torch::stack({t, t}, /*dim*/ 2);
10    TORCH_CHECK(a.sizes() == torch::ArrayRef<int64_t>({6, 5, 2}));
}
```

unbind

Listing 42: torch::unbind

```
1 static void TestUnbind() {
2     auto t = torch::empty({4, 6, 5}, torch::kInt);
3     std::vector<torch::Tensor> v = torch::unbind(t, /*dim*/ 1);
4     TORCH_CHECK(v.size() == t.size(1));
5     for (int32_t i = 0; i != v.size(); ++i) {
6         TORCH_CHECK(v[i].sizes() == torch::ArrayRef<int64_t>({4, 5}));
7     }
}
```

full

Listing 43: torch::full

```
1 static void TestFull() {
2     auto t = torch::full({2, 3}, 10, torch::kInt);
3     const int32_t *p = t.data_ptr<int32_t>();
4     for (int32_t i = 0; i != t.numel(); ++i) {
5         TORCH_CHECK(p[i] == 10);
6     }
```

split

Listing 44: torch::split

```
1 static void TestSplit() {
2     auto t = torch::arange(6).reshape({2, 3});
3     std::vector<torch::Tensor> s = t.split(1);
4     TORCH_CHECK(s.size() == 2);
5     TORCH_CHECK(s[0].sizes() == torch::ArrayRef<int64_t>({1, 3}));
6     TORCH_CHECK(s[1].sizes() == torch::ArrayRef<int64_t>({1, 3}));
7
8     s = t.split(1, /*dim*/ 1);
9     TORCH_CHECK(s.size() == 3);
10    TORCH_CHECK(s[0].sizes() == torch::ArrayRef<int64_t>({2, 1}));
11    TORCH_CHECK(s[1].sizes() == torch::ArrayRef<int64_t>({2, 1}));
12    TORCH_CHECK(s[2].sizes() == torch::ArrayRef<int64_t>({2, 1}));
```

zeros

Listing 45: torch::zeros

```
1 static void TestZeros() {  
2     auto t = torch::zeros({2, 3}, torch::kFloat);
```

cat

Listing 46: torch::cat

```
1 static void TestCat() {  
2     auto t = torch::arange(24).reshape({2, 3, 4});  
3     std::vector<torch::Tensor> v(5, t);  
4     auto p = torch::cat(v, /*dim*/ 1);  
5     TORCH_CHECK(p.sizes() == torch::ArrayRef<int64_t>({2, 3 * 5, 4}));
```

division

Listing 47: test division

```

1 static void TestDivision() {
2     auto t = torch::arange(4).to(torch::kInt);
3     auto b = t / 2;
4     TORCH_CHECK(b.scalar_type() == torch::kFloat);
5
6     const float *p = b.data_ptr<float>();
7     TORCH_CHECK(p[0] == 0 / 2.);
8     TORCH_CHECK(p[1] == 1 / 2.);
9     TORCH_CHECK(p[2] == 2 / 2.);
10    TORCH_CHECK(p[3] == 3 / 2.);
11
12    auto c = b.to(torch::kInt);
13
14    const int32_t *q = c.data_ptr<int32_t>();
15    TORCH_CHECK(q[0] == 0 / 2);
16    TORCH_CHECK(q[1] == 1 / 2);
17    TORCH_CHECK(q[2] == 2 / 2);
18    TORCH_CHECK(q[3] == 3 / 2);

```

default constructed

Listing 48: test default constructed

```

1 void TestDefaultConstructed() {
2     torch::Tensor t;

```

copy

Listing 49: test rowwise copy

```

1 void TestCopy() {
2     auto t0 = torch::tensor({1, 2, 3, 4, 5, 6}, torch::kFloat).reshape({2, 3});
3     auto t = torch::empty({4, 3}, torch::kFloat);
4
5     t.slice(/*dim*/ 0, 0, 1) = t0.slice(0, 0, 1);
6     t.slice(/*dim*/ 0, 1, 2) = t0.slice(0, 1, 2);
7     t.slice(/*dim*/ 0, 2, 3) = t0.slice(0, 0, 1) + 10;
8     t.slice(/*dim*/ 0, 3, 4) = t0.slice(0, 1, 2) + 10;
9

```

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```
10  std::cout << t << "\n";
```

default addmm

Listing 50: test default constructed

```
1  void TestAddmm() {
2      std::cout << "---TestAddmm---\n";
3      // 1 2 3
4      // 4 5 6
5      torch::Tensor m =
6          torch::tensor({1, 2, 3, 4, 5, 6}, torch::kFloat).reshape({2, 3});
7
8      torch::Tensor v = torch::tensor({1, 1, -1}, torch::kFloat).unsqueeze(1);
9
10     // 10 20 30
11     torch::Tensor a = torch::tensor({10, 20}, torch::kFloat).unsqueeze(1);
12     a.addmm_(m, v);
13     std::cout << a << "\n";
14     std::cout << a.squeeze(1) << "\n";
```

elementwise operation

Listing 51: test elementwise operation

```

1 void TestElementwiseOp() {
2     std::cout << "---TestElementwiseOp---\n";
3     torch::Tensor a = torch::tensor({1, 2, 3, 40}, torch::kFloat).reshape({2, 2});
4     torch::Tensor b =
5         torch::tensor({10, 20, 30, 4}, torch::kFloat).reshape({2, 2});
6     torch::Tensor c = a * b;
7     torch::Tensor d = a / b;
8     torch::Tensor e = 1.0 / a;
9     std::cout << c << "\n"; // [[10, 40], [90, 160]]
10    std::cout << d << "\n"; // [[0.1, 0.1], [0.1, 10]]
11    std::cout << e << "\n"; // [[1.0, 0.5], [0.3333, 0.0250]], float32

```

torch.roll

Listing 52: torch.roll

```

1 void TestRoll() {
2     // 1 2 3
3     // 4 5 6
4     torch::Tensor a =
5         torch::tensor({1, 2, 3, 4, 5, 6}, torch::kFloat).reshape({2, 3});
6     torch::Tensor b = a.roll(1 /*shift right 1 column*/, 1 /*dim*/);
7     // Now b is
8     // 3 2 1
9     // 6 4 5
10
11    // -----
12    // 1 2 3 4
13    // 5 6 7 8
14    //
15    // 9 10 11 12
16    // 13 14 15 16
17    a = torch::tensor({1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16},
18                    torch::kInt)
19        .reshape({2, 2, 4});
20    b = a.roll(1 /*shift right 1 column*/, 2 /*dim*/);
21    // now b is
22    // 4 1 2 3
23    // 8 5 6 7
24    //
25    // 12 9 10 11

```

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```

26 // 16 13 14 15
27 std::cout << b;

```

torch.mean

Listing 53: torch.mean

```

1  auto t = torch::arange(24).reshape({2, 3, 4});
2  std::vector<torch::Tensor> v(5, t);
3  auto p = torch::cat(v, /*dim*/ 1);
4  TORCH_CHECK(p.sizes() == torch::ArrayRef<int64_t>({2, 3 * 5, 4}));
5  }
6
7  static void TestDivision() {
8      auto t = torch::arange(4).to(torch::kInt);
9      auto b = t / 2;
10     TORCH_CHECK(b.scalar_type() == torch::kFloat);
11
12     const float *p = b.data_ptr<float>();
13     TORCH_CHECK(p[0] == 0 / 2.);
14     TORCH_CHECK(p[1] == 1 / 2.);
15     TORCH_CHECK(p[2] == 2 / 2.);
16     TORCH_CHECK(p[3] == 3 / 2.);
17
18     auto c = b.to(torch::kInt);
19
20     const int32_t *q = c.data_ptr<int32_t>();
21     TORCH_CHECK(q[0] == 0 / 2);
22     TORCH_CHECK(q[1] == 1 / 2);
23     TORCH_CHECK(q[2] == 2 / 2);
24     TORCH_CHECK(q[3] == 3 / 2);
25 }
26
27 void TestDefaultConstructed() {
28     torch::Tensor t;
29     TORCH_CHECK(t.size(0) == 0);
30 }
31
32 void TestCopy() {
33     auto t0 = torch::tensor({1, 2, 3, 4, 5, 6}, torch::kFloat).reshape({2, 3});
34     auto t = torch::empty({4, 3}, torch::kFloat);
35 }

```

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```

36 t.slice(/*dim*/ 0, 0, 1) = t0.slice(0, 0, 1);
37 t.slice(/*dim*/ 0, 1, 2) = t0.slice(0, 1, 2);
38 t.slice(/*dim*/ 0, 2, 3) = t0.slice(0, 0, 1) + 10;
39 t.slice(/*dim*/ 0, 3, 4) = t0.slice(0, 1, 2) + 10;
40
41 std::cout << t << "\n";
42 }
43
44 void TestAddmm() {
45     std::cout << "---TestAddmm---\n";
46     // 1 2 3
47     // 4 5 6
48     torch::Tensor m =
49         torch::tensor({1, 2, 3, 4, 5, 6}, torch::kFloat).reshape({2, 3});
50
51     torch::Tensor v = torch::tensor({1, 1, -1}, torch::kFloat).unsqueeze(1);
52
53     // 10 20 30
54     torch::Tensor a = torch::tensor({10, 20}, torch::kFloat).unsqueeze(1);
55     a.addmm_(m, v);
56     std::cout << a << "\n";
57     std::cout << a.squeeze(1) << "\n";
58 }
59
60 void TestElementwiseOp() {
61     std::cout << "---TestElementwiseOp---\n";
62     torch::Tensor a = torch::tensor({1, 2, 3, 40}, torch::kFloat).reshape({2, 2});
63     torch::Tensor b =
64         torch::tensor({10, 20, 30, 4}, torch::kFloat).reshape({2, 2});
65     torch::Tensor c = a * b;
66     torch::Tensor d = a / b;
67     torch::Tensor e = 1.0 / a;
68     std::cout << c << "\n"; // [[10, 40], [90, 160]]
69     std::cout << d << "\n"; // [[0.1, 0.1], [0.1, 10]]
70     std::cout << e << "\n"; // [[1.0, 0.5], [0.3333, 0.0250]], float32
71 }
72
73 void TestRoll() {
74     // 1 2 3
75     // 4 5 6
76     torch::Tensor a =
77         torch::tensor({1, 2, 3, 4, 5, 6}, torch::kFloat).reshape({2, 3});
78     torch::Tensor b = a.roll(1 /*shift right 1 column*/, 1 /*dim*/);

```

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```

79 // Now b is
80 // 3 2 1
81 // 6 4 5
82
83 // -----
84 // 1 2 3 4
85 // 5 6 7 8
86 //
87 // 9 10 11 12
88 // 13 14 15 16
89 a = torch::tensor({1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16},
90                   torch::kInt)
91     .reshape({2, 2, 4});
92 b = a.roll(1 /*shift right 1 column*/, 2 /*dim*/);
93 // now b is
94 // 4 1 2 3
95 // 8 5 6 7
96 //
97 // 12 9 10 11
98 // 16 13 14 15
99 std::cout << b;
100 }
101
102 void TestMean() {
103 // 1 2 3
104 // 4 5 6
105 torch::Tensor a =
106     torch::tensor({1, 2, 3, 4, 5, 6}, torch::kFloat).reshape({2, 3});
107 torch::Tensor b = a.mean(1 /*dim*/, true /*keep_dim*/);
108 std::cout << b;
109 // Now b is:
110 // 2
111 // 5
112 //-----
113 b = a.mean(1 /*dim*/, false /*keep_dim*/);
114 std::cout << b << "\n";

```

torch.slice

Listing 54: torch.slice

```

1 void TestSlice2() {
2     auto t = torch::full({2, 3}, -1);
3     std::cout << t << "\n";
4
5     // set the last column to 0
6     t.index({torch::indexing::Slice(), -1}) = 0;
7     std::cout << t << "\n";

```

torch.as_strided

Listing 55: torch.as_strided

```

1 void TestAsStrided() {
2     /*
3      0 1 2 3 4 5 6 7 8 9
4      */
5
6     /*
7      0 1 2
8      2 3 4
9      4 5 6
10     6 7 8
11     */
12     torch::Tensor a = torch::arange(0, 10).to(torch::kFloat);
13     // (10 - 3) // 2 + 1 = 4
14     torch::Tensor b = a.as_strided({4, 3}, {2, 1});
15     std::cout << a << "\n";
16     std::cout << b << "\n";

```

torch.argmax

Listing 56: torch.argmax

```

1 void TestArgMax() {
2     std::vector<float> v = {
3         //
4         0.2, 0.5, 0.1, 0.4,

```

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```

5      //
6      0.9, 0.2, 0.0, 0.3,
7      //
8      0.8, 0.99, 0.1, 0.3
9      //
10     };
11     torch::Tensor a = torch::from_blob(v.data(), {3, 4}, torch::kFloat);
12     std::cout << a << "\n"; // shape (3, 4)
13
14     torch::Tensor b = a.argmax(1);
15     std::cout << b << "\n"; // 1-d, shape: (3,)
16     // 1, 0, 1
17
18     // test 3-d
19     a = torch::from_blob(v.data(), {2, 3, 2}, torch::kFloat);
20     std::cout << a << "\n"; // shape (2, 3, 2)
21
22     b = a.argmax(-1);
23     std::cout << b << "\n"; // 1-d, shape: (2, 3)
24                             // 1, 1, 0
25                             // 1, 1, 1

```

torch.index

Listing 57: torch.index

```

1  void TestIndex() {
2      // see https://pytorch.org/cppdocs/notes/tensor_indexing.html
3      std::vector<float> v = {
4          //
5          0.2, 0.5, 0.1, 0.4,
6          //
7          0.9, 0.2, 0.0, 0.3,
8          //
9          0.8, 0.99, 0.1, 0.3
10         //
11     };
12     torch::Tensor a = torch::from_blob(v.data(), {3, 4}, torch::kFloat);
13     std::cout << a << "\n"; // shape (3, 4)
14
15     torch::Tensor b = a.index({0});
16     std::cout << b << "\n"; // 1-d, shape (4,) 0.2, 0.5, 0.1, 0.4

```

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```

17                                     //
18     b = a.index({2});
19     std::cout << b << "\n"; // 1-d, shape (4,) 0.8, 0.99, 0.1, 0.3

```

torch.nn.functional.pad

Listing 58: torch.nn.functional.pad

```

1  void TestPad() {
2      std::vector<float> v = {0, 1, 2, 3, 4, 5};
3      torch::Tensor a = torch::from_blob(v.data(), {2, 3}, torch::kFloat);
4
5      int32_t padding = 1;
6
7      // pad dim 1
8      #ifdef __ANDROID__
9          auto padding_value = torch::zeros(
10              {a.size(0), padding}, torch::dtype(torch::kFloat).device(a.device()));
11
12          torch::Tensor b = torch::cat({a, padding_value}, 1);
13      #else
14          torch::Tensor b = torch::nn::functional::pad(
15              a, torch::nn::functional::PadFuncOptions({0, padding})
16                  .mode(torch::kConstant)
17                  .value(0));
18      #endif
19      /*
20       0 1 2 0
21       3 4 5 0
22      */
23      std::cout << b << "\n";
24
25      // pad dim 1
26      #ifdef __ANDROID__
27          padding_value = torch::zeros({padding, a.size(1)},
28              torch::dtype(torch::kFloat).device(a.device()));
29
30          torch::Tensor c = torch::cat({a, padding_value}, 0);
31      #else
32          torch::Tensor c = torch::nn::functional::pad(
33              a, torch::nn::functional::PadFuncOptions({0, 0, 0, padding})
34                  .mode(torch::kConstant)

```

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```

35         .value(0));
36 #endif
37 /*
38     0 1 2
39     3 4 5
40     0 0 0
41 */
42     std::cout << c << "\n";
43 }

```

torch.index_put

Listing 59: *torch.index_put*

```

1 void TestIndexPut () {
2     // https://pytorch.org/cppdocs/notes/tensor\_indexing.html#setter
3     std::vector<float> v = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11};
4     torch::Tensor a = torch::from_blob(v.data(), {12}, torch::kFloat);
5     torch::Tensor indexes =
6         torch::tensor({0, 1, 3, 5}, torch::dtype(torch::kLong));
7
8     a.index_put_({indexes}, 0);
9     std::cout << a << "\n";
10    /*
11        0 0 2 0 4 0 6 7 8 9 10 11
12
13    */
14
15    // 2-d
16
17    v = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11};
18    /*
19        0 1 2
20        3 4 5
21        6 7 8
22        9 10 11
23    */
24    a = torch::from_blob(v.data(), {4, 3}, torch::kFloat);
25
26    indexes = torch::tensor({0, 2}, torch::dtype(torch::kLong));
27    a.index_put_({indexes}, 100);
28

```

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```

29  /*
30     100 100 100
31     3   4   5
32     100 100 100
33     9   10  11
34  */
35  std::cout << a << "\n";
36
37  v = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11};
38  a = torch::from_blob(v.data(), {4, 3}, torch::kFloat);
39
40  indexes = torch::tensor({0, 2}, torch::dtype(torch::kLong));
41  a.index_put_({"...", indexes}, 99);
42
43  /*
44     99 1 99
45     99 4 99
46     99 7 99
47     99 10 99
48  */
49  std::cout << a << "\n";
50
51  v = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11};
52  a = torch::from_blob(v.data(), {4, 3}, torch::kFloat);
53
54  indexes = torch::tensor({0, 2}, torch::dtype(torch::kLong));
55  a.index_put_({"...", 1}, 6666);
56  /*
57     0 6666 2
58     3 6666 5
59     6 6666 8
60     9 6666 11
61  */
62  std::cout << a << "\n";
63
64  // 3-d
65  v = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
66       12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23};
67  a = torch::from_blob(v.data(), {2, 3, 4}, torch::kFloat);
68
69  /*
70     0 1 2 3
71     4 5 6 7

```

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```

72     8 9 10 11
73
74     ---
75
76     12 13 14 15
77     16 17 18 19
78     20 21 22 23
79     */
80     std::cout << a << "\n";
81
82     indexes = torch::tensor({0}, torch::dtype(torch::kLong));
83     a.index_put_({indexes}, 88); // dim 0
84     std::cout << a << "\n";
85     /*
86         88 88 88 88
87         88 88 88 88
88         88 88 88 88
89
90     ---
91
92         12 13 14 15
93         16 17 18 19
94         20 21 22 23
95     */
96
97     v = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
98          12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23};
99     a = torch::from_blob(v.data(), {2, 3, 4}, torch::kFloat);
100    indexes = torch::tensor({0, 3}, torch::dtype(torch::kLong));
101    a.index_put_({"...", indexes}, 66); // the last dim, i.e., dim 2
102    std::cout << a << "\n";
103    /*
104        66 1 2 66
105        66 5 6 66
106        66 9 10 66
107
108    ---
109
110        66 13 14 66
111        66 17 18 66
112        66 21 22 66
113    */
114    v = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,

```

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```

115     12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23};
116 a = torch::from_blob(v.data(), {2, 3, 4}, torch::kFloat);
117 indexes = torch::tensor({1}, torch::dtype(torch::kLong));
118 a.index_put_({torch::indexing::None, indexes}, 55); // dim 1
119 std::cout << a << "\n";
120 /*
121     0 1 2 3
122     4 5 6 7
123     8 9 10 11
124
125     ---
126
127     55 55 55 55
128     55 55 55 55
129     55 55 55 55
130 */
131
132 v = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
133     12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23};
134 a = torch::from_blob(v.data(), {2, 3, 4}, torch::kFloat);
135 indexes = torch::tensor({1}, torch::dtype(torch::kLong));
136 a.index_put_(
137     {torch::indexing::Slice(torch::indexing::None, torch::indexing::None,
138         torch::indexing::None),
139     torch::indexing::Slice(torch::indexing::None, torch::indexing::None,
140         torch::indexing::None),
141     indexes},
142     33); // dim 2
143 std::cout << a << "\n";
144 /*
145     0 33 2 3
146     4 33 6 7
147     8 33 10 11
148
149     ---
150
151     12 33 14 15
152     16 33 18 19
153     20 33 22 23
154 */
155 }

```

torch.nonzero

Listing 60: torch.nonzero

```
1 void TestNonZero() {
2     auto t = torch::tensor({0, 2, 0, 0, 5, 0, 1}, torch::kInt);
3     std::cout << t << "\n";           // 1-d, shape (7,)
4     auto indexes = t.nonzero();        //
5     std::cout << indexes << "\n";     // 2-d, shape (3, 1) 1 4 6
6
7     indexes = indexes.squeeze();
8     auto v = t.index_select(0, indexes);
9     std::cout << v << "\n";
10 }
```

Listing 61: torch.nonzero

```
1 void TestNonZero2() {
2     auto a = torch::tensor({1, 5, 5}, torch::dtype(torch::kInt)).reshape({3, 1});
3     torch::Tensor b = (a == 5).nonzero();
4     // 1 0
5     // 2 0
6     std::cout << b << "\n"; // shape is (2, 2)
7
8     torch::Tensor c = (a.squeeze() == 5).nonzero();
9     // 1
10    // 2
11    std::cout << c << "\n"; // shape is (2, 1)
12
13    torch::Tensor d = (a == 50).nonzero();
14    std::cout << d.numel() << "\n";
15 }
```

9.4.11 intrusive_ptr

9.4.12 optional

9.4.13 PackedSequence

See

- <https://github.com/pytorch/pytorch/blob/master/torch/csrc/api/include/torch/nn/utils/rnn.h>
- <https://github.com/pytorch/pytorch/blob/master/torch/nn/utils/rnn.py>

pack_padded_sequence

Listing 62: ./code/packed-sequence/main.cc

```

1  static void TestPadPackedSequence() {
2      torch::Tensor t = torch::tensor({
3          {{10, 20, 30}, {0, 0, 0}, {0, 0, 0}},
4          {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}},
5          {{-1, 2, 3}, {-4, 5, 6}, {0, 0, 0}},
6      });
7      torch::Tensor lengths = torch::tensor({1, 3, 2});
8      torch::nn::utils::rnn::PackedSequence packed_seq =
9          torch::nn::utils::rnn::pack_padded_sequence(
10             t, lengths, /*batch_first*/ true, /*enforce_sorted*/ false);
11      std::cout << "data: " << packed_seq.data() << "\n";
12      std::cout << "batch_sizes: " << packed_seq.batch_sizes() << "\n";
13      std::cout << "sorted_indices: " << packed_seq.sorted_indices() << "\n";
14      std::cout << "unsorted_indices: " << packed_seq.unsorted_indices() << "\n";
15  }
16  /*
17  data:   1   2   3
18  -1   2   3
19  10  20  30
20   4   5   6
21  -4   5   6
22   7   8   9
23  [ CPULongType{6,3} ]
24  batch_sizes:  3
25   2
26   1
27  [ CPULongType{3} ]
28  sorted_indices:  1
29   2
30   0
31  [ CPULongType{3} ]
32  unsorted_indices:  2
33   0
34   1
35  [ CPULongType{3} ]
36  */

```

The output is

Listing 63: ./code/packed-sequence/main.cc

```

1  }
2  /*
3  data:   1   2   3
4    -1   2   3
5    10  20  30
6     4   5   6
7    -4   5   6
8     7   8   9
9  [ CPULongType{6,3} ]
10 batch_sizes:  3
11    2
12    1
13  [ CPULongType{3} ]
14 sorted_indices: 1
15    2

```

9.4.14 ivalue

Listing 64: ./code/ivalue/main.cc

```

1  #include "torch/script.h"
2
3  static void TestVectorOfTensor() {
4      torch::jit::Module m("m");
5      m.define(R"(
6          def forward(self, x, y):
7              return [x, y]
8      )");
9      auto x = torch::tensor({1, 2, 3});
10     auto y = torch::tensor({4, 5, 6});
11     auto i = m.run_method("forward", x, y);
12
13     assert(i.tagKind() == "GenericList");
14
15     torch::ArrayRef<torch::IValue> tensor_list = i.toListRef();
16     TORCH_CHECK(torch::allclose(x, tensor_list[0].toTensor()));
17     TORCH_CHECK(torch::allclose(y, tensor_list[1].toTensor()));
18
19     torch::List<torch::IValue> k = i.toList();
20

```

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```

21 torch::List<torch::Tensor> o =
22     c10::impl::toTypedList<torch::Tensor>(std::move(k));
23
24 TORCH_CHECK(torch::allclose(o[0], x));
25 TORCH_CHECK(torch::allclose(o[1], y));
26
27 std::vector<torch::Tensor> p = o.vec();
28 TORCH_CHECK(torch::allclose(p[0], x));
29 TORCH_CHECK(torch::allclose(p[1], y));
30 }
31
32 static void TestVectorOfTensor2() {
33     torch::jit::Module m("m");
34     m.define(R"(
35         def forward(self, x):
36             return [[x], [x,x]]
37     )");
38     auto x = torch::tensor({1, 2, 3});
39     auto i = m.run_method("forward", x);
40     TORCH_CHECK(i.tagKind() == "GenericList");
41
42     torch::List<torch::IValue> list = i.toList();
43     torch::Tensor a = list.get(0).toListRef()[0].toTensor();
44     TORCH_CHECK(torch::allclose(a, x));
45
46     std::vector<torch::Tensor> b =
47         c10::impl::toTypedList<torch::Tensor>(list.get(1).toList()).vec();
48     TORCH_CHECK(torch::allclose(b[0], x));
49     TORCH_CHECK(torch::allclose(b[1], x));
50 }
51
52 static void TestVectorOfTensor3() {
53     torch::jit::Module m("m");
54     m.define(R"(
55         def forward(self, x: List[torch.Tensor]):
56             return x[0] + x[1]
57     )");
58
59     std::vector<torch::Tensor> v;
60     v.push_back(torch::tensor({1, 2}));
61     v.push_back(torch::tensor({3, 4}));
62     c10::List<torch::Tensor> ilist(v);
63

```

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```

64  c10::impl::GenericList generic_list = c10::impl::toList(ilist);
65
66  c10::List<torch::Tensor> l2 =
67      c10::impl::toTypedList<torch::Tensor>(generic_list);
68
69  TORCH_CHECK(torch::allclose(l2[0], v[0]));
70  TORCH_CHECK(torch::allclose(l2[1], v[1]));
71
72  auto r = m.run_method("forward", generic_list);
73  TORCH_CHECK(torch::allclose(r.toTensor(), v[0] + v[1]));
74
75  // Note: We can pass a vector directly
76  r = m.run_method("forward", v);
77  TORCH_CHECK(torch::allclose(r.toTensor(), v[0] + v[1]));
78
79  r = m.run_method("forward", ilist); // also OK
80  TORCH_CHECK(torch::allclose(r.toTensor(), v[0] + v[1]));
81 }
82
83 static void TestVectorOfTensor4() {
84     torch::jit::Module m("m");
85     m.define(R"(
86         def forward(self, x: Tuple[List[torch.Tensor]]):
87             return x[0][0] + x[0][1]
88     )");
89
90     std::vector<torch::Tensor> v;
91     v.push_back(torch::tensor({1, 2}));
92     v.push_back(torch::tensor({3, 4}));
93     auto t = torch::ivalue::Tuple::create(v);
94
95     auto r = m.run_method("forward", t);
96     TORCH_CHECK(torch::allclose(r.toTensor(), v[0] + v[1]));
97 }
98
99 static void TestVectorOfTensor5() {
100     torch::jit::Module m("m");
101     m.define(R"(
102         def forward(self, x: Tuple[List[List[torch.Tensor]], List[torch.Tensor]]):
103             return x[0][0][0] + x[0][0][1] + x[1][0] + x[1][1]
104     )");
105
106     std::vector<torch::Tensor> v;

```

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```

107 v.push_back(torch::tensor({1, 2}));
108 v.push_back(torch::tensor({3, 4}));
109
110 std::vector<std::vector<torch::Tensor>> vv;
111 vv.push_back(v);
112 vv.push_back(v);
113
114 auto t = torch::IValue::Tuple::create(vv, v);
115
116 auto r = m.run_method("forward", t);
117 TORCH_CHECK(torch::allclose(r.toTensor(), v[0] + v[1] + v[0] + v[1]));
118 }
119
120 static void TestVectorOfTensor6() {
121     // List[List[Tensor]]
122     std::vector<torch::Tensor> v;
123     v.push_back(torch::tensor({1, 2}));
124     v.push_back(torch::tensor({3, 4}));
125
126     c10::List<torch::Tensor> ilist(v);
127     torch::IValue ivalue(ilist);
128     TORCH_CHECK(ivalue.tagKind() == "GenericList");
129
130     c10::List<c10::List<torch::Tensor>> ilist2(ilist);
131     ilist2.push_back(ilist);
132     ilist2.push_back(ilist);
133
134     torch::IValue ivalue2(ilist2);
135     TORCH_CHECK(ivalue2.tagKind() == "GenericList");
136
137     c10::List<torch::IValue> a0 = ivalue2.toList();
138     c10::List<c10::List<torch::Tensor>> a1 =
139         c10::impl::toTypedList<c10::List<torch::Tensor>>(a0);
140
141     c10::ArrayRef<torch::IValue> a = ivalue2.toListRef();
142
143     torch::List<torch::Tensor> b =
144         c10::impl::toTypedList<torch::Tensor>(a[0].toList());
145     for (int32_t i = 0; i != b.size(); ++i) {
146         std::cout << b[i] << "\n";
147     }
148     std::vector<std::vector<torch::Tensor>> v2{v};
149     torch::List<torch::List<torch::Tensor>> c;

```

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```

150   for (auto k : v2) {
151       c10::List<torch::Tensor> dd{torch::ArrayRef<torch::Tensor>(k)};
152       c.push_back(std::move(dd));
153   }
154 }
155
156 int main() {
157     TestVectorOfTensor();
158     TestVectorOfTensor2();
159     TestVectorOfTensor3();
160     TestVectorOfTensor4();
161     TestVectorOfTensor5();
162     TestVectorOfTensor6();
163     return 0;
164 }

```

9.4.15 method

See:

- <https://github.com/pytorch/pytorch/blob/master/torch/csrc/api/include/torch/imethod.h>

Listing 65: ./code/method/main.cc

```

1  #include "torch/script.h"
2
3  static void TestHello() {
4      torch::jit::Module m("m");
5      m.define(R"(
6          def forward(self, x: torch.Tensor, y: torch.Tensor):
7              return x + y
8      )");
9
10     torch::jit::Method method = m.get_method("forward");
11     TORCH_CHECK(method.name() == "forward");
12
13     const std::vector<std::string> &names = method.getArgumentNames();
14     TORCH_CHECK(names.size() == 2);
15     TORCH_CHECK(names[0] == "x");
16     TORCH_CHECK(names[1] == "y");
17
18     std::vector<torch::IValue> args;
19     auto x = torch::tensor({1, 2});

```

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```

20  auto y = torch::tensor({1, 2});
21  args.emplace_back(x);
22  args.emplace_back(y);
23  auto z = method(args).toTensor();
24
25  TORCH_CHECK(torch::equal(z, x + y));
26
27  std::shared_ptr<torch::jit::Graph> g = method.graph();
28  // see node/main.cc
29 }
30
31 int main() {
32     TestHello();
33     return 0;
34 }

```

9.4.16 type

See: - https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/core/jit_type_base.h - https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/core/jit_type.h - https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/core/type_ptr.h

`torch::Type` contains a member `torch::TypeKind`. `torch::SharedType` is a subclass of `torch::Type` and `std::enabled_shared_from_this<torch::SharedType>`.

```

// https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/core/jit_type_base.h
↪ #L637
using TypePtr = SingletonOrSharedTypePtr<Type>;

```

Listing 66: ./code/type/main.cc

```

1  #include "torch/script.h"
2
3  static void TestTypeKind() {
4      // https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/core/jit_type_base.h
5      torch::TypeKind k = torch::TypeKind::AnyType;
6      TORCH_CHECK(torch::typeKindToString(k) == std::string("AnyType"));
7
8      // NamedType is not a member of TypeKind
9  }
10
11 static void TestNumberType() {
12     // torch::NumberType::get() returns a static object!

```

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```

13  // so p and q are actually the same
14  torch::NumberTypePtr p = torch::NumberType::get();
15  torch::NumberTypePtr q = torch::NumberType::get();
16
17  TORCH_CHECK(p.get() == q.get());
18
19  TORCH_CHECK(p->str() == "Scalar");
20  TORCH_CHECK(p->kind() == torch::NumberType::Kind);
21  TORCH_CHECK(p->kind() == torch::TypeKind::NumberType);
22 }
23
24 static void TestIntType() {
25     torch::IntTypePtr p = torch::IntType::get();
26     TORCH_CHECK(p->str() == "int");
27     TORCH_CHECK(p->kind() == torch::TypeKind::IntType);
28     TORCH_CHECK(p->kind() == torch::IntType::Kind);
29     TORCH_CHECK(p->isSubtypeOf(torch::NumberType::get()) == true);
30 }
31
32 static void TestFloatType() {
33     torch::FloatTypePtr p = torch::FloatType::get();
34     TORCH_CHECK(p->str() == "float");
35     TORCH_CHECK(p->kind() == torch::TypeKind::FloatType);
36     TORCH_CHECK(p->kind() == torch::FloatType::Kind);
37     TORCH_CHECK(p->isSubtypeOf(torch::NumberType::get()) == true);
38     TORCH_CHECK(p->isSubtypeOf(torch::IntType::get()) == false);
39 }
40
41 static void TestBoolType() {
42     torch::BoolTypePtr p = torch::BoolType::get();
43     TORCH_CHECK(p->str() == "bool");
44     TORCH_CHECK(p->kind() == torch::TypeKind::BoolType);
45     TORCH_CHECK(p->kind() == torch::BoolType::Kind);
46     TORCH_CHECK(p->isSubtypeOf(torch::NumberType::get()) == true);
47     TORCH_CHECK(p->isSubtypeOf(torch::IntType::get()) == false);
48 }
49
50 static void TestNamedType() {
51     // torch::Type is an abstract class!
52     //
53     // torch::NamedType is an abstract class!
54     //
55     // torch::NamedType t(torch::TypeKind::AnyType, "foo.bar"); // error

```

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```

56 // TORCH_CHECK(t.name()->qualifiedName() == "foo.bar");
57 }
58
59 static void TestAnyType() {
60     torch::AnyTypePtr p = torch::AnyType::get();
61     TORCH_CHECK(p->Kind == torch::TypeKind::AnyType);
62     TORCH_CHECK(p->kind() == torch::TypeKind::AnyType);
63     TORCH_CHECK(p->str() == "Any");
64     TORCH_CHECK(p->requires_grad() == false);
65
66     TORCH_CHECK(p == torch::AnyType::get());
67
68     // available in newer versions of PyTorch
69     // TORCH_CHECK(p->equals(torch::AnyType::get()));
70
71     TORCH_CHECK(torch::toString(p) == "Any");
72 }
73
74 int main() {
75     TestTypeKind();
76     TestNumberType();
77     TestIntType();
78     TestFloatType();
79     TestNamedType();
80     TestAnyType();
81     return 0;
82 }

```

9.4.17 trace

Listing 67: ./code/trace/ex0.py

```

1  #!/usr/bin/env python3
2
3  import torch
4
5  import torch.nn as nn
6  from typing import List
7
8
9  class Foo(nn.Module):
10     def __init__(self):

```

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```

11     super().__init__()
12     self.relu = nn.ReLU()
13
14     def forward(self, x):
15         return self.relu(x)
16
17
18 def test_foo():
19     f = Foo()
20     m = torch.jit.trace(f, torch.rand(2, 3))
21
22     print(m(torch.rand(2)))
23     print(m(torch.rand(2, 3, 4)))
24     # Note: The input shape is dynamic, not fixed.
25
26
27 def simple(x: List[torch.Tensor], y: torch.Tensor):
28     x = x[0].item()
29     if x > 2:
30         return y + x + 1
31     elif x < 1:
32         return y
33     else:
34         return y + x
35
36
37 def test_simple():
38     f0 = torch.jit.trace(simple, ([torch.tensor([0])], torch.rand(2, 3)))
39     # print(dir(f0))
40     """
41     ['__call__', '__class__', '__delattr__', '__dict__', '__dir__', '__doc__',
42     '__eq__', '__format__', '__ge__', '__getattr__', '__gt__', '__hash__',
43     '__init__', '__init_subclass__', '__le__', '__lt__', '__module__', '__ne__',
44     '__new__', '__reduce__', '__reduce_ex__', '__repr__', '__setattr__',
45     '__sizeof__', '__str__', '__subclasshook__', '_debug_flush_compilation_cache',
46     'code', 'get_debug_state', 'graph', 'graph_for', 'inlined_graph', 'name',
47     'qualified_name', 'save', 'save_to_buffer', 'schema']
48     """
49     # print(f0.schema) # simple(Tensor[] x, Tensor y) -> (Tensor)
50     # print(f0.code)
51     """
52     def simple(x: List[Tensor],
53         y: Tensor) -> Tensor:

```

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```

54     return y
55     """
56     # print(f0.graph)
57     """
58     graph(%x : Tensor[],
59           %y : Float(2, 3, strides=[3, 1], requires_grad=0, device=cpu)):
60         return (%y)
61     """
62     # print(f0.inlined_graph) # same as the above one
63     # print(f0.name) # simple
64     print(f0.qualified_name) # __torch__.simple
65
66
67 def main():
68     # test_foo()
69     test_simple()
70
71
72 if __name__ == "__main__":
73     main()

```

Listing 68: ./code/trace/ex1.py

```

1  #!/usr/bin/env python3
2
3  import torch
4
5
6  def f(a, b):
7      c = a + b
8      d = c * c
9      e = torch.tanh(d * c)
10     return d + (e + e)
11
12
13 m = torch.jit.script(f)
14 print(m.graph)
15
16 """
17 graph(%a.1 : Tensor,
18       %b.1 : Tensor):
19     %4 : int = prim::Constant[value=1]()
20     %c.1 : Tensor = aten::add(%a.1, %b.1, %4) # ./ex1.py:7:8

```

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```

21  %d.1 : Tensor = aten::mul(%c.1, %c.1) # ./ex1.py:8:8
22  %11 : Tensor = aten::mul(%d.1, %c.1) # ./ex1.py:9:19
23  %e.1 : Tensor = aten::tanh(%11) # ./ex1.py:9:8
24  %17 : Tensor = aten::add(%e.1, %e.1, %4) # ./ex1.py:10:16
25  %19 : Tensor = aten::add(%d.1, %17, %4) # ./ex1.py:10:11
26  return (%19)
27  """
28
29  """
30  Note: for aten::add(a0, a1, a2), it does a0 + a2 * a1.
31  See torch/csrc/jit/codegen/fuser/codegen.cpp
32
33  """
34  assert isinstance(m.graph, torch._C.Graph)
35
36  # Every graph has inputs and outputs
37  # m.graph.inputs() returns an iterator
38  assert len(list(m.graph.inputs())) == 2, "It has two inputs: a, b, in our case"
39  it = m.graph.inputs()
40  a = next(it)
41  b = next(it)
42
43  assert isinstance(a, torch._C.Value)
44  assert isinstance(a.node(), torch._C.Node)
45
46  # every node has inputs and outputs
47  # a.node().inputs() is an iterator
48  assert list(a.node().inputs()) == []
49  assert a.node().kind() == "prim::Param"
50  assert a.node().inputsSize() == 0
51  assert a.node().outputsSize() == 2
52  print(next(a.node().outputs()))
53
54  oit = a.node().outputs()
55  assert next(oit) == a
56  assert next(oit) == b
57
58  assert next(a.node().outputs()) == a
59
60  assert a.node().outputsAt(0) == a
61  assert a.node().outputsAt(1) == b
62  assert a.node() == b.node()
63  assert a.node().attributeNames() == [], "this node has no attributes"

```

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```

64 assert a.debugName() == "a.1"
65 assert isinstance(a.type(), torch._C.TensorType)
66 assert a.type().kind() == "TensorType"
67 assert a.unique() == 0 # TODO(fangjun): what does it mean?
68 assert isinstance(a.uses(), list)
69 assert isinstance(a.uses()[0], torch._C.Use)
70 assert isinstance(a.uses()[0].user, torch._C.Node)
71
72 c_node = a.uses()[0].user
73 assert c_node.kind() == "aten::add"
74 assert c_node.attributeNames() == []
75 assert len(list(c_node.inputs())) == 3
76 c_it = c_node.inputs()
77 assert a == next(c_it)
78 assert b == next(c_it)
79 v4 = next(c_it)
80 assert v4.debugName() == "4"
81 assert c_node.hasAttributes() is False
82 assert c_node.hasMultipleOutputs() is False
83 assert c_node.hasUses() is True
84 assert (
85     c_node.schema()
86     == "aten::add.Tensor(Tensor self, Tensor other, *, Scalar alpha=1) -> (Tensor)"
87 )
88 print(c_node.schema())
89 print(type(c_node.schema()))
90 v4_node = v4.node()
91 assert v4_node.attributeNames() == ["value"]
92 assert v4_node.hasAttributes() is True
93 assert v4_node.hasAttribute("value") is True
94 # print(v4_node.t("value"))
95 print(dir(v4_node))

```

9.4.18 Node

Listing 69: ./code/node/main.cc

```

1  #include "torch/csrc/jit/passes/quantization/helper.h" // for removeTorchMangle
2  #include "torch/script.h"
3
4  static void TestRemoveTorchMangle() {
5      std::string s = torch::jit::removeTorchMangle("a.__torch_mangle_1.foo");

```

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```

6   TORCH_CHECK(s == "a.foo");
7
8   s = torch::jit::removeTorchMangle("a.__torch_mangle_123.foo");
9   TORCH_CHECK(s == "a.foo");
10  }
11
12  static void TestSimple() {
13      torch::jit::Module m("m");
14      m.define(R"(
15          def forward(self, x: torch.Tensor, y: torch.Tensor):
16              a = x + 2
17              b = y * 3
18              return a + b
19      )");
20      std::shared_ptr<torch::jit::Graph> graph = m.get_method("forward").graph();
21      std::cout << "graph string: \n" << graph->toString() << "\n";
22      // Or we can use graph->dump();
23      torch::jit::Block *block = graph->block();
24      for (auto it = block->nodes().begin(), end = block->nodes().end();
25           it != end;) {
26          torch::jit::Node *n = *it++;
27          torch::jit::NodeKind k = n->kind();
28          std::cout << "node kind: " << k << " " << k.toQualString() << "\n";
29      }
30  #if 0
31  graph string:
32  graph(%self : __torch__.m,
33        %x.1 : Tensor,
34        %y.1 : Tensor):
35      %5 : int = prim::Constant[value=1]()
36      %4 : int = prim::Constant[value=2]() # <string>:3:14
37      %8 : int = prim::Constant[value=3]() # <string>:4:14
38      %a.1 : Tensor = aten::add(%x.1, %4, %5) # <string>:3:10
39      %b.1 : Tensor = aten::mul(%y.1, %8) # <string>:4:10
40      %13 : Tensor = aten::add(%a.1, %b.1, %5) # <string>:5:13
41      return (%13)
42
43  node kind: 14 prim::Constant
44  node kind: 14 prim::Constant
45  node kind: 14 prim::Constant
46  node kind: 534 aten::add
47  node kind: 241 aten::mul
48  node kind: 534 aten::add

```

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```

49 #endif
50 }
51
52 static void TestFunctionCall() {
53     torch::jit::Module m("m");
54     m.define(R"(
55         def add(self, x: torch.Tensor, y: torch.Tensor):
56             '''my add doc'''
57             return x + y + 3
58
59         def forward(self, x: torch.Tensor, y: torch.Tensor):
60             c = self.add(x, y)
61             return c
62     )");
63     std::shared_ptr<torch::jit::Graph> graph = m.get_method("forward").graph();
64     std::cout << "graph string: \n" << graph->toString() << "\n";
65     torch::jit::Block *block = graph->block();
66     for (auto it = block->nodes().begin(), end = block->nodes().end();
67          it != end;) {
68         torch::jit::Node *n = *it++;
69         torch::jit::NodeKind k = n->kind();
70         std::cout << "node kind: " << k << " " << k.toQualString() << "\n";
71     }
72 #if 0
73 graph string:
74 graph(%self.1 : __torch__.m,
75       %x.1 : Tensor,
76       %y.1 : Tensor):
77   %c.1 : Tensor = prim::CallMethod[name="add"](%self.1, %x.1, %y.1) # <string>:6:10
78   return (%c.1)
79
80 node kind: 149 prim::CallMethod
81 #endif
82     for (auto it = block->nodes().begin(), end = block->nodes().end();
83          it != end;) {
84         torch::jit::Node *n = *it++;
85         torch::jit::NodeKind k = n->kind();
86         if (k == c10::prim::CallMethod) {
87             torch::ArrayRef<torch::jit::Value *> inputs = n->inputs();
88             TORCH_CHECK(inputs.size() == 3);
89
90             torch::jit::TypePtr type = inputs[0]->type();
91

```

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```

92     auto class_type = type->cast<torch::jit::ClassType>();
93     TORCH_CHECK(class_type->str() == "__torch__.m");
94     if (!class_type) {
95         std::cout << "Not a class type: " << type->str() << "\n";
96         continue;
97     }
98     // defined by the macro "CREATE_ACCESSOR()" in ir/ir.h
99     const std::string &function_name = n->s(c10::attr::name);
100    // const std::string &function_name = n->s(torch::jit::attr::name);
101    TORCH_CHECK(function_name == "add");
102
103    TORCH_CHECK(torch::jit::attr::name == c10::attr::name);
104
105    torch::jit::Function &function = class_type->getMethod(function_name);
106    if (!function.isGraphFunction()) {
107        std::cout << function_name << " is not a graph function"
108                << "\n";
109        continue;
110    }
111    std::string class_type_str =
112        torch::jit::removeTorchMangle(class_type->str());
113    // remove __torch__., which is 10 characters long
114    std::string no_torch_class_type_str = class_type_str.substr(10);
115    }
116    }
117    }
118
119    int main() {
120        // TestRemoveTorchMangle();
121        // TestSimple();
122        TestFunctionCall();
123        return 0;
124    }

```

9.4.19 symbol

See

- <https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/core/symbol.h>
- https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/core/interned_strings.h

Listing 70: ./code/symbol/main.cc

```

1  #include "torch/script.h"
2
3  static void TestHello() {
4      // defined in
5      // https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/core/interned_
6      ↪ strings.h
7      c10::Symbol s = c10::namespaces::prim;
8      TORCH_CHECK(int(s) == 0);
9
10     s = c10::namespaces::aten;
11     s = c10::namespaces::cuda;
12     s = c10::namespaces::onnx;
13     s = c10::namespaces::scope;
14     s = c10::namespaces::_caffe2;
15     s = c10::namespaces::namespaces;
16
17     s = c10::prim::Assign;
18     TORCH_CHECK(int(s) == 10);
19
20     s = c10::prim::Constant;
21     s = c10::prim::device;
22     s = c10::prim::dtype;
23
24     s = c10::aten::Bool;
25     s = c10::aten::sorted;
26
27     s = c10::onnx::Add;
28
29     // c10::_keys is an enum class.
30     int32_t max_num_symbols = int32_t(c10::_keys::num_symbols);
31
32     // max num symbols: 1535. (as of 2022-08-02)
33     std::cout << "max num symbols: " << max_num_symbols << "\n";
34
35     c10::_keys k = c10::_keys::namespaces_prim;
36     k = c10::_keys::namespaces_aten;
37     k = c10::_keys::prim_Constant;
38 }
39
40 static void TestSymbol() {
41     // Symbol contains only an integer, which is from an Enum
42     c10::Symbol s = c10::Symbol::fromQualString("prim::Constant");

```

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```

42 TORCH_CHECK(s == c10::prim::Constant);
43 TORCH_CHECK(s.is_prim() == true);
44 TORCH_CHECK(s.is_aten() == false);
45 TORCH_CHECK(s.ns() == c10::namespaces::prim);
46 TORCH_CHECK(s.toUnqualString() == std::string("Constant"));
47 TORCH_CHECK(s.toQualString() == std::string("prim::Constant"));
48
49 TORCH_CHECK(s == c10::Symbol::prim("Constant"));
50
51 // Register a new symbol
52 // It will first register the namespace k2 as the symbol "namespaces::k2"
53 s = c10::Symbol::fromQualString("k2::Foo");
54
55 TORCH_CHECK(s.ns() == c10::Symbol::fromQualString("namespaces::k2"));
56 TORCH_CHECK(s.toUnqualString() == std::string("Foo"));
57 TORCH_CHECK(s.toQualString() == std::string("k2::Foo"));
58 }
59
60 int main() {
61     TestHello();
62     TestSymbol();
63     return 0;
64 }

```

9.4.20 graph

Listing 71: ./code/graph/main.cc

```

1  #include "torch/script.h"
2
3  static void TestConv2d() {
4      torch::jit::Module m("m");
5      m.define(R"(
6          def __init__(self):
7              self.conv = torch.nn.Conv2d(2, 3)
8          def forward(self, x: torch.Tensor):
9              return self.conv(x)
10     )");
11     torch::jit::Method method = m.get_method("forward");
12     std::shared_ptr<torch::jit::Graph> g = method.graph();
13     torch::ArrayRef<torch::jit::Value*> inputs = g->inputs();
14     torch::ArrayRef<torch::jit::Value*> outputs = g->outputs();

```

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```

15 TORCH_CHECK(inputs.size() == 1);
16 TORCH_CHECK(outputs.size() == 1);
17
18 torch::jit::Value *in = inputs[0];
19 std::cout << in->type()->str() << "\n";
20 std::cout << in->debugName() << "\n";
21 }
22
23 int main() {
24     TestConv2d();
25     return 0;
26 }

```

Listing 72: ./code/graph/inline_calls.py

```

1  #!/usr/bin/env python3
2
3  from pathlib import Path
4
5  import torch
6  import torch.nn as nn
7
8
9  class Foo(nn.Module):
10     def __init__(self):
11         super().__init__()
12         self.linear = nn.Linear(2, 2)
13         self.linear2 = nn.Linear(2, 2)
14         self.relu = nn.ReLU()
15         self.t = torch.rand(2)
16
17     def forward(self, x: torch.Tensor):
18         y = self.linear(x + self.t)
19         y = self.linear2(y)
20         y = self.linear2(y)
21         # z = self.relu(y)
22         return nn.functional.elu(y)
23         return z
24
25
26 def generate_foo_pt():
27     f = Foo()
28     x = torch.rand(1, 2)

```

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```
29     m = torch.jit.trace(f, x)
30     m.save("foo.pt")
31
32
33 def test_foo_pt():
34     m = torch.jit.load("foo.pt")
35     assert isinstance(m.forward, torch._C.ScriptMethod)
36     assert isinstance(m.forward.graph, torch._C.Graph)
37     assert isinstance(m.forward.inlined_graph, torch._C.Graph)
38
39     print(m.linear.graph)
40     return
41
42     print(m.forward.graph)
43     # print(m.forward.inlined_graph)
44     g = m.forward.graph
45     nodes = g.nodes()
46
47     n = next(nodes)
48     print(dir(n))
49     assert n.kind() == "prim::GetAttr"
50     for i in n.inputs():
51         assert isinstance(i, torch._C.Value)
52         assert i.debugName() == "self.1"
53         assert isinstance(i.type(), torch._C.ClassType)
54         t = i.type()
55         assert t.str() == "__torch__.Foo"
56
57
58 def main():
59     generate_foo_pt()
60     # test_foo_pt()
61
62
63 if __name__ == "__main__":
64     main()
```


9.4.21 passes

`_jit_pass_fuse_add_relu`

See https://github.com/pytorch/pytorch/blob/master/torch/csrc/jit/passes/fuse_relu.cpp

Listing 73: `./code/passes/fuse_add_relu.py`

```

1  #!/usr/bin/env python3
2
3  import torch
4
5
6  class Foo(torch.nn.Module):
7     def forward(self, x: torch.Tensor, y: torch.Tensor):
8         a = torch.nn.functional.relu(x + y)
9         return a + 10
10
11
12 def main():
13     f = Foo()
14     m = torch.jit.trace(f, (torch.rand(3), torch.rand(3)))
15     g = m.graph
16
17     with open("fuse_add_relu-before.txt", "w") as f:
18         print(g, file=f)
19
20     torch._C._jit_pass_fuse_add_relu(g)
21
22     with open("fuse_add_relu-after.txt", "w") as f:
23         print(g, file=f)
24
25
26 if __name__ == "__main__":
27     main()

```

Listing 74: `./code/passes/fuse_add_relu-before.txt`

```

1  graph(%self : __torch__.Foo,
2     %x : Float(3, strides=[1], requires_grad=0, device=cpu),
3     %y : Float(3, strides=[1], requires_grad=0, device=cpu)):
4     %5 : int = prim::Constant[value=1]() # ./fuse_add_relu.py:8:0
5     %input : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::add(%x, %y, %5)
6     ↪ # ./fuse_add_relu.py:8:0

```

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```

6   %a : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::relu(%input) # /
↳Users/fangjun/py38/lib/python3.8/site-packages/torch/nn/functional.py:1457:0
7   %8 : Long(requires_grad=0, device=cpu) = prim::Constant[value={10}]() # ./fuse_add_
↳relu.py:9:0
8   %9 : int = prim::Constant[value=1]() # ./fuse_add_relu.py:9:0
9   %10 : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::add(%a, %8, %9) # .
↳./fuse_add_relu.py:9:0
10  return (%10)
11

```

Listing 75: ./code/passes/fuse_add_relu-after.txt

```

1 graph(%self : __torch__.Foo,
2       %x : Float(3, strides=[1], requires_grad=0, device=cpu),
3       %y : Float(3, strides=[1], requires_grad=0, device=cpu)):
4   %5 : int = prim::Constant[value=1]() # ./fuse_add_relu.py:8:0
5   %11 : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::_add_relu(%x, %y,
↳%5)
6   %8 : Long(requires_grad=0, device=cpu) = prim::Constant[value={10}]() # ./fuse_add_
↳relu.py:9:0
7   %9 : int = prim::Constant[value=1]() # ./fuse_add_relu.py:9:0
8   %10 : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::add(%11, %8, %9) #_
↳./fuse_add_relu.py:9:0
9   return (%10)
10

```

in_place relu

Listing 76: ./code/passes/fuse_add_relu_.py

```

1 #!/usr/bin/env python3
2
3 import torch
4
5
6 class Foo(torch.nn.Module):
7     def forward(self, x: torch.Tensor, y: torch.Tensor):
8         a = torch.nn.functional.relu(x + y, inplace=True)
9         return a + 10
10
11
12 def main():

```

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```

13     f = Foo()
14     m = torch.jit.trace(f, (torch.rand(3), torch.rand(3)))
15     g = m.graph
16
17     with open("fuse_add_relu-before.txt", "w") as f:
18         print(g, file=f)
19
20     torch._C._jit_pass_fuse_add_relu(g)
21
22     with open("fuse_add_relu-after.txt", "w") as f:
23         print(g, file=f)
24
25
26 if __name__ == "__main__":
27     main()

```

Listing 77: ./code/passes/fuse_add_relu-before.txt

```

1 graph(%self : __torch__.Foo,
2       %x : Float(3, strides=[1], requires_grad=0, device=cpu),
3       %y : Float(3, strides=[1], requires_grad=0, device=cpu)):
4   %5 : int = prim::Constant[value=1]() # ./fuse_add_relu.py:8:0
5   %input : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::add(%x, %y, %5)
6   ↪ # ./fuse_add_relu.py:8:0
7   %a : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::relu(%input) # /
8   ↪ Users/fangjun/py38/lib/python3.8/site-packages/torch/nn/functional.py:1455:0
9   %8 : Long(requires_grad=0, device=cpu) = prim::Constant[value={10}]() # ./fuse_add_
10  ↪ relu.py:9:0
11  %9 : int = prim::Constant[value=1]() # ./fuse_add_relu.py:9:0
12  %10 : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::add(%a, %8, %9) # .
13  ↪ ./fuse_add_relu.py:9:0
14  return (%10)

```

Listing 78: ./code/passes/fuse_add_relu-after.txt

```

1 graph(%self : __torch__.Foo,
2       %x : Float(3, strides=[1], requires_grad=0, device=cpu),
3       %y : Float(3, strides=[1], requires_grad=0, device=cpu)):
4   %5 : int = prim::Constant[value=1]() # ./fuse_add_relu.py:8:0
5   %11 : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::_add_relu(%x, %y,
6   ↪ %5)
7   %8 : Long(requires_grad=0, device=cpu) = prim::Constant[value={10}]() # ./fuse_add_

```

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```

↪relu_.py:9:0
7   %9 : int = prim::Constant[value=1]() # ./fuse_add_relu.py:9:0
8   %10 : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::add(%11, %8, %9) #_
↪./fuse_add_relu.py:9:0
9   return (%10)
10

```

`_jit_pass_fuse_linear`

See https://github.com/pytorch/pytorch/blob/master/torch/csrc/jit/passes/fuse_linear.cpp

Listing 79: `./code/passes/fuse_linear.py`

```

1  #!/usr/bin/env python3
2
3  import torch
4
5
6  class Foo(torch.nn.Module):
7      def forward(self, x: torch.Tensor, w: torch.Tensor, b: torch.Tensor):
8          return torch.matmul(x, w.t()) + b
9
10
11 def main():
12     f = Foo()
13     m = torch.jit.trace(f, (torch.rand(3), torch.rand(3, 3), torch.rand(3)))
14     g = m.graph
15
16     with open("fuse_linear-before.txt", "w") as f:
17         print(g, file=f)
18
19     torch._C._jit_pass_fuse_linear(g)
20
21     with open("fuse_linear-after.txt", "w") as f:
22         print(g, file=f)
23
24
25 if __name__ == "__main__":
26     main()

```

Listing 80: ./code/passes/fuse_linear-before.txt

```

1 graph(%self : __torch__.Foo,
2     %x : Float(3, strides=[1], requires_grad=0, device=cpu),
3     %w : Float(3, 3, strides=[3, 1], requires_grad=0, device=cpu),
4     %b : Float(3, strides=[1], requires_grad=0, device=cpu)):
5     %6 : Float(3, 3, strides=[1, 3], requires_grad=0, device=cpu) = aten::t(%w) # ./
↪fuse_linear.py:8:0
6     %7 : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::matmul(%x, %6) # ./
↪fuse_linear.py:8:0
7     %8 : int = prim::Constant[value=1]() # ./fuse_linear.py:8:0
8     %9 : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::add(%7, %b, %8) # ./
↪fuse_linear.py:8:0
9     return (%9)
10

```

Listing 81: ./code/passes/fuse_linear-after.txt

```

1 graph(%self : __torch__.Foo,
2     %x : Float(3, strides=[1], requires_grad=0, device=cpu),
3     %w : Float(3, 3, strides=[3, 1], requires_grad=0, device=cpu),
4     %b : Float(3, strides=[1], requires_grad=0, device=cpu)):
5     %11 : Tensor? = prim::Constant()
6     %13 : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::linear(%x, %w,
↪%11) # ./fuse_linear.py:8:0
7     %8 : int = prim::Constant[value=1]() # ./fuse_linear.py:8:0
8     %9 : Float(3, strides=[1], requires_grad=0, device=cpu) = aten::add(%13, %b, %8) # .
↪/fuse_linear.py:8:0
9     return (%9)
10

```

9.4.22 ops

aten::add

```

// add_res = a + b * alpha
%add_res = aten::add(%a, %b, %alpha)

// add_res = a + b * alpha
%add_res = aten::add(%a, %b, %alpha, %out)

```

aten::add_

```
// add_res = a * b * alpha
%add_res = aten::add_(%a, %b, %alpha)
```

aten::_add_relu

```
// res = relu(a + b * alpha)
%res = aten::_add_relu(%a, %b, %alpha)

// res = relu(a + b * alpha)
%res = aten::_add_relu(%a, %b, %alpha, %out)
```

aten::add_relu_

```
// res = relu(a + b * alpha)
%res = aten::add_relu_(%a, %b, %alpha)
```

aten::_add_relu_

```
// res = relu(a + b * alpha)
%res = aten::_add_relu_(%a, %b, %alpha)
```

aten::relu

```
%res = aten::relu_(%add_res)
```

9.4.23 Add extra files

Listing 82: ./code/extra-files/test.py

```
1  #!/usr/bin/env python3
2
3  import torch
4  import numpy as np
5
6
7  class Foo(torch.nn.Module):
```

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```

8     def __init__(self):
9         super().__init__()
10        self.x = torch.nn.Parameter(torch.tensor([2.5]))
11
12    def forward(self, y: torch.Tensor):
13        return self.x + y
14
15
16    @torch.inference_mode()
17    def main():
18        f = Foo()
19        print(f(torch.tensor([3.5, 2.0])))
20
21        mean = np.array([1.5, 2.25, -3.8], dtype=np.float32)
22        inv_std = np.array([2.5, 225, 380], dtype=np.float32)
23
24        extra_files = dict()
25        extra_files["window_size"] = "20"
26        extra_files["window_shift"] = "64"
27        extra_files["cmvn_dim"] = f"{mean.shape[0]}"
28        extra_files["cmvn_mean"] = mean.tobytes()
29        extra_files["cmvn_inv_std"] = inv_std.tobytes()
30
31        m = torch.jit.script(f)
32        m.save("m.pt", _extra_files=extra_files)
33
34        saved = dict()
35        saved["window_size"] = ""
36        saved["window_shift"] = ""
37
38        saved["cmvn_dim"] = ""
39        saved["cmvn_mean"] = ""
40        saved["cmvn_inv_std"] = ""
41
42        m2 = torch.jit.load("m.pt", _extra_files=saved)
43        saved["cmvn_mean"] = np.frombuffer(saved["cmvn_mean"], dtype=np.float32)
44        saved["cmvn_inv_std"] = np.frombuffer(saved["cmvn_inv_std"], dtype=np.float32)
45        print(saved)
46        print(type(saved["window_shift"])) # <class 'bytes'>
47
48
49    """
50    tensor([6.0000, 4.5000])

```

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```

51 {'window_size': b'20', 'window_shift': b'64', 'cmvn_dim': b'3', 'cmvn_mean': array([
    ↪ 1.5, 2.25, -3.8 ], dtype=float32), 'cmvn_inv_std': array([ 2.5, 225. , 380. ],
    ↪ dtype=float32)}
52 <class 'bytes'>
53 """
54
55
56 if __name__ == "__main__":
57     main()

```

Listing 83: ./code/extra-files/main.cc

```

1  #include "torch/script.h"
2
3  std::vector<float> ToFloat(const std::string &s) {
4      const float *p = reinterpret_cast<const float *>(s.data());
5      int32_t n = s.size() / 4;
6
7      // assume little endian
8      return {p, p + n};
9  }
10
11 int main() {
12     // see torch/csrc/jit/module.h
13     // see
14     // https://github.com/pytorch/pytorch/blob/main/test/cpp/jit/test_save_load.cpp#L76
15     torch::jit::ExtraFilesMap extra_files{{"window_size", ""},
16                                             {"cmvn_mean", ""},
17                                             {"cmvn_inv_std", ""},
18                                             {"cmvn_dim", ""}};
19     torch::jit::Module m = torch::jit::load("./m.pt", torch::kCPU, extra_files);
20
21     int32_t dim = atoi(extra_files["cmvn_dim"].c_str());
22     if (dim != extra_files["cmvn_mean"].size() / 4 ||
23         dim != extra_files["cmvn_inv_std"].size() / 4) {
24         std::cout << "corrupted data!\n";
25         return -1;
26     }
27
28     auto cmvn_mean = ToFloat(extra_files["cmvn_mean"]);
29     auto cmvn_inv_std = ToFloat(extra_files["cmvn_inv_std"]);
30     int32_t window_size = atoi(extra_files["window_size"].c_str());
31     std::cout << "int window size: " << window_size << "\n";

```

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```

32
33     for (const auto &p : extra_files) {
34         std::cout << p.first << ": " << p.second << "\n";
35     }
36     std::cout << "cmvn_mean: ";
37     for (auto f : cmvn_mean) {
38         std::cout << f << " ";
39     }
40     std::cout << "\n";
41
42     std::cout << "cmvn_inv_std: ";
43     for (auto f : cmvn_inv_std) {
44         std::cout << f << " ";
45     }
46     std::cout << "\n";
47
48     std::cout << "is training: " << m.is_training() << "\n";
49     m.eval();
50     std::cout << "after m.eval(): is training: " << m.is_training() << "\n";
51     torch::Tensor x = torch::tensor({1, 2, 3}, torch::kFloat);
52     torch::Tensor y = m.run_method("forward", x).toTensor();
53     std::cout << y << "\n";
54
55     return 0;
56 }
57 /*
58 int window_size: 20
59 cmvn_inv_std: @aCC
60 cmvn_mean: ?@33s
61 cmvn_dim: 3
62 window_size: 20
63 cmvn_mean: 1.5 2.25 -3.8
64 cmvn_inv_std: 2.5 225 380
65 is training: 1
66 after m.eval(): is training: 0
67     3.5000
68     4.5000
69     5.5000
70 [ CPUFloatType{3} ]
71 */

```

9.5 Logical operations

Listing 84: ./code/logical-op.py

```
1  #!/usr/bin/env python3
2
3  import torch
4
5  a = torch.tensor([float("inf")])
6  b = torch.tensor([float("nan")])
7  assert torch.isinf(a).item() is True
8  assert torch.isnan(a).item() is False
9
10 assert torch.isinf(b).item() is False
11 assert torch.isnan(b).item() is True
12
13 assert torch.logical_or(torch.isinf(a), torch.isnan(b)).item() is True
14
15 assert a.isinf().item() is True
16 assert a.isnan().item() is False
17
18 assert b.isinf().item() is False
19 assert b.isnan().item() is True
```

9.6 Note

To clip gradient, use:

```
tot_norm = torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=5, norm_type=2.
↪ 0)
if torch.logical_or(tot_norm.isnan(), tot_norm.isinf()):
    # skip this update
    continue
else:
    optimizer.step()
```

9.7 Quantization

9.7.1 Internals

<https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/core/QuantizerBase.h> defines the base class `Quantizer`.

<https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/quantized/Quantizer.h> defines the subclasses of `Quantizer`, such as

- `PerTensorAffineQuantizer` - `qscheme` is `kPerTensorAffine`.

QScheme

See <https://github.com/pytorch/pytorch/blob/master/c10/core/QScheme.h>

Listing 85: `./code/qscheme/main.cc`

```

1  #include "torch/script.h"
2
3  static void TestQScheme() {
4      TORCH_CHECK(torch::toString(torch::kPerTensorAffine) == "per_tensor_affine");
5
6      TORCH_CHECK(torch::toString(torch::kPerChannelAffine) ==
7                  "per_channel_affine");
8
9      TORCH_CHECK(torch::toString(torch::kPerTensorSymmetric) ==
10                 "per_tensor_symmetric");
11
12     TORCH_CHECK(torch::toString(torch::kPerChannelSymmetric) ==
13                 "per_channel_symmetric");
14
15     TORCH_CHECK(torch::toString(torch::kPerChannelAffineFloatQParams) ==
16                 "per_channel_affine_float_qparams");
17 }
18
19 int main() {
20     TestQScheme();
21     return 0;
22 }
```

PerTensorAffineQuantizer

It has 4 important methods:

- `QScheme qscheme() const`, always returns `kPerTensorAffine`.
- `double scale() const`
- `int64_t zero_point() const`
- `ScalarType scalar_type() const`

It uses `quantize_tensor_per_tensor_affine_cpu` when `FBGEMM` is available.

Otherwise, it uses <https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/native/quantized/cpu/kernels/QuantizedOpKernels.cpp#L3533>.

- For arm, it uses `quantize_tensor_arm`. It is a template with many specializations.
- For x86, it uses `quantize_val`
 - If `FBGEMM` is available, it uses `quantize_val`
 - Otherwise, it uses <https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/native/quantized/AffineQuantizerBase.cpp#L100>

```
template <typename T>
T quantize_val(double scale, int64_t zero_point, float value) {
    // std::nearbyint results in nearest integer value according to the current
    // rounding mode and the default rounding mode is rounds to even in half-way
    // cases in most popular processor architectures like x86 and ARM. This is
    // typically faster than an alternatives like std::round that rounds half-
    ↪way
    // cases away from zero, and can be consistent with SIMD implementations for
    // example in x86 using _mm512_cvtps_epi32 or mm512_round_ps with
    // _MM_FROUND_CUR_DIRECTION option that also follow the current rounding_
    ↪mode.
    int64_t qvalue;
    constexpr int64_t qmin = std::numeric_limits<typename T::underlying>::min();
    constexpr int64_t qmax = std::numeric_limits<typename T::underlying>::max();
    float inv_scale = 1.0f / static_cast<float>(scale);
    qvalue = static_cast<int64_t>(zero_point + Round(value * inv_scale));
    qvalue = std::max<int64_t>(qvalue, qmin);
    qvalue = std::min<int64_t>(qvalue, qmax);
    return static_cast<T>(qvalue);
}
```

`dequantize_val` is defined as:

```
template <typename T>
TORCH_API float dequantize_val(double scale, int64_t zero_point, T value) {
    return static_cast<float>(scale) * (value.val_ - static_cast<int32_t>(zero_point));
}
```

9.7.2 torch.quantize_per_tensor

See https://pytorch.org/docs/stable/generated/torch.quantize_per_tensor.html

```
def test_quantize_per_tensor():
    scale = 0.1
    zero_point = 1
    a = torch.tensor([10.0, 2.0], dtype=torch.float32)
    q = torch.quantize_per_tensor(
        input=a,
        scale=scale,
        zero_point=zero_point,
        dtype=torch.qint8,
    )
    assert isinstance(q, torch.Tensor)
    assert q.is_quantized is True

    assert q.q_scale() == scale
    assert q.q_zero_point() == zero_point
    assert str(q.qscheme()) == "torch.per_tensor_affine"
    assert q.dtype == torch.qint8

    c = q.int_repr()
    assert c[0] == a[0] / scale + zero_point
    assert c[1] == a[1] / scale + zero_point
    assert torch.all(torch.eq(c, torch.tensor([101, 21], dtype=torch.int8)))
    d = q.dequantize()
    assert d.dtype == torch.float32
    assert torch.all(torch.eq(d, a))

    f = torch.dequantize(q)
    assert torch.all(torch.eq(f, a))
    # print(q)
    """
    tensor([10.,  2.], size=(2,), dtype=torch.qint8,
           quantization_scheme=torch.per_tensor_affine, scale=0.1, zero_point=1)
    """
    assert q[0].item() == 10 # q[0].item() will dequantize() to a float
```

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```

assert q[1].item() == 2
print(type(q[0].item()))
q[0] = 2.5 # Note: it will quantize 2.5 and store it in q
print(q.int_repr())
"""
    tensor([26, 21], dtype=torch.int8)
"""

```

Compress ration

```

def test_size():
    r = torch.rand(100, 100, dtype=torch.float32)
    q = torch.quantize_per_tensor(r, scale=0.1, zero_point=0, dtype=torch.qint8)
    torch.save(r, "float32.pt")
    torch.save(q, "int8.pt")
    float_size = os.path.getsize("float32.pt")
    int8_size = os.path.getsize("int8.pt")
    print("float_size:", float_size)
    print("int8_size:", int8_size)
    print(f"ratio: {float_size}/{int8_size}: {float_size/int8_size:.3f}")
    os.remove("float32.pt")
    os.remove("int8.pt")
    """
        float_size: 40747
        int8_size: 10795
        ratio: 40747/10795: 3.775
    """

```

9.7.3 quantize_per_tensor_dynamic

Listing 86: ./code/quantize_per_tensor_dynamic/main.cc

```

1 // #include "ATen/native/quantized/cpu/QuantUtils.h" // for the latest pytorch
2
3 #include "ATen/native/quantized/cpu/quant_utils.h" // for torch 1.10
4 #include "ATen/ops/quantize_per_tensor_dynamic.h" // needs torch>=1.11
5 #include "torch/script.h"
6
7 // See
8 // https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/native/quantized/cpu/
  ↪ QuantUtils.h#L59

```

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```

9 static void TestChooseQuantizationParams() {
10     quant_utils::TensorQuantizationParams p;
11     p = quant_utils::ChooseQuantizationParams(-1 /*min*/, 2 /*max*/,
12                                               -128 /*qmin*/, 127 /*qmax*/);
13     std::cout << "zero_point: " << p.zero_point << "\n";
14     std::cout << "scale: " << p.scale << "\n";
15     /**
16      * scale = (max - min) / (qmax - qmin) = 3 / 255 = 0.0117647
17      * zero_point_min = qmin - min/scale = -128 - (-1)/scale = -43
18      * zero_point_max = qmax - max/scale = 127 - 2/scale = -43
19      *
20      * min_error = abs(qmin) - abs(min/scale) = 128 - 1/scale = 43
21      * max_error = abs(qmax) - abs(max/scale) = 127 - 2/scale = -43
22      *
23      * zero_point = (min_error < max_error) ? zero_point_min : zero_point_max
24      */
25 }
26
27 // See
28 // https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/native/quantized/
29 //   ↳ QTensor.cpp#L14
30 // and
31 static void TestQuantizePerTensorDynamic() {
32     torch::Tensor r = torch::tensor({-1, 0, 2}, torch::kFloat32);
33     torch::Tensor q = torch::quantize_per_tensor_dynamic(r, torch::kQInt8, false);
34     std::cout << "q: " << q << "\n";
35     #if 0
36     q: -1
37     0
38     2
39     [ QuantizedCPUQInt8Type{3}, qscheme: per_tensor_affine, scale: 0.0117647, zero_point: ↳
40     ↳ -43 ]
41     #endif
42     std::cout << "q.int_repr(): " << q.int_repr() << "\n";
43     #if 0
44     q.int_repr(): -128
45     -43
46     127
47     [ CPUCharType{3} ]
48     #endif
49 }
50
51 int main() {

```

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```
50 TestChooseQuantizationParams();
51 TestQuantizePerTensorDynamic();
52 return 0;
53 }
```

9.7.4 torch.quantize_per_channel

See https://pytorch.org/docs/stable/generated/torch.quantize_per_channel.html#torch.quantize_per_channel

```
def test_quantize_per_channel_2d():
    # (N, C)
    a = torch.tensor(
        [
            [1, 2, 3],
            [4, 5, 6],
        ],
        dtype=torch.float32,
    )
    assert a.shape == (2, 3)
    scales = torch.tensor([0.125, 0.25, 0.5])

    # It will be converted to torch.int64 internally
    zero_points = torch.tensor([10, 20, 30], dtype=torch.int32)
    q = torch.quantize_per_channel(
        input=a,
        scales=scales,
        zero_points=zero_points,
        axis=1,
        dtype=torch.qint8,
    )
    assert q.dtype == torch.qint8

    assert q.q_per_channel_scales().dtype == torch.float64
    assert torch.all(torch.eq(q.q_per_channel_scales(), scales))

    assert q.q_per_channel_zero_points().dtype == torch.int64
    assert torch.all(torch.eq(q.q_per_channel_zero_points(), zero_points))

    assert str(q.qscheme()) == "torch.per_channel_affine"

    assert q.q_per_channel_axis() == 1

    i = q.int_repr()
```

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```

expected_i = torch.tensor([[18, 28, 36], [42, 40, 42]], dtype=torch.int8)
assert i.dtype == torch.int8
assert torch.all(torch.eq(i, expected_i))

assert i[0][0].item() == a[0][0].item() / scales[0] + zero_points[0]
assert i[0][1].item() == a[0][1].item() / scales[1] + zero_points[1]
assert i[0][2].item() == a[0][2].item() / scales[2] + zero_points[2]

assert i[1][0].item() == a[1][0].item() / scales[0] + zero_points[0]
assert i[1][1].item() == a[1][1].item() / scales[1] + zero_points[1]
assert i[1][2].item() == a[1][2].item() / scales[2] + zero_points[2]

d = q.dequantize()
assert torch.all(torch.eq(d, a))

f = torch.dequantize(q)
assert torch.all(torch.eq(f, a))

# print(q)
"""
tensor([[1., 2., 3.],
        [4., 5., 6.]], size=(2, 3), dtype=torch.qint8,
        quantization_scheme=torch.per_channel_affine,
        scale=tensor([0.1250, 0.2500, 0.5000], dtype=torch.float64),
        zero_point=tensor([10, 20, 30]), axis=1)
"""

```

9.7.5 Observer

Listing 87: ./code/observer/ex0.py

```

1  #!/usr/bin/env python3
2
3  import torch
4  from torch.ao.quantization.observer import _with_args, MinMaxObserver
5
6
7  class Foo:
8      def __init__(self, a=1, b=2):
9          self.a = a
10         self.b = b
11

```

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```

12
13 def test_with_args():
14     Foo.with_args = classmethod(_with_args)
15     foo_builder = Foo.with_args(a=3).with_args(b=4).with_args(a=10)
16     f = foo_builder()
17     assert f.a == 10  # the last a=10 replaces the first a=3
18     assert f.b == 4
19
20     f2 = foo_builder()
21     assert id(f) != id(f2)
22
23
24 def test_min_max_observer():
25     ob = MinMaxObserver(dtype=torch.qint8)
26     print(ob)  # MinMaxObserver(min_val=inf, max_val=-inf)
27
28     ob(torch.tensor([1, 2, 3]))
29     print(ob)  # MinMaxObserver(min_val=1.0, max_val=3.0)
30
31     ob(torch.tensor([-1, 30]))
32     print(ob)  # MinMaxObserver(min_val=-1.0, max_val=30.0)
33     scale, zero_point = ob.calculate_qparams()
34     print("scale", scale)  # scale tensor([0.1216])
35     print("zero_point", zero_point)  # zero_point tensor([-120], dtype=torch.int32)
36
37
38 def main():
39     test_with_args()
40     test_min_max_observer()
41
42
43 if __name__ == "__main__":
44     main()

```

9.7.6 Hello

Listing 88: ./code/ex1.py

```

1  #!/usr/bin/env python3
2
3  import torch
4  import torch.nn as nn

```

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```

5
6
7 class Model(torch.nn.Module):
8     def __init__(self):
9         super().__init__()
10        self.fc = nn.Linear(1, 1)
11
12    def forward(self, x):
13        x = self.fc(x)
14        return x
15
16
17 def main():
18     m = Model()
19     model_int8 = torch.quantization.quantize_dynamic(
20         model=m,
21         qconfig_spec={torch.nn.Linear},
22         dtype=torch.qint8,
23     )
24     print(model_int8)
25     print(model_int8.fc)
26     assert model_int8.fc.weight().is_quantized
27     assert model_int8.fc.weight().dtype == torch.qint8
28
29     assert model_int8.fc.bias().is_quantized is False
30     assert model_int8.fc.bias().dtype == torch.float32
31     assert isinstance(model_int8.fc, torch.nn.quantized.dynamic.Linear)
32     print(type(model_int8.fc))
33
34     x = torch.tensor([[1.0]], dtype=torch.float32)
35     y = m(x)
36     print(x, y)  # tensor([[1.]]) tensor([[ -1.2900]], grad_fn=<AddmmBackward0>)
37
38     qy = model_int8(x)
39     print(qy)  # tensor([[ -1.2931]])
40
41
42 if __name__ == "__main__":
43     torch.manual_seed(20220723)
44     main()

```

9.7.7 References

The main implementation is in

<https://github.com/pytorch/pytorch/tree/master/aten/src/ATen/native/quantized>

- Introducing Quantized Tensor
<https://github.com/pytorch/pytorch/wiki/Introducing-Quantized-Tensor>
- Model Quantization for PyTorch (Proposal) #18318
<https://github.com/pytorch/pytorch/issues/18318>
- torch_quantization_design_proposal
https://github.com/pytorch/pytorch/wiki/torch_quantization_design_proposal

Links

- https://github.com/pytorch/pytorch/blob/master/test/quantization/core/test_quantized_tensor.py
- https://github.com/pytorch/pytorch/blob/master/test/quantization/core/experimental/test_quantized_tensor.py
- https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/test/quantized_test.cpp
- <https://github.com/pytorch/pytorch/blob/master/aten/src/ATen/native/quantized/README.md>
- <https://pytorch.org/blog/introduction-to-quantization-on-pytorch/>
- <https://pytorch.org/docs/stable/quantization.html>
- Deep Dive on PyTorch Quantization - Chris Gottbrath
YouTube: <https://www.youtube.com/watch?v=c3MT2qV5f9w>

9.8 android

9.8.1 References

- <https://zhuanlan.zhihu.com/p/54665674>
- Pytorch model to Caffe & ncnn
<https://github.com/starimeL/PytorchConverter>

9.9 onnx

9.9.1 Install

```
pip install onnx onnxruntime
pip install netron # for visualization
# Or go to https://netron.app/
```

API references

- <https://github.com/onnx/onnx/blob/main/docs/PythonAPIOverview.md>
- https://onnxruntime.ai/docs/api/python/api_summary.html#inferencesession

9.9.2 Simple OPs

```
#!/usr/bin/env python3

from typing import Any, Dict

import torch
import torch.nn as nn
from onnxruntime.quantization import QuantType, quantize_dynamic

import onnx

def add_meta_data(filename: str, meta_data: Dict[str, Any]):
    """Add meta data to an ONNX model. It is changed in-place.

    Args:
        filename:
            Filename of the ONNX model to be changed.
        meta_data:
            Key-value pairs.
    """
    model = onnx.load(filename)
    print(model)

    while len(model.metadata_props):
        model.metadata_props.pop()
```

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```
for key, value in meta_data.items():
    meta = model.metadata_props.add()
    meta.key = key
    meta.value = str(value)

onnx.save(model, filename)

class Model(torch.nn.Module):
    def __init__(self):
        super().__init__()
        self.my_linear = nn.Linear(3, 4)
        self.my_relu = torch.nn.ReLU()

    def forward(self, x):
        y = self.my_linear(x)
        y = self.my_relu(y)
        return y

def main():
    model = Model()
    print(list(model.named_parameters()))
    x = torch.rand(2, 5, 3)
    torch.onnx.export(
        model,
        x,
        "model.onnx",
        input_names=["x"],
        output_names=["y"],
        dynamic_axes={
            "x": {0: "N"},
            "y": {0: "N"},
        },
    )
    print(list(model.parameters()))
    meta_data = {
        "date": 20240822,
        "author": "me",
        "version": 10,
    }
    add_meta_data("model.onnx", meta_data)
```

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```

quantize_dynamic(
    model_input="model.onnx",
    model_output="model.int8.onnx",
    op_types_to_quantize=["MatMul", "Add"],
    weight_type=QuantType.QInt8,
)
w = model.my_linear.weight
max_w = w.abs().max().item()
scale = max_w * 2 / (255 - 1)
print(scale)
print((w / scale).to(torch.int8))

print("----")

m = onnx.load("model.onnx")
with open("model.onnx.txt", "w") as f:
    f.write(str(m))

m_int8 = onnx.load("model.int8.onnx")
with open("model.int8.onnx.txt", "w") as f:
    f.write(str(m_int8))

if __name__ == "__main__":
    torch.manual_seed(20240820)
    main()

```

linear

`nn.Linear` has two parameters `weight` and `bias`.

In PyTorch, `weight.shape` is `(out_channels, in_channels)`. But after exporting to onnx, `model.onnx` uses a shape `(in_channels, out_channels)`, which is a transpose of the PyTorch's weight.

To use int8 symmetric quantization for:

```

tensor([[ 0.0928, -0.0400,  0.0666],
        [-0.5535, -0.2698,  0.4867],
        [ 0.5245, -0.3856,  0.3486],
        [ 0.4714,  0.2035, -0.4349]], requires_grad=True)

w = model.my_linear.weight
max_w = w.abs().max().item()
scale = max_w * 2 / 255

```

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```
print(scale)
print((w / scale).to(torch.int8))
```

It prints:

```
0.0043414938683603325
tensor([[ 21,   -9,   15],
        [-127,  -62,  112],
        [ 120,  -88,   80],
        [ 108,   46, -100]], dtype=torch.int8)
```

Note that for symmetric quantization, zero point is 0.

In `model.onnx`, the int8 weights are saved as:

```
[
  [
    21,
    -127,
    120,
    108
  ],
  [
    -9,
    -62,
    -88,
    47
  ],
  [
    15,
    112,
    80,
    -100
  ]
]
```

Before quantization, the float32 weights in `model.onnx` are:

```
[
  [
    0.09278726577758789,
    -0.5535404682159424,
    0.5244883298873901,
    0.4713994264602661
  ],
  [
    0.09278726577758789,
    -0.5535404682159424,
    0.5244883298873901,
    0.4713994264602661
  ],
  [
    0.09278726577758789,
    -0.5535404682159424,
    0.5244883298873901,
    0.4713994264602661
  ]
]
```

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```
[
    -0.04004639387130737,
    -0.26981082558631897,
    -0.3855680227279663,
    0.2035449743270874
],
[
    0.06659042835235596,
    0.48669636249542236,
    0.34855782985687256,
    -0.43489107489585876
]
]
```

The file `model.onnx.txt` is given below:

The file `model.int8.onnx.txt` is given below:

Note that `onnxruntime` uses:

```
val_fp32 = scale * (val_quantized - zero_point)
```

The CPU implementation for `DynamicLinearQuantizer` can be found at

- `DynamicQuantizeLinear<T>::Compute`: <https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/providers/cpu/quantization/dynamicquantizelinear.cc#L23>
- `GetQuantizationParameter`: <https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/util/qmath.h>

It shows how to compute the scale and `zero_point`. Note that it uses asymmetric quantization by default.

```
- scale = (float_max - float_min) / (int_max - int_min)
- zero_point = RoundHalfToEven(int_min - float_min / scale)
```

- `MlasQuantizeLinearKernel`: <https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/mlas/lib/power/QuantizePower.cpp>

It does the actual quantization using MLAS.

9.9.3 Hello

Listing 89: ./code/hello/ex0.py

```
1  #!/usr/bin/env python3
2
3  import torch
4  import torch.nn as nn
5
6
7  class Foo(nn.Module):
8      def __init__(self, i):
9          super().__init__()
10         self.relu = nn.ReLU()
11         self.i = 1
12
13     def forward(self, x):
14         if x.sum().item() > 0:
15             return self.relu(x + 1)
16         else:
17             return self.relu(x + 2)
18
19
20 def main():
21     f = Foo(1)
22     f.eval()  # f.train(False)
23     f = torch.jit.script(f)
24
25     x = torch.rand(2, 3, 4)
26     # [N, T, C]
27     torch.onnx.export(
28         f,
29         x,
30         "f.onnx",
31         verbose=False,
32         input_names=["x"],
33         output_names=["y"],
34         dynamic_axes={"x": {0: "batch_size", 1: "T"}, "y": [0, 1]},
35         # dynamic_axes={"x": [0, 1], "y": [0, 1]},
36     )
37
38
39 if __name__ == "__main__":
40     main()
```

Listing 90: ./code/hello/ex0-1.py

```
1  #!/usr/bin/env python3
2
3  import onnx
4
5
6  def main():
7      model = onnx.load("f.onnx")
8      # print(model)
9      # Check that the model is well formed
10     onnx.checker.check_model(model)
11     # Print a human readable representation of the graph
12     print(onnx.helper.printable_graph(model.graph))
13     onnx.save(model, "f2.onnx")
14
15
16  if __name__ == "__main__":
17     main()
```

Listing 91: ./code/hello/ex0-2.py

```
1  #!/usr/bin/env python3
2
3  import onnxruntime as ort
4  import numpy as np
5
6
7  def main():
8      # https://github.com/microsoft/onnxruntime/issues/10113
9      options = ort.SessionOptions()
10     options.inter_op_num_threads = 1
11     options.intra_op_num_threads = 1
12
13     ort_session = ort.InferenceSession("f.onnx", sess_options=options)
14
15     x = np.arange(24).reshape(2, 3, 4).astype(np.float32)
16     ortvalue = ort.OrtValue.ortvalue_from_numpy(x)
17     assert ortvalue.device_name() == "cpu"
18     assert list(ortvalue.shape()) == list(x.shape)
19     assert ortvalue.data_type() == "tensor(float)"
20     assert ortvalue.is_tensor() is True
21
```

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```

22     results = ort_session.run(["y"], {"x": ortvalue})
23     print(results)
24
25     ort_inputs = {ort_session.get_inputs()[0].name: x}
26     results = ort_session.run(["y"], ort_inputs)
27     print(results)
28
29     results = ort_session.run(["y"], {"x": x})
30     print(results)
31
32     # https://onnxruntime.ai/docs/api/python/api\_summary.html#onnxruntime.NodeArg
33     inputs = ort_session.get_inputs()
34     assert isinstance(inputs, list)
35     assert len(inputs) == 1
36     assert isinstance(inputs[0], ort.NodeArg)
37     print(inputs[0].name, inputs[0].type, inputs[0].shape)
38     assert inputs[0].name == "x"
39     assert inputs[0].type == "tensor(float)"
40     assert inputs[0].shape == ["batch_size", "T", 4]
41
42     outputs = ort_session.get_outputs()
43     assert isinstance(outputs, list)
44     assert isinstance(outputs[0], ort.NodeArg)
45     assert len(outputs) == 1
46     assert outputs[0].name == "y"
47     assert outputs[0].type == "tensor(float)"
48     assert outputs[0].shape == ["y_dynamic_axes_1", "y_dynamic_axes_2", 4]
49
50
51 if __name__ == "__main__":
52     main()

```

9.9.4 Multiple models

Listing 92: ./code/multiple-models/ex.py

```

1  #!/usr/bin/env python3
2
3  import torch
4  import torch.nn as nn
5  import onnx
6  import onnxruntime as ort

```

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```
7 import numpy as np
8 import os
9
10
11 class Foo(nn.Module):
12     def forward(self, x):
13         return x + 1
14
15
16 class Bar(nn.Module):
17     def forward(self, x):
18         return x - 1
19
20
21 def export_to_onnx():
22     x = torch.rand(2, 3, dtype=torch.float32)
23     f = Foo()
24     torch.onnx.export(
25         f,
26         x,
27         "f.onnx",
28         verbose=False,
29         input_names=["x1"],
30         output_names=["y1"],
31         dynamic_axes={
32             "x1": {0: "N", 1: "T"},
33             "y1": {0: "N", 1: "T"},
34         },
35     )
36
37     x = torch.rand(1, dtype=torch.float32)
38     b = Bar()
39     torch.onnx.export(
40         b,
41         x,
42         "b.onnx",
43         verbose=False,
44         input_names=["x2"],
45         output_names=["y2"],
46         dynamic_axes={
47             "x2": {0: "N"},
48             "y2": {0: "N"},
49         },
```

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```
50     )
51
52
53 def merge_models():
54     f = onnx.load("f.onnx")
55     f = onnx.compose.add_prefix(f, prefix="f/")
56     b = onnx.load("b.onnx")
57     combined_model = onnx.compose.merge_models(f, b, io_map={})
58     onnx.save(combined_model, "all.onnx")
59
60
61 def test_merged_model():
62     # https://github.com/microsoft/onnxruntime/issues/10113
63     options = ort.SessionOptions()
64     options.inter_op_num_threads = 1
65     options.intra_op_num_threads = 1
66
67     all_model = onnx.load("all.onnx")
68
69     extractor = onnx.utils.Extractor(all_model)
70
71     f = extractor.extract_model(input_names=["f/x1"], output_names=["f/y1"])
72     f_session = ort.InferenceSession(f.SerializeToString(), sess_options=options)
73     f_inputs = f_session.get_inputs()
74     f_out = f_session.run(["f/y1"], {"f/x1": np.array([[1, 3]], dtype=np.float32)})
75     print(f_out[0]) # [[2. 4.]]
76
77     b = extractor.extract_model(input_names=["x2"], output_names=["y2"])
78     b_session = ort.InferenceSession(b.SerializeToString(), sess_options=options)
79     b_inputs = b_session.get_inputs()
80     b_out = b_session.run(["y2"], {"x2": np.array([1, 3], dtype=np.float32)})
81     print(b_out[0]) # [0. 2.]
82
83
84 def main():
85     export_to_onnx()
86     merge_models()
87     test_merged_model()
88     os.remove("f.onnx")
89     os.remove("b.onnx")
90     os.remove("all.onnx")
91
92
```

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```

93 if __name__ == "__main__":
94     main()

```

We can first merge multiple models into one and then extract them.

9.9.5 References

- (OPTIONAL) EXPORTING A MODEL FROM PYTORCH TO ONNX AND RUNNING IT USING ONNX RUNTIME

https://pytorch.org/tutorials/advanced/super_resolution_with_onnxruntime.html

- Dynamic dummy input when exporting a PyTorch model? #654

<https://github.com/onnx/onnx/issues/654>

- onnxruntime latest version segment fault #10113

<https://github.com/microsoft/onnxruntime/issues/10113>

9.9.6 Quantization

TO READ

- 2017-Mixed precision training, See [MNA+17].
- Achieving FP32 Accuracy for INT8 Inference Using Quantization Aware Training with NVIDIA TensorRT

A blog, <https://developer.nvidia.com/blog/achieving-fp32-accuracy-for-int8-inference-using-quantization-aware-training-with-tensorrt>

- 2019-Neural network distiller: A python package for dnn compression research

See :cite:p`zmora2019neural`. It is from Intel.

It also has a github repo at <https://github.com/IntelLabs/distiller>

Read its code about quantization! It has 4.3k stars!

- 2011-Improving the speed of neural networks on CPUs

See [VSM+11]. It is from Google.

Has stuff about SSE.

2020-Integer quantization for deep learning inference: Principles and empirical evaluation

See [WJZ+20].

This paper is from NVidia. It lists a table comparing throughput of fp16, int8, int4, int1 on NVIDIA Turing GPUs.

An example:

```
#!/usr/bin/env python3

import math

def test_asymmetric_quant():
    # We use the notation from
    # https://arxiv.org/pdf/2004.09602
    # equation 1 and 2
    #
    #  $f(x) = sx + z$ 
    # it is called asymmetric because
    # float_low != -float_high
    # and
    # int_low != -int_high
    #
    # It is also called affine quantization in this paper

    float_low = -3
    float_high = 4

    int_low = -128
    int_high = 127

    s = (int_high - int_low) / (float_high - float_low) # 36.4285
    z1 = -float_low * s - 127 # -17.71
    z = -round(float_low * s) - 127 # -18
    print(s, z1, z)

    f = lambda x: round(s * x) + z

    print(f(float_low), f(float_high), f(1)) # -127, 128, 18

def test_symmetric_quant():
    # We use the notation from
    # https://arxiv.org/pdf/2004.09602
    # equation 1 and 2
```

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```
#
#  $f(x) = sx + z$ , where  $z$  is 0
# it is called symmetric because
#  $\text{float\_low} == -\text{float\_high}$ 
# and
#  $\text{int\_low} == -\text{int\_high}$ 
#
# It is also called affine quantization in this paper

float_low = -4
float_high = 4

int_low = -127
int_high = 127

s = (int_high - int_low) / (float_high - float_low) # 36.4285
z1 = -float_low * s - 127 # -17.71
z = -round(float_low * s) - 127 # -18
print(s, z1, z) # 31.75 0.0 0

f = lambda x: round(s * x) + z

print(f(float_low), f(float_high), f(1)) # -127, 127, 32

def main():
    print("---asymmetric---")
    test_asymmetric_quant()
    print("---symmetric---")
    test_symmetric_quant()

if __name__ == "__main__":
    main()
```

2022-A survey of quantization methods for efficient neural network inference

See [GKD+22]

1998-Quantization

See [GN98].

Quantization

9.9.7 Quantization

- FP8 FORMATS FOR DEEP LEARNING
<https://arxiv.org/pdf/2209.05433>
- A study of BFLOAT16 for deep learning training
<https://arxiv.org/abs/1905.12322>

9.9.8 TODOs

- <https://github.com/microsoft/onnxruntime-extensions>
- <https://github.com/onnx/tutorials/blob/main/PyTorchCustomOperator/README.md>
- <https://github.com/microsoft/onnxscript>

9.10 triton

9.10.1 Hello

test-cuda

Listing 93: ./code/test-cuda.py

```
1  #!/usr/bin/env python3
2
3  import torch
4
5  """
6  L2_cache_size: 6291456
7  gcnArchName: Tesla V100-PCIE-32GB
```

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```

8  is_integrated: 0
9  is_multi_gpu_board: 0
10 major: 7
11 max_threads_per_multi_processor: 2048
12 minor: 0
13 multi_processor_count: 80
14 name: Tesla V100-PCIE-32GB
15 regs_per_multiprocessor: 65536
16 shared_memory_per_block: 49152
17 shared_memory_per_block_optin: 98304
18 shared_memory_per_multiprocessor: 98304
19 total_memory: 34079899648
20 uuid: 2e9d29fc-608b-1348-6c9d-c190dc3bddbe
21 warp_size: 32
22 """
23
24
25 def test():
26     # print(help(torch.cuda.get_device_properties("cuda:0")))
27     props = torch.cuda.get_device_properties(0)
28     for attr in dir(props):
29         # skip internal/private attributes
30         if not attr.startswith("_"):
31             print(f"{attr}: {getattr(props, attr)}")
32
33
34 def main():
35     test()
36
37
38 if __name__ == "__main__":
39     main()

```

vector additon

Listing 94: ./code/vector_addition.py

```

1  #!/usr/bin/env python3
2
3  import torch
4  import triton
5  import triton.language as tl

```

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```

6
7 # e.g., cuda:0
8 DEVICE = triton.runtime.driver.active.get_active_torch_device()
9 assert isinstance(DEVICE, torch.device), type(DEVICE)
10
11
12 @triton.jit
13 def add_kernel(x_ptr, y_ptr, output_ptr, n_elements, BLOCK_SIZE: tl.constexpr):
14     pid = tl.program_id(axis=0)
15     block_start = pid * BLOCK_SIZE
16     offsets = block_start + tl.arange(0, BLOCK_SIZE)
17     mask = offsets < n_elements
18     x = tl.load(x_ptr + offsets, mask=mask)
19     y = tl.load(y_ptr + offsets, mask=mask)
20     output = x + y
21     tl.store(output_ptr + offsets, output, mask=mask)
22
23
24 def add(x: torch.Tensor, y: torch.Tensor):
25     output = torch.empty_like(x)
26     assert x.device == DEVICE and y.device == DEVICE and output.device == DEVICE, (
27         x.device,
28         y.device,
29         output.device,
30     )
31     n_elements = output.numel()
32     grid = lambda meta: (triton.cdiv(n_elements, meta["BLOCK_SIZE"]),)
33     add_kernel[grid](x, y, output, n_elements, BLOCK_SIZE=1024)
34     return output
35
36
37 torch.manual_seed(0)
38 size = 98432
39 x = torch.rand(size, device=DEVICE)
40 y = torch.rand(size, device=DEVICE)
41 output_torch = x + y
42 output_triton = add(x, y)
43 print(output_torch[:10])
44 print(output_triton[:10])
45 print(
46     f"The maximum difference between torch and triton is "
47     f"{torch.max(torch.abs(output_torch - output_triton))}"
48 )

```

vector subtraction

Listing 95: ./code/vector_sub.py

```

1  #!/usr/bin/env python3
2  import torch
3  import triton
4  import triton.language as tl
5
6  device = torch.device("cuda", 0)
7
8
9  @triton.jit
10 def sub_kernel(x_ptr, y_ptr, output_ptr, n_elements, BLOCK_SIZE: tl.constexpr):
11     start = tl.program_id(0)
12     offset = start * BLOCK_SIZE + tl.arange(0, BLOCK_SIZE)
13     mask = offset < n_elements
14     x = tl.load(x_ptr + offset, mask=mask)
15     y = tl.load(y_ptr + offset, mask=mask)
16     z = x - y
17     tl.store(output_ptr + offset, z, mask=mask)
18
19
20 def sub(x: torch.Tensor, y: torch.Tensor) -> torch.Tensor:
21     assert x.device == device, (x.device, device)
22     assert y.device == device, (y.device, device)
23     n = x.nelement()
24     grid = lambda meta: (triton.cdiv(n, meta["BLOCK_SIZE"]),)
25     z = torch.empty_like(x)
26     sub_kernel[grid](x, y, z, n, BLOCK_SIZE=1024)
27     return z
28
29
30 def main():
31     n = 10
32     x = torch.rand(n, device=device)
33     y = torch.rand(n, device=device)
34     z0 = x - y
35     z1 = sub(x, y)
36     print(z0)
37     print(z1)
38     print((z0 - z1).abs().max())
39
40

```

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```
41 if __name__ == "__main__":  
42     main()
```

vector add scalar

Listing 96: ./code/vector_add_scalar.py

```
1  #!/usr/bin/env python3  
2  import torch  
3  import triton  
4  import triton.language as tl  
5  
6  device = torch.device("cuda", 0)  
7  
8  
9  @triton.jit  
10 def vector_add_scalar_kernel(x_ptr, y_ptr, scalar, n, BLOCK_SIZE: tl.constexpr):  
11     start = tl.program_id(0)  
12     offset = start * BLOCK_SIZE + tl.arange(0, BLOCK_SIZE)  
13     mask = offset < n  
14     x = tl.load(x_ptr + offset, mask=mask)  
15     y = x + scalar  
16     tl.store(y_ptr + offset, y, mask=mask)  
17  
18  
19 def vector_add_scalar(x: torch.Tensor, scalar: float):  
20     assert x.device == device, (x.device, device)  
21     y = torch.empty_like(x)  
22     n = x.nelement()  
23     grid = lambda meta: (triton.cdiv(n, meta["BLOCK_SIZE"]),)  
24     vector_add_scalar_kernel[grid](x, y, scalar, n, BLOCK_SIZE=4096)  
25     return y  
26  
27  
28 def main():  
29     x = torch.randn(10240, device=device)  
30     scalar = 2.25  
31     y0 = x + scalar  
32     y1 = vector_add_scalar(x, scalar)  
33     # print(x)  
34     # print(y0)  
35     # print(y1)
```

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```

36     print((y0 - y1).abs().max())
37
38
39 if __name__ == "__main__":
40     main()

```

matrix add matrix

Listing 97: ./code/matrix_add_matrix.py

```

1  #!/usr/bin/env python3
2  import torch
3  import triton
4  import triton.language as tl
5
6  device = torch.device("cuda", 0)
7
8
9  @triton.jit
10 def matrix_add_matrix_kernel(
11     x_ptr,
12     y_ptr,
13     out_ptr,
14     num_rows,
15     num_cols,
16     BLOCK_SIZE_ROW: tl.constexpr,
17     BLOCK_SIZE_COL: tl.constexpr,
18 ):
19     row_start = tl.program_id(0)
20     col_start = tl.program_id(1)
21
22     row_offset = row_start * BLOCK_SIZE_ROW + tl.arange(0, BLOCK_SIZE_ROW)[: , None]
23     col_offset = col_start * BLOCK_SIZE_COL + tl.arange(0, BLOCK_SIZE_COL)[None, :]
24     mask = (row_offset < num_rows) & (col_offset < num_cols)
25
26     offset = row_offset * num_cols + col_offset
27
28     x = tl.load(x_ptr + offset, mask=mask)
29     y = tl.load(y_ptr + offset, mask=mask)
30     out = x + y
31     tl.store(out_ptr + offset, out, mask=mask)
32

```

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```

33
34 def matrix_add_matrix(x: torch.Tensor, y: torch.Tensor):
35     assert x.device == device, (x.device, device)
36     assert y.device == device, (y.device, device)
37     z = torch.empty_like(x)
38     num_rows = x.shape[0]
39     num_cols = x.shape[1]
40     grid = lambda meta: (
41         triton.cdiv(num_rows, meta["BLOCK_SIZE_ROW"]),
42         triton.cdiv(num_cols, meta["BLOCK_SIZE_COL"]),
43     )
44     matrix_add_matrix_kernel[grid](
45         x, y, z, num_rows, num_cols, BLOCK_SIZE_ROW=16, BLOCK_SIZE_COL=32
46     )
47     return z
48
49
50 def main():
51     x = torch.randn(301, 529, device=device)
52     y = torch.randn(301, 529, device=device)
53     z0 = x + y
54     z1 = matrix_add_matrix(x, y)
55     # print(x)
56     # print(y)
57     # print(z0)
58     # print(z1)
59     print((z0 - z1).abs().max())
60
61
62 if __name__ == "__main__":
63     torch.manual_seed(20250810)
64     main()

```

batched matrix add matrix

Listing 98: ./code/batched_matrix_add.py

```

1  #!/usr/bin/env python3
2  import torch
3  import triton
4  import triton.language as tl
5

```

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```

6 device = torch.device("cuda", 0)
7
8
9 @triton.jit
10 def add_kernel(
11     a_ptr,
12     b_ptr,
13     c_ptr,
14     num_rows,
15     num_cols,
16     BLOCK_SIZE_ROW: tl.constexpr,
17     BLOCK_SIZE_COL: tl.constexpr,
18 ):
19     batch_start = tl.program_id(0)
20     row_start = tl.program_id(1)
21     col_start = tl.program_id(2)
22
23     batch_offset = batch_start * num_rows * num_cols
24
25     row_offset = row_start * BLOCK_SIZE_ROW + tl.arange(0, BLOCK_SIZE_ROW)[:, None]
26     col_offset = col_start * BLOCK_SIZE_COL + tl.arange(0, BLOCK_SIZE_COL)[None, :]
27
28     mask = (row_offset < num_rows) & (col_offset < num_cols)
29
30     offset = batch_offset + row_offset * num_cols + col_offset
31     a = tl.load(a_ptr + offset, mask=mask)
32     b = tl.load(b_ptr + offset, mask=mask)
33     c = a + b
34     tl.store(c_ptr + offset, c, mask=mask)
35
36
37 def batched_add(a: torch.Tensor, b: torch.Tensor):
38     assert a.device == device, (a.device, device)
39     assert b.device == device, (b.device, device)
40
41     n, r, c = a.shape
42     BLOCK_SIZE_ROW = 32
43     BLOCK_SIZE_COL = 32
44     grid = (n, triton.cdiv(r, BLOCK_SIZE_ROW), triton.cdiv(c, BLOCK_SIZE_COL))
45     out = torch.empty_like(a)
46     add_kernel[grid](a, b, out, r, c, BLOCK_SIZE_ROW, BLOCK_SIZE_COL)
47     return out
48

```

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```
49
50 def main():
51     a = torch.randn(5, 10, 20, device=device)
52     b = torch.randn(5, 10, 20, device=device)
53     c0 = a + b
54     c1 = batched_add(a, b)
55     print((c0 - c1).abs().max())
56
57
58 if __name__ == "__main__":
59     main()
```

matrix multiplication

Listing 99: ./code/matrix_mul.py

```
1  #!/usr/bin/env python3
2
3  import torch
4  import triton.language as tl
5  import triton
6
7
8  def is_cuda():
9      return triton.runtime.driver.active.get_current_target().backend == "cuda"
10
11
12 def get_cuda_autotune_config():
13     return [
14         triton.Config(
15             {
16                 "BLOCK_SIZE_M": 128,
17                 "BLOCK_SIZE_N": 256,
18                 "BLOCK_SIZE_K": 64,
19                 "GROUP_SIZE_M": 8,
20             },
21             num_stages=3,
22             num_warps=8,
23         ),
24         triton.Config(
25             {
26                 "BLOCK_SIZE_M": 64,
```

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```
27         "BLOCK_SIZE_N": 256,  
28         "BLOCK_SIZE_K": 32,  
29         "GROUP_SIZE_M": 8,  
30     },  
31     num_stages=4,  
32     num_warps=4,  
33 ),  
34 triton.Config(  
35     {  
36         "BLOCK_SIZE_M": 128,  
37         "BLOCK_SIZE_N": 128,  
38         "BLOCK_SIZE_K": 32,  
39         "GROUP_SIZE_M": 8,  
40     },  
41     num_stages=4,  
42     num_warps=4,  
43 ),  
44 triton.Config(  
45     {  
46         "BLOCK_SIZE_M": 128,  
47         "BLOCK_SIZE_N": 64,  
48         "BLOCK_SIZE_K": 32,  
49         "GROUP_SIZE_M": 8,  
50     },  
51     num_stages=4,  
52     num_warps=4,  
53 ),  
54 triton.Config(  
55     {  
56         "BLOCK_SIZE_M": 64,  
57         "BLOCK_SIZE_N": 128,  
58         "BLOCK_SIZE_K": 32,  
59         "GROUP_SIZE_M": 8,  
60     },  
61     num_stages=4,  
62     num_warps=4,  
63 ),  
64 triton.Config(  
65     {  
66         "BLOCK_SIZE_M": 128,  
67         "BLOCK_SIZE_N": 32,  
68         "BLOCK_SIZE_K": 32,  
69         "GROUP_SIZE_M": 8,
```

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```
70         },
71         num_stages=4,
72         num_warps=4,
73     ),
74     triton.Config(
75         {
76             "BLOCK_SIZE_M": 64,
77             "BLOCK_SIZE_N": 32,
78             "BLOCK_SIZE_K": 32,
79             "GROUP_SIZE_M": 8,
80         },
81         num_stages=5,
82         num_warps=2,
83     ),
84     triton.Config(
85         {
86             "BLOCK_SIZE_M": 32,
87             "BLOCK_SIZE_N": 64,
88             "BLOCK_SIZE_K": 32,
89             "GROUP_SIZE_M": 8,
90         },
91         num_stages=5,
92         num_warps=2,
93     ),
94     # Good config for fp8 inputs.
95     triton.Config(
96         {
97             "BLOCK_SIZE_M": 128,
98             "BLOCK_SIZE_N": 256,
99             "BLOCK_SIZE_K": 128,
100             "GROUP_SIZE_M": 8,
101         },
102         num_stages=3,
103         num_warps=8,
104     ),
105     triton.Config(
106         {
107             "BLOCK_SIZE_M": 256,
108             "BLOCK_SIZE_N": 128,
109             "BLOCK_SIZE_K": 128,
110             "GROUP_SIZE_M": 8,
111         },
112         num_stages=3,
```

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```
113         num_warps=8,
114     ),
115     triton.Config(
116         {
117             "BLOCK_SIZE_M": 256,
118             "BLOCK_SIZE_N": 64,
119             "BLOCK_SIZE_K": 128,
120             "GROUP_SIZE_M": 8,
121         },
122         num_stages=4,
123         num_warps=4,
124     ),
125     triton.Config(
126         {
127             "BLOCK_SIZE_M": 64,
128             "BLOCK_SIZE_N": 256,
129             "BLOCK_SIZE_K": 128,
130             "GROUP_SIZE_M": 8,
131         },
132         num_stages=4,
133         num_warps=4,
134     ),
135     triton.Config(
136         {
137             "BLOCK_SIZE_M": 128,
138             "BLOCK_SIZE_N": 128,
139             "BLOCK_SIZE_K": 128,
140             "GROUP_SIZE_M": 8,
141         },
142         num_stages=4,
143         num_warps=4,
144     ),
145     triton.Config(
146         {
147             "BLOCK_SIZE_M": 128,
148             "BLOCK_SIZE_N": 64,
149             "BLOCK_SIZE_K": 64,
150             "GROUP_SIZE_M": 8,
151         },
152         num_stages=4,
153         num_warps=4,
154     ),
155     triton.Config(
```

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```
156         {
157             "BLOCK_SIZE_M": 64,
158             "BLOCK_SIZE_N": 128,
159             "BLOCK_SIZE_K": 64,
160             "GROUP_SIZE_M": 8,
161         },
162         num_stages=4,
163         num_warps=4,
164     ),
165     triton.Config(
166         {
167             "BLOCK_SIZE_M": 128,
168             "BLOCK_SIZE_N": 32,
169             "BLOCK_SIZE_K": 64,
170             "GROUP_SIZE_M": 8,
171         },
172         num_stages=4,
173         num_warps=4,
174     ),
175 ][:2]
176
177
178 device = torch.device("cuda", 0)
179
180
181 def get_autotune_config():
182     assert is_cuda()
183     return get_cuda_autotune_config()
184
185
186 @triton.autotune(
187     configs=get_autotune_config(),
188     key=["M", "N", "K"],
189 )
190 @triton.jit
191 def matmul_kernel(
192     a_ptr,
193     b_ptr,
194     c_ptr,
195     M,
196     N,
197     K,
198     stride_am,
```

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```

199     stride_ak,
200     stride_bk,
201     stride_bn,
202     stride_cm,
203     stride_cn,
204     BLOCK_SIZE_M: tl.constexpr,
205     BLOCK_SIZE_N: tl.constexpr,
206     BLOCK_SIZE_K: tl.constexpr,
207     GROUP_SIZE_M: tl.constexpr,
208     ACTIVATION: tl.constexpr,
209 ):
210     pid = tl.program_id(axis=0)
211     num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
212     num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
213     num_pid_in_group = GROUP_SIZE_M * num_pid_n
214     group_id = pid // num_pid_in_group
215     first_pid_m = group_id * GROUP_SIZE_M
216     group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
217     pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)
218     pid_n = (pid % num_pid_in_group) // group_size_m
219
220     tl.assume(pid_m >= 0)
221     tl.assume(pid_n >= 0)
222     tl.assume(stride_am >= 0)
223     tl.assume(stride_ak >= 0)
224     tl.assume(stride_bk >= 0)
225     tl.assume(stride_bn >= 0)
226     tl.assume(stride_cm >= 0)
227     tl.assume(stride_cn >= 0)
228
229     offs_am = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M
230     offs_bn = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
231     offs_k = tl.arange(0, BLOCK_SIZE_K)
232
233     a_ptrs = a_ptr + (offs_am[:, None] * stride_am + offs_k[None, :] * stride_ak)
234     b_ptrs = b_ptr + (offs_k[:, None] * stride_bk + offs_bn[None, :] * stride_bn)
235     accumulator = tl.zeros((BLOCK_SIZE_M, BLOCK_SIZE_N), dtype=tl.float32)
236
237     for k in range(0, tl.cdiv(K, BLOCK_SIZE_K)):
238         a = tl.load(a_ptrs, mask=offs_k[None, :] < K - k * BLOCK_SIZE_K, other=0.0)
239         b = tl.load(b_ptrs, mask=offs_k[:, None] < K - k * BLOCK_SIZE_K, other=0.0)
240         accumulator = tl.dot(a, b, accumulator)
241

```

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```

242     a_ptrs += BLOCK_SIZE_K * stride_ak
243     b_ptrs += BLOCK_SIZE_K * stride_bk
244     if ACTIVATION == "leaky_relu":
245         accumulator = leaky_relu(accumulator)
246
247     c = accumulator.to(tl.float16)
248
249     offs_cm = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
250     offs_cn = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
251     c_ptrs = c_ptr + (offs_cm[:, None] * stride_cm + offs_cn[None, :] * stride_cn)
252     c_mask = (offs_cm[:, None] < M) & (offs_cn[None, :] < N)
253     tl.store(c_ptrs, c, mask=c_mask)
254
255
256 @triton.jit
257 def leaky_relu(x):
258     return tl.where(x >= 0, x, 0.01 * x)
259
260
261 def matmul(a, b, activation=""):
262     assert a.device == device, (a.device, device)
263     assert b.device == device, (b.device, device)
264     assert a.shape[1] == b.shape[0], (a.shape, b.shape)
265     assert a.is_contiguous()
266
267     M, K = a.shape
268     _, N = b.shape
269
270     c = torch.empty((M, N), device=device, dtype=torch.float16)
271     grid = lambda meta: (
272         triton.cdiv(M, meta["BLOCK_SIZE_M"]) * triton.cdiv(N, meta["BLOCK_SIZE_N"]),
273     )
274     matmul_kernel[grid](
275         a,
276         b,
277         c,
278         M,
279         N,
280         K,
281         a.stride(0),
282         a.stride(1),
283         b.stride(0),
284         b.stride(1),

```

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```

285         c.stride(0),
286         c.stride(1),
287         ACTIVATION=activation,
288     )
289     return c
290
291
292 def main():
293     a = torch.randn(512, 512, device=device, dtype=torch.float16) - 0.5
294     b = torch.randn(512, 512, device=device, dtype=torch.float16) - 0.5
295     c0 = torch.matmul(a, b)
296     # c1 = matmul(a, b, activation="leaky_relu")
297     c1 = matmul(a, b)
298     print((c0 - c1).abs().max())
299
300
301 if __name__ == "__main__":
302
303     torch.manual_seed(20250811)
304     main()

```

9.10.2 References

- https://github.com/ModelTC/LightLLM/tree/main/lightllm/common/basemodel/triton_kernel
- <https://github.com/unslothai/unsloth/tree/main/unsloth/kernels>
- GPU Teaching Kit - Accelerated Computing
<http://gputeachingkit.hwu.crhc.illinois.edu/>
It has videos, slides
- Simplifying CUDA kernels with Triton: A Pythonic Approach to GPU Programming
- NVIDIA Ampere Architecture In-Depth
- The Deep Learning Compiler: A Comprehensive Survey
<https://arxiv.org/pdf/2002.03794>
- <https://github.com/tugot17/pmpp>
Complete solutions to the Programming Massively Parallel Processors Edition 4
- Lecture 1 How to profile CUDA kernels in PyTorch
https://www.youtube.com/watch?v=LuhJEEJQgUM&t=2200s&ab_channel=GPUMODE

- <https://github.com/gpu-mode/lectures>
- Programming Massively Parallel Processors (PMPP)
<https://stevengong.co/notes/Programming-Massively-Parallel-Processors>
It has lots of notes

9.10.3 vector addition

9.11 nn.LSTM

See <https://pytorch.org/docs/stable/generated/torch.nn.LSTM.html>

Listing 100: ./code/lstm-test.py

```
1  #!/usr/bin/env python3
2
3  import torch
4  import torch.nn as nn
5
6
7  """
8  self.lstm = LSTM(
9          input_size=2,
10         hidden_size=5,
11         num_layers=1,
12         bias=True,
13         proj_size=2,
14 )
15
16 lstm.weight_ih_l0 [20, 2]
17 lstm.weight_hh_l0 [20, 2]
18 lstm.bias_ih_l0 [20]
19 lstm.bias_hh_l0 [20]
20 lstm.weight_hr_l0 [2, 5]
21 """
22
23
24 class Foo(nn.Module):
25     def __init__(self):
26         super().__init__()
27         self.lstm = nn.LSTM(
28                 input_size=3,
29                 hidden_size=5,
```

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```

30         num_layers=1,
31         bias=True,
32         proj_size=4,
33     )
34
35     def forward(self, x, h0, c0):
36         """
37         Args:
38             x:
39                 (T, N, H_in), H_in is input dimension of x
40             h0:
41                 (num_layers, N, H_out), H_out is proj_size
42             c0:
43                 (num_layers, N, H_cell), H_cell is hidden_dim
44         """
45         y, (hx, cx) = self.lstm(x, (h0, c0))
46         return y, hx, cx
47
48
49 @torch.no_grad()
50 def main():
51     f = Foo()
52     dim_in = 3
53     dim_proj = 4
54     dim_hidden = 5
55     x = torch.rand(1, 1, dim_in)
56     h0 = torch.rand(1, 1, dim_proj)
57     c0 = torch.rand(1, 1, dim_hidden)
58     y, hx, cx = f(x, h0, c0)
59
60     w_ih = f.state_dict()["lstm.weight_ih_l0"]
61     w_hh = f.state_dict()["lstm.weight_hh_l0"]
62
63     b_ih = f.state_dict()["lstm.bias_ih_l0"]
64     b_hh = f.state_dict()["lstm.bias_hh_l0"]
65
66     w_hr = f.state_dict()["lstm.weight_hr_l0"]
67
68     w_ii, w_if, w_ig, w_io = w_ih.split(5, dim=0)
69     w_hi, w_hf, w_hg, w_ho = w_hh.split(5, dim=0)
70
71     b_ii, b_if, b_ig, b_io = b_ih.split(5, dim=0)
72     b_hi, b_hf, b_hg, b_ho = b_hh.split(5, dim=0)

```

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```

73
74     print(y, hx, cx)
75     print(y.shape)
76     print(hx.shape)
77     print(cx.shape)
78
79     i_gate = (x @ w_ii.t() + b_ii + h0 @ w_hi.t() + b_hi).sigmoid()
80     f_gate = (x @ w_if.t() + b_if + h0 @ w_hf.t() + b_hf).sigmoid()
81     g_gate = (x @ w_ig.t() + b_ig + h0 @ w_hg.t() + b_hg).tanh()
82     o_gate = (x @ w_io.t() + b_io + h0 @ w_ho.t() + b_ho).sigmoid()
83     c = f_gate * c0 + i_gate * g_gate
84
85     h = o_gate * c.tanh()
86     h = h @ w_hr.t()
87
88     print(h, h, c)
89
90
91 if __name__ == "__main__":
92     torch.manual_seed(20220903)
93     main()

```

9.12 torchvision

1. How is the C++ code compiled and how is it loaded in PyTorch.

from torchvision.io import read_file to read a binary file and it returns a 1-d tensor of dtype torch.uint8.

There is also a write_file to write a 1-d tensor of dtype torch.uint8.

from torchvision.io import read_image to read an image. Supported formats are jpg, png, gif.

Color images are represented as a uint8 tensor of shape (channels, height, width).

transforms.v2._utils contains functions for checking and parsing sizes, paddings, fill.

Use v2.ToDtype(torch.float32, scale=True) to convert dtype.

- RandomAffine calls F.affine. RandomAffine is in transforms/v2/_geometry.py. Affine has 4 parameters: angle for rotation, translate, scale, shear. RandomAffine._get_params() just return the 4 parameters. Note that we can call the class method RandomAffine.get_params(), which is defined in transform/transform.py
- F.affine() is defined in transforms/v2/functional/_geometry.py with name def affine_image_get_inverse_affine_matrix() is used to get the matrix for affine transform.

Inside `F.affine()`, it calls `_affine_grid` to get a grid from the affine transform matrix and then use `_apply_grid_transform` to transform the input.

TODO: look at `torch.nn.functional.affine_grid()` and `grid_sample()`. See also <https://github.com/wuneng/WarpAffine2GridSample/blob/master/main.py#L56>

Deep Learning Paper Implementations: Spatial Transformer Networks - Part I

9.13 autograd

See

- <https://pytorch.org/docs/stable/notes/autograd.html>
- <https://dev-discuss.pytorch.org/t/how-to-read-the-autograd-codebase/383>
- <https://pytorch.org/docs/stable/notes/extending.html>

FAQ:

- pack and unpack hooks for saved tensors. See <https://pytorch.org/docs/stable/notes/autograd.html#hooks-for-saved-tensors>

9.13.1 `_saved_self`, `_saved_other`, `_saved_result`, `_saved_exponent`

1. `x*2`, multiply with a scalar

```
x = torch.tensor([1.2, 3], requires_grad=True)
y = x*2
print(y.grad_fn.__class__) # '<MulBackward0>'
print(y.grad_fn._saved_self) # None
print(y.grad_fn._saved_other) # tensor(2)
```

2. `x * a`, multiply with a tensor that requires grad

```
x = torch.tensor([1.2, 3], requires_grad=True)
a = torch.tensor([10.2], requires_grad=True)
y = x*a
print(y.grad_fn.__class__) # '<MulBackward0>'
print(y.grad_fn._saved_self) # tensor([1.2, 3], requires_grad=True)
print(y.grad_fn._saved_other) # tensor([10.2], requires_grad=True)

print(y.grad_fn._saved_self is x) # True
print(y.grad_fn._saved_other is a) # True
```

3. `x * a`, multiply with a tensor that does not require grad

```
x = torch.tensor([1.2, 3], requires_grad=True)
a = torch.tensor([10.2])
y = x*a
print(y.grad_fn.__class__) # '<MulBackward0>'
print(y.grad_fn._saved_self) # None
print(y.grad_fn._saved_other) # tensor([10.2])

print(y.grad_fn._saved_other is a) # True
```

4. `sin(x)`

```
x = torch.tensor([1.2, 3], requires_grad=True)
y = x.sin()
print(y.grad_fn.__class__) # '<SinBackward0>'
print(y.grad_fn._saved_self is x) # True
```

5. `x.relu`

```
x = torch.tensor([1.2, 3], requires_grad=True)
y = x.relu()
print(y.grad_fn.__class__) # '<ReluBackward0>'
print(y.grad_fn._saved_result.equal(y)) # True
print(y.grad_fn._saved_result is y) # False
```

6. `x.log()`

```
x = torch.tensor([1.2, 3], requires_grad=True)
y = x.log()
print(y.grad_fn.__class__) # '<LogBackward0>'
print(y.grad_fn._saved_self is x) # True
```

9.13.2 Example 1

9.13.3 Where is Mulbackward0 defined

- `tools/autograd/derivatives.yaml`

See <https://github.com/pytorch/pytorch/blob/main/tools/autograd/derivatives.yaml#L1197>

After building PyTorch, it generates several files

- `torch/csrc/autograd/generated/Functions.h`
- `torch/csrc/autograd/generated/Functions.cpp`

9.14 seed

To fix the seed, use:

```
import random
import numpy as np
import torch

seed = 20240809
random.seed(seed)
np.random.seed(seed)
torch.manual_seed(seed)
```

9.15 float16

```
#!/usr/bin/env python3

import warnings

warnings.filterwarnings("ignore")

import torch

# sign: 1
# exp: 5
# significand: 10
def test_f16_subnormal():
    # For subnormal numbers, the exponent is always 0
    #
    #  $(-1)^{\text{sign\_bit}} * 2^{(-14)} * 0.\text{significantbits}$ 

    # use little endian
    b = b"\x00\x00"
    t = torch.frombuffer(b, dtype=torch.float16)

    # the smallest positive subnormal number
    b = b"\x01\x00"
    t = torch.frombuffer(b, dtype=torch.float16).item()
    expected = 2 ** (-14) * (0 + 1 / 1024)
    assert t == expected, (t, expected, t - expected)
    # t is about 0.000000059604645 = 5.9604645*1e-8
```

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```

# Use little endian, \x80 means the sign bit is 1
b = b"\x01\x80"
t = torch.frombuffer(b, dtype=torch.float16).item()
expected = -1 * 2 ** (-14) * (0 + 1 / 1024)
assert t == expected, (t, expected, t - expected)

b = b"\xff\x03"
t = torch.frombuffer(b, dtype=torch.float16).item()
expected = 2 ** (-14) * (0 + 1023 / 1024)
assert t == expected, (t, expected, t - expected)
# t is about 0.000060975552 = 6.0975552*1e-5

b = b"\xff\x83"
t = torch.frombuffer(b, dtype=torch.float16).item()
expected = -1 * 2 ** (-14) * (0 + 1023 / 1024)
assert t == expected, (t, expected, t - expected)
# -6.097555160522461e-05

# sign: 1
# exp: 5
# significand: 10
def test_f16_normal():
    # smallest positive normal number
    b = b"\x00\x04"
    t = torch.frombuffer(b, dtype=torch.float16).item()
    expected = 2 ** (-14) * (1 + 0 / 1024) # 6.103515625e-05
    assert t == expected, (t, expected, t - expected)

    # largest number less than 1
    b = b"\xff\x3b"
    t = torch.frombuffer(b, dtype=torch.float16).item()
    expected = 2 ** (-1) * (1 + 1023 / 1024) # 0.99951171875
    assert t == expected, (t, expected, t - expected)

    # 1
    # 0b 0 01111 0000000000
    # 0b 0011 1100 0000 0000
    # 0x 3c00
    b = b"\x00\x3c"
    t = torch.frombuffer(b, dtype=torch.float16).item()
    expected = 1

```

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```

assert t == expected, (t, expected, t - expected)

# smallest number larger than 1
b = b"\x01\x3c"
t = torch.frombuffer(b, dtype=torch.float16).item()
expected = 2 ** (0) * (1 + 1 / 1024) # 1.0009765625
assert t == expected, (t, expected, t - expected)

# largest normal number
# 0b 0 11110 1111111111
# 0b 0111 1011 1111 1111
# 0x7c ff
b = b"\xff\x7b"
t = torch.frombuffer(b, dtype=torch.float16).item()
expected = 2 ** (15) * (1 + 1023 / 1024) # 65504
assert t == expected, (t, expected, t - expected)

# -2
# 0b 1 10000 0000000000
# 0b 1100 0000 0000 0000
# 0xc0 00
b = b"\x00\xc0"
t = torch.frombuffer(b, dtype=torch.float16).item()
expected = -2
assert t == expected, (t, expected, t - expected)

b = b"\xff\xfb"
t = torch.frombuffer(b, dtype=torch.float16).item()
expected = (-1) * 2 ** (15) * (1 + 1023 / 1024) # -65504
assert t == expected, (t, expected, t - expected)

def main():
    test_f16_subnormal()
    test_f16_normal()

if __name__ == "__main__":
    main()

```

9.15.1 IEEE 754 float16

Please see https://en.wikipedia.org/wiki/Half-precision_floating-point_format

- sign: 1
- exponent: 5
- significand: 10

exponent offset is 15.

9.15.2 bfloat16 (brain float16)

- sign: 1
- exponent: 8
- significand: 7

exponent offset is 127.

See https://en.wikipedia.org/wiki/Bfloat16_floating-point_format

9.16 Dynamo

9.17 random

Note that, we have to save the random generator state for the CPU and GPU.

Listing 101: ./code/random_test.py

```
1  #!/usr/bin/env python3
2
3  import torch
4
5
6  class Foo(torch.nn.Module):
7     def __init__(self):
8         super().__init__()
9
10    def forward(self, x):
11        y = torch.rand(*x.shape).to(x.device)
12        return torch.nn.functional.dropout(x + y, p=0.3)
13
14
```

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```
15 def test_cpu():
16     f = Foo()
17
18     x1 = torch.rand(3, 5)
19     x2 = x1.clone()
20     x3 = x1.clone()
21
22     cpu_state = torch.get_rng_state()
23     y1 = f(x1)
24     print("y1", y1, y1.sum(), y1.mean())
25     with torch.random.fork_rng(devices=[]):
26         y2 = f(x2)
27
28     with torch.random.fork_rng(devices=[]):
29         torch.set_rng_state(cpu_state)
30         y3 = f(x3)
31
32     print("y2", y2, y2.sum(), y2.mean())
33     print("y3", y3, y3.sum(), y3.mean())
34
35
36 def test_cuda():
37     f = Foo()
38     device = torch.device("cuda", 0)
39
40     x1 = torch.rand(3, 5).to(device)
41     x2 = x1.clone()
42     x3 = x1.clone()
43
44     cpu_state = torch.get_rng_state()
45     cuda_state = torch.cuda.get_rng_state(device)
46     print(
47         "cuda_state",
48         type(cuda_state),
49         cuda_state.device,
50         cuda_state.dtype,
51         cuda_state.shape,
52     )
53
54     y1 = f(x1)
55     print("y1", y1, y1.sum(), y1.mean())
56     with torch.random.fork_rng(devices=[]):
57         y2 = f(x2)
```

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```

58
59     with torch.random.fork_rng(devices=[device]):
60         torch.set_rng_state(cpu_state)
61         torch.cuda.set_rng_state(cuda_state, device)
62         y3 = f(x3)
63
64     print("y2", y2, y2.sum(), y2.mean())
65     print("y3", y3, y3.sum(), y3.mean())
66
67
68 def main():
69     test_cpu()
70     print(torch.cuda.is_available())
71     if torch.cuda.is_available():
72         test_cuda()
73
74
75 if __name__ == "__main__":
76     torch.manual_seed(20241030)
77     main()
78
79 """
80 -----macos-----
81 y1 tensor([[1.8172, 0.9755, 1.4394, 1.3970, 0.0000],
82           [1.0299, 2.4723, 1.1365, 0.0000, 0.7647],
83           [2.0160, 1.8454, 1.9144, 1.8337, 1.5052]]) tensor(20.1471) tensor(1.3431)
84 y2 tensor([[2.1100, 0.6028, 0.9254, 0.0000, 1.3935],
85           [1.9948, 0.0000, 1.4811, 1.2179, 0.8196],
86           [2.1118, 1.3885, 1.5176, 1.2972, 2.2623]]) tensor(19.1227) tensor(1.2748)
87 y3 tensor([[1.8172, 0.9755, 1.4394, 1.3970, 0.0000],
88           [1.0299, 2.4723, 1.1365, 0.0000, 0.7647],
89           [2.0160, 1.8454, 1.9144, 1.8337, 1.5052]]) tensor(20.1471) tensor(1.3431)
90 False
91 -----Linux-----
92 y1 tensor([[1.8172, 0.9755, 1.4394, 1.3970, 0.0000],
93           [1.0299, 2.4723, 1.1365, 0.0000, 0.7647],
94           [2.0160, 1.8454, 1.9144, 1.8337, 1.5052]]) tensor(20.1471) tensor(1.3431)
95 y2 tensor([[2.1100, 0.6028, 0.9254, 0.0000, 1.3935],
96           [1.9948, 0.0000, 1.4811, 1.2179, 0.8196],
97           [2.1118, 1.3885, 1.5176, 1.2972, 2.2623]]) tensor(19.1227) tensor(1.2748)
98 y3 tensor([[1.8172, 0.9755, 1.4394, 1.3970, 0.0000],
99           [1.0299, 2.4723, 1.1365, 0.0000, 0.7647],
100          [2.0160, 1.8454, 1.9144, 1.8337, 1.5052]]) tensor(20.1471) tensor(1.3431)

```

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```

101 True
102 cuda_state <class 'torch.Tensor'> cpu torch.uint8 torch.Size([16])
103 y1 tensor([[1.2276, 0.0716, 0.0000, 0.5980, 1.4526],
104           [1.5889, 0.5063, 0.0000, 1.0267, 1.5081],
105           [1.7808, 1.3360, 1.5424, 1.8120, 0.0000]], device='cuda:0') tensor(14.4510,
↳ device='cuda:0') tensor(0.9634, device='cuda:0')
106 y2 tensor([[1.8274, 0.0000, 1.1841, 0.6805, 1.0811],
107           [1.4730, 1.7636, 1.4561, 1.1214, 0.0000],
108           [1.6906, 1.0212, 1.7333, 1.2885, 2.6000]], device='cuda:0') tensor(18.9209,
↳ device='cuda:0') tensor(1.2614, device='cuda:0')
109 y3 tensor([[1.2276, 0.0716, 0.0000, 0.5980, 1.4526],
110           [1.5889, 0.5063, 0.0000, 1.0267, 1.5081],
111           [1.7808, 1.3360, 1.5424, 1.8120, 0.0000]], device='cuda:0') tensor(14.4510,
↳ device='cuda:0') tensor(0.9634, device='cuda:0')
112 """

```

9.18 compile

On my mac:

```

>>> torch.__version__
'2.2.2'
>>> torch.compiler.list_backends()
['cudagraphs', 'inductor', 'onnxrt', 'openxla', 'openxla_eval', 'tvm']

```

On Linux:

```

>>> torch.__version__
'2.6.0+cu118'
>>> torch.compiler.list_backends()
['cudagraphs', 'inductor', 'onnxrt', 'openxla', 'tvm']

```

9.18.1 TODO

- https://pytorch.org/docs/stable/torch.compiler_dynamic_shapes.html
- https://pytorch.org/docs/stable/torch.compiler_dynamo_deepdive.html

9.19 benchmark

See <https://pytorch.org/tutorials/recipes/recipes/benchmark.html>

9.20 cv2

Convert a tensor to an image.

```
pip install opencv-python
```

```
import cv2
cv2.imwrite('test.png', bool_tensor.to(torch.int)*255)
```

9.21 einsum

Listing 102: ./code/einsum-test.py

```
1  #!/usr/bin/env python3
2
3  import torch
4
5
6  # mat multiply
7  def test_case1():
8      a = torch.rand(2, 3)
9      b = torch.rand(3, 5)
10     c = torch.matmul(a, b)
11     d = torch.einsum("ij,jk", a, b)
12     assert torch.allclose(c, d)
13
14     # ->ik is optional
15     e = torch.einsum("ij,jk->ik", a, b)
16     assert torch.allclose(c, e)
17
18     # also transpose the output
19     f = torch.einsum("ij,jk->ki", a, b)
20     assert torch.allclose(c.t(), f)
21     print(c)
22     print(d)
23     print(e)
```

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```

24     print(f)
25
26
27 # extract diagonal
28 def test_case2():
29     a = torch.arange(25).reshape(5, 5)
30
31     # extract diagonal of a square matrix
32     b = torch.einsum("ii->i", a)
33     c = torch.diag(a, diagonal=0)
34     assert torch.equal(b, c)
35     print(a)
36     print(b)
37     print(c)
38     print(a.shape, b.shape, c.shape, a.dtype, b.dtype, c.dtype)
39
40
41 # element-wise product
42 def test_case3():
43     a = torch.rand(3, 5)
44     b = torch.rand(3, 5)
45
46     # element-wise product
47     c = torch.einsum("ij,ij->ij", a, b)
48
49     assert a.shape == b.shape == c.shape, (a.shape, b.shape, c.shape)
50
51     d = a * b
52
53     assert torch.allclose(c, d)
54
55     # a**3
56     e = torch.einsum("ij,ij,ij->ij", a, a, a)
57     f = a**3
58     assert torch.allclose(e, f)
59
60
61 def my_einsum(a, b, c):
62     return torch.einsum("ijkl, ijlm, ikml -> ijk", a, b, c)
63
64
65 def test_einsum():
66     i = 2

```

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```
67     j = 3
68     k = 4
69     l = 5
70     m = 6
71     a = torch.rand(i, j, k, l)
72     b = torch.rand(i, j, l, m)
73     c = torch.rand(i, k, m, l)
74
75     scores = torch.einsum("ijkl, ijlm, ikml -> ijk", a, b, c)
76     m = torch.compile(my_einsum)
77     scores_2 = m(a, b, c)
78
79     assert torch.allclose(scores, scores_2)
80
81     m3 = torch.compile(torch.einsum)
82     scores_3 = m3("ijkl, ijlm, ikml -> ijk", a, b, c)
83     assert torch.allclose(scores, scores_3)
84
85
86 def main():
87     # test_case1()
88     # test_case2()
89     # test_case3()
90     test_einsum()
91
92
93 if __name__ == "__main__":
94     torch.manual_seed(20250623)
95     main()
```


10.1 Install from Source

Go to <https://www.python.org/ftp/python/> to download python, e.g.,

```
wget https://www.python.org/ftp/python/3.8.0/Python-3.8.0.tar.xz
wget https://www.python.org/ftp/python/3.9.0/Python-3.9.0.tar.xz
wget https://www.python.org/ftp/python/3.10.0/Python-3.10.0.tar.xz

mkdir -p $HOME/software/python
tar xvf Python-3.8.0.tar.xz
cd Python-3.8.0
./configure --help
./configure --prefix=$HOME/software/python/3.8.0 --enable-shared --enable-
↳ optimizations
```

10.2 asyncio

10.2.1 asyncio.Future

10.2.2 iterator

See <https://peps.python.org/pep-0234/>

10.2.3 yield

10.2.4 Hello World

Exercise 1

Listing 1: ./code/hello_world/ex1.py

```
1 import asyncio
2
3
4 async def hello():
5     print("hello world")
6
7
8 asyncio.run(hello())
```

Exercise 2

Listing 2: ./code/hello_world/ex2.py

```
1 import asyncio
2 import time
3
4
5 loop = asyncio.get_event_loop()
6
7
8 @asyncio.coroutine
9 def hello():
10     print(f"hello {time.strftime('%X')}")
11     yield from asyncio.sleep(1)
12     print(f"world {time.strftime('%X')}")
13
14
15 if __name__ == "__main__":
16     loop.run_until_complete(hello())
```

10.2.5 References

- PEP 234 –Iterators

<https://peps.python.org/pep-0234/>

- Why does defining `__getitem__` on a class make it iterable in python?

<https://localcoder.org/why-does-defining-getitem-on-a-class-make-it-iterable-in-python>

- PEP 255 –Simple Generators

<https://peps.python.org/pep-0255/>

- Curious Course on Coroutines and Concurrency

https://www.youtube.com/watch?v=Z_OAIhXziw&ab_channel=DavidBeazley

By David Beazley.

- Generator Tricks for Systems Programmers

<https://www.dabeaz.com/generators2/>

- Generators: The Final Frontier

<https://www.youtube.com/watch?v=5-qadlG7tWo&ab_channel=DavidBeazley>

By David Beazley.

10.2.6 TODOs

`asyncio.to_thread()` runs the function in an executor, where the default executor is a threadpool executor, which invokes `loop.run_in_executor()` indirectly.

How to set the executor of a loop? Maybe something related to *set_default_executor*?

If we want to schedule a callback to run in the loop from the C++ code, we can use *loop.call_soon_safe()* method.

10.3 argv

From the doc <https://docs.python.org/3/library/sys.html>:

```
The list of command line arguments passed to a Python script.
argv[0] is the script name (it is operating system dependent whether
this is a full pathname or not). If the command was executed using the
-c command line option to the interpreter, argv[0] is set to the string
'-c'. If no script name was passed to the Python interpreter,
argv[0] is the empty string.
```

Note that `argv` is at least of size 1, though `argv[0]` may be an empty string.

```
import sys
print(sys.argv)
```

10.4 TODO

Python with zeroMQ (c extension)

10.5 time

```
import time
print(f'Started at {time.strftime("%X")}')
# do something
print(f'Finished at {time.strftime("%X")}')
```

10.6 Numbers

10.6.1 binary representation

Listing 3: ./code/numbers/representations.py

```
1 print(bin(1))    # 0b1
2 print(bin(3))    # 0b11
3 print(bin(255))  # 0b11111111
4 print(bin(256))  # 0b100000000
5 assert isinstance(bin(1), str)
6 assert int("11", base=2) == 3
7 assert int("0b11", base=0) == 3
8 assert hex(2) == "0x2"
9 assert hex(10) == "0xa"
10
11 assert oct(10) == "0o12"
12 assert int("12", base=8) == 10
13 assert int("0o12", base=0) == 10
14
15 assert 1_000 == 1000
16 assert 1_000_000 == 1000000
```

10.7 str

10.7.1 format

See <https://docs.python.org/3/library/string.html#formatspec> and <https://peps.python.org/pep-3101/>

Listing 4: ./code/str/format.py

```
1 a = 1
2 b = 2
3 c = 3
4 assert "{}".format(a) == "1"
5 assert "{}".format(b) == "2"
6 assert "{0} {1} {foo}".format(a, b, foo=c) == "1 2 3"
7
8 # 1 - the first positional argument (counting from 0)
9 # foo - it is a keyword argument
10 # 0 - the zeros positional
11 assert "{1} {foo} {0}".format(a, b, foo=c) == "2 3 1"
```

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```
12
13 assert "{0} {1} {0} {0}".format(a, b) == "1 2 1 1"
14
15 assert "skip braces {0} {}".format(a) == "skip braces 1 {}"
16 print("{}") # {}
17 try:
18     print("{} {}".format(a))
19 except IndexError as e:
20     assert str(e) == "Replacement index 1 out of range for positional args tuple"
21
22 assert "{0:2}".format(a) == " 1"
23 assert "{0:02}".format(a) == "01"
24 assert "{0:03}".format(a) == "001"
25 assert "{0:1}".format(-1) == "-1"
26 assert "{0:2}".format(-1) == "-1"
27 assert "{0:3}".format(-1) == " -1"
28 assert "{0:03}".format(-1) == "-01"
29
30 assert "{0:.2f}".format(0.5) == "0.50"
31 assert "{0:.3f}".format(0.5) == "0.500"
```

10.8 enum

10.8.1 Hello

See <https://docs.python.org/3.11/howto/enum.html>

Enum

Note:

- It is iterable, i.e., supports `__iter__`
- name and value
- alias and `@unique`.
- `__members__`.
- `str` and `repr`.
- `auto`

Listing 5: ./code/hello/ex1.py

```

1  from enum import Enum
2
3
4  class Color(Enum):
5      RED = 1
6      GREEN = 2
7      BLUE = 3
8      # BLUE = 4 # TypeError: Attempted to reuse key: 'BLUE'
9      ALIAS_FOR_RED = 1 # Use @unique to disallow this
10     MAX_COLOR = 4 # Note the naming convention
11
12
13  assert isinstance(Color.RED, Color)
14
15  assert str(Color(1)) == "Color.RED"
16  assert str(repr(Color(1))) == "<Color.RED: 1>"
17
18  assert Color.RED.name == "RED"
19  assert Color.BLUE.value == 3
20
21  print(list(Color))
22  print(type(list(Color)[0]))
23  for c in Color:
24      print(c, type(c))
25
26  """
27  [<Color.RED: 1>, <Color.GREEN: 2>, <Color.BLUE: 3>, <Color.MAX_COLOR: 4>]
28  <enum 'Color'>
29  Color.RED <enum 'Color'>
30  Color.GREEN <enum 'Color'>
31  Color.BLUE <enum 'Color'>
32  Color.MAX_COLOR <enum 'Color'>
33  """
34
35  assert Color(1) == Color.RED
36  assert Color["RED"] == Color.RED
37  assert Color["ALIAS_FOR_RED"] == Color.RED
38
39  print(Color.__members__)
40  """
41  {'RED': <Color.RED: 1>, 'GREEN': <Color.GREEN: 2>, 'BLUE': <Color.BLUE: 3>, 'ALIAS_
  ↳FOR_RED': <Color.RED: 1>, 'MAX_COLOR': <Color.MAX_COLOR: 4>}

```

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42 """

Flag

Listing 6: ./code/hello/ex2.py

```
1 from enum import Flag
2
3
4 class Weekday(Flag):
5     MONDAY = 1 << 0
6     TUESDAY = 1 << 1
7     WEDNESDAY = 1 << 2
8     THURSDAY = 1 << 3
9     FRIDAY = 1 << 4
10    SATURDAY = 1 << 5
11    SUNDAY = 1 << 6
12
13
14 assert Weekday.MONDAY.value == 1
15 assert Weekday.TUESDAY.value == 2
16 assert Weekday.WEDNESDAY.value == 4
17 assert Weekday.THURSDAY.value == 8
18 assert Weekday.FRIDAY.value == 16
19 assert Weekday.SATURDAY.value == 32
20 assert Weekday.SUNDAY.value == 64
21
22 weekend = Weekday.SATURDAY | Weekday.SUNDAY
23 print(weekend) # Weekday.SUNDAY|SATURDAY
24 print(repr(weekend)) # <Weekday.SUNDAY|SATURDAY: 96>
25 assert Weekday.SATURDAY in weekend
26 assert Weekday.SUNDAY in weekend
27 assert Weekday.MONDAY not in weekend
```

auto

Listing 7: ./code/hello/ex3.py

```
1 from enum import Enum, Flag, auto
2
3
```

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```
4 class Weekday(Flag):
5     MONDAY = auto() # start from 1
6     TUESDAY = auto()
7     WEDNESDAY = auto()
8     THURSDAY = auto()
9     FRIDAY = auto()
10    SATURDAY = 128
11    SUNDAY = auto()
12
13
14    assert Weekday.MONDAY.value == 1
15    assert Weekday.TUESDAY.value == 2
16    assert Weekday.WEDNESDAY.value == 4
17    assert Weekday.THURSDAY.value == 8
18    assert Weekday.FRIDAY.value == 16
19    assert Weekday.SATURDAY.value == 128
20    assert Weekday.SUNDAY.value == 256
21
22
23 class Color(Enum):
24     RED = auto() # start from 1
25     GREEN = auto()
26     BLUE = auto()
27     YELLOW = 10
28     WHITE = auto()
29
30
31    assert Color.RED.value == 1
32    assert Color.GREEN.value == 2
33    assert Color.BLUE.value == 3
34    assert Color.YELLOW.value == 10
35    assert Color.WHITE.value == 11
```

10.9 socket

10.9.1 AddressFamily

It is an IntEnum and all of its members are exported to socket.

Listing 8: ./code/address-family.py

```

1 import socket
2
3 print(list(socket.AddressFamily))
4
5 """
6 [<AddressFamily.AF_UNSPEC: 0>, <AddressFamily.AF_UNIX: 1>,
7 <AddressFamily.AF_INET: 2>, <AddressFamily.AF_AX25: 3>,
8 <AddressFamily.AF_IPX: 4>, <AddressFamily.AF_APPLETALK: 5>,
9 <AddressFamily.AF_NETROM: 6>, <AddressFamily.AF_BRIDGE: 7>,
10 <AddressFamily.AF_ATMPVC: 8>, <AddressFamily.AF_X25: 9>,
11 <AddressFamily.AF_INET6: 10>, <AddressFamily.AF_ROSE: 11>,
12 <AddressFamily.AF_NETBEUI: 13>, <AddressFamily.AF_SECURITY: 14>,
13 <AddressFamily.AF_KEY: 15>, <AddressFamily.AF_NETLINK: 16>,
14 <AddressFamily.AF_PACKET: 17>, <AddressFamily.AF_ASH: 18>,
15 <AddressFamily.AF_ECONET: 19>, <AddressFamily.AF_ATMSVC: 20>,
16 <AddressFamily.AF_RDS: 21>, <AddressFamily.AF_SNA: 22>,
17 <AddressFamily.AF_IRDA: 23>, <AddressFamily.AF_PPPOX: 24>,
18 <AddressFamily.AF_WANPIPE: 25>, <AddressFamily.AF_LLC: 26>,
19 <AddressFamily.AF_CAN: 29>, <AddressFamily.AF_TIPC: 30>,
20 <AddressFamily.AF_BLUETOOTH: 31>, <AddressFamily.AF_ALG: 38>,
21 <AddressFamily.AF_VSOCK: 40>, <AddressFamily.AF_QIPCRT: 42>]
22 """
23
24 assert socket.AF_UNIX == socket.AddressFamily.AF_UNIX
25 assert socket.AF_INET == socket.AddressFamily.AF_INET

```

10.9.2 SocketKind

It is an IntEnum and all of its members are exported to socket.

Listing 9: ./code/socket-kind.py

```

1 import socket
2
3 print(list(socket.SocketKind))
4
5 """
6 [<SocketKind.SOCK_STREAM: 1>, <SocketKind.SOCK_DGRAM: 2>,
7 <SocketKind.SOCK_RAW: 3>, <SocketKind.SOCK_RDM: 4>,
8 <SocketKind.SOCK_SEQPACKET: 5>, <SocketKind.SOCK_NONBLOCK: 2048>,
9 <SocketKind.SOCK_CLOEXEC: 524288>]
10 """

```

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```

11 assert socket.SOCK_STREAM == socket.SocketKind.SOCK_STREAM
12 assert socket.SOCK_DGRAM == socket.SocketKind.SOCK_DGRAM

```

10.9.3 struct sockaddr_in

See also

- https://www.gta.ufrj.br/ensino/eel878/sockets/sockaddr_inman.html
- <https://man7.org/linux/man-pages/man7/ip.7.html>

Listing 10: ./code/sockaddr_in.h

```

1  // https://github.com/lattera/glibc/blob/master/bits/sockaddr.h
2  /* POSIX.1g specifies this type name for the `sa_family' member.  */
3  typedef unsigned short int sa_family_t;
4
5  #define __SOCKADDR_COMMON(sa_prefix) sa_family_t sa_prefix##family
6
7  // https://github.com/lattera/glibc/blob/master/bits/socket.h
8
9  struct sockaddr {
10     __SOCKADDR_COMMON(sa_); /* Common data: address family and length.  */
11     char sa_data[14];        /* Address data.  */
12 };
13
14 // https://github.com/lattera/glibc/blob/master/inet/netinet/in.h
15 struct sockaddr_in {
16     __SOCKADDR_COMMON(sin_);
17     in_port_t sin_port;      /* Port number.  */
18     struct in_addr sin_addr; /* Internet address.  */
19
20     /* Pad to size of `struct sockaddr'.  */
21     unsigned char sin_zero[sizeof(struct sockaddr) - __SOCKADDR_COMMON_SIZE -
22                               sizeof(in_port_t) - sizeof(struct in_addr)];
23 };
24
25 typedef uint32_t in_addr_t;
26 struct in_addr {
27     in_addr_t s_addr;
28 };
29
30 /* Address to accept any incoming messages.  */
31 #define INADDR_ANY ((in_addr_t)0x00000000)

```

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```

32  /* Address to send to all hosts.  */
33  #define INADDR_BROADCAST ((in_addr_t)0xffffffff)
34  /* Address indicating an error return.  */
35  #define INADDR_NONE ((in_addr_t)0xffffffff)
36
37  /* Network number for local host loopback.  */
38  #define IN_LOOPBACKNET 127
39  /* Address to loopback in software to local host.  */
40  #ifndef INADDR_LOOPBACK
41  #define INADDR_LOOPBACK ((in_addr_t)0x7f000001) /* Inet 127.0.0.1.  */
42  #endif

```

10.9.4 AddressInfo

Listing 11: ./code/address-info.py

```

1  import socket
2
3  print(list(socket.AddressInfo))
4  """
5  [<AddressInfo.AI_PASSIVE: 1>, <AddressInfo.AI_CANONNAME: 2>,
6  <AddressInfo.AI_NUMERICHOST: 4>, <AddressInfo.AI_V4MAPPED: 8>,
7  <AddressInfo.AI_ALL: 16>, <AddressInfo.AI_ADDRCONFIG: 32>,
8  <AddressInfo.AI_NUMERICSERV: 1024>]
9  """
10 assert socket.AI_PASSIVE == socket.AddressInfo.AI_PASSIVE

```

10.9.5 inet_pton

https://man7.org/linux/man-pages/man3/inet_pton.3.html

Representation format to network address.

The resulting network address is in network order, i.e., big endian.

Listing 12: ./code/inet_pton.c

```

1  #include <arpa/inet.h>
2  #include <stdio.h>
3
4  int main() {
5      struct in_addr addr;

```

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```

6  int res = inet_pton(AF_INET, "192.168.1.2", &addr);
7  printf("%08x\n", addr.s_addr);
8  printf("192: %x\n", 192);
9  printf("168: %x\n", 168);
10 printf("1: %x\n", 1);
11 printf("2: %x\n", 2);
12 return 0;
13 }
14 #if 0
15 ./inet_pton
16 0201a8c0
17 192: c0
18 168: a8
19 1: 1
20 2: 2
21 #endif

```

Its implementation can be found at https://github.com/bminor/glibc/blob/master/resolv/inet_pton.c

Listing 13: ./code/inet_pton_impl.c

```

1  // See https://github.com/bminor/glibc/blob/master/resolv/inet_pton.c
2  //
3  /* Copyright (C) 1996-2022 Free Software Foundation, Inc.
4     This file is part of the GNU C Library.
5
6     The GNU C Library is free software; you can redistribute it and/or
7     modify it under the terms of the GNU Lesser General Public
8     License as published by the Free Software Foundation; either
9     version 2.1 of the License, or (at your option) any later version.
10
11    The GNU C Library is distributed in the hope that it will be useful,
12    but WITHOUT ANY WARRANTY; without even the implied warranty of
13    MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the GNU
14    Lesser General Public License for more details.
15
16    You should have received a copy of the GNU Lesser General Public
17    License along with the GNU C Library; if not, see
18    <https://www.gnu.org/licenses/>.  */
19
20 /*
21  * Copyright (c) 1996,1999 by Internet Software Consortium.
22  *
23  * Permission to use, copy, modify, and distribute this software for any

```

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```

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31  * DAMAGES OR ANY DAMAGES WHATSOEVER RESULTING FROM LOSS OF USE, DATA OR
32  * PROFITS, WHETHER IN AN ACTION OF CONTRACT, NEGLIGENCE OR OTHER TORTIOUS
33  * ACTION, ARISING OUT OF OR IN CONNECTION WITH THE USE OR PERFORMANCE OF THIS
34  * SOFTWARE.
35  */
36
37 #include <arpa/inet.h>
38 #include <arpa/nameser.h>
39 #include <ctype.h>
40 #include <errno.h>
41 #include <netinet/in.h>
42 #include <resolv/resolv-internal.h>
43 #include <string.h>
44 #include <sys/socket.h>
45 #include <sys/types.h>
46
47 static int inet_pton4 (const char *src, const char *src_end, u_char *dst);
48 static int inet_pton6 (const char *src, const char *src_end, u_char *dst);
49
50 int
51 __inet_pton_length (int af, const char *src, size_t srclen, void *dst)
52 {
53     switch (af)
54     {
55     case AF_INET:
56         return inet_pton4 (src, src + srclen, dst);
57     case AF_INET6:
58         return inet_pton6 (src, src + srclen, dst);
59     default:
60         __set_errno (EAFNOSUPPORT);
61         return -1;
62     }
63 }
64 libc_hidden_def (__inet_pton_length)
65
66 /* Like __inet_pton_length, but use strlen (SRC) as the length of

```

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```

67  SRC.  */
68  int
69  __inet_pton (int af, const char *src, void *dst)
70  {
71      return __inet_pton_length (af, src, strlen (src), dst);
72  }
73  libc_hidden_def (__inet_pton)
74  weak_alias (__inet_pton, inet_pton)
75  libc_hidden_weak (inet_pton)
76
77  /* Like inet_aton but without all the hexadecimal, octal and shorthand
78     (and trailing garbage is not ignored). Return 1 if SRC is a valid
79     dotted quad, else 0. This function does not touch DST unless it's
80     returning 1.
81     Author: Paul Vixie, 1996.  */
82  static int
83  inet_pton4 (const char *src, const char *end, unsigned char *dst)
84  {
85      int saw_digit, octets, ch;
86      unsigned char tmp[NS_INADDRSZ], *tp;
87
88      saw_digit = 0;
89      octets = 0;
90      *(tp = tmp) = 0;
91      while (src < end)
92      {
93          ch = *src++;
94          if (ch >= '0' && ch <= '9')
95          {
96              unsigned int new = *tp * 10 + (ch - '0');
97
98              if (saw_digit && *tp == 0)
99                  return 0;
100             if (new > 255)
101                 return 0;
102             *tp = new;
103             if (! saw_digit)
104             {
105                 if (++octets > 4)
106                     return 0;
107                 saw_digit = 1;
108             }
109         }

```

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```

110     else if (ch == '.' && saw_digit)
111     {
112         if (octets == 4)
113             return 0;
114         *++tp = 0;
115         saw_digit = 0;
116     }
117     else
118         return 0;
119 }
120 if (octets < 4)
121     return 0;
122 memcpy (dst, tmp, NS_INADDRSZ);
123 return 1;
124 }
125
126 /* Return the value of CH as a hexademical digit, or -1 if it is a
127    different type of character.  */
128 static int
129 hex_digit_value (char ch)
130 {
131     if ('0' <= ch && ch <= '9')
132         return ch - '0';
133     if ('a' <= ch && ch <= 'f')
134         return ch - 'a' + 10;
135     if ('A' <= ch && ch <= 'F')
136         return ch - 'A' + 10;
137     return -1;
138 }
139
140 /* Convert presentation-level IPv6 address to network order binary
141    form.  Return 1 if SRC is a valid [RFC1884 2.2] address, else 0.
142    This function does not touch DST unless it's returning 1.
143    Author: Paul Vixie, 1996.  Inspired by Mark Andrews.  */
144 static int
145 inet_pton6 (const char *src, const char *src_endp, unsigned char *dst)
146 {
147     unsigned char tmp[NS_IN6ADDRSZ], *tp, *endp, *colonp;
148     const char *curtok;
149     int ch;
150     size_t xdigits_seen;    /* Number of hex digits since colon.  */
151     unsigned int val;
152

```

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```
153 tp = memset (tmp, '\0', NS_IN6ADDRSZ);
154 endp = tp + NS_IN6ADDRSZ;
155 colonp = NULL;
156
157 /* Leading :: requires some special handling. */
158 if (src == src_endp)
159     return 0;
160 if (*src == ':')
161     {
162         ++src;
163         if (src == src_endp || *src != ':')
164             return 0;
165     }
166
167 curtok = src;
168 xdigits_seen = 0;
169 val = 0;
170 while (src < src_endp)
171     {
172         ch = *src++;
173         int digit = hex_digit_value (ch);
174         if (digit >= 0)
175             {
176                 if (xdigits_seen == 4)
177                     return 0;
178                 val <= 4;
179                 val |= digit;
180                 if (val > 0xffff)
181                     return 0;
182                 ++xdigits_seen;
183                 continue;
184             }
185         if (ch == ':')
186             {
187                 curtok = src;
188                 if (xdigits_seen == 0)
189                     {
190                         if (colonp)
191                             return 0;
192                         colonp = tp;
193                         continue;
194                     }
195                 else if (src == src_endp)
```

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```

196     return 0;
197     if (tp + NS_INT16SZ > endp)
198         return 0;
199     *tp++ = (unsigned char) (val >> 8) & 0xff;
200     *tp++ = (unsigned char) val & 0xff;
201     xdigits_seen = 0;
202     val = 0;
203     continue;
204 }
205 if (ch == '.' && ((tp + NS_INADDRSZ) <= endp)
206     && inet_pton4 (curtok, src_endp, tp) > 0)
207 {
208     tp += NS_INADDRSZ;
209     xdigits_seen = 0;
210     break; /* '\0' was seen by inet_pton4. */
211 }
212 return 0;
213 }
214 if (xdigits_seen > 0)
215 {
216     if (tp + NS_INT16SZ > endp)
217         return 0;
218     *tp++ = (unsigned char) (val >> 8) & 0xff;
219     *tp++ = (unsigned char) val & 0xff;
220 }
221 if (colonp != NULL)
222 {
223     /* Replace :: with zeros. */
224     if (tp == endp)
225         /* :: would expand to a zero-width field. */
226         return 0;
227     size_t n = tp - colonp;
228     memmove (endp - n, colonp, n);
229     memset (colonp, 0, endp - n - colonp);
230     tp = endp;
231 }
232 if (tp != endp)
233     return 0;
234 memcpy (dst, tmp, NS_IN6ADDRSZ);
235 return 1;
236 }

```

10.9.6 inet_ntop

Network address to representation format.

See https://man7.org/linux/man-pages/man3/inet_ntop.3.html

Listing 14: ./code/inet_ntop.c

```

1  #include <arpa/inet.h>
2  #include <stdio.h>
3
4  int main() {
5      struct in_addr addr;
6      uint8_t *p = (uint8_t *)&addr.s_addr;
7      p[0] = 192;
8      p[1] = 168;
9      p[2] = 1;
10     p[3] = 2;
11     char buf[INET_ADDRSTRLEN];
12     const char *ret = inet_ntop(AF_INET, &addr.s_addr, buf, sizeof(buf));
13     printf("%s\n", buf);
14     printf("%p, %p\n", buf, ret);
15     return 0;
16 }
17 #if 0
18 192.168.1.2
19 0x7ffc808b5e80, 0x7ffc808b5e80
20 #endif

```

Its implementation can be found at https://github.com/bminor/glibc/blob/master/resolv/inet_ntop.c

Listing 15: ./code/inet_ntop_impl.c

```

1  // https://github.com/bminor/glibc/blob/master/resolv/inet_ntop.c
2  /*
3   * Copyright (c) 1996-1999 by Internet Software Consortium.
4   *
5   * Permission to use, copy, modify, and distribute this software for any
6   * purpose with or without fee is hereby granted, provided that the above
7   * copyright notice and this permission notice appear in all copies.
8   *
9   * THE SOFTWARE IS PROVIDED "AS IS" AND INTERNET SOFTWARE CONSORTIUM DISCLAIMS
10  * ALL WARRANTIES WITH REGARD TO THIS SOFTWARE INCLUDING ALL IMPLIED WARRANTIES
11  * OF MERCHANTABILITY AND FITNESS. IN NO EVENT SHALL INTERNET SOFTWARE
12  * CONSORTIUM BE LIABLE FOR ANY SPECIAL, DIRECT, INDIRECT, OR CONSEQUENTIAL
13  * DAMAGES OR ANY DAMAGES WHATSOEVER RESULTING FROM LOSS OF USE, DATA OR

```

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```
14  * PROFITS, WHETHER IN AN ACTION OF CONTRACT, NEGLIGENCE OR OTHER TORTIOUS
15  * ACTION, ARISING OUT OF OR IN CONNECTION WITH THE USE OR PERFORMANCE OF THIS
16  * SOFTWARE.
17  */
18
19  #include <sys/param.h>
20  #include <sys/types.h>
21  #include <sys/socket.h>
22
23  #include <netinet/in.h>
24  #include <arpa/inet.h>
25  #include <arpa/nameser.h>
26
27  #include <errno.h>
28  #include <stdio.h>
29  #include <string.h>
30
31  #ifdef SPRINTF_CHAR
32  # define SPRINTF(x) strlen(sprintf/**/x)
33  #else
34  # define SPRINTF(x) ((size_t)sprintf x)
35  #endif
36
37  /*
38   * WARNING: Don't even consider trying to compile this on a system where
39   * sizeof(int) < 4.  sizeof(int) > 4 is fine; all the world's not a VAX.
40   */
41
42  static const char *inet_ntop4 (const u_char *src, char *dst, socklen_t size);
43  static const char *inet_ntop6 (const u_char *src, char *dst, socklen_t size);
44
45  /* char *
46   * inet_ntop(af, src, dst, size)
47   *      convert a network format address to presentation format.
48   * return:
49   *      pointer to presentation format address (`dst'), or NULL (see errno).
50   * author:
51   *      Paul Vixie, 1996.
52   */
53  const char *
54  inet_ntop (int af, const void *src, char *dst, socklen_t size)
55  {
56      switch (af) {
```

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```

57     case AF_INET:
58         return (inet_ntop4(src, dst, size));
59     case AF_INET6:
60         return (inet_ntop6(src, dst, size));
61     default:
62         __set_errno (EAFNOSUPPORT);
63         return (NULL);
64     }
65     /* NOTREACHED */
66 }
67 libc_hidden_def (inet_ntop)
68
69 /* const char *
70  * inet_ntop4(src, dst, size)
71  *     format an IPv4 address
72  * return:
73  *     `dst' (as a const)
74  * notes:
75  *     (1) uses no statics
76  *     (2) takes a u_char* not an in_addr as input
77  * author:
78  *     Paul Vixie, 1996.
79  */
80 static const char *
81 inet_ntop4 (const u_char *src, char *dst, socklen_t size)
82 {
83     static const char fmt[] = "%u.%u.%u.%u";
84     char tmp[sizeof "255.255.255.255"];
85
86     if (SPRINTF((tmp, fmt, src[0], src[1], src[2], src[3])) >= size) {
87         __set_errno (ENOSPC);
88         return (NULL);
89     }
90     return strcpy(dst, tmp);
91 }
92
93 /* const char *
94  * inet_ntop6(src, dst, size)
95  *     convert IPv6 binary address into presentation (printable) format
96  * author:
97  *     Paul Vixie, 1996.
98  */
99 static const char *

```

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```

100 inet_ntop6 (const u_char *src, char *dst, socklen_t size)
101 {
102     /*
103      * Note that int32_t and int16_t need only be "at least" large enough
104      * to contain a value of the specified size. On some systems, like
105      * Crays, there is no such thing as an integer variable with 16 bits.
106      * Keep this in mind if you think this function should have been coded
107      * to use pointer overlays. All the world's not a VAX.
108      */
109     char tmp[sizeof "ffff:ffff:ffff:ffff:ffff:ffff:255.255.255.255"], *tp;
110     struct { int base, len; } best, cur;
111     u_int words[NS_IN6ADDRSZ / NS_INT16SZ];
112     int i;
113
114     /*
115      * Preprocess:
116      *     Copy the input (bytewise) array into a wordwise array.
117      *     Find the longest run of 0x00's in src[] for :: shorthanding.
118      */
119     memset(words, '\\0', sizeof words);
120     for (i = 0; i < NS_IN6ADDRSZ; i += 2)
121         words[i / 2] = (src[i] << 8) | src[i + 1];
122     best.base = -1;
123     cur.base = -1;
124     best.len = 0;
125     cur.len = 0;
126     for (i = 0; i < (NS_IN6ADDRSZ / NS_INT16SZ); i++) {
127         if (words[i] == 0) {
128             if (cur.base == -1)
129                 cur.base = i, cur.len = 1;
130             else
131                 cur.len++;
132         } else {
133             if (cur.base != -1) {
134                 if (best.base == -1 || cur.len > best.len)
135                     best = cur;
136                 cur.base = -1;
137             }
138         }
139     }
140     if (cur.base != -1) {
141         if (best.base == -1 || cur.len > best.len)
142             best = cur;

```

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```

143     }
144     if (best.base != -1 && best.len < 2)
145         best.base = -1;
146
147     /*
148      * Format the result.
149      */
150     tp = tmp;
151     for (i = 0; i < (NS_IN6ADDRSZ / NS_INT16SZ); i++) {
152         /* Are we inside the best run of 0x00's? */
153         if (best.base != -1 && i >= best.base &&
154             i < (best.base + best.len)) {
155             if (i == best.base)
156                 *tp++ = ':';
157             continue;
158         }
159         /* Are we following an initial run of 0x00s or any real hex? */
160         if (i != 0)
161             *tp++ = ':';
162         /* Is this address an encapsulated IPv4? */
163         if (i == 6 && best.base == 0 &&
164             (best.len == 6 || (best.len == 5 && words[5] == 0xffff))) {
165             if (!inet_ntop4(src+12, tp, sizeof tmp - (tp - tmp)))
166                 return (NULL);
167             tp += strlen(tp);
168             break;
169         }
170         tp += SPRINTF((tp, "%x", words[i]));
171     }
172     /* Was it a trailing run of 0x00's? */
173     if (best.base != -1 && (best.base + best.len) ==
174         (NS_IN6ADDRSZ / NS_INT16SZ))
175         *tp++ = ':';
176     *tp++ = '\0';
177
178     /*
179      * Check for overflow, copy, and we're done.
180      */
181     if ((socklen_t)(tp - tmp) > size) {
182         __set_errno (ENOSPC);
183         return (NULL);
184     }
185     return strcpy(dst, tmp);
186 }

```

10.9.7 Echo Server and Client

Server

Listing 16: ./code/echo-hello/server.py

```
1  #!/usr/bin/env python3
2  import socket
3  import threading
4
5  # nc localhost 6006
6
7
8  def run_server():
9      sock = socket.socket(family=socket.AF_INET, type=socket.SOCK_STREAM)
10     sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
11     sock.bind(("", 6006))
12     # sock.setblocking(False) # It returns socket.BlockingIOError
13     sock.listen(2) # backlog is 2
14     while True:
15         client_sock, addr = sock.accept()
16         assert isinstance(client_sock, socket.socket)
17         assert isinstance(addr, tuple)
18         assert isinstance(addr[0], str)
19         assert isinstance(addr[1], int)
20         print("Connected from", addr) # Connected from ('127.0.0.1', 54266)
21         threading.Thread(target=handle_client, args=(client_sock,)).start()
22
23
24 def handle_client(sock: socket.socket):
25     while True:
26         data = sock.recv(1024)
27         if not data:
28             break
29         sock.sendall(data.decode("utf-8").upper().encode())
30     print("Disconnected from", sock.getpeername())
31     sock.close()
32
33
34 if __name__ == "__main__":
35     run_server()
```

To test the server, use `nc localhost 6006` or use the following client.

Client

Listing 17: ./code/echo-hello/client.py

```

1 def main():
2     sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
3     sock.connect(("localhost", 6006))
4     for i in range(5):
5         sock.send("hello".encode())
6         b = sock.recv(1024)
7         print(b.decode())
8         time.sleep(0.5)

```

Server2

With `concurrent.futures.ThreadPoolExecutor`.

Listing 18: ./code/echo-hello/server2.py

```

1 #!/usr/bin/env python3
2  import socket
3  import threading
4  from concurrent.futures import ThreadPoolExecutor
5
6  # nc localhost 6006
7  pool = ThreadPoolExecutor(max_workers=3)
8
9
10 def run_server():
11     sock = socket.socket(family=socket.AF_INET, type=socket.SOCK_STREAM)
12     sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
13     sock.bind("", 6006)
14     # sock.setblocking(False) # It returns socket.BlockingIOError
15     sock.listen(2) # backlog is 2
16     while True:
17         client_sock, addr = sock.accept()
18         assert isinstance(client_sock, socket.socket)
19         assert isinstance(addr, tuple)
20         assert isinstance(addr[0], str)
21         assert isinstance(addr[1], int)
22         print("Connected from", addr) # Connected from ('127.0.0.1', 54266)
23         pool.submit(handle_client, client_sock)
24
25

```

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```
26 def handle_client(sock: socket.socket):
27     while True:
28         data = sock.recv(1024)
29         if not data:
30             break
31         sock.sendall(data.decode("utf-8").upper().encode())
32     print("Disconnected from", sock.getpeername())
33     sock.close()
34
35
36 if __name__ == "__main__":
37     run_server()
```

10.9.8 TODOs

- Beej's Guide to Network Programming Using Internet Sockets

<https://www.gta.ufrj.br/ensino/eel878/sockets/index.html>

- LWN.net Weekly Edition Archives

<https://lwn.net/Archives/>

10.10 numpy

10.10.1 Basics

Listing 19: ./code/basics.py

```
#!/usr/bin/env python3
import numpy as np

def test_attr():
    a = np.array([[1, 2], [3, 4]], dtype=np.float32)
    assert isinstance(a, np.ndarray)

    assert a.dtype == np.float32
    assert isinstance(a.dtype, np.dtype)
    assert a.dtype.name == "float32"

    # number of axes
```

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```
assert a.ndim == 2

assert a.shape == (2, 2)
assert len(a.shape) == a.ndim

# total number of elements is a.size
assert a.size == a.shape[0] * a.shape[1]

# each float occupies 4 bytes
assert a.itemsize == 4
assert isinstance(a.data, memoryview)

def test_array_creation():
    a = np.array([2, 3, 4])
    assert a.dtype == np.int64
    assert a.ndim == 1

    b = np.array([2.0, 3, 4])
    assert b.dtype == np.float64

    c = np.array([[1, 2], [3, 4]])
    assert c.ndim == 2
    assert c.shape == (2, 2)
    assert c.dtype == np.int64

    d = np.zeros((2, 3))
    assert d.dtype == np.float64
    assert d.shape == (2, 3)

    f = np.ones((2, 2))
    assert f.dtype == np.float64

    # uninitialized content; containing garbage data.
    e = np.empty((5, 5))
    assert e.dtype == np.float64

    g = np.empty((1, 2), dtype=np.int32)
    assert g.dtype == np.int32

    h = np.arange(3)
    assert h.dtype == np.int64
```

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```
# start=0, end=3
i = np.arange(3, dtype=np.int32)
assert i.dtype == np.int32
np.testing.assert_equal(i, np.array([0, 1, 2]))

# start=1, end=3
k = np.arange(1, 3)
np.testing.assert_equal(k, np.array([1, 2]))

# start=1, end=8, step=3
m = np.arange(1, 8, 3)
np.testing.assert_equal(m, np.array([1, 4, 7]))

# 4 numbers in the range [2, 5]
n = np.linspace(2, 5, 4)
assert n.dtype == np.float64
np.testing.assert_equal(n, np.array([2, 3, 4, 5]))

# note: The last number is included
m = np.linspace(2, 5, 3)
np.testing.assert_equal(m, np.array([2, 3.5, 5]))

def main():
    test_attr()
    test_array_creation()

if __name__ == "__main__":
    main()
```

Write to a binary file in Python and read it from C/C++

```
import numpy as np

a = np.array([0.25, 0.35, 0.45], dtype=np.float32)
with open("a.bin", "wb") as f:
    f.write(a.tobytes())

# or
a.tofile("a.bin")
```

```

#include <stdio.h>
#include <vector>

int main() {
    FILE *fp = fopen("./a.bin", "rb");
    fseek(fp, 0, SEEK_END);
    int32_t n = ftell(fp);
    fseek(fp, 0, SEEK_SET);
    printf("n: %d\n", n);

    std::vector<float> d(n / sizeof(float));
    fread(d.data(), sizeof(float), n, fp);
    fclose(fp);
    for (auto f : d) {
        printf("%f ", f);
    }
    printf("\n");

    return 0;
}

```

Write to a binary file in C++ and read it from Python

```

std::vector<float> v(100*560);
FILE *fp = fopen("b.bin", "wb");
fwrite(v.data(), sizeof(float), v.size(), fp);
fclose(fp);

```

```

features = np.fromfile('./b.bin', dtype='uint8').view(np.float32).reshape(-1, 560)

```

10.11 pip

install pip using:

```
python3 -m ensurepip --default-pip
```

To update pip and setuptools and wheel:

```
python3 -m pip install --upgrade pip setuptools wheel
```

After creating a virtual environment with:

```
python3 -m venv my_env  
source ./my_env/bin/active
```

We can use:

```
curl https://bootstrap.pypa.io/get-pip.py  
python3 ./get-pip.py
```

to install pip

Refer to <https://packaging.python.org/en/latest/tutorials/installing-packages/#optionally-create-a-virtual-environment>

10.12 os

Check file extension:

```
a = 'a.jpg'  
assert a.endswith('jpg') is True  
  
# Note we can also pass a tuple to endswith  
assert a.endswith(('jpg', 'jpeg')) is True
```

Split extension:

```
os.path.splitext('a.txt.exe') == ('a.txt', '.exe')
```

10.13 matplotlib

10.13.1 Colab notebook examples

- Aman's AI Journal | Primers | Matplotlib Tutorial

<https://colab.research.google.com/drive/1DEUmJquT15LX3aXE4hi9fl8hvkU2fc51>

10.14 struct

See <https://docs.python.org/3/library/struct.html>

10.15 contextmanager

Listing 20: ./code/context_manager/main.py

```
1  #!/usr/bin/env python3
2
3  import contextlib
4  from contextlib import contextmanager
5
6  number = 10
7
8
9  def get_number():
10     return number
11
12
13  def set_number(n):
14     global number
15     number = n
16
17
18  @contextmanager
19  def process(new_n):
20     old_n = get_number()
21     set_number(new_n)
22     try:
23         yield
24     finally:
25         set_number(old_n)
26
27
28  print(process(1))  # <contextlib._GeneratorContextManager object at 0x108a2ccd0>
29  print(type(process(1)))  # <class 'contextlib._GeneratorContextManager'>
30  assert isinstance(process(1), contextlib._GeneratorContextManager), type(process(1))
31
32  with process(100):
33     # inside this block ,number is changed to 100
34     assert number == 100, number
35
36  # outside of the above block, number is reset to its original value
37  assert number == 10, number
38
39
40  class MyProcess:
```

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```
41     def __init__(self, new_n):
42         self.new_n = new_n
43
44     def __enter__(self):
45         self.old_n = get_number()
46         set_number(self.new_n)
47
48     def __exit__(self, type, value, traceback):
49         set_number(self.old_n)
50
51
52 with MyProcess(-1):
53     assert number == -1, number
54
55 assert number == 10, number
```

See also <https://docs.python.org/3/library/contextlib.html#contextlib.contextmanager>

To manipulate a lock:

```
@contextmanager
def locked(lock):
    lock.acquire()
    try:
        yield
    finally:
        lock.release()

with locked(lock):
    pass
```

Or use a class:

```
class locked:
    def __init__(self, lock):
        self.lock = lock

    def __enter__(self):
        self.lock.acquire()

    def __exit__(self, type, value, traceback):
        self.lock.release()
```


10.16 iter

It is a builtin function.

```
help(iter)

Help on built-in function iter in module builtins:

iter(...)
    iter(iterable) -> iterator
    iter(callable, sentinel) -> iterator

    Get an iterator from an object.  In the first form, the argument must
    supply its own iterator, or be a sequence.
    In the second form, the callable is called until it returns the sentinel.
```

See also

- [PEP 234 –Iterators](#)
- [PEP 255 –Simple Generators](#)
- [PEP 342 –Coroutines via Enhanced Generators](#)

10.17 for loop

10.18 sequence

- mutable sequences:
- immutable sequences:

`PySequence_Check`

```
int
PySequence_Check(PyObject *s)
{
    if (PyDict_Check(s))
        return 0;
    return Py_TYPE(s)->tp_as_sequence &&
        Py_TYPE(s)->tp_as_sequence->sq_item != NULL;
}
```

We can see that if it is not a dict and if it has a method `__getitem__`, then it returns 1.

11.1 Install

11.1.1 formatter

Install <https://github.com/google/google-java-format>

```
wget https://github.com/google/google-java-format/releases/download/v1.15.0/google-  
→java-format-1.15.0-all-deps.jar
```

Create a script with filename `google-java-format`:

```
#!/usr/bin/env bash  
  
java -jar /ceph-sh0/fangjun/download/google-java-format-1.15.0-all-deps.jar $@
```

`chmod +x google-java-format` and add the path to PATH.

11.1.2 JDK

Go to <https://www.oracle.com/java/technologies/downloads/#java17> and download

```
wget https://download.oracle.com/java/17/latest/jdk-17_linux-x64_bin.tar.gz  
mkdir /ceph-fj/fangjun/software/  
tar xvf jdk-17_linux-x64_bin.tar.gz -C /ceph-fj/fangjun/software
```

And then set the following environment variables:

```
export JAVA_HOME=/ceph-fj/fangjun/software/jdk-17.0.3  
export PATH=$JAVA_HOME/bin:$JAVA_HOME
```

The source code of JDK can be download from

```
git clone https://github.com/openjdk/jdk.git
```

11.2 Hello world

Look at the code <https://github.com/openjdk/jdk/blob/master/src/java.base/share/classes/java/lang/System.java> to learn the coding style of java.

All primitive types, such as `int`, `char`, and `double` begin with a lowercase. All other types, such as `String` and `Math`, begin with an uppercase.

```
package java.lang;

// Class names looks like: MyClass
public final class System {}

// method name
private static native void registerNatives();

// constructor
private System() {}

// member variables
public static final InputStream in = null;
private static String notSupportedJnuEncoding;

// constants
private static final int NEVER = 1;
```

It using `//` and `/**/` as comments, same as C++.

Listing 1: Hello.java

```
// Usage 1:
//  java Hello.java
// Usage 2:
//  javac Hello.java
//  java Hello
//
// Note:
//  - "javac Hello.java" generates a file "Hello.class"
//  - "java Hello" takes as input "Hello.class" and executes it
//
// The class name Hello must match the filename Hello.java
```

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```
// By convention, the class name looks like `MyClass`.
class Hello {
    // Note: How is array defined in java
    public static void main(String[] args) {
        System.out.println("hello world");
    }
} // There is no ';' here
// Every line of runnable code must be in some `class`
```

Listing 2: EqualTest.java

```
class EqualTest {
    public int i;

    public EqualTest(int a) {
        this.i = a;
    }

    public boolean equals(Object anObject) {
        if (this == anObject) {
            return true;
        }
        if (anObject instanceof EqualTest) {
            return this.i == ((EqualTest) anObject).i;
        }
        return false;
    }

    public static void main(String[] args) {
        EqualTest e1 = new EqualTest(10);
        EqualTest e2 = new EqualTest(10);

        System.out.println(e1 == e2); // false, compare the reference
        System.out.println(e1 != e2); // true
        System.out.println(e1.equals(e2)); // true, compare the contained value
    }
}
```

Listing 3: Variables.java

```
class Variables {
    public static void main(String[] args) {
        String msg = "Hello world";
    }
}
```

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```
int i = 10;
i++; // ok
++i; // ok
// i-- // error, should be --i
--i;
int ii = i % 2;
i = 10;
ii = i / 3; // truncated
System.out.println(ii); // 3
ii += 2;
System.out.println(ii); // 5
// also has:
// -=, &=, |=, >>=, <<=, ^=, /=, *=
// Similar to C/C++

/* float f = 1.25; // error: possible lossy conversion from double to float */
// Caution: We have to use `1.25f` instead of `1.25` to assign a float variable
float f = 1.25f;
double d = 1.25;
d = 1.25d; // also ok

// f = d; // error: possible lossy conversion from double to float
f = (float) d; // ok, explicit cast
d = f; // ok

char c = 'h'; // a character

i = 10;
System.out.println(msg);
System.out.println(i); // 10
System.out.println(f); // 1.25
System.out.println(d); // 1.25
System.out.println(c); // h

// define multiple variable in the same line
int i1 = 1, i2;
// System.out.println(i2); // variable i2 might not have been initialized
i2 = 3;
System.out.println(i2); // 3
boolean b = false;
System.out.println(b); // false
b = true;
// b = 1; // error: int cannot be converted to boolean
```

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```

System.out.println(b); // true

// byte b0 = 128; // error: lossy conversion from int to byte
byte b0 = 127; // ok
/* b0 = -129; // error: lossy conversion from int to type */
b0 = -128; // ok

// short s0 = 32768; // error: lossy conversion from int to short
short s0 = 32767; // ok
// s0 = -32769; // error: lossy conversion from int to short
s0 = -32768; // ok

// other types: int, long, float, double, boolean, char
// Note: char has 2-byte
// boolean has 1-bit
// There are no unsigned integers !!

// ternary operator :?
System.out.println(2 > 3 ? "yes" : "no"); // no
}
}

```

Listing 4: Strings.java

```

class Strings {
    public static void main(String[] args) {
        String s = "abc";
        System.out.println(s.length()); // 3
        System.out.println(s.toUpperCase()); // ABC
        System.out.println(s.toUpperCase().toLowerCase()); // abc
        System.out.println(s.indexOf("ab")); // 0
        System.out.println(s.indexOf("a")); // 0
        System.out.println(s.indexOf("bc")); // 1
        System.out.println(s.indexOf("bca")); // when not found, it is -1

        System.out.println(s.charAt(0) == 'a'); // true
        System.out.println(s + "123" == "abc123"); // false, == compares the reference
        System.out.println((s + "123").equals("abc123")); // true, compare the content
        System.out.println(s.concat("123").equals("abc123")); // true, compare the content
        System.out.println((s + 123).equals("abc123")); // true, compare the content
        System.out.println((123 + s).equals("123abc")); // true, compare the content
        s = "a\b";
    }
}

```

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```
}  
;
```

Listing 5: MathTest.java

```
class MathTest {  
    public static void main(String[] args) {  
        System.out.println(Math.max(2, 3)); // 3  
        System.out.println(Math.max(2, 3.5)); // 3.5  
        System.out.println(Math.min(2, 3.5)); // 2  
        System.out.println(Math.sqrt(6.25)); // 2.5  
        System.out.println(Math.sqrt(9)); // 3.0, note it is a double  
        System.out.println(Math.abs(-9)); // -9  
        System.out.println(Math.abs(-9.0)); // -9.0  
        System.out.println(Math.random()); // a number in the range [0, 1)  
    }  
}
```

Listing 6: ArrayTest.java

```
class ArrayTest {  
    public static void main(String[] args) {  
        int[] a = {1, 2, 3};  
        for (int i : a) {  
            System.out.println(i);  
        }  
        /*  
        1  
        2  
        3  
        */  
        System.out.println(a[0]); // 1  
        System.out.println(a.length); // 3  
        for (int i = 0; i != a.length; ++i) {  
            System.out.println(a[i]);  
        }  
        /*  
        1  
        2  
        3  
        */  
    }  
}
```


11.2.1 TODOs

1. How to define an array? How to iterate an array?
2. What methods does the `String` class have
3. What are final variables?
4. How to define a package?
5. What is static import?
6. What is a lambda expression?
7. How to write comments using javadoc?
8. How to generate documentation from javadoc?
9. How to use a for-each loop?
10. How to import a class from another file?

11.3 Reference

- <https://www.w3schools.com/java/default.asp>
- <https://docs.oracle.com/javase/tutorial/>
- <https://docs.oracle.com/en/java/javase/17/docs/api/index.html>
- <https://github.com/openjdk/jdk.git>

Clone it and you can find the source code in `src/java.base/share/classes/java/lang/System.java` for `java.lang.System`.

JAVASCRIPT

12.1 Hello world

strict mode:

```
"use strict";
```

- case sensitive
- numbers (all numbers are 64-bit floating type)
- strings (' ' and " "), there is no character type
- Like python, strings are immutable
- strings uses utf-16, like c#
- array
- boolean, true and false
- null and undefined
- string interpolation

```
`hello ${someVariable}`
```

- semicolon is optional. Better to always add it
- garbage collector
- == supports type conversion
- === does not support type conversion. (Recommended to use this one)

```
console.log('hello world')  
console.log(eval('3 + 5'))
```

To write multi-line javascript, use shift + Enter for a new line.

```
(function(){  
  "use strict";  
  /* Start of your code */  
  function greetMe(yourName) {  
    alert('Hello ' + yourName);  
  }  
  
  greetMe('World');  
  /* End of your code */  
})();
```

It is case sensitive. Statements are separated by `;`. Comments are the same as in C/C++.

12.1.1 variables

Listing 1: `./code/hello_world/variables.js`

```
1  // use const to define a constant  
2  const pizza = true;  
3  console.log(pizza); // true  
4  
5  var foo = 'foo';  
6  if (foo) {  
7    var foo = 'fooz'  
8    console.log('foo is', foo); // foo is fooz  
9  }  
10  
11 // note it outputs foo is fooz below since  
12 // we use `var` inside the `if` statement  
13 console.log('foo is', foo); // foo is fooz  
14  
15 var bar = 'bar';  
16  
17 if (bar) {  
18   let bar = 'barz'  
19   console.log('bar is', bar); // foo is barz  
20 }  
21  
22 // note it outputs bar is bar below since  
23 // we use `let` inside the `if` statement  
24 console.log('bar is', bar); // foo is bar  
25  
26 // we should use let as much as possible
```

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```
27 // variables defined by `let` is scoped inside a block {}
28
29 array = [];
30 for (var i = 0; i < 3; i++) {
31
32     array.push(function() { console.log(i); });
33 }
34 array[0](); // 3
35 array[1](); // 3
36 array[2](); // 3
37 // since we are using var in the for loop, the closures capture the same i
38
39 array2 = [];
40 for (let k = 0; k < 3; k++) {
41
42     array2.push(function() { console.log(k); });
43 }
44 array2[0](); // 0
45 array2[1](); // 1
46 array2[2](); // 2
```

12.1.2 array

Listing 2: ./code/hello_world/array.js

```
1 let a = [ 1, 2, 3 ];
2 function sum(arr) {
3     let s = 0;
4     for (let x of arr) {
5         s += x;
6     }
7     return s;
8 }
9 // Sum of the array [1,2,3] is 6
10 console.log('Sum of the array [' + a + '] is ' + sum(a));
11
12 function sum2(arr) {
13     let s = 0;
14     for (let i = 0; i != arr.length; ++i) {
15         s += arr[i];
16     }
17     return s;
```

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```
18 }  
19 console.log(sum2(a)); // 6
```

Note that there are two ways to iterate an array:

- `for(let x of array)`
- `for(let i = 0; i != array.length; ++i) { ... }`

To run the above code, use:

```
node array.js
```

12.1.3 class

Listing 3: ./code/hello_world/class.js

```
1 class Point {  
2   constructor(x, y) {  
3     this.x = x;  
4     this.y = y;  
5   }  
6  
7   distance() { return Math.sqrt(this.x * this.x + this.y * this.y); }  
8 }  
9  
10 let p = new Point(1, 1);  
11 console.log(p.distance()); // 1.4142135623730951
```

It defines a `Point` class with two fields `x`, `y`. `Point` has two methods: a constructor and a method `distance()`.

Note that class names are by convention capitalized.

12.1.4 template strings

Listing 4: ./code/hello_world/template_strings.js

```
1 let a = "a";  
2 let b = "b";  
3 console.log("a is " + a + ", b is " + b); // a is a, b is b  
4  
5 // using string interpolation  
6 console.log(`a is ${a}, b is ${b}`); // a is a, b is b
```

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```

7
8 // Note that it uses `${}`.
9 // It preserves spaces and line breaks
10
11 // multiline examples
12 let s = `
13 a is ${a}
14 b is ${b}
15 `
16 console.log(s);

```

Note that it is similar to the F-string, `f" ", f' ', f""" """, f''' '''` in Python.

12.1.5 functions

Listing 5: ./code/hello_world/functions.js

```

1 // note that is supports default arguments
2 function hello(msg = "hi") { console.log(msg); }
3 // the above is a function declaration, it is hoisted to the top
4 // in other words, we can invoke it before seeing it.
5
6 hello();          // hi
7 hello('world');  // world
8
9 // this is a function expression.
10 const hi = function(msg = 'hi') { console.log(msg); };
11 hi();             // hi
12 hi('world');     // world
13
14 const add = function(a, b = 1) { return a + b; };
15 console.log(add(1, 2)); // 3
16 console.log(add(1));    // 2
17
18 // arrow functions
19 // there is no keyword `function` and no `return`.
20 const inc = (a, b = 1) => a + b;
21 console.log(inc(1, 2)); // 3
22 console.log(inc(1));    // 2
23
24 const dec = a => a - 1;
25 console.log(dec(10));  // 9
26

```

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```
27 // for multiline statement, use {}
28 const sub = a => {
29     let b = a - 1;
30     return b;
31 };
32 console.log(sub(100)); // 99
```

Note that it does not have keyword arguments.

12.2 Numbers in javascript

Listing 6: ./code/numbers.js

```
1 // all numbers are of 64-bit floating point type
2
3 let a = 32;    // decimal integer
4 let b = 0x20; // hexadecimal integer
5 let c = 0X20;
6 console.log(a, b, c); // 32 32 32
7
8 let d = 0b00010000; // binary integer
9 let e = 0o20;
10 console.log(d, e); // 16 16
11
12 console.log(a / 5); // 6.4. Note that it is not an integer!
13
14 d = 0b1000_0000; // use _ to separate digit for readability
15 console.log(d); // 128
16
17 console.log(2 ** 3); // 8, ** means power
18
19 // Note that we don't need to import Math
20 //
21 // Round to the nearest integer
22 console.log(Math.round(0.6)); // 1
23 console.log(Math.round(0.5)); // 1
24 console.log(Math.round(0.2)); // 0
25
26 console.log(Math.trunc(3.9)); // 3
27
28 // Round to the nearest integer
29 console.log(Math.round(-0.6)); // -1
```

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```
30
31 console.log(Math.ceil(-0.6)); // -0
32 console.log(Math.ceil(0.6)); // 1
33 console.log(Math.random()); // uniform distribution in the range 0 <= x < 1.0
34
35 console.log(Math.PI); // 3.141592653589793
36 console.log(Math.E); // 2.718281828459045
37
38 // max accepts arbitrary number of arguments
39 console.log(Math.max(2, 3)); // 3
40 console.log(Math.max(2, 3, 10)); // 10
41 console.log(Math.max(2, 3, 10, 100)); // 100
42
43 console.log(Math.pow(2, 5)); // 32
44
45 // note that sin() is not in degrees
46 console.log(Math.sin(30)); // -0.988
47 console.log(Math.sin(Math.PI / 6)); // 0.4999999
48
49 // natural log. base in 2.718
50 console.log(Math.log(Math.E)); // 1
51
52 console.log(Math.log(100) / Math.LN10); // 2
53 console.log(Math.log10(100)); // 2
54 console.log(Math.log2(1024)); // 10
55
56 // log1p(x) == log(1 + x)
57 console.log(Math.log1p(3), Math.log(1 + 3)); // 1.38629, 1.38629
58
59 console.log(Math.exp(2)); // e^2, 7.38905609893065
60
61 // division by 0 is not an error!
62 console.log(1 / 0); // Infinity
63 console.log(-1 / 0); // -Infinity
64 console.log(0 / 0); // NaN
65 console.log(Infinity / Infinity); // NaN
66 console.log(Number.isNaN(NaN)); // true
67
68 // a string to a number
69 console.log(Number("123") === 123); // true
70
71 console.log(parseInt("20") === 20); // true
72
```

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```
73 // parseInt supports hexadecimal!
74 // It does not support octal or binary.
75 console.log(parseInt("0x20") === 32); // true
76
77 console.log(parseInt("20", 16) === 32); // true
78
79 // octal
80 console.log(parseInt("20", 8) === 16); // true
```

12.3 Boolean in javascript

Listing 7: ./code/boolean.js

```
1  /*
2  null
3  undefined
4  0
5  -0
6  NaN
7  ""
8
9  The above 6 values are automatically converted to false.
10
11 All other values are converted to true. Even empty array is converted to true
12 */
13
14 console.log(true && null);      // null
15 console.log(true && undefined); // undefined
16 console.log([] && true);       // true
17 console.log([] || false);     // []
```

12.4 Strings in javascript

Listing 8: ./code/strings.js

```
1  let s = "abcdefg";
2  console.log(s.length);    // 7
3  console.log(`s is ${s}`); // s is abcdefg
4
```

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```
5 // start 1 (inclusive)
6 // end 4 (exclusive)
7 console.log(s.substring(1, 4)); // bcd
8 console.log(s.slice(1, 4));    // bcd
9
10 // last 3 characters
11 console.log(s.slice(-3)); // efg
12
13 // start 3 (inclusive)
14 // to the end of the string
15 console.log(s.slice(3)); // defg
16
17 // split using delimiter 'd'
18 console.log(s.split('d')) // ['abc', 'efg']
19
20 // split using delimiter 'de'
21 console.log(s.split('de')) // ['abc', 'fg']
22
23 s = "1232526314";
24 // note: the delimiter is the whole string '23'
25 console.log(s.split('23')) // ['1', '2526314']
26
27 console.log(s.split('2')) // ['1', '3', '5', '6314']
28
29 s = "abcdefga";
30 console.log(s.indexOf('a')); // 0, only the first occurrence
31 console.log(s.indexOf('a', 2)); // 7, search starting from the 2nd position
32 console.log(s.indexOf('c')); // 2
33 console.log(s.indexOf('c', 1)); // 2
34 console.log(s.indexOf('c', 2)); // 2
35 console.log(s.indexOf('c', 3)); // -1, not found
36 console.log(s.indexOf('x')); // -1
37
38 console.log(s.lastIndexOf('a')); // 7
39 console.log(s.indexOf('bc')); // 1
40
41 console.log(s.startsWith('a')); // true
42 console.log(s.startsWith('ab')); // true
43 console.log(s.startsWith('ac')); // false
44
45 console.log(s.endsWith('a')); // true
46 console.log(s.endsWith('ga')); // true
47 console.log(s.endsWith('da')); // false
```

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```
48
49 // include a substring
50 console.log(s.includes('de')); // true
51 console.log(s.includes('ded')); // false
52
53 // s is not changed. It return a new string
54 console.log(s.replace('ab', 'AB')); // ABcdefga
55 console.log(s.toUpperCase());      // ABCDEFGA
56 console.log(s.toLowerCase());      // abcdefga
57
58 console.log(s[0]);                  // a
59 console.log(s[1]);                  // b
60 console.log(s.charAt(0));           // a
61 console.log(s.charAt(1));           // b
62
63 // pad spaces to the left to make the length 3
64 console.log('a' +
65             'x'.padStart(3)) // "a  x"
66
67 console.log('a' +
68             'x'.padStart(3, '-')) // "a--x"
69
70 // pad with '-' to the left to make the total length 5
71 console.log('a' +
72             'xy'.padStart(5, '-')) // "a---xy"
73
74 console.log('a' +
75             '  b  '.trim() + 'c'); // abc
76
77 console.log('a' +
78             '  b  '.trimStart() + 'c'); // ab  c
79
80 console.log('a' +
81             '  b  '.trimEnd() + 'c'); // a  bc
82
83 console.log('ab'.repeat(3)); // ababab
84
85 console.log(String(123) === "123"); // true, a number to a string
86
87 let k = 20
88 console.log(k.toString() === "20"); // true
89 console.log("0x" + k.toString(16)); // 0x14, hexadecimal
90
```

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```
91 k = 12.3456;  
92 console.log(k.toFixed(0)); // 12  
93  
94 k = 12.5;  
95 console.log(k.toFixed(0)); // 13, note that it performs rounding  
96  
97 k = 12.3456;  
98 console.log(k.toFixed(1)); // 12.3  
99 console.log(k.toFixed(2)); // 12.35, note that it performs rounding  
100 console.log(k.toFixed(5)); // 12.34560  
101  
102 console.log(k.toExponential(1)); // 1.2e+1  
103 console.log(k.toExponential(2)); // 1.23e+1  
104 console.log(k.toExponential(3)); // 1.235e+1  
105  
106 console.log(k.toPrecision(1)); // 1e+1  
107 console.log(k.toPrecision(2)); // 12  
108 console.log(k.toPrecision(3)); // 12.3  
109 console.log(k.toPrecision(4)); // 12.35  
110 console.log(k.toPrecision(5)); // 12.346
```

String interpolation example:

```
const myName = "world";  
console.log(`hello ${myName}`);
```

12.5 array in javascript

All typed array are zero initialized.

Listing 9: ./code/array2.js

```
1 // how to iterate an array  
2 const a = [ 10, 20, 30 ];  
3 a.forEach((val, index) => {console.log(index, val)});  
4 /*  
5 0 10  
6 1 20  
7 2 30  
8 */
```

Listing 10: ./code/array.js

```
1  let a = [ 1, 2, 3 ];
2  let b = a; // a reference
3  b[0] = 100;
4  console.log(a[0] === 100); // true
5
6  a = [ 1, 2, 3 ];
7  let c = [];
8  for (let i = 0; i < a.length; ++i) {
9    c[i] = a[i]; // note: no need to pre-allocate space for c
10 }
11 console.log(c); // [1, 2, 3]
12
13 c[10] = 20;
14 console.log(c); // [1, 2, 3, <7 empty items>, 20]
15 console.log(c[4]); // undefined
16 console.log(c[5]); // undefined
17 console.log(c.length); // 11, note is 1 larger than 10
18
19 let s = "";
20 for (let i in c) {
21   s += ` ${i}`;
22 }
23 // note that indexes with undefined are not printed
24 console.log(s); // 0 1 2 10
25
26 s = "";
27 for (let i of c) {
28   s += ` ${i}`;
29 }
30
31 // there are seven undefined below
32 console.log(s); // 1 2 3 undefined undefined .... 20
33
34 a = [ 1, 2, 3 ]
35 c = Array.from(a); // return a copy of the array a
36 c[0] = 100;
37 console.log(a[0] == 1); // true
38
39 function equalArrays(a, b) {
40   if (a === b) {
41     return true;
42   }
```

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```
43
44   if (a.length !== b.length) {
45       return false;
46   }
47
48   for (let i = 0; i < a.length; i++) {
49       if (a[i] !== b[i]) {
50           return false;
51       }
52   }
53
54   return true;
55 }
56
57 console.log(equalArrays(a, c));           // false
58 console.log(equalArrays(a, a));           // true
59 console.log(equalArrays(a, [ 1, 2, 3 ])); // true
60 c[0] = 1;
61 console.log(equalArrays(a, c));           // true
62
63 a = [ 1, 2, 3 ];
64 a.push(4); // use push to append()
65 a.push(5);
66 console.log(a); // [1, 2, 3, 4, 5]
67 a.push(6, 'ten'); // push two elements
68 console.log(a); // [1, 2, 3, 4, 5, 6, 'ten']
```

12.6 typed array in javascript

Typed array is a view into a buffer.

When creating from an array, the array is copied to a newly created buffer.

When creating from a length, a buffer is allocated, which is initialized with 0s.

When creating from a buffer, the buffer is shared.

See

- https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/TypedArray
- https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Typed_arrays

Listing 11: ./code/typed_array.js

```
1  // https://nodejs.org/api/assert.html
2  const assert = require('node:assert');
3
4  assert.equal(Int8Array.BYTES_PER_ELEMENT, 1);
5  assert.equal(Uint8Array.BYTES_PER_ELEMENT, 1);
6
7  assert.equal(Int16Array.BYTES_PER_ELEMENT, 2);
8  assert.equal(Uint16Array.BYTES_PER_ELEMENT, 2);
9
10 assert.equal(Int32Array.BYTES_PER_ELEMENT, 4);
11 assert.equal(Uint32Array.BYTES_PER_ELEMENT, 4);
12
13 assert.equal(Float32Array.BYTES_PER_ELEMENT, 4);
14 assert.equal(Float64Array.BYTES_PER_ELEMENT, 8);
15
16 function test_uint32_array() {
17   let a = new Uint32Array(2);
18   assert.equal(a.length, 2);
19   assert.equal(a[1], 0);
20   a[0] = 0x12345678
21   assert.equal(a[0], 0x12345678);
22
23   // type of buf: ArrayBuffer
24   let buf = a.buffer;
25   assert.equal(buf.byteLength, a.length * Uint32Array.BYTES_PER_ELEMENT);
26   let b = new Uint8Array(buf)
27   assert.equal(b[0], 0x78); // little endian
28   b[1] = 0x23;             // also changes a
29   assert.equal(a[0], 0x12342378);
30 }
31
32 function test_int32_array() {
33   let a = new Int32Array(2);
34   // all typed arrays are 0 initialized
35   assert.equal(a[0], 0);
36   assert.equal(a[1], 0);
37
38   let b = new Int32Array(a); // copy
39   b[0] = 10;
40   assert.equal(a[0], 0);
41
42   let c = Int32Array.from(a);
```

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```
43 c[0] = 1;
44 assert.equal(a[0], 0);
45
46 let d = Int32Array.of(1, 2, 3);
47 assert.equal(d.length, 3);
48 assert.equal(d[0], 1);
49 assert.equal(d[1], 2);
50 assert.equal(d[2], 3);
51
52 let e = new Int32Array(a.buffer); // share the underlying buffer
53 e[0] = 11;
54 assert.equal(a[0], 11);
55
56 // the from method always copys
57 let f = Int32Array.from(a.buffer); // copy!
58 f[0] = 0;
59 assert.equal(a[0], 11); // a is not changed
60
61 let i = [ 5, 1 ];
62 let g = Int32Array.from(i); // from() always copy
63 assert.equal(g.length, 2);
64 assert.equal(g[0], 5);
65 i[0] = 10; // g is not changed
66 assert.equal(g[0], 5);
67 }
68
69 function test_int32_array2() {
70   let a = new Int32Array(10);
71   let b = [ 1, 2, 3 ];
72   a.set(b); // copy b to a, start from offset 0
73   assert.equal(a[0], b[0]);
74   assert.equal(a[1], b[1]);
75   assert.equal(a[2], b[2]);
76
77   a.set(b, 2); // copy b to a, start from offset 2
78   assert.equal(a[2], b[0]);
79   assert.equal(a[3], b[1]);
80   assert.equal(a[4], b[2]);
81
82   let c = Int32Array.of(100, 200);
83   a.set(c); // copy c to a, at offset 0
84   assert.equal(a[0], c[0]);
85   assert.equal(a[1], c[1]);
```

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```
86
87   a = Int32Array.of(0, 1, 2, 3, 4, 5, 6, 7, 8, 9);
88   b = a.slice(1, 3); // copy. start index 1 (included), stop index 3 (excluded)
89   assert.equal(b.length, 2);
90   assert.equal(b[0], a[1]);
91   assert.equal(b[1], a[2]);
92   b[0] = 100;
93   assert.equal(a[1], 1); // a is not changed
94
95   // share memory. start index 1 (included), stop index 3 (excluded)
96   c = a.subarray(1, 3);
97   assert.equal(c.length, 2);
98   assert.equal(c[0], a[1]);
99   assert.equal(c[1], a[2]);
100  c[0] = 100;
101  assert.equal(a[1], 100);
102
103  // slice() copies data!
104  let d = a.slice(1, 3);
105  assert.equal(d.length, 2);
106  assert.equal(d[0], a[1]);
107  assert.equal(d[1], a[2]);
108  d[0] = 1000;
109  assert.equal(a[1], 100); // a is not changed
110 }
111
112 test_uint32_array();
113 test_int32_array();
114 test_int32_array2();
```

Listing 12: ./code/int8_array.js

```
1  // https://nodejs.org/api/assert.html
2  const assert = require('node:assert');
3
4  // From an array
5  let a = [ 1, 2, 3 ];
6  let b = new Int8Array(a); // copy a into b
7  b[0] = 10
8  assert.equal(a[0], 1);
9  assert.equal(b[0], 10);
10 assert(b.length, 3);
11
```

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```
12 // From a length. 0 initialized
13 c = new Int8Array(2);
14 assert.equal(c.length, 2);
15 for (let i of c) {
16     assert.equal(i, 0);
17 }
18
19 assert.equal(c.BYTES_PER_ELEMENT, 1);
```

Listing 13: ./code/int32_array.js

```
1 // https://nodejs.org/api/assert.html
2 const assert = require('node:assert');
3
4 // 8 bytes buffer
5 let buffer = new ArrayBuffer(8);
6
7 let a = new Int32Array(buffer);
8 assert.equal(a.length, 2); // 2 ints
9 assert.equal(a.byteLength, 8); // 8 bytes
10 assert.equal(a.byteOffset, 0);
11
12 // a and b share the same buffer
13 let b = new Int8Array(buffer);
14
15 a[0] = 0x12345678;
16 assert.equal(b[0], 0x78); // little endian
17 assert.equal(b[1], 0x56); // little endian
18 assert.equal(b[2], 0x34); // little endian
19 assert.equal(b[3], 0x12); // little endian
20
21 // subarray(start, end) -> No copy
22 let c = b.subarray(1, 3); // view to the same buffer, no copy
23 assert.equal(c.length, 2);
24 assert.equal(c.byteOffset, 1);
25 b[1] = 0x23;
26 b[2] = 0x32;
27 assert.equal(c[0], 0x23);
28 assert.equal(c[1], 0x32);
29
30 // slice(start, end) -> copy
31 let d = b.slice(1, 3); // copy
32 assert.equal(d.length, 2);
```

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```
33 assert.equal(d[0], 0x23);
34 assert.equal(d[1], 0x32);
35 b[0] = 0x11;
36 b[1] = 0x22;
37 assert.equal(d[0], 0x23);
38 assert.equal(d[1], 0x32);
39
40 // share the same buffer
41 let e = new Int8Array(b.buffer);
42 b[0] = 0x30;
43 assert.equal(e[0], 0x30);
```

12.7 Regular expressions in javascript

Listing 14: ./code/regular_expression.js

```
1  /*
2  /^HTML/   Matches the letters HTML at the start of a string
3
4  /^HTML/i  Matches the letters HTML (ignore cases) at the start of a string
5
6  */
7
8  // /[0-9][0-9]*/   Match a non-zero digit, followed by any \# of digits
9  //
10 // /\bjavascript\b/i  Match "javascript" as a word, case-insensitive
11
12 console.log(/^HTML/.test("HTML ABC")); // true
13 console.log(/^HTML/.test("Html ABC")); // false
14 console.log(/^HTML/i.test("Html ABC")); // true, ignore cases
15
16 console.log(/[0-9][0-9]*/.test("0")); // true
17 console.log(/[0-9][0-9]*/.test("01234")); // true
18
19 console.log(/\bjavascript\b/i.test("hello javascript")); // true
20 console.log(/\bjavascript\b/i.test("hello_javascript")); // false
21
22 let text = "testing: 1, 2, 3"
23 let pattern = /\d+/g // matches all instances of one or more digits
24
25 console.log(pattern.test(text)); // true, a match exists
```

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```
26 // return the first matched position
27 console.log(text.search(pattern)); // 9
28 console.log(text.match(pattern)); // ['1', '2', '3'], array of all matches
29 console.log(text.replace(pattern, '#')); // testing: #, #, #
30
31
32 console.log(text.split(/\D+/)) // split on non-digit, ['', '1', '2', '3']
```

12.8 node

Go to <https://nodejs.org/en/download/> to download pre-built binaries:

```
wget https://nodejs.org/dist/v16.15.1/node-v16.15.1-linux-x64.tar.xz
tar xvf node-v16.15.1-linux-x64.tar.xz
```

and then add `/path/to/node-v16.15.1-linux-x64/bin/` to `PATH`.

```
node -v
```

to show the version, e.g., `v18.16.0`.

It also contains the executable `npm`. In python we have `requirements.txt` and we can use `pip install -r ./requirements.txt`. Similarly, there is `package.json` in javascript and we can use `npm install` in the folder containing `package.json`.

To create a new project, run `npm init -y`, it will generate `package.json`.

To install a local package, use `npm install package-name`. To remove a local package, use `npm remove package-name` or `npm uninstall package-name`.

To install a global package, use `npm install -g package-name`. To remove a global package, use `npm remove -g package-name` or `npm uninstall -g package-name`.

An alternative for `npm` is `yarn`: `npm install -g yarn`.

We can run `yarn` inside the directory containing `package.json`.

If a project contains a file `yarn.lock`, then it is using the tool `yarn`.

```
which node
which npm
```

```
/Users/fangjun/software/node-v21.1.0-darwin-x64/bin/node
/Users/fangjun/software/node-v21.1.0-darwin-x64/bin/npm
```

Globally installed npm packages are placed in `prefix/lib/node_modules` and executable files are placed in `prefix/bin`. Use `-g` or `--global` in `npm install`.

Locally installed npm packages are placed in `pwd/node_modules` and executable files are placed in `pwd/node_modules/.bin`.

```
npm install -g eslint
wchi eslint
```

prints:

```
/Users/fangjun/software/node-v21.1.0-darwin-x64/bin/eslint
```

To install a specific version, use:

```
npm install ejs@2.7.4
```

`npm i == npm install`.

`npm install --global == npm install -g`.

versions:

- `1.2.33`, an exact version
- `>1.2.33`
- `>=1.2.33``
- `<1.2.33``
- `<=1.2.33``
- `^1.2.33``, the latest compatible major version, i.e., `<2.0.0`.
- `~1.2.33`, the latest version `<1.3.0`. `~` means latest minor number.
- `*`, any version

To list locally installed packages:

```
npm list
```

To list globally installed packages:

```
npm list --global
npm list -g
```

12.8.1 glibc 2.32 on ubuntu 18.04

```
# see also https://github.com/nodesource/distributions/issues/1392
sudo apt install -y gawk

wget https://github.com/bminor/glibc/archive/refs/tags/glibc-2.32.tar.gz
tar xvf glibc-2.32.tar.gz
mkdir glibc-build
cd glibc-build
cd glibc-glibc-2.32/
unset LD_LIBRARY_PATH
unset CFLAGS
unset CPPFLAGS
unset CXXFLAGS
../configure --prefix=/star-fj/fangjun/software/glibc-2.32
make install
apt install -y patchelf
patchelf --set-interpreter /star-fj/fangjun/software/glibc-2.32/lib/ld-linux-x86-64.
↳so.2 --set-rpath /star-fj/fangjun/software/glibc-2.32/lib:/usr/lib/x86_64-linux-
↳gnu:/lib/x86_64-linux-gnu ./node
```

12.8.2 process

12.9 TODOs

1. This page https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/Installing_basic_software lists some tools to minify code:
 - WebPack: <https://webpack.js.org/>
 - Grunt: <https://gruntjs.com/>
 - Gulp: <https://gulpjs.com/>
2. Color picker tool: https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Colors/Color_picker_tool
3. Google font: <https://fonts.google.com/> and https://developers.google.com/fonts/docs/getting_started
4. <https://github.com/WiseLibs/better-sqlite3/blob/master/lib/database.js> study it!

12.10 npm

All packages can be found at <https://npmjs.org/>.

12.10.1 TODO

Read the following packages:

- <https://github.com/martinandert/date-names>
- <https://github.com/dcousens/ordinal.js>
- <https://github.com/TooTallNate/node-wav>

12.10.2 npm init -y

```
mkdir t
cd t
npm init -y
```

will generate a file `package.json`:

```
{
  "name": "t",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

12.10.3 mirror

```
npm config set registry https://registry.npmmirror.com
npm config get registry
```


12.10.4 To publish a new package for the first time

Available files:

```
/Users/fangjun/open-source/piper-phonemize/scripts/npm
(py312) fangjuns-MacBook-Pro:~$ ls
index.js                                package.json-2                        piper-phonemize-
↪wasm-nodejs.wasm README.md
package.json                            piper-phonemize-wasm-nodejs.js      piper-phonemize.
↪node.js
```

Now run:

```
cd scripts/npm

npm pack --dry-run

npm login
# It prints a URL, click it to open a browser

npm publish --access public
# Now you are ready to configure to publish it via github workflows
```

12.10.5 To pack a folder as a package and install it later

```
cd scripts/npm
npm pack
```

It will generate `<package-name>-<version>.tgz` inside the current package directory.

```
npm install ./xxx.tgz
```

12.11 module

Nodejs supports both CommonJS and ESmodules. Browsers supports only ESmodules.

12.11.1 CommonJS modules

Export a single function:

```
# mod1.js
const mod1Function = () => console.log('Mod1 is alive!');
module.exports = mod1Function;

# main.js
mod1Function = require('./mod1.js')
mod1Function()
# Since we used "module.exports = mod1Function;", we can use mod1Function() here.
↳ directly.
#
# If we use "module.exports = {mod1Function};", then we need to use "mod1Function.
↳ mod1Function();"

```

Export more function:

```
# mod1.js
const mod1Function = () => console.log('Mod1 is alive!');
const mod1Function2 = () => console.log('Mod1 is rolling, baby!');

module.exports = { mod1Function, mod1Function2 };

# main.js
const { mod1Function, mod1Function2 } = require('./mod1.js');
mod1Function();
mod1Function2();

# or main.js
const lib = require('./mod1.js');
lib.mod1Function();
lib.mod1Function2();

```

12.11.2 ESmodules

Use `npm init -y` to create `package.json`:

```
{
  "name": "modulestestapp",
  "version": "1.0.0",
  "description": "",

```

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```
"main": "index.js",
"scripts": {
  "test": "echo \\\"Error: no test specified\\\" && exit 1"
},
"keywords": [],
"author": "",
"license": "ISC",
"type": "module"
}
```

We need to manually add `"type": "module"`.

Export a single function:

```
// mod1.js
const mod1Function = () => console.log('Mod1 is alive!')
export { mod1Function }

// main.js
import { mod1Function } from './mod1.js'
// import { mod1Function } from 'mod1' // or use this one
mod1Function()
```

Export two functions:

```
// mod1.js
const mod1Function = () => console.log('Mod1 is alive!')
const mod1Function2 = () => console.log('Mod1 is rolling, baby!')

export { mod1Function, mod1Function2 }

// main.js
import { mod1Function, mod1Function2 } from './mod1.js'
// import { mod1Function, mod1Function2 } from 'mod1' // or use this one
mod1Function()
mod1Function2()

// main.js (rename imported functions)
import { mod1Function as funct1, mod1Function2 as funct2 } from './mod1.js'
funct1()
funct2()

// main.js (rename2)
import * as mod1 from './mod1.js'
```

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```
mod1.mod1Function()  
mod1.mod1Function2()
```

Default export:

```
// mod1.js  
const mod1Function = () => console.log('Mod1 is alive!')  
const mod1Function2 = () => console.log('Mod1 is rolling, baby!')  
  
export default mod1Function  
export { mod1Function2 }  
  
// main.js  
import mod1Function, { mod1Function2 } from './mod1.js'  
// import mod1Function, { mod1Function2 } from 'mod1' // or use this one  
mod1Function()  
mod1Function2()  
  
// main.js (rename it to lalala for the default export)  
import lalala, { mod1Function2 } from './mod1.js'  
lalala()  
mod1Function2()
```

To use a module in html, use

```
<script src="./main.js" type="module"></script>
```

12.11.3 webpack

To bundle all files of a module into a single file:

```
npm install --save-dev webpack webpack-cli
```

```
/* webpack.config.js */  
const path = require('path');  
  
module.exports = {  
  entry: './main.js',  
  output: {  
    path: path.resolve(__dirname, 'dist'),  
    filename: 'bundle.js',  
  },  
};
```

```
// package.json
{
  "name": "testappv2",
  "version": "1.0.0",
  "description": "",
  "main": "main.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "build": "webpack"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "devDependencies": {
    "webpack": "^5.72.0",
    "webpack-cli": "^4.9.2"
  }
}
```

After running `npm run build`, it will generate `dist/bundle.js`. We can use

```
<script src="./dist/bundle.js" type="module"></script>
```

See more at <https://www.freecodecamp.org/news/modules-in-javascript/>

12.12 object

Listing 15: ./code/object.js

```
1  const o = {  
2    "hello" : "world"  
3  };  
4  console.log(o.hello);           // world  
5  console.log(o["hello"]);       // world  
6  console.log('hello' in o);     // true  
7  console.log(!('hello' in o));  // false  
8  console.log(o.foo);            // undefined  
9  console.log(o.foo || 'bar');   // bar  
10 console.log(undefined || 'foobar'); // foobar
```

12.13 node-addon-api

See also <https://nodejs.github.io/node-addon-examples/getting-started/first/>

GYP documentation: <https://gyp.gsrc.io/docs/UserDocumentation.md>

- <https://github.com/mapbox/node-pre-gyp>
- <https://github.com/bchr02/node-pre-gyp-github>
- How to write a c++ node addon api with cmake-js and node-addon-api module
- Unlocking the Power of Native Modules in JavaScript: An Introductory Guide
- Mastering Native Node.js Addons with node-addon-api: A Comprehensive Guide
- <https://github.com/nodejs/node-addon-api/blob/main/doc/object.md>

```
# See https://www.npmjs.com/package/generator-napi-module  
npm install -g yo  
npm install -g generator-napi-module
```

```
mkdir first  
cd first  
yo napi-module  
npm test
```

To install node-gyp:

```
npm install node-gyp
```

You will find the executable inside `./node_modules/.bin/node-gyp`.

You can run:

```
./node_modules/.bin/node-gyp configure  
./node_modules/.bin/node-gyp build
```

```
node -p "require('node-addon-api')"
```

prints:

```
{  
  include: '/Users/fangjun/open-source/notes/docs/source/javascript/code/node-addon-  
↪api/first/node_modules/node-addon-api',  
  gyp: '/Users/fangjun/open-source/notes/docs/source/javascript/code/node-addon-api/  
↪first/node_modules/node-addon-api/src/node_api.gyp:nothing',  
  isNodeApiBuiltin: true,  
  needsFlag: false  
}
```

12.14 vue

See

- <https://vuejs.org/guide/introduction.html>
- <https://vuejs.org/tutorial/#step-1>
- <https://vuejs.org/guide/quick-start.html>

13.1 Hello world

Listing 1: hello_world.html

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Hello World</title>
  </head>
  <body>
    <p>Hello world</p>
  </body>
</html>
```

13.1.1 comments

```
<!-- this is a comment -->
```

13.1.2 images

```
</img>
</img>
</img>
```

13.1.3 ordered lists

```
<p> The following points </p>

<ol>
  <li> First </li>
  <li> Second </li>
</ol>
```

13.1.4 unordered lists

```
<p> The following points </p>

<ul>
  <li> foo </li>
  <li> bar </li>
</ul>
```

13.1.5 links

```
<a href="https://www.google.com">some text</a>
```

13.1.6 div

```
<label for="quota">Number of primes:</label>
<input type="text" id="quota" name="quota" value="1000000" />

<button id="generate">Generate primes</button>
<button id="reload">Reload</button>

<div id="output"></div>

<pre readonly class="event-log"></pre>

// js
const quota = document.querySelector("#quota");
const output = document.querySelector("#output");
document.querySelector("#generate").addEventListener("click", () => {
```

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```
const primes = generatePrimes(quota.value);

// note we use output.textContent for div
output.textContent = `Finished generating ${quota.value} primes!`;
});

document.querySelector("#reload").addEventListener("click", () => {
  document.location.reload();
});

const log = document.querySelector(".event-log");

// textContent for pre
log.textContent = "";
```

13.1.7 span

```
<span id="age"></span>

// js
var ageEl = document.getElementById("age")
ageEl.innerText = "22"
```

13.2 References

- Structuring the web with HTML

<https://developer.mozilla.org/en-US/docs/Learn/HTML>

14.1 Hello world

14.1.1 comment

```
/* this is a comment */
```

```
p { color: red; }
```

Then, in some html file, use:

```
<link href="abc/foo.css" rel="stylesheet">
```

14.1.2 Selector

- tag name or element name: e.g., `p` selects `<p>`; `h1` selects `<h1>`.
- ID: e.g., `#my-id` selects `<a id="my-id">` or `<p id="my-id">`
- class: e.g., `.my-class` selects `<a class="my-class">` and `<p class="my-class">`
- attribute: e.g., `img[src]` selects `` but not `<img>`

See https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/CSS_basics#different_types_of_selectors and https://developer.mozilla.org/en-US/docs/Learn/CSS/Building_blocks/Selectors for more.

Example with multiple rules:

Listing 1: Example with multiple rules

```
p {  
  color: red;  
  width: 500px;
```

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```
border: 1px solid black;
}
```

Example with multiple selectors:

Listing 2: Example with multiple selectors

```
p, li, h1 { color: red; }
```

14.2 References

- CSS basics

https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/CSS_basics

15.1 GIL

PROTOCOL BUFFERS

16.1 Installation

16.1.1 C++

See <https://github.com/protocolbuffers/protobuf/blob/main/src/README.md>.

```
make protocol-buffers
cd protocol-buffers
wget https://github.com/protocolbuffers/protobuf/releases/download/v3.20.1/protobuf-
↪all-3.20.1.tar.gz
tar xvf protobuf-all-3.20.1.tar.gz
cd protobuf-all-3.20.1
./configure --prefix=$HOME/software/protobuf-3.20.1
make -j 20
make -j 10 check
make install 2>&1 | tee my-log.txt
cd $HOME/software/protobuf-3.20.1
tree . > tree-log.txt
```

```
$ export PKG_CONFIG_PATH=$HOME/software/protobuf-3.20.1:$PKG_CONFIG_PATH

$ pkg-config --cflags protobuf
-I/root/fangjun/software/protobuf-3.20.1/include

$ pkg-config --libs protobuf
-L/root/fangjun/software/protobuf-3.20.1/lib -lprotobuf

$ pkg-config --cflags --libs  protobuf
-I/root/fangjun/software/protobuf-3.20.1/include -L/root/fangjun/software/protobuf-3.
↪20.1/lib -lprotobuf
```

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```
$ pkg-config --libs-only-L protobuf
-L/root/fangjun/software/protobuf-3.20.1/lib

$ pkg-config --libs-only-l protobuf
-lprotobuf
```

```
$ export PATH=$HOME/software/protobuf-3.20.1/bin:$PATH
$ protoc --version
libprotoc 3.20.1
```

Listing 1: ./code/my-log.txt (Installation logs)

```
1 Making install in .
2 make[1]: Entering directory '/ceph-fj/fangjun/open-source-2/protocol-buffers/protobuf-
  ↳ 3.20.1'
3 make[2]: Entering directory '/ceph-fj/fangjun/open-source-2/protocol-buffers/protobuf-
  ↳ 3.20.1'
4 make[2]: Nothing to be done for 'install-exec-am'.
5 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/lib/pkgconfig'
6 /usr/bin/install -c -m 644 protobuf.pc protobuf-lite.pc '/root/fangjun/software/
  ↳ protobuf-3.20.1/lib/pkgconfig'
7 make[2]: Leaving directory '/ceph-fj/fangjun/open-source-2/protocol-buffers/protobuf-
  ↳ 3.20.1'
8 make[1]: Leaving directory '/ceph-fj/fangjun/open-source-2/protocol-buffers/protobuf-
  ↳ 3.20.1'
9 Making install in src
10 make[1]: Entering directory '/ceph-fj/fangjun/open-source-2/protocol-buffers/protobuf-
  ↳ 3.20.1/src'
11 make[2]: Entering directory '/ceph-fj/fangjun/open-source-2/protocol-buffers/protobuf-
  ↳ 3.20.1/src'
12 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/lib'
13 /bin/bash ../libtool --mode=install /usr/bin/install -c libprotobuf-lite.la
  ↳ libprotobuf.la libprotoc.la '/root/fangjun/software/protobuf-3.20.1/lib'
14 libtool: install: /usr/bin/install -c .libs/libprotobuf-lite.so.31.0.1 /root/fangjun/
  ↳ software/protobuf-3.20.1/lib/libprotobuf-lite.so.31.0.1
15 libtool: install: (cd /root/fangjun/software/protobuf-3.20.1/lib && { ln -s -f
  ↳ libprotobuf-lite.so.31.0.1 libprotobuf-lite.so.31 || { rm -f libprotobuf-lite.so.31
  ↳ && ln -s libprotobuf-lite.so.31.0.1 libprotobuf-lite.so.31; }; })
16 libtool: install: (cd /root/fangjun/software/protobuf-3.20.1/lib && { ln -s -f
  ↳ libprotobuf-lite.so.31.0.1 libprotobuf-lite.so || { rm -f libprotobuf-lite.so && ln
  ↳ -s libprotobuf-lite.so.31.0.1 libprotobuf-lite.so; }; })
17 libtool: install: /usr/bin/install -c .libs/libprotobuf-lite.lai /root/fangjun/
  ↳ software/protobuf-3.20.1/lib/libprotobuf-lite.la
```

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```

18 libtool: install: /usr/bin/install -c .libs/libprotobuf.so.31.0.1 /root/fangjun/
    ↳software/protobuf-3.20.1/lib/libprotobuf.so.31.0.1
19 libtool: install: (cd /root/fangjun/software/protobuf-3.20.1/lib && { ln -s -f_
    ↳libprotobuf.so.31.0.1 libprotobuf.so.31 || { rm -f libprotobuf.so.31 && ln -s_
    ↳libprotobuf.so.31.0.1 libprotobuf.so.31; }; })
20 libtool: install: (cd /root/fangjun/software/protobuf-3.20.1/lib && { ln -s -f_
    ↳libprotobuf.so.31.0.1 libprotobuf.so || { rm -f libprotobuf.so && ln -s libprotobuf.
    ↳so.31.0.1 libprotobuf.so; }; })
21 libtool: install: /usr/bin/install -c .libs/libprotobuf.lai /root/fangjun/software/
    ↳protobuf-3.20.1/lib/libprotobuf.la
22 libtool: warning: relinking 'libprotoc.la'
23 libtool: install: (cd /root/fangjun/open-source-2/protocol-buffers/protobuf-3.20.1/
    ↳src; /bin/bash "/root/fangjun/open-source-2/protocol-buffers/protobuf-3.20.1/libtool
    ↳" --silent --tag CXX --mode=relink g++ -pthread -DHAVE_PTHREAD=1 -DHAVE_ZLIB=1 -
    ↳Wall -Wno-sign-compare -O2 -g -std=c++11 -DNDEBUG -version-info 31:1:0 -export-
    ↳dynamic -no-undefined -Wl,--version-script=./libprotoc.map -o libprotoc.la -rpath /
    ↳root/fangjun/software/protobuf-3.20.1/lib google/protobuf/compiler/code_generator.
    ↳lo google/protobuf/compiler/command_line_interface.lo google/protobuf/compiler/cpp/
    ↳cpp_enum.lo google/protobuf/compiler/cpp/cpp_enum_field.lo google/protobuf/compiler/
    ↳cpp/cpp_extension.lo google/protobuf/compiler/cpp/cpp_field.lo google/protobuf/
    ↳compiler/cpp/cpp_file.lo google/protobuf/compiler/cpp/cpp_generator.lo google/
    ↳protobuf/compiler/cpp/cpp_helpers.lo google/protobuf/compiler/cpp/cpp_map_field.lo_
    ↳google/protobuf/compiler/cpp/cpp_message.lo google/protobuf/compiler/cpp/cpp_
    ↳message_field.lo google/protobuf/compiler/cpp/cpp_padding_optimizer.lo google/
    ↳protobuf/compiler/cpp/cpp_parse_function_generator.lo google/protobuf/compiler/cpp/
    ↳cpp_primitive_field.lo google/protobuf/compiler/cpp/cpp_service.lo google/protobuf/
    ↳compiler/cpp/cpp_string_field.lo google/protobuf/compiler/csharp/csharp_doc_comment.
    ↳lo google/protobuf/compiler/csharp/csharp_enum.lo google/protobuf/compiler/csharp/
    ↳csharp_enum_field.lo google/protobuf/compiler/csharp/csharp_field_base.lo google/
    ↳protobuf/compiler/csharp/csharp_generator.lo google/protobuf/compiler/csharp/csharp_
    ↳helpers.lo google/protobuf/compiler/csharp/csharp_map_field.lo google/protobuf/
    ↳compiler/csharp/csharp_message.lo google/protobuf/compiler/csharp/csharp_message_
    ↳field.lo google/protobuf/compiler/csharp/csharp_primitive_field.lo google/protobuf/
    ↳compiler/csharp/csharp_reflection_class.lo google/protobuf/compiler/csharp/csharp_
    ↳repeated_enum_field.lo google/protobuf/compiler/csharp/csharp_repeated_message_
    ↳field.lo google/protobuf/compiler/csharp/csharp_repeated_primitive_field.lo google/
    ↳protobuf/compiler/csharp/csharp_source_generator_base.lo google/protobuf/compiler/
    ↳csharp/csharp_wrapper_field.lo google/protobuf/compiler/java/java_context.lo google/
    ↳protobuf/compiler/java/java_doc_comment.lo google/protobuf/compiler/java/java_enum.
    ↳lo google/protobuf/compiler/java/java_enum_field.lo google/protobuf/compiler/java/
    ↳java_enum_field_lite.lo google/protobuf/compiler/java/java_enum_lite.lo google/
    ↳protobuf/compiler/java/java_extension.lo google/protobuf/compiler/java/java_
    ↳extension_lite.lo google/protobuf/compiler/java/java_field.lo google/protobuf/

```

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```

↪compiler/java/java_file.lo google/protobuf/compiler/java/java_generator.lo google/
↪protobuf/compiler/java/java_generator_factory.lo google/protobuf/compiler/java/java_
↪helpers.lo google/protobuf/compiler/java/java_kotlin_generator.lo google/protobuf/
↪compiler/java/java_map_field.lo google/protobuf/compiler/java/java_map_field_lite.
↪lo google/protobuf/compiler/java/java_message.lo google/protobuf/compiler/java/java_
↪message_builder.lo google/protobuf/compiler/java/java_message_builder_lite.lo
↪google/protobuf/compiler/java/java_message_field.lo google/protobuf/compiler/java/
↪java_message_field_lite.lo google/protobuf/compiler/java/java_message_lite.lo
↪google/protobuf/compiler/java/java_name_resolver.lo google/protobuf/compiler/java/
↪java_primitive_field.lo google/protobuf/compiler/java/java_primitive_field_lite.lo
↪google/protobuf/compiler/java/java_service.lo google/protobuf/compiler/java/java_
↪shared_code_generator.lo google/protobuf/compiler/java/java_string_field.lo google/
↪protobuf/compiler/java/java_string_field_lite.lo google/protobuf/compiler/js/js_
↪generator.lo google/protobuf/compiler/js/well_known_types_embed.lo google/protobuf/
↪compiler/objectivec/objectivec_enum.lo google/protobuf/compiler/objectivec/
↪objectivec_enum_field.lo google/protobuf/compiler/objectivec/objectivec_extension.
↪lo google/protobuf/compiler/objectivec/objectivec_field.lo google/protobuf/compiler/
↪objectivec/objectivec_file.lo google/protobuf/compiler/objectivec/objectivec_
↪generator.lo google/protobuf/compiler/objectivec/objectivec_helpers.lo google/
↪protobuf/compiler/objectivec/objectivec_map_field.lo google/protobuf/compiler/
↪objectivec/objectivec_message.lo google/protobuf/compiler/objectivec/objectivec_
↪message_field.lo google/protobuf/compiler/objectivec/objectivec_oneof.lo google/
↪protobuf/compiler/objectivec/objectivec_primitive_field.lo google/protobuf/compiler/
↪php/php_generator.lo google/protobuf/compiler/plugin.lo google/protobuf/compiler/
↪plugin.pb.lo google/protobuf/compiler/python/python_generator.lo google/protobuf/
↪compiler/python/python_helpers.lo google/protobuf/compiler/python/python_pyi_
↪generator.lo google/protobuf/compiler/ruby/ruby_generator.lo google/protobuf/
↪compiler/subprocess.lo google/protobuf/compiler/zip_writer.lo -lpthread libprotobuf.
↪la -lz )
24 libtool: install: /usr/bin/install -c .libs/libprotoc.so.31.0.1T /root/fangjun/
↪software/protobuf-3.20.1/lib/libprotoc.so.31.0.1
25 libtool: install: (cd /root/fangjun/software/protobuf-3.20.1/lib && { ln -s -f
↪libprotoc.so.31.0.1 libprotoc.so.31 || { rm -f libprotoc.so.31 && ln -s libprotoc.
↪so.31.0.1 libprotoc.so.31; }; })
26 libtool: install: (cd /root/fangjun/software/protobuf-3.20.1/lib && { ln -s -f
↪libprotoc.so.31.0.1 libprotoc.so || { rm -f libprotoc.so && ln -s libprotoc.so.31.0.
↪1 libprotoc.so; }; })
27 libtool: install: /usr/bin/install -c .libs/libprotoc.lai /root/fangjun/software/
↪protobuf-3.20.1/lib/libprotoc.la
28 libtool: install: /usr/bin/install -c .libs/libprotobuf-lite.a /root/fangjun/software/
↪protobuf-3.20.1/lib/libprotobuf-lite.a
29 libtool: install: chmod 644 /root/fangjun/software/protobuf-3.20.1/lib/libprotobuf-
↪lite.a

```

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```

30 libtool: install: ranlib /root/fangjun/software/protobuf-3.20.1/lib/libprotobuf-lite.a
31 libtool: install: /usr/bin/install -c .libs/libprotobuf.a /root/fangjun/software/
↳protobuf-3.20.1/lib/libprotobuf.a
32 libtool: install: chmod 644 /root/fangjun/software/protobuf-3.20.1/lib/libprotobuf.a
33 libtool: install: ranlib /root/fangjun/software/protobuf-3.20.1/lib/libprotobuf.a
34 libtool: install: /usr/bin/install -c .libs/libprotoc.a /root/fangjun/software/
↳protobuf-3.20.1/lib/libprotoc.a
35 libtool: install: chmod 644 /root/fangjun/software/protobuf-3.20.1/lib/libprotoc.a
36 libtool: install: ranlib /root/fangjun/software/protobuf-3.20.1/lib/libprotoc.a
37 libtool: finish: PATH="/ceph-fj/fangjun/software/py38/bin:/ceph-fj/fangjun/software/
↳jdk-17.0.3/bin:/ceph-fj/fangjun/software/cmake/bin:/ceph-fj/fangjun/software/
↳texlive2021-20210325/bin/x86_64-linux:/ceph-sh1/fangjun/software/cuda-10.2.89/bin:/
↳ceph-fj/fangjun/software/bin:/ceph-sh1/fangjun/software/bin:/ceph-sh1/fangjun/
↳software/nvim-linux64/bin:/ceph-fj/fangjun/software/py38/bin:/ceph-fj/fangjun/
↳software/cmake/bin:/ceph-fj/fangjun/software/texlive2021-20210325/bin/x86_64-linux:/
↳ceph-sh1/fangjun/software/cuda-10.2.89/bin:/ceph-sh1/fangjun/software/nvim-linux64/
↳bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/sbin"
↳ldconfig -n /root/fangjun/software/protobuf-3.20.1/lib
38 -----
39 Libraries have been installed in:
40     /root/fangjun/software/protobuf-3.20.1/lib
41
42 If you ever happen to want to link against installed libraries
43 in a given directory, LIBDIR, you must either use libtool, and
44 specify the full pathname of the library, or use the '-LLIBDIR'
45 flag during linking and do at least one of the following:
46     - add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
47       during execution
48     - add LIBDIR to the 'LD_RUN_PATH' environment variable
49       during linking
50     - use the '-Wl,-rpath -Wl,LIBDIR' linker flag
51     - have your system administrator add LIBDIR to '/etc/ld.so.conf'
52
53 See any operating system documentation about shared libraries for
54 more information, such as the ld(1) and ld.so(8) manual pages.
55 -----
56 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/bin'
57 /bin/bash ../libtool --mode=install /usr/bin/install -c protoc '/root/fangjun/
↳software/protobuf-3.20.1/bin'
58 libtool: install: /usr/bin/install -c .libs/protoc /root/fangjun/software/protobuf-3.
↳20.1/bin/protoc
59 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include'
60 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf'

```

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```

61 /usr/bin/install -c -m 644 google/protobuf/any.proto google/protobuf/api.proto_
↳ google/protobuf/descriptor.proto google/protobuf/duration.proto google/protobuf/
↳ empty.proto google/protobuf/field_mask.proto google/protobuf/source_context.proto_
↳ google/protobuf/struct.proto google/protobuf/timestamp.proto google/protobuf/type.
↳ proto google/protobuf/wrappers.proto '/root/fangjun/software/protobuf-3.20.1/
↳ include/google/protobuf'
62 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/
↳ compiler'
63 /usr/bin/install -c -m 644 google/protobuf/compiler/plugin.proto '/root/fangjun/
↳ software/protobuf-3.20.1/include/google/protobuf/compiler'
64 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include'
65 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf'
66 /usr/bin/install -c -m 644 google/protobuf/any.h google/protobuf/any.pb.h google/
↳ protobuf/api.pb.h google/protobuf/arena.h google/protobuf/arena_impl.h google/
↳ protobuf/arenastring.h google/protobuf/arenaz_sampler.h google/protobuf/descriptor.
↳ h google/protobuf/descriptor.pb.h google/protobuf/descriptor_database.h google/
↳ protobuf/duration.pb.h google/protobuf/dynamic_message.h google/protobuf/empty.pb.h_
↳ google/protobuf/explicitly_constructed.h google/protobuf/extension_set.h google/
↳ protobuf/extension_set_inl.h google/protobuf/field_access_listener.h google/
↳ protobuf/field_mask.pb.h google/protobuf/generated_enum_reflection.h google/
↳ protobuf/generated_enum_util.h google/protobuf/generated_message_bases.h google/
↳ protobuf/generated_message_reflection.h google/protobuf/generated_message_tctable_
↳ decl.h google/protobuf/generated_message_tctable_impl.h google/protobuf/generated_
↳ message_util.h google/protobuf/has_bits.h google/protobuf/implicit_weak_message.h_
↳ google/protobuf/inlined_string_field.h google/protobuf/map.h google/protobuf/map_
↳ entry.h google/protobuf/map_entry_lite.h google/protobuf/map_field.h google/
↳ protobuf/map_field_inl.h google/protobuf/map_field_lite.h google/protobuf/map_type_
↳ handler.h google/protobuf/message.h google/protobuf/message_lite.h google/protobuf/
↳ metadata.h google/protobuf/metadata_lite.h google/protobuf/parse_context.h '/root/
↳ fangjun/software/protobuf-3.20.1/include/google/protobuf'
67 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/
↳ compiler/js'
68 /usr/bin/install -c -m 644 google/protobuf/compiler/js/js_generator.h '/root/
↳ fangjun/software/protobuf-3.20.1/include/google/protobuf/compiler/js'
69 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/
↳ compiler/cpp'
70 /usr/bin/install -c -m 644 google/protobuf/compiler/cpp/cpp_file.h google/protobuf/
↳ compiler/cpp/cpp_generator.h google/protobuf/compiler/cpp/cpp_helpers.h google/
↳ protobuf/compiler/cpp/cpp_names.h '/root/fangjun/software/protobuf-3.20.1/include/
↳ google/protobuf/compiler/cpp'
71 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/
↳ compiler/ruby'
72 /usr/bin/install -c -m 644 google/protobuf/compiler/ruby/ruby_generator.h '/root/

```

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```

↪fangjun/software/protobuf-3.20.1/include/google/protobuf/compiler/ruby'
73 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/
↪compiler/python'
74 /usr/bin/install -c -m 644 google/protobuf/compiler/python/python_generator.h_
↪google/protobuf/compiler/python/python_pyi_generator.h '/root/fangjun/software/
↪protobuf-3.20.1/include/google/protobuf/compiler/python'
75 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/util'
76 /usr/bin/install -c -m 644 google/protobuf/util/delimited_message_util.h google/
↪protobuf/util/field_comparator.h google/protobuf/util/field_mask_util.h google/
↪protobuf/util/json_util.h google/protobuf/util/message_differencer.h google/
↪protobuf/util/time_util.h google/protobuf/util/type_resolver.h google/protobuf/util/
↪type_resolver_util.h '/root/fangjun/software/protobuf-3.20.1/include/google/
↪protobuf/util'
77 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/io'
78 /usr/bin/install -c -m 644 google/protobuf/io/coded_stream.h google/protobuf/io/
↪gzip_stream.h google/protobuf/io/io_win32.h google/protobuf/io/printer.h google/
↪protobuf/io/strtod.h google/protobuf/io/tokenizer.h google/protobuf/io/zero_copy_
↪stream.h google/protobuf/io/zero_copy_stream_impl.h google/protobuf/io/zero_copy_
↪stream_impl_lite.h '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/
↪io'
79 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/
↪compiler/csharp'
80 /usr/bin/install -c -m 644 google/protobuf/compiler/csharp/csharp_doc_comment.h_
↪google/protobuf/compiler/csharp/csharp_generator.h google/protobuf/compiler/csharp/
↪csharp_names.h google/protobuf/compiler/csharp/csharp_options.h '/root/fangjun/
↪software/protobuf-3.20.1/include/google/protobuf/compiler/csharp'
81 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/
↪compiler/php'
82 /usr/bin/install -c -m 644 google/protobuf/compiler/php/php_generator.h '/root/
↪fangjun/software/protobuf-3.20.1/include/google/protobuf/compiler/php'
83 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/stubs'
84 /usr/bin/install -c -m 644 google/protobuf/stubs/bytestream.h google/protobuf/stubs/
↪callback.h google/protobuf/stubs/casts.h google/protobuf/stubs/common.h google/
↪protobuf/stubs/hash.h google/protobuf/stubs/logging.h google/protobuf/stubs/macros.
↪h google/protobuf/stubs/map_util.h google/protobuf/stubs/mutex.h google/protobuf/
↪stubs/once.h google/protobuf/stubs/platform_macros.h google/protobuf/stubs/port.h_
↪google/protobuf/stubs/status.h google/protobuf/stubs/stl_util.h google/protobuf/
↪stubs/stringpiece.h google/protobuf/stubs/strutil.h google/protobuf/stubs/template_
↪util.h '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/stubs'
85 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/
↪compiler/java'
86 /usr/bin/install -c -m 644 google/protobuf/compiler/java/java_generator.h google/
↪protobuf/compiler/java/java_kotlin_generator.h google/protobuf/compiler/java/java_

```

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```

↪names.h '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/compiler/
↪java'
87 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf'
88 /usr/bin/install -c -m 644 google/protobuf/port.h google/protobuf/port_def.inc_
↪google/protobuf/port_undef.inc google/protobuf/reflection.h google/protobuf/
↪reflection_ops.h google/protobuf/repeated_field.h google/protobuf/repeated_ptr_
↪field.h google/protobuf/service.h google/protobuf/source_context.pb.h google/
↪protobuf/struct.pb.h google/protobuf/text_format.h google/protobuf/timestamp.pb.h_
↪google/protobuf/type.pb.h google/protobuf/unknown_field_set.h google/protobuf/wire_
↪format.h google/protobuf/wire_format_lite.h google/protobuf/wrappers.pb.h '/root/
↪fangjun/software/protobuf-3.20.1/include/google/protobuf'
89 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/
↪compiler/objectivec'
90 /usr/bin/install -c -m 644 google/protobuf/compiler/objectivec/objectivec_generator.
↪h google/protobuf/compiler/objectivec/objectivec_helpers.h '/root/fangjun/software/
↪protobuf-3.20.1/include/google/protobuf/compiler/objectivec'
91 /bin/mkdir -p '/root/fangjun/software/protobuf-3.20.1/include/google/protobuf/
↪compiler'
92 /usr/bin/install -c -m 644 google/protobuf/compiler/code_generator.h google/
↪protobuf/compiler/command_line_interface.h google/protobuf/compiler/importer.h_
↪google/protobuf/compiler/parser.h google/protobuf/compiler/plugin.h google/protobuf/
↪compiler/plugin.pb.h '/root/fangjun/software/protobuf-3.20.1/include/google/
↪protobuf/compiler'
93 make[2]: Leaving directory '/ceph-fj/fangjun/open-source-2/protocol-buffers/protobuf-
↪3.20.1/src'
94 make[1]: Leaving directory '/ceph-fj/fangjun/open-source-2/protocol-buffers/protobuf-
↪3.20.1/src'

```

Listing 2: ./code/tree-log.txt (Installed files)

```

1 .
2 |-- bin
3 |   `-- protoc
4 |-- include
5 |   `-- google
6 |       `-- protobuf
7 |           |-- any.h
8 |           |-- any.pb.h
9 |           |-- any.proto
10 |          |-- api.pb.h
11 |          |-- api.proto
12 |          |-- arena.h
13 |          |-- arena_impl.h

```

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```
14 |         |-- arenastring.h
15 |         |-- arenaz_sampler.h
16 |         |-- compiler
17 |         |   |-- code_generator.h
18 |         |   |-- command_line_interface.h
19 |         |   |-- cpp
20 |         |   |   |-- cpp_file.h
21 |         |   |   |-- cpp_generator.h
22 |         |   |   |-- cpp_helpers.h
23 |         |   |   `-- cpp_names.h
24 |         |-- csharp
25 |         |   |-- csharp_doc_comment.h
26 |         |   |-- csharp_generator.h
27 |         |   |-- csharp_names.h
28 |         |   `-- csharp_options.h
29 |         |-- importer.h
30 |         |-- java
31 |         |   |-- java_generator.h
32 |         |   |-- java_kotlin_generator.h
33 |         |   `-- java_names.h
34 |         |-- js
35 |         |   `-- js_generator.h
36 |         |-- objectivec
37 |         |   |-- objectivec_generator.h
38 |         |   `-- objectivec_helpers.h
39 |         |-- parser.h
40 |         |-- php
41 |         |   `-- php_generator.h
42 |         |-- plugin.h
43 |         |-- plugin.pb.h
44 |         |-- plugin.proto
45 |         |-- python
46 |         |   |-- python_generator.h
47 |         |   `-- python_pyi_generator.h
48 |         `-- ruby
49 |             `-- ruby_generator.h
50 |     |-- descriptor.h
51 |     |-- descriptor.pb.h
52 |     |-- descriptor.proto
53 |     |-- descriptor_database.h
54 |     |-- duration.pb.h
55 |     |-- duration.proto
56 |     |-- dynamic_message.h
```

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```
57 |         |-- empty.pb.h
58 |         |-- empty.proto
59 |         |-- explicitly_constructed.h
60 |         |-- extension_set.h
61 |         |-- extension_set_inl.h
62 |         |-- field_access_listener.h
63 |         |-- field_mask.pb.h
64 |         |-- field_mask.proto
65 |         |-- generated_enum_reflection.h
66 |         |-- generated_enum_util.h
67 |         |-- generated_message_bases.h
68 |         |-- generated_message_reflection.h
69 |         |-- generated_message_tctable_decl.h
70 |         |-- generated_message_tctable_impl.h
71 |         |-- generated_message_util.h
72 |         |-- has_bits.h
73 |         |-- implicit_weak_message.h
74 |         |-- inlined_string_field.h
75 |         |-- io
76 |         |   |-- coded_stream.h
77 |         |   |-- gzip_stream.h
78 |         |   |-- io_win32.h
79 |         |   |-- printer.h
80 |         |   |-- strtod.h
81 |         |   |-- tokenizer.h
82 |         |   |-- zero_copy_stream.h
83 |         |   |-- zero_copy_stream_impl.h
84 |         |   |-- zero_copy_stream_impl_lite.h
85 |         |-- map.h
86 |         |-- map_entry.h
87 |         |-- map_entry_lite.h
88 |         |-- map_field.h
89 |         |-- map_field_inl.h
90 |         |-- map_field_lite.h
91 |         |-- map_type_handler.h
92 |         |-- message.h
93 |         |-- message_lite.h
94 |         |-- metadata.h
95 |         |-- metadata_lite.h
96 |         |-- parse_context.h
97 |         |-- port.h
98 |         |-- port_def.inc
99 |         |-- port_undef.inc
```

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```
100 |         |-- reflection.h
101 |         |-- reflection_ops.h
102 |         |-- repeated_field.h
103 |         |-- repeated_ptr_field.h
104 |         |-- service.h
105 |         |-- source_context.pb.h
106 |         |-- source_context.proto
107 |         |-- struct.pb.h
108 |         |-- struct.proto
109 |         |-- stubs
110 |         |   |-- bytestream.h
111 |         |   |-- callback.h
112 |         |   |-- casts.h
113 |         |   |-- common.h
114 |         |   |-- hash.h
115 |         |   |-- logging.h
116 |         |   |-- macros.h
117 |         |   |-- map_util.h
118 |         |   |-- mutex.h
119 |         |   |-- once.h
120 |         |   |-- platform_macros.h
121 |         |   |-- port.h
122 |         |   |-- status.h
123 |         |   |-- stl_util.h
124 |         |   |-- stringpiece.h
125 |         |   |-- strutil.h
126 |         |   |-- template_util.h
127 |         |-- text_format.h
128 |         |-- timestamp.pb.h
129 |         |-- timestamp.proto
130 |         |-- type.pb.h
131 |         |-- type.proto
132 |         |-- unknown_field_set.h
133 |         |-- util
134 |         |   |-- delimited_message_util.h
135 |         |   |-- field_comparator.h
136 |         |   |-- field_mask_util.h
137 |         |   |-- json_util.h
138 |         |   |-- message_differencer.h
139 |         |   |-- time_util.h
140 |         |   |-- type_resolver.h
141 |         |   |-- type_resolver_util.h
142 |         |-- wire_format.h
```

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```

143 |         |-- wire_format_lite.h
144 |         |-- wrappers.pb.h
145 |         `-- wrappers.proto
146 |-- lib
147 |   |-- libprotobuf-lite.a
148 |   |-- libprotobuf-lite.la
149 |   |-- libprotobuf-lite.so -> libprotobuf-lite.so.31.0.1
150 |   |-- libprotobuf-lite.so.31 -> libprotobuf-lite.so.31.0.1
151 |   |-- libprotobuf-lite.so.31.0.1
152 |   |-- libprotobuf.a
153 |   |-- libprotobuf.la
154 |   |-- libprotobuf.so -> libprotobuf.so.31.0.1
155 |   |-- libprotobuf.so.31 -> libprotobuf.so.31.0.1
156 |   |-- libprotobuf.so.31.0.1
157 |   |-- libprotoc.a
158 |   |-- libprotoc.la
159 |   |-- libprotoc.so -> libprotoc.so.31.0.1
160 |   |-- libprotoc.so.31 -> libprotoc.so.31.0.1
161 |   |-- libprotoc.so.31.0.1
162 |   `-- pkgconfig
163 |       |-- protobuf-lite.pc
164 |       `-- protobuf.pc
165 `-- tree-log.txt
166
167 18 directories, 146 files

```

16.1.2 Install with cmake

```

make protocol-buffers
cd protocol-buffers
wget https://github.com/protocolbuffers/protobuf/releases/download/v3.20.1/protobuf-
↪all-3.20.1.tar.gz
tar xvf protobuf-all-3.20.1.tar.gz
cd protobuf-all-3.20.1
mkdir my-build
cd my-build
cmake -Dprotobuf_BUILD_SHARED_LIBS=ON -DCMAKE_INSTALL_PREFIX=/ceph-fj/fangjun/
↪software/protobuf-3.20.1-cmake/ ../cmake 2>&1 | tee cmake-configure-shared-lib.log
make -j10 2>&1 | tee make-shared.log
make install 2>&1 | tee make-shared-install.log

cd $HOME/software/protobuf-3.20.1-cmake

```

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```
tree . > tree-cmake-log.txt
```

16.2 Hello

16.2.1 hello.proto

See

- <https://developers.google.com/protocol-buffers/docs/cpptutorial>
- <https://developers.google.com/protocol-buffers/docs/pythontutorial>

Listing 3: ./code/hello.proto

```
1 syntax = "proto2";
2
3 package tutorial;
4
5 // available types:
6 // bool, int32, float, double, string
7 message Person {
8     optional string name = 1;
9     optional int32 id = 2;
10    optional string email = 3;
11
12    enum PhoneType {
13        MOBILE = 0;
14        HOME = 1;
15        WORK = 2;
16    }
17
18    message PhoneNumber {
19        optional string number = 1;
20        optional PhoneType type = 2 [ default = HOME ];
21    }
22
23    repeated PhoneNumber phones = 4;
24 }
25
26 message AddressBook { repeated Person people = 1; }
```

16.2.2 makefile

Listing 4: ./code/Makefile

```
1 .PHONY: all clean
2
3 all: hello.pb.cc hello.pb.h hello_pb2.py
4
5 hello.pb.cc hello.pb.h: hello.proto
6     protoc -I=./ --cpp_out=./ ./hello.proto
7
8 hello_pb2.py: hello.proto
9     protoc -I=./ --python_out=./ ./hello.proto
10
11 clean:
12     $(RM) hello.pb.cc hello.pb.h hello_pb2.py
```

16.2.3 hello.pb.h

Listing 5: ./code/hello.pb.h

```
1 // Generated by the protocol buffer compiler.  DO NOT EDIT!
2 // source: hello.proto
3
4 #ifndef GOOGLE_PROTOBUF_INCLUDED_hello_2eproto
5 #define GOOGLE_PROTOBUF_INCLUDED_hello_2eproto
6
7 #include <limits>
8 #include <string>
9
10 #include <google/protobuf/port_def.inc>
11 #if PROTOBUF_VERSION < 3020000
12 #error This file was generated by a newer version of protoc which is
13 #error incompatible with your Protocol Buffer headers. Please update
14 #error your headers.
15 #endif
16 #if 3020001 < PROTOBUF_MIN_PROTOC_VERSION
17 #error This file was generated by an older version of protoc which is
18 #error incompatible with your Protocol Buffer headers. Please
19 #error regenerate this file with a newer version of protoc.
20 #endif
21
22 #include <google/protobuf/port_undef.inc>
```

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```

23 #include <google/protobuf/io/coded_stream.h>
24 #include <google/protobuf/arena.h>
25 #include <google/protobuf/arenastring.h>
26 #include <google/protobuf/generated_message_util.h>
27 #include <google/protobuf/metadata_lite.h>
28 #include <google/protobuf/generated_message_reflection.h>
29 #include <google/protobuf/message.h>
30 #include <google/protobuf/repeated_field.h> // IWYU pragma: export
31 #include <google/protobuf/extension_set.h> // IWYU pragma: export
32 #include <google/protobuf/generated_enum_reflection.h>
33 #include <google/protobuf/unknown_field_set.h>
34 // @@protoc_insertion_point(includes)
35 #include <google/protobuf/port_def.inc>
36 #define PROTOBUF_INTERNAL_EXPORT_hello_2eproto
37 PROTOBUF_NAMESPACE_OPEN
38 namespace internal {
39 class AnyMetadata;
40 } // namespace internal
41 PROTOBUF_NAMESPACE_CLOSE
42
43 // Internal implementation detail -- do not use these members.
44 struct TableStruct_hello_2eproto {
45     static const uint32_t offsets[];
46 };
47 extern const ::PROTOBUF_NAMESPACE_ID::internal::DescriptorTable descriptor_table_
48 ↪ hello_2eproto;
49 namespace tutorial {
50 class AddressBook;
51 struct AddressBookDefaultTypeInternal;
52 extern AddressBookDefaultTypeInternal _AddressBook_default_instance_;
53 class Person;
54 struct PersonDefaultTypeInternal;
55 extern PersonDefaultTypeInternal _Person_default_instance_;
56 class Person_PhoneNumber;
57 struct Person_PhoneNumberDefaultTypeInternal;
58 extern Person_PhoneNumberDefaultTypeInternal _Person_PhoneNumber_default_instance_;
59 } // namespace tutorial
60 PROTOBUF_NAMESPACE_OPEN
61 template<> ::tutorial::AddressBook* Arena::CreateMaybeMessage<::tutorial::AddressBook>
62 ↪ (Arena*);
63 template<> ::tutorial::Person* Arena::CreateMaybeMessage<::tutorial::Person> (Arena*);
64 template<> ::tutorial::Person_PhoneNumber* Arena::CreateMaybeMessage
65 ↪ <::tutorial::Person_PhoneNumber> (Arena*);

```

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```

63 PROTOBUF_NAMESPACE_CLOSE
64 namespace tutorial {
65
66 enum Person_PhoneType : int {
67     Person_PhoneType_MOBILE = 0,
68     Person_PhoneType_HOME = 1,
69     Person_PhoneType_WORK = 2
70 };
71 bool Person_PhoneType_IsValid(int value);
72 constexpr Person_PhoneType Person_PhoneType_PhoneType_MIN = Person_PhoneType_MOBILE;
73 constexpr Person_PhoneType Person_PhoneType_PhoneType_MAX = Person_PhoneType_WORK;
74 constexpr int Person_PhoneType_PhoneType_ARRAYSIZE = Person_PhoneType_PhoneType_MAX +
↳ 1;
75
76 const ::PROTOBUF_NAMESPACE_ID::EnumDescriptor* Person_PhoneType_descriptor();
77 template<typename T>
78 inline const std::string& Person_PhoneType_Name(T enum_t_value) {
79     static_assert(::std::is_same<T, Person_PhoneType>::value ||
80         ::std::is_integral<T>::value,
81         "Incorrect type passed to function Person_PhoneType_Name.");
82     return ::PROTOBUF_NAMESPACE_ID::internal::NameOfEnum(
83         Person_PhoneType_descriptor(), enum_t_value);
84 }
85 inline bool Person_PhoneType_Parse(
86     ::PROTOBUF_NAMESPACE_ID::ConstStringParam name, Person_PhoneType* value) {
87     return ::PROTOBUF_NAMESPACE_ID::internal::ParseNamedEnum<Person_PhoneType>(
88         Person_PhoneType_descriptor(), name, value);
89 }
90 // =====
91
92 class Person_PhoneNumber final :
93     public ::PROTOBUF_NAMESPACE_ID::Message /* @@protoc_insertion_point(class_
↳ definition:tutorial.Person.PhoneNumber) */ {
94 public:
95     inline Person_PhoneNumber() : Person_PhoneNumber(nullptr) {}
96     ~Person_PhoneNumber() override;
97     explicit PROTOBUF_CONSTEXPR Person_PhoneNumber(::PROTOBUF_NAMESPACE_
↳ ID::internal::ConstantInitialized);
98
99     Person_PhoneNumber(const Person_PhoneNumber& from);
100     Person_PhoneNumber(Person_PhoneNumber&& from) noexcept
101         : Person_PhoneNumber() {
102         *this = ::std::move(from);

```

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```

103     }
104
105     inline Person_PhoneNumber& operator=(const Person_PhoneNumber& from) {
106         CopyFrom(from);
107         return *this;
108     }
109     inline Person_PhoneNumber& operator=(Person_PhoneNumber&& from) noexcept {
110         if (this == &from) return *this;
111         if (GetOwningArena() == from.GetOwningArena())
112             #ifdef PROTOBUF_FORCE_COPY_IN_MOVE
113                 && GetOwningArena() != nullptr
114             #endif // !PROTOBUF_FORCE_COPY_IN_MOVE
115         ) {
116             InternalSwap(&from);
117         } else {
118             CopyFrom(from);
119         }
120         return *this;
121     }
122
123     inline const ::PROTOBUF_NAMESPACE_ID::UnknownFieldSet& unknown_fields() const {
124         return _internal_metadata_.unknown_fields<::PROTOBUF_NAMESPACE_
↵ID::UnknownFieldSet> (::PROTOBUF_NAMESPACE_ID::UnknownFieldSet::default_instance);
125     }
126     inline ::PROTOBUF_NAMESPACE_ID::UnknownFieldSet* mutable_unknown_fields() {
127         return _internal_metadata_.mutable_unknown_fields<::PROTOBUF_NAMESPACE_
↵ID::UnknownFieldSet> ();
128     }
129
130     static const ::PROTOBUF_NAMESPACE_ID::Descriptor* descriptor() {
131         return GetDescriptor();
132     }
133     static const ::PROTOBUF_NAMESPACE_ID::Descriptor* GetDescriptor() {
134         return default_instance().GetMetadata().descriptor;
135     }
136     static const ::PROTOBUF_NAMESPACE_ID::Reflection* GetReflection() {
137         return default_instance().GetMetadata().reflection;
138     }
139     static const Person_PhoneNumber& default_instance() {
140         return *internal_default_instance();
141     }
142     static inline const Person_PhoneNumber* internal_default_instance() {
143         return reinterpret_cast<const Person_PhoneNumber*>(

```

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```

144         &_Person_PhoneNumber_default_instance_);
145     }
146     static constexpr int kIndexInFileMessages =
147         0;
148
149     friend void swap(Person_PhoneNumber& a, Person_PhoneNumber& b) {
150         a.Swap(&b);
151     }
152     inline void Swap(Person_PhoneNumber* other) {
153         if (other == this) return;
154 #ifdef PROTOBUF_FORCE_COPY_IN_SWAP
155         if (GetOwningArena() != nullptr &&
156             GetOwningArena() == other->GetOwningArena()) {
157             #else // PROTOBUF_FORCE_COPY_IN_SWAP
158             if (GetOwningArena() == other->GetOwningArena()) {
159 #endif // !PROTOBUF_FORCE_COPY_IN_SWAP
160                 InternalSwap(other);
161             } else {
162                 ::PROTOBUF_NAMESPACE_ID::internal::GenericSwap(this, other);
163             }
164         }
165     void UnsafeArenaSwap(Person_PhoneNumber* other) {
166         if (other == this) return;
167         GOOGLE_DCHECK(GetOwningArena() == other->GetOwningArena());
168         InternalSwap(other);
169     }
170
171     // implements Message -----
172
173     Person_PhoneNumber* New(::PROTOBUF_NAMESPACE_ID::Arena* arena = nullptr) const_
174     ↪ final {
175         return CreateMaybeMessage<Person_PhoneNumber>(arena);
176     }
177     using ::PROTOBUF_NAMESPACE_ID::Message::CopyFrom;
178     void CopyFrom(const Person_PhoneNumber& from);
179     using ::PROTOBUF_NAMESPACE_ID::Message::MergeFrom;
180     void MergeFrom(const Person_PhoneNumber& from);
181     private:
182     static void MergeImpl(::PROTOBUF_NAMESPACE_ID::Message* to, const ::PROTOBUF_
183     ↪ NAMESPACE_ID::Message& from);
184     public:
185     PROTOBUF_ATTRIBUTE_REINITIALIZES void Clear() final;
186     bool IsInitialized() const final;

```

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```

185
186     size_t ByteSizeLong() const final;
187     const char* _InternalParse(const char* ptr, ::PROTOBUF_NAMESPACE_
↪ID::internal::ParseContext* ctx) final;
188     uint8_t* _InternalSerialize(
189         uint8_t* target, ::PROTOBUF_NAMESPACE_ID::io::EpsCopyOutputStream* stream) ↪
↪const final;
190     int GetCachedSize() const final { return _cached_size_.Get(); }
191
192     private:
193     void SharedCtor();
194     void SharedDtor();
195     void SetCachedSize(int size) const final;
196     void InternalSwap(Person_PhoneNumber* other);
197
198     private:
199     friend class ::PROTOBUF_NAMESPACE_ID::internal::AnyMetadata;
200     static ::PROTOBUF_NAMESPACE_ID::StringPiece FullMessageName() {
201         return "tutorial.Person.PhoneNumber";
202     }
203     protected:
204     explicit Person_PhoneNumber(::PROTOBUF_NAMESPACE_ID::Arena* arena,
205                                 bool is_message_owned = false);
206     public:
207
208     static const ClassData _class_data_;
209     const ::PROTOBUF_NAMESPACE_ID::Message::ClassData*GetClassData() const final;
210
211     ::PROTOBUF_NAMESPACE_ID::Metadata GetMetadata() const final;
212
213     // nested types -----
214
215     // accessors -----
216
217     enum : int {
218         kNumberFieldNumber = 1,
219         kTypeFieldNumber = 2,
220     };
221     // optional string number = 1;
222     bool has_number() const;
223     private:
224     bool _internal_has_number() const;
225     public:

```

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```

226 void clear_number();
227 const std::string& number() const;
228 template <typename ArgT0 = const std::string&, typename... ArgT>
229 void set_number(ArgT0&& arg0, ArgT... args);
230 std::string* mutable_number();
231 PROTOBUF_NODISCARD std::string* release_number();
232 void set_allocated_number(std::string* number);
233 private:
234 const std::string& _internal_number() const;
235 inline PROTOBUF_ALWAYS_INLINE void _internal_set_number(const std::string& value);
236 std::string* _internal_mutable_number();
237 public:
238
239 // optional .tutorial.Person.PhoneType type = 2 [default = HOME];
240 bool has_type() const;
241 private:
242 bool _internal_has_type() const;
243 public:
244 void clear_type();
245 ::tutorial::Person_PhoneType type() const;
246 void set_type(::tutorial::Person_PhoneType value);
247 private:
248 ::tutorial::Person_PhoneType _internal_type() const;
249 void _internal_set_type(::tutorial::Person_PhoneType value);
250 public:
251
252 // @@protoc_insertion_point(class_scope:tutorial.Person.PhoneNumber)
253 private:
254 class _Internal;
255
256 template <typename T> friend class ::PROTOBUF_NAMESPACE_ID::Arena::InternalHelper;
257 typedef void InternalArenaConstructable_;
258 typedef void DestructorSkippable_;
259 ::PROTOBUF_NAMESPACE_ID::internal::HasBits<1> _has_bits_;
260 mutable ::PROTOBUF_NAMESPACE_ID::internal::CachedSize _cached_size_;
261 ::PROTOBUF_NAMESPACE_ID::internal::ArenaStringPtr number_;
262 int type_;
263 friend struct ::TableStruct_hello_2eproto;
264 };
265 // -----
266
267 class Person final :
268     public ::PROTOBUF_NAMESPACE_ID::Message /* @@protoc_insertion_point(class_

```

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```

↪definition:tutorial.Person) */ {
269 public:
270     inline Person() : Person(nullptr) {}
271     ~Person() override;
272     explicit PROTOBUF_CONSTEXPR Person(::PROTOBUF_NAMESPACE_
↪ID::internal::ConstantInitialized);

273
274     Person(const Person& from);
275     Person(Person&& from) noexcept
276         : Person() {
277         *this = ::std::move(from);
278     }
279
280     inline Person& operator=(const Person& from) {
281         CopyFrom(from);
282         return *this;
283     }
284     inline Person& operator=(Person&& from) noexcept {
285         if (this == &from) return *this;
286         if (GetOwningArena() == from.GetOwningArena())
287             #ifdef PROTOBUF_FORCE_COPY_IN_MOVE
288                 && GetOwningArena() != nullptr
289             #endif // !PROTOBUF_FORCE_COPY_IN_MOVE
290         ) {
291             InternalSwap(&from);
292         } else {
293             CopyFrom(from);
294         }
295         return *this;
296     }
297
298     inline const ::PROTOBUF_NAMESPACE_ID::UnknownFieldSet& unknown_fields() const {
299         return _internal_metadata_.unknown_fields<::PROTOBUF_NAMESPACE_
↪ID::UnknownFieldSet> (::PROTOBUF_NAMESPACE_ID::UnknownFieldSet::default_instance);
300     }
301     inline ::PROTOBUF_NAMESPACE_ID::UnknownFieldSet* mutable_unknown_fields() {
302         return _internal_metadata_.mutable_unknown_fields<::PROTOBUF_NAMESPACE_
↪ID::UnknownFieldSet> ();
303     }
304
305     static const ::PROTOBUF_NAMESPACE_ID::Descriptor* descriptor() {
306         return GetDescriptor();
307     }

```

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```

308 static const ::PROTOBUF_NAMESPACE_ID::Descriptor* GetDescriptor() {
309     return default_instance().GetMetadata().descriptor;
310 }
311 static const ::PROTOBUF_NAMESPACE_ID::Reflection* GetReflection() {
312     return default_instance().GetMetadata().reflection;
313 }
314 static const Person& default_instance() {
315     return *internal_default_instance();
316 }
317 static inline const Person* internal_default_instance() {
318     return reinterpret_cast<const Person*>(
319         &_Person_default_instance_);
320 }
321 static constexpr int kIndexInFileMessages =
322     1;
323
324 friend void swap(Person& a, Person& b) {
325     a.Swap(&b);
326 }
327 inline void Swap(Person* other) {
328     if (other == this) return;
329 #ifdef PROTOBUF_FORCE_COPY_IN_SWAP
330     if (GetOwningArena() != nullptr &&
331         GetOwningArena() == other->GetOwningArena()) {
332         #else // PROTOBUF_FORCE_COPY_IN_SWAP
333         if (GetOwningArena() == other->GetOwningArena()) {
334         #endif // !PROTOBUF_FORCE_COPY_IN_SWAP
335             InternalSwap(other);
336         } else {
337             ::PROTOBUF_NAMESPACE_ID::internal::GenericSwap(this, other);
338         }
339     }
340 void UnsafeArenaSwap(Person* other) {
341     if (other == this) return;
342     GOOGLE_DCHECK(GetOwningArena() == other->GetOwningArena());
343     InternalSwap(other);
344 }
345
346 // implements Message -----
347
348 Person* New(::PROTOBUF_NAMESPACE_ID::Arena* arena = nullptr) const final {
349     return CreateMaybeMessage<Person>(arena);
350 }

```

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```

351 using ::PROTOBUF_NAMESPACE_ID::Message::CopyFrom;
352 void CopyFrom(const Person& from);
353 using ::PROTOBUF_NAMESPACE_ID::Message::MergeFrom;
354 void MergeFrom(const Person& from);
355 private:
356 static void MergeImpl(::PROTOBUF_NAMESPACE_ID::Message* to, const ::PROTOBUF_
↵NAMESPACE_ID::Message& from);
357 public:
358 PROTOBUF_ATTRIBUTE_REINITIALIZES void Clear() final;
359 bool IsInitialized() const final;
360
361 size_t ByteSizeLong() const final;
362 const char* _InternalParse(const char* ptr, ::PROTOBUF_NAMESPACE_
↵ID::internal::ParseContext* ctx) final;
363 uint8_t* _InternalSerialize(
364     uint8_t* target, ::PROTOBUF_NAMESPACE_ID::io::EpsCopyOutputStream* stream) ↵
↵const final;
365 int GetCachedSize() const final { return _cached_size_.Get(); }
366
367 private:
368 void SharedCtor();
369 void SharedDtor();
370 void SetCachedSize(int size) const final;
371 void InternalSwap(Person* other);
372
373 private:
374 friend class ::PROTOBUF_NAMESPACE_ID::internal::AnyMetadata;
375 static ::PROTOBUF_NAMESPACE_ID::StringPiece FullMessageName() {
376     return "tutorial.Person";
377 }
378 protected:
379 explicit Person(::PROTOBUF_NAMESPACE_ID::Arena* arena,
380                bool is_message_owned = false);
381 public:
382
383 static const ClassData _class_data_;
384 const ::PROTOBUF_NAMESPACE_ID::Message::ClassData* GetClassData() const final;
385
386 ::PROTOBUF_NAMESPACE_ID::Metadata GetMetadata() const final;
387
388 // nested types -----
389
390 typedef Person_PhoneNumber PhoneNumber;

```

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```

391
392 typedef Person_PhoneType PhoneType;
393 static constexpr PhoneType MOBILE =
394     Person_PhoneType_MOBILE;
395 static constexpr PhoneType HOME =
396     Person_PhoneType_HOME;
397 static constexpr PhoneType WORK =
398     Person_PhoneType_WORK;
399 static inline bool PhoneType_IsValid(int value) {
400     return Person_PhoneType_IsValid(value);
401 }
402 static constexpr PhoneType PhoneType_MIN =
403     Person_PhoneType_PhoneType_MIN;
404 static constexpr PhoneType PhoneType_MAX =
405     Person_PhoneType_PhoneType_MAX;
406 static constexpr int PhoneType_ARRAYSIZE =
407     Person_PhoneType_PhoneType_ARRAYSIZE;
408 static inline const ::PROTOBUF_NAMESPACE_ID::EnumDescriptor*
409 PhoneType_descriptor() {
410     return Person_PhoneType_descriptor();
411 }
412 template<typename T>
413 static inline const std::string& PhoneType_Name(T enum_t_value) {
414     static_assert(::std::is_same<T, PhoneType>::value ||
415         ::std::is_integral<T>::value,
416         "Incorrect type passed to function PhoneType_Name.");
417     return Person_PhoneType_Name(enum_t_value);
418 }
419 static inline bool PhoneType_Parse(::PROTOBUF_NAMESPACE_ID::ConstStringParam name,
420     PhoneType* value) {
421     return Person_PhoneType_Parse(name, value);
422 }
423
424 // accessors -----
425
426 enum : int {
427     kPhonesFieldNumber = 4,
428     kNameFieldNumber = 1,
429     kEmailFieldNumber = 3,
430     kIdFieldNumber = 2,
431 };
432 // repeated .tutorial.Person.PhoneNumber phones = 4;
433 int phones_size() const;

```

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```

434 private:
435     int _internal_phones_size() const;
436 public:
437     void clear_phones();
438     ::tutorial::Person_PhoneNumber* mutable_phones(int index);
439     ::PROTOBUF_NAMESPACE_ID::RepeatedPtrField< ::tutorial::Person_PhoneNumber >*
440         mutable_phones();
441 private:
442     const ::tutorial::Person_PhoneNumber& _internal_phones(int index) const;
443     ::tutorial::Person_PhoneNumber* _internal_add_phones();
444 public:
445     const ::tutorial::Person_PhoneNumber& phones(int index) const;
446     ::tutorial::Person_PhoneNumber* add_phones();
447     const ::PROTOBUF_NAMESPACE_ID::RepeatedPtrField< ::tutorial::Person_PhoneNumber >&
448         phones() const;
449
450     // optional string name = 1;
451     bool has_name() const;
452 private:
453     bool _internal_has_name() const;
454 public:
455     void clear_name();
456     const std::string& name() const;
457     template <typename ArgT0 = const std::string&, typename... ArgT>
458     void set_name(ArgT0&& arg0, ArgT... args);
459     std::string* mutable_name();
460     PROTOBUF_NODISCARD std::string* release_name();
461     void set_allocated_name(std::string* name);
462 private:
463     const std::string& _internal_name() const;
464     inline PROTOBUF_ALWAYS_INLINE void _internal_set_name(const std::string& value);
465     std::string* _internal_mutable_name();
466 public:
467
468     // optional string email = 3;
469     bool has_email() const;
470 private:
471     bool _internal_has_email() const;
472 public:
473     void clear_email();
474     const std::string& email() const;
475     template <typename ArgT0 = const std::string&, typename... ArgT>
476     void set_email(ArgT0&& arg0, ArgT... args);

```

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```

477     std::string* mutable_email();
478     PROTOBUF_NODISCARD std::string* release_email();
479     void set_allocated_email(std::string* email);
480     private:
481     const std::string& _internal_email() const;
482     inline PROTOBUF_ALWAYS_INLINE void _internal_set_email(const std::string& value);
483     std::string* _internal_mutable_email();
484     public:
485
486     // optional int32 id = 2;
487     bool has_id() const;
488     private:
489     bool _internal_has_id() const;
490     public:
491     void clear_id();
492     int32_t id() const;
493     void set_id(int32_t value);
494     private:
495     int32_t _internal_id() const;
496     void _internal_set_id(int32_t value);
497     public:
498
499     // @@protoc_insertion_point(class_scope:tutorial.Person)
500     private:
501     class _Internal;
502
503     template <typename T> friend class ::PROTOBUF_NAMESPACE_ID::Arena::InternalHelper;
504     typedef void InternalArenaConstructable_;
505     typedef void DestructorSkippable_;
506     ::PROTOBUF_NAMESPACE_ID::internal::HasBits<1> _has_bits_;
507     mutable ::PROTOBUF_NAMESPACE_ID::internal::CachedSize _cached_size_;
508     ::PROTOBUF_NAMESPACE_ID::RepeatedPtrField< ::tutorial::Person_PhoneNumber > phones_;
509     ::PROTOBUF_NAMESPACE_ID::internal::ArenaStringPtr name_;
510     ::PROTOBUF_NAMESPACE_ID::internal::ArenaStringPtr email_;
511     int32_t id_;
512     friend struct ::TableStruct_hello_2eproto;
513 };
514 // -----
515
516 class AddressBook final :
517     public ::PROTOBUF_NAMESPACE_ID::Message /* @@protoc_insertion_point(class_
↳ definition:tutorial.AddressBook) */ {
518     public:

```

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```

519 inline AddressBook() : AddressBook(nullptr) {}
520 ~AddressBook() override;
521 explicit PROTOBUF_CONSTEXPR AddressBook(::PROTOBUF_NAMESPACE_
↪ID::internal::ConstantInitialized);
522
523 AddressBook(const AddressBook& from);
524 AddressBook(AddressBook&& from) noexcept
525     : AddressBook() {
526     *this = ::std::move(from);
527 }
528
529 inline AddressBook& operator=(const AddressBook& from) {
530     CopyFrom(from);
531     return *this;
532 }
533 inline AddressBook& operator=(AddressBook&& from) noexcept {
534     if (this == &from) return *this;
535     if (GetOwningArena() == from.GetOwningArena())
536 #ifdef PROTOBUF_FORCE_COPY_IN_MOVE
537         && GetOwningArena() != nullptr
538 #endif // !PROTOBUF_FORCE_COPY_IN_MOVE
539     ) {
540         InternalSwap(&from);
541     } else {
542         CopyFrom(from);
543     }
544     return *this;
545 }
546
547 inline const ::PROTOBUF_NAMESPACE_ID::UnknownFieldSet& unknown_fields() const {
548     return _internal_metadata_.unknown_fields<::PROTOBUF_NAMESPACE_
↪ID::UnknownFieldSet> (::PROTOBUF_NAMESPACE_ID::UnknownFieldSet::default_instance);
549 }
550 inline ::PROTOBUF_NAMESPACE_ID::UnknownFieldSet* mutable_unknown_fields() {
551     return _internal_metadata_.mutable_unknown_fields<::PROTOBUF_NAMESPACE_
↪ID::UnknownFieldSet> ();
552 }
553
554 static const ::PROTOBUF_NAMESPACE_ID::Descriptor* descriptor() {
555     return GetDescriptor();
556 }
557 static const ::PROTOBUF_NAMESPACE_ID::Descriptor* GetDescriptor() {
558     return default_instance().GetMetadata().descriptor;

```

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```

559     }
560     static const ::PROTOBUF_NAMESPACE_ID::Reflection* GetReflection() {
561         return default_instance().GetMetadata().reflection;
562     }
563     static const AddressBook& default_instance() {
564         return *internal_default_instance();
565     }
566     static inline const AddressBook* internal_default_instance() {
567         return reinterpret_cast<const AddressBook*>(
568             &_AddressBook_default_instance_);
569     }
570     static constexpr int kIndexInFileMessages =
571         2;
572
573     friend void swap(AddressBook& a, AddressBook& b) {
574         a.Swap(&b);
575     }
576     inline void Swap(AddressBook* other) {
577         if (other == this) return;
578 #ifdef PROTOBUF_FORCE_COPY_IN_SWAP
579         if (GetOwningArena() != nullptr &&
580             GetOwningArena() == other->GetOwningArena()) {
581             // PROTOBUF_FORCE_COPY_IN_SWAP
582             if (GetOwningArena() == other->GetOwningArena()) {
583 #endif // !PROTOBUF_FORCE_COPY_IN_SWAP
584                 InternalSwap(other);
585             } else {
586                 ::PROTOBUF_NAMESPACE_ID::internal::GenericSwap(this, other);
587             }
588         }
589     void UnsafeArenaSwap(AddressBook* other) {
590         if (other == this) return;
591         GOOGLE_DCHECK(GetOwningArena() == other->GetOwningArena());
592         InternalSwap(other);
593     }
594
595     // implements Message -----
596
597     AddressBook* New(::PROTOBUF_NAMESPACE_ID::Arena* arena = nullptr) const final {
598         return CreateMaybeMessage<AddressBook>(arena);
599     }
600     using ::PROTOBUF_NAMESPACE_ID::Message::CopyFrom;
601     void CopyFrom(const AddressBook& from);

```

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```

602     using ::PROTOBUF_NAMESPACE_ID::Message::MergeFrom;
603     void MergeFrom(const AddressBook& from);
604     private:
605     static void MergeImpl(::PROTOBUF_NAMESPACE_ID::Message* to, const ::PROTOBUF_
↳NAMESPACE_ID::Message& from);
606     public:
607     PROTOBUF_ATTRIBUTE_REINITIALIZES void Clear() final;
608     bool IsInitialized() const final;
609
610     size_t ByteSizeLong() const final;
611     const char* _InternalParse(const char* ptr, ::PROTOBUF_NAMESPACE_
↳ID::internal::ParseContext* ctx) final;
612     uint8_t* _InternalSerialize(
613         uint8_t* target, ::PROTOBUF_NAMESPACE_ID::io::EpsCopyOutputStream* stream) _
↳const final;
614     int GetCachedSize() const final { return _cached_size_.Get(); }
615
616     private:
617     void SharedCtor();
618     void SharedDtor();
619     void SetCachedSize(int size) const final;
620     void InternalSwap(AddressBook* other);
621
622     private:
623     friend class ::PROTOBUF_NAMESPACE_ID::internal::AnyMetadata;
624     static ::PROTOBUF_NAMESPACE_ID::StringPiece FullMessageName() {
625         return "tutorial.AddressBook";
626     }
627     protected:
628     explicit AddressBook(::PROTOBUF_NAMESPACE_ID::Arena* arena,
629                         bool is_message_owned = false);
630     public:
631
632     static const ClassData _class_data_;
633     const ::PROTOBUF_NAMESPACE_ID::Message::ClassData*GetClassData() const final;
634
635     ::PROTOBUF_NAMESPACE_ID::Metadata GetMetadata() const final;
636
637     // nested types -----
638
639     // accessors -----
640
641     enum : int {

```

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```

642     kPeopleFieldName = 1,
643 };
644 // repeated .tutorial.Person people = 1;
645 int people_size() const;
646 private:
647     int _internal_people_size() const;
648 public:
649     void clear_people();
650     ::tutorial::Person* mutable_people(int index);
651     ::PROTOBUF_NAMESPACE_ID::RepeatedPtrField< ::tutorial::Person >*
652         mutable_people();
653 private:
654     const ::tutorial::Person& _internal_people(int index) const;
655     ::tutorial::Person* _internal_add_people();
656 public:
657     const ::tutorial::Person& people(int index) const;
658     ::tutorial::Person* add_people();
659     const ::PROTOBUF_NAMESPACE_ID::RepeatedPtrField< ::tutorial::Person >&
660         people() const;
661
662     // @@protoc_insertion_point(class_scope:tutorial.AddressBook)
663 private:
664     class _Internal;
665
666     template <typename T> friend class ::PROTOBUF_NAMESPACE_ID::Arena::InternalHelper;
667     typedef void InternalArenaConstructable_;
668     typedef void DestructorSkippable_;
669     ::PROTOBUF_NAMESPACE_ID::RepeatedPtrField< ::tutorial::Person > people_;
670     mutable ::PROTOBUF_NAMESPACE_ID::internal::CachedSize _cached_size_;
671     friend struct ::TableStruct_hello_2eproto;
672 };
673 // =====
674
675 // =====
676
677
678 #ifdef __GNUC__
679     #pragma GCC diagnostic push
680     #pragma GCC diagnostic ignored "-Wstrict-aliasing"
681 #endif // __GNUC__
682 // Person_PhoneNumber
683
684 // optional string number = 1;

```

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```

685 inline bool Person_PhoneNumber::_internal_has_number() const {
686     bool value = (_has_bits_[0] & 0x00000001u) != 0;
687     return value;
688 }
689 inline bool Person_PhoneNumber::has_number() const {
690     return _internal_has_number();
691 }
692 inline void Person_PhoneNumber::clear_number() {
693     number_.ClearToEmpty();
694     _has_bits_[0] &= ~0x00000001u;
695 }
696 inline const std::string& Person_PhoneNumber::number() const {
697     // @@protoc_insertion_point(field_get:tutorial.Person.PhoneNumber.number)
698     return _internal_number();
699 }
700 template <typename ArgT0, typename... ArgT>
701 inline PROTOBUF_ALWAYS_INLINE
702 void Person_PhoneNumber::set_number(ArgT0&& arg0, ArgT... args) {
703     _has_bits_[0] |= 0x00000001u;
704     number_.Set(static_cast<ArgT0 &&>(arg0), args..., GetArenaForAllocation());
705     // @@protoc_insertion_point(field_set:tutorial.Person.PhoneNumber.number)
706 }
707 inline std::string* Person_PhoneNumber::mutable_number() {
708     std::string* _s = _internal_mutable_number();
709     // @@protoc_insertion_point(field_mutable:tutorial.Person.PhoneNumber.number)
710     return _s;
711 }
712 inline const std::string& Person_PhoneNumber::_internal_number() const {
713     return number_.Get();
714 }
715 inline void Person_PhoneNumber::_internal_set_number(const std::string& value) {
716     _has_bits_[0] |= 0x00000001u;
717     number_.Set(value, GetArenaForAllocation());
718 }
719 inline std::string* Person_PhoneNumber::_internal_mutable_number() {
720     _has_bits_[0] |= 0x00000001u;
721     return number_.Mutable(GetArenaForAllocation());
722 }
723 inline std::string* Person_PhoneNumber::release_number() {
724     // @@protoc_insertion_point(field_release:tutorial.Person.PhoneNumber.number)
725     if (!_internal_has_number()) {
726         return nullptr;
727     }

```

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```

728     _has_bits_[0] &= ~0x00000001u;
729     auto* p = number_.Release();
730 #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING
731     if (number_.IsDefault()) {
732         number_.Set("", GetArenaForAllocation());
733     }
734 #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING
735     return p;
736 }
737 inline void Person_PhoneNumber::set_allocated_number(std::string* number) {
738     if (number != nullptr) {
739         _has_bits_[0] |= 0x00000001u;
740     } else {
741         _has_bits_[0] &= ~0x00000001u;
742     }
743     number_.SetAllocated(number, GetArenaForAllocation());
744 #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING
745     if (number_.IsDefault()) {
746         number_.Set("", GetArenaForAllocation());
747     }
748 #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING
749     // @@protoc_insertion_point(field_set_allocated:tutorial.Person.PhoneNumber.number)
750 }
751
752 // optional .tutorial.Person.PhoneType type = 2 [default = HOME];
753 inline bool Person_PhoneNumber::_internal_has_type() const {
754     bool value = (_has_bits_[0] & 0x00000002u) != 0;
755     return value;
756 }
757 inline bool Person_PhoneNumber::has_type() const {
758     return _internal_has_type();
759 }
760 inline void Person_PhoneNumber::clear_type() {
761     type_ = 1;
762     _has_bits_[0] &= ~0x00000002u;
763 }
764 inline ::tutorial::Person_PhoneType Person_PhoneNumber::_internal_type() const {
765     return static_cast< ::tutorial::Person_PhoneType >(type_);
766 }
767 inline ::tutorial::Person_PhoneType Person_PhoneNumber::type() const {
768     // @@protoc_insertion_point(field_get:tutorial.Person.PhoneNumber.type)
769     return _internal_type();
770 }

```

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```

771 inline void Person_PhoneNumber::_internal_set_type(::tutorial::Person_PhoneType_
    ↪value) {
772     assert(::tutorial::Person_PhoneType_IsValid(value));
773     _has_bits_[0] |= 0x00000002u;
774     type_ = value;
775 }
776 inline void Person_PhoneNumber::set_type(::tutorial::Person_PhoneType value) {
777     _internal_set_type(value);
778     // @@protoc_insertion_point(field_set:tutorial.Person.PhoneNumber.type)
779 }
780
781 // -----
782
783 // Person
784
785 // optional string name = 1;
786 inline bool Person::_internal_has_name() const {
787     bool value = (_has_bits_[0] & 0x00000001u) != 0;
788     return value;
789 }
790 inline bool Person::has_name() const {
791     return _internal_has_name();
792 }
793 inline void Person::clear_name() {
794     name_.ClearToEmpty();
795     _has_bits_[0] &= ~0x00000001u;
796 }
797 inline const std::string& Person::name() const {
798     // @@protoc_insertion_point(field_get:tutorial.Person.name)
799     return _internal_name();
800 }
801 template <typename ArgT0, typename... ArgT>
802 inline PROTOBUF_ALWAYS_INLINE
803 void Person::set_name(ArgT0&& arg0, ArgT... args) {
804     _has_bits_[0] |= 0x00000001u;
805     name_.Set(static_cast<ArgT0 &&>(arg0), args..., GetArenaForAllocation());
806     // @@protoc_insertion_point(field_set:tutorial.Person.name)
807 }
808 inline std::string* Person::mutable_name() {
809     std::string* _s = _internal_mutable_name();
810     // @@protoc_insertion_point(field_mutable:tutorial.Person.name)
811     return _s;
812 }

```

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```

813 inline const std::string& Person::_internal_name() const {
814     return name_.Get();
815 }
816 inline void Person::_internal_set_name(const std::string& value) {
817     _has_bits_[0] |= 0x00000001u;
818     name_.Set(value, GetArenaForAllocation());
819 }
820 inline std::string* Person::_internal_mutable_name() {
821     _has_bits_[0] |= 0x00000001u;
822     return name_.Mutable(GetArenaForAllocation());
823 }
824 inline std::string* Person::release_name() {
825     // @@protoc_insertion_point(field_release:tutorial.Person.name)
826     if (!_internal_has_name()) {
827         return nullptr;
828     }
829     _has_bits_[0] &= ~0x00000001u;
830     auto* p = name_.Release();
831 #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING
832     if (name_.IsDefault()) {
833         name_.Set("", GetArenaForAllocation());
834     }
835 #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING
836     return p;
837 }
838 inline void Person::set_allocated_name(std::string* name) {
839     if (name != nullptr) {
840         _has_bits_[0] |= 0x00000001u;
841     } else {
842         _has_bits_[0] &= ~0x00000001u;
843     }
844     name_.SetAllocated(name, GetArenaForAllocation());
845 #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING
846     if (name_.IsDefault()) {
847         name_.Set("", GetArenaForAllocation());
848     }
849 #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING
850     // @@protoc_insertion_point(field_set_allocated:tutorial.Person.name)
851 }
852
853 // optional int32 id = 2;
854 inline bool Person::_internal_has_id() const {
855     bool value = (_has_bits_[0] & 0x00000004u) != 0;

```

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```

856     return value;
857 }
858 inline bool Person::has_id() const {
859     return _internal_has_id();
860 }
861 inline void Person::clear_id() {
862     id_ = 0;
863     _has_bits_[0] &= ~0x00000004u;
864 }
865 inline int32_t Person::_internal_id() const {
866     return id_;
867 }
868 inline int32_t Person::id() const {
869     // @@protoc_insertion_point(field_get:tutorial.Person.id)
870     return _internal_id();
871 }
872 inline void Person::_internal_set_id(int32_t value) {
873     _has_bits_[0] |= 0x00000004u;
874     id_ = value;
875 }
876 inline void Person::set_id(int32_t value) {
877     _internal_set_id(value);
878     // @@protoc_insertion_point(field_set:tutorial.Person.id)
879 }
880
881 // optional string email = 3;
882 inline bool Person::_internal_has_email() const {
883     bool value = (_has_bits_[0] & 0x00000002u) != 0;
884     return value;
885 }
886 inline bool Person::has_email() const {
887     return _internal_has_email();
888 }
889 inline void Person::clear_email() {
890     email_.ClearToEmpty();
891     _has_bits_[0] &= ~0x00000002u;
892 }
893 inline const std::string& Person::email() const {
894     // @@protoc_insertion_point(field_get:tutorial.Person.email)
895     return _internal_email();
896 }
897 template <typename ArgT0, typename... ArgT>
898 inline PROTOBUF_ALWAYS_INLINE

```

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```

899 void Person::set_email(ArgT0&& arg0, ArgT... args) {
900     _has_bits_[0] |= 0x00000002u;
901     email_.Set(static_cast<ArgT0 &&>(arg0), args..., GetArenaForAllocation());
902     // @@protoc_insertion_point(field_set:tutorial.Person.email)
903 }
904 inline std::string* Person::mutable_email() {
905     std::string* _s = _internal_mutable_email();
906     // @@protoc_insertion_point(field_mutable:tutorial.Person.email)
907     return _s;
908 }
909 inline const std::string& Person::_internal_email() const {
910     return email_.Get();
911 }
912 inline void Person::_internal_set_email(const std::string& value) {
913     _has_bits_[0] |= 0x00000002u;
914     email_.Set(value, GetArenaForAllocation());
915 }
916 inline std::string* Person::_internal_mutable_email() {
917     _has_bits_[0] |= 0x00000002u;
918     return email_.Mutable(GetArenaForAllocation());
919 }
920 inline std::string* Person::release_email() {
921     // @@protoc_insertion_point(field_release:tutorial.Person.email)
922     if (!_internal_has_email()) {
923         return nullptr;
924     }
925     _has_bits_[0] &= ~0x00000002u;
926     auto* p = email_.Release();
927 #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING
928     if (email_.IsDefault()) {
929         email_.Set("", GetArenaForAllocation());
930     }
931 #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING
932     return p;
933 }
934 inline void Person::set_allocated_email(std::string* email) {
935     if (email != nullptr) {
936         _has_bits_[0] |= 0x00000002u;
937     } else {
938         _has_bits_[0] &= ~0x00000002u;
939     }
940     email_.SetAllocated(email, GetArenaForAllocation());
941 #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING

```

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```

942     if (email_.IsDefault()) {
943         email_.Set("", GetArenaForAllocation());
944     }
945 #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING
946     // @@protoc_insertion_point(field_set_allocated:tutorial.Person.email)
947 }
948
949 // repeated .tutorial.Person.PhoneNumber phones = 4;
950 inline int Person::_internal_phones_size() const {
951     return phones_.size();
952 }
953 inline int Person::phones_size() const {
954     return _internal_phones_size();
955 }
956 inline void Person::clear_phones() {
957     phones_.Clear();
958 }
959 inline ::tutorial::Person_PhoneNumber* Person::mutable_phones(int index) {
960     // @@protoc_insertion_point(field_mutable:tutorial.Person.phones)
961     return phones_.Mutable(index);
962 }
963 inline ::PROTOBUF_NAMESPACE_ID::RepeatedPtrField< ::tutorial::Person_PhoneNumber >*
964 Person::mutable_phones() {
965     // @@protoc_insertion_point(field_mutable_list:tutorial.Person.phones)
966     return &phones_;
967 }
968 inline const ::tutorial::Person_PhoneNumber& Person::_internal_phones(int index)
969 ↪const {
970     return phones_.Get(index);
971 }
972 inline const ::tutorial::Person_PhoneNumber& Person::phones(int index) const {
973     // @@protoc_insertion_point(field_get:tutorial.Person.phones)
974     return _internal_phones(index);
975 }
976 inline ::tutorial::Person_PhoneNumber* Person::_internal_add_phones() {
977     return phones_.Add();
978 }
979 inline ::tutorial::Person_PhoneNumber* Person::add_phones() {
980     ::tutorial::Person_PhoneNumber* _add = _internal_add_phones();
981     // @@protoc_insertion_point(field_add:tutorial.Person.phones)
982     return _add;
983 }
984 inline const ::PROTOBUF_NAMESPACE_ID::RepeatedPtrField< ::tutorial::Person_

```

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```

↪PhoneNumber >&
984 Person::phones() const {
985     // @@protoc_insertion_point(field_list:tutorial.Person.phones)
986     return phones_;
987 }
988
989 // -----
990
991 // AddressBook
992
993 // repeated .tutorial.Person people = 1;
994 inline int AddressBook::_internal_people_size() const {
995     return people_.size();
996 }
997 inline int AddressBook::people_size() const {
998     return _internal_people_size();
999 }
1000 inline void AddressBook::clear_people() {
1001     people_.Clear();
1002 }
1003 inline ::tutorial::Person* AddressBook::mutable_people(int index) {
1004     // @@protoc_insertion_point(field_mutable:tutorial.AddressBook.people)
1005     return people_.Mutable(index);
1006 }
1007 inline ::PROTOBUF_NAMESPACE_ID::RepeatedPtrField< ::tutorial::Person >*
1008 AddressBook::mutable_people() {
1009     // @@protoc_insertion_point(field_mutable_list:tutorial.AddressBook.people)
1010     return &people_;
1011 }
1012 inline const ::tutorial::Person& AddressBook::_internal_people(int index) const {
1013     return people_.Get(index);
1014 }
1015 inline const ::tutorial::Person& AddressBook::people(int index) const {
1016     // @@protoc_insertion_point(field_get:tutorial.AddressBook.people)
1017     return _internal_people(index);
1018 }
1019 inline ::tutorial::Person* AddressBook::_internal_add_people() {
1020     return people_.Add();
1021 }
1022 inline ::tutorial::Person* AddressBook::add_people() {
1023     ::tutorial::Person* _add = _internal_add_people();
1024     // @@protoc_insertion_point(field_add:tutorial.AddressBook.people)
1025     return _add;

```

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```

1026 }
1027 inline const ::PROTOBUF_NAMESPACE_ID::RepeatedPtrField< ::tutorial::Person >&
1028 AddressBook::people() const {
1029     // @@protoc_insertion_point(field_list:tutorial.AddressBook.people)
1030     return people_;
1031 }
1032
1033 #ifdef __GNUC__
1034     #pragma GCC diagnostic pop
1035 #endif // __GNUC__
1036 // -----
1037
1038 // -----
1039
1040
1041 // @@protoc_insertion_point(namespace_scope)
1042
1043 } // namespace tutorial
1044
1045 PROTOBUF_NAMESPACE_OPEN
1046
1047 template <> struct is_proto_enum< ::tutorial::Person_PhoneType> : ::std::true_type {};
1048 template <>
1049 inline const EnumDescriptor* GetEnumDescriptor< ::tutorial::Person_PhoneType>() {
1050     return ::tutorial::Person_PhoneType_descriptor();
1051 }
1052
1053 PROTOBUF_NAMESPACE_CLOSE
1054
1055 // @@protoc_insertion_point(global_scope)
1056
1057 #include <google/protobuf/port_undef.inc>
1058 #endif // GOOGLE_PROTOBUF_INCLUDED_GOOGLE_PROTOBUF_INCLUDED_hello_2eproto

```

16.2.4 hello.pb.cc

Listing 6: ./code/hello.pb.cc

```

1 // Generated by the protocol buffer compiler.  DO NOT EDIT!
2 // source: hello.proto
3
4 #include "hello.pb.h"

```

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```

5
6 #include <algorithm>
7
8 #include <google/protobuf/io/coded_stream.h>
9 #include <google/protobuf/extension_set.h>
10 #include <google/protobuf/wire_format_lite.h>
11 #include <google/protobuf/descriptor.h>
12 #include <google/protobuf/generated_message_reflection.h>
13 #include <google/protobuf/reflection_ops.h>
14 #include <google/protobuf/wire_format.h>
15 // @@protoc_insertion_point(includes)
16 #include <google/protobuf/port_def.inc>
17
18 PROTOBUF_PRAGMA_INIT_SEG
19
20 namespace _pb = ::PROTOBUF_NAMESPACE_ID;
21 namespace _pbi = _pb::internal;
22
23 namespace tutorial {
24 PROTOBUF_CONSTEXPR Person_PhoneNumber::Person_PhoneNumber(
25     ::_pbi::ConstantInitialized)
26     : number_(&::_pbi::fixed_address_empty_string, ::_pbi::ConstantInitialized{})
27     , type_(1)
28 {}
29 struct Person_PhoneNumberDefaultTypeInternal {
30     PROTOBUF_CONSTEXPR Person_PhoneNumberDefaultTypeInternal()
31         : _instance(::_pbi::ConstantInitialized{}) {}
32     ~Person_PhoneNumberDefaultTypeInternal() {}
33     union {
34         Person_PhoneNumber _instance;
35     };
36 };
37 PROTOBUF_ATTRIBUTE_NO_DESTROY PROTOBUF_CONSTINIT PROTOBUF_ATTRIBUTE_INIT_PRIORITY1
38 ↪Person_PhoneNumberDefaultTypeInternal _Person_PhoneNumber_default_instance_;
39 PROTOBUF_CONSTEXPR Person::Person(
40     ::_pbi::ConstantInitialized)
41     : phones_()
42     , name_(&::_pbi::fixed_address_empty_string, ::_pbi::ConstantInitialized{})
43     , email_(&::_pbi::fixed_address_empty_string, ::_pbi::ConstantInitialized{})
44     , id_(0){}
45 struct PersonDefaultTypeInternal {
46     PROTOBUF_CONSTEXPR PersonDefaultTypeInternal()
47         : _instance(::_pbi::ConstantInitialized{}) {}

```

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```

47 ~PersonDefaultTypeInternal() {}
48 union {
49     Person _instance;
50 };
51 };
52 PROTOBUF_ATTRIBUTE_NO_DESTROY PROTOBUF_CONSTINIT PROTOBUF_ATTRIBUTE_INIT_PRIORITY1_
↪PersonDefaultTypeInternal _Person_default_instance_;
53 PROTOBUF_CONSTEXPR AddressBook::AddressBook(
54     ::_pbi::ConstantInitialized)
55     : people_() {}
56 struct AddressBookDefaultTypeInternal {
57     PROTOBUF_CONSTEXPR AddressBookDefaultTypeInternal()
58         : _instance(::_pbi::ConstantInitialized{}) {}
59     ~AddressBookDefaultTypeInternal() {}
60     union {
61         AddressBook _instance;
62     };
63 };
64 PROTOBUF_ATTRIBUTE_NO_DESTROY PROTOBUF_CONSTINIT PROTOBUF_ATTRIBUTE_INIT_PRIORITY1_
↪AddressBookDefaultTypeInternal _AddressBook_default_instance_;
65 } // namespace tutorial
66 static ::_pb::Metadata file_level_metadata_hello_2eproto[3];
67 static const ::_pb::EnumDescriptor* file_level_enum_descriptors_hello_2eproto[1];
68 static constexpr ::_pb::ServiceDescriptor const** file_level_service_descriptors_
↪hello_2eproto = nullptr;
69
70 const uint32_t TableStruct_hello_2eproto::offsets[] PROTOBUF_SECTION_
↪VARIABLE(protodesc_cold) = {
71     PROTOBUF_FIELD_OFFSET(::tutorial::Person_PhoneNumber, _has_bits_),
72     PROTOBUF_FIELD_OFFSET(::tutorial::Person_PhoneNumber, _internal_metadata_),
73     ~0u, // no _extensions_
74     ~0u, // no _oneof_case_
75     ~0u, // no _weak_field_map_
76     ~0u, // no _inlined_string_donated_
77     PROTOBUF_FIELD_OFFSET(::tutorial::Person_PhoneNumber, number_),
78     PROTOBUF_FIELD_OFFSET(::tutorial::Person_PhoneNumber, type_),
79     0,
80     1,
81     PROTOBUF_FIELD_OFFSET(::tutorial::Person, _has_bits_),
82     PROTOBUF_FIELD_OFFSET(::tutorial::Person, _internal_metadata_),
83     ~0u, // no _extensions_
84     ~0u, // no _oneof_case_
85     ~0u, // no _weak_field_map_

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86 ~0u, // no _inlined_string_donated_
87 PROTOBUF_FIELD_OFFSET(::tutorial::Person, name_),
88 PROTOBUF_FIELD_OFFSET(::tutorial::Person, id_),
89 PROTOBUF_FIELD_OFFSET(::tutorial::Person, email_),
90 PROTOBUF_FIELD_OFFSET(::tutorial::Person, phones_),
91 0,
92 2,
93 1,
94 ~0u,
95 ~0u, // no _has_bits_
96 PROTOBUF_FIELD_OFFSET(::tutorial::AddressBook, _internal_metadata_),
97 ~0u, // no _extensions_
98 ~0u, // no _oneof_case_
99 ~0u, // no _weak_field_map_
100 ~0u, // no _inlined_string_donated_
101 PROTOBUF_FIELD_OFFSET(::tutorial::AddressBook, people_),
102 };
103 static const ::_pbi::MigrationSchema schemas[] PROTOBUF_SECTION_VARIABLE(protodesc_
↪cold) = {
104 { 0, 8, -1, sizeof(::tutorial::Person_PhoneNumber)},
105 { 10, 20, -1, sizeof(::tutorial::Person)},
106 { 24, -1, -1, sizeof(::tutorial::AddressBook)},
107 };
108
109 static const ::_pb::Message* const file_default_instances[] = {
110 &::tutorial::Person_PhoneNumber_default_instance_.instance,
111 &::tutorial::Person_default_instance_.instance,
112 &::tutorial::AddressBook_default_instance_.instance,
113 };
114
115 const char descriptor_table_protodef_hello_2eproto[] PROTOBUF_SECTION_
↪VARIABLE(protodesc_cold) =
116 "\n\013hello.proto\022\010tutorial\"333\001\n\006Person\022\014\n\004na"
117 "me\030\001 \001(\t\022\n\n\002id\030\002 \001(\005\022\r\n\005email\030\003 \001(\t\022,\n\006p"
↪t\022,\n\006p"
118 "hones\030\004 \003(\0132\034.tutorial.Person.PhoneNumbe"
119 "r\032M\n\013PhoneNumber\022\016\n\006number\030\001 \001(\t\022.\n\004type"
120 "\030\002 \001(\0162\032.tutorial.Person.PhoneType:\004HOME"
121 "\"+\n\tPhoneType\022\n\n\006MOBILE\020\000\022\010\n\004HOME\020\001\022\010\n\004W"
↪"
122 "ORK\020\002\"/\n\013AddressBook\022 \n\006people\030\001 \003(\0132\020.t"
123 "utorial.Person"
124 ;

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```

125 static ::_pbi::once_flag descriptor_table_hello_2eproto_once;
126 const ::_pbi::DescriptorTable descriptor_table_hello_2eproto = {
127     false, false, 294, descriptor_table_protodef_hello_2eproto,
128     "hello.proto",
129     &descriptor_table_hello_2eproto_once, nullptr, 0, 3,
130     schemas, file_default_instances, TableStruct_hello_2eproto::offsets,
131     file_level_metadata_hello_2eproto, file_level_enum_descriptors_hello_2eproto,
132     file_level_service_descriptors_hello_2eproto,
133 };
134 PROTOBUF_ATTRIBUTE_WEAK const ::_pbi::DescriptorTable* descriptor_table_hello_2eproto_
135 ↪getter() {
136     return &descriptor_table_hello_2eproto;
137 }
138 // Force running AddDescriptors() at dynamic initialization time.
139 PROTOBUF_ATTRIBUTE_INIT_PRIORITY2 static ::_pbi::AddDescriptorsRunner dynamic_init_
140 ↪dummy_hello_2eproto(&descriptor_table_hello_2eproto);
141 namespace tutorial {
142 const ::PROTOBUF_NAMESPACE_ID::EnumDescriptor* Person_PhoneType_descriptor() {
143     ::PROTOBUF_NAMESPACE_ID::internal::AssignDescriptors(&descriptor_table_hello_
144 ↪2eproto);
145     return file_level_enum_descriptors_hello_2eproto[0];
146 }
147 bool Person_PhoneType_IsValid(int value) {
148     switch (value) {
149     case 0:
150     case 1:
151     case 2:
152         return true;
153     default:
154         return false;
155     }
156 }
157 #if (__cplusplus < 201703) && (!defined(_MSC_VER) || (_MSC_VER >= 1900 && _MSC_VER <
158 ↪1912))
159 constexpr Person_PhoneType Person::MOBILE;
160 constexpr Person_PhoneType Person::HOME;
161 constexpr Person_PhoneType Person::WORK;
162 constexpr Person_PhoneType Person::PhoneType_MIN;
163 constexpr Person_PhoneType Person::PhoneType_MAX;
164 constexpr int Person::PhoneType_ARRAYSIZE;
165 #endif // (__cplusplus < 201703) && (!defined(_MSC_VER) || (_MSC_VER >= 1900 && _MSC_

```

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```

↪VER < 1912))

164
165 // =====
166
167 class Person_PhoneNumber::_Internal {
168 public:
169     using HasBits = decltype(std::declval<Person_PhoneNumber>()._has_bits_);
170     static void set_has_number(HasBits* has_bits) {
171         (*has_bits)[0] |= 1u;
172     }
173     static void set_has_type(HasBits* has_bits) {
174         (*has_bits)[0] |= 2u;
175     }
176 };
177
178 Person_PhoneNumber::Person_PhoneNumber(::PROTOBUF_NAMESPACE_ID::Arena* arena,
179                                         bool is_message_owned)
180 : ::PROTOBUF_NAMESPACE_ID::Message(arena, is_message_owned) {
181     SharedCtor();
182     // @@protoc_insertion_point(arena_constructor:tutorial.Person.PhoneNumber)
183 }
184 Person_PhoneNumber::Person_PhoneNumber(const Person_PhoneNumber& from)
185 : ::PROTOBUF_NAMESPACE_ID::Message(),
186   _has_bits_(from._has_bits_) {
187     _internal_metadata_.MergeFrom<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>(from._
↪internal_metadata_);
188     number_.InitDefault();
189     #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING
190     number_.Set("", GetArenaForAllocation());
191     #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING
192     if (from._internal_has_number()) {
193         number_.Set(from._internal_number(),
194                     GetArenaForAllocation());
195     }
196     type_ = from.type_;
197     // @@protoc_insertion_point(copy_constructor:tutorial.Person.PhoneNumber)
198 }
199
200 inline void Person_PhoneNumber::SharedCtor() {
201     number_.InitDefault();
202     #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING
203     number_.Set("", GetArenaForAllocation());
204     #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING

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205 type_ = 1;
206 }
207
208 Person_PhoneNumber::~~Person_PhoneNumber() {
209     // @@protoc_insertion_point(destructor:tutorial.Person.PhoneNumber)
210     if (auto *arena = _internal_metadata_.DeleteReturnArena<::PROTOBUF_NAMESPACE_
↳ ID::UnknownFieldSet>()) {
211         (void)arena;
212         return;
213     }
214     SharedDtor();
215 }
216
217 inline void Person_PhoneNumber::SharedDtor() {
218     GOOGLE_DCHECK(GetArenaForAllocation() == nullptr);
219     number_.Destroy();
220 }
221
222 void Person_PhoneNumber::SetCachedSize(int size) const {
223     _cached_size_.Set(size);
224 }
225
226 void Person_PhoneNumber::Clear() {
227     // @@protoc_insertion_point(message_clear_start:tutorial.Person.PhoneNumber)
228     uint32_t cached_has_bits = 0;
229     // Prevent compiler warnings about cached_has_bits being unused
230     (void) cached_has_bits;
231
232     cached_has_bits = _has_bits_[0];
233     if (cached_has_bits & 0x00000003u) {
234         if (cached_has_bits & 0x00000001u) {
235             number_.ClearNonDefaultToEmpty();
236         }
237         type_ = 1;
238     }
239     _has_bits_.Clear();
240     _internal_metadata_.Clear<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>();
241 }
242
243 const char* Person_PhoneNumber::_InternalParse(const char* ptr, ::_pbi::ParseContext* _
↳ ctx) {
244     #define CHK_(x) if (PROTOBUF_PREDICT_FALSE(!(x))) goto failure
245     _Internal::HasBits has_bits{};

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246 while (!ctx->Done(&ptr)) {
247     uint32_t tag;
248     ptr = ::_pbi::ReadTag(ptr, &tag);
249     switch (tag >> 3) {
250         // optional string number = 1;
251         case 1:
252             if (PROTOBUF_PREDICT_TRUE(static_cast<uint8_t>(tag) == 10)) {
253                 auto str = _internal_mutable_number();
254                 ptr = ::_pbi::InlineGreedyStringParser(str, ptr, ctx);
255                 CHK_(ptr);
256                 #ifndef NDEBUG
257                 ::_pbi::VerifyUTF8(str, "tutorial.Person.PhoneNumber.number");
258                 #endif // !NDEBUG
259             } else
260                 goto handle_unusual;
261             continue;
262         // optional .tutorial.Person.PhoneType type = 2 [default = HOME];
263         case 2:
264             if (PROTOBUF_PREDICT_TRUE(static_cast<uint8_t>(tag) == 16)) {
265                 uint64_t val = ::PROTOBUF_NAMESPACE_ID::internal::ReadVarint64(&ptr);
266                 CHK_(ptr);
267                 if (PROTOBUF_PREDICT_TRUE(::tutorial::Person_PhoneType_IsValid(val))) {
268                     _internal_set_type(static_cast<::tutorial::Person_PhoneType>(val));
269                 } else {
270                     ::PROTOBUF_NAMESPACE_ID::internal::WriteVarint(2, val, mutable_unknown_
↪fields());
271                 }
272             } else
273                 goto handle_unusual;
274             continue;
275         default:
276             goto handle_unusual;
277     } // switch
278     handle_unusual:
279     if ((tag == 0) || ((tag & 7) == 4)) {
280         CHK_(ptr);
281         ctx->SetLastTag(tag);
282         goto message_done;
283     }
284     ptr = UnknownFieldParse(
285         tag,
286         _internal_metadata_.mutable_unknown_fields<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>(),

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```

287     ptr, ctx);
288     CHK_(ptr != nullptr);
289 } // while
290 message_done:
291     _has_bits_.Or(has_bits);
292     return ptr;
293 failure:
294     ptr = nullptr;
295     goto message_done;
296 #undef CHK_
297 }
298
299 uint8_t* Person_PhoneNumber::_InternalSerialize(
300     uint8_t* target, ::PROTOBUF_NAMESPACE_ID::io::EpsCopyOutputStream* stream) const {
301     // @@protoc_insertion_point(serialize_to_array_start:tutorial.Person.PhoneNumber)
302     uint32_t cached_has_bits = 0;
303     (void) cached_has_bits;
304
305     cached_has_bits = _has_bits_[0];
306     // optional string number = 1;
307     if (cached_has_bits & 0x00000001u) {
308         ::PROTOBUF_NAMESPACE_ID::internal::WireFormat::VerifyUTF8StringNamedField(
309             this->_internal_number().data(), static_cast<int>(this->_internal_number().
310 ↪length()),
311             ::PROTOBUF_NAMESPACE_ID::internal::WireFormat::SERIALIZE,
312             "tutorial.Person.PhoneNumber.number");
313         target = stream->WriteStringMaybeAliased(
314             1, this->_internal_number(), target);
315     }
316
317     // optional .tutorial.Person.PhoneType type = 2 [default = HOME];
318     if (cached_has_bits & 0x00000002u) {
319         target = stream->EnsureSpace(target);
320         target = ::_pbi::WireFormatLite::WriteEnumToArray(
321             2, this->_internal_type(), target);
322     }
323
324     if (PROTOBUF_PREDICT_FALSE(_internal_metadata_.have_unknown_fields())) {
325         target = ::_pbi::WireFormat::InternalSerializeUnknownFieldsToArray(
326             _internal_metadata_.unknown_fields<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>
327 ↪(::PROTOBUF_NAMESPACE_ID::UnknownFieldSet::default_instance), target, stream);
328     }
329     // @@protoc_insertion_point(serialize_to_array_end:tutorial.Person.PhoneNumber)

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```

328     return target;
329 }
330
331 size_t Person_PhoneNumber::ByteSizeLong() const {
332     // @@protoc_insertion_point(message_byte_size_start:tutorial.Person.PhoneNumber)
333     size_t total_size = 0;
334
335     uint32_t cached_has_bits = 0;
336     // Prevent compiler warnings about cached_has_bits being unused
337     (void) cached_has_bits;
338
339     cached_has_bits = _has_bits_[0];
340     if (cached_has_bits & 0x00000003u) {
341         // optional string number = 1;
342         if (cached_has_bits & 0x00000001u) {
343             total_size += 1 +
344                 ::PROTOBUF_NAMESPACE_ID::internal::WireFormatLite::StringSize(
345                     this->_internal_number());
346         }
347
348         // optional .tutorial.Person.PhoneType type = 2 [default = HOME];
349         if (cached_has_bits & 0x00000002u) {
350             total_size += 1 +
351                 ::_pbi::WireFormatLite::EnumSize(this->_internal_type());
352         }
353     }
354 }
355 return MaybeComputeUnknownFieldsSize(total_size, &_cached_size_);
356 }
357
358 const ::PROTOBUF_NAMESPACE_ID::Message::ClassData Person_PhoneNumber::_class_data_ = {
359     ::PROTOBUF_NAMESPACE_ID::Message::CopyWithSizeCheck,
360     Person_PhoneNumber::MergeImpl
361 };
362 const ::PROTOBUF_NAMESPACE_ID::Message::ClassData*Person_PhoneNumber::GetClassData() {
363     ↪const { return &_class_data_; }
364
365 void Person_PhoneNumber::MergeImpl(::PROTOBUF_NAMESPACE_ID::Message* to,
366     const ::PROTOBUF_NAMESPACE_ID::Message& from) {
367     static_cast<Person_PhoneNumber*>(to)->MergeFrom(
368         static_cast<const Person_PhoneNumber&>(from));
369 }

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370
371 void Person_PhoneNumber::MergeFrom(const Person_PhoneNumber& from) {
372 // @@protoc_insertion_point(class_specific_merge_from_start:tutorial.Person.
    ↪PhoneNumber)
373     GOOGLE_DCHECK_NE(&from, this);
374     uint32_t cached_has_bits = 0;
375     (void) cached_has_bits;
376
377     cached_has_bits = from._has_bits_[0];
378     if (cached_has_bits & 0x00000003u) {
379         if (cached_has_bits & 0x00000001u) {
380             _internal_set_number(from._internal_number());
381         }
382         if (cached_has_bits & 0x00000002u) {
383             type_ = from.type_;
384         }
385         _has_bits_[0] |= cached_has_bits;
386     }
387     _internal_metadata_.MergeFrom<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>(from._
    ↪internal_metadata_);
388 }
389
390 void Person_PhoneNumber::CopyFrom(const Person_PhoneNumber& from) {
391 // @@protoc_insertion_point(class_specific_copy_from_start:tutorial.Person.
    ↪PhoneNumber)
392     if (&from == this) return;
393     Clear();
394     MergeFrom(from);
395 }
396
397 bool Person_PhoneNumber::IsInitialized() const {
398     return true;
399 }
400
401 void Person_PhoneNumber::InternalSwap(Person_PhoneNumber* other) {
402     using std::swap;
403     auto* lhs_arena = GetArenaForAllocation();
404     auto* rhs_arena = other->GetArenaForAllocation();
405     _internal_metadata_.InternalSwap(&other->_internal_metadata_);
406     swap(_has_bits_[0], other->_has_bits_[0]);
407     ::PROTOBUF_NAMESPACE_ID::internal::ArenaStringPtr::InternalSwap(
408         &number_, lhs_arena,
409         &other->number_, rhs_arena

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410 );
411 swap(type_, other->type_);
412 }
413
414 ::PROTOBUF_NAMESPACE_ID::Metadata Person_PhoneNumber::GetMetadata() const {
415     return ::_pbi::AssignDescriptors(
416         &descriptor_table_hello_2eproto_getter, &descriptor_table_hello_2eproto_once,
417         file_level_metadata_hello_2eproto[0]);
418 }
419
420 // =====
421
422 class Person::_Internal {
423 public:
424     using HasBits = decltype(std::declval<Person>()._has_bits_);
425     static void set_has_name(HasBits* has_bits) {
426         (*has_bits)[0] |= 1u;
427     }
428     static void set_has_id(HasBits* has_bits) {
429         (*has_bits)[0] |= 4u;
430     }
431     static void set_has_email(HasBits* has_bits) {
432         (*has_bits)[0] |= 2u;
433     }
434 };
435
436 Person::Person(::PROTOBUF_NAMESPACE_ID::Arena* arena,
437                bool is_message_owned)
438     : ::PROTOBUF_NAMESPACE_ID::Message(arena, is_message_owned),
439     phones_(arena) {
440     SharedCtor();
441     // @@protoc_insertion_point(arena_constructor:tutorial.Person)
442 }
443
444 Person::Person(const Person& from)
445     : ::PROTOBUF_NAMESPACE_ID::Message(),
446     _has_bits_(from._has_bits_),
447     phones_(from.phones_) {
448     _internal_metadata_.MergeFrom<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>(from._
449 ↪internal_metadata_);
450     name_.InitDefault();
451     #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING
452         name_.Set("", GetArenaForAllocation());
453     #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING

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452     if (from._internal_has_name()) {
453         name_.Set(from._internal_name(),
454             GetArenaForAllocation());
455     }
456     email_.InitDefault();
457     #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING
458         email_.Set("", GetArenaForAllocation());
459     #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING
460     if (from._internal_has_email()) {
461         email_.Set(from._internal_email(),
462             GetArenaForAllocation());
463     }
464     id_ = from.id_;
465     // @@protoc_insertion_point(copy_constructor:tutorial.Person)
466 }
467
468 inline void Person::SharedCtor() {
469     name_.InitDefault();
470     #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING
471         name_.Set("", GetArenaForAllocation());
472     #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING
473     email_.InitDefault();
474     #ifdef PROTOBUF_FORCE_COPY_DEFAULT_STRING
475         email_.Set("", GetArenaForAllocation());
476     #endif // PROTOBUF_FORCE_COPY_DEFAULT_STRING
477     id_ = 0;
478 }
479
480 Person::~Person() {
481     // @@protoc_insertion_point(destructor:tutorial.Person)
482     if (auto *arena = _internal_metadata_.DeleteReturnArena<:PROTOBUF_NAMESPACE_
483 ↪ ID::UnknownFieldSet>()) {
484         (void)arena;
485         return;
486     }
487     SharedDtor();
488 }
489
490 inline void Person::SharedDtor() {
491     GOOGLE_DCHECK(GetArenaForAllocation() == nullptr);
492     name_.Destroy();
493     email_.Destroy();
494 }

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```

494
495 void Person::SetCachedSize(int size) const {
496     _cached_size_.Set(size);
497 }
498
499 void Person::Clear() {
500     // @@protoc_insertion_point(message_clear_start:tutorial.Person)
501     uint32_t cached_has_bits = 0;
502     // Prevent compiler warnings about cached_has_bits being unused
503     (void) cached_has_bits;
504
505     phones_.Clear();
506     cached_has_bits = _has_bits_[0];
507     if (cached_has_bits & 0x00000003u) {
508         if (cached_has_bits & 0x00000001u) {
509             name_.ClearNonDefaultToEmpty();
510         }
511         if (cached_has_bits & 0x00000002u) {
512             email_.ClearNonDefaultToEmpty();
513         }
514     }
515     id_ = 0;
516     _has_bits_.Clear();
517     _internal_metadata_.Clear<:PROTOBUF_NAMESPACE_ID::UnknownFieldSet>();
518 }
519
520 const char* Person::_InternalParse(const char* ptr, ::_pbi::ParseContext* ctx) {
521     #define CHK_(x) if (PROTOBUF_PREDICT_FALSE(!(x))) goto failure
522     _Internal::HasBits has_bits{};
523     while (!ctx->Done(&ptr)) {
524         uint32_t tag;
525         ptr = ::_pbi::ReadTag(ptr, &tag);
526         switch (tag >> 3) {
527             // optional string name = 1;
528             case 1:
529                 if (PROTOBUF_PREDICT_TRUE(static_cast<uint8_t>(tag) == 10)) {
530                     auto str = _internal_mutable_name();
531                     ptr = ::_pbi::InlineGreedyStringParser(str, ptr, ctx);
532                     CHK_(ptr);
533                     #ifndef NDEBUG
534                     ::_pbi::VerifyUTF8(str, "tutorial.Person.name");
535                     #endif // !NDEBUG
536                 } else

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537     goto handle_unusual;
538     continue;
539     // optional int32 id = 2;
540     case 2:
541         if (PROTOBUF_PREDICT_TRUE(static_cast<uint8_t>(tag) == 16)) {
542             _Internal::set_has_id(&has_bits);
543             id_ = ::PROTOBUF_NAMESPACE_ID::internal::ReadVarint32(&ptr);
544             CHK_(ptr);
545         } else
546             goto handle_unusual;
547         continue;
548     // optional string email = 3;
549     case 3:
550         if (PROTOBUF_PREDICT_TRUE(static_cast<uint8_t>(tag) == 26)) {
551             auto str = _internal_mutable_email();
552             ptr = ::_pbi::InlineGreedyStringParser(str, ptr, ctx);
553             CHK_(ptr);
554             #ifndef NDEBUB
555             ::_pbi::VerifyUTF8(str, "tutorial.Person.email");
556             #endif // !NDEBUB
557         } else
558             goto handle_unusual;
559         continue;
560     // repeated .tutorial.Person.PhoneNumber phones = 4;
561     case 4:
562         if (PROTOBUF_PREDICT_TRUE(static_cast<uint8_t>(tag) == 34)) {
563             ptr -= 1;
564             do {
565                 ptr += 1;
566                 ptr = ctx->ParseMessage(_internal_add_phones(), ptr);
567                 CHK_(ptr);
568                 if (!ctx->DataAvailable(ptr)) break;
569             } while (::PROTOBUF_NAMESPACE_ID::internal::ExpectTag<34>(ptr));
570         } else
571             goto handle_unusual;
572         continue;
573     default:
574         goto handle_unusual;
575 } // switch
576 handle_unusual:
577 if ((tag == 0) || ((tag & 7) == 4)) {
578     CHK_(ptr);
579     ctx->SetLastTag(tag);

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```

580     goto message_done;
581 }
582 ptr = UnknownFieldParse(
583     tag,
584     _internal_metadata_.mutable_unknown_fields<::PROTOBUF_NAMESPACE_
↪ID::UnknownFieldSet>(),
585     ptr, ctx);
586 CHK_(ptr != nullptr);
587 } // while
588 message_done:
589     _has_bits_.Or(has_bits);
590     return ptr;
591 failure:
592     ptr = nullptr;
593     goto message_done;
594 #undef CHK_
595 }
596
597 uint8_t* Person::_InternalSerialize(
598     uint8_t* target, ::PROTOBUF_NAMESPACE_ID::io::EpsCopyOutputStream* stream) const {
599     // @@protoc_insertion_point(serialize_to_array_start:tutorial.Person)
600     uint32_t cached_has_bits = 0;
601     (void) cached_has_bits;
602
603     cached_has_bits = _has_bits_[0];
604     // optional string name = 1;
605     if (cached_has_bits & 0x00000001u) {
606         ::PROTOBUF_NAMESPACE_ID::internal::WireFormat::VerifyUTF8StringNamedField(
607             this->_internal_name().data(), static_cast<int>(this->_internal_name().
↪length()),
608             ::PROTOBUF_NAMESPACE_ID::internal::WireFormat::SERIALIZE,
609             "tutorial.Person.name");
610         target = stream->WriteStringMaybeAliased(
611             1, this->_internal_name(), target);
612     }
613
614     // optional int32 id = 2;
615     if (cached_has_bits & 0x00000004u) {
616         target = stream->EnsureSpace(target);
617         target = ::_pbi::WireFormatLite::WriteInt32ToArray(2, this->_internal_id(), ↪
↪target);
618     }
619

```

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```

620 // optional string email = 3;
621 if (cached_has_bits & 0x00000002u) {
622     ::PROTOBUF_NAMESPACE_ID::internal::WireFormat::VerifyUTF8StringNamedField(
623         this->_internal_email().data(), static_cast<int>(this->_internal_email().
↪length()),
624         ::PROTOBUF_NAMESPACE_ID::internal::WireFormat::SERIALIZE,
625         "tutorial.Person.email");
626     target = stream->WriteStringMaybeAliased(
627         3, this->_internal_email(), target);
628 }
629
630 // repeated .tutorial.Person.PhoneNumber phones = 4;
631 for (unsigned i = 0,
632      n = static_cast<unsigned>(this->_internal_phones_size()); i < n; i++) {
633     const auto& repfield = this->_internal_phones(i);
634     target = ::PROTOBUF_NAMESPACE_ID::internal::WireFormatLite::
635         InternalWriteMessage(4, repfield, repfield.GetCachedSize(), target, stream);
636 }
637
638 if (PROTOBUF_PREDICT_FALSE(_internal_metadata_.have_unknown_fields())) {
639     target = ::_pbi::WireFormat::InternalSerializeUnknownFieldsToArray(
640         _internal_metadata_.unknown_fields<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>
↪(::PROTOBUF_NAMESPACE_ID::UnknownFieldSet::default_instance), target, stream);
641 }
642 // @@protoc_insertion_point(serialize_to_array_end:tutorial.Person)
643 return target;
644 }
645
646 size_t Person::ByteSizeLong() const {
647     // @@protoc_insertion_point(message_byte_size_start:tutorial.Person)
648     size_t total_size = 0;
649
650     uint32_t cached_has_bits = 0;
651     // Prevent compiler warnings about cached_has_bits being unused
652     (void) cached_has_bits;
653
654     // repeated .tutorial.Person.PhoneNumber phones = 4;
655     total_size += 1UL * this->_internal_phones_size();
656     for (const auto& msg : this->phones_) {
657         total_size +=
658             ::PROTOBUF_NAMESPACE_ID::internal::WireFormatLite::MessageSize(msg);
659     }
660

```

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```

661 cached_has_bits = _has_bits_[0];
662 if (cached_has_bits & 0x00000007u) {
663     // optional string name = 1;
664     if (cached_has_bits & 0x00000001u) {
665         total_size += 1 +
666             ::PROTOBUF_NAMESPACE_ID::internal::WireFormatLite::StringSize(
667                 this->_internal_name());
668     }
669
670     // optional string email = 3;
671     if (cached_has_bits & 0x00000002u) {
672         total_size += 1 +
673             ::PROTOBUF_NAMESPACE_ID::internal::WireFormatLite::StringSize(
674                 this->_internal_email());
675     }
676
677     // optional int32 id = 2;
678     if (cached_has_bits & 0x00000004u) {
679         total_size += ::_pbi::WireFormatLite::Int32SizePlusOne(this->_internal_id());
680     }
681 }
682
683 return MaybeComputeUnknownFieldsSize(total_size, &_cached_size_);
684 }
685
686 const ::PROTOBUF_NAMESPACE_ID::Message::ClassData Person::_class_data_ = {
687     ::PROTOBUF_NAMESPACE_ID::Message::CopyWithSizeCheck,
688     Person::MergeImpl
689 };
690 const ::PROTOBUF_NAMESPACE_ID::Message::ClassData*Person::GetClassData() const {
691     ↪return &_class_data_; }
692
693 void Person::MergeImpl(::PROTOBUF_NAMESPACE_ID::Message* to,
694     const ::PROTOBUF_NAMESPACE_ID::Message& from) {
695     static_cast<Person*>(to)->MergeFrom(
696         static_cast<const Person&>(from));
697 }
698
699 void Person::MergeFrom(const Person& from) {
700     // @@protoc_insertion_point(class_specific_merge_from_start:tutorial.Person)
701     GOOGLE_DCHECK_NE(&from, this);
702     uint32_t cached_has_bits = 0;

```

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```

703     (void) cached_has_bits;
704
705     phones_.MergeFrom(from.phones_);
706     cached_has_bits = from._has_bits_[0];
707     if (cached_has_bits & 0x00000007u) {
708         if (cached_has_bits & 0x00000001u) {
709             _internal_set_name(from._internal_name());
710         }
711         if (cached_has_bits & 0x00000002u) {
712             _internal_set_email(from._internal_email());
713         }
714         if (cached_has_bits & 0x00000004u) {
715             id_ = from.id_;
716         }
717         _has_bits_[0] |= cached_has_bits;
718     }
719     _internal_metadata_.MergeFrom<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>(from._
↪internal_metadata_);
720 }
721
722 void Person::CopyFrom(const Person& from) {
723     // @@protoc_insertion_point(class_specific_copy_from_start:tutorial.Person)
724     if (&from == this) return;
725     Clear();
726     MergeFrom(from);
727 }
728
729 bool Person::IsInitialized() const {
730     return true;
731 }
732
733 void Person::InternalSwap(Person* other) {
734     using std::swap;
735     auto* lhs_arena = GetArenaForAllocation();
736     auto* rhs_arena = other->GetArenaForAllocation();
737     _internal_metadata_.InternalSwap(&other->_internal_metadata_);
738     swap(_has_bits_[0], other->_has_bits_[0]);
739     phones_.InternalSwap(&other->phones_);
740     ::PROTOBUF_NAMESPACE_ID::internal::ArenaStringPtr::InternalSwap(
741         &name_, lhs_arena,
742         &other->name_, rhs_arena
743     );
744     ::PROTOBUF_NAMESPACE_ID::internal::ArenaStringPtr::InternalSwap(

```

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```

745     &email_, lhs_arena,
746     &other->email_, rhs_arena
747 );
748 swap(id_, other->id_);
749 }
750
751 ::PROTOBUF_NAMESPACE_ID::Metadata Person::GetMetadata() const {
752     return ::_pbi::AssignDescriptors(
753         &descriptor_table_hello_2eproto_getter, &descriptor_table_hello_2eproto_once,
754         file_level_metadata_hello_2eproto[1]);
755 }
756
757 // =====
758
759 class AddressBook::_Internal {
760 public:
761 };
762
763 AddressBook::AddressBook(::PROTOBUF_NAMESPACE_ID::Arena* arena,
764                          bool is_message_owned)
765     : ::PROTOBUF_NAMESPACE_ID::Message(arena, is_message_owned),
766     people_(arena) {
767     SharedCtor();
768     // @@protoc_insertion_point(arena_constructor:tutorial.AddressBook)
769 }
770
771 AddressBook::AddressBook(const AddressBook& from)
772     : ::PROTOBUF_NAMESPACE_ID::Message(),
773     people_(from.people_) {
774     _internal_metadata_.MergeFrom<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>(from._
775 ↪_internal_metadata_);
776     // @@protoc_insertion_point(copy_constructor:tutorial.AddressBook)
777 }
778
779 inline void AddressBook::SharedCtor() {
780 }
781
782 AddressBook::~AddressBook() {
783     // @@protoc_insertion_point(destructor:tutorial.AddressBook)
784     if (auto *arena = _internal_metadata_.DeleteReturnArena<::PROTOBUF_NAMESPACE_
785 ↪ID::UnknownFieldSet>()) {
786         (void)arena;
787         return;
788     }
789 }

```

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```

786     SharedDtor();
787 }
788
789 inline void AddressBook::SharedDtor() {
790     GOOGLE_DCHECK(GetArenaForAllocation() == nullptr);
791 }
792
793 void AddressBook::SetCachedSize(int size) const {
794     _cached_size_.Set(size);
795 }
796
797 void AddressBook::Clear() {
798     // @@protoc_insertion_point(message_clear_start:tutorial.AddressBook)
799     uint32_t cached_has_bits = 0;
800     // Prevent compiler warnings about cached_has_bits being unused
801     (void) cached_has_bits;
802
803     people_.Clear();
804     _internal_metadata_.Clear<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>();
805 }
806
807 const char* AddressBook::_InternalParse(const char* ptr, ::_pbi::ParseContext* ctx) {
808     #define CHK_(x) if (PROTOBUF_PREDICT_FALSE(!(x))) goto failure
809     while (!ctx->Done(&ptr)) {
810         uint32_t tag;
811         ptr = ::_pbi::ReadTag(ptr, &tag);
812         switch (tag >> 3) {
813             // repeated .tutorial.Person people = 1;
814             case 1:
815                 if (PROTOBUF_PREDICT_TRUE(static_cast<uint8_t>(tag) == 10)) {
816                     ptr -= 1;
817                     do {
818                         ptr += 1;
819                         ptr = ctx->ParseMessage(_internal_add_people(), ptr);
820                         CHK_(ptr);
821                         if (!ctx->DataAvailable(ptr)) break;
822                     } while (::PROTOBUF_NAMESPACE_ID::internal::ExpectTag<10>(ptr));
823                 } else
824                     goto handle_unusual;
825                 continue;
826             default:
827                 goto handle_unusual;
828         } // switch

```

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```

829 handle_unusual:
830     if ((tag == 0) || ((tag & 7) == 4)) {
831         CHK_(ptr);
832         ctx->SetLastTag(tag);
833         goto message_done;
834     }
835     ptr = UnknownFieldParse(
836         tag,
837         _internal_metadata_.mutable_unknown_fields<::PROTOBUF_NAMESPACE_
↪ID::UnknownFieldSet>(),
838         ptr, ctx);
839     CHK_(ptr != nullptr);
840 } // while
841 message_done:
842     return ptr;
843 failure:
844     ptr = nullptr;
845     goto message_done;
846 #undef CHK_
847 }
848
849 uint8_t* AddressBook::_InternalSerialize(
850     uint8_t* target, ::PROTOBUF_NAMESPACE_ID::io::EpsCopyOutputStream* stream) const {
851     // @@protoc_insertion_point(serialize_to_array_start:tutorial.AddressBook)
852     uint32_t cached_has_bits = 0;
853     (void) cached_has_bits;
854
855     // repeated .tutorial.Person people = 1;
856     for (unsigned i = 0,
857         n = static_cast<unsigned>(this->_internal_people_size()); i < n; i++) {
858         const auto& repfield = this->_internal_people(i);
859         target = ::PROTOBUF_NAMESPACE_ID::internal::WireFormatLite::
860             InternalWriteMessage(1, repfield, repfield.GetCachedSize(), target, stream);
861     }
862
863     if (PROTOBUF_PREDICT_FALSE(_internal_metadata_.have_unknown_fields())) {
864         target = ::_pbi::WireFormat::InternalSerializeUnknownFieldsToArray(
865             _internal_metadata_.unknown_fields<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>
↪(::PROTOBUF_NAMESPACE_ID::UnknownFieldSet::default_instance), target, stream);
866     }
867     // @@protoc_insertion_point(serialize_to_array_end:tutorial.AddressBook)
868     return target;
869 }

```

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```

870
871 size_t AddressBook::ByteSizeLong() const {
872     // @@protoc_insertion_point(message_byte_size_start:tutorial.AddressBook)
873     size_t total_size = 0;
874
875     uint32_t cached_has_bits = 0;
876     // Prevent compiler warnings about cached_has_bits being unused
877     (void) cached_has_bits;
878
879     // repeated .tutorial.Person people = 1;
880     total_size += 1UL * this->_internal_people_size();
881     for (const auto& msg : this->people_) {
882         total_size +=
883             ::PROTOBUF_NAMESPACE_ID::internal::WireFormatLite::MessageSize(msg);
884     }
885
886     return MaybeComputeUnknownFieldsSize(total_size, &_cached_size_);
887 }
888
889 const ::PROTOBUF_NAMESPACE_ID::Message::ClassData AddressBook::_class_data_ = {
890     ::PROTOBUF_NAMESPACE_ID::Message::CopyWithSizeCheck,
891     AddressBook::MergeImpl
892 };
893 const ::PROTOBUF_NAMESPACE_ID::Message::ClassData*AddressBook::GetClassData() const {
894     ↪ return &_class_data_; }
895
896 void AddressBook::MergeImpl(::PROTOBUF_NAMESPACE_ID::Message* to,
897     const ::PROTOBUF_NAMESPACE_ID::Message& from) {
898     static_cast<AddressBook*>(to)->MergeFrom(
899         static_cast<const AddressBook&>(from));
900 }
901
902 void AddressBook::MergeFrom(const AddressBook& from) {
903     // @@protoc_insertion_point(class_specific_merge_from_start:tutorial.AddressBook)
904     GOOGLE_DCHECK_NE(&from, this);
905     uint32_t cached_has_bits = 0;
906     (void) cached_has_bits;
907
908     people_.MergeFrom(from.people_);
909     _internal_metadata_.MergeFrom<::PROTOBUF_NAMESPACE_ID::UnknownFieldSet>(from._
910     ↪ internal_metadata_);

```

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```

911
912 void AddressBook::CopyFrom(const AddressBook& from) {
913     // @@protoc_insertion_point(class_specific_copy_from_start:tutorial.AddressBook)
914     if (&from == this) return;
915     Clear();
916     MergeFrom(from);
917 }
918
919 bool AddressBook::IsInitialized() const {
920     return true;
921 }
922
923 void AddressBook::InternalSwap(AddressBook* other) {
924     using std::swap;
925     _internal_metadata_.InternalSwap(&other->_internal_metadata_);
926     people_.InternalSwap(&other->people_);
927 }
928
929 ::PROTOBUF_NAMESPACE_ID::Metadata AddressBook::GetMetadata() const {
930     return ::_pbi::AssignDescriptors(
931         &descriptor_table_hello_2eproto_getter, &descriptor_table_hello_2eproto_once,
932         file_level_metadata_hello_2eproto[2]);
933 }
934
935 // @@protoc_insertion_point(namespace_scope)
936 } // namespace tutorial
937 PROTOBUF_NAMESPACE_OPEN
938 template<> PROTOBUF_NOINLINE ::tutorial::Person_PhoneNumber*
939 Arena::CreateMaybeMessage< ::tutorial::Person_PhoneNumber >(Arena* arena) {
940     return Arena::CreateMessageInternal< ::tutorial::Person_PhoneNumber >(arena);
941 }
942 template<> PROTOBUF_NOINLINE ::tutorial::Person*
943 Arena::CreateMaybeMessage< ::tutorial::Person >(Arena* arena) {
944     return Arena::CreateMessageInternal< ::tutorial::Person >(arena);
945 }
946 template<> PROTOBUF_NOINLINE ::tutorial::AddressBook*
947 Arena::CreateMaybeMessage< ::tutorial::AddressBook >(Arena* arena) {
948     return Arena::CreateMessageInternal< ::tutorial::AddressBook >(arena);
949 }
950 PROTOBUF_NAMESPACE_CLOSE
951
952 // @@protoc_insertion_point(global_scope)
953 #include <google/protobuf/port_undef.inc>

```

17.1 Install

See <https://grpc.io/docs/languages/cpp/quickstart/>

```
git clone --recurse-submodules -b v1.46.3 --depth 1 --shallow-submodules https://  
→github.com/grpc/grpc  
mkdir build  
cd build  
cmake -DgRPC_INSTALL=ON -DgRPC_BUILD_TESTS=OFF -DCMAKE_INSTALL_PREFIX=/ceph-fj/  
→fangjun/software/grpc-1.46.3 .. 2>&1 | tee cmake-configure-1.log  
make -j20 2>&1 | tee make-1.log  
make install 2>&1 | tee make-2.log
```


18.1 TODOs

- Striking gold in binutils
<https://lwn.net/Articles/274859/>
- A ToC of the 20 part linker essay
<https://lwn.net/Articles/276782/>

There are other resources for linkers and loaders, see

- Executables linking and loading reading
<http://research.tedneward.com/reading/software/linking-loading/index.html>
- Optimizing real-world applications with GCC Link Time Optimization
<https://pdfs.semanticscholar.org/6adf/872e3533f40a607f39cdeaf264585efde9af.pdf>
by Honza Hubicka, whose scholar page is <https://scholar.google.cz/citations?user=vhXJ0JEAAAAJ&hl=en>

LINKER AND LOADER

19.1 References

- A ToC of the 20 part linker essay
<https://lwn.net/Articles/276782/>, which is written by Ian Lance Taylor
 1. Introduction, personal history, first half of what's-a-linker
 2. What's-a-linker: Dynamic linking, linker data types, linker operation
 3. Address spaces, Object file formats
 4. Shared Libraries
 5. More Shared Libraries -- specifically, linker implementation; ELF Symbols
- <https://www.ucw.cz/~hubicka/>, author of the gold linker
- Rod Evans: Surfing With a Linker Alien <http://www.linker-aliens.org/blogs/rie/>
 1. Hello there
- Michael Walker's Weblog <http://www.linker-aliens.org/blogs/msw/>
 1. Hello World
 2. How to build a Shared Library
 3. Library Bindings - let's be a little bit more precise shall we

Note: It shows the usage of `LD_DEBUG`, `pldd`, `ldd`, `pgrep` `elfdump`.

- Solaris Linking Blogs (Combined Index) <http://www.linker-aliens.org/blogs/>
- `LD_LIBRARY_PATH` - just say no
http://www.linker-aliens.org/blogs/rie/entry/tt_ld_library_path_tt/

- <https://github.com/berkus/odin/blob/master/tools/sjofn/sjofn.c>

An ELF linker. Read its source code!

19.2 Questions

1. How to view PLT?
2. How to view the relocation information? How many types of relocation are there?
3. What PIC code and non-PIC code look like?
4. What is lazy binding and how to use `LD_BIND_NOW`?
5. What is PLT and GOT?

20.1 aishell

20.1.1 AM training

The first one was added on 2019-02-01.

`asr_train.py` is in `espnet/bin/asr_train.py`, which invokes `espnet.asr.pytorch_backend.asr.train`.

The model is from `espnet.nets.pytorch_backend.e2e_asr.E2E`.

The encoder type *vggblstm*, 3 layers, hidden dim, 1024, proj dim 1024, subsampling 1_2_2_1_1.

Command is:

```
asr_train.py \  
  --config conf/train.yaml \  
  --preprocess-conf \  
  --ngpu 1 \  
  --backend pytorch \  
  --outdir exp/xxx \  
  --debugmode 1 \  
  --dict data/lang_char/train_sp_units.txt \  
  --minibatches 0 \  
  --verbose 0 \  
  --resume \  
  --train-json xxx/data.json \  
  --valid-json yyy/data.json
```


21.1 Tutorials

- <https://cmake.org/cmake/help/latest/guide/tutorial/index.html>

21.2 Install

Go to <https://github.com/Kitware/CMake/releases> for download.

```
wget https://github.com/Kitware/CMake/releases/download/v3.10.3/cmake-3.10.3-Linux-  
x86_64.sh  
chmod +x ./cmake-3.10.3-Linux-x86_64.sh  
./cmake-3.10.3-Linux-x86_64.sh --help  
mkdir /path/to/software/cmake-3.10.3  
./cmake-3.10.3-Linux-x86_64.sh --prefix=/path/to/software/cmake-3.10.3 --skip-license  
export PATH=/path/to/software/cmake-3.10.3/bin:$PATH
```


HUGGINGFACE

22.1 spaces

22.1.1 Install client API

```
pip install huggingface_hub
```

```
(py38) kuangfangjun:t$ python3
Python 3.8.0 (default, Oct 28 2019, 16:14:01)
[GCC 8.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> from huggingface_hub import hf_hub_download
>>> hf_hub_download(repo_id="google/pegasus-xsum", filename="config.json")
Downloading: 100%|_____
↪_____| 1.39k/1.39k [00:00<00:00, 1.12MB/s]
'/root/fangjun/.cache/huggingface/hub/models--google--pegasus-xsum/snapshots/
↪a0aa5531c00f59a32a167b75130805098b046f9c/config.json'
>>>
```

22.1.2 gradio

- <https://huggingface.co/spaces/alphacep/asr>
- <https://huggingface.co/spaces/jonatasgrosmann/asr>
- <https://github.com/gradio-app/gradio/issues/1359>
- <https://huggingface.co/spaces/Gradio-Blocks/neon-tts-plugin-coqui/blob/main/app.py>, css styles

22.1.3 Mirrors

- <https://hf.qhduan.com/spaces/k2-fsa/automatic-speech-recognition>
- <https://hf.qhduan.com/spaces/k2-fsa/automatic-speech-recognition-with-whisper>
- <https://hf.qhduan.com/spaces/k2-fsa/text-to-speech>

EECS E6870 SPEECH RECOGNITION

23.1 Notes

- <https://www.ee.columbia.edu/~stanchen/spring16/e6870/outline.html>
 - username: speech
 - password: pythonrules

24.1 Hello

```
git clone https://github.com/tencent/ncnn
cd ncnn
git checkout 7b4e77671a4457a414b60cee5425758212e725cf
mkdir build
cd build
cmake -DCMAKE_PREFIX_PATH=/ceph-fj/fangjun/software/protobuf-3.20.1-cmake ..
```

We have to make the following changes:

Listing 1: ./code/hello/7b4e77.diff

```
1 diff --git a/tools/CMakeLists.txt b/tools/CMakeLists.txt
2 index 0b710050..e1a5b3d0 100644
3 --- a/tools/CMakeLists.txt
4 +++ b/tools/CMakeLists.txt
5 @@ -8,6 +8,7 @@ include_directories(${CMAKE_CURRENT_BINARY_DIR})
6   protobuf_generate_cpp(CAFFE_PROTO_SRCS CAFFE_PROTO_HDRS caffe.proto)
7
8   add_executable(caffe2ncnn caffe2ncnn.cpp ${CAFFE_PROTO_SRCS} ${CAFFE_PROTO_HDRS})
9 +include_directories(${Protobuf_INCLUDE_DIR})
10
11   target_link_libraries(caffe2ncnn ${PROTOBUF_LIBRARIES})
12
13 diff --git a/tools/caffe2ncnn.cpp b/tools/caffe2ncnn.cpp
14 index 0eff756a..229cc653 100644
15 --- a/tools/caffe2ncnn.cpp
16 +++ b/tools/caffe2ncnn.cpp
17 @@ -193,7 +193,7 @@ static bool read_proto_from_binary(const char* filepath,
18   ↪google::protobuf::Message
19     google::protobuf::io::IstreamInputStream input(&fs);
```

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```

19     google::protobuf::io::CodedInputStream codedstr(&input);
20
21 -     codedstr.SetTotalBytesLimit(INT_MAX, INT_MAX / 2);
22 +     codedstr.SetTotalBytesLimit(INT_MAX);
23
24     bool success = message->ParseFromCodedStream(&codedstr);

```

To install the Python package:

```

cd ncnn
mkdir build
cd build
cmake ..
make -j
cd ..
pip install .

```

24.1.1 test-net

Listing 2: ./code/hello/test-net.cc

```

#include "datareader.h"
#include "layer/relu.h"
#include "layer_type.h"
#include "net.h"
#include <fstream>
#include <iostream>

#ifdef MY_CHECK
#define MY_CHECK(x, y) \
    do { \
        if (x != y) { \
            NCNN_LOGE("#x " != " #y); \
            exit(EXIT_FAILURE); \
        } \
    } while (0)
#endif

class DataReaderFromEmpty : public ncnn::DataReader {
public:
    virtual int scan(const char *format, void *p) const { return 0; }
    virtual size_t read(void *buf, size_t size) const {

```

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```

    memset(buf, 0, size);
    return size;
}
};

ncnn::Layer *MyLayerCreator(void * /*userdata*/) { return new ncnn::ReLU(); }
void MyLayerDestoryer(ncnn::Layer *layer, void * /*userdata*/) { delete layer; }

static void TestCustomLayer() {
    static const char *s = R"(
7767517
2 2
Input      data                0 1 data
MyLayer    my_layer           1 1 data out
)";

    std::ofstream of("filename.param");
    of << s;
    of.close();

    ncnn::Net net;
    net.register_custom_layer("MyLayer", MyLayerCreator, MyLayerDestoryer,
                              nullptr);

    // This is the first custom layer, so its index is 0
    MY_CHECK(net.custom_layer_to_index("MyLayer"), 0);

    DataReaderFromEmpty dr;

    // return 0 on success
    // What does load_param() do?
    // If we need to set options, we have to set it before calling load_param()
    // (1) Read the magic number
    // (2) Allocate space for layers and blobs
    auto ret = net.load_param("filename.param");
    MY_CHECK(ret, 0);

    // load_model() is optional if there are no parameters to read
    // ret = net.load_model(dr);
    // MY_CHECK(ret, 0);

    ncnn::Mat m(2, 3);
    float *p = m;
    for (int32_t i = 0; i != m.total(); ++i) {
        p[i] = i - float(m.total()) / 2;
    }
}

```

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```

}

for (int32_t i = 0; i != m.total(); ++i) {
    std::cout << p[i] << ", ";
}
std::cout << "\n";

ncnn::Extractor ex = net.create_extractor();
ex.input("data", m);
ncnn::Mat out;
ex.extract("out", out);
p = out;
for (int32_t i = 0; i != out.total(); ++i) {
    std::cout << p[i] << ", ";
}
std::cout << "\n";
}

/*
ncnn::Net uses pimpl

std::vector<Layer*> layers;
std::vector<Blob> blobs;
*/
void TestNet() {
    TestCustomLayer();
    NCNN_LOGE("Test net\n");
    /*
    77675157 -> magic number
    3 3 -> layer_count, blob_count

Input          data          0 1 data
->
layer type: Input
Layer name: data
bottom count: 0
top count: 1
It uses ncnn::create_layer(layer_type) to create a layer

If create_layer() returns a nullptr, it invokes
create_custom_layer(layer_type)

note: create_layer() is a global function, while create_custom_layer()

```

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is a method of `ncnn::Net`.

It will also invoke `layer->load_param(paramdict);`

Input layer has index 0 since it is the first row

0 1 data -> data is the output blob of Input layer. It has index 0 since it is the first blob

```

    */
    static const char *s = R"(
7767517
3 3
Input          data          0 1 data
Convolution     conv0_fwd     1 1 data conv0_fwd 0=3 1=3 11=3 2=1
↪12=1 3=1 13=1 4=0 14=0 5=1 6=81
InnerProduct     dense0_fwd     1 1 conv0_fwd output 0=1 1=1
↪2=151875
)";
    std::ofstream of("filename.param");
    of << s;
    of.close();

    ncnn::Net net;
    DataReaderFromEmpty dr;

    // return 0 on success
    // What does load_param() do?
    // If we need to set options, we have to set it before calling load_param()
    // (1) Read the magic number
    // (2) Allocate space for layers and blobs
    auto ret = net.load_param("filename.param");
    MY_CHECK(ret, 0);

    ret = net.load_model(dr);
    MY_CHECK(ret, 0);

    MY_CHECK(net.blobs().size(), 3);
    MY_CHECK(net.layers().size(), 3);

    // test layer 0
    // every layer has a type
    MY_CHECK(net.layers()[0]->type, "Input");

```

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```

MY_CHECK(net.layers()[0]->typeid, ncnn::LayerType::Input);

// every layer has a name
MY_CHECK(net.layers()[0]->name, "data");

// every layer has an input vector
MY_CHECK(net.layers()[0]->bottoms.size(), 0);

// every layer has an output vector
MY_CHECK(net.layers()[0]->tops.size(), 1);
// indexes in tops are indexes into the global blobs vector
MY_CHECK(net.layers()[0]->tops[0], 0);

// every blob has a name
MY_CHECK(net.blobs()[0].name, "data");

// every blob has a producer (layer index)
MY_CHECK(net.blobs()[0].producer, 0); // producer is layer 0

// every blob has a consumer (layer index)
MY_CHECK(net.blobs()[0].consumer, 1); // consumer is layer 1
}

```

24.1.2 test-param-dict

Listing 3: ./code/hello/test-param-dict.cc

```

#include "paramdict.h"
#ifdef MY_CHECK
#define MY_CHECK(x, y)                                     \
do {                                                       \
    if (x != y) {                                         \
        NCNN_LOGE("#x " != " #y");                     \
        exit(EXIT_FAILURE);                             \
    }                                                     \
} while (0)
#endif

/*
It has an array of struct.
The size of the array is 32, so the maximum number of parameters is 32
*/

```

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```

void TestParamDict() {
    NCNN_LOGE("Test param dict\n");
    ncnn::ParamDict pd;

    // int 2
    pd.set(0, 10);
    MY_CHECK(pd.type(0), 2);
    MY_CHECK(pd.get(0, 0), 10);

    // float 3
    pd.set(1, 10.5f);
    MY_CHECK(pd.type(1), 3);
    // Caution: The type of the default value (the second argument) is very
    // important. It determines which overload to invoke
    MY_CHECK(pd.get(1, 0.f), 10.5);

    // mat 4
    ncnn::Mat m(1);
    m[0] = 10.25;
    pd.set(2, m);
    MY_CHECK(pd.type(2), 4);
    MY_CHECK(pd.get(2, ncnn::Mat()) [0], 10.25);
}

```

24.1.3 test-data-reader

Listing 4: ./code/hello/test-data-reader.cc

```

#include "datareader.h"
#include "paramdict.h"
#include <iostream>

#ifdef MY_CHECK
#define MY_CHECK(x, y) \
    do { \
        if (x != y) { \
            NCNN_LOGE("#x " != " #y); \
            exit(EXIT_FAILURE); \
        } \
    } while (0)
#endif

```

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```

class ParamDict2 : public ncnn::ParamDict {
public:
    int load_param2(const ncnn::DataReader &dr) { return load_param(dr); }

    int load_param_bin2(const ncnn::DataReader &dr) { return load_param_bin(dr); }
};

void TestDataReader() {
    NCNN_LOGE("Test data reader\n");
    int32_t a = 4;
    FILE *fp = fopen("a.bin", "wb");
    fwrite(&a, sizeof(a), 1, fp);

    float f = 1.25;
    fwrite(&f, sizeof(f), 1, fp);

    fclose(fp);

    fp = fopen("a.bin", "rb");

    ncnn::DataReaderFromStdio dr(fp);
    int32_t p;
    float q;
    dr.read(&p, sizeof(p));
    dr.read(&q, sizeof(q));
    MY_CHECK(p, a);
    MY_CHECK(q, f);
    fclose(fp);

    // datareader with param dict
    fp = fopen("a.txt", "w");
    fprintf(fp, "0=100 1=1.250000 -23303=5,0.125,0.25,0.50,0.625,1.0");
    fclose(fp);

    fp = fopen("a.txt", "r");
    ncnn::DataReaderFromStdio dr2(fp);
    ParamDict2 pd;
    pd.load_param2(dr2);
    MY_CHECK(pd.get(0, 0), 100);
    MY_CHECK(pd.get(1, 0.f), 1.25);
    ncnn::Mat m = pd.get(3, ncnn::Mat());
    MY_CHECK(m.w, 5);
    MY_CHECK(m[0], 0.125);

```

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```
MY_CHECK(m[1], 0.25);
MY_CHECK(m[2], 0.50);
MY_CHECK(m[3], 0.625);
MY_CHECK(m[4], 1.0);

fclose(fp);

// for binary
//
//      binary 0
//      binary 100
//      binary 1
//      binary 1.250000
//      binary 3 | array_bit
//      binary 5
//      binary 0.1
//      binary 0.2
//      binary 0.4
//      binary 0.8
//      binary 1.0
//      binary -233 (EOP)
fp = fopen("a.bin", "wb");
int32_t id = 0;
fwrite(&id, sizeof(id), 1, fp);
int32_t val = 100;
fwrite(&val, sizeof(val), 1, fp);

id = 1;
float val2 = 1.25;
fwrite(&id, sizeof(id), 1, fp);
fwrite(&val, sizeof(val2), 1, fp);

id = -23303;
fwrite(&id, sizeof(id), 1, fp);
int32_t len = m.w;
fwrite(&len, sizeof(len), 1, fp);
fwrite(m.data, sizeof(float), m.w, fp);

int32_t eop = -233;
fwrite(&eop, sizeof(eop), 1, fp);
fclose(fp);

fp = fopen("a.bin", "rb");
```

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```
ncnn::DataReaderFromStdio dr3(fp);
ParamDict2 pd3;
pd3.load_param_bin2(dr3);
fclose(fp);
MY_CHECK(pd.get(0, 0), 100);

// ncnn does not know that param 1 is a float. The user has to specify that!
MY_CHECK(pd.get(1, 0.f), 1.25);

ncnn::Mat m2 = pd.get(3, ncnn::Mat());
MY_CHECK(m2.w, 5);
MY_CHECK(m2[0], 0.125);
MY_CHECK(m2[1], 0.25);
MY_CHECK(m2[2], 0.50);
MY_CHECK(m2[3], 0.625);
MY_CHECK(m2[4], 1.0);
}
```

25.1 Installation

Refer to <https://llvm.org/docs/GettingStarted.html#requirements>

```
git clone --depth 1 https://github.com/llvm/llvm-project.git
cd llvm-project
mkdir build
cd build
cmake -DCMAKE_BUILD_TYPE=Release -G Ninja ../llvm
# cmake -DCMAKE_BUILD_TYPE=Release -DLLVM_ENABLE_PROJECTS="clang;lld" -G Ninja ../llvm

cmake --build . --target <target>
# or
export PATH=$PWD/build/bin:$PATH
export LD_LIBRARY_PATH=$PWD/build/lib:$LD_LIBRARY_PATH
```

<https://llvm.org/docs/GettingStarted.html#getting-a-modern-host-c-toolchain> describes how to install GCC from source.

Use of a user provided GCC:

```
mkdir build
cd build
CC=$HOME/toolchains/bin/gcc CXX=$HOME/toolchains/bin/g++ \
  cmake .. -DCMAKE_CXX_LINK_FLAGS="-Wl,-rpath,$HOME/toolchains/lib64 -L$HOME/
↳ toolchains/lib64"
```

Useful tools that can be found in *build/bin*:

```
$ llvm-config --cxxflags
-I/ceph-fj/fangjun/open-source-2/llvm-project/llvm/include -I/ceph-fj/fangjun/open-
↳ source-2/llvm-project/build/include -std=c++14 -fno-exceptions -fno-rtti -D_GNU_
↳ SOURCE -D__STDC_CONSTANT_MACROS -D__STDC_FORMAT_MACROS -D__STDC_LIMIT_MACROS
```

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```
$ llvm-config --libdir
/ceph-fj/fangjun/open-source-2/llvm-project/build/lib

$ llvm-config --cflags
-I/ceph-fj/fangjun/open-source-2/llvm-project/llvm/include -I/ceph-fj/fangjun/open-
↪source-2/llvm-project/build/include -D__GNU_SOURCE -D__STDC_CONSTANT_MACROS -D__
↪STDC_FORMAT_MACROS -D__STDC_LIMIT_MACROS

$ llvm-config --ldflags
-L/ceph-fj/fangjun/open-source-2/llvm-project/build/lib

$ llvm-config --src-root
/ceph-fj/fangjun/open-source-2/llvm-project/llvm

$ llvm-config --obj-root
/ceph-fj/fangjun/open-source-2/llvm-project/build

$ llvm-config --version
15.0.0git

$ llvm-config --bindir
/ceph-fj/fangjun/open-source-2/llvm-project/build/bin
```

25.2 ninja

```
pip install ninja

ninja
```

will look for the file `$PWD/build.ninja`

```
ninja --help
ninja -C build -j 20
ninja -t targets
ninja -t clean
ninja -t cleandead

ninja -v # be verbose while compiling files

# suppose hello is a target
ninja -v hello
```

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```
ninja -v -t clean hello

ninja -n -v hello # dry run
```

25.3 Documentation

- Coding standard

<https://llvm.org/docs/CodingStandards.html>

- LLVM Developer Policy

<https://llvm.org/docs/DeveloperPolicy.html>

- doxygen doc

<https://llvm.org/doxygen/>

- <http://www.aosabook.org/en/llvm.html>

25.4 Intermediate representation

```
clang --help

-emit-llvm  Use the LLVM representation for assembler and object files
-S          Only run preprocess and compilation steps
```

- `clang -S -emit-llvm ex.c` generates a text file `ex.ll`.
- `clang -c -emit-llvm ex.c` generates a binary file `ex.bc`.
- `llvm-dis ex.bc` generates a file `ex.ll`, which is identical with the file generated using `clang -S -emit-llvm ex.c`.`
- `llvm-as ex.ll` generates a file `ex.bc`, which is identical with the file generated using `clang -c -emit-llvm ex.c`.
- `llc ex.ll` generates the assembly file `ex.s`
- `lli ex.ll` can run this file. Use `echo $?` to see the return value.

See https://llvm.org/devmtg/2019-04/slides/Tutorial-Bridgers-LLVM_IR_tutorial.pdf.

25.5 Install GCC

25.5.1 12.2.0

```
tar xvf gcc-12.2.0.tar.gz
cd gcc-12.2.0
mkdir build
cd build

unset C_INCLUDE_PATH
unset CPLUS_INCLUDE_PATH
unset LD_LIBRARY_PATH
unset LIBRARY_PATH

../configure --prefix=/ceph-fj/fangjun/software/gcc-12.2.0
make -j 5
make install
```

```
gcc_dir=/ceph-fj/fangjun/software/gcc-12.2.0
export PATH=$gcc_dir/bin:$PATH
export CC=$gcc_dir/bin/gcc
export CXX=$gcc_dir/bin/g++
export LIBRARY_PATH=$gcc_dir/lib64:$LIBRARY_PATH
export LD_LIBRARY_PATH=$gcc_dir/lib64:$LD_LIBRARY_PATH
export C_INCLUDE_PATH=$gcc_dir/include
export CPLUS_INCLUDE_PATH=$gcc_dir/include
```

25.5.2 11.4.0

```
sudo apt-get install -y libgmp-dev libmpfr-dev libmpc-dev
sudo apt-get install -y flex bison

tar xvf gcc-11.4.0.tar.gz
cd gcc-11.4.0
mkdir build
cd build

unset C_INCLUDE_PATH
unset CPLUS_INCLUDE_PATH
unset LD_LIBRARY_PATH
unset LIBRARY_PATH
```

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```
../configure \
  --prefix=/star-fj/fangjun/software/gcc-11.4.0 \
  --enable-languages=c,c++ \
  --disable-multilib \
  --disable-bootstrap \
  --disable-nls

make -j1
make install
```

```
gcc_dir=/star-fj/fangjun/software/gcc-11.4.0
export PATH=$gcc_dir/bin:$PATH
export CC=$gcc_dir/bin/gcc
export CXX=$gcc_dir/bin/g++
export LIBRARY_PATH=$gcc_dir/lib64:$LIBRARY_PATH
export LD_LIBRARY_PATH=$gcc_dir/lib64:$LD_LIBRARY_PATH
export C_INCLUDE_PATH=$gcc_dir/include
export CPLUS_INCLUDE_PATH=$gcc_dir/include
```

25.5.3 Using conda

```
conda install -c conda-forge gcc==9.5.0
conda install -c conda-forge gxx==9.5.0
```


26.1 kotlin

26.1.1 Install

Kotlin to java online converter: <https://www.codeconvert.ai/kotlin-to-java-converter>.

See <https://kotlinlang.org/docs/command-line.html#run-scripts>

```
wget https://github.com/JetBrains/kotlin/releases/download/v1.7.22/kotlin-compiler-1.7.22.zip
cd $HOME/software
mkdir kotlin
cd kotlin
unzip /path/to/kotlin-compiler-1.7.22.zip
mv kotlinc 1.7.22
export PATH=$HOME/software/kotlin/1.7.22/bin:$PATH
```

Hello world

Create a file `hello.kt`:

Listing 1: ./code/install/hello.kt

```
fun main() {  
    println("hello world")  
}
```

Usage 1

```
kotlinc hello.kt
```

It will generate a binary file `HelloKt.class` and a folder `META-INF`.

```
$ javap HelloKt.class
```

The above command prints the following:

```
Compiled from "hello.kt"  
public final class HelloKt {  
    public static final void main();  
    public static void main(java.lang.String[]);  
}
```

We can use:

```
kotlin HelloKt.class
```

to execute it.

Usage 2

```
kotlinc hello.kt -d hello.jar  
# or  
kotlinc hello.kt -include-runtime -d hello.jar  
  
# It can be run using:  
java -jar hello.jar  
kotlin -classpath ./hello.jar HelloKt
```

Note that `hello.jar` is actually a zip file.

kotlinc-jvm

To run kotlin inside an interactive shell, we can run the command:

```
kotlinc-jvm
```

or the command:

```
kotlin
```

Naming convention

- class: `ClassName`
- function: `funcName`
- variable: `variableName`
- filename extension: `kt`

26.1.2 Basic types

See <https://kotlinlang.org/docs/basic-types.html>

Note `String.length` and `Array.size`. Both of them have `count()`, i.e., `String.count()` and `Array.count()`.

`String.isNotBlank()` and `String.isNotEmpty()`.

`String.isBlank()` and `String.isEmpty()` returns true for an empty string "".

Differences between `Array<Float>` and `FloatArray`:

- `Array<Float>` stores objects
- `FloatArray` is a primitive array
- `Array<Float>` to `FloatArray`: `.toFloatArray()`
- `FloatArray` to `Array<Float>`: `.toTypedArray()`

ranges:

- `start..end`: `[start, end]`
- `start until end`: `[start, end)`

```
kotlinc -include-runtime -d hello.jar hello.kt
```

Listing 2: ./code/basic-types/hello.kt

```
fun testInt() {
    val a = 1
    check(a is Int)
    check(a == 1)

    val b = 1L
    check(b is Long)
    check(b == 1L)

    val c :Byte = -1
    check(c is Byte)
    check(c == (-1).toByte())

    val d: Short = 1
    check(d is Short)
    check(d == 1.toShort())

    val g = 0x0a
    check(g is Int)
    check(g == 10)

    val f = 0b00001000
    check(f is Int)
    check(f == 8)

    val h = 0b0000_1000
    check(h is Int)
    check(h == 8)

    val k = 1_000
    check(k is Int)
    check(k == 1000)

    val m = 1 / 2
    check(m is Int)
    check(m == 0)

    val n = 1 / 2.toFloat()
    check(n is Float)
    check(n == 0.5f)
}
```

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```
fun testFloat() {
    val a = 1.0
    check(a is Double)
    check(a == 1.0)

    val b = 2.0F
    check(b is Float)
    check(b == 2.0F)
    check(b == (2.0).toFloat())

    // either f or F is ok
    val c = 2.0f
    check(c is Float)
    check(c == 2.0F)

    val d = (2.2F).toInt()
    check(d is Int)
    check(d == 2)
}

fun testBitwise() {
    // support only Int and Long
    val a = 0b01
    val b = 0b10

    val c = a or b // in C++: c = a | b

    check(c == 3)

    val d = a and b // in C++: d = a & b
    check(d == 0)

    val f = a shl 3 // in C++: a << 3
    check(f == 8)

    val g = b shr 1 // in C++: g = b >> 1
    check(g == 1)

    val h = a xor b // in C++: h = a ^ b
    check(h == 3)

    val k :Byte = 0b0100_0001
```

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```
        // test inv()
    }

    fun testJavaClass() {
        val a = 1
        println(a.javaClass) // int
    }

    fun testBoolean() {
        val a = true
        val b = false
        val c: Boolean = true
        check(a)
        check(!b)
        check(c)

        val d = a && b
        check(d == false)

        val e = a || b
        check(e)
    }

    fun testChar() {
        // a char is 2-byte, utf-16
        val a = 'a'
        check(a is Char)
        check(a.isDigit() == false)
        check(a.isLetter() == true)

        val b = '\uff00'
        check(b is Char)

        val c = '9'
        check(c.isDigit() == true)
        check(c.isLetter() == false)

        check(Char.SIZE_BITS == 16)
        check('0'.code == 0x30)
        check('A'.lowercaseChar() == 'a')
        check('a'.uppercaseChar() == 'A')
    }
}
```

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```
// note String is immutable, like Python.
fun testString() {
    // a sequence of Char, utf16
    // immutable
    val a = "Abc"
    check(a + 1 == "Abc1")
    check(a.count() == 3)
    check(a.length == 3)
    check(a.lastIndex == a.length - 1)
    check(a.uppercase() == "ABC")
    check(a.lowercase() == "abc")

    // remove the first two characters
    check(a.drop(2) == "c")

    // remove the last two characters
    check(a.dropLast(2) == "A")

    check(a.indexOf("bc") == 1)

    // multi-line strings, like python
    val b = """
line 1
line 2
"""
    println(b)
/*

line 1
line 2

*/

    val c = """
|line 1
|line 2
|""".trimMargin()
    println(c)
/*
line 1
line 2
*/
}
```

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```
val d = ""
>line 1
>line 2
"".trimMargin(">")
println(d)
/*
line 1
line 2
*/

// format
val k = String.format("%x %02x", 10, 10)
check(k == "a 0a")
}

fun testArray() {
    val a = Array(3) {it}
    check(a[0] == 0)
    check(a[1] == 1)
    check(a[2] == 2)

    // val b = arrayOf(0, 1, 2)
    val b = arrayOf<Int>(0, 1, 2)
    check(a.contentEquals(b)) // we cannot use == since they save references

    a[0] = 10 // assign a value
    check(a[0] == 10)

    check(a.sum() == 10 + 1 + 2)

    check(a.joinToString() == "10, 1, 2")
}

fun shortArrayToByteArray() {
    val a: Array<Short> = arrayOf(0x0102, 0x0304)
    val b: Array<Byte> = Array(a.size * 2) {0}
    println(a.contentToString()) // [258, 772]
    println(b.contentToString()) // [0, 0, 0, 0]

    for (i in a.indices) {
        b[2*i] = (a[i].toInt() and 0xff).toByte()
        b[2*i+1] = ((a[i].toInt() shr 8) and 0xff).toByte()
    }
}
```

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```
println(b.contentToString()) // [2, 1, 4, 3]
}

fun byteArrayToShortArray() {
    val a: Array<Byte> = arrayOf(0x01, 0x02, 0x03, 0x04)
    val b: Array<Short> = Array(a.size / 2) {0}

    println(a.contentToString()) // [258, 772]
    println(b.contentToString()) // [0, 0, 0, 0]

    for (i in b.indices) {
        b[i] = (a[2*i].toInt() + (a[2*i+1].toInt() shl 8)).toShort()
    }
    check(b[0].toInt() == 0x0201)
    check(b[1].toInt() == 0x0403)
}

fun main() {
    testInt()
    testFloat()
    testBitwise()
        testJavaClass()
        testBoolean()
        testChar()
        testString()
    testArray()
    shortArrayToByteArray()
    byteArrayToShortArray()
}
```

CharSequence

See <https://github.com/JetBrains/kotlin/blob/0938b46726b9c6938df309098316ce741815bb55/core/builtins/native/kotlin/CharSequence.kt#L22>

It is an interface.

26.1.3 Collections

See:

- <https://kotlinlang.org/docs/kotlin-tour-collections.html>
- <https://kotlinlang.org/docs/collections-overview.html>

```
fun testList() {
    val a = mutableListOf(1, 2, 3)
    println(a.javaClass) // class java.util.ArrayList

    /*
    for((i, v) in a.withIndex()) {
        check(i + 1 == v)
    }
    */

    val b: List<Int> = a // view of a, no data is copied
    a[0] = 10 // b is also changed since b is a view of a
    check(b[0] == 10)

    check(a.first() == a[0])
    check(a.last() == a[a.size - 1])

    check(a.count() == a.size)
    check(10 in a)
    check(11 !in a)

    // append
    a.add(20)
    check(a.size == 4) // now there are 4 elements
    check(b.size == 4) // size of b is also changed
    check(a.last() == 20)

    a.add(1, 100) // at index 1, insert 100
    check(a[1] == 100)
    check(a[2] == 2)

    a.add(2)
    a.remove(2) // remove the first occurrence of 2
    check(a.last() == 2)
}

fun testSet() {
    val a = mutableSetOf(1, 0, 2)
```

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```
println(a.javaClass) // class java.util.LinkedHashSet

println(a)
check(0 in a)
check(3 !in a)

check(a.count() == 3)
check(a.size == 3)

// remove a element
a.remove(0)
check(0 !in a)

a.add(0)
check(0 in a)
}

fun testMap() {
    val a: MutableMap<Int, String> = mutableMapOf(0 to "zero", 1 to "one", 2 to "two")
    println(a.javaClass) // class java.util.LinkedHashMap

    check(a.count() == 3)
    check(a[0] == "zero")
    check(a.containsKey(0))

    a.remove(0)
    check(0 !in a)
    check(a.containsKey(0) == false)

    a.put(0, "zero") // equivalent to a[0] = "zero"
    check(0 in a)
    check(a.containsKey(0))
}

fun main() {
    testList()
    testSet()
    testMap()
}
```

26.1.4 jni

To get the method signature, we first get the `.class` file of the kotlin or java code, and then use:

```
javap -s /path/to/xxx.class
```

26.1.5 Hello jni

References:

- <https://matt-moore.medium.com/kotlin-jni-for-native-code-835e93af7ddf>
- <https://docs.oracle.com/javase/7/docs/technotes/guides/jni/spec/functions.html>
- <https://developer.android.com/training/articles/perf-jni.html#primitive-arrays>

Listing 3: `./code/jni/hello/Makefile`

```
CXXFLAGS := -I $(JAVA_HOME)/include
CXXFLAGS += -I $(JAVA_HOME)/include/linux

all: NativeSample.jar libhello.so

NativeSample.jar: NativeSample.kt Main.kt
    kotlinc-jvm -include-runtime -d NativeSample.jar NativeSample.kt Main.kt

libhello.so: hello.cc hello.h
    $(CXX) -o $@ -shared -fPIC $(CXXFLAGS) hello.cc

run: all
    java -jar NativeSample.jar

clean:
    $(RM) *.jar *.so
```

Note: We have to disable name mangling here.

<https://docs.oracle.com/javase/7/docs/technotes/guides/jni/spec/design.html> describe the specification:

- `Java_NativeSample_sayHello`: The prefix is fixed with `Java`. `NativeSample` is the class name in kotlin. `sayHello` is the method in kotlin.

Listing 4: `./code/jni/hello/hello.h`

```
#ifndef HELLO_H
#define HELLO_H
```

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```

#include "jni.h"

#ifdef __cplusplus
extern "C" {
#endif

JNIEXPORT void JNICALL Java_NativeSample_sayHello(JNIEnv *env, jobject obj);

JNIEXPORT jfloat JNICALL Java_NativeSample_sum(JNIEnv *env, jobject obj,
                                              jfloatArray array);

#ifdef __cplusplus
}
#endif

#endif // HELLO_H

```

The first argument must be `JNIEnv *env`. Since it is a non-static kotlin method, the second argument is a reference to the object. Otherwise, it is a reference to the class.

Listing 5: ./code/jni/hello/hello.cc

```

#include "hello.h"

JNIEXPORT void JNICALL Java_NativeSample_sayHello(JNIEnv *env, jobject obj) {
    std::cout << "hello world\n";
}

JNIEXPORT jfloat JNICALL Java_NativeSample_sum(JNIEnv *env, jobject obj,
                                              jfloatArray array) {
    // the last arg is set to nullptr, which means we are not interested
    // in whether the returned pointer points to a copied region or not.
    jfloat *p = env->GetFloatArrayElements(array, nullptr);
    jsize n = env->GetArrayLength(array);

    jfloat s = 0;
    for (int32_t i = 0; i != n; ++i) {
        s += p[i];
    }
    // See
    // https://docs.oracle.com/javase/7/docs/technotes/guides/jni/spec/functions.html
    // JNI_ABORT: means to free the buffer pointed by p but don't copy the
    // buffer to array
    env->ReleaseFloatArrayElements(array, p, JNI_ABORT);
}

```

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```
    return s;
}
```

```
System.loadLibrary("hello");
```

- On Linux, it looks for `libhello.so`
- On Windows, it looks for `hello.dll`

```
external fun sayHello():
```

- It will look for the function with name `Java_NativeSample_sayHello`.

Listing 6: `./code/jni/hello/NativeSample.kt`

```
class NativeSample {
    companion object {
        init {
            System.loadLibrary("hello")
        }
    }

    external fun sayHello()
    external fun sum(array: FloatArray): Float
}
```

Listing 7: ./code/jni/hello/Main.kt

```

fun main() {
    var s = NativeSample()
    s.sayHello()
    var a : FloatArray = listOf(1.2f, 3.0f).toFloatArray()
    println(s.sum(a)) // 4.2

    a = floatArrayOf(3.0f, 4.0f, 5.0f)
    println(s.sum(a)) // 12.0

    a = FloatArray(100) {it.toFloat()}
    println(s.sum(a)) // 4950.0
}

```

26.1.6 Binding kaldi-native-fbank for kotlin

This note shows how to bind <https://github.com/csukuangfj/kaldi-native-fbank> for kotlin.

Listing 8: ./code/jni/feat-extractor/Makefile

```

SHERPA_NCNN_INSTALL_DIR := /ceph-fj/fangjun/open-source/sherpa-ncnn/build/install

CXXFLAGS := -I $(JAVA_HOME)/include
CXXFLAGS += -I $(JAVA_HOME)/include/linux
CXXFLAGS += -I $(SHERPA_NCNN_INSTALL_DIR)/include
CXXFLAGS += -I $(SHERPA_NCNN_INSTALL_DIR)/include/ncnn
CXXFLAGS += -Wall

LDFLAGS := -L $(SHERPA_NCNN_INSTALL_DIR)/lib -lkaldi-native-fbank-core -lsherpa-ncnn-
↳ core -lncnn
LDFLAGS += -Wl,-rpath,$(SHERPA_NCNN_INSTALL_DIR)/lib

all: main.jar libsherpa-ncnn.so

main.jar: Main.kt OnlineFeature.kt WaveReader.kt Model.kt
    kotlinc-jvm -include-runtime -d main.jar Main.kt OnlineFeature.kt WaveReader.
↳ kt Model.kt

libsherpa-ncnn.so: online-feature.cc sherpa-ncnn.cc online-feature.h sherpa-ncnn.h
    $(CXX) -o $@ -shared -fPIC $(CXXFLAGS) online-feature.cc sherpa-ncnn.cc
↳ $(LDFLAGS)

```

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```
run: all
    java -jar main.jar

clean:
    $(RM) main.jar libsherpa-ncnn.so
```

Listing 9: ./code/jni/feat-extractor/online-feature.h

```
#ifndef ONLINE_FEATURE_H_
#define ONLINE_FEATURE_H_
#include "jni.h"

#ifdef __cplusplus
extern "C" {
#endif

JNIEXPORT jlong JNICALL Java_OnlineFbank_new(JNIEnv *env, jobject obj,
                                              jobject opts);

JNIEXPORT void JNICALL Java_OnlineFbank_delete(JNIEnv *env, jobject obj,
                                              jlong ptr);

JNIEXPORT jint JNICALL Java_OnlineFbank_dim(JNIEnv *env, jobject obj,
                                              jlong ptr);

JNIEXPORT jfloat JNICALL Java_OnlineFbank_frameShiftInSeconds(JNIEnv *env,
                                                                jobject obj,
                                                                jlong ptr);

JNIEXPORT jint JNICALL Java_OnlineFbank_numFramesReady(JNIEnv *env, jobject obj,
                                                         jlong ptr);

JNIEXPORT jboolean JNICALL Java_OnlineFbank_isLastFrame(JNIEnv *env,
                                                         jobject obj, jlong ptr,
                                                         jint i);

JNIEXPORT void JNICALL Java_OnlineFbank_inputFinished(JNIEnv *env, jobject obj,
                                                         jlong ptr);

JNIEXPORT void JNICALL Java_OnlineFbank_acceptWaveform(JNIEnv *env, jobject obj,
                                                         jlong ptr,
                                                         jfloatArray samples,
                                                         jfloat sample_rate);
```

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```

JNIEXPORT jfloatArray JNICALL Java_OnlineFbank_getFrame(JNIEnv *env,
                                                         jobject obj, jlong ptr,
                                                         jint i);

JNIEXPORT jfloatArray JNICALL Java_OnlineFbank_getFrames(JNIEnv *env,
                                                          jobject /*obj*/,
                                                          jlong ptr, jint start,
                                                          jint n);

#ifdef __cplusplus
}
#endif

#endif // ONLINE_FEATURE_H_

```

Listing 10: ./code/jni/feat-extractor/online-feature.cc

```

#include "online-feature.h"
#include "kaldi-native-fbank/csrc/online-feature.h"

JNIEXPORT jlong JNICALL Java_OnlineFbank_new(JNIEnv *env, jobject /*obj*/,
                                              jobject opts) {
    jclass cls = env->GetObjectClass(opts);
    jfieldID fid;

    // https://docs.oracle.com/javase/7/docs/technotes/guides/jni/spec/types.html
    // https://courses.cs.washington.edu/courses/cse341/99wi/java/tutorial/native1.1/
    ↪ implementing/field.html

    knf::FbankOptions fbank_opts;

    fid = env->GetFieldID(cls, "use_energy", "Z");
    fbank_opts.use_energy = env->GetBooleanField(opts, fid);

    fid = env->GetFieldID(cls, "energy_floor", "F");
    fbank_opts.energy_floor = env->GetFloatField(opts, fid);

    fid = env->GetFieldID(cls, "raw_energy", "Z");
    fbank_opts.raw_energy = env->GetBooleanField(opts, fid);

    fid = env->GetFieldID(cls, "htk_compat", "Z");
    fbank_opts.htk_compat = env->GetBooleanField(opts, fid);
}

```

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```

fid = env->GetFieldID(cls, "use_log_fbank", "Z");
fbank_opts.use_log_fbank = env->GetBooleanField(opts, fid);

fid = env->GetFieldID(cls, "use_power", "Z");
fbank_opts.use_power = env->GetBooleanField(opts, fid);

fid = env->GetFieldID(cls, "frame_opts", "LFrameExtractionOptions;");

jobject frame_opts = env->GetObjectField(opts, fid);
jclass frame_opts_cls = env->GetObjectClass(frame_opts);

fid = env->GetFieldID(frame_opts_cls, "samp_freq", "F");
fbank_opts.frame_opts.samp_freq = env->GetFloatField(frame_opts, fid);

fid = env->GetFieldID(frame_opts_cls, "frame_shift_ms", "F");
fbank_opts.frame_opts.frame_shift_ms = env->GetFloatField(frame_opts, fid);

fid = env->GetFieldID(frame_opts_cls, "frame_length_ms", "F");
fbank_opts.frame_opts.frame_length_ms = env->GetFloatField(frame_opts, fid);

fid = env->GetFieldID(frame_opts_cls, "dither", "F");
fbank_opts.frame_opts.dither = env->GetFloatField(frame_opts, fid);

fid = env->GetFieldID(frame_opts_cls, "preemph_coeff", "F");
fbank_opts.frame_opts.preemph_coeff = env->GetFloatField(frame_opts, fid);

fid = env->GetFieldID(frame_opts_cls, "remove_dc_offset", "Z");
fbank_opts.frame_opts.remove_dc_offset =
    env->GetBooleanField(frame_opts, fid);

fid = env->GetFieldID(frame_opts_cls, "window_type", "Ljava/lang/String;");
jstring window_type = (jstring)env->GetObjectField(frame_opts, fid);
const char *p_window_type = env->GetStringUTFChars(window_type, nullptr);
fbank_opts.frame_opts.window_type = p_window_type;
env->ReleaseStringUTFChars(window_type, p_window_type);

fid = env->GetFieldID(frame_opts_cls, "round_to_power_of_two", "Z");
fbank_opts.frame_opts.round_to_power_of_two =
    env->GetBooleanField(frame_opts, fid);

fid = env->GetFieldID(frame_opts_cls, "blackman_coeff", "F");
fbank_opts.frame_opts.blackman_coeff = env->GetFloatField(frame_opts, fid);

```

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```

fid = env->GetFieldID(frame_opts_cls, "snip_edges", "Z");
fbank_opts.frame_opts.snip_edges = env->GetBooleanField(frame_opts, fid);

fid = env->GetFieldID(frame_opts_cls, "max_feature_vectors", "I");
fbank_opts.frame_opts.max_feature_vectors = env->GetIntField(frame_opts, fid);

fid = env->GetFieldID(cls, "mel_opts", "LMelBanksOptions;");
jobject mel_opts = env->GetObjectField(opts, fid);
jclass mel_opts_cls = env->GetObjectClass(mel_opts);

fid = env->GetFieldID(mel_opts_cls, "num_bins", "I");
fbank_opts.mel_opts.num_bins = env->GetIntField(mel_opts, fid);

fid = env->GetFieldID(mel_opts_cls, "low_freq", "F");
fbank_opts.mel_opts.low_freq = env->GetFloatField(mel_opts, fid);

fid = env->GetFieldID(mel_opts_cls, "high_freq", "F");
fbank_opts.mel_opts.high_freq = env->GetFloatField(mel_opts, fid);

fid = env->GetFieldID(mel_opts_cls, "vtln_low", "F");
fbank_opts.mel_opts.vtln_low = env->GetFloatField(mel_opts, fid);

fid = env->GetFieldID(mel_opts_cls, "vtln_high", "F");
fbank_opts.mel_opts.vtln_high = env->GetFloatField(mel_opts, fid);

fid = env->GetFieldID(mel_opts_cls, "debug_mel", "Z");
fbank_opts.mel_opts.debug_mel = env->GetBooleanField(mel_opts, fid);

fid = env->GetFieldID(mel_opts_cls, "htk_mode", "Z");
fbank_opts.mel_opts.htk_mode = env->GetBooleanField(mel_opts, fid);

auto online_fbank = new knf::OnlineFbank(fbank_opts);

return (jlong)online_fbank;
}

JNIEXPORT void JNICALL Java_OnlineFbank_delete(JNIEnv *env, jobject obj,
                                              jlong ptr) {
    delete reinterpret_cast<knf::OnlineFbank *>(ptr);
}

JNIEXPORT jint JNICALL Java_OnlineFbank_dim(JNIEnv *env, jobject obj,

```

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```

        jlong ptr) {
    return reinterpret_cast<const knf::OnlineFbank *>(ptr)->Dim();
}

JNIEXPORT jfloat JNICALL Java_OnlineFbank_frameShiftInSeconds(JNIEnv *env,
                                                                jobject obj,
                                                                jlong ptr) {
    return reinterpret_cast<const knf::OnlineFbank *>(ptr)->FrameShiftInSeconds();
}

JNIEXPORT jint JNICALL Java_OnlineFbank_numFramesReady(JNIEnv *env, jobject obj,
                                                         jlong ptr) {
    return reinterpret_cast<const knf::OnlineFbank *>(ptr)->NumFramesReady();
}

JNIEXPORT jboolean JNICALL Java_OnlineFbank_isLastFrame(JNIEnv *env,
                                                         jobject obj, jlong ptr,
                                                         jint i) {
    return reinterpret_cast<const knf::OnlineFbank *>(ptr)->IsLastFrame(i);
}

JNIEXPORT void JNICALL Java_OnlineFbank_inputFinished(JNIEnv *env, jobject obj,
                                                       jlong ptr) {
    reinterpret_cast<knf::OnlineFbank *>(ptr)->InputFinished();
}

JNIEXPORT void JNICALL Java_OnlineFbank_acceptWaveform(JNIEnv *env, jobject obj,
                                                         jlong ptr,
                                                         jfloatArray samples,
                                                         jfloat sample_rate) {
    jfloat *p = env->GetFloatArrayElements(samples, nullptr);
    jsize n = env->GetArrayLength(samples);

    reinterpret_cast<knf::OnlineFbank *>(ptr)->AcceptWaveform(sample_rate, p, n);

    env->ReleaseFloatArrayElements(samples, p, JNI_ABORT);
}

JNIEXPORT jfloatArray JNICALL Java_OnlineFbank_getFrame(JNIEnv *env,
                                                         jobject obj, jlong ptr,
                                                         jint i) {
    auto online_fbank = reinterpret_cast<const knf::OnlineFbank *>(ptr);
    auto frame = online_fbank->GetFrame(i);

```

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```

    auto dim = online_fbank->Dim();

    jfloatArray ans = env->NewFloatArray(dim);
    env->SetFloatArrayRegion(ans, 0, dim, frame);

    return ans;
}

JNIEXPORT jfloatArray JNICALL Java_OnlineFbank_getFrames(JNIEnv *env,
                                                         jobject /*obj*/,
                                                         jlong ptr, jint start,
                                                         jint n) {
    auto online_fbank = reinterpret_cast<const knf::OnlineFbank *>(ptr);
    auto dim = online_fbank->Dim();

    if (start + n > online_fbank->NumFramesReady()) {
        return nullptr;
    }

    jfloatArray ans = env->NewFloatArray(n * dim);
    for (int32_t i = 0; i != n; ++i) {
        auto frame = online_fbank->GetFrame(start + i);
        env->SetFloatArrayRegion(ans, i * dim, dim, frame);
    }

    return ans;
}

```

Listing 11: ./code/jni/feat-extractor/sherpa-ncnn.h

```

#ifndef SHERPA_NCNN_H_
#define SHERPA_NCNN_H_
#include "jni.h"

#ifdef __cplusplus
extern "C" {
#endif

JNIEXPORT jfloatArray JNICALL Java_WaveReader_00024Companion_readWave(
    JNIEnv *env, jclass cls, jstring filename, jfloat expected_sample_rate);

JNIEXPORT jlong JNICALL Java_Model_new(JNIEnv *env, jobject /*obj*/,
                                         jobject config);

```

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```

JNIEXPORT void JNICALL Java_Model_delete(JNIEnv *env, jobject /*obj*/,
                                           jlong ptr);

JNIEXPORT jint JNICALL Java_Model_segment(JNIEnv *env, jobject /*obj*/,
                                           jlong ptr);

JNIEXPORT jint JNICALL Java_Model_offset(JNIEnv *env, jobject /*obj*/,
                                           jlong ptr);

#ifdef __cplusplus
}
#endif

#endif // SHERPA_NCNN_H_

```

Listing 12: ./code/jni/feat-extractor/sherpa-ncnn.cc

```

#include "sherpa-ncnn.h"
#include "sherpa-ncnn/csrc/model.h"
#include "sherpa-ncnn/csrc/wave-reader.h"
#include <iostream>

JNIEXPORT jfloatArray JNICALL Java_WaveReader_00024Companion_readWave(
    JNIEnv *env, jclass cls, jstring filename, jfloat expected_sample_rate) {

    const char *p_filename = env->GetStringUTFChars(filename, nullptr);

    bool is_ok = false;
    std::vector<float> samples =
        sherpa_ncnn::ReadWave(p_filename, expected_sample_rate, &is_ok);
    env->ReleaseStringUTFChars(filename, p_filename);

    if (!is_ok) {
        return nullptr;
    }

    jfloatArray ans = env->NewFloatArray(samples.size());
    env->SetFloatArrayRegion(ans, 0, samples.size(), samples.data());
    return ans;
}

JNIEXPORT jlong JNICALL Java_Model_new(JNIEnv *env, jobject /*obj*/,

```

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```

        jobject config) {
    sherpa_ncnn::ModelConfig model_config;

    jclass cls = env->GetObjectClass(config);

    jfieldID fid = env->GetFieldID(cls, "encoderParam", "Ljava/lang/String;");
    jstring s = (jstring)env->GetObjectField(config, fid);
    const char *p = env->GetStringUTFChars(s, nullptr);
    model_config.encoder_param = p;
    env->ReleaseStringUTFChars(s, p);

    fid = env->GetFieldID(cls, "encoderBin", "Ljava/lang/String;");
    s = (jstring)env->GetObjectField(config, fid);
    p = env->GetStringUTFChars(s, nullptr);
    model_config.encoder_bin = p;
    env->ReleaseStringUTFChars(s, p);

    fid = env->GetFieldID(cls, "decoderParam", "Ljava/lang/String;");
    s = (jstring)env->GetObjectField(config, fid);
    p = env->GetStringUTFChars(s, nullptr);
    model_config.decoder_param = p;
    env->ReleaseStringUTFChars(s, p);

    fid = env->GetFieldID(cls, "decoderBin", "Ljava/lang/String;");
    s = (jstring)env->GetObjectField(config, fid);
    p = env->GetStringUTFChars(s, nullptr);
    model_config.decoder_bin = p;
    env->ReleaseStringUTFChars(s, p);

    fid = env->GetFieldID(cls, "joinerParam", "Ljava/lang/String;");
    s = (jstring)env->GetObjectField(config, fid);
    p = env->GetStringUTFChars(s, nullptr);
    model_config.joiner_param = p;
    env->ReleaseStringUTFChars(s, p);

    fid = env->GetFieldID(cls, "joinerBin", "Ljava/lang/String;");
    s = (jstring)env->GetObjectField(config, fid);
    p = env->GetStringUTFChars(s, nullptr);
    model_config.joiner_bin = p;
    env->ReleaseStringUTFChars(s, p);

    fid = env->GetFieldID(cls, "numThreads", "I");
    model_config.num_threads = env->GetIntField(config, fid);

```

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```

std::cout << model_config.ToString() << "\n";

auto model = sherpa_ncnn::Model::Create(model_config);

return (jlong) (model.release());
}

JNIEXPORT void JNICALL Java_Model_delete(JNIEnv *env, jobject /*obj*/,
                                           jlong ptr) {
    delete reinterpret_cast<sherpa_ncnn::Model *>(ptr);
}

JNIEXPORT jint JNICALL Java_Model_segment(JNIEnv *env, jobject /*obj*/,
                                           jlong ptr) {
    return reinterpret_cast<const sherpa_ncnn::Model *>(ptr)->Segment();
}

JNIEXPORT jint JNICALL Java_Model_offset(JNIEnv *env, jobject /*obj*/,
                                           jlong ptr) {
    return reinterpret_cast<const sherpa_ncnn::Model *>(ptr)->Offset();
}

```

Listing 13: ./code/jni/feat-extractor/Main.kt

```

fun main() {
    var fbank_opts = FbankOptions()
    fbank_opts.mel_opts.num_bins = 80

    var online_fbank = OnlineFbank(fbank_opts)

    var samples = WaveReader.readWave("./1089-134686-0001.wav", 16000.0f)
    if(samples != null) {
        online_fbank.acceptWaveform(samples)
    }
    online_fbank.inputFinished()

    var modelConfig = ModelConfig(
        encoderParam="./sherpa-ncnn-conv-emformer-transducer-2022-12-
↪04/encoder_jit_trace-epoch-30-avg-10-pnnx.ncnn.param",
        encoderBin="./sherpa-ncnn-conv-emformer-transducer-2022-12-04/
↪encoder_jit_trace-epoch-30-avg-10-pnnx.ncnn.bin",
        decoderParam="./sherpa-ncnn-conv-emformer-transducer-2022-12-
↪04/decoder_jit_trace-epoch-30-avg-10-pnnx.ncnn.param",

```

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```

        decoderBin="./sherpa-ncnn-conv-emformer-transducer-2022-12-04/
↪decoder_jit_trace-epoch-30-avg-10-pnnx.ncnn.bin",
        joinerParam="./sherpa-ncnn-conv-emformer-transducer-2022-12-
↪04/joiner_jit_trace-epoch-30-avg-10-pnnx.ncnn.param",
        joinerBin="./sherpa-ncnn-conv-emformer-transducer-2022-12-04/
↪joiner_jit_trace-epoch-30-avg-10-pnnx.ncnn.bin",
        numThreads=4,
    )
    var model = Model(modelConfig)
    println("segment: ${model.segment}")
    println("offset: ${model.offset}")
}

```

Listing 14: ./code/jni/feat-extractor/OnlineFeature.kt

```

data class FrameExtractionOptions(
    var samp_freq: Float = 16000.0f,
    var frame_shift_ms: Float = 10.0f,
    var frame_length_ms: Float = 25.0f,
    var dither: Float = 0.0f,
    var preemph_coeff: Float = 0.97f,
    var remove_dc_offset: Boolean = true,
    var window_type: String = "povey",
    var round_to_power_of_two: Boolean = true,
    var blackman_coeff: Float = 0.42f,
    var snip_edges: Boolean = true,
    var max_feature_vectors: Int = -1
)

data class MelBanksOptions(
    var num_bins : Int = 25,
    var low_freq : Float = 20.0f,
    var high_freq : Float = 0.0f,
    var vtln_low : Float = 100.0f,
    var vtln_high : Float = -500.0f,
    var debug_mel : Boolean = false,
    var htk_mode : Boolean = false,
)

data class FbankOptions(
    var frame_opts: FrameExtractionOptions = FrameExtractionOptions(),
    var mel_opts: MelBanksOptions = MelBanksOptions(),
    var use_energy: Boolean = false,
)

```

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```

    var energy_floor: Float = 0.0f,
    var raw_energy: Boolean = true,
    var htk_compat: Boolean = false,
    var use_log_fbank: Boolean = true,
    var use_power: Boolean = true,
)

class OnlineFbank(var opts: FbankOptions) {
    private var ptr: Long

    init {
        ptr = new(opts)
    }

    protected fun finalize() {
        delete(ptr)
    }

    val dim: Int
        get() = dim(ptr)

    val frameShiftInSeconds: Float
        get() = frameShiftInSeconds(ptr)

    val numFramesReady: Int
        get() = numFramesReady(ptr)

    fun isLastFrame(i: Int) : Boolean = isLastFrame(ptr, i)
    fun inputFinished() = inputFinished(ptr)
    fun acceptWaveform(samples: FloatArray) = acceptWaveform(ptr, samples, opts.frame_
↳ opts.samp_freq)
    fun getFrame(i: Int): FloatArray = getFrame(ptr, i)
    fun getFrames(start: Int, n: Int): FloatArray = getFrames(ptr, start, n)

    private external fun new(opts: FbankOptions): Long
    private external fun delete(ptr: Long)
    private external fun dim(ptr: Long): Int
    private external fun frameShiftInSeconds(ptr: Long): Float
    private external fun numFramesReady(ptr: Long): Int
    private external fun isLastFrame(ptr: Long, i: Int): Boolean
    private external fun inputFinished(ptr: Long)
    private external fun acceptWaveform(ptr: Long, samples: FloatArray, sample_rate:

```

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```

↪Float)
private external fun getFrame(ptr: Long, i: Int): FloatArray
private external fun getFrames(ptr: Long, start: Int, n: Int): FloatArray

companion object {
    init {
        System.loadLibrary("sherpa-ncnn")
    }
}
}

```

Listing 15: ./code/jni/feat-extractor/WaveReader.kt

```

class WaveReader {

    companion object {
        // Read a mono wave file.
        // No resampling is made.
        external fun readWave(filename: String, expected_sample_rate: Float = 16000.0f) : ↪
        ↪FloatArray?

        init {
            System.loadLibrary("sherpa-ncnn")
        }
    }
}

```

Listing 16: ./code/jni/feat-extractor/Model.kt

```

data class ModelConfig(
    var encoderParam: String,
    var encoderBin: String,
    var decoderParam: String,
    var decoderBin: String,
    var joinerParam: String,
    var joinerBin: String,
    var numThreads: Int = 4,
)

class Model(var config: ModelConfig) {
    private var ptr: Long

    init {

```

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```
        ptr = new(config)
    }

    protected fun finalize() {
        delete(ptr)
    }

    val segment: Int
        get() = segment(ptr)

    val offset: Int
        get() = offset(ptr)

    private external fun new(config: ModelConfig): Long
    private external fun delete(ptr: Long)
    private external fun segment(ptr: Long): Int
    private external fun offset(ptr: Long): Int
}
```

26.1.7 References

- <https://github.com/JetBrains/kotlin>

Source code of Kotlin.

- <https://kotlin-quick-reference.com/>

An online book

26.1.8 Control flows

See <https://kotlinlang.org/docs/control-flow.html>

```
kotlinc -include-runtime -d hello.jar hello.kt
```

Listing 17: ./code/control-flows/hello.kt

```
fun testIf() {
    val a = 1
    val b = 2
    var max = -1
    if (a > b) {
```

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```
    max = a
  } else {
    max = b
  }
  check(max == b)

  // if can be used in an expression
  val min = if (a < b) {a} else {b}
  check(min == a)

  //
  if (true && true || false) {
    check(true)
  }
}

fun testWhen() {
  val a = 1
  var c: Int
  when(a) {
    1 -> c = 10
    2 -> c = 20
    else -> c = 30
  }
  check(c == 10)

  val d = when {
    a < 0 -> 3
    a == 0 -> 10
    a == 1 -> 30
    else -> 5
  }
  check(d == 30)

  val k: Any = 1
  val b = when (k) {
    is Int -> 10
    is Float -> 20
    else -> 30
  }
  check(b == 10)
}
```

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```
@OptIn(kotlin.ExperimentalStdlibApi::class)
fun testRange() {
    check(5 in 0..5) // [0,5]
    check(5 !in 0..<5) // [0,5)

    check(4 in 4 downTo 1) // [4,3,2,1]
    check(1 in 4 downTo 1) // [4,3,2,1]

    check(0 in 0..5 step 3)
    check(1 !in 0..5 step 3)
    check(2 !in 0..5 step 3)
    check(3 in 0..5 step 3)
    check(4 !in 0..5 step 3)
    check(5 !in 0..5 step 3)

    check(5 in 5 downTo 0 step 3)
    check(4 !in 5 downTo 0 step 3)
    check(3 !in 5 downTo 0 step 3)
    check(2 in 5 downTo 0 step 3)
    check(1 !in 5 downTo 0 step 3)
    check(0 !in 5 downTo 0 step 3)

    val c = 'a'
    check(c in 'a' .. 'z' || c in 'A' .. 'Z')
    check(c !in '0' .. '9')

    val d = '0'
    check(d in '0' .. '9')
    check(d !in 'a'..'z' && d !in 'A'..'Z')
}

fun testFor() {
    var a: Array<Int> = Array(5) {0}

    for (i in 0..4) {
        a[i] = i
    }

    check(a[0] == 0)
    check(a[1] == 1)
    check(a[2] == 2)
    check(a[3] == 3)
```

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```
for(i in 4 downTo 0) {
    a[i] = 4 - i
}

check(a[4] == 0);
check(a[3] == 1);
check(a[2] == 2);
check(a[1] == 3);
check(a[0] == 4);

for (i in a.indices) {
    when (i) {
        0 -> check(a[i] == 4)
        1 -> check(a[i] == 3)
        2 -> check(a[i] == 2)
        3 -> check(a[i] == 1)
        4 -> check(a[i] == 0)
    }
}

/*
for ((i, value) in a.withIndexes()) {
    when (i) {
        0 -> check(value == 4)
        1 -> check(value == 3)
        2 -> check(value == 2)
        3 -> check(value == 1)
        4 -> check(value == 0)
    }
}

*/
}

fun main() {
    testIf()
    testWhen()
    testRange()
    testFor()
}
```

26.1.9 Functions

See <https://kotlinlang.org/docs/functions.html>

typealias:

```
// https://kotlinlang.org/docs/type-aliases.html
typealias ClickHandler = (Button, ClickEvent) -> Unit
```

```
kotlinc -include-runtime -d hello.jar hello.kt
```

Listing 18: ./code/functions/hello.kt

```
fun sum1(x: Int, y: Int): Int {
    return x + y
}

// ignore the {} and return
fun sum2(x: Int, y: Int): Int = x + y

// ignore the return type hint
fun sum3(x: Int, y: Int) = x + y

fun testSum() {
    check(sum1(1, 2) == 3)
    check(sum2(1, 2) == 3)
    check(sum3(1, 2) == 3)
}

fun double(x: Double): Double {
    return x * 2
}

// we can use trailing comma in the function parameters
//
@OptIn(kotlin.ExperimentalStdlibApi::class)
fun powerOf(
    x: Int,
    exponent: Int,
): Int {
    var n = 1
    for (i in 0 ..< exponent) {
        n *= x
    }
    return n
}
```

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```
}

fun default(a: Int, b: Int = 2): Int {
    return a + b
}

// we can call a.size in setting the default value for b!!!
fun default2(a: Array<Float>, b: Int = a.size): Float {
    return a[0] + b
}

fun testDouble() {
    val x = double(2.5)
    check(x == 5.0)
}

fun testPowerOf() {
    val x = powerOf(2, 3)
    check(x == 8)
}

fun testDefault() {
    val x = default(1)
    check(x == 3)

    val y = default(1, 3)
    check(y == 4)
}

fun testDefault2() {
    val a: Array<Float> = arrayOf(1.5f, 2.0f, 3.0f)
    val b = default2(a)
    check(b == 1.5f+3)

    val c = default2(a, 10)
    check(c == 1.5f+10)
}

fun testLambda() {
    val makeUppercase = {x: String -> x.uppercase()}
    check(makeUppercase("ab") == "AB")
}
```

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```
fun testLambda2(a: Int, b: Int, f: (Int, Int)->Int): Int {
    return f(a, b)
}

fun testLambda3(a: Int, f: (Int)->Int): Int {
    return f(a);
}

fun main() {
    testSum()
    testDouble()
    testPowerOf()
    testDefault()
    testDefault2()
    testLambda()
    var i = testLambda2(2, 3) {a, b -> a + b}
    check(i == 5)

    i = testLambda2(2, 3) {a, b -> a - b}
    check(i == -1)

    // the default argument is named as it
    i = testLambda3(2) {it + 1}
    check(i == 2 + 1)
}
```

26.1.10 Package

```
// a.kt
package geometry.shape

class Rectangle
fun createUnitRectangle() = Rectangle()

// b.kt
import geometry.shape.Rectangle
import geometry.shape.createUnitRectangle

// c.kt
import geometry.shape.*
```

26.1.11 for

```
val numbers = arrayOf(1, 2, 3)
for (n in numbers) {
    println(n)
}

for (i in numbers.indices) {
    println(numbers[i])
}

for ((i, value) in numbers.withIndex()) {
    check(numbers[i] == value)
}

// 0 <= i < numbers.size
for (i in 0 until numbers.size) {
    println(numbers[i])
}

// forEach
val a = arrayOf(1, 2, 3)
a.forEach(::println)
```

26.1.12 enum

```
enum class Weekday {
    Monday,
    Tuesday,
    Wednesday,
    Thursday,
}

for(day in Weekday.values()) {
    // day.ordinal: 0, 1, 2, 3, 4
    // day.name: Monday, Tuesday, Wednesday, Thursday
}

Weekday.valueOf("Monday") == Weekday.Monday
```

26.1.13 Coroutine

Setup from commandline

Refer to <https://stackoverflow.com/questions/79103733/how-to-use-kotlinx-with-the-command-line-compiler-kotlin>

Visit <https://mvnrepository.com/artifact/org.jetbrains.kotlinx/kotlinx-coroutines-core-jvm>, select a version, e.g., 1.10.1, and it goes to <https://mvnrepository.com/artifact/org.jetbrains.kotlinx/kotlinx-coroutines-core-jvm/1.10.1>. Click the link <https://repo1.maven.org/maven2/org/jetbrains/kotlinx/kotlinx-coroutines-core-jvm/1.10.1/kotlinx-coroutines-core-jvm-1.10.1.jar>.

Hint: For android, it is `kotlinx-coroutines-android`. For js, it is `kotlinx-coroutines-core-js`.

Listing 19: `./code/coroutine/Hello.kt`

```
import kotlinx.coroutines.delay
import kotlinx.coroutines.launch
import kotlinx.coroutines.runBlocking

fun main() =
    runBlocking {
        launch {
            delay(1000)
            println("Kotlin Coroutines World!")
        }
        println("Hello")
    }
```

Run:

```
kotlinc -cp ./kotlinx-coroutines-core-jvm-1.10.1.jar Hello.kt -d hello.jar
```

It gives the following error:

```
kotlinx-coroutines-core-jvm-1.10.1.jar!/META-INF/kotlinx-coroutines-core.kotlin_
↪module:
error: module was compiled with an incompatible version of Kotlin.
The binary version of its metadata is 2.1.0, expected version is 1.8.0.
```

`kotlinc --version` prints:

```
info: kotlinc-jvm 1.8.0 (JRE 17.0.5+9-LTS-191)
```

so we need to update our kotlin:


```
cd $HOME/software/
mkdir -p kotlin
cd kotlin
wget https://github.com/JetBrains/kotlin/releases/download/v2.1.0/kotlin-compiler-2.1.0.zip
unzip kotlin-compiler-2.1.0.zip

mv kotlinc 2.1.0
export PATH=$HOME/software/kotlin/2.1.0/bin:$PATH
```

Then we can run:

```
kotlinc -cp ./kotlinx-coroutines-core-jvm-1.10.1.jar Hello.kt -d hello.jar
```

It will generate a file `hello.jar`.

```
ls -lh hello.jar
-rw-r--r--  1 fangjun  staff   4.0K Jan  1 16:20 hello.jar
```

We can run it with:

```
kotlin -classpath ./hello.jar:./kotlinx-coroutines-core-jvm-1.10.1.jar  HelloKt
```

Alternatively, we can use:

```
kotlinc -cp ./kotlinx-coroutines-core-jvm-1.10.1.jar Hello.kt -include-runtime -d hello.jar
```

`hello.jar` is much larger than before:

```
ls -lh hello.jar
-rw-r--r--  1 fangjun  staff   4.6M Jan  1 16:22 hello.jar
```

Then we can use:

```
java -classpath ./hello.jar:./kotlinx-coroutines-core-jvm-1.10.1.jar  HelloKt
java -cp ./hello.jar:./kotlinx-coroutines-core-jvm-1.10.1.jar  HelloKt
kotlin -cp ./hello.jar:./kotlinx-coroutines-core-jvm-1.10.1.jar  HelloKt
kotlin -classpath ./hello.jar:./kotlinx-coroutines-core-jvm-1.10.1.jar  HelloKt
```

thread related functions

```
Thread.sleep(1000L) # sleep 1000ms, i.e., 1s
```

common suspendable functions

```
delay(1000L) # sleep 1000m, i.e., 1s
```

Predefined coroutine dispatcher

- `Dispatchers.Default`: Compute-intensive
- `Dispatchers.IO`: File I/O, blocking socket I/O, etc
- `Dispatchers.Main`: For UI thread

functions without a scope

- `runBlocking`, it is not a suspend function, coroutine builder, scope builder
- `coroutineScope()`, it is a suspend function, scope builder
- `withContext()`, it is a suspend function

functions requiring a scope

- `launch()`, e.g., `GlobalScope.launch`, it is not a suspend function. coroutine builder
- `async()`, e.g., `GlobalScope.async`, it is not a suspend function. coroutine builder

26.1.14 Kotlin Multiplatform

`project_root/build.gradle.kts` contains `compose.desktop`, which contains:

```
compose.desktop {
    application {
        mainClass = "MainKt"

        nativeDistributions {
            targetFormats(TargetFormat.Dmg, TargetFormat.Msi, TargetFormat.Deb)
            packageName = "demo"
            packageVersion = "1.0.0"
        }
    }
}
```

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```

    }
}
}

```

We can also add `TargetFormat.Exe`.

At the commandline, run `./gradlew packageMsi`. It generates `project_root/build/compose/binaries/main/msi`.

26.1.15 Kotlin Native

See

- <https://kotlinlang.org/docs/native-get-started.html>
- <https://github.com/kgit2/c-interop-klib>
- <https://github.com/Kotlin/kmp-native-wizard>

Go to <https://github.com/JetBrains/kotlin/releases/tag/v2.1.20>, download `kotlin-native-prebuilt-macos-x86_64-2.1.20.tar.gz`. Unzip it and put its bin to the PATH:

```
export PATH=/Users/fangjun/software/
```

Listing 20: `./code/native/hello.kt`

```

fun main() {
    println("Hello, Kotlin/Native!")
}

```

```
kotlinc-native ./hello.kt -o hello
```

Next, we can run `./hello.kexe`.

sherpa-onnx

The `sherpa-onnx-c-api.def` file:

```

headers = sherpa-onnx/c-api/c-api.h
headerFilter = sherpa-onnx/c-api/c-api.h
compilerOpts = -I /Users/fangjun/open-source/sherpa-onnx
package = sherpa_onnx
linkerOpts = -L /Users/fangjun/open-source/sherpa-onnx/build/lib -lsherpa-onnx-c-api

```

The `main.kt`:

```
import kotlinx.cinterop.*
import sherpa_onnx.*

fun main() {
    print("hello")
}
```

The command:

```
cinterop -def ./sherpa-onnx-c-api.def -o libsherpa_onnx_c_api.klib
kotlinc-native ./main.kt -l libsherpa_onnx_c_api.klib

# a.txt contains the kotlin code for our c api
klib dump-metadata ./libsherpa_onnx_c_api.klib -print-signatures true > a.txt
```

char

In C extern unsigned char my_c_byte.

In kotlin, there is a implicit extern var my_c_byte: ByteVar. The variable type in Kotlin is ByteVar.

```
var bytePtr: NativePtr = my_c_byte.ptr

// Note: bytePtr.pointed == my_c_byte

// Read it
val byteValue: Byte = bytePtr.pointed.value

// Change it
bytePtr.pointed.value = 0x42

// Read it again
val newByteValue: Byte = bytePtr.pointed.value
```

26.2 Basics

There are two tools: NDK and SDK. Each tool has a version.

Android has a concept of Android native API level.

There are three environment variables to set:

- ANDROID_NDK_ROOT
- ANDROID_SDK_ROOT

- ANDROID_HOME

The name of the app is determined by `<application android:label` in `AndroidManifest.xml`. We can use `@string/app_name`.

The title of the nav bar is determined by `<activity android:label` in `AndroidManifest.xml`.

Button click:

```
val button: Button = findViewById(R.id.button1)
button.setOnClickListener {
    Toast.makeText(this, "Clicked me!", Toast.LengthShort).show()
}
```

The menu is added in `res/menu`. The directory `res/menu` is created by us manually. After creating the directory, we can right click it, and select `menu resource file`. If we name it `main`, it will create a file `main.xml` inside `res/menu`. Inside `main.xml`, each menu item contains something like below:

```
<item android:id="@+id/add_item" android:title="Add" />
```

After creating `main.xml`, click the menu `Code` -> `Override Method`, select `onOptionsItemSelected`:

```
menuInflater.inflate(R.menu.main, menu)
return true
```

To define a function when a menu item is clicked, click the menu `Code`->`Override Method`, select `onOptionsItemSelected`.

```
when (item.itemId) {
    R.id.add_item -> { xxx }
    R.id.remove_item -> { xxx }
}
return true
```

To destroy an activity, call `finish()`.

26.2.1 TextView

It is a label in other GUI.

```
<TextView
    android:id="@+id/textView"
    android:text="hello world"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
/>
```

To change the alignment, use `android:gravity`.

```
android:gravity="center"
// top, bottom, center, start, end
// center_vertical|center_horizontal
```

To change the color, use `android:textColor`

```
android:textColor="#00ff00"
```

To change the font size, use `android:textSize="24sp"`

26.2.2 Button

```
<Button
    android:id="@+id/button"
    android:text="Button"
    android:textAllCaps="false"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
/>
```

26.2.3 EditText

```
<EditText
    android:id="@+id/editText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="this is some hint"
    android:maxLines="10"
/>
```

To get the text of `EditText` in the kotlin code, use:

```
val inputText = editText.text.toString()
```

26.2.4 LinearLayout

```
<LinearLayout
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
/>
```

orientation can also be horizontal.

26.2.5 view binding

Edit app/build.gradle, add the following:

```
android { ...
    kotlinOptions {
        jvmTarget = '1.8'
    }
    buildFeatures {
        viewBinding true
    }
}
```

Then in the main activity, use:

```
// activity_main.xml -> ActivityMainBinding
import com.bignerdranch.android.geoquiz.databinding.ActivityMainBinding

private lateinit var binding: ActivityMainBinding
binding = ActivityMainBinding.inflate(layoutInflater)
setContentView(binding.root)

binding.trueButton.setOnClickListener { view: View -> }

binding.falseButton.setOnClickListener { view: View -> }
```

26.2.6 directories

```
val nativeDirPath = context.applicationInfo.nativeLibraryDir
Log.i(TAG, "nativeDirPath: $nativeDirPath")

// nativeDirPath: /data/app/~~cqPVTPOsjtVv50X9wWpr1g==/com.k2fsa.sherpa.onnx.simulate.
↳streaming.asr-LWDzOBpzdRoEuYbbcqyjTg==/lib/arm64
```

```
val assetFilePath = context.filesDir.absolutePath
Log.i(TAG, "assetFilePath: $assetFilePath")

// assetFilePath: /data/user/0/com.k2fsa.sherpa.onnx.simulate.streaming.asr/files
```

```
context.dataDir

// /data/user/0/com.k2fsa.sherpa.onnx.simulate.streaming.asr
```

26.3 Installation

26.3.1 Install NDK on Linux (not recommended)

(Use the following SDK to install NDK)

See <https://developer.android.com/ndk/downloads/index.html>

See [https://www.cryptopp.com/wiki/Android_Setup_\(Command_Line\)](https://www.cryptopp.com/wiki/Android_Setup_(Command_Line)) for details.

```
wget https://dl.google.com/android/repository/android-ndk-r25-linux.zip
unzip -d /ceph-fj/fangjun/software android-ndk-r25-linux.zip
# It will create /ceph-fj/fangjun/software/android-ndk-r25
```

Other versions can be downloaded from <https://github.com/android/ndk/wiki/Unsupported-Downloads>

```
wget https://dl.google.com/android/repository/android-ndk-r24-linux.zip
unzip -d /ceph-fj/fangjun/software android-ndk-r24-linux.zip

# It will create /ceph-fj/fangjun/software/android-ndk-r24
```

Now create a symlink in /ceph-fj/fangjun/software:

```
cd /ceph-fj/fangjun/software
ln -s android-ndk-r25 android-ndk
```

Set the following environment variable:


```
export ANDROID_NDK_ROOT=/ceph-fj/fangjun/software/android-ndk
export PATH=$ANDROID_NDK_ROOT:$PATH
```

26.3.2 Install SDK on Linux

Download the commandline tools only from <https://developer.android.com/studio#downloads>

```
wget https://dl.google.com/android/repository/commandlinetools-linux-8512546_latest.
↪zip
unzip -d /ceph-fj/fangjun/software/android-sdk ./commandlinetools-linux-8512546_
↪latest.zip
# Everything is inside /ceph-fj/fangjun/software/android-sdk/cmdline-tools/
cd /ceph-fj/fangjun/software/android-sdk
mv cmdline-tools latest
mkdir cmdline-tools
mv latest ./cmdline-tools/
```

If we don't run `mv cmdline-tools latest`, it will throw the following error

```
$ /ceph-fj/fangjun/software/android-sdk/cmdline-tools/bin/sdkmanager --list

Error: Could not determine SDK root.
Error: Either specify it explicitly with --sdk_root= or move this package into its
↪expected location: <sdk>/cmdline-tools/latest/
```

```
sdkmanager --update
sdkmanager --list

# Install the build tools
sdkmanager "platforms;android-28" "build-tools;28.0.3"
# It will create the following directories inside /ceph-fj/fangjun/software/android-
↪sdk/
#
# build-tools, emulator, licenses, patcher, platform-tools, platforms, tools
```

```
$ sdkmanager --list_installed
Installed packages:=====] 100% Fetch remote repository...

  Path                  | Version | Description                  | Location
  -----              | -       | -                             | -
  build-tools;28.0.3    | 28.0.3  | Android SDK Build-Tools 28.0.3 | build-tools/28.0.3
  emulator              | 31.3.10 | Android Emulator              | emulator
  patcher;v4            | 1       | SDK Patch Applier v4         | patcher/v4
  platform-tools        | 33.0.2  | Android SDK Platform-Tools   | platform-tools
```

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```
platforms;android-28 | 6 | Android SDK Platform 28 | platforms/android-  
↳28
```

```
sdkmanager --help  
yes | sdkmanager --licenses # to accept or licenses, the decision is saved in a  
↳cache file.
```

Set the following environment variables:

```
export ANDROID_SDK_ROOT=/ceph-fj/fangjun/software/android-sdk  
export PATH=$ANDROID_SDK_ROOT/cmdline-tools/latest/bin:$PATH  
  
export ANDROID_HOME=/ceph-fj/fangjun/software/android-sdk  
  
# for emulator  
export PATH=$ANDROID_SDK_ROOT/emulator:$PATH  
  
# for adb  
export PATH=$ANDROID_SDK_ROOT/platform-tools:$PATH  
  
# We installed 28.0.3 before  
export PATH=$ANDROID_SDK_ROOT/build-tools/28.0.3:$PATH # change it for different  
↳versions
```

Now install NDK using sdkmanager:

```
# sdkmanager --list | grep ndk  
sdkmanager "ndk;21.0.6113669"  
# it will download android-ndk-r21 and will generate  
# android-sdk/ndk/21.0.6113669  
  
export ANDROID_NDK_HOME=$ANDROID_SDK_ROOT/ndk/21.0.6113669  
export PATH=$ANDROID_NDK_ROOT:$PATH
```

26.3.3 emulator

```
find /Users/fangjun/software/my-android -name emulator  
/Users/fangjun/software/my-android/emulator -list-avds  
/Users/fangjun/software/my-android/emulator -avd Pixel_4_API_31 -dns-server 8.8.8.8
```

Refer to <https://developer.android.com/studio/run/emulator-networking#proxy> to setup the proxy.

26.4 cmake

See

- <https://cmake.org/cmake/help/latest/manual/cmake-toolchains.7.html>
- <https://developer.android.com/ndk/guides/cmake#command-line>
- `/ceph-fj/fangjun/software/android-ndk/build/cmake/android.toolchain.cmake`

User provided:

- `ANDROID_NDK`: Set to the path of android-ndk
- `ANDROID_ABI`: armeabi-v7a, arm64-v8a, x86, x86_64,
- `ANDROID_PLATFORM`
- `ANDROID_NATIVE_API_LEVEL`
- `ANDROID_TOOLCHAIN`
- `ANDROID`: TRUE
- `CMAKE_SYSTEM_NAME`: Android
- `ANDROID_STL`
- `ANDROID_HOST_TAG`

Auto generated:

- `ANDROID_NDK_MAJOR` - see android-ndk/source.properties
- `ANDROID_NDK_MINOR` - see android-ndk/source.properties
- `ANDROID_NDK_BUILD` - see android-ndk/source.properties
- `ANDROID_NDK_REVISION` - see android-ndk/source.properties
- `ANDROID_TOOLCHAIN_ROOT`
- `ANDROID_C_COMPILER`

26.5 hello

In this note, we describe how to build an executable, how to create an emulator, and how to run the executable in the emulator via `adb push`, `adb shell`.

Listing 21: ./code/hello/hello.cc

```

1  #include <iostream>
2
3  int main() {
4      std::cout << "hello world\n";
5      return 0;
6  }

```

Listing 22: ./code/hello/CMakeLists

```

1  cmake_minimum_required(VERSION 3.8)
2
3  project(hello)
4
5  add_executable(hello hello.cc)

```

26.5.1 x86

```

export ANDROID_NDK_ROOT=/ceph-fj/fangjun/software/android-ndk
cmake -DCMAKE_TOOLCHAIN_FILE=$ANDROID_NDK_ROOT/build/cmake/android.toolchain.cmake -
↳DANDROID_ABI=x86 ..

```

Other values:

- -DANDROID_ABI="arm64-v8a"
- -DANDROID_ABI="armeabi-v7a"
- -DANDROID_ARM_NEON=ON
- -DANDROID_PLATFORM=android-21
- -DANDROID_PLATFORM=android-24

```

$ sdkmanager --list | grep system-images | grep x86 | grep android-28
$ sdkmanager "system-images;android-28;default;x86"

$ which avdmanager
/ceph-fj/fangjun/software/android-sdk/cmdline-tools/latest/bin/avdmanage

$ avdmanager create avd --help
$ -k --package : Package path of the system image for this AVD
#               (e.g., 'system-images;android-19;google_apis;x86').
# -n --name     : Name of the new AVD [required]

```

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```
# -b --abi      : The ABI to use for the AVD. The default is to auto-select
#                the ABI if the platform has only one ABI for its system images
# -g --tag      : The sys-img tag to use for the AVD. The default is to
#                auto-select if the platform has only one tag for its system
#                images
$ avdmanager create avd -k "system-images;android-28;default;x86" -n hello -b x86 -g
↳ default
# Use the default option [no] when it prompts:
# Do you wish to create a custom hardware profile? [no]
#
$ avdmanager delete avd -n hello # to delete it
```

```
$ avdmanager list avd
Available Android Virtual Devices:
  Name: hello
  Path: /root/fangjun/.android/avd/hello.avd
  Target: Default Android System Image
         Based on: Android 9.0 (Pie) Tag/ABI: default/x86
  Sdcard: 512 MB
```

```
emulator -avd hello -no-window -no-accel # then, open a new terminal
```

```
$ adb devices

* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
emulator-5554    offline
```

```
# push the binary from code/hello/build/hello

adb push ./code/hello/build/hello /sdcard # not able to use chmod +x in it
adb push ./code/hello/build/hello /data/local
adb shell /data/local/hello
adb shell
generic_x86:/ #
```

```
generic_x86:/ # ./data/local/hello
hello world
```

26.6 Android.mk

See https://developer.android.com/ndk/guides/android_mk

26.6.1 hello

Listing 23: ./code/android_mk/hello/jni/foo.cc

```
1 #include <iostream>
2 int main() { std::cout << "hello world\n"; }
```

Listing 24: ./code/android_mk/hello/jni/Android.mk

```
1 LOCAL_PATH := $(call my-dir)
2 include $(CLEAR_VARS)
3 LOCAL_MODULE := foo
4 LOCAL_SRC_FILES := foo.cc
5 # include $(BUILD_SHARED_LIBRARY)
6 include $(BUILD_EXECUTABLE)
```

Listing 25: ./code/android_mk/hello/jni/Application.mk

```
1 APP_ABI := x86
2 APP_STL := c++_shared
```

APP_STL := c++_shared is to fix the following errors:

```
ld: error: undefined symbol: std::__ndk1::cout
```

To compile:

```
cd code/android_mk/hello
ndk-build
```

It will generate two directories in hello: libs and obj..

```
adb push libs/x86/foo /data/local
adb push libs/x86/libc++_shared.so /data/local
adb shell
cd /data/local
export LD_LIBRARY_PATH=.
./foo
```

26.7 adb

26.7.1 install on macos

```
wget https://dl.google.com/android/repository/platform-tools-latest-darwin.zip
# unzip it and you will find the binary `adb`
```

26.7.2 install on windows

Go to <https://adbshell.com/downloads> to download it.

26.7.3 install on Linux

```
wget https://dl.google.com/android/repository/platform-tools-latest-linux.zip
# unzip it
```

26.7.4 Frequently used commands

```
adb kill-server
adb start-server
adb devices
adb devices -l
```

adb devices:

```
List of devices attached
192.168.65.204:44959    device
adb-RFARA0X7Z9Y-qzseiX._adb-tls-connect._tcp.  device
```

adb devices -l:

```
List of devices attached
192.168.65.204:44959    device product:freshbscn model:SM_R860 device:freshbs_
↳transport_id:3
adb-RFARA0X7Z9Y-qzseiX._adb-tls-connect._tcp.  device product:freshbscn model:SM_R860_
↳device:freshbs transport_id:1
```

26.7.5 pair a device via wireless

```
# the port is different from the port when connecting to the device
```

```
adb pair 192.168.65.204:36029
```

It will show:

```
Enter pairing code: 156066
```

and:

```
Successfully paired to 192.168.65.204:36029 [guid=adb-RFARA0X7Z9Y-qzseiX]
```

26.7.6 connect a device

```
# the port is different from the port when pairing with the device
```

```
ad connect 192.168.65.204:44959
```

It will show:

```
connected to 192.168.65.204:44959
```

26.7.7 run binaries

```
adb push ./hello-world /data/local/tmp/
```

26.7.8 paste text to android emulator

```
adb shell "input text 'hello world'"
```

26.7.9 Install an apk

```
adb -e install path/to/app.apk
```

```
# Install to a specific device
```

```
adb -s <DEVICE ID> install <PATH TO APK>
```


26.8 tts

Install Android Accessibility Suite via Google Play.

- <https://github.com/Miserlou/Android-SDK-Samples/tree/master/TtsEngine>
- <https://github.com/benjaminwan/ChineseTtsTflite>

26.9 jetpack

See also <https://github.com/android/codelab-android-compose>

26.9.1 material design

```
MaterialTheme(
    colorScheme = MyAppsColorScheme,
    typography = MyAppsTypography,
    shapes = MyAppsShapes,
) {
}
```

26.9.2 state

```
// var selectedAnswer: MutableState<Answer?> = mutableStateOf(null)
var selectedAnswer: MutableState<Answer?> = remember { mutableStateOf(null) }
var selectedAnswer: MutableState<Answer?> = rememberSaveable { mutableStateOf(null) }

// note that we use the member ``value`` here
isSelected = (selectedAnswer.value == answer)
```

```
var selectedAnswer: Answer? by rememberSaveable { mutableStateOf(null) }

isSelected = (selectedAnswer == answer)
```

```
val mutableState = remember { mutableStateOf(default) }

var value by remember { mutableStateOf(default) }

val (value, setValue) = remember { mutableStateOf(default) }
```

26.9.3 bom

<https://developer.android.com/jetpack/compose/bom/bom-mapping>

26.9.4 Arrangement

<https://vitor-ramos.medium.com/understand-arrangement-and-alignment-in-jetpack-compose-7633f2ed5b39>

- `Arrangement.Start`, `Arrangement.Top`
- `Arrangement.End`, `Arrangement.Bottom`
- `Arrangement.Center`
- `Arrangement.SpaceBetween`
- `Arrangement.SpaceAround`
- `Arrangement.SpaceEvenly`
- `Arrangement.spacedBy(8.dp)`

26.9.5 Alignment

- `Alignment.Start`
- `Alignment.End`
- `Alignment.Center`

26.9.6 Surface

```
Surface(  
    modifier = Modifier.fillMaxSize(),  
    color = MaterialTheme.colorscheme.background  
)  
  
Surface(  
    color = MaterialTheme.colorscheme.primary  
)  
  
Surface(  
    color = MaterialTheme.colorscheme.surface,  
    shape = RoundedCornerShape(8.dp),  
    border = BorderStroke(2.dp, MaterialTheme.colorscheme.surfaceVariant),  
    shadowElevation = 8.dp,
```

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```
        tonalElevation = 8.dp,  
    )
```

26.9.7 Preview

```
@Preview(showBackground=true, showSystemUi=true, name="My name")  
  
// dark mode  
@Preview(  
    showBackground = true,  
    widthDp = 320,  
    uiMode = UI_MODE_NIGHT_YES,  
    name = "GreetingPreviewDark"  
)
```

26.9.8 Column

```
Column(  
    verticalArrangement = Arrangement.Center,  
    horizontalAlignment = Alignment.CenterHorizontally,  
    modifier = modifier.padding(8.dp)  
)
```

26.9.9 LazyColumn

```
import androidx.compose.foundation.lazy.LazyColumn  
import androidx.compose.foundation.lazy.items  
  
@Composable  
private fun Greetings(  
    modifier: Modifier = Modifier,  
    names: List<String> = List(1000) { "$it" }  
) {  
    LazyColumn(modifier = modifier.padding(vertical = 4.dp)) {  
        items(items = names) { name ->  
            Greeting(name = name)  
        }  
    }  
}
```

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```
}  
}
```

26.9.10 Row

See [https://developer.android.com/reference/kotlin/androidx/compose/foundation/layout/package-summary#Row\(androidx.compose.ui.Modifier,androidx.compose.foundation.layout.Arrangement.Horizontal,androidx.compose.ui.Alignment.Vertical,kotlin.Function1\)](https://developer.android.com/reference/kotlin/androidx/compose/foundation/layout/package-summary#Row(androidx.compose.ui.Modifier,androidx.compose.foundation.layout.Arrangement.Horizontal,androidx.compose.ui.Alignment.Vertical,kotlin.Function1))

```
Row(modifier = Modifier.padding(all = 8.dp))  
  
Row(  
    modifier = Modifier.fillMaxSize(),  
    horizontalArrangement = Arrangement.Center,  
    verticalAlignment = Alignment.Top  
) {  
}  
  
Row(modifier = Modifier.padding(all = 8.dp)) {  
    Column(modifier = Modifier.weight(1f)) {  
        Text(text = "Hello")  
        Text(text = "World")  
    }  
  
    ElevatedButton(  
        onClick = {}  
    ) {  
        Text("Show more")  
    }  
}  
  
Row(  
    modifier = Modifier  
        .fillMaxSize()  
        .border(width = 2.dp, color = Color.Green)  
) {
```

26.9.11 LazyRow

```
LazyRow(  
    horizontalArrangement = Arrangement.spacedBy(8.dp)  
)  
  
@Composable  
fun AlignYourBodyRow(  
    modifier: Modifier = Modifier  
) {  
    LazyRow(  
        horizontalArrangement = Arrangement.spacedBy(8.dp),  
        contentPadding = PaddingValues(horizontal = 16.dp),  
        modifier = modifier  
    ) {  
        items(alignYourBodyData) { item ->  
            AlignYourBodyElement(item.drawable, item.text)  
        }  
    }  
}
```

26.9.12 Spacer

```
// within a row  
Spacer(modifier = Modifier.width(8.dp))  
  
// within a column  
Spacer(modifier = Modifier.height(4.dp))  
  
Spacer(Modifier.height(16.dp))  
  
Spacker(  
    modifier = Modifier  
        .matchParentSize()  
        .background(color = Color.Gray.copy(alpha = .7f))  
)
```

26.9.13 Text

```
Text (
    text = "Happy Birthday Sam!",
    fontSize = 100.sp,
    lineHeight = 116.sp
    textAlign = TextAlign.Center
)

Text (
    text = "From Emma",
    modifier = Modifier.padding(16.dp).align(alignment = Alignment.End)
)

Text(text = name, style = MaterialTheme.typography.headlineMedium)

Text (
    text = name,
    style = MaterialTheme.typography.headlineMedium.copy(
        fontWeight = FontWeight.ExtraBold
    )
)

Text(text = stringResource(R.string.ab1_inversions))

Text (
    "Hello compose",
    Modifier.background(Color.Magenta)
        .size(200.dp, 30.dp)
        .padding(5.dp)
        .alpha(0.5f)
        .align(Alignment.BottomEnd)
        .clickable {}
)
```

26.9.14 Button

```
Button(
    onclick = {}
) {
    Text("Show less")
}
```

26.9.15 IconButton

```
IconButton(onClick = {...}) {  
    Icon(Icons.Filled.Close, contentDescription = "Close")  
}
```

26.9.16 Checkbox

```
Checkbox(  
    checked = true,  
    onCheckedChange = {it->}  
)
```

26.9.17 Modifier

See <https://developer.android.com/jetpack/compose/modifiers>

```
modifier = Modifier.fillMaxSize()  
  
modifier = modifier.padding(8.dp)  
  
modifier = modifier.padding(vertical = 4.dp, horizontal = 8.dp)  
  
modifier = Modifier.padding(start = 4.dp, end = 8.dp)  
  
modifier = modifier.padding(bottom = 4.dp)  
  
modifier = modifier.size(40.dp)  
  
modifier = modifier.weight(1f)  
  
modifier = modifier.heightIn(min = 56.dp)  
  
modifier = Modifier.align(Arrangement.CenterVertically)  
  
modifier.clip(CircleShape)  
  
modifier.clickable {  
    // some code  
}  
  
// For a function, it'd better to set the last argument as modifier
```

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```
@Composable
fun MyApp(modifier: Modifier = Modifier) {}
```

26.9.18 Box

```
Box(
    modifier = Modifier.fillMaxSize(),
    contentAlignment = Alignment.Center,
)

Box(
    modifier = Modifier
        .padding(all = 2.dp)
        .size(56.dp)
        .background(color = Color.Magenta)
) {
}
```

26.9.19 Scaffold

```
Scaffold(
    topBar = {
        InfoHeader(...)
    },
    bottomBar = {
        Button(
            onClick = {...}
        ) {
            Text(...)
        }
    }
) {
    //
}
```


26.9.20 TextField

See [https://developer.android.com/reference/kotlin/androidx/compose/material3/package-summary#TextField\(androidx.compose.ui.text.input.TextFieldValue,kotlin.Function1,androidx.compose.ui.Modifier,kotlin.Boolean,kotlin.Boolean,androidx.compose.ui.text.TextStyle,kotlin.Function0,kotlin.Function0,kotlin.Function0,kotlin.Function0,kotlin.Function0,kotlin.Boolean,androidx.compose.ui.text.input.VisualTransformation,androidx.compose.foundation.text.KeyboardOptions,androidx.compose.foundation.text.KeyboardActions,kotlin.Boolean,kotlin.Int,kotlin.Int,androidx.compose.foundation.interaction.MutableInteractionSource,androidx.compose.ui.graphics.Shape,androidx.compose.material3.TextFieldColors\)](https://developer.android.com/reference/kotlin/androidx/compose/material3/package-summary#TextField(androidx.compose.ui.text.input.TextFieldValue,kotlin.Function1,androidx.compose.ui.Modifier,kotlin.Boolean,kotlin.Boolean,androidx.compose.ui.text.TextStyle,kotlin.Function0,kotlin.Function0,kotlin.Function0,kotlin.Function0,kotlin.Function0,kotlin.Boolean,androidx.compose.ui.text.input.VisualTransformation,androidx.compose.foundation.text.KeyboardOptions,androidx.compose.foundation.text.KeyboardActions,kotlin.Boolean,kotlin.Int,kotlin.Int,androidx.compose.foundation.interaction.MutableInteractionSource,androidx.compose.ui.graphics.Shape,androidx.compose.material3.TextFieldColors))

```
var text by rememberSaveable(stateSaver = TextFieldValue.Saver) {
    mutableStateOf(TextFieldValue("example", TextRange(0, 7)))
}

TextField(
    value = text,
    onChange = { text = it },
    label = { Text("Label") }
)

TextField(
    value = "",
    onChange = {},
    modifier = modifier
        .fillMaxWidth()
        .heightIn(min = 56.dp)
)
```

26.9.21 remember

```
// Note we use val here with "= remember"
val expanded = remember { mutableStateOf(false) }

expanded.value = !expanded.value

val extraPadding = if(expanded.value) 48.dp else 0.dp
```

```
// Note we use var here with "by remember"
var expanded by remember { mutableStateOf(false) }

expanded = !expanded

val extraPadding = if(expanded) 48.dp else 0.dp
```

```
var expanded by rememberSaveable { mutableStateOf(false) }
val extraPadding by animateDpAsState {
    if (expanded) 48.dp else 0.dp
}
```

26.9.22 Example 1

See <https://developer.android.com/codelabs/jetpack-compose-basics?continue=https%3A%2F%2Fdeveloper.android.com%2Fcourses%2Fpathways%2Fjetpack-compose-for-android-developers-1%23codelab-https%3A%2F%2Fdeveloper.android.com%2Fcodelabs%2Fjetpack-compose-basics#4>

Note how Surface is used.

```
class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            BasicsCodelabTheme {
                MyApp(modifier = Modifier.fillMaxSize())
            }
        }
    }
}

@Composable
fun MyApp(modifier: Modifier = Modifier) {
    Surface(
        modifier = modifier,
        color = MaterialTheme.colorScheme.background
    ) {
        Greeting("Android")
    }
}

@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Surface(color = MaterialTheme.colorScheme.primary) {
        Text(
            text = "Hello $name!",
            modifier = modifier.padding(24.dp)
        )
    }
}
```

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```

@Preview(showBackground = true)
@Composable
fun GreetingPreview() {
    BasicsCodelabTheme {
        MyApp()
    }
}

```

26.9.23 Example 2

See <https://developer.android.com/codelabs/jetpack-compose-basics?continue=https%3A%2F%2Fdeveloper.android.com%2Fcourses%2Fpathways%2Fjetpack-compose-for-android-developers-1%23codelab-https%3A%2F%2Fdeveloper.android.com%2Fcodelabs%2Fjetpack-compose-basics#7>

Note that we pass a lambda function `onContinueClicked` to change the state.

```

@Composable
fun MyApp(modifier: Modifier = Modifier) {

    var shouldShowOnboarding by remember { mutableStateOf(true) }

    Surface(modifier) {
        if (shouldShowOnboarding) {
            OnboardingScreen(onContinueClicked = { shouldShowOnboarding = false })
        } else {
            Greetings()
        }
    }
}

@Composable
fun OnboardingScreen(
    onContinueClicked: () -> Unit,
    modifier: Modifier = Modifier
) {

    Column(
        modifier = modifier.fillMaxSize(),
        verticalArrangement = Arrangement.Center,
        horizontalAlignment = Alignment.CenterHorizontally
    ) {

```

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```

        Text("Welcome to the Basics Codelab!")
        Button(
            modifier = Modifier
                .padding(vertical = 24.dp),
            onClick = onContinueClicked
        ) {
            Text("Continue")
        }
    }
}

@Preview(showBackground = true, widthDp = 320, heightDp = 320)
@Composable
fun OnboardingPreview() {
    BasicsCodelabTheme {
        OnboardingScreen(onContinueClicked = {}) // Do nothing on click.
    }
}

```

26.9.24 Example 3

See <https://developer.android.com/codelabs/jetpack-compose-basics?continue=https%3A%2F%2Fdeveloper.android.com%2Fcourses%2Fpathways%2Fjetpack-compose-for-android-developers-1%23codelab-https%3A%2F%2Fdeveloper.android.com%2Fcodelabs%2Fjetpack-compose-basics#11>

Change the colors.

```

// Color.kt

val Navy = Color(0xFF073042)
val Blue = Color(0xFF4285F4)
val LightBlue = Color(0xFFD7EFFE)
val Chartreuse = Color(0xFFE7FF7CF)

```

```

// Theme.kt

private val LightColorScheme = lightColorScheme(
    surface = Blue,
    onSurface = Color.White,
    primary = LightBlue,
    onPrimary = Navy
)

```

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```

)

private val DarkColorScheme = darkColorScheme(
    surface = Blue,
    onSurface = Navy,
    primary = Navy,
    onPrimary = Chartreuse
)

```

26.9.25 Example 4

```

@Composable
fun FavoriteCollectionCard(
    @DrawableRes drawable: Int,
    @StringRes text: Int,
    modifier: Modifier = Modifier
) {
    Surface(
        shape = MaterialTheme.shapes.medium,
        color = MaterialTheme.colorScheme.surfaceVariant,
        modifier = modifier
    ) {
        Row(
            verticalAlignment = Alignment.CenterVertically,
            modifier = Modifier.width(255.dp)
        ) {
            Image(
                painter = painterResource(drawable),
                contentDescription = null,
                contentScale = ContentScale.Crop,
                modifier = Modifier.size(80.dp)
            )
            Text(
                text = stringResource(text),
                style = MaterialTheme.typography.titleMedium,
                modifier = Modifier.padding(horizontal = 16.dp)
            )
        }
    }
}

@Preview(showBackground = true, backgroundColor = 0xFFFF5F0EE)

```

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```
@Composable
fun FavoriteCollectionCardPreview() {
    MySootheTheme {
        FavoriteCollectionCard(
            text = R.string.fc2_nature_meditations,
            drawable = R.drawable.fc2_nature_meditations,
            modifier = Modifier.padding(8.dp)
        )
    }
}
```

26.9.26 Example 5

scroll

```
@Composable
fun HomeScreen(modifier: Modifier = Modifier) {
    Column(
        modifier
            .verticalScroll(rememberScrollState())
    ) {
        Spacer(Modifier.height(16.dp))
        SearchBar(Modifier.padding(horizontal = 16.dp))
        HomeSection(title = R.string.align_your_body) {
            AlignYourBodyRow()
        }
        HomeSection(title = R.string.favorite_collections) {
            FavoriteCollectionsGrid()
        }
        Spacer(Modifier.height(16.dp))
    }
}
```

26.9.27 Example 6

```
@Composable
fun SingleChoiceQuestion(answers: List<Answer>) {
    Column {
        if(answers.isEmpty()) {
            Text("...")
        } else {
```

(continues on next page)

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```
        answers.forEach {answer ->
            Text("...")
        }
    }
}
```

26.9.28 Example 7

```
@Composable
fun MyAppTopAppBar(topAppBarText: String, onBackPressed: () -> Unit) {
    TopAppBar(
        title = {
            Text(
                text = topAppBarText,
                textAlign = TextAlign.Center,
                modifier = Modifier
                    .fillMaxSize()
                    .wrapContentSize(Alignment.Center)
            )
        },
        navigationIcon = {
            IconButton(onClick = onBackPressed) {
                Icon(
                    Icons.Filled.ArrowBack,
                    contentDescription = localizedString
                )
            }
        },
        // ...
    )
}
```

26.10 Samples

- <https://developer.android.com/courses/android-basics-compose/course>
 - <https://developer.android.com/courses/android-basics-compose/unit-2>
- <https://developer.android.com/modern-android-development>
- <https://developer.android.com/courses/pathways/android-basics-compose-unit-1-pathway-3>

- <https://developer.android.com/courses/pathways/jetpack-compose-for-android-developers-1>
- <https://github.com/android/compose-samples>
- <https://developer.android.com/courses/android-basics-compose/course>

26.11 Resources

```
import androidx.annotation.StringRes

data class Question(@StringRes val textResId: Int, val answer: Boolean)
```

Recommended to use @StringRes decoration, though it is not required.

To add a new id, use:

```
android:id="@+id/my_button"
```

To reference a string, use:

```
android:text="@string/my_string"
```

26.12 Release

See also

- <https://developer.android.com/build/building-commandline>
- https://developer.android.com/build/building-commandline#sign_commandline

```
cd android/SherpaOnnxTtsEngine
./gradlew assembleRelease
ls -lh app/build/outputs/apk/release/*.apk
```

```
-rw-r--r--@ 1 fangjun  staff   257M Mar 27 15:39 app/build/outputs/apk/release/app-
↳ release-unsigned.apk
```

```
which keytool
```

```
/Users/fangjun/software/jdk-19.0.1.jdk/Contents/Home/bin/keytool
```

```
keytool -genkey -v -keystore my-release-key.jks -keyalg RSA -keysize 2048 -validity_
↳ 10000 -alias my-alias
```



```

Enter keystore password: <enter the password, see my email>
Re-enter new password: <enter the password, see my email>
Enter the distinguished name. Provide a single dot (.) to leave a sub-component empty.
↳ or press ENTER to use the default value in braces.
What is your first and last name?
    [Unknown]: k2-fsa
What is the name of your organizational unit?
    [Unknown]: k2-fsa
What is the name of your organization?
    [Unknown]: k2-fsa
What is the name of your City or Locality?
    [Unknown]: Peking
What is the name of your State or Province?
    [Unknown]: Peking
What is the two-letter country code for this unit?
    [Unknown]: cn
Is CN=k2-fsa, OU=k2-fsa, O=k2-fsa, L=Peking, ST=Peking, C=cn correct?
    [no]: yes
Generating 2,048 bit RSA key pair and self-signed certificate (SHA384withRSA) with a
↳ validity of 10,000 days
    for: CN=k2-fsa, OU=k2-fsa, O=k2-fsa, L=Peking, ST=Unknown, C=cn
[Storing my-release-key.js]

```

```
ls -lh my-release-key
```

```
-rw-r--r--  1 fangjun  staff   2.7K Mar 27 16:10 my-release-key
```

```
file my-release-key
```

```
my-release-key: data
```

```
ls -l my-release-key
```

```
-rw-r--r--  1 fangjun  staff  2732 Mar 27 16:10 my-release-key
```

```
cd android/SherpaOnnxTtsEngine/app/build/outputs/apk/release
cp ../../../../my-release-key ./
ls -l app-release-unsigned.apk
```

```
-rw-r--r--@ 1 fangjun  staff 269073239 Mar 27 15:39 app-release-unsigned.apk
```

```
/Users/fangjun/software/my-android/build-tools/34.0.0/zipalign -v -p 4 ./app-release-  
↪unsigned.apk ./app-release-unsigned-aligned.apk
```

```
ls -l app-release-unsigned-aligned.apk
```

```
-rw-r--r--@ 1 fangjun  staff  269073239 Mar 27 16:16 app-release-unsigned-aligned.apk
```

```
/Users/fangjun/software/my-android/build-tools/34.0.0/apksigner sign --ks ./my-  
↪release-key --out app-release.apk ./app-release-unsigned-aligned.apk
```

```
Keystore password for signer #1: <enter the password for the keystore>
```

```
ls -l ./*.apk
```

```
-rw-r--r--@ 1 fangjun  staff  269073239 Mar 27 16:16 app-release-unsigned-aligned.apk  
-rw-r--r--@ 1 fangjun  staff  269073239 Mar 27 15:39 app-release-unsigned.apk  
-rw-r--r--@ 1 fangjun  staff  269181994 Mar 27 16:19 app-release.apk
```

```
which jarsigner
```

```
/Users/fangjun/software/jdk-19.0.1.jdk/Contents/Home/bin/jarsigner
```

```
jarsigner --verify -certs -verbose ./app-release.apk > a.txt
```

To use it with GitHub actions, see <https://github.com/marketplace/actions/sign-android-release>

```
openssl base64 < ./my-release-key.js | tr -d '\n' > ./my-release-key.base64.txt
```

Set the following secrets:

- ANDROID_SIGNING_KEY: the content base64.txt
- ANDROID_SIGNING_KEY_ALIAS: my-alias
- ANDROID_SIGNING_KEY_STORE_PASSWORD: see my email

26.13 Coroutine

- <https://github.com/JetBrains/kotlin/blob/e747a5a9a6070fd2ee0b2fd1fda1de3230c8307f/compiler/backend/src/org/jetbrains/kotlin/codegen/coroutines/coroutines-codegen.md>

26.14 Run CPP on Android

```
#include <iostream>
```

```
int main() {  
    std::cout << "main!\n";  
    return 0;  
}
```

```
if (CMAKE_VERSION VERSION_GREATER_EQUAL "4.0.0")  
    set(CMAKE_POLICY_VERSION_MINIMUM 3.5)  
endif()  
  
cmake_minimum_required(VERSION 3.13 FATAL_ERROR)  
  
project(qnn-demo)  
include_directories(/Users/fangjun/Downloads/qairt/2.39.0.250926/include/QNN/  
add_executable(qnn-demo ./main.cc)
```

```
#!/usr/bin/env bash  
set -ex  
  
dir=$PWD/build-android-arm64-v8a  
  
mkdir -p $dir  
cd $dir  
  
if [ -z $ANDROID_NDK ]; then  
    # For macOS, I have installed Android Studio, select the menu  
    # Tools -> SDK manager -> Android SDK  
    # and set "Android SDK location" to /Users/fangjun/software/my-android/sdk  
    ANDROID_NDK=/Users/fangjun/software/my-android/sdk/ndk/29.0.14206865  
fi  
  
if [ ! -d $ANDROID_NDK ]; then
```

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```

    echo Please set the environment variable ANDROID_NDK before you run this script
    exit 1
fi
$ANDROID_NDK/ndk-build --help

cmake -DCMAKE_TOOLCHAIN_FILE="$ANDROID_NDK/build/cmake/android.toolchain.cmake" \
      -DCMAKE_INSTALL_PREFIX=./install \
      -DANDROID_ABI="arm64-v8a" \
      -DANDROID_PLATFORM=android-28 ..
make

```

- 1. On PC, Run `./build-android-arm64-v8a.sh`
- 2. On PC, run `adb push build-android-arm64-v8a/qnn-demo /data/local/tmp/`
- 3. On PC, run `adb shell`
- 4. On phone, run:
`cd /data/local/tmp ./qnn-demo`

26.15 gradle

26.15.1 Install

```

brew install gradle

which grade

# /opt/homebrew/bin/gradle

```

```

cat $(which gradle)

#!/bin/bash
JAVA_HOME="${JAVA_HOME:-/opt/homebrew/opt/openjdk/libexec/openjdk.jdk/Contents/Home}"
exec "/opt/homebrew/Cellar/gradle/9.6.1/libexec/bin/gradle" "$@"

```

Content of `/opt/homebrew/Cellar/gradle/9.6.1/libexec/bin/gradle`:

```

#!/bin/sh

#
# Copyright © 2015 the original authors.

```

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```

#
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
#     https://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.
#
# SPDX-License-Identifier: Apache-2.0
#

#####
#
# Gradle start up script for POSIX generated by Gradle.
#
# Important for running:
#
# (1) You need a POSIX-compliant shell to run this script. If your /bin/sh is
#     noncompliant, but you have some other compliant shell such as ksh or
#     bash, then to run this script, type that shell name before the whole
#     command line, like:
#
#         ksh Gradle
#
#     Busybox and similar reduced shells will NOT work, because this script
#     requires all of these POSIX shell features:
#
#         * functions;
#         * expansions «$var», «${var}», «${var:-default}», «${var+SET}»,
#           «${var#prefix}», «${var%suffix}», and «$( cmd )»;
#         * compound commands having a testable exit status, especially «case»;
#         * various built-in commands including «command», «set», and «ulimit».
#
# Important for patching:
#
# (2) This script targets any POSIX shell, so it avoids extensions provided
#     by Bash, Ksh, etc; in particular arrays are avoided.
#
#     The "traditional" practice of packing multiple parameters into a

```

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```

#      space-separated string is a well documented source of bugs and security
#      problems, so this is (mostly) avoided, by progressively accumulating
#      options in "$@" , and eventually passing that to Java.
#
#      Where the inherited environment variables (DEFAULT_JVM_OPTS, JAVA_OPTS,
#      and GRADLE_OPTS) rely on word-splitting, this is performed explicitly;
#      see the in-line comments for details.
#
#      There are tweaks for specific operating systems such as AIX, CygWin,
#      Darwin, MinGW, and NonStop.
#
#      (3) This script is generated from the Groovy template
#      https://github.com/gradle/gradle/blob/
#      ↪3d91ce3b8caaf77ad09f381f43615b715b53f72c/platforms/jvm/plugins-application/src/main/
#      ↪resources/org/gradle/api/internal/plugins/unixStartScript.txt
#      within the Gradle project.
#
#      You can find Gradle at https://github.com/gradle/gradle/.
#
#####

# Attempt to set APP_HOME

# Resolve links: $0 may be a link
app_path=$0

# Need this for daisy-chained symlinks.
while
    APP_HOME=${app_path%"${app_path##*/}"} # leaves a trailing /; empty if no
    ↪leading path
do
    [ -h "$app_path" ]
done
    ls=$( ls -ld "$app_path" )
    link=${ls##* } -> ' '
    case $link in
        # (
        /*)  app_path=$link ;; # (
        *)   app_path=$APP_HOME$link ;;
    esac
done

# This is normally unused
# shellcheck disable=SC2034
APP_BASE_NAME=${0##*/}

```

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```

# Discard cd standard output in case $CDPATH is set (https://github.com/gradle/gradle/
↪issues/25036)
APP_HOME=$( cd -P "${APP_HOME:-./}.." > /dev/null && printf '%s\n' "$PWD" ) || exit

# Use the maximum available, or set MAX_FD != -1 to use that value.
MAX_FD=maximum

warn () {
    echo "$*"
} >&2

die () {
    echo
    echo "$*"
    echo
    exit 1
} >&2

# OS specific support (must be 'true' or 'false').
cygwin=false
msys=false
darwin=false
nonstop=false
case "$( uname )" in
    CYGWIN* )      cygwin=true ;;
    Darwin* )      darwin=true ;;
    MSYS* | MINGW* ) msys=true ;;
    NONSTOP* )     nonstop=true ;;
esac

# Determine the Java command to use to start the JVM.
if [ -n "$JAVA_HOME" ] ; then
    if [ -x "$JAVA_HOME/jre/sh/java" ] ; then
        # IBM's JDK on AIX uses strange locations for the executables
        JAVACMD=$JAVA_HOME/jre/sh/java
    else
        JAVACMD=$JAVA_HOME/bin/java
    fi
    if [ ! -x "$JAVACMD" ] ; then
        die "ERROR: JAVA_HOME is set to an invalid directory: $JAVA_HOME

```

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```

Please set the JAVA_HOME variable in your environment to match the
location of your Java installation."
    fi
else
    JAVACMD=java
    if ! command -v java >/dev/null 2>&1
    then
        die "ERROR: JAVA_HOME is not set and no 'java' command could be found in your_
↪PATH.

Please set the JAVA_HOME variable in your environment to match the
location of your Java installation."
    fi
fi

# Increase the maximum file descriptors if we can.
if ! "$cygwin" && ! "$darwin" && ! "$nonstop" ; then
    case $MAX_FD in #(
        max*)
            # In POSIX sh, ulimit -H is undefined. That's why the result is checked to_
↪see if it worked.
            # shellcheck disable=SC2039,SC3045
            MAX_FD=$( ulimit -H -n ) ||
                warn "Could not query maximum file descriptor limit"
        esac
        case $MAX_FD in #(
            '' | soft) :;; #(
            *)
                # In POSIX sh, ulimit -n is undefined. That's why the result is checked to_
↪see if it worked.
                # shellcheck disable=SC2039,SC3045
                ulimit -n "$MAX_FD" ||
                    warn "Could not set maximum file descriptor limit to $MAX_FD"
        esac
    fi

# Collect all arguments for the java command, stacking in reverse order:
#   * args from the command line
#   * the main class name
#   * -classpath
#   * -D...appname settings
#   * --module-path (only if needed)
#   * DEFAULT_JVM_OPTS, JAVA_OPTS, and GRADLE_OPTS environment variables.

```

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```

# For Cygwin or MSYS, switch paths to Windows format before running java
if "$cygwin" || "$msys" ; then
    APP_HOME=$( cygpath --path --mixed "$APP_HOME" )

    JAVACMD=$( cygpath --unix "$JAVACMD" )

    # Now convert the arguments - kludge to limit ourselves to /bin/sh
    for arg do
        if
            case $arg in
                -*)      false ;;
                /**)      t=${arg#/} t=/${t%%/*}
                           [ -e "$t" ] ;;
                *)        false ;;
            esac
        then
            arg=$( cygpath --path --ignore --mixed "$arg" )
        fi
        # Roll the args list around exactly as many times as the number of
        # args, so each arg winds up back in the position where it started, but
        # possibly modified.
        #
        # NB: a `for` loop captures its iteration list before it begins, so
        # changing the positional parameters here affects neither the number of
        # iterations, nor the values presented in `arg`.
        shift                    # remove old arg
        set -- "$@" "$arg"       # push replacement arg
    done
fi

# Add default JVM options here. You can also use JAVA_OPTS and GRADLE_OPTS to pass
# JVM options to this script.
DEFAULT_JVM_OPTS='"-Xmx64m" "-Xms64m"' "\"-javaagent:$APP_HOME/lib/agents/gradle-
instrumentation-agent-9.6.1.jar\""

# Collect all arguments for the java command:
#
# * DEFAULT_JVM_OPTS, JAVA_OPTS, and optsEnvironmentVar are not allowed to contain
#   shell fragments,
#
#   and any embedded shellness will be escaped.
#
# * For example: A user cannot expect ${Hostname} to be expanded, as it is an
#   environment variable and will be

```

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```

#      treated as '${Hostname}' itself on the command line.

set -- \
    "-Dorg.gradle.appname=$APP_BASE_NAME" \
    -jar "$APP_HOME/lib/gradle-gradle-cli-main-9.6.1.jar" \
    "$@"

# Stop when "xargs" is not available.
if ! command -v xargs >/dev/null 2>&1
then
    die "xargs is not available"
fi

# Use "xargs" to parse quoted args.
#
# With -n1 it outputs one arg per line, with the quotes and backslashes removed.
#
# In Bash we could simply go:
#
#   readarray ARGS < <( xargs -n1 <<<"$var" ) &&
#   set -- "${ARGS[@]}" "$@"
#
# but POSIX shell has neither arrays nor command substitution, so instead we
# post-process each arg (as a line of input to sed) to backslash-escape any
# character that might be a shell metacharacter, then use eval to reverse
# that process (while maintaining the separation between arguments), and wrap
# the whole thing up as a single "set" statement.
#
# This will of course break if any of these variables contains a newline or
# an unmatched quote.
#

eval "set -- $(
    printf '%s\n' "$DEFAULT_JVM_OPTS $JAVA_OPTS $GRADLE_OPTS" |
    xargs -n1 |
    sed ' s~[^-[:alnum:]+,./:=@_]~\\&~g; ' |
    tr '\n' ' '
)" "$@"

exec "$JAVACMD" "$@"

```

26.15.2 A book: Get going with gradle

A book <https://assets.tomgregory.com/get-going-with-gradle/get-going-with-gradle-book.pdf>

26.15.3 Basics

If a project uses gradle, run the following command to build it:

```
./gradlew build

# on windows, use
./gradlew.bat build
```

`./build.gradle.kts` in the project contains the build script

- `kts`: kotlin script
- DSL: domain specific language.

```
# show a list of available tasks
./gradlew tasks

# show version
./gradlew --version
```

On macos, it prints:

```
./gradlew --version

-----
Gradle 9.1.0
-----

Build time:      2025-09-18 13:05:56 UTC
Revision:        e45a8dbf2470c2e2474ccc25be9f49331406a07e

Kotlin:          2.2.0
Groovy:          4.0.28
Ant:             Apache Ant(TM) version 1.10.15 compiled on August 25 2024
Launcher JVM:    25.0.1 (Oracle Corporation 25.0.1+8-LTS-27)
Daemon JVM:      Compatible with Java 21, Azul Zulu, nativeImageCapable=false (from
↳ gradle/gradle-daemon-jvm.properties)
OS:              Mac OS X 15.5 aarch64
```

On windows, it prints:

```
.\gradlew.bat --version
```

```
-----  
Gradle 9.6.1  
-----
```

```
Build time: 2026-06-26 14:25:50 UTC
```

```
Revision: 309d128bd9fe8c0b71311878fc660b9cbaa07c51
```

```
Kotlin: 2.3.21
```

```
Groovy: 4.0.32
```

```
Ant: Apache Ant(TM) version 1.10.17 compiled on April 6 2026
```

```
Launcher JVM: 21.0.7 (Oracle Corporation 21.0.7+8-LTS-245)
```

```
Daemon JVM: D:\software\jdk-21.0.7 (no Daemon JVM specified, using current JavaU  
↪home)
```

```
OS: Windows 10 10.0 amd64
```

27.1 Install

```
git clone --depth 1 https://github.com/qemu/qemu
cd qemu
```

27.1.1 qemu-arm

Refer to <https://github.com/Tencent/ncnn/blob/master/.github/workflows/linux-arm-cpu-gcc.yml>

```
# To generate only qemu-arm
./configure --prefix=/ceph-fj/fangjun/software/qemu/ --target-list=arm-linux-user,
↪aarch64-linux-user --disable-system
make -j10

# It genertates the executable: ./build/qemu-arm and ./build/qemu-aarch64

# If we run `make install`, it will generate
#
# /ceph-fj/fangjun/software/qemu/bin/qemu-arm
# /ceph-fj/fangjun/software/qemu/bin/qemu-aarch64
#
# Add it to PATH and use it!
```

To run it, we have to download some cross-compile toochain, e.g,

Go to <https://developer.arm.com/tools-and-software/open-source-software/developer-tools/gnu-toolchain/gnu-a/downloads/8-3-2019-03> to download the toolchain.

```
mkdir /ceph-fj/fangjun/software
cd /ceph-fj/fangjun/software
tar xvf /path/to/gcc-arm-8.3-2019.03-x86_64-arm-linux-gnueabihf.tar.xz
```

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```
export PATH=/ceph-fj/fangjun/software/gcc-arm-8.3-2019.03-x86_64-arm-linux-gnueabi/
↪bin:$PATH
```

If we have built an executable using the above toolchain, we can run it with `qemu-arm`:

```
./build/qemu-arm /path/to/sherpa-ncnn
```

It throws the following error:

```
qemu-arm: Unable to reserve 0xfffff000 bytes of virtual address space at
0x1000 (Success) for use as guest address space (check your virtual memory
ulimit setting, min_mmap_addr or reserve less using -R option)
```

We can use

```
./build/qemu-arm -B 0x10000000 /path/to/sherpa-ncnn
```

which throws the following new error:

```
(py38) kuangfangjun:qemu$ find /ceph-fj/fangjun/software/gcc-arm-8.3-2019.03-x86_64-
↪arm-linux-gnueabi/ -name "ld-linux-armhf.so.3"
/ceph-fj/fangjun/software/gcc-arm-8.3-2019.03-x86_64-arm-linux-gnueabi/arm-linux-
↪gnueabi/libc/lib/ld-linux-armhf.so.3
(py38) kuangfangjun:qemu$ export QEMU_LD_PREFIX=/ceph-fj/fangjun/software/gcc-arm-8.3-
↪2019.03-x86_64-arm-linux-gnueabi/arm-linux-gnueabi/libc
```

Now we can restart:

```
./build/qemu-arm -B 0x10000000 /path/to/sherpa-ncnn
```

```
export PATH=/ceph-fj/fangjun/software/qemu/bin:$PATH
export QEMU_LD_PREFIX=/ceph-fj/fangjun/software/gcc-arm-8.3-2019.03-x86_64-arm-linux-
↪gnueabi/arm-linux-gnueabi/libc

qemu-arm -B 0x10000000 ./build-arm-linux-gnueabi/bin/sherpa-ncnn \
./sherpa-ncnn-2022-09-05/tokens.txt \
./sherpa-ncnn-2022-09-05/bar/encoder_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.
↪param \
./sherpa-ncnn-2022-09-05/bar/encoder_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.bin \
./sherpa-ncnn-2022-09-05/bar/decoder_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.
↪param \
./sherpa-ncnn-2022-09-05/bar/decoder_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.bin \
./sherpa-ncnn-2022-09-05/bar/joiner_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.param
```

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```

↪ \
./sherpa-ncnn-2022-09-05/bar/joiner_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.bin \
./test_wavs/1221-135766-0002.wav

```

27.1.2 qemu-aarch64

```

wget https://releases.linaro.org/components/toolchain/binaries/latest-7/aarch64-linux-
↪gnu/gcc-linaro-7.5.0-2019.12-x86_64_aarch64-linux-gnu.tar.xz
cd /ceph-fj/fangjun/software
tar xvf /path/to/gcc-linaro-7.5.0-2019.12-x86_64_aarch64-linux-gnu.tar.xz

```

```

export PATH=/ceph-fj/fangjun/software/qemu/bin:$PATH
export QEMU_LD_PREFIX=/ceph-fj/fangjun/software/gcc-linaro-7.5.0-2019.12-x86_64_
↪aarch64-linux-gnu/aarch64-linux-gnu/libc

qemu-aarch64 -B 0x10000000 ./build-aarch64-linux-gnu/bin/sherpa-ncnn \
./sherpa-ncnn-2022-09-05/tokens.txt \
./sherpa-ncnn-2022-09-05/bar/encoder_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.
↪param \
./sherpa-ncnn-2022-09-05/bar/encoder_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.bin \
./sherpa-ncnn-2022-09-05/bar/decoder_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.
↪param \
./sherpa-ncnn-2022-09-05/bar/decoder_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.bin \
./sherpa-ncnn-2022-09-05/bar/joiner_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.param_
↪\
./sherpa-ncnn-2022-09-05/bar/joiner_jit_trace-v2-iter-468000-avg-16-pnnx.ncnn.bin \
./test_wavs/1221-135766-0002.wav

```


28.1 Basics

28.1.1 Extract part of a wave

```
# offset 2 seconds, length 0.195 seconds  
sox in.mp3 out.mp3 trim 2 0.195
```

28.1.2 Fix broken waves

```
sox --ignore-length corrupted.wav fixed.wav
```

28.1.3 pad

```
# add 2 seconds of silence at the begining and 3 seconds of silence  
# at the end.  
sox in.wav out.wav pad 2 3
```

28.1.4 raw

```
sox -t raw -r 16000 -e signed -b 16 -c 1 test.raw -r 48000 out.wav
```

28.1.5 Generate

```
sox -r 16000 -n -b 16 -c 1 a.wav synth 10 sin 1000
```

28.1.6 remix

Split multi-channel wave to single channels.

```
sox stereo_inputfile.wav leftchannel.wav remix 1
sox stereo_inputfile.wav rightchannel.wav remix 2
```

28.2 base64

```
import base64
encode_string = base64.b64encode(open("audio.wav", "rb").read())
wav_file = open("temp.wav", "wb")
decode_string = base64.b64decode(encode_string)
wav_file.write(decode_string)
```

29.1 Install

Refer to <https://mnn-docs.readthedocs.io/en/latest/compile/engine.html>

```
./schema/generate.sh
mkdir build
cd build
cmake -DMNN_BUILD_CONVERTER=ON -DMNN_SUPPORT_DEPRECATED_OP=OFF -DMNN_BUILD_TORCH=ON .
↪.
make -j 10
```

Note: Use `CPLUS_INCLUDE_PATH` and `C_INCLUDE_PATH` to add additional paths for searching.

29.1.1 Python

See <https://mnn-docs.readthedocs.io/en/latest/compile/pymnn.html>

To build a python package, use:

```
cd pymnn/pip_package/
python3 ./build_deps.py
python3 setup.py bdist_wheel --version 2.1.1
# We can get the version from include/MNN/MNNDefine.h
# or we can give it an arbitrary version string.
pip install ./dist/MNN-2.1.1-cp38-cp38-linux_x86_64.whl
cd $HOME
python3 -c "import MNN; print(dir(MNN))"
```

29.2 Hello

Create a simple torchscript model:

Listing 1: ./code/hello/ex1.py

```
1  #!/usr/bin/env python3
2
3  import torch
4
5
6  class Foo(torch.nn.Module):
7      def forward(self, x):
8          return torch.nn.functional.relu(x)
9
10
11 f = Foo()
12 x = torch.rand(2)
13 m = torch.jit.trace(f, x)
14 m.save("ex1.pt")
15 print(m.graph)
```

```
MNNConvert -f TORCH --modelFile ./ex1.pt --MNNModel ex1.mnn --bizCode MNN
```

It prints:

```
Start to Convert Other Model Format To MNN Model...
Start to Optimize the MNN Net...
inputTensors : [ x.1, ]
outputTensors: [ 2, ]
Converted Success!
```

We can use:

```
netron ex1.mnn --port 6006
```

to view it:

```
Serving 'ex1.mnn' at http://localhost:6006
```

To convert the model to json, use:

```
MNNDump2Json ./ex1.mnn ./ex1.json
```

which generates:

Listing 2: ./code/hello/ex1.json

```

1 { "bizCode": "MNN", "extraInfo":
2 { "version": "2.1.1" }
3 , "oplists":
4 [
5 { "main_type": "Input", "main":
6 { "dtype": "DT_FLOAT", "dformat": "NCHW" }
7 , "name": "x.1", "outputIndexes":
8 [ 0 ]
9 , "type": "Input", "defaultDimentionFormat": "NHWC" }
10 ,
11 { "inputIndexes":
12 [ 0 ]
13 , "main_type": "Relu", "main":
14 { "slope": 0.0 }
15 , "name": "2", "outputIndexes":
16 [ 1 ]
17 , "type": "ReLU", "defaultDimentionFormat": "NCHW" }
18 ]
19 , "outputName":
20 [ "2" ]
21 , "preferForwardType": "CPU", "sourceType": "TORCH", "tensorName":
22 [ "x.1", "2" ]
23 , "tensorNumber": 0, "usage": "INFERENCE", "mnn_uuid": "d24b26b1-14c8-42cb-9ce1-
↪2a9b867714e9" }

```

To convert ex1.json back to a .mnn file, use:

```
MNNRevert2Buffer ex1.json ex1.mnn
```

To show the information of the model:

```
MNNConvert --framework MNN --modelFile ./ex1.mnn --info
```

It prints:

```

Model default dimensionFormat is NCHW
Model Inputs:
[ x.1 ]: dimensionFormat: NCHW, size: [ ], type is float
Model Outputs:
[ 2 ]
Model Version: 2.1.1

```

To run it with MNN in Python:

Listing 3: ./code/hello/test-ex1-mnn.py

```

1  #!/usr/bin/env python3
2
3  import numpy as np
4  import MNN
5  import torch
6
7
8  def main():
9      interpreter = MNN.Interpreter("ex1.mnn")
10
11      config = {}
12      config["precision"] = "low" # low, high, normal
13      config["backend"] = "CPU"
14      # config["thread"] = 1
15      session = interpreter.createSession(config)
16
17      # a = torch.tensor([1, -2.5, 3.2], dtype=torch.float32)
18      # b = a.numpy()
19
20      input_tensor = interpreter.getSessionInput(session)
21      interpreter.resizeTensor(input_tensor, (3,))
22      interpreter.resizeSession(session)
23      print("input_tensor.getShape()", input_tensor.getShape())
24
25      input_data = MNN.expr.const(
26          np.array([1, -2.5, 3.2], dtype=np.float32),
27          (3,),
28          MNN.expr.NCHW,
29          MNN.expr.float,
30      )
31      input_tensor.copyFrom(MNN.Tensor(input_data))
32      print("input_tensor.getNumpyData()", input_tensor.getNumpyData())
33
34      interpreter.runSession(session)
35
36      output_tensor = interpreter.getSessionOutput(session)
37      print("output_tensor.getNumpyData()", output_tensor.getNumpyData())
38
39      output_data = MNN.Tensor(
40          output_tensor.getShape(), MNN.Halide_Type_Float, MNN.Tensor_DimensionType_
41      ↪Caffe
42      )

```

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```
42     output_tensor.copyToHostTensor(output_data)
43     print("output_data.getNumpyData()", output_data.getNumpyData())
44
45
46 if __name__ == "__main__":
47     main()
```

It prints:

```
input_tensor.getShape() (3,)
input_tensor.getNumpyData() [ 1. -2.5  3.2]
output_tensor.getNumpyData() [1.  0.  3.2]
output_data.getNumpyData() [1.  0.  3.2]
```


30.1 References

- <https://www.intel.com/content/www/us/en/docs/intrinsics-guide/index.html>
All APIs for Intel intrinsics with examples
- <https://developer.arm.com/architectures/instruction-sets/intrinsics>
All APIs for ARM intrinsics.
- 15-418/15-618: Parallel Computer Architecture and Programming, Spring 2018: Schedule
<https://www.cs.cmu.edu/afs/cs.cmu.edu/academic/class/15418-s18/www/schedule.html>
- How to Write Fast Code 18-645 (CMU, ECE)
<https://users.ece.cmu.edu/~pueschel/teaching/18-645-CMU-spring08/course.html>
- <http://spcl.inf.ethz.ch/Teaching/2018-dphpc/lectures/lecture8-simd.pdf>
- Crunching Numbers with AVX and AVX2

SSE:

- 128-bit
- `__m128`, `__m128d`
- `_mm_load_ps`, `_mm_add_pd`

AVX:

- 256-bit
- `__m256`, `__mm256d`
- `_mm256_load_ps`, `_mm256_add_pd`

AVX512:

- 512-bit

30.2 Headers

- `mmintrin.h` MMX
- `xmmintrin.h` SSE
- `emmintrin.h`, SSE2
- `pmmmintrin.h`, SSE3
- `tmmintrin.h`, SSSE3
- `smmintrin.h`, SSE4.1
- `nmmintrin.h`, SSE4.2
- `ammintrin.h`, SSE4A
- `wmmmintrin.h`, AES
- `immintrin.h`, AVX

30.3 SSE

The header file is `xmmintrin.h`, it is in `/usr/lib/gcc/x86_64-linux-gnu/7/include`. There is also a file `mmintrin.h`.

30.4 avx

SSE4 data types:

- `__m128`, 4 floats
- `__m128d`, 2 doubles
- `__m128i`, it depends, can be 16 8-bit, 8 16-bit, 4 32-bit, 2 64-bit

AVX2 data types:

- `__m256`, 8 floats
- `__m256d`, 4 doubles
- `__m256i`, 32 8-bit, 16 16-bit, 8 32-bit, 4 64-bit

Listing 1: ./code/avx/main.cc

```

1  #include <cassert>
2  #include <immintrin.h>
3
4  // ps means packed single precision
5  static void TestLoadStore() {
6      alignas(16) float a[4] = {1, 2, 3, 4};
7      alignas(16) float b[4];
8      __m128 f = _mm_load_ps(a);
9      // f = _mm_loadu_ps(a); // if a not aligned
10     _mm_store_ps(b, f);
11     // _mm_storeu_ps(b, f); // if b is not aligned
12     assert(b[0] == a[0]);
13     assert(b[1] == a[1]);
14     assert(b[2] == a[2]);
15     assert(b[3] == a[3]);
16
17     // set manually
18     f = _mm_set_ps(a[3], a[2], a[1], a[0]);
19     _mm_store_ps(b, f);
20     assert(b[0] == a[0]);
21     assert(b[1] == a[1]);
22     assert(b[2] == a[2]);
23     assert(b[3] == a[3]);
24
25     // for double
26     alignas(32) double k[4] = {1, 2, 3, 4};
27     __m256d d = _mm256_load_pd(k);
28     // d = _mm256_loadu_pd(k); // if k is not aligned
29     alignas(32) double m[4];
30     _mm256_store_pd(m, d);
31     // _mm256_storeu_pd(m, d); // if m is not aligned
32     assert(m[0] == k[0]);
33     assert(m[1] == k[1]);
34     assert(m[2] == k[2]);
35     assert(m[3] == k[3]);
36
37     d = _mm256_set_pd(k[3], k[2], k[1], k[0]);
38     _mm256_store_pd(m, d);
39     assert(m[0] == k[0]);
40     assert(m[1] == k[1]);
41     assert(m[2] == k[2]);
42     assert(m[3] == k[3]);
43 }

```

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```
44
45 static void TestLoadStore1() {
46     float a = 10;
47     float b[4];
48     __m128 f = _mm_load_ps1(&a);
49     _mm_store_ps(b, f);
50     assert(b[0] == a);
51     assert(b[1] == a);
52     assert(b[2] == a);
53     assert(b[3] == a);
54 }
55
56 static void TestAdd() {
57     float a[4] = {1, 2, 3, 4};
58     float b[4] = {10, 20, 30, 40};
59     __m128 f = _mm_load_ps(a);
60     __m128 g = _mm_load_ps(b);
61     __m128 h = _mm_add_ps(f, g);
62     float c[4];
63     _mm_store_ps(c, h);
64     assert(c[0] == a[0] + b[0]);
65     assert(c[1] == a[1] + b[1]);
66     assert(c[2] == a[2] + b[2]);
67     assert(c[3] == a[3] + b[3]);
68 }
69
70 static void AddIndex1(double *x, int32_t n) {
71     for (int32_t i = 0; i < n; ++i) {
72         x[i] = x[i] + i;
73     }
74 }
75
76 // assume n % 4 == 0
77 static void AddIndex2(double *x, int32_t n) {
78     assert(n % 4 == 0);
79     __m256d index, x_vec;
80     for (int32_t i = 0; i < n; i += 4) {
81         x_vec = _mm256_load_pd(x + i);
82         // x_vec[0] = x[i]
83         // x_vec[1] = x[i+1]
84         // x_vec[2] = x[i+2]
85         // x_vec[3] = x[i+3]
86
```

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```
87     index = _mm256_set_pd(i + 3, i + 2, i + 1, i);
88     // index[0] = i
89     // index[1] = i+1
90     // index[2] = i+2
91     // index[3] = i+3
92
93     x_vec = _mm256_add_pd(x_vec, index);
94     // x_vec[0] = x_vec[0] + index[0]
95     // x_vec[1] = x_vec[1] + index[1]
96     // x_vec[2] = x_vec[2] + index[2]
97     // x_vec[3] = x_vec[3] + index[3]
98
99     _mm256_store_pd(x + i, x_vec);
100    // (x+i)[0] = x_vec[0]
101    // (x+i)[1] = x_vec[1]
102    // (x+i)[2] = x_vec[2]
103    // (x+i)[3] = x_vec[3]
104    }
105 }
106
107 static void TestAddIndex() {
108     alignas(32) double a[64];
109     alignas(32) double b[64];
110     for (int32_t i = 0; i != 64; ++i) {
111         a[i] = b[i] = i;
112     }
113     AddIndex1(a, 64);
114     AddIndex2(b, 64);
115     for (int32_t i = 0; i != 64; ++i) {
116         assert(a[i] == b[i]);
117     }
118 }
119
120 int main() {
121     TestLoadStore();
122     TestLoadStore1();
123     TestAdd();
124     TestAddIndex();
125     return 0;
126 }
```


31.1 Install

```
git clone https://github.com/chriskohlhoff/asio/  
cd asio/asio  
./autogen.sh  
./configure --prefix=/ceph-fj/fangjun/software/asio  
make -j 10  
make install
```

It will create `include` and `lib/pkgconfig/asio.pc` inside `/ceph-fj/fangjun/software/asio`. Note that it is a header only library.

31.2 Hello

<https://think-async.com/Asio/asio-1.24.0/doc/asio/tutorial/tuttimer1.html>

Listing 1: `./code/Makefile`

```
1 CXXFLAGS := -I /ceph-fj/fangjun/software/asio/include -pthread  
2  
3 srcs := $(shell find . -type f -name "*.cc" -printf "%f ")  
4 targets :=$(srcs:%.cc=%)  
5  
6 all: $(targets)  
7  
8 %: %.cc  
9     $(CXX) $(CXXFLAGS) -o $@ $<  
10  
11  
12 .PHONY: clean
```

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```
13 clean:
14     $(RM) $(targets)
```

Listing 2: ./code/1-timer-synchronous.cc

```
1  #include "asio.hpp"
2  #include <iostream>
3
4  int main() {
5      asio::io_context io;
6      std::cout << "sleep for 1 second\n";
7      asio::steady_timer t(io, asio::chrono::seconds(1));
8      t.wait();
9      std::cout << "hello world\n";
10     return 0;
11 }
```

Listing 3: ./code/2-timer-asynchronous.cc

```
1  #include "asio.hpp"
2  #include <iostream>
3
4  void print(const asio::error_code &) { std::cout << "done\n"; }
5
6  int main() {
7      asio::io_context io;
8      std::cout << "sleep for 1 seconds\n";
9      asio::steady_timer t(io, asio::chrono::seconds(1));
10
11     t.async_wait(&print);
12     t.async_wait(&print);
13     io.run();
14     return 0;
15 }
```

Listing 4: ./code/3-timer-async-bind-argument.cc

```
1  #include "asio.hpp"
2  #include <iostream>
3
4  void print(const asio::error_code &ec, asio::steady_timer *t, int *counter) {
5      if (*counter < 3) {
6          std::cout << *counter << "\n";
```

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```

7      *counter += 1;
8      t->expires_at(t->expiry() + asio::chrono::seconds(1));
9      t->async_wait(
10         [t, counter](const asio::error_code &e) { print(e, t, counter); });
11     }
12 }
13
14 int main() {
15     asio::io_context io;
16     asio::steady_timer t(io, asio::chrono::seconds(1));
17     int counter = 0;
18
19     // use a lambda to pass extra parameters
20     t.async_wait(
21         [&t, &counter](const asio::error_code &ec) { print(ec, &t, &counter); });
22     io.run();
23     return 0;
24 }

```

Listing 5: ./code/4-timer-async-member-function.cc

```

1  #include "asio.hpp"
2  #include <iostream>
3
4  class printer {
5  public:
6      printer(asio::io_context &io)
7          : t_(io, asio::chrono::seconds(1)), counter_(0) {
8          t_.async_wait([this](const asio::error_code &ec) { print(ec); });
9      }
10
11     void print(const asio::error_code &ec) {
12         if (counter_ < 6) {
13             std::cout << counter_ << "\n";
14             counter_ += 1;
15             t_.async_wait([this](const asio::error_code &e) { this->print(e); });
16         }
17     }
18
19 private:
20     asio::steady_timer t_;
21     int counter_;
22 };

```

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```

23 int main() {
24     asio::io_context io;
25     printer p(io);
26     io.run();
27     return 0;
28 }

```

Listing 6: ./code/5-timer-async-thread-non-synchronization.cc

```

1  #include "asio.hpp"
2  #include <iostream>
3  #include <thread>
4
5  class printer {
6  public:
7      printer(asio::io_context &io)
8          : t1_(io, asio::chrono::seconds(1)), t2_(io, asio::chrono::seconds(1)) {
9          t1_.async_wait([this](const asio::error_code &e) { print1(e); });
10         t2_.async_wait([this](const asio::error_code &e) { print2(e); });
11     }
12
13     void print1(const asio::error_code &ec) {
14         if (counter_ < 8) {
15             std::cout << "print1 id: " << std::this_thread::get_id() << ", counter "
16                 << counter_ << "\n";
17             counter_ += 1;
18             t1_.expires_at(t1_.expiry() + asio::chrono::seconds(1));
19             t1_.async_wait([this](const asio::error_code &e) { print1(e); });
20         }
21     }
22
23     void print2(const asio::error_code &ec) {
24         if (counter_ < 8) {
25             std::cout << "print2 id: " << std::this_thread::get_id() << ", counter "
26                 << counter_ << "\n";
27             counter_ += 1;
28             t2_.expires_at(t2_.expiry() + asio::chrono::seconds(1));
29             t2_.async_wait([this](const asio::error_code &e) { print2(e); });
30         }
31     }
32
33 private:
34     asio::steady_timer t1_;

```

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```

35     asio::steady_timer t2_;
36     int counter_ = 0;
37 };
38
39 int main() {
40     asio::io_context io;
41     printer p(io);
42     std::thread t([&io]() { io.run(); }); // NOTE: print1() and print2() can run
43                                           // either in t or in the main thread
44     io.run();
45     t.join();
46 }
47
48 #if 0
49 (py38) kuangfangjun:code$ ./5-timer-async-thread-non-synchronization
50 print1 id: print2 id: 140164266059584, counter 0
51 140164247860992, counter 1
52 print1 id: 140164266059584, counter 2
53 print2 id: 140164247860992, counter 3
54 print1 id: 140164266059584, counter 4
55 print2 id: 140164247860992, counter 5
56 print1 id: 140164266059584, counter 6
57 print2 id: 140164247860992, counter 7
58 (py38) kuangfangjun:code$ ./5-timer-async-thread-non-synchronization
59 print2 id: print1 id: 140191018206976, counter 140191036405568, counter 0
60 0
61 print1 id: 140191018206976, counter print2 id: 2
62 140191036405568, counter 3
63 print1 id: 140191018206976, counter 4
64 print2 id: 140191036405568, counter 5
65 print1 id: 140191018206976, counter 6
66 print2 id: 140191036405568, counter 7
67 #endif

```

Listing 7: ./code/6-timer-async-thread-with-synchronization.cc

```

1  #include "asio.hpp"
2  #include <iostream>
3  #include <thread>
4
5  class printer {
6  public:
7      printer(asio::io_context &io)

```

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```

8      : strand_(asio::make_strand(io)), t1_(io, asio::chrono::seconds(1)),
9      t2_(io, asio::chrono::seconds(1)) {
10     t1_.async_wait(asio::bind_executor(
11         strand_, [this](const asio::error_code &e) { print1(e); }));
12     t2_.async_wait(asio::bind_executor(
13         strand_, [this](const asio::error_code &e) { print2(e); }));
14 }
15
16 void print1(const asio::error_code &ec) {
17     if (counter_ < 8) {
18         std::cout << "print1 id: " << std::this_thread::get_id() << ", counter "
19             << counter_ << "\n";
20         counter_ += 1;
21         t1_.expires_at(t1_.expiry() + asio::chrono::seconds(1));
22         t1_.async_wait([this](const asio::error_code &e) { print1(e); });
23     }
24 }
25
26 void print2(const asio::error_code &ec) {
27     if (counter_ < 8) {
28         std::cout << "print2 id: " << std::this_thread::get_id() << ", counter "
29             << counter_ << "\n";
30         counter_ += 1;
31         t2_.expires_at(t2_.expiry() + asio::chrono::seconds(1));
32         t2_.async_wait([this](const asio::error_code &e) { print2(e); });
33     }
34 }
35
36 private:
37     asio::strand<asio::io_context::executor_type> strand_;
38     asio::steady_timer t1_;
39     asio::steady_timer t2_;
40     int counter_ = 0;
41 };
42
43 int main() {
44     asio::io_context io;
45     printer p(io);
46     std::thread t([&io]() { io.run(); }); // NOTE: print1() and print2() can run
47                                         // either in t or in the main thread
48     io.run();
49     t.join();
50 }

```

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```

51 #if 0
52 (py38) kuangfangjun:code$ ./6-timer-async-thread-with-synchronization
53 print1 id: 140648598263616, counter 0
54 print2 id: 140648598263616, counter 1
55 print1 id: 140648580065024, counter 2
56 print2 id: 140648598263616, counter 3
57 print1 id: 140648580065024, counter 4
58 print2 id: 140648598263616, counter 5
59 print1 id: 140648580065024, counter 6
60 print2 id: 140648598263616, counter 7
61 (py38) kuangfangjun:code$ ./6-timer-async-thread-with-synchronization
62 print1 id: 139671509145408, counter 0
63 print2 id: 139671509145408, counter 1
64 print1 id: 139671490946816, counter 2
65 print2 id: 139671509145408, counter 3
66 print1 id: 139671490946816, counter 4
67 print2 id: 139671509145408, counter 5
68 print1 id: 139671490946816, counter 6
69 print2 id: 139671509145408, counter 7
70 #endif

```

Listing 8: ./code/7-daytime-sync-client.cc

```

1  #include "asio.hpp"
2  #include <array>
3  #include <iostream>
4
5  using asio::ip::tcp;
6  int main(int argc, char *argv[]) {
7      try {
8          if (argc != 2) {
9              std::cerr << "Usage: client <host>" << std::endl;
10             return 1;
11         }
12
13         asio::io_context io;
14         tcp::resolver resolver(io);
15
16         tcp::resolver::results_type endpoints =
17             resolver.resolve(argv[1], "daytime");
18
19         tcp::socket socket(io);
20         asio::connect(socket, endpoints);

```

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```

21
22     for (;;) {
23         std::array<char, 128> buf;
24         asio::error_code ec;
25         size_t len = socket.read_some(asio::buffer(buf), ec);
26         if (ec == asio::error::eof) {
27             break;
28         } else if (ec) {
29             throw asio::system_error(ec);
30         }
31         std::cout.write(buf.data(), len);
32     }
33 } catch (std::exception &e) {
34     std::cerr << "exception: " << e.what() << "\n";
35 }
36 return 0;
37 }

```

Listing 9: ./code/8-daytime-sync-server.cc

```

1  #include "asio.hpp"
2  #include <ctime>
3  #include <iostream>
4  #include <string>
5
6  using asio::ip::tcp;
7  std::string make_daytime_string() {
8      using namespace std;
9      time_t now = time(0);
10     return ctime(&now);
11 }
12
13 int main() {
14     try {
15         asio::io_context io;
16         // need sudo permission to bind port 13
17         tcp::acceptor acceptor(io, tcp::endpoint(tcp::v4(), 13));
18         std::cout << "started\n";
19         for (;;) {
20             tcp::socket socket(io);
21             acceptor.accept(socket);
22
23             auto message = make_daytime_string();

```

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```

24     asio::error_code ec;
25     asio::write(socket, asio::buffer(message), ec);
26 }
27 } catch (std::exception &e) {
28     std::cout << "Exception: " << e.what() << "\n";
29 }
30 return 0;
31 }

```

Listing 10: ./code/9-daytime-async-server.cc

```

1  #include "asio.hpp"
2  #include <ctime>
3  #include <iostream>
4  #include <memory>
5
6  using asio::ip::tcp;
7
8  std::string make_daytime_string() {
9      using namespace std;
10     time_t now = time(0);
11     return ctime(&now);
12 }
13
14 class tcp_connection : public std::enable_shared_from_this<tcp_connection> {
15 public:
16     using pointer = std::shared_ptr<tcp_connection>;
17
18     static pointer create(asio::io_context &io) {
19         return pointer(new tcp_connection(io));
20         // return std::make_shared<tcp_connection>(io);
21     }
22
23     tcp::socket &socket() { return socket_; }
24
25     void start() {
26         message_ = make_daytime_string();
27         auto p = shared_from_this();
28         asio::async_write(
29             socket_, asio::buffer(message_),
30             [p](const asio::error_code &ec, size_t bytes_transferred) {
31                 p->handle_write(ec, bytes_transferred);
32             });

```

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```

33     }
34
35 private:
36     tcp_connection(asio::io_context &io) : socket_(io) {}
37
38     void handle_write(const asio::error_code & /*ec*/,
39                      size_t /*bytes_transferred*/) {}
40
41     tcp::socket socket_;
42     std::string message_;
43 };
44
45 class tcp_server {
46 public:
47     tcp_server(asio::io_context &io)
48         : io_(io), acceptor_(io, tcp::endpoint(tcp::v4(), 13)) {
49         start_accept();
50     }
51
52 private:
53     void start_accept() {
54         tcp_connection::pointer new_connection = tcp_connection::create(io_);
55         acceptor_.async_accept(new_connection->socket(),
56                               [this, new_connection](const asio::error_code &ec) {
57                                   this->handle_accept(new_connection, ec);
58                               });
59     }
60     void handle_accept(tcp_connection::pointer new_connection,
61                       const asio::error_code &ec) {
62         if (!ec) {
63             new_connection->start();
64         }
65         start_accept();
66     }
67
68     asio::io_context &io_;
69     tcp::acceptor acceptor_;
70 };
71
72 int main() {
73     try {
74         asio::io_context io;
75         tcp_server server(io);

```

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```
76     io.run();  
77 } catch (std::exception &e) {  
78     std::cout << "Exception: " << e.what();  
79 }  
80 return 0;  
81 }
```


WEBSOCKETPP

32.1 Install

```
git clone https://github.com/zaphoyd/websocketpp
cd websocketpp
mkdir build
cmake -DCMAKE_INSTALL_PREFIX=/ceph-fj/fangjun/software/websocketpp ..
make -j install
```

32.2 hello

Listing 1: ./code/hello/Makefile

```
1 CXXFLAGS := -DASIO_STANDALONE
2 CXXFLAGS += -I/ceph-fj/fangjun/software/asio/include
3 CXXFLAGS += -I/ceph-fj/fangjun/software/websocketpp/include
4 CXXFLAGS += -std=c++14
5 LDFLAGS += -pthread
6
7 main: main.o uri.o
8     $(CXX) -o $@ $^ $(LDFLAGS)
9
10 %.o: %.cc
11     $(CXX) -c $(CXXFLAGS) -o $@ $<
12
13 clean:
14     $(RM) *.o main
```

Listing 2: ./code/hello/main.cc

```

1  #include "websocketpp/client.hpp"
2  #include "websocketpp/config/asio_no_tls_client.hpp"
3  #include <assert.h>
4  #include <iostream>
5  #include <memory>
6  #include <string>
7  #include <thread>
8
9  using client = websocketpp::client<websocketpp::config::asio_client>;
10
11 class connection_metadata {
12 public:
13     using ptr = std::shared_ptr<connection_metadata>;
14     connection_metadata(int32_t id, websocketpp::connection_hdl hdl,
15                        std::string uri)
16         : m_id(id), m_hdl(hdl), m_status("Connecting"), m_uri(uri),
17           m_server("N/A") {}
18     void on_open(client *c, websocketpp::connection_hdl hdl) {
19         m_status = "Open";
20         client::connection_ptr con = c->get_con_from_hdl(hdl);
21         m_server = con->get_response_header("Server");
22
23         // the type of con->get_response() is defined in
24         // websocketpp/http/response.hpp
25         assert(con->get_response().get_version() == "HTTP/1.1");
26     #if 0
27         // header_list is a std::map<std::string, std::string>
28         auto header_list = con->get_response().get_headers();
29         for (const auto &it : header_list) {
30             std::cout << it.first << ": " << it.second << "\n";
31         }
32         /*
33         Connection: upgrade
34         Date: Wed, 05 Oct 2022 11:33:39 GMT
35         Sec-WebSocket-Accept: QvLQVaiZn5YWerU3s15SuDWypjo=
36         Server: nginx
37         Upgrade: websocket
38         */
39     #endif
40     assert(con->get_response().get_body() == "");
41 }
42

```

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```

43 void on_fail(client *c, websocketpp::connection_hdl hdl) {
44     m_status = "Failed";
45     client::connection_ptr con = c->get_con_from_hdl(hdl);
46     m_server = con->get_response_header("Server");
47     m_error_reason = con->get_ec().message();
48 }
49
50 void on_close(client *c, websocketpp::connection_hdl hdl) {
51     m_status = "Closed";
52     client::connection_ptr con = c->get_con_from_hdl(hdl);
53     std::ostringstream os;
54     os << "close code: " << con->get_remote_close_code() << " ("
55         << websocketpp::close::status::get_string(con->get_remote_close_code())
56         << "), close reason: " << con->get_remote_close_reason();
57     m_error_reason = os.str();
58 }
59
60 websocketpp::connection_hdl get_hdl() const { return m_hdl; }
61 int32_t get_id() const { return m_id; }
62 const std::string &get_status() const { return m_status; }
63
64 friend std::ostream &operator<<(std::ostream &out,
65                                 const connection_metadata &data);
66
67 private:
68     int32_t m_id;
69     websocketpp::connection_hdl m_hdl;
70     std::string m_status;
71     std::string m_uri;
72     std::string m_server;
73     std::string m_error_reason;
74 };
75
76 std::ostream &operator<<(std::ostream &out, const connection_metadata &data) {
77     out << "> URI: " << data.m_uri << "\n"
78         << "> Status: " << data.m_status << "> Remote Server: "
79         << (data.m_server.empty() ? "None Specified" : data.m_server) << "\n"
80         << "> Error/close reason: "
81         << (data.m_error_reason.empty() ? "N/A" : data.m_error_reason);
82     return out;
83 }
84
85 class websocket_endpoint {

```

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```

86 public:
87     websocket_endpoint() : m_next_id(0) {
88         m_endpoint.clear_access_channels(websocketpp::log::alevel::all);
89         m_endpoint.clear_error_channels(websocketpp::log::elevel::all);
90
91         m_endpoint.init_asio();
92         m_endpoint.start_perpetual();
93
94         m_thread.reset(new std::thread(&client::run, &m_endpoint));
95     }
96
97     ~websocket_endpoint() {
98         m_endpoint.stop_perpetual();
99         for (auto it = m_connection_list.begin(); it != m_connection_list.end();
100             ++it) {
101             if (it->second->get_status() != "Open") {
102                 continue;
103             }
104
105             std::cout << "> Closing connection " << it->second->get_id() << std::endl;
106             websocketpp::lib::error_code ec;
107             m_endpoint.close(it->second->get_hdl(),
108                             websocketpp::close::status::going_away, "", ec);
109
110             if (ec) {
111                 std::cout << "> Error closing connection " << it->second->get_id()
112                     << ": " << ec.message() << std::endl;
113             }
114             m_thread->join();
115         }
116
117     int32_t connect(std::string uri) {
118         websocketpp::lib::error_code ec;
119         client::connection_ptr con = m_endpoint.get_connection(uri, ec);
120         if (ec) {
121             std::cout << "> Connect initialization error: " << ec.message()
122                 << std::endl;
123             return -1;
124         }
125         int32_t new_id = m_next_id++;
126         connection_metadata::ptr metadata_ptr(
127             new connection_metadata(new_id, con->get_handle(), uri));
128         m_connection_list[new_id] = metadata_ptr;

```

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```

129
130 // see websocketpp/connection.hpp
131 con->set_open_handler(
132     [metadata_ptr = metadata_ptr, this](websocketpp::connection_hdl hdl) {
133         metadata_ptr->on_open(&m_endpoint, hdl);
134     });
135
136 // see websocketpp/connection.hpp
137 con->set_fail_handler(
138     [metadata_ptr = metadata_ptr, this](websocketpp::connection_hdl hdl) {
139         metadata_ptr->on_fail(&m_endpoint, hdl);
140     });
141
142 con->set_close_handler(
143     [metadata_ptr = metadata_ptr, this](websocketpp::connection_hdl hdl) {
144         metadata_ptr->on_close(&m_endpoint, hdl);
145     });
146
147 m_endpoint.connect(con);
148
149 return new_id;
150 }
151
152 connection_metadata::ptr get_metadata(int32_t id) const {
153     auto metadata_it = m_connection_list.find(id);
154     if (metadata_it == m_connection_list.end()) {
155         return connection_metadata::ptr();
156     } else {
157         return metadata_it->second;
158     }
159 }
160
161 void close(int32_t id, websocketpp::close::status::value code,
162     const std::string &reason = "") {
163     websocketpp::lib::error_code ec;
164     auto metadata_it = m_connection_list.find(id);
165     if (metadata_it == m_connection_list.end()) {
166         std::cout << "> No connection found with id " << id << std::endl;
167         return;
168     }
169
170     m_endpoint.close(metadata_it->second->get_hdl(), code, reason, ec);
171     if (ec) {

```

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```

172     std::cout << "> Error initiating close: " << ec.message() << std::endl;
173 }
174 }
175
176 private:
177     client m_endpoint;
178     std::shared_ptr<std::thread> m_thread;
179     std::map<int32_t, connection_metadata::ptr> m_connection_list;
180     int32_t m_next_id;
181 };
182
183 void test_uri();
184
185 int main() {
186     test_uri();
187     // return 0;
188
189     bool done = false;
190     std::string input;
191     websocket_endpoint endponit;
192
193     while (true) {
194         std::cout << "Enter command: ";
195         std::getline(std::cin, input);
196         if (input == "quit") {
197             done = true;
198         } else if (input == "help") {
199             std::cout << "\nCommand List:\n"
200                 << "connect <ws uri>\n"
201                 << "show <connection_id>\n"
202                 << "close <connection_id> <close_code> <close_reason>\n"
203                 << "help: Display the help and exit\n"
204                 << "quit: Exit the program\n"
205                 << "\n";
206         } else if (input.substr(0, 7) == "connect") {
207             int32_t id = endponit.connect(input.substr(8));
208             if (id != -1) {
209                 std::cout << "Created connection with id " << id << "\n";
210             }
211         } else if (input.substr(0, 5) == "close") {
212             std::stringstream ss(input);
213             std::string cmd;
214             int32_t id;

```

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```

215     int32_t close_code = websocketpp::close::status::normal;
216     std::string reason;
217     ss >> cmd >> id >> close_code;
218     std::getline(ss, reason);
219     endponit.close(id, close_code, reason);
220 } else if (input.substr(0, 4) == "show") {
221     int32_t id = atoi(input.substr(5).c_str());
222     connection_metadata::ptr metadata = endponit.get_metadata(id);
223     if (metadata) {
224         std::cout << *metadata << std::endl;
225     } else {
226         std::cout << "Unknown connection id " << id << std::endl;
227     }
228 } else {
229     std::cout << "Unrecognized command\n";
230 }
231 if (done)
232     break;
233 }
234 return 0;
235 }
236 // connect ws://websocket-echo.com
237 // close 0 1001 example message

```

Listing 3: ./code/hello/uri.cc

```

1  #include "websocketpp/uri.hpp"
2  #include <assert.h>
3  #include <iostream>
4
5  static void check_valid_ipv4() {
6      std::string s = "10.192.168.1";
7      bool b = websocketpp::uri_helper::ipv4_literal(s.begin(), s.end());
8      assert(b == true);
9
10     s = "256.192.168.1";
11     b = websocketpp::uri_helper::ipv4_literal(s.begin(), s.end());
12     assert(b == false);
13 }
14
15 static void check_uri() {
16     websocketpp::uri uri("ws://localhost:81");
17     assert(uri.str() == "ws://localhost:81/");

```

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```
18  assert(uri.get_valid() == true);
19  assert(uri.is_ip_v6_literal() == false);
20  assert(uri.get_secure() == false);
21  assert(uri.get_scheme() == "ws");
22  assert(uri.get_host() == "localhost");
23  assert(uri.get_port() == 81);
24  assert(uri.get_port_str() == "81");
25  assert(uri.get_host_port() == "localhost:81");
26  assert(uri.get_authority() == "localhost:81");
27  assert(uri.get_resource() == "/");
28  assert(uri.get_query() == "");
29  }
30
31  static void check_uri2() {
32      websocketpp::uri uri("wss://localhost/foo/bar?hello=12");
33      assert(uri.str() == "wss://localhost/foo/bar?hello=12");
34      assert(uri.get_valid() == true);
35      assert(uri.is_ip_v6_literal() == false);
36      assert(uri.get_secure() == true);
37      assert(uri.get_scheme() == "wss");
38      assert(uri.get_host() == "localhost");
39      assert(uri.get_port() == 443);
40      assert(uri.get_port_str() == "443");
41      assert(uri.get_host_port() == "localhost");
42      assert(uri.get_authority() == "localhost:443");
43      assert(uri.get_resource() == "/foo/bar?hello=12");
44      assert(uri.get_query() == "hello=12");
45  }
46
47  void test_uri() {
48      check_valid_ip_v4();
49      check_uri();
50      check_uri2();
51  }
```

OPERATING SYSTEMS

33.1 macos

Copy content to the clipboard:

```
cat result.txt | pbcopy
```

33.1.1 Port scan

```
brew install masscan  
sudo masscan -p 1-10000 137.184.92.35
```

33.1.2 Update GCC

```
brew upgrade gcc  
  
export CC=/usr/local/Cellar/gcc/13.1.0/bin/gcc-13  
export CXX=/usr/local/Cellar/gcc/13.1.0/bin/g++-13  
export CMAKE_C_COMPILER=$CC  
export CMAKE_CXX_COMPILER=$CXX
```

33.2 lldb

- GDB to LLDB command map <https://lldb.llvm.org/use/map.html>

In gdb, we use `frame 1` to switch to frame 1, but in lldb, we use `frame select 1`.

34.1 C++ String

Note: `std::string` contains an array of bytes. Its length is the number of bytes in the array.

Listing 1: `./code/cpp_string.cc`

```
#include <iostream>
#include <string>
int main() {
    std::string s = "nihaohello";
    auto p = s.c_str();
    fprintf(stderr, "s.size: %d\n", (int)s.size());
    for (int i = 0; i != s.size(); ++i) {
        fprintf(stderr, "%d: %02x, %c\n", i, p[i], p[i]);
    }
    return 0;
    // UTF8 encoding of s is
    // \xe4\xbd\xa0\xe5\xa5\xbd\x68\x65\x6c\x6c\x6f
    //
    // UTF8 encoding of ni is 0xe4 0xbd 0xa0
    // UTF8 encoding of hao is 0xe5 0xa5 0xbd
    // UTF8 encoding of h is 0x68
    // UTF8 encoding of e is 0x65
    // UTF8 encoding of l is 0x6c
    // UTF8 encoding of o is 0x6f
}
/*
It prints:
s.size: 11

0: fffffffe4, x
1: fffffffbd, x
```

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```

2: fffffffa0, x
3: fffffffe5, x
4: fffffffa5, x
5: fffffffbd, x
6: 68, h
7: 65, e
8: 6c, l
9: 6c, l
10: 6f, o
*/

```

34.2 kotlin String

Note: The length kotlin String equals to the number of characters, not number of bytes.

Listing 2: ./code/kotlin_string.kt

```

fun main() {
    var s = "nihaohello"
    println("s.length: ${s.length}")
    var bytes : ByteArray = "nihaohello".toByteArray()
    // for(b in bytes) {
    //     println(b.toUByte().toString(radix = 16))
    // }
    bytes.forEachIndexed{i, b ->
        println("$i, ${b.toUByte().toString(radix=16)}")
    }
}
/*
s.length: 7

0, e4
1, bd
2, a0
3, e5
4, a5
5, bd
6, 68
7, 65
8, 6c
9, 6c
10, 6f
*/

```

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```
}
```

34.3 Basics

If the terminal displays chinese characters as ???, then use the following environment variable to fix it.

```
export LC_ALL=en_US.UTF-8
```

34.4 TODOs

- Unicode Characters –What Every Developer Must Know About Encoding

<https://www.freecodecamp.org/news/everything-you-need-to-know-about-encoding/>

35.1 Call cpp

1. Start xcode
2. Create a new project, macOS -> Command Line Tool
3. Product name: TestCpp
4. Language Swift
5. Edit main.swift, keep only `println("hello world")` and remove other lines
6. Product -> Run
7. Create a c++ shared library

Listing 1: ./code/call_cpp/hello.h

```
class A {  
public:  
    A(int);  
    int getInt() const;  
  
private:  
    int i_;  
};
```

Listing 2: ./code/call_cpp/hello.cc

```
#include "hello.h"  
  
A::A(int k) : i_(k) {}  
int A::getInt() const { return i_; }
```

Listing 3: ./code/call_cpp/Makefile

```
all: libhello.a

libhello.a: hello.h hello.cc
    g++ -c hello.cc -o hello.o
    ar r libhello.a hello.o

clean:
    $(RM) libhello.a hello.o
```

8. In xcode, project->build phases->frameworks and libraries, click +, and select libhello.a. Then, modify build settings to change the library search paths (in search paths)
9. Add hello.h to the same folder of main.swift. File->Add files to TestCpp.
10. Add a wrapper. File->New->File->C++ file, next, choose an arbitrary name, e.g., wrapper.cc. Uncheck Also create a header file. We only need the .cc file. In the popped-up dialog, select Create bridging header.
11. If we don't select Create bridging header, we have to go to build settings, swift compiler, objective-c bridging header, and enter a header name.
12. Content of wrapper.cc

```
#include "hello.h"
extern "C" int getIntFromCpp() {
    return A(10).getInt();
}
```

13. Content of the bridging header TestCpp-Briding-Header.h:

```
int getIntFromCpp();
```

14. In main.swift, use print(getIntFromCpp())

35.2 Swift

35.2.1 Data Types

- Int, Float, Double, String

There are no implicit conversions between types.

Integer to string:

```
let a = 3
let b = "a is " + String(a)
```

String interpolation, use `\(value)`. `value` can be an expression:: To get a lowercased string, use the method `.lowercased()` of a string variable.

```
let s = 10 let s2 = "s is (s)"
```

Array of string:

```
var arr = [1, 2, 3]
# The type of arr is [Int]

# to append a new value to arr
arr.append(10)

# to make it empty
arr = []

# Create an empty array
var arr2: [Int] = []
```

Dictionary (string to int):

```
var d = ["one": 1, "two": 2]

# type of d is [String: Int]

# make d empty
d = [:]

# create an empty dict that maps string to int
var d2: [String: Int] = [:]
```

To iterate an array:

```
let values = [1, 2, 3]
var s = 0
for i in values {
    s += i
}
print(s)
# s is 6
```

For loop (1):

```
var k = 0
for i in 0..<4 {
    k += i
}
print(k)
# k is 6
# note: 0..<4 produces 0, 1, 2, 3
# 4 is not included
```

For loop (2):

```
var k = 0
for i in 0...4 {
    k += i
}
print(k)
# k is 10
# note: 0...4 produces 0, 1, 2, 3, 4
# 4 is included
```

Optional types:

```
var i: Int? = 1
print(i == nil) // false

// if i is not nil, then assign i to b
// if i is nil, then assign 10 to b
//
var b = i ?? 10
print(b) // b is 1

i = nil
b = i ?? 10
print(b) // b is 10
```

35.2.2 The basics

See also <https://docs.swift.org/swift-book/documentation/the-swift-programming-language/thebasics/>

- Semicolon is not required at the end of a single statement, like python
- For two statements at the same line, you need to use a semicolon to separate them

Integer types:

- Int8, Int16, Int32, Int64

- UInt8, UInt16, UInt32, UInt64
- Int, UInt

Real floating point types:

- Float, Double

Listing 4: ./code/the-basics.swift

```
// swiftc -suppress-warnings ./the-basics.swift

if true {
    let maximumNumberOfLoginAttempts = 10 // a constant
    var currentLoginAttempt = 0 // a variable
}

if true {
    var environment = "development"
    let maximumNumberOfLoginAttempts: Int

    if environment == "development" {
        maximumNumberOfLoginAttempts = 100
    } else {
        maximumNumberOfLoginAttempts = 10
    }

    // We don't need to initialize the const variable on declaration.
    // Before accessing it , we need to initialize it

    precondition(maximumNumberOfLoginAttempts == 100)
}

if true {
    // declare multiple variables in the same line
    var x = 0.0, y = 0.0, z = 0.0

    var welcomeMessage: String

    var red, green, blue: Int
}

if true {
    print() // output a new line
    print(1, 2, 3) // output: 1 2 3
    print(1, 2, 3, separator: "-") // output: 1-2-3
}
```

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```
}

if true {
    let minValue = UInt8.min
    let maxValue = UInt8.max
    precondition(minValue == 0)
    precondition(maxValue == 255)

    let a: Int16 = 100
    let b: Int = Int(a) // need explicit conversion
}

if true {
    // type alias
    typealias AudioSample = Int16
    var a: AudioSample = 10
    var b: Int16 = a
}

if true {
    // boolean
    var a = true
    var b = false
}

if true {
    // tuples
    var a = (1, "ok")
    precondition(a.0 == 1)
    precondition(a.1 == "ok")
    a.0 = 10 // since a is defined using var, not let, we can change its elements

    let (i, s) = a
    precondition(i == 10)
    precondition(s == "ok")
    print(a) // (1, "ok")

    let (k, _) = a // ignore the second entry
    precondition(k == 10)

    let (_, s1) = a // ignore the first entry
    precondition(s1 == "ok")
}
```

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```
let b = (index: 11, msg: "hello") // name the elements
precondition(b.0 == 11)
precondition(b.index == 11)

precondition(b.1 == "hello")
precondition(b.msg == "hello")
}

if true {
    // note that i is of type Int?
    let i = Int("hello")
    precondition(i == nil)

    // write it explicitly
    let b: Int? = Int("hello")

    let c: Int = Int("10")! // unwrap with !
}

if true {
    let a = Int("1");
    let b = Int("2");
    let c = Int("10");

    if let a {
        print("a \(a) is an integer")
    }

    // we can also use var which means we can change a inside if
    if var a {
        a += 1
        print("a is an integer \(a)")
    }

    // true if and only if all conditions hold
    if let a, let b, let c, c > 2 {
        print("here")
    }
}
```

35.2.3 Basics

See

- A Swift Tour

https://docs.swift.org/swift-book/GuidedTour/GuidedTour.html#apple_ref/doc/uid/TP40014097-CH2-ID1

- Getting Started

<https://www.swift.org/getting-started>

- Size, Stride, Alignment

<https://swiftunboxed.com/internals/size-stride-alignment/>

- Swift And C: Everything You Need to Know on Types, Pointers and more

<https://www.uraimo.com/2016/04/07/swift-and-c-everything-you-need-to-know/>

```
swift --version
swift-driver version: 1.62.15 Apple Swift version 5.7.2 (swiftlang-5.7.2.135.5 clang-
↳ 1400.0.29.51)
Target: x86_64-apple-macosx13.0
```

To start the REPL commandline:

```
swift repl
```

Listing 5: ./code/1-hello.swift

```
// swiftc ./1-hello.swift

print("Hello, world!")
// Note:
// 1. It is not println but it still prints a new line
// 2. There is no semicolon
// 3. There is no main(), like Python
```

```
swiftc ./1-hello.swift
```

It will generate an executable ./1-hello.

```
otool -L ./1-hello

./1-hello:
  /usr/lib/libobjc.A.dylib (compatibility version 1.0.0, current version 228.0.0)
  /usr/lib/libSystem.B.dylib (compatibility version 1.0.0, current version 1319.0.0)
```

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```

/usr/lib/swift/libswiftCore.dylib (compatibility version 1.0.0, current version 5.7.
↪1)

ls -lh 1-hello

-rwxr-xr-x  1 fangjun  staff   33K Jan 21 11:55 1-hello

```

Listing 6: ./code/2-variables.swift

```

var myVariable = 42
myVariable = 50
let myConstant = 43
print("myVariable is \(myVariable)")
print("myConstant is \(myConstant)")

let implicitInteger = 1
let implicitDouble = 1.0
let explicitDouble: Double = 1
let myFloat: Float = 1.2

let label = "The width is "
let width = 100
var widthLabel = label + String(width)
widthLabel = "\(label)\(width)"
print(widthLabel)

// """
let s = """
    abc
    def
    """
// Note: there are leading spaces before the ending """
print(s)

/*
myVariable is 50
myConstant is 43
The width is 100
abc
def
*/

```

For arrays, see also <https://docs.swift.org/swift-book/documentation/the-swift-programming-language/collectiontypes>

Listing 7: ./code/3-array-dict.swift

```
var fruits = ["strawberries", "limes", "apples"]
fruits[1] = "grapes"

print(fruits) // ["strawberries", "grapes", "apples"]
precondition(fruits.isEmpty == false)
precondition(fruits.count == 3)

// resized automatically
fruits.append("blueberries")
precondition(fruits.count == 4)
precondition(fruits.first == "strawberries")
precondition(fruits.last == "blueberries")

// assign by copy!
var fruits2 = fruits
fruits2.append("banana")
precondition(fruits.last == "blueberries")
precondition(fruits2.last == "banana")

print(fruits) // ["strawberries", "grapes", "apples", "blueberries"]
print(fruits2) // ["strawberries", "grapes", "apples", "blueberries", "banana"]

var intArray: [Int] = [1, 2, 3]
precondition(intArray.count == 3)
precondition(intArray[0] == 1)
precondition(intArray[1] == 2)
precondition(intArray[2] == 3)
intArray = []
precondition(intArray.isEmpty)

var floatArray = Array(repeating: Float(0), count: 3)
precondition(floatArray.count == 3)
precondition(floatArray[0] == 0)
precondition(floatArray[1] == 0)
precondition(floatArray[2] == 0)

var floatArray2 = Array<Float>(repeating: 0, count: 3)
precondition(floatArray2.count == 3)
precondition(floatArray2[0] == 0)
precondition(floatArray2[1] == 0)
precondition(floatArray2[2] == 0)
```

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```

var occupations = [
    "Tom": "Captain",
    "Jerry": "Mechanic",
]
print(occupations) // ["Tom": "Captain", "Jerry": "Mechanic"]
occupations["Tom"] = "Teacher"
precondition(occupations["Dan"] == nil) // not exist, so return nil

let emptyArray: [String] = []
let emptyDict: [String: Float] = [:]

```

Listing 8: ./code/4-if.swift

```

let individualScores = [75, 43, 103, 87, 12]
var teamScore = 0
for score in individualScores {
    if score > 50 {
        teamScore += 3
    } else {
        teamScore += 1
    }
}
// 11
print(teamScore)

var optionalString: String? = "Hello"
// false
print(optionalString == nil)

var optionalName: String? = "Tom"
var greeting = "Hello!"

// if optionalName is nil, then the condition is false
// if optionalName is not nil, then the condition is true and optionalName is
↳unwrapped and assigned to name
if let name = optionalName {
    greeting = "Hello, \(name)"
}
// Hello, Tom
print(greeting)

let nickname: String? = nil
let fullName: String = "Tom Green"

```

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```
// Hi, Tom Green
let informalGreeting = "Hi, \(nickname ?? fullName)"
// if nickname is not nil, then nickname is used
// if nickname is nil, then fullName is used

print(informalGreeting)

if let nickname {
    print("Hey, \(nickname)")
}

// if can be used as an expression!
let i = 3
let b = if i >= 0 {
    1
} else {
    -1
}

// 1
print(b)

var s: String? // initialized to nil by default
print(s == nil) // true
```

Listing 9: ./code/5-switch.swift

```
let vegetable = "red pepper"
switch vegetable {
    case "celery":
        print("celery")
    case "cucumber", "watercress":
        print("cucumber or watercress")
    case let x where x.hasSuffix("pepper"):
        print("x")
    default:
        print("Everything tastes good in soup.")
}

// 1. no need to use break
// 2. default is mandatory so that it is exhaustive
```

Listing 10: ./code/6-for.swift

```
let interestingNumbers = [
    "Prime": [2, 3, 5, 7, 11, 13],
    "Fibonacci": [1, 1, 2, 3, 5, 8],
    "Square": [1, 4, 9, 16, 25],
]
var largest = 0
for (_, numbers) in interestingNumbers {
    for number in numbers {
        if number > largest {
            largest = number
        }
    }
}
print(largest) // 25

var total = 0
for i in 0..<4 {
    total += i
}
print(total) // 6

total = 0
for i in 0...4 {
    total += i
}
print(total) // 10
```

Listing 11: ./code/7-while.swift

```
var n = 2
while n < 100 {
    n *= 2;
}
print(n) // 128

var m = 2
repeat {
    m *= 2;
} while (m < 100)

print(m) // 128
```

Listing 12: ./code/8-func.swift

```
func greet(person: String, day: String) -> String {
    return "Hello \(person), today is \(day)"
}

print(greet(person: "Bob", day: "Tuesday"))
// print(greet("Bob", day: "Tuesday")) // error: missing argument label 'person:' in_
↳ call

func greet2(_ person: String, on day: String) -> String {
    return "Hello \(person), today is \(day)"
}
// print(greet2(person: "Bob", day: "Tuesday")) // error: incorrect argument labels_
↳ in call (have 'person:day:', expected '_:on:')
print(greet2("Bob", on: "Tuesday"))

func calculateStatistics(scores: [Int]) -> (min: Int, max: Int, sum: Int) {
    var min = scores[0]
    var max = scores[1]
    var sum = 0

    for score in scores {
        if score > max {
            max = score
        } else if score < min {
            min = score
        }

        sum += score
    }

    return (min, max, sum)
}

let statistics = calculateStatistics(scores: [5, 3, 100, 3, 9])
print(statistics.sum) // 120
print(statistics.2) // 120

func returnFifteen() -> Int {
    var y = 10
    func add() {
        y += 5
    }
    add()
}
```

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```

    return y
}
print(returnFifteen()) // 15

func makeIncrementer() -> ((Int) -> Int) {
    func addOne(number: Int) -> Int {
        return number + 1
    }

    return addOne
}

var increment = makeIncrementer()
print(increment(7)) // 8

func hasAnyElement(list: [Int], condition: (Int) -> Bool) -> Bool {
    for item in list {
        if condition(item) {
            return true
        }
    }
    return false
}

func lessThanTen(number: Int) -> Bool {
    return number < 10
}

var numbers = [20, 19, 7, 12]
print(hasAnyElement(list: numbers, condition: lessThanTen)) // true

// the closure has to be put in {}
print(hasAnyElement(list: numbers, condition: {(number: Int) -> Bool in number < 1}))
↪ // false

// we can omit the input argument type and return type
print(hasAnyElement(list: numbers, condition: {number in number < 1})) // false

print(numbers.map({ (number: Int) -> Int in
    let result = 3 * number
    return result
})) // [60, 57, 21, 36]

let mappedNumber = numbers.map({number in 3 * number})

```

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```
print(mappedNumber) // [60, 57, 21, 36]

var sortedNumbers = numbers.sorted {$0 > $1}
print(sortedNumbers) // [20, 19, 12, 7]

sortedNumbers = numbers.sorted {$0 < $1} // also ok
print(sortedNumbers) // [7, 12, 19, 20]
```

Listing 13: ./code/9-class.swift

```
class Shape {
    var numberOfSides = 0
    func simpleDescription() -> String {
        return "A shape with \(numberOfSides) sides."
    }
}

var shape = Shape()
print(shape.numberOfSides) // 0
shape.numberOfSides = 4
print(shape.numberOfSides) // 4
var simpleDescription = shape.simpleDescription()
print(simpleDescription) // A shape with 4 sides

// with constructor
class NamedShape {
    var numberOfSides = 0
    var name: String

    init(name: String, numberOfSides: Int) {
        self.name = name
        self.numberOfSides = numberOfSides
    }

    func simpleDescription() -> String {
        return "\(name): A shape with \(numberOfSides) sides"
    }
}

var namedShape = NamedShape(name: "Hello", numberOfSides: 10)
print(namedShape.simpleDescription())

class Square: NamedShape {
```

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```
var sideLength: Double

init(sideLength: Double, name: String) {
    self.sideLength = sideLength
    super.init(name: name, numberOfSides: 4)
}

func area() -> Double {
    return sideLength * sideLength
}

override func simpleDescription() -> String {
    return "A square with side length \(sideLength)"
}
}

var square = Square(sideLength: 10, name: "MySquare")
print(square.simpleDescription()) // A square with side length 10.0
print(square.area()) // 100.0

// property getter/setter

class EquilateralTriangle: NamedShape {
    var sideLength: Double = 0.0

    init(sideLength: Double, name: String) {
        self.sideLength = sideLength
        super.init(name: name, numberOfSides: 3)
    }

    var perimeter: Double {
        get {
            return 3.0 * sideLength
        }
        set {
            // note: the new value has the implicit name newValue
            sideLength = newValue / 3.0
        }
    }

    override func simpleDescription() -> String {
        return "An equilateral triangle with side of length \(sideLength)."
    }
}
```

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```
var triangle = EquilateralTriangle(sideLength: 10, name: "MyTriangle")
print(triangle.perimeter) // 30.0
triangle.perimeter = 15
print(triangle.perimeter) // 15.0
print(triangle.sideLength) // 5.0
```

Listing 14: ./code/10-enum.swift

```
enum Rank: Int {
    case ace = 1
    case two, three, four, five, six, seven, eight, nine, ten
    case jack, queen, king

    func simpleDescription() -> String {
        switch self {
            case .ace:
                return "ace"
            case .jack:
                return "jack"
            case .queen:
                return "queen"
            case .king:
                return "king"
            default:
                return String(self.rawValue)
        }
    }
}

let ace = Rank.ace
let aceRawValue = ace.rawValue
print(ace) // ace
print(aceRawValue) // 1

let two = Rank(rawValue: 2)
print(two ?? "zwei") // two
```

Listing 15: ./code/11-memory-layout.swift

```
assert(MemoryLayout<Bool>.size == 1)
assert(MemoryLayout<Bool>.stride == 1)
assert(MemoryLayout<Bool>.alignment == 1)

assert(MemoryLayout<Int>.size == 8)
```

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```

assert (MemoryLayout<Int>.stride == 8)
assert (MemoryLayout<Int>.alignment == 8)

assert (MemoryLayout<Int32>.size == 4)
assert (MemoryLayout<Int32>.stride == 4)
assert (MemoryLayout<Int32>.alignment == 4)

// Like C/C++
struct Example {
    let foo: Int // 8
    let bar: Bool // 1
}

assert (MemoryLayout<Example>.size == 9)
assert (MemoryLayout<Example>.stride == 16)
assert (MemoryLayout<Example>.alignment == 8)

let ex = Example(foo: 10, bar: true)
assert (MemoryLayout.size(ofValue: ex) == 9)
assert (MemoryLayout.stride(ofValue: ex) == 16)
assert (MemoryLayout.alignment(ofValue: ex) == 8)

struct Example2 {
    let bar: Bool // 1
    let foo: Int // 8
}

assert (MemoryLayout<Example2>.size == 16)
assert (MemoryLayout<Example2>.stride == 16)
assert (MemoryLayout<Example2>.alignment == 8)

```

Listing 16: ./code/12-weak-reference.swift

```

class Author {
    let name: String
    // weak reference is required
    weak var post: Post?

    init(name: String) {self.name = name}
    deinit {print("Author deinit")}
}

class Post {
    let title: String

```

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```

var author: Author?
init(title: String) {self.title = title}
deinit {print("Post deinit")}
}

var author: Author? = Author(name: "John Snow")
var post: Post? = Post(title: "foo bar")
post?.author = author
author?.post = post

print(author?.post) // Optional(main.Post)
post = nil // Post deinit
print(author?.post) // nil
author = nil // Author deinit

```

- <https://developer.apple.com/documentation/swift/unsafeMutableRawPointer>

Listing 17: ./code/13-pointers.swift

```

// Unsafe[Mutable][Raw][Buffer]Pointer[Type]
//
// Mutable: means you can change the value
// Raw: means it points to a blob of bytes
// Buffer: means it works like a collection
// Type: means generic typed pointers
//
// UnsafePointer<T>
// UnsafeMutablePointer<T>
//
// UnsafeRawPointer
// UnsafeMutableRawPointer
//
// UnsafeRawBufferPointer
// UnsafeMutableRawBufferPointer
//
// UnsafeBufferPointer<T>
// UnsafeMutableBufferPointer<T>

let count = 2
let stride = MemoryLayout<Int>.stride
let alignment = MemoryLayout<Int>.alignment
let byteCount = count * stride

```

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```

let pointer = UnsafeMutableRawPointer.allocate(byteCount: byteCount, alignment:
↳alignment)

defer {
    pointer.deallocate()
}

pointer.storeBytes(of: 30, as: Int.self)
pointer.advanced(by: stride).storeBytes(of: 3, as: Int.self)
assert(pointer.load(as: Int.self) == 30)
assert(pointer.load(fromByteOffset: stride, as: Int.self) == 3)
assert(pointer.advanced(by: stride).load(as: Int.self) == 3)

let bufferPointer = UnsafeRawBufferPointer(start: pointer, count: byteCount)
for (index, byte) in bufferPointer.enumerated() {
    print("byte \(index) -> \(byte)")
}
/*
byte 0 -> 30
byte 1 -> 0
byte 2 -> 0
byte 3 -> 0
byte 4 -> 0
byte 5 -> 0
byte 6 -> 0
byte 7 -> 0
byte 8 -> 3
byte 9 -> 0
byte 10 -> 0
byte 11 -> 0
byte 12 -> 0
byte 13 -> 0
byte 14 -> 0
byte 15 -> 0
*/

```

- <https://developer.apple.com/documentation/swift/unsafemutablepointer>

Listing 18: ./code/14-typed-pointers.swift

```

let count = 2
let stride = MemoryLayout<Int>.stride
let pointer = UnsafeMutablePointer<Int>.allocate(capacity: count)
pointer.initialize(repeating: 0, count: count)

```

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```

defer {
    pointer.deinitialize(count: count)
    pointer.deallocate()
}

pointer.pointee = 42
pointer.advanced(by: 1).pointee = 6

let bufferPointer = UnsafeBufferPointer(start: pointer, count: count)

for (index, value) in bufferPointer.enumerated() {
    print("value \(index) -> \(value)")
}
/*
value 0 -> 42
value 1 -> 6
*/

```

Listing 19: ./code/15-async.swift

```

import Foundation // for sleep

func fetchUserID(from server: String) async -> Int {
    let tid = pthread_self()
    print("fetch userID from \(server) with thread ID \(tid)")
    if server == "primary" {
        return 97
    }

    return 501
}

func fetchUserName(from server: String) async -> String {
    print("fetch userName from \(server) with thread ID \(pthread_self())")
    let userID = await fetchUserID(from: server)
    print("after await fetch userName from \(server) with thread ID \(pthread_self()),
    ↪ user id \(userID)")
    if userID == 501 {
        return "John Appleseed"
    }

    return "Guest"
}

```

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```

}

func connectUser(to server: String) async {
    print("here0")
    async let userID = fetchUserID(from: server)
    print("here1")
    async let userName = fetchUserName(from: server)
    print("here2")
    let greeting = await "Hello \(userName), user ID \(userID)"
    print(greeting)
}

if false {
    Task {
        await connectUser(to: "primary")
    }

    sleep(1)
}

if true {
    let userIDs = await withTaskGroup(of: Int.self) { group in
        for server in ["primary", "secondary", "development"] {
            group.addTask {
                return await fetchUserID(from: server)
            }
        }

        var results: [Int] = []

        // for await returns result in completion order, not in adding order
        for await result in group {
            results.append(result)
        }

        return results
    }

    print(userIDs)
}

if true {
    let userNames = await withTaskGroup(of: String.self) { group in

```

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```
    for server in ["primary", "secondary", "development"] {
        group.addTask {
            return await fetchUserName(from: server)
        }
    }

    var results: [String] = []
    for await result in group {
        results.append(result)
    }

    return results
}

print(userNames)
}
```

35.2.4 Package

See

- Using the Package Manager

<https://www.swift.org/getting-started/#using-the-package-manager>

- Package Manager

<https://www.swift.org/package-manager/>

- How Swift imports C APIs

<https://github.com/apple/swift/blob/main/docs/HowSwiftImportsC APIs.md>

hello

```
mkdir hello
cd hello
swift package init
```

```
.
├─ Package.swift
├─ README.md
├─ Sources
└─ hello
```

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```
|      └─ hello.swift
└─ Tests
    └─ helloTests
        └─ helloTests.swift
```

4 directories, 4 files

Listing 20: ./code/package/hello/Package.swift

```
// swift-tools-version: 5.7
// The swift-tools-version declares the minimum version of Swift required to build
↳ this package.

import PackageDescription

let package = Package(
    name: "hello",
    products: [
        // Products define the executables and libraries a package produces, and make
↳ them visible to other packages.
        .library(
            name: "hello",
            targets: ["hello"]),
    ],
    dependencies: [
        // Dependencies declare other packages that this package depends on.
        // .package(url: /* package url */, from: "1.0.0"),
    ],
    targets: [
        // Targets are the basic building blocks of a package. A target can define a
↳ module or a test suite.
        // Targets can depend on other targets in this package, and on products in
↳ packages this package depends on.
        .target(
            name: "hello",
            dependencies: []),
        .testTarget(
            name: "helloTests",
            dependencies: ["hello"]),
    ]
)
```

Listing 21: ./code/package/hello/README

```
# hello

A description of this package.
```

Listing 22: ./code/package/hello/Sources/hello/hello.swift

```
public struct hello {
    public private(set) var text = "Hello, World!"

    public init() {
    }
}
```

Listing 23: ./code/package/hello/Tests/helloTests/helloTests.swift

```
import XCTest
@testable import hello

final class helloTests: XCTestCase {
    func testExample() throws {
        // This is an example of a functional test case.
        // Use XCTAssert and related functions to verify your tests produce the
        ↪ correct
        // results.
        XCTAssertEqual(hello().text, "Hello, World!")
    }
}
```

hello2

```
mkdir hello2
cd hello2
swift package init --type executable
```

```
.
├─ Package.swift
├─ README.md
├─ Sources
│   └─ hello2
│       └─ hello2.swift
```

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```
└─ Tests
  └─ hello2Tests
    └─ hello2Tests.swift
```

4 directories, 4 files

Listing 24: ./code/package/hello2/Package.swift

```
// swift-tools-version: 5.7
// The swift-tools-version declares the minimum version of Swift required to build
↳ this package.

import PackageDescription

let package = Package(
    name: "hello2",
    dependencies: [
        // Dependencies declare other packages that this package depends on.
        // .package(url: /* package url */, from: "1.0.0"),
    ],
    targets: [
        // Targets are the basic building blocks of a package. A target can define a
↳ module or a test suite.
        // Targets can depend on other targets in this package, and on products in
↳ packages this package depends on.
        .executableTarget(
            name: "hello2",
            dependencies: []),
        .testTarget(
            name: "hello2Tests",
            dependencies: ["hello2"]),
    ]
)
```

Listing 25: ./code/package/hello2/README

```
# hello2
```

A description of this package.

Listing 26: ./code/package/hello2/Sources/hello2/hello2.swift

```
@main
public struct hello2 {
    public private(set) var text = "Hello, World!"

    public static func main() {
        print(hello2().text)
    }
}
```

Listing 27: ./code/package/hello2/Tests/hello2Tests/hello2Tests.swift

```
import XCTest
@testable import hello2

final class hello2Tests: XCTestCase {
    func testExample() throws {
        // This is an example of a functional test case.
        // Use XCTAssert and related functions to verify your tests produce the
        ↪correct
        // results.
        XCTAssertEqual(hello2().text, "Hello, World!")
    }
}
```

```
swift run hello2

Building for debugging...
[3/3] Linking hello2
Build complete! (0.88s)
Hello, World!
```

35.2.5 Pointer

35.2.6 callback

Pass a function from swift to c and call this swift function in c.

- Swift + C: Callback Interoperability

35.2.7 struct

struct, arrays, strings, dictionaries, are value types.

Listing 28: ./code/struct.swift

```
struct Resolution {
    var width = 0
    var height = 0
}

let r = Resolution()
// r.width = 100 // since r is a constant, we cannot change its members
print(r)

// we can pass initial values
var r2 = Resolution(width: 20, height: 10)
print(r2)

// we can pass initial values
r2 = Resolution(width: 20)
print(r2) // r2.height is default to 0

r2 = Resolution(height: 20)
print(r2) // r2.width is default to 0

r2 = Resolution(width: 10, height: 20)
var r3 = r2 // by copy
r3.width = 0
precondition(r2.width == 10) // r2 is not affected
```

35.3 SwiftUI

35.3.1 TTS

- Extend Speech Synthesis with personal and custom voices

<https://developer.apple.com/videos/play/wwdc2023/10033/>

- Creating an audio unit extension

https://developer.apple.com/documentation/avfaudio/audio_engine/audio_units/creating_an_audio_unit_extension

It describes how to set a parameter from the host app

- <https://appmakers.dev/bcp-47-language-codes-list/>

xcode, create project, select ios or macos, select Audio Unit Extension App

1. Product name: sherpa-onnx-tts
2. Team: None
3. Organization identifier: org.k2-fsa
4. Organization name: Next-gen Kaldi
5. Audio unit type: Speech synthesizer
6. Subtype code: k2k2
7. Manufacturer code: k2k2
8. User interface: No user interface
9. Include tests: Don't check it

example:

```
public override var internalRenderBlock: AUInternalRenderBlock {
    return { [weak self]
        actionFlags, timestamp, frameCount, outputBusNumber, outputAudioBufferList,
↪ _, _ in
        guard let self else { return kAudio_ParamError }

        // This is the audio buffer we are going to fill up
        var unsafeBuffer = ↪
↪ UnsafeMutableAudioBufferListPointer(outputAudioBufferList)[0]
        let frames = unsafeBuffer.mData!.assumingMemoryBound(to: Float32.self)

        var sourceBuffer = UnsafeMutableAudioBufferListPointer(self.currentBuffer!.
↪ mutableAudioBufferList)[0]
        let sourceFrames = sourceBuffer.mData!.assumingMemoryBound(to: Float32.
↪ self)

        for frame in 0..
```

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```
        return noErr
    }
```

35.3.2 Basics-0

- Build an iOS app with SwiftUI

<https://www.swift.org/getting-started/swiftui/>

After creating the project WhyNotTry and saved in ~/t, it creates a directory WhyNotTry inside ~/t.

Inside ~/t/WhyNotTry, there are two directories:

- WhyNotTry

Inside WhyNotTry, there are several .swift files:

- ContentView.swift
- WhyNotTryApp.swift

- WhyNotTry.xcodeproj

Inside WhyNotTry.xcodeproj, there is a file project.xproj

Content of WhyNotTryApp.swift:

```
//
//  WhyNotTryApp.swift
//  WhyNotTry
//
//  Created by fangjun on 2024/7/25.
//

import SwiftUI

@main
struct WhyNotTryApp: App {
    var body: some Scene {
        WindowGroup {
            ContentView()
        }
    }
}
```

Content of ContentView.swift:

```
//  
// ContentView.swift  
// WhyNotTry  
//  
// Created by fangjun on 2024/7/25.  
//  
  
import SwiftUI  
  
struct ContentView: View {  
    var body: some View {  
        VStack {  
            Image(systemName: "globe")  
                .imageScale(.large)  
                .foregroundColor(.accentColor)  
            Text("Hello, world!")  
        }  
        .padding()  
    }  
}  
  
struct ContentView_Previews: PreviewProvider {  
    static var previews: some View {  
        ContentView()  
    }  
}
```

To draw a circle:

```
Circle()  
    .fill(Color.blue)  
    .padding(.all)
```

Use overlay to place one view over another, e.g.:

```
Circle()  
    .fill(Color.blue)  
    .padding(.all)  
    .overlay(  
        Image(systemName: "figure.archery")  
            .font(.system(size: 144))  
            .foregroundColor(.white)  
    )
```

To add Text:


```
Text("Archery")
    .font(.title)

Text("Archery")
    .font(.largeTitle.bold())
```

To arrange view vertically, use:

```
VStack {
    Circle()
    Text("hello")
}
```

My first example ContentView.swift:

```
//
//  ContentView.swift
//  WhyNotTry
//
//  Created by fangjun on 2024/7/25.
//

import SwiftUI

struct ContentView: View {
    let activities = ["Archery", "Baseball", "Basketball"]
    let colors: [Color] = [.blue, .cyan, .gray, .green, .indigo, .mint, .orange, .
    ↪pink, .purple, .red]
    @State private var selected = "Archery"
    @State private var id = 1
    var body: some View {
        VStack {
            Text("Why not try...")
                .font(.largeTitle.bold())
            Spacer()
            VStack {
                Circle()
                    .fill(colors.randomElement() ?? .blue)
                    .padding(*@START_MENU_TOKEN@*/.all/*@END_MENU_TOKEN@*/)
                    .overlay(
                        Image(systemName: "figure.\(selected.lowercased())")
                            .font(.system(size: 144))
                            .foregroundColor(Color.white)
                    )
            }
        }
    }
}
```

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```
        Text("Archery")
            .font(.title)
    }
    .transition(.slide)
    .id(id)

    Spacer()

    Button("Try again") {
        withAnimation(.easeInOut(duration: 1)) {
            selected = activities.randomElement() ?? "Archery"
            id += 1
        }
    }
    .buttonStyle(.borderedProminent)
}

struct ContentView_Previews: PreviewProvider {
    static var previews: some View {
        ContentView()
    }
}
```

35.3.3 Basics 1

- Creating and combining views

<https://developer.apple.com/tutorials/swiftui/creating-and-combining-views>

The `CircleImage.swift` file:

```
//
//  CircleImage.swift
//  WhyNotTry
//
//  Created by fangjun on 2024/7/25.
//

import SwiftUI
```

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```

struct CircleImage: View {
    var body: some View {
        Image("turtlerock")
            .clipShape(Circle())
            .overlay {
                Circle().stroke(.white, lineWidth: 4)
            }
            .shadow(radius: 7)
    }
}

struct CircleImage_Previews: PreviewProvider {
    static var previews: some View {
        CircleImage()
    }
}

```

The ContentView.swift file:

```

//
//  ContentView.swift
//  WhyNotTry
//
//  Created by fangjun on 2024/7/25.
//

import SwiftUI

struct ContentView: View {
    var body: some View {
        VStack {
            CircleImage()

            VStack(alignment: .leading) {
                Text("Turtle Rock")
                    .font(.title)
                HStack {
                    Text("Joshua Tree National Park")
                    Spacer()
                    Text("California")
                }
            }
            .font(.subheadline)
            .foregroundColor(.secondary)
        }
    }
}

```

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```
        Divider()

        Text("About Turtle Rock")
            .font(.title2)

        Text("Descriptive text goes here")

    }
    .padding()

    Spacer()
}
}

struct ContentView_Previews: PreviewProvider {
    static var previews: some View {
        ContentView()
    }
}
```

35.4 Objective C

35.4.1 Basics

Where to find Foundation.h:

```
cd /Applications/Xcode.app/
find . -name "Foundation.h"
```

```
./Contents/Developer/Platforms/AppleTVOS.platform/Developer/SDKs/AppleTVOS.sdk/System/
↪Library/Frameworks/Foundation.framework/Headers/Foundation.h
./Contents/Developer/Platforms/iPhoneOS.platform/Developer/SDKs/iPhoneOS.sdk/System/
↪Library/Frameworks/Foundation.framework/Headers/Foundation.h
./Contents/Developer/Platforms/WatchOS.platform/Developer/SDKs/WatchOS.sdk/System/
↪Library/Frameworks/Foundation.framework/Headers/Foundation.h
./Contents/Developer/Platforms/MacOSX.platform/Developer/SDKs/MacOSX.sdk/System/
↪Library/Frameworks/Foundation.framework/Versions/C/Headers/Foundation.h
```

EMBEDDED SYSTEMS

36.1 m3axpi

- https://wiki.sipeed.com/hardware/zh/maixIII/ax-pi/basic_usage.html

36.1.1 How to setup date

```
# If network is not available, then setup the http proxy
#
# export http_proxy=http://localhost:6009
# export HTTP_PROXY=http://localhost:6009
sudo date -s "$(wget --no-cache -S -O /dev/null google.com 2>&1 | \
  sed -n -e '/ *Date: */ {' -e s///p -e q -e '})'"

# or use
sudo date -s "$(wget --no-cache -S -O /dev/null bing.com 2>&1 | \
  sed -n -e '/ *Date: */ {' -e s///p -e q -e '})'"
```

36.1.2 Python proxy server

```
https://github.com/csukuangfj/python-proxy-server
```

36.2 Raspberry PI

36.2.1 How to connect it to Windows with direct ethernet connection

1. Change the firewall on Windows to allow port for DHCP. The DHCP server uses port 67, supporting both TCP and UDP
2. Download a DHCP server from <https://www.dhcpserver.de/cms/download/>. I have uploaded it to <https://huggingface.co/csukuangfj/dhcp-server-for-windows>
3. Change the IP address of the network interface on Windows to 192.168.1.2.
4. Download and install wireshark. Start it to capture the traffic of the network interface, on which we will run the DHCP server. We can use the filter dhcp to filter out packets.
5. Start the dhcpwiz.exe to configure the DHCP server. We set the starting address to 192.168.1.100. Note: Don't put 192.168.1.2 into the pool since it has already been occupied.
6. Use a network cable to connect the windows machine with Raspberry pi.
7. We can see the IP address of the Raspberry pi from the packets captured by wireshark.

36.2.2 Default IP of Raspberry pi

```
pi@raspberrypi:~ $ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 169.254.49.182 netmask 255.255.0.0 broadcast 169.254.255.255
    inet6 fe80::11f:baaf:1325:9f1e prefixlen 64 scopeid 0x20<link>
    ether dc:a6:32:cc:76:1c txqueuelen 1000 (Ethernet)
    RX packets 275806 bytes 67146923 (64.0 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5359 bytes 454586 (443.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

We can use Wireshark to capture its network traffic and find its address. Then we can assign a static address, e.g., 169.254.49.181 to our own computer.

169.254.0.0/16 IPv4 link local addresses.

36.2.3 Headless setup

1. Download the 64-bit image from <https://www.raspberrypi.com/software/operating-systems/>. The current version is <https://www.raspberrypi.com/software/operating-systems/#raspberrypi-os-64-bit>
2. After flashing to a SD card with BalenaEtcher, create two files in the drive `bootfs`
 - A empty file with name `ssh`, not it has no extension.
 - **`userconf.txt`** containing

```
pi:$6$45qY/WWAHw7hR5Fk
↪$30VJU1878J3iQhc2WczQhOtjmiqWijSgUE9oY92v1BI14n9PiF2TPzxRjlf/
↪wMshZZJ0Oee/cdDxYoQeJvtKR0
```

`pi`` is the user name while the second part is the encrypted password
``pi` generated by `openssl passwd -6`.

36.3 ALSA

Advanced Linux Sound Architecture (ALSA).

36.3.1 USB microphone for raspberry pi

```
pi@raspberrypi:~ $ arecord -l
**** List of CAPTURE Hardware Devices ****
card 3: UACDemoV10 [UACDemoV1.0], device 0: USB Audio [USB Audio]
  Subdevices: 1/1
  Subdevice #0: subdevice #0
```

36.3.2 alsa-utils

See <https://github.com/alsa-project/alsa-utils>

```
sudo apt-get install alsa-utils
```

It contains source code for `arecord`, `areplay`, etc.

aplay

List playback devices:

```
aplay -l

**** List of PLAYBACK Hardware Devices ****
card 1: vc4hdmi0 [vc4-hdmi-0], device 0: MAI PCM i2s-hifi-0 [MAI PCM i2s-hifi-0]
  Subdevices: 1/1
  Subdevice #0: subdevice #0
card 2: vc4hdmi1 [vc4-hdmi-1], device 0: MAI PCM i2s-hifi-0 [MAI PCM i2s-hifi-0]
  Subdevices: 1/1
  Subdevice #0: subdevice #0
```

To play `foo.wav` on card 1, device 0, use:

```
aplay -D hw:1,0 foo.wav
```

arecord

```
cat /proc/asound/pcm
00-00: bcm2835 Headphones : bcm2835 Headphones : playback 8
01-00: MAI PCM i2s-hifi-0 : MAI PCM i2s-hifi-0 : playback 1
02-00: MAI PCM i2s-hifi-0 : MAI PCM i2s-hifi-0 : playback 1
03-00: USB Audio : USB Audio : capture 1
```

```
arecord -l

**** List of CAPTURE Hardware Devices ****
card 3: UACDemoV10 [UACDemoV1.0], device 0: USB Audio [USB Audio]
  Subdevices: 1/1
  Subdevice #0: subdevice #0
```

```
cat /proc/asound/card3/stream0

Jieli Technology UACDemoV1.0 at usb-0000:01:00.0-1.4, full speed : USB Audio

Capture:
  Status: Stop
  Interface 1
    Altset 1
    Format: S16_LE
    Channels: 1
    Endpoint: 0x83 (3 IN) (ASYNC)
```

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```
Rates: 48000
Bits: 16
Channel map: MONO
```

36.3.3 Cross compiling alsa-lib

See <https://github.com/alsa-project/alsa-lib/blob/master/INSTALL>

```
git clone --depth 1 https://github.com/alsa-project/alsa-lib
cd alsa-lib
CC=aarch64-linux-gnu-gcc ./gitcompile --host=aarch64-linux-gnu
```

```
/bin/bash: plantuml: command not found:sudo apt-get install plantuml
```

```
alsa-lib$ find . -name "*.so"

./src/.libs/libasound.so
./src/topology/.libs/libatopology.so
```

```
alsa-lib$ find . -name asoundlib.h
./include/asoundlib.h
```

Then copy the include directory to raspberry pi and rename include to alsa. Also, copy libasound.so to raspberry pi.

We can pass `-I/path/to/alsa/parent` and `-lasound` to `g++` on raspberry pi.

36.3.4 Introduction to Sound Programming with ALSA

```
sudo apt-get install libasound2-dev libasound2
```

See

- <https://www.linuxjournal.com/article/6735>.
- <https://www.alsa-project.org/alsa-doc/alsa-lib/index.html>
- <https://vovkos.github.io/doxyrest/samples/alsa/index.html>, which has more detailed doc for APIs.
- A Tutorial on Using the ALSA Audio API
- PCM (digital audio) interface
- `snd_pcm_open`
- `snd_strerror`

- `snd_device_name_hint`
- `snd_pcm_hw_params_malloc`
- `snd_pcm_hw_params_free`
- `snd_pcm_hw_params_any`
- `snd_pcm_hw_params_set_access`
- `snd_pcm_hw_params_set_format`
- `snd_pcm_hw_params_set_rate_near`
- `snd_pcm_hw_params_set_channels`
- `snd_pcm_hw_params`
- `snd_async_add_pcm_handler`, which calls `snd_pcm_prepare` automatically.
- `snd_pcm_prepare`
- `snd_pcm_start`
- `snd_pcm_drop`
- `snd_pcm_drain`
- `snd_pcm_readi`
- `snd_pcm_readn`
- `snd_pcm_format_t`
- `snd_pcm_hw_params_alloca`

36.4 nano-pi

- user: pi, password pi
- user: root, password: fa

How to assign a static IP on Ubuntu 16.04:

```
ifconfig # it shows eth0

sudo vim /etc/network/interfaces
```

The default content is:

```
auto lo
iface lo inet loopback
```

Add the following:

```
# this is a comment line
# iface eth0 inet dhcp
iface eth0 inet static
    address 192.168.1.100
    netmask 255.255.255.0
```

```
sudo /etc/init.d/networking restart
```

Recording:

```
arecord -l
arecord -Dhw:0,1 -f S16_LE -r 16000 -d 10 -c 2 -t wav test.wav
```

Install cmake from source:

```
./configure -- -DCMAKE_USE_OPENSSL=OFF -DBUILD_TESTING=OFF
```

```
sudo apt-get install cmake
sudo apt-get install libasound2-dev alsa-utils
```

ssh config on macos:

```
Host nano-pi
    Hostname 192.168.1.3
    User pi
    IdentityFile ~/.ssh/pi
    RemoteForward 8080 localhost:8080
```

```
sudo ifup eth0
```

36.5 star visionfive 2

boot mode settings:

- https://doc-en.rvspace.org/VisionFive2/Quick_Start_Guide/VisionFive2_SDK_QSG/boot_mode_settings.html
- RGPI0_1: low, RGPI0_0: high for sd card

Note: The network interface name of the board is end1 and end0

- QR CODE -> end0
- FLY_CORE FC-11Q280 2307 -> end1, I have set static IP address for it: 192.168.1.100

Login:

notes

- username: user
- password: starfive (the same password for root)

Install toolchains:

```
sudo apt-get install -y gcc-riscv64-linux-gnu
```

Setup proxy:

```
# on my macOS
Host starfive
  Hostname 192.168.1.100
  User user
  IdentityFile ~/.ssh/starfive
  RemoteForward 8080 localhost:8080
```

```
# on the board
export http_proxy=http://localhost:8080
export HTTP_PROXY=http://localhost:8080

sudo touch /etc/apt/apt.conf.d/proxy.conf
sudo vi /etc/apt/apt.conf.d/proxy.conf

# Add one line to proxy.conf
Acquire::http::Proxy "http://user:password@proxy.server:port/";
Acquire::https::Proxy "http://user:password@proxy.server:port/";
```

Python proxy server

```
https://github.com/csukuangfj/python-proxy-server
```

Free space:

```
df -h
cd /usr/share
sudo rm -rf help
sudo rm -rf doc
```

install usb:

```
wget https://snapshot.debian.org/archive/debian-ports/20221225T084846Z/pool-riscv64/
↪main/u/usbutils/usbutils_014-1+b1_riscv64.deb

sudo apt-get install ./usbutils*.deb
```

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```
wget https://snapshot.debian.org/archive/debian-ports/20221225T084846Z/pool-riscv64/
↪main/n/net-tools/net-tools_2.10-0.1_riscv64.deb
sudo apt-get install ./net-tools*.deb
```

```
lsusb
cat /proc/asound/cards
lsmod
```

Compile the kernel by ourselves:

See <https://github.com/starfive-tech/VisionFive2>

```
git clone https://github.com/starfive-tech/VisionFive2

sudo apt-get install build-essential g++ git autoconf automake autotools-dev texinfo
↪bison xxd curl flex gawk gdisk gperf libgmp-dev libmpfr-dev libmpc-dev libz-dev
↪libssl-dev libncurses-dev libtool patchutils python3 screen texinfo unzip zlib1g-
↪dev libyaml-dev wget cpio bc dosfstools mtools device-tree-compiler libglib2.0-dev
↪libpixman-1-dev kpartx

sudo apt-get install git-lfs

cd ~/VisionFive2/
git checkout JH7110_VisionFive2_devel
git submodule update --init --recursive
cd buildroot && git checkout --track origin/JH7110_VisionFive2_devel && cd ..
cd u-boot && git checkout --track origin/JH7110_VisionFive2_devel && cd ..
cd linux && git checkout --track origin/JH7110_VisionFive2_devel && cd ..
cd opensbi && git checkout master && cd ..
cd soft_3rdpart && git checkout JH7110_VisionFive2_devel && cd ..

cd ~/VisionFive2/soft_3rdpart/IMG_GPU/out
git lfs pull

cd ~/VisionFive2/
make -j$(nproc)
make buildroot_rootfs -j$(nproc)
make img

# The generated file is work/sdcard.img
# username: root
# password: starfive
# The inner network interface is eth1, i.e., the interface with name FLY_CORE
# The outer network interface is eth0, i.e., the interface with a QR code
```

```
# after login with root
vi /etc/network/interfaces

# iface eth1 inet dhcp
#   pre-up /etc/network/nfs_check
#   wait-delay 15
#   hostname $(hostname)
iface eth1 inet static
    address 192.168.1.100
    netmask 255.255.255.0
```

```
/etc/init.d/S40network restart
```

To show the current directory in the terminal:

```
PS1='${debian_chroot:+($debian_chroot)}\[\033[01;32m\]\u@\h\[\033[00m\]:\[\033[01;34m\]
↪\W\[\033[00m\]\$ '
```

36.6 Lichee Pi 4A

Refer to *Raspberry PI* to setup a DHCP server

After the board gets an IP, ssh to it.

```
/sbin/ifconfig
```

It shows two interfaces: end0 and end1. We don't use lo and wlan0.

end0 is for Ehternet0 and end1 for Ethernet1.

Now we assign a static address to end0.

```
sudo vim /etc/network/interfaces
```

The default content is:

```
source /etc/network/interfaces.d/*
```

We append the following to /etc/network/interfaces:

```
# this is a comment line
# iface eth0 inet dhcp
auto end0
iface eth0 inet static
```

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```
address 192.168.1.8
netmask 255.255.255.0
```

```
sudo /etc/init.d/networking restart
```

Power off the board and connect end0 to my mac.

Create a user:

```
sudo adduser fangjun
sudo usermod -a -G sudo fangjun
```

36.6.1 openKylin

https://wiki.sipeed.com/hardware/zh/lichee/th1520/lpi4a/3_images.html#openKylin

Download address <https://www.openkylin.top/downloads/index-cn.html>. Select openKylin 2.0 alpha RISC-V for LiCheePi4A.

```
-rw-rw-r-- 1 fangjun fangjun 2.8G Mar  7 04:06 openKylin-2.0-alpha-licheepi4a.tar.xz
```

Power off the board, press the BOOT key and don't release it, power on, and then release it. In virtual box, select the USB device T-HEAD USB download gadget [0223].

scp openKylin-2.0-alpha-licheepi4a.tar.xz to my virtual machine.

```
scp -P 1234 openKylin-2.0-alpha-licheepi4a.tar.xz localhost:/mnt/sdb/shared
```

Refer to <https://docs.openkylin.top/zh/%E7%A4%BE%E5%8C%BA%E5%BC%80%E5%8F%91%E6%8C%87%E5%8D%97/riscv%E4%B8%8A%E5%AE%89%E8%A3%85openKylin>

On virtual machine:

```
mkdir t
cd t
tar xf ../openKylin-2.0-alpha-licheepi4a.tar.xz
cd openkylin-2.0-alpha-licheepi4a/fastboot/linux/
```

```
$ lsusb
Bus 002 Device 003: ID 2345:7654 T-HEAD USB download gadget
```

```
sudo ./fastboot/linux/fastboot flash ram ./images/8G/u-boot-nonsec-2020.10-r0-noswap.
↪bin
```

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```
Warning: skip copying ram image avb footer (ram partition size: 0, ram image size:↵
↵957544).
Sending 'ram' (935 KB)                                OKAY [ 0.384s]
Writing 'ram'                                          OKAY [ 0.008s]
Finished. Total time: 0.444s
```

```
sudo ./fastboot/linux/fastboot reboot
```

```
Rebooting                                             OKAY [ 0.031s]
Finished. Total time: 1.451s
```

```
sleep 1
```

```
sudo ./fastboot/linux/fastboot flash uboot ./images/8G/u-boot-nonsec-2020.10-r0-
↵noswap.bin
```

```
Warning: skip copying uboot image avb footer (uboot partition size: 0, uboot image↵
↵size: 957544).
Sending 'uboot' (935 KB)                                OKAY [ 0.259s]
Writing 'uboot'                                          OKAY [ 0.081s]
Finished. Total time: 0.403s
```

```
sudo ./fastboot/linux/fastboot flash boot ./images/8G/boot.ext4
```

```
Warning: skip copying boot image avb footer (boot partition size: 409600, boot image↵
↵size: 61440000).
Sending 'boot' (60000 KB)                                OKAY [ 14.813s]
Writing 'boot'                                          OKAY [ 0.462s]
Finished. Total time: 15.335s
```

```
sudo ./fastboot/linux/fastboot flash root ./images/openkylin-2.0-alpha-licheepi4a-
↵riscv64.ext4
```

```
Sending sparse 'root' 1/64 (113621 KB)                OKAY [ 28.396s]
Writing 'root'                                          OKAY [ 1.437s]
Sending sparse 'root' 2/64 (114573 KB)                OKAY [ 28.678s]
Writing 'root'                                          OKAY [ 1.119s]
Sending sparse 'root' 3/64 (114465 KB)                OKAY [ 31.803s]
Writing 'root'                                          OKAY [ 3.088s]
Sending sparse 'root' 4/64 (114687 KB)                OKAY [ 29.581s]
Writing 'root'                                          OKAY [ 1.332s]
Sending sparse 'root' 5/64 (114686 KB)                OKAY [ 29.088s]
```

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Writing 'root'	OKAY [1.243s]
Sending sparse 'root' 6/64 (114684 KB)	OKAY [28.922s]
Writing 'root'	OKAY [0.926s]
Sending sparse 'root' 7/64 (110724 KB)	OKAY [27.861s]
Writing 'root'	OKAY [0.868s]
Sending sparse 'root' 8/64 (106020 KB)	OKAY [26.523s]
Writing 'root'	OKAY [0.905s]
Sending sparse 'root' 9/64 (114685 KB)	OKAY [29.080s]
Writing 'root'	OKAY [0.990s]
Sending sparse 'root' 10/64 (113701 KB)	OKAY [28.919s]
Writing 'root'	OKAY [1.000s]
Sending sparse 'root' 11/64 (114152 KB)	OKAY [29.780s]
Writing 'root'	OKAY [1.916s]
Sending sparse 'root' 12/64 (113537 KB)	OKAY [29.624s]
Writing 'root'	OKAY [1.600s]
Sending sparse 'root' 13/64 (113881 KB)	OKAY [29.739s]
Writing 'root'	OKAY [1.634s]
Sending sparse 'root' 14/64 (113118 KB)	OKAY [28.989s]
Writing 'root'	OKAY [1.169s]
Sending sparse 'root' 15/64 (106670 KB)	OKAY [27.178s]
Writing 'root'	OKAY [1.157s]
Sending sparse 'root' 16/64 (114685 KB)	OKAY [29.728s]
Writing 'root'	OKAY [1.656s]
Sending sparse 'root' 17/64 (112204 KB)	OKAY [28.763s]
Writing 'root'	OKAY [0.990s]
Sending sparse 'root' 18/64 (114547 KB)	OKAY [30.126s]
Writing 'root'	OKAY [1.495s]
Sending sparse 'root' 19/64 (111451 KB)	OKAY [28.567s]
Writing 'root'	OKAY [1.276s]
Sending sparse 'root' 20/64 (114651 KB)	OKAY [29.283s]
Writing 'root'	OKAY [1.380s]
Sending sparse 'root' 21/64 (104981 KB)	OKAY [27.260s]
Writing 'root'	OKAY [1.565s]
Sending sparse 'root' 22/64 (108444 KB)	OKAY [28.772s]
Writing 'root'	OKAY [1.958s]
Sending sparse 'root' 23/64 (112916 KB)	OKAY [27.882s]
Writing 'root'	OKAY [0.964s]
Sending sparse 'root' 24/64 (110055 KB)	OKAY [28.402s]
Writing 'root'	OKAY [1.373s]
Sending sparse 'root' 25/64 (114686 KB)	OKAY [29.349s]
Writing 'root'	OKAY [1.308s]
Sending sparse 'root' 26/64 (109020 KB)	OKAY [28.892s]
Writing 'root'	OKAY [1.883s]

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Sending sparse 'root' 27/64 (113813 KB)	OKAY [30.411s]
Writing 'root'	OKAY [2.091s]
Sending sparse 'root' 27/64 (113813 KB)	OKAY [30.411s]
Writing 'root'	OKAY [2.091s]
Sending sparse 'root' 28/64 (112541 KB)	OKAY [28.959s]
Writing 'root'	OKAY [1.048s]
Sending sparse 'root' 29/64 (108317 KB)	OKAY [27.795s]
Writing 'root'	OKAY [0.927s]
Sending sparse 'root' 30/64 (103881 KB)	OKAY [26.063s]
Writing 'root'	OKAY [0.883s]
Sending sparse 'root' 31/64 (112380 KB)	OKAY [28.106s]
Writing 'root'	OKAY [0.923s]
Sending sparse 'root' 32/64 (113912 KB)	OKAY [28.863s]
Writing 'root'	OKAY [0.887s]
Sending sparse 'root' 33/64 (107545 KB)	OKAY [27.507s]
Writing 'root'	OKAY [0.901s]
Sending sparse 'root' 34/64 (108964 KB)	OKAY [27.396s]
Writing 'root'	OKAY [0.924s]
Sending sparse 'root' 35/64 (114685 KB)	OKAY [28.831s]
Writing 'root'	OKAY [1.052s]
Sending sparse 'root' 36/64 (114635 KB)	OKAY [29.927s]
Writing 'root'	OKAY [1.857s]
Sending sparse 'root' 37/64 (106582 KB)	OKAY [34.001s]
Writing 'root'	OKAY [4.941s]
Sending sparse 'root' 38/64 (112349 KB)	OKAY [28.087s]
Writing 'root'	OKAY [1.019s]
Sending sparse 'root' 39/64 (114597 KB)	OKAY [28.444s]
Writing 'root'	OKAY [1.003s]
Sending sparse 'root' 40/64 (113833 KB)	OKAY [28.063s]
Writing 'root'	OKAY [0.955s]
Sending sparse 'root' 41/64 (111932 KB)	OKAY [27.982s]
Writing 'root'	OKAY [0.852s]
Sending sparse 'root' 42/64 (114656 KB)	OKAY [29.274s]
Writing 'root'	OKAY [0.919s]
Sending sparse 'root' 43/64 (114409 KB)	OKAY [28.753s]
Writing 'root'	OKAY [1.011s]
Sending sparse 'root' 44/64 (112007 KB)	OKAY [28.708s]
Writing 'root'	OKAY [1.232s]
Sending sparse 'root' 45/64 (114687 KB)	OKAY [29.922s]
Writing 'root'	OKAY [1.978s]
Sending sparse 'root' 46/64 (114684 KB)	OKAY [28.482s]
Writing 'root'	OKAY [0.903s]
Sending sparse 'root' 47/64 (111932 KB)	OKAY [28.518s]

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```

Writing 'root'                                OKAY [ 0.917s]
Sending sparse 'root' 48/64 (106984 KB)        OKAY [ 26.597s]
Writing 'root'                                OKAY [ 0.869s]
Sending sparse 'root' 49/64 (114684 KB)        OKAY [ 28.722s]
Writing 'root'                                OKAY [ 0.888s]
Sending sparse 'root' 50/64 (113296 KB)        OKAY [ 27.990s]
Writing 'root'                                OKAY [ 0.901s]
Sending sparse 'root' 51/64 (114685 KB)        OKAY [ 28.955s]
Writing 'root'                                OKAY [ 1.035s]
Sending sparse 'root' 52/64 (114684 KB)        OKAY [ 28.549s]
Writing 'root'                                OKAY [ 0.906s]
Sending sparse 'root' 53/64 (114685 KB)        OKAY [ 29.215s]
Writing 'root'                                OKAY [ 1.025s]
Sending sparse 'root' 54/64 (102348 KB)        OKAY [ 24.596s]
Writing 'root'                                OKAY [ 1.030s]
Sending sparse 'root' 55/64 (113644 KB)        OKAY [ 28.859s]
Writing 'root'                                OKAY [ 0.899s]
Sending sparse 'root' 56/64 (111648 KB)        OKAY [ 28.154s]
Writing 'root'                                OKAY [ 0.940s]
Sending sparse 'root' 57/64 (105900 KB)        OKAY [ 26.766s]
Writing 'root'                                OKAY [ 0.832s]
Sending sparse 'root' 58/64 (104752 KB)        OKAY [ 26.516s]
Writing 'root'                                OKAY [ 0.832s]
Sending sparse 'root' 59/64 (114684 KB)        OKAY [ 28.399s]
Writing 'root'                                OKAY [ 0.963s]
Sending sparse 'root' 60/64 (114684 KB)        OKAY [ 28.732s]
Writing 'root'                                OKAY [ 0.912s]
Sending sparse 'root' 61/64 (110325 KB)        OKAY [ 27.705s]
Writing 'root'                                OKAY [ 0.991s]
Sending sparse 'root' 62/64 (114684 KB)        OKAY [ 28.856s]
Writing 'root'                                OKAY [ 0.897s]
Sending sparse 'root' 63/64 (109388 KB)        OKAY [ 27.435s]
Writing 'root'                                OKAY [ 0.860s]
Sending sparse 'root' 64/64 (24856 KB)         OKAY [ 6.347s]
Writing 'root'                                OKAY [ 0.205s]
Finished. Total time: 1929.583s

```

```

username: openkylin
password: openkylin

```

The network interfaces on openKylin is eth0 and eth1. The original content of /etc/network/interfaces are:

```
auto lo iface lo inet loopback
auto eth0 iface eth0 inet dhcp
```

Change it to:

```
auto lo iface lo inet loopback
# auto eth0 iface eth0 inet dhcp
iface eth0 inet static
    address 192.168.1.8
    netmask 255.255.255.0
```

36.7 ai-pro

- <http://www.orangepi.cn/html/hardWare/computerAndMicrocontrollers/service-and-support/Orange-Pi-AIpro.html>
- [http://www.orangepi.org/html/hardWare/computerAndMicrocontrollers/service-and-support/Orange-Pi-AIpro\(8-12t\).html](http://www.orangepi.org/html/hardWare/computerAndMicrocontrollers/service-and-support/Orange-Pi-AIpro(8-12t).html)

36.7.1 user1

- username: root
- password: Mind@123

36.7.2 user2

- username: HwHiAiUser
- password: Mind@123

36.7.3 static ip

```
ifconfig
```

shows eth0.

Now we assign a static address to eth0.

Use `nmcli connection show` to list current connections, which displays:

NAME	UUID	TYPE	DEVICE
Wired connection 1	xxxx	ethernet	eth0

Then run `nmcli connection mod "Wired connection 1" ipv4.addresses "192.168.1.10/24" ipv4.method "manual"`

Then run `sudo reboot`.

Power off the board and connect it to my mac.

36.7.4 npu

- ACL: Ascend computing language
- CANN: Compute architecture for neural networks

37.1 Basics

Use only password:

```
ssh -o PubkeyAuthentication=no -o PreferredAuthentications=password example.com
```

37.1.1 Remote forward

```
# Run it on host2
ssh -R 6006:localhost:22 foo@host1
# We can access the port 6006 on host1, which is forwarded to host2's localhost:22
```

```
# host2's ~/.ssh/config Host host1
```

```
Hostname ip_of_host1 User user_at_host1 RemoteForward 6006 localhost:22
```

37.1.2 Local forward

```
# Run it on host2
ssh -L 6006:localhost:22 foo@host1
# We can access the 6006 on host2, which is forwarded to host1's localhost:22
```

```
# host2's ~/.ssh/config Host host1
```

```
Hostname ip_of_host1 User user_at_host1 LocalForward 6006 localhost:22 # 6006 is the port on this com-
puter, i.e., host2 # Access 6006 on host2 will be forward to # localhost:22 on host1
```


38.1 Basics

The documentation is at <https://onnx.ai/onnx/intro/index.html>.

38.1.1 TypeProto

If it is a tensor type, we need to provide element type and shape of the tensor, where the shape is of type `TensorShapeProto`.

38.1.2 TensorProto

It defines a enum for data type.

We need to specify its data type, dims, and the actual data.

38.2 Python notes

See <https://onnx.ai/onnx/api/index.html#l-python-onnx-api>

```
pip install onnx
```

The data structure is defined in a protocol buffer file

<https://github.com/onnx/onnx/blob/main/onnx/onnx.proto>

See <https://protobuf.dev/reference/python/python-generated/> for Python APIs for protocol buffers.

38.2.1 TensorProto

It is defined in <https://github.com/onnx/onnx/blob/main/onnx/onnx.proto#L483>

<https://github.com/onnx/onnx/blob/main/onnx/mapping.py> defines a mapping to map datatypes from onnx to numpy.

Listing 1: ./code/tensor-proto.py

```

1  #!/usr/bin/env python3
2  import onnx
3  import numpy as np
4
5
6  def test_dtype():
7      # the following line is not portable as it uses the internal implementation detail
8      # assert onnx.mapping.TENSOR_TYPE_TO_NP_TYPE[onnx.TensorProto.FLOAT] == np.
9      ↪float32
10     assert onnx.helper.tensor_dtype_to_np_dtype(onnx.TensorProto.FLOAT) == np.float32
11     assert onnx.helper.tensor_dtype_to_np_dtype(onnx.TensorProto.INT8) == np.int8
12
13     # We can convert a dtype to a string
14     assert (
15         onnx.helper.tensor_dtype_to_string(onnx.TensorProto.FLOAT)
16         == "TensorProto.FLOAT"
17     )
18     assert onnx.TensorProto.DataType.Name(onnx.TensorProto.FLOAT) == "FLOAT"
19
20 def test_make_tensor():
21     v = np.array([[1, 2], [3, 4]])
22     t = onnx.helper.make_tensor(
23         name="my",
24         data_type=onnx.helper.np_dtype_to_tensor_dtype(v.dtype),
25         dims=v.shape,
26         vals=v,
27     )
28     print(t)
29
30     """
31     dims: 2
32     dims: 2
33     data_type: 7
34     int64_data: 1
35     int64_data: 2
36     int64_data: 3

```

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```

37     int64_data: 4
38     name: "my"
39     """
40
41
42 def test_build_tensor_manually():
43     n = np.arange(5, dtype=np.float32)
44     t = onnx.TensorProto()
45     t.name = "my-tensor"
46     t.data_type = onnx.TensorProto.FLOAT
47     t.dims.extend(n.shape)
48     t.float_data.extend(n)
49     print(t)
50     """
51     dims: 5
52     data_type: 1
53     float_data: 0.0
54     float_data: 1.0
55     float_data: 2.0
56     float_data: 3.0
57     float_data: 4.0
58     name: "my-tensor"
59     """
60
61
62 def main():
63     test_dtype()
64     test_make_tensor()
65     test_build_tensor_manually()
66
67
68 if __name__ == "__main__":
69     main()

```

38.2.2 TensorShapeProto

It is defined in <https://github.com/onnx/onnx/blob/main/onnx/onnx.proto#L661>.

Listing 2: ./code/tensor-shape-proto.py

```

1  #!/usr/bin/env python3
2  import onnx
3  import numpy as np

```

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```

4
5
6 def main():
7     shape_proto = onnx.TensorShapeProto()
8
9     # shape_proto.dim is a list of messages, so we need to use add()
10    dim = shape_proto.dim.add()
11    dim.dim_value = 10
12
13    dim2 = shape_proto.dim.add()
14    dim2.dim_param = "N"
15    print(shape_proto)
16    """
17    dim {
18        dim_value: 10
19    }
20    dim {
21        dim_param: "N"
22    }
23    """
24
25    # Only one of dim_value and dim_param can be set
26    for d in shape_proto.dim:
27        which = d.WhichOneof("value")
28        if which == "dim_value":
29            print(d.dim_value) # 10
30        elif which == "dim_param":
31            print(d.dim_param) # N
32        else:
33            assert which is None
34
35
36 if __name__ == "__main__":
37     main()

```

One thing to note is that it contains a `oneof` field. Also note it can contain symbolic names for shapes.

```

message TensorShapeProto {
    message Dimension {
        oneof value {
            int64 dim_value = 1;
            string dim_param = 2;    // namespace Shape
        };
        // Standard denotation can optionally be used to denote tensor

```

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```

// dimensions with standard semantic descriptions to ensure
// that operations are applied to the correct axis of a tensor.
// Refer to https://github.com/onnx/onnx/blob/main/docs/DimensionDenotation.md
↪ #denotation-definition
// for pre-defined dimension denotations.
optional string denotation = 3;
};
repeated Dimension dim = 1;
}

```

38.3 ValueInfoProto

38.4 onnxruntime

38.4.1 Install

Official installation doc: <https://onnxruntime.ai/docs/install/>.

One thing to note is that the so-called pre-built mobile packages contain only a subset of the operators, while the pre-built full packages contain all the operators.

See <https://onnxruntime.ai/docs/build/custom.html> for a custom build.

See <https://onnxruntime.ai/docs/build/inferencing.html> for building for inferencing.

38.4.2 Basics

Listing 3: ./code/Makefile

```

1 CXXFLAGS := -std=c++17
2 CXXFLAGS += -I /Users/fangjun/Downloads/onnxruntime-osx-x86_64-1.18.0/include
3 LDFLAGS := -L /Users/fangjun/Downloads/onnxruntime-osx-x86_64-1.18.0/lib
4 LDFLAGS += -l onnxruntime
5 LDFLAGS += -Wl,-rpath,/Users/fangjun/Downloads/onnxruntime-osx-x86_64-1.18.0/lib
6
7 main: main.cc c-api-test.cc cpp-api-test.cc ./custom-op.cc ./custom-op-2.cc ./custom-
↪ op-3.cc
8      $(CXX) $(CXXFLAGS) -o $@ $^ $(LDFLAGS)

```

Listing 4: ./code/main.cc

```
1  #include <iostream>
2
3  #include "onnxruntime_cxx_api.h" // NOLINT
4
5  void TestCApi();
6  void TestCppApi();
7  void TestCustomModel();
8  void TestCustomModel2();
9  void TestCustomModel3();
10
11 int main() {
12     TestCApi();
13     TestCppApi();
14
15     std::cout << "---test custom model---\n";
16     TestCustomModel();
17
18     std::cout << "---test custom model2---\n";
19     TestCustomModel2();
20
21     std::cout << "---test custom model3---\n";
22     TestCustomModel3();
23
24     std::cout << "ORT_API_VERSION: " << ORT_API_VERSION << "\n";
25     return 0;
26 }
27 /*
28 GetVersionString(): 1.18.0
29 Available providers: CoreMLExecutionProvider, CPUExecutionProvider
30 allocator name: Cpu
31 ---test custom model---
32 110
33 220
34 330
35 ---test custom model2---
36 11.5
37 2.5
38 3.5
39 44.5
40 ---test custom model3---
41 11
42 22
```

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```

43  ORT_API_VERSION: 18
44  */

```

Listing 5: ./code/c-api-test.cc

```

1  #include "onnxruntime_c_api.h" // NOLINT
2  #include <cassert>
3  #include <stdio.h>
4
5  static void TestOrtStatus() {
6      const OrtApiBase *api_base = OrtGetApiBase();
7      const OrtApi *api = api_base->GetApi(ORT_API_VERSION);
8      OrtErrorCode code = ORT_OK;
9      const char *msg = "this is a message";
10
11     OrtStatus *status = api->CreateStatus(code, msg);
12     assert(api->GetErrorCode(status) == code);
13
14     const char *msg2 = api->GetErrorMessage(status);
15     assert(strcmp(msg, msg2) == 0);
16
17     // status addr: 0x600001e54040, msg2 addr: 0x600001e54044
18     fprintf(stderr, "status addr: %p, msg2 addr: %p\n", status, msg2);
19
20     // note that sizeof(code) is 4 in my test
21     assert((intptr_t)status + sizeof(code) == (intptr_t)msg2);
22
23     // we have to free the status to avoid memory leak
24     api->ReleaseStatus(status);
25 }
26
27 static void TestOrtApiBase() {
28
29     // OrtApiBase only has two method
30     const OrtApiBase *api_base = OrtGetApiBase();
31     fprintf(stderr, "GetVersionString(): %s\n", api_base->GetVersionString());
32
33     const OrtApi *api = api_base->GetApi(ORT_API_VERSION);
34     fprintf(stderr, "OrtApi: %p\n", api);
35
36     const char *info = api->GetBuildInfoString();
37     fprintf(stderr, "info: %s\n", info);
38 }

```

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```

39
40 void TestCApi() {
41     TestOrtApiBase();
42     TestOrtStatus();
43 }

```

Listing 6: ./code/cpp-api-test.cc

```

1  #include "onnxruntime_cxx_api.h" // NOLINT
2  #include <assert.h>
3  #include <iostream>
4  #include <sstream>
5
6  static void TestOrtGetApi() {
7      const OrtApi &api = Ort::GetApi(); // it returns a const reference
8
9      std::string version = Ort::GetVersionString();
10     std::cout << "version: " << version << "\n";
11 }
12
13 static void PrintAvailableProviders() {
14     std::vector<std::string> providers = Ort::GetAvailableProviders();
15     std::ostringstream os;
16     os << "Available providers: ";
17     std::string sep = "";
18     for (const auto &p : providers) {
19         os << sep << p;
20         sep = ", ";
21     }
22     std::cout << os.str() << "\n";
23 }
24
25 static void TestCreateTensorFromBuffer() {
26     std::vector<int32_t> v = {1, 2, 3, 4, 5, 6};
27     std::array<int64_t, 2> shape = {2, 3};
28     auto memory_info =
29         Ort::MemoryInfo::CreateCpu(OrtDeviceAllocator, OrtMemTypeDefault);
30
31     Ort::Value x = Ort::Value::CreateTensor<int32_t>(
32         memory_info, v.data(), v.size(), shape.data(), shape.size());
33
34     // memory is shared between x and v
35     int32_t *p = x.GetTensorMutableData<int32_t>();

```

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```

36  p[0] = 10;
37  assert(v[0] == 10);
38
39  v[1] = 20;
40  assert(p[1] == 20);
41 }
42
43 static void TestCreateTensor() {
44     Ort::AllocatorWithDefaultOptions allocator;
45
46     std::array<int64_t, 2> shape = {2, 3};
47     auto memory_info =
48         Ort::MemoryInfo::CreateCpu(OrtDeviceAllocator, OrtMemTypeDefault);
49
50     Ort::Value x =
51         Ort::Value::CreateTensor<int32_t>(allocator, shape.data(), shape.size());
52     assert(x.IsTensor());
53     assert(x.HasValue());
54     Ort::TypeInfo type_info = x.GetTypeInfo();
55     auto tensor_type_and_shape_info = type_info.GetTensorTypeAndShapeInfo();
56     assert(tensor_type_and_shape_info.GetElementCount() == 2 * 3);
57     assert(tensor_type_and_shape_info.GetDimensionsCount() == 2);
58     std::vector<int64_t> x_shape = tensor_type_and_shape_info.GetShape();
59     assert(x_shape.size() == shape.size());
60     assert(x_shape[0] == shape[0]);
61     assert(x_shape[1] == shape[1]);
62
63     assert(tensor_type_and_shape_info.GetElementType() ==
64         ONNX_TENSOR_ELEMENT_DATA_TYPE_INT32);
65
66     Ort::ConstMemoryInfo memory_info2 = x.GetTensorMemoryInfo();
67     std::cout << "allocator name: " << memory_info2.GetAllocatorName() << "\n";
68 }
69
70 static void TestDataType() {
71     static_assert(Ort::TypeToTensorType<float>::type ==
72         ONNX_TENSOR_ELEMENT_DATA_TYPE_FLOAT);
73
74     static_assert(Ort::TypeToTensorType<double>::type ==
75         ONNX_TENSOR_ELEMENT_DATA_TYPE_DOUBLE);
76
77     static_assert(Ort::TypeToTensorType<int8_t>::type ==
78         ONNX_TENSOR_ELEMENT_DATA_TYPE_INT8);

```

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```

79  static_assert(Ort::TypeToTensorType<int16_t>::type ==
80              ONNX_TENSOR_ELEMENT_DATA_TYPE_INT16);
81  static_assert(Ort::TypeToTensorType<int32_t>::type ==
82              ONNX_TENSOR_ELEMENT_DATA_TYPE_INT32);
83  static_assert(Ort::TypeToTensorType<int64_t>::type ==
84              ONNX_TENSOR_ELEMENT_DATA_TYPE_INT64);
85  static_assert(Ort::TypeToTensorType<uint8_t>::type ==
86              ONNX_TENSOR_ELEMENT_DATA_TYPE_UINT8);
87  static_assert(Ort::TypeToTensorType<uint16_t>::type ==
88              ONNX_TENSOR_ELEMENT_DATA_TYPE_UINT16);
89  static_assert(Ort::TypeToTensorType<uint32_t>::type ==
90              ONNX_TENSOR_ELEMENT_DATA_TYPE_UINT32);
91  static_assert(Ort::TypeToTensorType<uint64_t>::type ==
92              ONNX_TENSOR_ELEMENT_DATA_TYPE_UINT64);
93  static_assert(Ort::TypeToTensorType<bool>::type ==
94              ONNX_TENSOR_ELEMENT_DATA_TYPE_BOOL);
95  }
96
97  void TestCppApi() {
98      TestOrtGetApi();
99      PrintAvailableProviders();
100     TestCreateTensorFromBuffer();
101     TestCreateTensor();
102     TestDataTypes();
103 }

```

Listing 7: ./code/custom-op.cc

```

1  /*
2  references:
3  https://onnxruntime.ai/docs/reference/operators/add-custom-op.html
4
5  */
6  #include "onnxruntime_lite_custom_op.h"
7  #include <iostream>
8  #include <utility>
9  #include <vector>
10
11 static void KernelOne(const Ort::Custom::Tensor<float> &X,
12                     const Ort::Custom::Tensor<float> &Y,
13                     Ort::Custom::Tensor<float> &Z) {
14     auto input_shape = X.Shape();
15     auto x_raw = X.Data();

```

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```

16  auto y_raw = Y.Data();
17  auto z_raw = Z.Allocate(input_shape);
18  for (int64_t i = 0; i < Z.NumberOfElement(); ++i) {
19      z_raw[i] = x_raw[i] + y_raw[i];
20  }
21 }
22
23 static Ort::CustomOpDomain TestCustomOp() {
24     Ort::CustomOpDomain v1_domain{"com.k2fsa.org"};
25     // please make sure that custom_op_one has the same lifetime as the consuming
26     // session
27     //
28     // Here we use a static variable so it is never released.
29     // in practice, we can move it to a member variable of a class
30     static std::unique_ptr<Ort::Custom::OrtLiteCustomOp> custom_op_one{
31         Ort::Custom::CreateLiteCustomOp("CustomOpOne", "CPUExecutionProvider",
32                                         KernelOne)};
33     v1_domain.Add(custom_op_one.get());
34
35     return v1_domain;
36 }
37
38 void TestCustomModel() {
39     Ort::Env env;
40     Ort::SessionOptions sess_opts;
41     sess_opts.SetIntraOpNumThreads(1);
42     sess_opts.SetInterOpNumThreads(1);
43
44     Ort::CustomOpDomain v1_domain = TestCustomOp();
45
46     Ort::SessionOptions session_options;
47     sess_opts.Add(v1_domain);
48     // create a session with the session_options ...
49
50     std::unique_ptr<Ort::Session> sess =
51         std::make_unique<Ort::Session>(env, "./e.onnx", sess_opts);
52
53     auto memory_info =
54         Ort::MemoryInfo::CreateCpu(OrtDeviceAllocator, OrtMemTypeDefault);
55
56     std::vector<float> x = {10, 20, 30};
57     std::vector<float> y = {100, 200, 300};
58

```

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```

59     std::array<int64_t, 1> shape = {3};
60
61     Ort::Value x_tensor = Ort::Value::CreateTensor(
62         memory_info, x.data(), x.size(), shape.data(), shape.size());
63
64     Ort::Value y_tensor = Ort::Value::CreateTensor(
65         memory_info, y.data(), y.size(), shape.data(), shape.size());
66
67     std::vector<Ort::Value> inputs;
68     inputs.push_back(std::move(x_tensor));
69     inputs.push_back(std::move(y_tensor));
70
71     std::vector<const char*> input_names = {"l_x", "l_y"};
72     std::vector<const char*> output_names = {"my_add_op"};
73     auto out = sess->Run({}, input_names.data(), inputs.data(), inputs.size(),
74         output_names.data(), output_names.size());
75     const float *p = out[0].GetTensorData<float>();
76     for (int i = 0; i < 3; ++i) {
77         std::cout << p[i] << "\n ";
78     }
79 }

```

Listing 8: ./code/e.py

```

1  #!/usr/bin/env python3
2
3  from torch._custom_op import impl as custom_op
4  import torch
5  import onnx
6  import onnxscript
7  from onnxscript import opset18
8
9  import warnings
10 warnings.filterwarnings("ignore")
11
12 @custom_op.custom_op("mylibrary::my_add_op")
13 def my_add_op(x: torch.Tensor, y: torch.Tensor) -> torch.Tensor:
14     # Since we are using mylibrary::my_add_op, so the function
15     # name must be my_add_op; otherwise, it will throw an error
16     # when this script is run
17     pass
18
19 @my_add_op.impl_abstract()

```

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```

20 def my_add_op_impl_abstract_any_name_is_ok(x, y):
21     return torch.empty_like(x)
22
23 @my_add_op.impl("cpu")
24 def my_add_op_impl_any_name_is_ok(tensor_x):
25     return torch.round(tensor_x + tensor_x) # add x to itself, and round the result
26
27 class CustomFoo(torch.nn.Module):
28     def forward(self, x, y):
29         return my_add_op(x, y)
30
31
32 custom_opset = onnxscript.values.Opset(domain="com.k2fsa.org", version=1)
33
34 @onnxscript.script(custom_opset)
35 def custom_my_add(x, y):
36     return custom_opset.CustomOpOne(x, y)
37
38 def main():
39     torch._dynamo.allow_in_graph(my_add_op)
40     x = torch.randn(3)
41     y = torch.randn(3)
42     custom_addandround_model = CustomFoo()
43     onnx_registry = torch.onnx.OnnxRegistry()
44     onnx_registry.register_op(
45         namespace="mylibrary", op_name="my_add_op", overload="default", ↵
↵function=custom_my_add
46     )
47
48     export_options = torch.onnx.ExportOptions(onnx_registry=onnx_registry)
49     onnx_program = torch.onnx.dynamo_export(
50         custom_addandround_model, x, y, export_options=export_options
51     )
52     onnx_program.save("./e.onnx")
53     with open('e.txt', 'w') as f:
54         f.write(str(onnx_program.model_proto))
55     onnx_model = onnx.load("e.onnx")
56     onnx.checker.check_model(onnx_model)
57
58
59 if __name__ == '__main__':
60     main()

```

Listing 9: ./code/e.txt

```
1  ir_version: 8
2  opset_import {
3    domain: "com.k2fsa.org"
4    version: 1
5  }
6  opset_import {
7    domain: ""
8    version: 18
9  }
10 opset_import {
11   domain: "pkg.onnxscript.torch_lib.common"
12   version: 1
13 }
14 producer_name: "pytorch"
15 producer_version: "2.4.0"
16 graph {
17   node {
18     input: "l_x_"
19     input: "l_y_"
20     output: "my_add_op"
21     name: "custom_my_add_0_n0"
22     op_type: "CustomOpOne"
23     domain: "com.k2fsa.org"
24   }
25   name: "main_graph"
26   input {
27     name: "l_x_"
28     type {
29       tensor_type {
30         elem_type: 1
31         shape {
32           dim {
33             dim_value: 3
34           }
35         }
36       }
37     }
38   }
39   input {
40     name: "l_y_"
41     type {
42       tensor_type {
```

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```

43     elem_type: 1
44     shape {
45         dim {
46             dim_value: 3
47         }
48     }
49 }
50 }
51 }
52 output {
53     name: "my_add_op"
54     type {
55         tensor_type {
56             elem_type: 1
57             shape {
58                 dim {
59                     dim_value: 3
60                 }
61             }
62         }
63     }
64 }
65 }

```

Listing 10: ./code/custom-op-2.cc

```

1  /*
2  references:
3  https://onnxruntime.ai/docs/reference/operators/add-custom-op.html
4
5  */
6  #include "onnxruntime_lite_custom_op.h"
7  #include <iostream>
8  #include <utility>
9  #include <vector>
10
11 static void KernelOne(const Ort::Custom::Tensor<float> &X,
12                      const Ort::Custom::Tensor<float> &Y,
13                      Ort::Custom::Tensor<float> &Z) {
14     auto input_shape = X.Shape();
15     auto x_raw = X.Data();
16     auto y_raw = Y.Data();
17     auto z_raw = Z.Allocate(input_shape);

```

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```

18   for (int64_t i = 0; i < Z.NumberOfElement(); ++i) {
19       z_raw[i] = x_raw[i] + y_raw[i];
20   }
21 }
22
23 static Ort::CustomOpDomain TestCustomOp2() {
24     Ort::CustomOpDomain v1_domain{"com.k2fsa.org"};
25     // please make sure that custom_op_one has the same lifetime as the consuming
26     // session
27     //
28     // Here we use a static variable so it is never released.
29     // in practice, we can move it to a member variable of a class
30     static std::unique_ptr<Ort::Custom::OrtLiteCustomOp> custom_op_one{
31         Ort::Custom::CreateLiteCustomOp("CustomOpOne2", "CPUExecutionProvider",
32                                         KernelOne)};
33     v1_domain.Add(custom_op_one.get());
34
35     return v1_domain;
36 }
37
38 void TestCustomModel2() {
39     Ort::Env env;
40     Ort::SessionOptions sess_opts;
41     sess_opts.SetIntraOpNumThreads(1);
42     sess_opts.SetInterOpNumThreads(1);
43
44     Ort::CustomOpDomain v1_domain = TestCustomOp2();
45
46     Ort::SessionOptions session_options;
47     sess_opts.Add(v1_domain);
48     // create a session with the session_options ...
49
50     std::unique_ptr<Ort::Session> sess =
51         std::make_unique<Ort::Session>(env, "./f.onnx", sess_opts);
52
53     auto memory_info =
54         Ort::MemoryInfo::CreateCpu(OrtDeviceAllocator, OrtMemTypeDefault);
55     // foo is [1.5, 2.5, 3.5, 4.5]
56     //
57
58     std::vector<float> x = {10, -20, -30, 40};
59
60     std::array<int64_t, 1> shape = {4};

```

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```

61
62 Ort::Value x_tensor = Ort::Value::CreateTensor(
63     memory_info, x.data(), x.size(), shape.data(), shape.size());
64
65 std::vector<Ort::Value> inputs;
66 inputs.push_back(std::move(x_tensor));
67
68 std::vector<const char*> input_names = {"1_x_"};
69 std::vector<const char*> output_names = {"my_add_op2"};
70 auto out = sess->Run({}, input_names.data(), inputs.data(), inputs.size(),
71     output_names.data(), output_names.size());
72 const float *p = out[0].GetTensorData<float>();
73 for (int i = 0; i < 4; ++i) {
74     std::cout << p[i] << "\n ";
75 }
76 }

```

Listing 11: ./code/f.py

```

1  #!/usr/bin/env python3
2
3  from torch._custom_op import impl as custom_op
4  import torch
5  import onnx
6  import onnxscript
7  from onnxscript import opset18
8
9  import warnings
10 warnings.filterwarnings("ignore")
11
12 @custom_op.custom_op("mylibrary::my_add_op2")
13 def my_add_op2(x: torch.Tensor, y: torch.Tensor) -> torch.Tensor:
14     # Since we are using mylibrary::my_add_op2, so the function
15     # name must be my_add_op2; otherwise, it will throw an error
16     # when this script is run
17     pass
18
19 @my_add_op2.impl_abstract()
20 def my_add_op_impl_abstract_any_name_is_ok(x, y):
21     return torch.empty_like(x)
22
23 @my_add_op2.impl("cpu")
24 def my_add_op_impl_any_name_is_ok(x, y):

```

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```

25     x = torch.nn.functional.relu(x)
26     return x+y    # add x to itself, and round the result
27
28 class CustomFoo(torch.nn.Module):
29     def __init__(self):
30         super().__init__()
31         self.register_parameter('foo', torch.nn.Parameter(torch.tensor([1.5, 2.5, 3.5,
↪ 4.5])))
32         self.relu = torch.nn.ReLU()
33     def forward(self, x):
34         x = self.relu(x)
35         return my_add_op2(x, self.foo)
36
37
38 custom_opset = onnxscript.values.Opset(domain="com.k2fsa.org", version=1)
39
40 @onnxscript.script(custom_opset)
41 def custom_my_add(x, y):
42     return custom_opset.CustomOpOne2(x, y)
43
44 @torch.no_grad()
45 def main():
46     torch._dynamo.allow_in_graph(my_add_op2)
47     x = torch.randn(4)
48     custom_addandround_model = CustomFoo()
49     onnx_registry = torch.onnx.OnnxRegistry()
50     onnx_registry.register_op(
51         namespace="mylibrary", op_name="my_add_op2", overload="default", ↪
↪function=custom_my_add
52     )
53
54     export_options = torch.onnx.ExportOptions(onnx_registry=onnx_registry)
55     onnx_program = torch.onnx.dynamo_export(
56         custom_addandround_model, x, export_options=export_options
57     )
58     onnx_program.save("./f.onnx")
59     with open('f.txt', 'w') as f:
60         f.write(str(onnx_program.model_proto))
61     onnx_model = onnx.load("f.onnx")
62     onnx.checker.check_model(onnx_model)
63
64
65 if __name__ == '__main__':
66     main()

```

Listing 12: ./code/f.txt

```
1  ir_version: 8
2  opset_import {
3    domain: "pkg.onnxscript.torch_lib"
4    version: 1
5  }
6  opset_import {
7    domain: "pkg.torch.2.4.0+cpu"
8    version: 1
9  }
10 opset_import {
11   domain: "com.k2fsa.org"
12   version: 1
13 }
14 opset_import {
15   domain: ""
16   version: 18
17 }
18 opset_import {
19   domain: "pkg.onnxscript.torch_lib.common"
20   version: 1
21 }
22 producer_name: "pytorch"
23 producer_version: "2.4.0"
24 graph {
25   node {
26     input: "l_x_"
27     output: "relu_1"
28     name: "torch.nn.modules.activation.ReLU_relu_1_0_aten_relu_0_n0"
29     op_type: "Relu"
30   }
31   node {
32     input: "relu_1"
33     input: "foo"
34     output: "my_add_op2"
35     name: "custom_my_add_1_n0"
36     op_type: "CustomOpOne2"
37     domain: "com.k2fsa.org"
38   }
39   name: "main_graph"
40   initializer {
41     dims: 4
42     data_type: 1
```

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```
43   name: "foo"
44   raw_data: "\000\000\300?\000\000 @\000\000`@\000\000\220@"
45 }
46 input {
47   name: "l_x_"
48   type {
49     tensor_type {
50       elem_type: 1
51       shape {
52         dim {
53           dim_value: 4
54         }
55       }
56     }
57   }
58 }
59 output {
60   name: "my_add_op2"
61   type {
62     tensor_type {
63       elem_type: 1
64       shape {
65         dim {
66           dim_value: 4
67         }
68       }
69     }
70   }
71 }
72 value_info {
73   name: "relu_1"
74   type {
75     tensor_type {
76       elem_type: 1
77       shape {
78         dim {
79           dim_value: 4
80         }
81       }
82     }
83   }
84 }
85 }
```

Listing 13: ./code/custom-op-3.cc

```

1  /*
2  references:
3  https://onnxruntime.ai/docs/reference/operators/add-custom-op.html
4
5  */
6  #include "onnxruntime_lite_custom_op.h"
7  #include <iostream>
8  #include <utility>
9  #include <vector>
10
11 static void KernelOne(const Ort::Custom::Tensor<uint8_t> &X,
12                      const Ort::Custom::Tensor<float> &scale_tensor,
13                      Ort::Custom::Tensor<float> &Y) {
14     auto input_shape = X.Shape();
15     auto x_raw = X.Data();
16     auto scale = scale_tensor.Data()[0];
17     auto y_raw = Y.Allocate(input_shape);
18     for (int64_t i = 0; i < Y.NumberOfElement(); ++i) {
19
20         // scale each uint8 number
21         y_raw[i] = x_raw[i] * scale;
22     }
23 }
24
25 static Ort::CustomOpDomain TestCustomOp3() {
26     Ort::CustomOpDomain v1_domain{"com.k2fsa.org"};
27     // please make sure that custom_op_one has the same lifetime as the consuming
28     // session
29     //
30     // Here we use a static variable so it is never released.
31     // in practice, we can move it to a member variable of a class
32     static std::unique_ptr<Ort::Custom::OrtLiteCustomOp> custom_op_one{
33         Ort::Custom::CreateLiteCustomOp("MyCast", "CPUExecutionProvider",
34                                         KernelOne)};
35     v1_domain.Add(custom_op_one.get());
36
37     return v1_domain;
38 }
39
40 void TestCustomModel3() {
41     Ort::Env env;
42     Ort::SessionOptions sess_opts;

```

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```

43 sess_opts.SetIntraOpNumThreads(1);
44 sess_opts.SetInterOpNumThreads(1);
45
46 Ort::CustomOpDomain v1_domain = TestCustomOp3();
47
48 Ort::SessionOptions session_options;
49 sess_opts.Add(v1_domain);
50 // create a session with the session_options ...
51
52 std::unique_ptr<Ort::Session> sess =
53     std::make_unique<Ort::Session>(env, "./g.onnx", sess_opts);
54
55 auto memory_info =
56     Ort::MemoryInfo::CreateCpu(OrtDeviceAllocator, OrtMemTypeDefault);
57 // foo is [10, 20]
58 //
59
60 std::vector<float> x = {1, 2};
61
62 std::array<int64_t, 1> shape = {2};
63
64 Ort::Value x_tensor = Ort::Value::CreateTensor(
65     memory_info, x.data(), x.size(), shape.data(), shape.size());
66
67 std::vector<Ort::Value> inputs;
68 inputs.push_back(std::move(x_tensor));
69
70 std::vector<const char*> input_names = {"l_x_"};
71 std::vector<const char*> output_names = {"add"};
72 auto out = sess->Run({}, input_names.data(), inputs.data(), inputs.size(),
73     output_names.data(), output_names.size());
74 const float *p = out[0].GetTensorData<float>();
75 for (int i = 0; i < 2; ++i) {
76     std::cout << p[i] << "\n ";
77 }
78 }

```

Listing 14: ./code/g.py

```

1 #!/usr/bin/env python3
2
3 from torch._custom_op import impl as custom_op
4 import torch

```

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```

5 import onnx
6 import onnxscript
7 from onnxscript import opset18
8
9 import warnings
10 warnings.filterwarnings("ignore")
11
12 @custom_op.custom_op("mylibrary::my_cast")
13 def my_cast(x: torch.Tensor, scale: float = 0.25) -> torch.Tensor:
14     # Since we are using mylibrary::my_cast, so the function
15     # name must be my_cast; otherwise, it will throw an error
16     # when this script is run
17     return x.to(torch.float32)
18
19 @my_cast.impl_abstract()
20 def my_cast_impl_abstract_any_name_is_ok(x, scale: float = 0.25):
21     return x.to(torch.float32)
22
23 @my_cast.impl("cpu")
24 def my_cast_impl_any_name_is_ok(x, scale: float = 0.25):
25     return x.to(torch.float32) # add x to itself, and round the result
26
27 class CustomFoo(torch.nn.Module):
28     def __init__(self):
29         super().__init__()
30         self.register_buffer('foo', torch.tensor([10, 20], dtype=torch.uint8))
31         self.scale = 0.125
32
33     def forward(self, x):
34         return x + my_cast(self.foo, self.scale)
35
36
37 custom_opset = onnxscript.values.Opset(domain="com.k2fsa.org", version=1)
38
39 @onnxscript.script(custom_opset, default_opset=opset18)
40 def custom_my_cast(x, scale: float = 0.5):
41     return custom_opset.MyCast(x, scale)
42
43 @torch.no_grad()
44 def main():
45     torch._dynamo.allow_in_graph(my_cast)
46     x = torch.randn(2)
47     custom_addandround_model = CustomFoo()

```

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```

48     onnx_registry = torch.onnx.OnnxRegistry()
49     onnx_registry.register_op(
50         namespace="mylibrary", op_name="my_cast", overload="default", function=custom_
↪my_cast
51     )
52
53     export_options = torch.onnx.ExportOptions(onnx_registry=onnx_registry)
54     onnx_program = torch.onnx.dynamo_export(
55         custom_addandround_model, x, export_options=export_options
56     )
57     onnx_program.save("./g.onnx")
58     with open('g.txt', 'w') as f:
59         f.write(str(onnx_program.model_proto))
60     onnx_model = onnx.load("g.onnx")
61     onnx.checker.check_model(onnx_model)
62
63
64 if __name__ == '__main__':
65     main()

```

Listing 15: ./code/g.txt

```

1  ir_version: 8
2  opset_import {
3      domain: "com.k2fsa.org"
4      version: 1
5  }
6  opset_import {
7      domain: "pkg.onnxscript.torch_lib"
8      version: 1
9  }
10 opset_import {
11     domain: ""
12     version: 18
13 }
14 opset_import {
15     domain: "pkg.onnxscript.torch_lib.common"
16     version: 1
17 }
18 producer_name: "pytorch"
19 producer_version: "2.4.0"
20 graph {
21     node {

```

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```

22     output: "custom_my_cast_0_scale"
23     name: "custom_my_cast_0_n0"
24     op_type: "Constant"
25     attribute {
26         name: "value_float"
27         type: FLOAT
28         f: 0.125
29     }
30 }
31 node {
32     input: "foo"
33     input: "custom_my_cast_0_scale"
34     output: "my_cast"
35     name: "custom_my_cast_0_n1"
36     op_type: "MyCast"
37     domain: "com.k2fsa.org"
38 }
39 node {
40     input: "l_x_"
41     input: "my_cast"
42     output: "add"
43     name: "aten_add_1"
44     op_type: "aten_add"
45     domain: "pkg.onnxscript.torch_lib"
46     attribute {
47         name: "alpha"
48         type: FLOAT
49         f: 1
50     }
51 }
52 name: "main_graph"
53 initializer {
54     dims: 2
55     data_type: 2
56     name: "foo"
57     raw_data: "\n\024"
58 }
59 input {
60     name: "l_x_"
61     type {
62         tensor_type {
63             elem_type: 1
64             shape {

```

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```
65         dim {
66             dim_value: 2
67         }
68     }
69 }
70 }
71 }
72 output {
73     name: "add"
74     type {
75         tensor_type {
76             elem_type: 1
77             shape {
78                 dim {
79                     dim_value: 2
80                 }
81             }
82         }
83     }
84 }
85 value_info {
86     name: "custom_my_cast_0_scale"
87     type {
88         tensor_type {
89             elem_type: 1
90             shape {
91             }
92         }
93     }
94 }
95 value_info {
96     name: "my_cast"
97     type {
98         tensor_type {
99             elem_type: 1
100             shape {
101                 dim {
102                     dim_value: 2
103                 }
104             }
105         }
106     }
107 }
```

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```
108 value_info {
109   name: "pkg.onnxscript.torch_lib::aten_add/self"
110   type {
111     tensor_type {
112       elem_type: 1
113       shape {
114         dim {
115           dim_value: 2
116         }
117       }
118     }
119   }
120 }
121 value_info {
122   name: "pkg.onnxscript.torch_lib::aten_add/other"
123   type {
124     tensor_type {
125       elem_type: 1
126       shape {
127         dim {
128           dim_value: 2
129         }
130       }
131     }
132   }
133 }
134 value_info {
135   name: "pkg.onnxscript.torch_lib::aten_add/alpha"
136   type {
137     tensor_type {
138       elem_type: 1
139       shape {
140       }
141     }
142   }
143 }
144 value_info {
145   name: "pkg.onnxscript.torch_lib::aten_add/other_1"
146   type {
147     tensor_type {
148       elem_type: 1
149       shape {
150         dim {
```

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```
151         dim_value: 2
152     }
153 }
154 }
155 }
156 }
157 value_info {
158     name: "pkg.onnxscript.torch_lib::aten_add/return_val"
159     type {
160         tensor_type {
161             elem_type: 1
162             shape {
163                 dim {
164                     dim_value: 2
165                 }
166             }
167         }
168     }
169 }
170 }
171 functions {
172     name: "aten_add"
173     input: "self"
174     input: "other"
175     output: "return_val"
176     attribute_proto {
177         name: "alpha"
178         type: FLOAT
179         f: 1
180     }
181     node {
182         output: "alpha"
183         name: "n0"
184         op_type: "Constant"
185         attribute {
186             name: "value_float"
187             ref_attr_name: "alpha"
188             type: FLOAT
189         }
190     }
191     node {
192         input: "other"
193         input: "alpha"
```

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```

194     output: "other_1"
195     name: "n2"
196     op_type: "Mul"
197 }
198 node {
199     input: "self"
200     input: "other_1"
201     output: "return_val"
202     name: "n3"
203     op_type: "Add"
204 }
205 doc_string: "add.Tensor(Tensor self, Tensor other, *, Scalar alpha=1) -> Tensor"
206 opset_import {
207     domain: ""
208     version: 18
209 }
210 domain: "pkg.onnxscript.torch_lib"
211 }

```

38.4.3 Internals

OrtApiBase

It is a struct containing two function pointers.

```

// https://github.com/microsoft/onnxruntime/blob/main/include/onnxruntime/core/
↪session/onnxruntime_c_api.h
struct OrtApiBase {
    const OrtApi*(ORT_API_CALL* GetApi)(uint32_t version) NO_EXCEPTION;
    const char*(ORT_API_CALL* GetVersionString)(void) NO_EXCEPTION;
};

```

In https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/session/onnxruntime_c_api.cc, it uses a lot of static assertions

```

// https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/session/
↪onnxruntime_c_api.cc

// OrtApiBase can never change as there is no way to know what version of OrtApiBase_
↪is returned by OrtGetApiBase.
static_assert(sizeof(OrtApiBase) == sizeof(void*) * 2, "New methods can't be added to_
↪OrtApiBase as it is not versioned");

```

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```
static_assert(offsetof(OrtApiBase, GetApi) / sizeof(void*) == 0, "These functions_
↳cannot be reordered");
static_assert(offsetof(OrtApiBase, GetVersionString) / sizeof(void*) == 1, "These_
↳functions cannot be reordered");
static_assert(std::is_same_v<decltype(OrtApiBase::GetApi), const OrtApi*(ORT_API_
↳CALL*) (uint32_t)NO_EXCEPTION>, "This function's signature can never change");
static_assert(std::is_same_v<decltype(OrtApiBase::GetVersionString), const char*(ORT_
↳API_CALL*) (void)NO_EXCEPTION>, "This function's signature can never change");
```

It defines a global variable in https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/session/onnxruntime_c_api.cc

```
static constexpr OrtApiBase ort_api_base = {
    &OrtApis::GetApi,
    &OrtApis::GetVersionString};

const OrtApiBase* ORT_API_CALL OrtGetApiBase(void) NO_EXCEPTION {
    return &ort_api_base;
}

ORT_API(const OrtApi*, OrtApis::GetApi, uint32_t version) {
    if (version >= 1 && version <= ORT_API_VERSION)
        return &ort_api_1_to_20;

    fprintf(stderr,
        "The requested API version [%u] is not available, only API versions [1, %u]_
↳are supported in this build."
        " Current ORT Version is: %s\n",
        version, ORT_API_VERSION, ORT_VERSION);

    return nullptr; // Unsupported version
}
```

OrtApi

It is a struct containing lots of function pointers. It is initialized in https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/session/onnxruntime_c_api.cc#L2399

```
static constexpr OrtApi ort_api_1_to_20 = {
    // NOTE: The ordering of these fields MUST not change after that version has_
↳shipped since existing binaries depend on this ordering.

    // Shipped as version 1 - DO NOT MODIFY (see above text for more information)
```

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```

    &OrtApis::CreateStatus,
    &OrtApis::GetErrorCode,
    &OrtApis::GetErrorMessage,
    ...
};

```

The initialized function pointers are declared in https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/session/ort_apis.h

OrtStatus

It is defined in https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/framework/error_code.cc#L12

```

struct OrtStatus {
    OrtErrorCode code;
    char msg[1]; // a null-terminated string
};

```

So usually a pointer to it is used. The way to create a pointer to it is given in https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/framework/error_code.cc#L24

```

inline OrtStatus* NewStatus(size_t clen) {
    auto* buf = new (std::nothrow) uint8_t[sizeof(OrtStatus) + clen];
    if (buf == nullptr) return nullptr; // OOM. What we can do here? abort()?
    return new (buf) OrtStatus;
}

```

Note that it uses placement new.

Its usage is given below:

CreateStatus

```

// https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/framework/
error_code.cc

// Even we say it may not return NULL, indeed it may.
_Check_return_ _Ret_notnull_ OrtStatus* ORT_API_CALL
OrtApis::CreateStatus(OrtErrorCode code,
                                                                _In_z_
const char* msg) NO_EXCEPTION {
    assert(!(code == 0 && msg != nullptr));

```

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```

SafeInt<size_t> clen(nullptr == msg ? 0 : strlen(msg, onnxruntime::kMaxStrLen));
OrtStatus* p = NewStatus(clen);
if (p == nullptr)
    return nullptr;
p->code = code;
memcpy(p->msg, msg, clen);
p->msg[clen] = '\\0';
return p;
}

```

ReleaseStatus

We need to free the pointer after using it to avoid memory leak.

struct OrtApi provides two function pointers to create and release OrtStatus:

```

OrtStatus*(ORT_API_CALL* CreateStatus)(OrtErrorCode code, _In_ const char* msg)NO_
↳ EXCEPTION ORT_ALL_ARGS_NONNULL;

ORT_CLASS_RELEASE(Status);

```

where ORT_CLASS_RELEASE is a macro and is defined as:

```

#define ORT_CLASS_RELEASE(X) void(ORT_API_CALL * Release##X)(_Frees_ptr_opt_ Ort##X *↳
↳ input)

```

```

// https://github.com/microsoft/onnxruntime/blob/main/onnxruntime/core/framework/
↳ error_code.cc

```

```

ORT_API(void, OrtApis::ReleaseStatus, _Frees_ptr_opt_ OrtStatus* value) { delete[]↳
↳ reinterpret_cast<uint8_t*>(value); }

```

OrtErrorCode

OrtErrorCode is defined in https://github.com/microsoft/onnxruntime/blob/main/include/onnxruntime/core/session/onnxruntime_c_api.h#L245

```

typedef enum OrtErrorCode {
    ORT_OK,
    ORT_FAIL,
    ORT_INVALID_ARGUMENT,
    ORT_NO_SUCHFILE,

```

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```

    ORT_NO_MODEL,
    ORT_ENGINE_ERROR,
    ORT_RUNTIME_EXCEPTION,
    ORT_INVALID_PROTOBUF,
    ORT_MODEL_LOADED,
    ORT_NOT_IMPLEMENTED,
    ORT_INVALID_GRAPH,
    ORT_EP_FAIL,
} OrtErrorCode;

```

GetApi

It is defined in `onnxruntime_cxx_api.h`. The function returns a global variable. We can use a macro to control how this global variable is initialized.

If the macro `ORT_API_MANUAL_INIT` is defined, then we have to invoke `InitApi()` by ourselves. We can also invoke an overload version `InitApi(const OrtApi* api) noexcept`.

By default, `ORT_API_MANUAL_INIT` is not defined and we have

```

template <typename T>
const OrtApi* Global<T>::api_ = OrtGetApiBase()->GetApi(ORT_API_VERSION);

```

```

/// This returns a reference to the OrtApi interface in use
inline const OrtApi& GetApi() noexcept { return *Global<void>::api_; }

```

Note that it uses a global variable template.

OrtAllocator

In the C++ code of `onnxruntime`, there is a macro `USE_MIMALLOC`. If it is defined, then <https://github.com/microsoft/mimalloc> is used.

During allocation, the return pointer is ensured to be `xx` bytes aligned, where `xx` is a constant.

38.5 protobuf notes

38.5.1 repeated

See

- [<https://protobuf.dev/reference/python/python-generated/#repeated-fields>](https://protobuf.dev/reference/python/python-generated/#repeated-fields)

Two examples below:

- 1. the repeated field contains messages
- 2. the repeated field contain primitive fields

```
message NodeProto {
  repeated string input = 1;    // namespace Value
}

message TensorShapeProto {
  message Dimension {
    oneof value {
      int64 dim_value = 1;
      string dim_param = 2;    // namespace Shape
    };
    // Standard denotation can optionally be used to denote tensor
    // dimensions with standard semantic descriptions to ensure
    // that operations are applied to the correct axis of a tensor.
    // Refer to https://github.com/onnx/onnx/blob/main/docs/DimensionDenotation.md
    ↪ #denotation-definition
    // for pre-defined dimension denotations.
    optional string denotation = 3;
  };
  repeated Dimension dim = 1;
}
```

Usage in Python:

```
#!/usr/bin/env python3

import onnx

# Works as a python list, but it is not a Python list

def test1():
    n = onnx.NodeProto()
    assert len(n.input) == 0, len(n.input)
    n.input.append("i0")
    n.input.append("i1")

    n.input.extend(["i2", "i3"])
    print(n)
    """
    input: "i0"
```

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```
input: "i1"
input: "i2"
input: "i3"
"""

assert len(n.input) == 4
assert n.input[0] == "i0"
assert n.input[1] == "i1"
assert n.input[2] == "i2"
assert n.input[3] == "i3"

del n.input[0]
assert len(n.input) == 3
assert n.input == ["i1", "i2", "i3"], n.input

# There are at least 3 methods to clear a repeated fields

# Method 1: Use ClearField()

n.ClearField("input")
assert len(n.input) == 0

# Method 2: Use del
n.input.append("ii")
assert len(n.input) == 1
del n.input[:]

assert len(n.input) == 0

# Method 3, use pop
n.input.append("1")
n.input.append("2")
n.input.extend(["3", "4", "5"])

while len(n.input) > 0:
    n.input.pop()

assert len(n.input) == 0

# Method 4
n.input.append("10")
assert len(n.input) == 1
n.input[:] = []
```

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```
assert len(n.input) == 0

def test2():
    s = onnx.TensorShapeProto()
    assert len(s.dim) == 0, len(s.dim)

    d = s.dim.add()
    assert id(d) == id(s.dim[0]), (id(d), id(s.dim[0]))

    d.dim_value = 1

    p = s.dim.add()
    p.dim_param = "N"

    # Copy an existing one
    k = onnx.TensorShapeProto.Dimension()
    k.dim_value = 10

    s.dim.append(k)
    assert s.dim[2] == k, (s.dim[2], k)

    # It is a copy, so id is different
    assert id(s.dim[2]) != id(k), (id(s.dim[2]), id(k))

    print(s)
    """
    dim {
      dim_value: 1
    }
    dim {
      dim_param: "N"
    }
    dim {
      dim_value: 10
    }
    """

def main():
    test1()
    test2()
```

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```
if __name__ == "__main__":  
    main()
```

38.6 onnxscript

38.6.1 Basics

```
#!/usr/bin/env python3  
  
from onnxscript import opset15 as op  
from onnxscript import script, FLOAT  
import onnx  
import torch  
import onnxruntime as ort  
  
class Model:  
    def __init__(self, filename):  
        session_opts = ort.SessionOptions()  
        self.session_opts = session_opts  
        self.model = ort.InferenceSession(  
            filename,  
            sess_options=self.session_opts,  
            providers=["CPUExecutionProvider"],  
        )  
  
    def run(  
        self,  
        x: torch.Tensor,  
    ) -> torch.Tensor:  
        y = self.model.run(  
            [  
                self.model.get_outputs()[0].name,  
            ],  
            {  
                self.model.get_inputs()[0].name: x.numpy(),  
            },  
        )[0]  
        return torch.from_numpy(y)  
  
@script()
```

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```

def Softplus(x: FLOAT):
    return op.Log(op.Exp(x) + 1.0)

def main():
    # fp = Softplus.to_function_proto()
    # print(fp)
    mp = Softplus.to_model_proto()
    print('type(mp)', type(mp)) # <class 'onnx.onnx_ml_pb2.ModelProto'>
    mp.producer_name = 'next-gen kaldi'
    mp.producer_version = '1.10.22'
    mp.model_version = 99
    mp.doc_string = 'my first example model with onnxscript'

    # add meta data
    p = mp.metadata_props.add()
    p.key = 'my-numbers'
    p.value = '1,2,3'

    onnx.save_model(mp, 'a.onnx')
    print(mp)

    x = torch.tensor([1, 0.5])
    m = Model('a.onnx')
    y = m.run(x)
    print(x, y)

    y2 = Softplus(x.numpy())
    print(y2)

if __name__ == '__main__':
    main()

```

```

type(mp) <class 'onnx.onnx_ml_pb2.ModelProto'>
ir_version: 8
opset_import {
  domain: ""
  version: 15
}
producer_name: "next-gen kaldi"
producer_version: "1.10.22"
model_version: 99
doc_string: "my first example model with onnxscript"

```

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```
graph {
  node {
    input: "x"
    output: "tmp"
    name: "n0"
    op_type: "Exp"
    domain: ""
  }
  node {
    output: "const"
    name: "n1"
    op_type: "Constant"
    domain: ""
    attribute {
      name: "value"
      type: TENSOR
      t {
        data_type: 1
        float_data: 1
        name: "const"
      }
    }
  }
  node {
    input: "const"
    input: "tmp"
    output: "const_cast"
    name: "n2"
    op_type: "CastLike"
    domain: ""
  }
  node {
    input: "tmp"
    input: "const_cast"
    output: "tmp_0"
    name: "n3"
    op_type: "Add"
    domain: ""
  }
  node {
    input: "tmp_0"
    output: "return_val"
    name: "n4"
  }
}
```

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```
    op_type: "Log"
    domain: ""
  }
  name: "Softplus"
  input {
    name: "x"
    type {
      tensor_type {
        elem_type: 1
        shape {
        }
      }
    }
  }
  output {
    name: "return_val"
  }
}
metadata_props {
  key: "my-numbers"
  value: "1,2,3"
}

tensor([1.0000, 0.5000]) tensor([1.3133, 0.9741])
[1.3132616 0.974077 ]
```


39.1 Basics

- [Interop with Native Libraries](#)
- [Default marshalling for strings](#)
- [cbrumme's WebLog](#)
- [.NET Standard](#)
- [Preprocessor symbols](#)
- [Packaging Icon within the nupkg](#)
- [MSBuild reference for .NET SDK projects](#)
- [Visual Studio for Mac documentation](#)
- [DllImportAttribute Class](#)
- [A Closer Look at Platform Invoke](#)
- [Identifying Functions in DLLs](#)
- [DllImportAttribute.CallingConvention Field](#)
- [C# Coding Conventions](#)
- [IDisposable.Dispose Method](#)
- [Quickstart: Create and publish a package with the dotnet CLI](#)
- [Creating a NuGet package for a library with platform-specific API](#)
- [.NET RID Catalog](#)

39.1.1 Use the commandline

```
# To activate the developer commandline
"C:\Program Files (x86)\Microsoft Visual Studio\2019\Community\VC\Auxiliary\Build\
↪vcvars64.bat"
```

```
# it generates ./a.exe
csc ./a.cs
```

```
# it generates b.dll
csc /target:library b.cs

# it generates a.exe
csc /r:b.dll a.cs
```

```
dumbpin -exports sherpa-onnx-c-api.dll
```

If the name in the C++ function is Hello, we can use either:

```
[DllImport("dllname")]
void Hello();
```

or

```
[DllImport("dllname", EntryPoint="Hello")]
void HelloANewName();
```

39.1.2 Visual Studio for Mac

After installation, start it and click Visual Studio -> About Visual studio -> Show details, it will show the locations:

- Runtime: /usr/local/share/dotnet/dotnet
- SDK: /usr/local/share/dotnet/sdk/7.0.203/Sdks

```
$ /usr/local/share/dotnet/dotnet

Usage: dotnet [options]
Usage: dotnet [path-to-application]

Options:
  -h|--help          Display help.
```

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```
--info          Display .NET information.
--list-sdks      Display the installed SDKs.
--list-runtimes  Display the installed runtimes.

path-to-application:
    The path to an application .dll file to execute.
```

39.1.3 global usings

Requires C# >= 10.0

Create a file, e.g., GlobalUsings.cs:

39.1.4 nullable types

```
Product? p = products[0];
string val;
if (p != null) {
    val = p.Name;
} else {
    val = "No value";
}
return View(new string[] { val });

string? val = products[0]?.Name;
if (val != null) {
    return View(new string[] { val });
}
return View(new string[] { "No Value" });

return View(new string[] { products[0]?.Name ?? "No Value" });

return View(new string[] { products[0]!.Name });
```

39.1.5 Dictionary

```
Dictionary<string, Product> products = new Dictionary<string, Product> {
    { "Kayak", new Product { Name = "Kayak", Price = 275M } },
    { "Lifejacket", new Product { Name = "Lifejacket", Price = 48.95M } }
};

Dictionary<string, Product> products = new Dictionary<string, Product> {
    ["Kayak"] = new Product { Name = "Kayak", Price = 275M },
    ["Lifejacket"] = new Product { Name = "Lifejacket", Price = 48.95M }
};

Dictionary<string, Product> products = new() {
    ["Kayak"] = new Product { Name = "Kayak", Price = 275M },
    ["Lifejacket"] = new Product { Name = "Lifejacket", Price = 48.95M }
};
```

39.1.6 foreach

```
public class ShoppingCart : IEnumerable<Product?> {
    public IEnumerable<Product?> Products { get; set; }

    public IEnumerator<Product?> GetEnumerator() => Products?.GetEnumerator() ??
↳ Enumerable.Empty<Product?>().GetEnumerator();

    IEnumerator IEnumerable.GetEnumerator() => GetEnumerator();
}
```

39.1.7 my first page

```
@model MyMessage
@{
    Layout = null;
}
<!DOCTYPE html>

<html>
<head>
    <meta name="viewport" content="width=device-width"/>
    <title>Hello my page</title>
    <style>
```

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```

</style>
</head>
<body>
  <h1> please fill the form</h1>
  <form asp-action="Index" method="post">
    <div>
      <label asp-for="Id">Id</label>
      <input asp-for="Id"/>
    </div>
    <div>
      <label asp-for="Name">Name</label>
      <input asp-for="Name"/>
    </div>
    <button type="submit">submit</button>
    <a asp-action="Index">clear</a>
  </form>
</body>
</html>

```

39.1.8 EF

- Microsoft.EntityFrameworkCore.SQLite
- Microsoft.VisualStudio.Web.CodeGeneration.Design
- Microsoft.EntityFrameworkCore.SqlServer
- Microsoft.EntityFrameworkCore.Design
- Microsoft.EntityFrameworkCore.Tools

```

dotnet tool uninstall --global dotnet-aspnet-codegenerator
dotnet tool install --global dotnet-aspnet-codegenerator
dotnet tool uninstall --global dotnet-ef
dotnet tool install --global dotnet-ef

export PATH=$HOME/.dotnet/tools:$PATH

```

```

dotnet aspnet-codegenerator controller -name MoviesController -m Movie -dc MvcMovie.
↪Data.MvcMovieContext --relativeFolderPath Controllers --useDefaultLayout --
↪referenceScriptLibraries -sqlite

```

```

Building project ...
Finding the generator 'controller'...

```

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```
Running the generator 'controller'...
--useSqlite|-sqlite option is obsolete now. Use --databaseProvider|-dbProvider
↳ instead in the future.

Minimal hosting scenario!
Generating a new DbContext class 'MvcMovie.Data.MvcMovieContext'
Attempting to compile the application in memory with the added DbContext.
Attempting to figure out the EntityFramework metadata for the model and DbContext:
↳ 'Movie'

Using database provider 'Microsoft.EntityFrameworkCore.Sqlite'!

Added DbContext : '/Data/MvcMovieContext.cs'
Added Controller : '/Controllers/MoviesController.cs'.
Added View : /Views/Movies/Create.cshtml
Added View : /Views/Movies/Edit.cshtml
Added View : /Views/Movies/Details.cshtml
Added View : /Views/Movies/Delete.cshtml
Added View : /Views/Movies/Index.cshtml
```

```
diff --git a/MvcMovie/Program.cs b/MvcMovie/Program.cs
index 9fbb57d..b96f671 100644
--- a/MvcMovie/Program.cs
+++ b/MvcMovie/Program.cs
@@ -1,4 +1,9 @@
-var builder = WebApplication.CreateBuilder(args);
+using Microsoft.EntityFrameworkCore;
+using Microsoft.Extensions.DependencyInjection;
+using MvcMovie.Data;
+var builder = WebApplication.CreateBuilder(args);
+builder.Services.AddDbContext<MvcMovieContext>(options =>
+    options.UseSqlite(builder.Configuration.GetConnectionString("MvcMovieContext") ??
+↳ throw new InvalidOperationException("Connection string 'MvcMovieContext' not found.
+↳ ")));

// Add services to the container.
builder.Services.AddControllersWithViews();
```

```
diff --git a/MvcMovie/appsettings.json b/MvcMovie/appsettings.json
index af0538f..d8b9276 100644
--- a/MvcMovie/appsettings.json
+++ b/MvcMovie/appsettings.json
@@ -1,10 +1,12 @@
```

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```

- {
+ {
    "Logging": {
        "LogLevel": {
            "Default": "Information",
            "Microsoft.AspNetCore": "Warning"
        }
    },
-   "AllowedHosts": "*"
- }
-
+   "AllowedHosts": "*",
+   "ConnectionStrings": {
+       "MvcMovieContext": "Data Source=MvcMovieContext-95c663f1-d863-4557-a405-
↪ 0d9cf818bb16.db"
+   }
+ }

```

```
dotnet ef migrations add InitialCreate
```

39.2 nuget

vosk is using non-SDK style, while the latest .NET doc uses SDK style.

```
msbuild ./t.csproj -t:targetName -p:Configuration=Release -verbosity:detail
```

```
dotnet msbuild -property:TargetFramework=netcoreapp2.0 -preprocess:output.xml
```

39.2.1 References

- Walkthrough: Use MSBuild
- Build items
- MSBuild items
- ItemGroup element (MSBuild)
- Use wildcards to specify items
- How to: Exclude files from the build
- MSBuild well-known item metadata

- How to: Select the files to build
- .NET project SDKs

Select file

```
// All cs files in the current directory
<CSFile Include="*.cs"/>

// All files in the D drive
<VBFile Include="D:/**/*.*.vb"/>

// Include all cs files except the file DoNotBuild.cs
<ItemGroup>
  <CSFile Include="*.cs" Exclude="DoNotBuild.cs"/>
</ItemGroup>

<Compile Include="*.cs" />
<Compile Include="*.res" Exclude="Form1.cs"> // note: Form1.cs is included in the
↪first statement

<CSFile Include="Form1.cs"/>
<CSFile Include="File1.cs;File2.cs"/>
<CSFile Include="*.cs"/>
<JPGFile Include="Images\**\*.jpg"/>

// All *.cs except Form2.cs and Form3.cs
<CSFile Include="*.cs" Exclude="Form2.cs;Form3.cs"/>

// all *.jpg files except the version2 directory
<JPGFile
  Include="Images\**\*.jpg"
  Exclude="Images\**\Version2\*.jpg"/>

// Include only in Release build
<Compile
  Include="Formula.vb"
  Condition="'$(Configuration)' == 'Release' " />
```

- To reference an item metadata: %(<ItemMetadataName>) or %(<ItemType.ItemMetaName>)
- To reference a property, use \$(property)
- To reference an item, use @(item)


```
<Project xmlns="http://schemas.microsoft.com/developer/msbuild/2003">
  <ItemGroup>
    <Stuff Include="One.cs" >
      <Display>false</Display>
    </Stuff>
    <Stuff Include="Two.cs">
      <Display>true</Display>
    </Stuff>
  </ItemGroup>
  <Target Name="Batching">
    <Message Text="@ (Stuff) " Condition=" '%(Display)' == 'true' "/>
  </Target>
</Project>

// Stuff is an item, so we use @(Stuff)
// Display is an item metadata, so we use %(Display)
// @ (Stuff) is "One.cs;Two.cs"
```

Some well-known item metadata %(FullPath), %(Filename)

39.2.2 publish a new package

It supports trusted publishing, like pub.dev for flutter. In addition, it can create and publish a new package from github actions, while flutter needs to first publish a new package locally.

39.3 dotnet

```
dotnet --list-sdks

# Create a directory and create a file global.json in it
# global.json contains the default sdk version to use
dotnet new globaljson --sdk-version 6.0.100 --output FirstProject

# Create an ASP.Net Core MVC web project, and put it inside the directory
# FirstProject.
dotnet new mvc --no-https --output FirstProject --framework net6.0

# Create a solution named FirstProject and put it inside ./FirstProject
dotnet new sln -o FirstProject

# Add the project "FirstProject" to the solution "FirstProject"
dotnet sln FirstProject add FirstProject
```

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```
# list supported projects
dotnet new --list
```

39.4 Blazor

Create a page:

```
dotnet new razorcomponent -n Todo -o Pages
```

First TODO page:

```
public class TodoItem {
    public string? Title {get; set;}
    public bool IsDone {get; set;} = false;
}
```

```
@page "/todo"

<h3>Todo (@todos.Count(todo => !todo.IsDone))</h3>

<ul>
    @foreach (var todo in todos)
    {
        <li>
            <input type="checkbox" @bind="todo.IsDone" />
            <input @bind="todo.Title" />
        </li>
    }
</ul>

<input placeholder="something to do" @bind="newTodo" @bind:event="oninput" />
<button @onclick="AddItem">Add</button>
<p> todo is @newTodo </p>

@code {
    private List<TodoItem> todos = new();
    private string? newTodo;

    private void AddItem() {
        todos.Add(new TodoItem { Title = newTodo});
    }
}
```

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```
newTodo = string.Empty;
}

}
```

On parameter set:

```
@page "/counter/{startingValue}"
@code {

    private int currentCount = 0;

    [Parameter]
    private int startingValue { get; set; } = 0

    protected override void OnParametersSet()
    {
        currentCount = startingValue;
        base.OnParametersSet();
    }
}
```

39.5 Hello

```
dotnet new console -h Hello
```

39.5.1 unsafe

```
using System.Diagnostics;
using System.Runtime.InteropServices;

/*
We have to add to ./Hello.csproj
  <AllowUnsafeBlocks>true</AllowUnsafeBlocks>
*/

public class UnsafeTest {
```

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```
class MyStruct {
    public int A;
    public char B;
}

public static void TestPointer() {
    // for a primitive (unmanaged type)
    int a = 3;
    unsafe {
        int*p = &a;
        *p = 10;
        Debug.Assert(a == 10);
        a = 20;
        Debug.Assert(p[0] == 20); // like C/C++
    }

    // now for an array. We have to ensure the address of the array
    // is not moved by using fixed. This is also called a pin operation.
    unsafe {
        byte[] b = new byte[5];
        fixed(byte*p = &b[0]) {
            // p is a constant inside this block due to the usage of fixed.
            // That is, p is a fixed variable
            //
            // Note: We can also use (byte*p = b)
            p[0] = 10;
            Debug.Assert(b[0] == 10);

            b[1] = 20;
            Debug.Assert(*(p+1) == 20);
            Debug.Assert(p[1] == 20);
            Console.WriteLine($"address of p: 0x{(long)p:X}");
            Console.WriteLine($"address of p+1: 0x{(long)(p+1):X}");
        }
    }

    // test struct
    MyStruct s = new MyStruct {A=10, B='B'};
    unsafe {
        MyStruct* p = &s;
        Debug.Assert(p->A == 10); // similar to C/C++
        Debug.Assert(p->B == 'B');
        p->B = 'C';
    }
}
```

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```
    Debug.Assert(s.B == 'C');
}

// from stackalloc
unsafe {
    int*p = stackalloc int[3];
    // the content is uninitialized
    p[0] = 10;
    ++p;
    p[0] = 20;
    ++p;
    Debug.Assert(p[-2] == 10);
    Debug.Assert(p[-1] == 20);
}

IntPtr pp = Marshal.AllocHGlobal(sizeof(int));
unsafe {
    int* p = (int*)pp.ToPointer();
    p[0] = 10;
}
int i = Marshal.ReadInt32(pp);
Debug.Assert(i == 10);

Marshal.FreeHGlobal(pp);

}
}
```

References:

- [Unmanaged types \(C# reference\)](#)
- [fixed statement - pin a variable for pointer operations](#)
- [stackalloc](#)
- [Pointer related operators - take the address of variables, dereference storage locations, and access memory locations](#)
- [Unsafe code, pointer types, and function pointers](#)
- [23 Unsafe code](#)
- [Dos and Don'ts of stackalloc](#)

39.5.2 Span

```
using System.Diagnostics;
using System.Runtime.InteropServices;

class SpanTest {
    public static void Test() {
        TestByteArray();
        TestStackAlloc();
        TestGlobalAlloc();
    }

    public static void TestByteArray() {
        byte[] b = new byte[10];
        Span<byte> s = new Span<byte>(b);
        // s points to the memory holds by b
        s[0] = 10;
        Debug.Assert(b[0] == 10);

        // s2 = b[3:5]
        Span<byte> s2 = s.Slice(start:3, length: 2);
        s2[0] = 12;
        s2[1] = 13;
        Debug.Assert(b[3] == 12);
        Debug.Assert(b[4] == 13);
    }

    public static void TestStackAlloc() {
        Span<int> s = stackalloc int[10];
        s.Clear(); // set all contents to default(int);
        Debug.Assert(s.Length == 10);
    }

    public static void TestGlobalAlloc() {
        IntPtr p = Marshal.AllocHGlobal(sizeof(int) * 2);
        Span<int> s;
        unsafe {
            // p.ToPointer() returns void*, so it has to be used in void*
            s = new Span<int>(p.ToPointer(), 2);
            s[0] = 10;
            s[1] = 11;
        }

        int a = Marshal.ReadInt32(p);
        int b = Marshal.ReadInt32(p, sizeof(int));
    }
}
```

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```
    Debug.Assert(a == s[0]);  
    Debug.Assert(b == s[1]);  
  
    Marshal.FreeHGlobal(p);  
}  
}
```

References:

-

39.5.3 File

39.5.4 TODO

- stackalloc
- unsafe code
- unit test
- Marshal, read/write memory
- MemoryStream, FileStream
- FileReader, BinaryWriter
- directory exists, file exists

40.1 Install

```
python3 -m venv flask
source flask/bin/activate
pip3 install flask
```

40.2 Hello

```
#!/usr/bin/env python3

# flask --app hello run

from flask import Flask
from markupsafe import escape

print("__name__", __name__)

app = Flask(__name__)

@app.route("/")
def hello_world():
    app.logger.debug("A value for debugging")
    app.logger.warning("A warning occurred (%d apples)", 42)
    app.logger.error("An error occurred")

    return "<p>Hello world!</p>"
```

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```
@app.route("/<name>")
def hello(name):
    return f"Hello, {escape(name)}!"

@app.route("/user/<username>")
def show_user_profile(username):
    # show the user profile for that user
    return f"User {escape(username)}"

# If a user uses /post/ab, it will show Not Found
@app.route("/post/<int:post_id>")
def show_post(post_id):
    print("type(post_id)", type(post_id)) # type(pos_id) <class 'int'>
    # show the post with the given id, the id is an integer
    return f"Post {post_id}"

@app.route("/path/<path:subpath>")
def show_subpath(subpath):
    print("type(subpath)", type(subpath)) # type(subpath) <class 'str'>
    # show the subpath after /path/
    return f"Subpath {escape(subpath)}"
```

```
flask --app hello run
```

The default filename is `app.py`. Since ours is `hello.py` we have to use `--app hello`.

To listen on all public IP addresses, use:

```
flask --app hello run --host=0.0.0.0
```

To enable debug mode, use `flask --app hello run --debug`

40.3 MySQL

Use the community edition.

40.3.1 Installation on macOS

See <https://dev.mysql.com/doc/mysql-macos-excerpt/5.7/en/macos-installation.html>

```
tar xvf mysql-8.0.33-macos13-x86_64.tar.gz

2023-05-28T15:14:16.627733Z 0 [System] [MY-013169] [Server] /Users/fangjun/software/
↳mysql/mysql-8.0.33-macos13-x86_64/bin/mysqld (mysqld 8.0.33) initializing of server
↳in progress as process 11553
2023-05-28T15:14:16.629538Z 0 [Warning] [MY-010159] [Server] Setting lower_case_table_
↳names=2 because file system for /Users/fangjun/software/mysql/mysql-8.0.33-macos13-
↳x86_64/data/ is case insensitive
2023-05-28T15:14:16.633467Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has
↳started.
2023-05-28T15:14:16.800011Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has
↳ended.
2023-05-28T15:14:17.600930Z 6 [Note] [MY-010454] [Server] A temporary password is
↳generated for root@localhost: ;;Geh%7QmF%<
2023-05-28T15:14:18.316658Z 0 [System] [MY-013172] [Server] Received SHUTDOWN from
↳user <via user signal>. Shutting down mysqld (Version: 8.0.33).

(py38) fangjuns-MacBook-Pro:mysql-8.0.33-macos13-x86_64 fangjun$ bin/mysql_ssl_rsa_
↳setup --datadir ./data

(py38) fangjuns-MacBook-Pro:mysql-8.0.33-macos13-x86_64 fangjun$ ./bin/mysql -u root -
↳p
# Enter the password generated in the above logs
# After logging, change the password to test

mysql> ALTER USER 'root'@localhost IDENTIFIED BY 'test';
mysql> EXIT
```

40.4 Node

40.4.1 Installation on macos

<https://nodejs.org/en/download>

```
wget https://nodejs.org/dist/v18.16.0/node-v18.16.0-darwin-x64.tar.gz
tar xvf node-v18.16.0-darwin-x64.tar.gz
export PATH=/Users/fangjun/software/node-v18.16.0-darwin-x64/bin:$PATH
```

41.1 Hello ARMv8

Starting from armv8, it has hardware floating point unit as well as NEON.

41.2 Registers

- Integer registers

<https://learn.microsoft.com/en-us/cpp/build/arm64-windows-abi-conventions?view=msvc-170#integer-registers>

41.3 Intrinsics

42.1 Add new disk

```
sudo fdisk -l  
sudo fdisk /dev/sdb  
sudo mkfs.ext4 /dev/sdb1  
sudo mkdir /mnt/sdb  
sudo mount /dev/sdb1 /mnt/sdb
```


43.1 TODOs

- The `path` package
- The `strconv` package
- The `strings` package
- The `sort` package
- The `io` package
- The `bytes` package
- The `strings` package
- The `os` package
- <https://go101.org/>

43.2 Install

See <https://go.dev/doc/install>

43.2.1 macos

```
cd ~/software
wget https://go.dev/dl/go1.20.5.darwin-amd64.tar.gz
tar xvf go1.20.5.darwin-amd64.tar.gz
export PATH=~/software/go/bin:$PATH

go install golang.org/x/tools/cmd/godoc
```

43.3 Cross compile

```
go env
```

```
GO111MODULE=""
GOARCH="amd64"
GOBIN=""
GOCACHE="/Users/fangjun/Library/Caches/go-build"
GOENV="/Users/fangjun/Library/Application Support/go/env"
GOEXE=""
GOEXPERIMENT=""
GOFLAGS=""
GOHOSTARCH="amd64"
GOHOSTOS="darwin"
GOINSECURE=""
GOMODCACHE="/Users/fangjun/go/pkg/mod"
GONOPROXY=""
GONOSUMDB=""
GOOS="darwin"
GOPATH="/Users/fangjun/go"
GOPRIVATE=""
GOPROXY="https://proxy.golang.org,direct"
GOROOT="/Users/fangjun/software/go"
GOSUMDB="sum.golang.org"
GOTMPDIR=""
GOTOOLDIR="/Users/fangjun/software/go/pkg/tool/darwin_amd64"
GOVCS=""
GOVERSION="go1.20.5"
GCCGO="gccgo"
GOAMD64="v1"
AR="ar"
CC="clang"
CXX="clang++"
CGO_ENABLED="1"
GOMOD="/dev/null"
GOWORK=""
CGO_CFLAGS="-O2 -g"
CGO_CPPFLAGS=""
CGO_CXXFLAGS="-O2 -g"
CGO_FFLAGS="-O2 -g"
CGO_LDFLAGS="-O2 -g"
PKG_CONFIG="pkg-config"
GOGCCFLAGS="-fPIC -arch x86_64 -m64 -pthread -fno-caret-diagnostics -Qunused-
```

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```
↪arguments -fmessage-length=0 -fdebug-prefix-map=/var/folders/fw/cg40_07j7cb18_p_
↪ymf0zhkr0000gn/T/go-build1403459817=/tmp/go-build -gno-record-gcc-switches -fno-
↪common"
```

43.3.1 Build for windows

```
GOOS=windows go build
```

43.3.2 Build for Linux

```
GOOS=linux go build
```

43.3.3 Build for macOS

```
GOOS=darwin go build
```

43.3.4 Build for Raspberry Pi (arm64)

```
GOOS=linux GOARCH=arm64 go build
```

43.3.5 Build for Raspberry Pi (arm32)

```
GOOS=linux GOARCH=arm go build
```

43.4 Basics

43.4.1 Environment variables

go env VARIABLE_NAME to view the value of the go environment variable with name VARIABLE_NAME.

```
go env GOPATH # example value: /Users/fangjun/go
go env GOARCH # example value: amd64, 386, arm
go env GOOS # example value: linux, darwin, windows
```

go help environment

43.4.2 Hello world

See <https://go.dev/doc/tutorial/getting-started>

```
go mod init example/hello
```

The above command will create a file `go.mod` in the current directory.

```
# content of go.mod
module example/hello

go 1.20
```

Note that `go mod init` does not create any directories. It only creates a file `go.mod` with the above content.

43.4.3 verbose build

```
go build -x -v
```

43.4.4 exported

For non-builtin functions and variables, if the name begins with an uppercase, then it is exported. Otherwise, it is not exported.

Note that for builtin functions, the above rule does not apply. For instance, both `print` and `println` are exported.

43.4.5 go vet

Use `go vet` to quick check the correctness of the code.

43.5 hello

Listing 1: `./code/hello2/ex.go`

```
package main

import "fmt"

func main() {
    fmt.Println("hello world")
}
```

To run the program, use:

```
go run ex.go
```

For a recent version of go, go build will throw the following error:

```
$ go build
go: cannot find main module, but found .git/config in /Users/fangjun/open-source/notes
    to create a module there, run:
    cd ../../../../.. && go mod init
```

But we can use go build ex.go, which works fine and will generate an executable ex.

To format the file in-place, use go fmt ex.go.

To see the help for a command, for instance the command for go fmt, use go help fmt.

To view the doc, run:

```
gofmt
# The above command is equivalent to
gofmt -http localhost:6060
# Then start your browser and visit http://localhost:6060
```

To install a tour of go, run

```
# see https://go.dev/tour/list
go install golang.org/x/website/tour@latest
```

```
import "fmt"
import "math/rand"
```

is equivalent to

```
import (
    "fmt"
    "math/rand"
)
```

Format specifier:

- %v, for any value
- %T, print the type
- %x, same as C/C++
- %s, for string and byte slice

```
# to view the help information of the function fmt.Printf
go doc fmt.Printf
```

```
go doc builtin.append
go doc builtin.make
go doc builtin.copy
go doc builtin.new
go doc builtin.len
go doc builtin.cap
```

43.6 array

An array is of fixed size and cannot be resized, like C/C++.

- `len(a) == cap(a)`, return the capacity of an array

Listing 2: `./code/array/main.go`

```
// go mod tidy
// go build ./main.go

package main

import (
    "fmt"
    "unsafe"
)

// array has value semantics. Assignment is done by value copy
// array cannot be resized
func test1() {
    fmt.Println("---test1---")
    var a [2]int           // zero initialized
    fmt.Println(a)         // [0 0]
    fmt.Println(cap(a))    // 2
    fmt.Println(len(a))    // length

    for i := 0; i < len(a); i++ {
        fmt.Println(a[i])
    }
    // 0
    // 0
}
```

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```

// i is the index
// v is the value
for i, v := range a {
    fmt.Println(i, v)
}
// 0 0
// 1 0

// to ignore the index, use _
for _, v := range a {
    fmt.Println(v)
}

// to ignore v
for i := range a {
    fmt.Println(a[i])
}

// infer the array size automatically
b := [...]int{1, 2, 3}
fmt.Println(b) // [1 2 3]

// c is a slice
// b[start:end], end is exclusive
// c contains b[0] and b[1]
var c []int = b[0:2]
fmt.Println(c, len(c)) // [1 2] 2

// creates an array, and return a slice that references the array
d := []int{1, 2}
fmt.Println(d) // [1 2]
}

func test2() {
    fmt.Println("---test2---")
    // func make([]T, len, cap) []T
    a := make([]int, 0, 10)
    b := a

    c := []int{0, 1, 2, 3}
    fmt.Println("len(c)", len(c), "cap(c)", cap(c)) // len(c) 4 cap(c) 4

    d := []int{4, 5, 6, 7}

```

(continues on next page)

(continued from previous page)

```

    e := []int{8, 9, 10, 11}
    // func append(s []T, x ...T) []T
    a = append(a, c...)
    a = append(a, d...)
    fmt.Println(a) // [0 1 2 3 4 5 6 7]
    b = b[0:10]
    fmt.Println(b) // [0 1 2 3 4 5 6 7 8 0 0]

    a = append(a, e...) // allocated new array
    fmt.Println(a)      // [0 1 2 3 4 5 6 7 8 9 10 11]
    fmt.Println(b)      // [0 1 2 3 4 5 6 7 8 0 0]
}

func test3() {
    fmt.Println("--test3--")

    // a slice is initialized to nil
    var a []int
    fmt.Println(a == nil) // true
    fmt.Println(len(a), cap(a)) // 0, 0
    fmt.Println(a) // []
    fmt.Printf("%#v\n", a) // []int(nil)
    // fmt.Println(a[0]) // panic: runtime error: index out of range
    ↪ [0] with length 0

    b := new([]int)
    fmt.Println(b == nil) // false
    fmt.Println(*b == nil) // true

    // the underlying array for c is of length 0 and capacity 0
    c := []int{}
    fmt.Println(c == nil) // false
    fmt.Println(len(c), cap(c)) // 0, 0
    fmt.Println(c) // []
    fmt.Printf("%#v\n", c) // []int{}
    // fmt.Println(c[0]) // panic: runtime error: index out of range
    ↪ [0] with length 0

    d := []int(nil)
    fmt.Println(d == nil) // true
}

func test4() {

```

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```

fmt.Println("--test4--")
var a []int
fmt.Println(a == nil) // true
a = append(a, 1)
fmt.Println(a[0] == 1) // true

var b []int = []int{}
fmt.Println(b != nil) // true
b = append(b, 2)
fmt.Println(b[0] == 2) // true

// the first argument of append() can be nil

c := append([]int(nil), 1)
fmt.Println(c[0] == 1) // true

d := append([]int{}, 1)
fmt.Println(d[0] == 1) // true

f := []int{1, 2, 3}
g := []int{10}
for _, v := range f {
    g = append(g, v)
}
fmt.Println(g) // [10 1 2 3]

h := []int{10}
h = append(h, f...) // recommended approach to append a slice
fmt.Println(h)      // [10 1 2 3]
}

func test5() {
    fmt.Println("---test5---")
    var a []int
    fmt.Println(a == nil) // true

    // pointer, length, capacity, 8 + 8 + 8 = 24 on a 64-bit computer
    fmt.Println(unsafe.Sizeof(a)) // 24
}

func test6() {
    fmt.Println("---test6---")
    var a []int

```

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```
b := []int{} // it is an error to use b := []int, []int (type) is not an
↳ expression

c := make([]int, 0)
fmt.Printf("%#v\n", a) // []int(nil)
fmt.Printf("%#v\n", b) // []int{}
fmt.Printf("%#v\n", c) // []int{}

d := make([]int, 1, 3)
fmt.Println(len(d)) // 1
fmt.Println(cap(d)) // 3
// fmt.Println(d[1]) // panic: runtime error: index out of range [1] with
↳ length 1

}

func main() {
    test1()
    test2()
    test3()
    test4()
    test5()
    test6()
}
```

43.6.1 slice

Differences between:

```
a := []int
```

and:

```
b := []int{}
```

43.7 Integers

Like C++, it uses UTF-8 encoding for strings in Go.

Listing 3: ./code/integers/main.go

```

package main

import (
    "fmt"
    "unsafe"
)

// int8 int16 int32 int64 int
// uint8 uint16 uint32 uint64 uint
//
// int != int32
// uint != uint32
//
// On 32-bit systems, int has 32 bits
// On 64-bit systems, int has 64 bits
func main() {
    var i8 int8 = 1
    fmt.Printf("i8 %[1]d %[1]T, %d\n", i8, unsafe.Sizeof(i8)) // int8 1 int8, 1

    i8 += 127 // wrap around, same like c/c++
    fmt.Printf("i8 %[1]d %[1]T\n", i8) // int8 -128 int8

    // Note i has 8 bytes on my 64-bit macOS
    i := 10
    fmt.Printf("i %[1]d %[1]T, %d\n", i, unsafe.Sizeof(i)) // i 10 int, 8
    fmt.Printf("i/3 %[1]d %[1]T\n", i/3) // i/3 3 int, same as
    ↪ c/c++

    var i32 int32 = 100
    fmt.Printf("i32 %[1]d %[1]T, %d\n", i32, unsafe.Sizeof(i32)) // i32 100 int32,
    ↪ 4

    fmt.Printf("%5d\n", i32) // ' 100'
    fmt.Printf("%05d\n", i32) // '00100'
    fmt.Printf("%-5d\n", i32) // '100 '
}

```

Two kinds of for loops for strings:

- iterating it byte by byte
- use for-range loop to iterate utf-8 encoded character by character

43.8 string

A string is immutable in Go, like Python.

Listing 4: ./code/string/main.go

```
package main

import "fmt"

func testChar() {
    c := 'a'
    fmt.Printf("%[1]d %[1]c %[1]T\n", c) // 97 a int32

    var a rune = 'a'
    fmt.Printf("%[1]d %[1]c %[1]T\n", a) // 97 a int32

    // rune is an alias to int32
}

func main() {
    s := "hello"
    s2 := fmt.Sprintf("%s world", s) // hello world

    fmt.Println(s, s2) // hello hello world

    // use %q to surround the output with double quotes ""
    fmt.Printf("%q, %s\n", s, s) // "hello", hello

    // raw string inside ``, like R"()" in C++
    // can contain multiple lines
    s3 := `how\a\nyou`
    fmt.Println(s3) // how\a\nyou

    // for loop

    // utf-8 encoding of ä is 0xC3 0xA4
    // utf-32 encoding of ä is 0x000000E4

    // iterating the utf-8 encoded bytes array
    s = "aäa"
    for i := 0; i < len(s); i++ {
        fmt.Printf("%d 0x%x, %T\n", i, s[i], s[i])
    }
}
```

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```

    // 0 0x61, uint8
    // 1 0xc3, uint8
    // 2 0xa4, uint8
    // 3 0x61, uint8
    fmt.Println("----")

    // use a range for loop;
    // returns a utf-8 encoded character in each iteration
    //
    // the type of i is int
    // note the type of c is int32
    for i, c := range s {
        fmt.Printf("%d 0x%x, %v, %s, %T\n", i, c, c, string(c), c)
    }
    // 0 0x61, 97, a, int32
    // 1 0xe4, 228, ä, int32
    // 3 0x61, 97, a, int32

    testChar()
}

```

43.9 for

There is only `i++` in go. `++i` is invalid.

There is no `while` in go. The `for` loop has many variants.

Listing 5: `./code/for/main.go`

```

package main

import (
    "fmt"
)

// iterate a slice
func test1() {
    fmt.Println("---test1---")
    a := []int{1, 2}
    for i := 0; i < len(a); i++ {
        fmt.Println(a[i])
    }
}

```

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```
    }
    // 1
    // 2

    // fmt.Println(i) // undefined: i
}

// iterate a slice
func test2() {
    fmt.Println("---test2---")
    a := []int{1, 2}

    // i is the index
    // v is the value
    // similar to enumerate() in Python
    for i, v := range a {
        fmt.Println(i, v)
    }
    // 0 1
    // 1 2
    // fmt.Println(i) // undefined: i
    // fmt.Println(v) // undefined: v
}

func test3() {
    fmt.Println("---test3---")
    a := []int{1, 2}

    // i is the index
    for i := range a {
        fmt.Println(i, a[i])
    }
    // 0 1
    // 1 2
}

func test4() {
    fmt.Println("---test4---")
    a := []int{1, 2}

    // i is the index
    for _, v := range a {
        fmt.Println(v)
    }
}
```

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```
    }
    // 1
    // 2
    // fmt.Println(i) // undefined: i
}

func test5() {
    fmt.Println("---test5---")
    a := []int{1, 2}

    i := 0
    for i < len(a) {
        fmt.Println(i, a[i])
        i += 1
    }
    // 0 1
    // 1 2
}

func test6() {
    fmt.Println("---test6---")
    a := []int{1, 2}

    i := 0
    for ; i < len(a); i += 1 {
        fmt.Println(i, a[i])
    }
    // 0 1
    // 1 2
    fmt.Println(i) // 2
}

func test7() {
    fmt.Println("---test7---")
    a := []int{1, 2}

    i := 0

    // like a while loop
    for {
        if i >= len(a) {
            break
        }
    }
}
```

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```
        fmt.Println(i, a[i])
        i += 1
    }
    // 0 1
    // 1 2
    fmt.Println(i) // 2
}

func test8() {
    // for go >= 1.22
    fmt.Println("---test8---")
    for i := range 3 {
        fmt.Println(i)
    }
    // 0
    // 1
    // 2
}

func main() {
    test1()
    test2()
    test3()
    test4()
    test5()
    test6()
    test7()
    test8()
}
```

43.10 map

Listing 6: ./code/map/main.go

```
package main

import "fmt"
import "sort"

func test1() {
    fmt.Println("---test1---")
}
```

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```

var a map[string]string

// a zero initialized map equals to nil
fmt.Println(a == nil) // true
fmt.Println(len(a))   // 0
fmt.Printf("%#v\n", a) // map[string]string(nil)

b := map[string]string{}
fmt.Println(b == nil) // false
fmt.Println(len(b))   // 0
fmt.Printf("%#v\n", b) // map[string]string{}

// If the map does not contain the key it returns a zero initialized value.
// Since the value type is string, the zero value is an empty string ""
c := a["foo"]
fmt.Printf("%q\n", c) // ""

// If the left side has two values, the second element is false if the key
// does not exist
_, ok := a["foo"]
fmt.Printf("%v %T\n", ok, ok) // false, bool

if g, ok := a["bar"]; ok {
    fmt.Println("bar exists", g)
} else {
    fmt.Println("bar does not exist") // bar does not exist
}

// a["bar"] = "zoo" // panic: assignment to entry in nil map

// insert a new entry into the map
b["bar"] = "zoo"

if g, ok := b["bar"]; ok {
    fmt.Println("bar exists", g) // bar exists zoo
} else {
    fmt.Println("bar does not exist")
}
}

func test2() {
    fmt.Println("---test2---")
    a := map[string]string{

```

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```
        "hello": "world",
        "foo":   "bar",
    }

    for key, value := range a {
        fmt.Printf("%s %q\n", key, value)
    }
    // Note: the output is not deterministic.
    //
    // hello "world"
    // foo "bar"

    b := make([]string, 0, len(a))
    for key := range a {
        b = append(b, key)
    }

    sort.Strings(b)
    for _, key := range b {
        fmt.Printf("%s %q\n", key, a[key])
    }
    // foo "bar"
    // hello "world"
}

func main() {
    test1()
    test2()
}
```

43.11 switch

No need to use break in switch in go.

Need to use fallthrough explicitly.

Listing 7: ./code/switch/main.go

```
package main

import "fmt"
```

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```
func test1() {
    fmt.Println("---test1---")
    a := 1

    switch {
    case a == 1:
        fmt.Println("a is 1")
    case a == 2:
        fmt.Println("a is 2")
    default:
        fmt.Println("a is default")
    }
    // a is 1

    a = 11
    switch {
    case a == 11:
        fmt.Println("a is 11")
        fallthrough
    case a == 22:
        fmt.Println("a is 22")
    default:
        fmt.Println("a is default")
    }
    // a is 11
    // a is 12
}
```

```
func test2() {
    fmt.Println("---test2---")
    a := 10
    switch a {
    case 10:
        fmt.Println("a is 10")
    case 11:
        fmt.Println("a is 11")
    }

    // a is 10

    // no need to use default, like c/c++
    // no need to use break
}
```

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```
func main() {  
    test1()  
    test2()  
}
```

43.12 Proxy

43.12.1 <https://goproxy.cn/>

```
go env -w GO111MODULE=on  
go env -w GOPROXY=https://goproxy.cn,direct
```

44.1 Basics

<https://github.com/microsoft/onnxruntime/issues/15235#issuecomment-1487609280>

<https://medium.com/microsoftazure/build-and-deploy-fast-and-portable-speech-recognition-applications-with-onnx-runtime-and-whisper>

44.1.1 transforms

```
from transformers import AutoProcessor, pipeline
model_path = "optimum/whisper-tiny.en"
processor = WhisperProcessor.from_pretrained(model_path)
```

https://huggingface.co/optimum/whisper-tiny.en/blob/main/preprocessor_config.json defines the preprocessor.

44.1.2 optimum

45.1 Scripts

45.1.1 Activate Visual studio

```
"C:\Program Files (x86)\Microsoft Visual Studio\2019\Community\VC\Auxiliary\Build\  
↪vcvars64.bat"
```

45.1.2 View DLL dependencies

```
dumpbin /dependents .\onnxruntime_providers_cuda.dll
```

45.1.3 Remove folder

```
del /s /q .*
```

45.1.4 Change default shell to powershell for OpenSSH

See <https://superuser.com/questions/1799896/how-to-configure-powershell-for-openssh-instead-of-cmd>

45.1.5 re-enable cmd after uninstalling anaconda

```
# Run it in powershell  
C:\Windows\System32\reg.exe DELETE "HKCU\Software\Microsoft\Command Processor" /v  
↪AutoRun /f
```

45.2 Visual studio

Activate it in the command line:

```
"C:\Program Files (x86)\Microsoft Visual Studio\2019\Community\VC\Auxiliary\Build\  
↪vcvars64.bat"
```

45.2.1 Property files

- Managing dependencies in Visual Studio C++ projects with property files

<https://blog.conan.io/2021/02/10/Dependencies-Visual-Studio-C++-property-files.html>

- .vcxproj and .props file structure

<https://learn.microsoft.com/en-us/cpp/build/reference/vcxproj-file-structure?view=msvc-160>

View symbols:

```
dumpbin -linkermember .\xxx.lib
```

45.3 Firewall

1. Run `wf.msc` after pressing `win + R`.

45.4 CRT

See also <https://www.amd.com/en/developer/resources/technical-articles/2026/a-practical-approach-to-using-sherpa-onnx-production-re.html>

- <https://learn.microsoft.com/en-us/cpp/build/reference/md-mt-ld-use-run-time-library?view=msvc-170>

It describes MT, MTd, MD, MDd

- C runtime (CRT) and C++ standard library (STL) .lib files

<https://learn.microsoft.com/en-us/cpp/c-runtime-library/crt-library-features?view=msvc-170>

```
dumpbin /directives lib\Release\sherpa-onnx-c-api.lib | findstr RuntimeLibrary
```

```
for %f in (*.lib) do @echo %f & dumpbin /directives "%f" | findstr RuntimeLibrary
```


46.1 Install

```
pip install aqtinstall
aqt list-qt mac desktop --arch 6.2.0

# It prints:
clang_64 wasm_32
```

```
aqt list-qt mac desktop --modules 6.2.0 clang_64

# It prints:
debug_info qt3d qt5compat qtcharts qtconnectivity qtdatavis3d qtimageformats qtlottie_
→ qtmultimedia qtnetworkauth qtpositioning qtquick3d qtquicktimeline qtremoteobjects_
→ qtscxml qtsensors qtserialbus qtserialport qtshadertools qtvirtualkeyboard_
→ qtwebchannel qtwebengine qtwebsockets qtwebview
```

```
aqt install-qt mac desktop 6.2.0 clang_64 -m all
aqt install-qt mac desktop 6.2.0 -m multimedia
```

It creates `./6.2.0` and installs everything inside.

```
aqt install-qt mac desktop 6.2.0 --archives qtbase
```

46.2 Hello

```
mkdir -p code/hello  
cd code/hello  
touch hello.cc  
qmake -project # it will create hello.pro
```

WEBASSEMBLY

47.1 hello

<https://webassembly.github.io/wabt/demo/wat2wasm/index.html>

47.2 emsdk

47.2.1 Install

See https://emscripten.org/docs/getting_started/downloads.html

```
git clone https://github.com/emscripten-core/emsdk.git

cd emsdk

git pull

./emsdk install latest

./emsdk activate latest

source ./emsdk_env.sh

emcc -v

# See https://emscripten.org/docs/getting_started/downloads.html
```

47.2.2 hello.cc

```
emcc ./hello.cc
# It generates a.out.js, a.out.wasm

(py38) kuangfangjun:test-wasm$ ls
a.out.js  a.out.wasm  hello.cc

(py38) kuangfangjun:test-wasm$ file a.out.wasm
a.out.wasm: WebAssembly (wasm) binary module version 0x1 (MVP)

(py38) kuangfangjun:test-wasm$ ls -lh a.out.wasm
-rwxr-xr-x 1 kuangfangjun root 156K Jul 27 15:10 a.out.wasm
```

```
emcc ./hello.cc -o hello.js
# It generates hello.js, hello.wasm
```

Create a file `test.html` containing:

```
<!doctype html>

<html>
  <head>
    <meta charset="utf-8">
    <title>Emscripten:Hello world</title>
  </head>

  <body>
    <script src="hello.js"></script>
  </body>
</html>
```

Start a python http web server `python3 -m http.server 6006`, the log is printed to the browser console.

Or use `emrun ./hello.html`.

Also, we can use node to run it: `node hello.js`.

Now delete `hello.html` and use

```
emcc hello.cc -o hello.html
# It will generate: hello.html, hello.js, hello.wasm
```

47.2.3 emscripten.h

https://emscripten.org/docs/api_reference/emscripten.h.html

48.1 Basics

The GitHub repo is <https://github.com/deezer/spleeter>

Configuration is at <https://github.com/deezer/spleeter/tree/master/spleeter/resources>.

49.1 Basics

```
python3 -m pip install Django
python3 -m django --version

django-admin startproject mysite

cd mysite
python3 manage.py runserver

python3 manage.py runserver 8080
python3 manage.py runserver 0.0.0.0:8000

python3 manage.py diffsettings

python3 manage.py startapp polls
```

https://github.com/django/django/blob/main/django/conf/global_settings.py

50.1 Basics

- <https://github.com/moonhighway/learning-react>

React developer tools: a browser extension

- <https://github.com/facebook/react>
- <https://chrome.google.com/webstore/detail/react-developer-tools/fmkadmapgofadopljbjfkapdkoienihi>

See *node*

51.1 vits

https://github.com/jaywalnut310/vits/blob/main/filelists/ljs_audio_text_test_filelist.txt.cleaned is processed by phonemizer

<https://github.com/FENRIR/MB-iSTFT-VITS2> has more detailed documentation.

<https://github.com/rhasspy/piper> is a C++ implementation of vits.

51.2 unicode

<https://github.com/avian2/unicode>

```
pip install unicode
```

```
(py38) fangjuns-MacBook-Pro:~ fangjun$ echo heiße |unicode  
heisse  
(py38) fangjuns-MacBook-Pro:~ fangjun$ echo 'hello hanzi nihao!' | unicode  
hello Ni Hao !
```

51.3 phonemizer

Its documentation is at <https://bootphon.github.io/phonemizer/>.

See <https://bootphon.github.io/phonemizer/install.html> for installation.

On macOS, it is:

```
brew install espeak  
pip install phonemizer
```

```
DUMMY1/LJ049-0022.wav|The Secret Service believed
```

```
# cleaned
```

```
(py38) fangjuns-MacBook-Pro:~ fangjun$ echo "The Secret Service believed" | phonemize
```

51.4 books

Visit

- <https://libgen.is/>
- <http://gen.lib.rus.ec/>
- NeuralText-to-SpeechSynthesis, 2023

51.5 papers

- A Survey on Neural Speech Synthesis
<https://browse.arxiv.org/pdf/2106.15561.pdf>
- WaveNet: A generative model for raw audio
<https://arxiv.org/pdf/1609.03499.pdf>

51.6 pypinyin

- doc: https://pypinyin.readthedocs.io/zh_CN/master/
- pypi: <https://pypi.org/project/pypinyin/>

51.7 todo

- <https://github.com/PABanner/bark.cpp>, c++
- <https://github.com/rhasspy/piper>, c++
- <https://github.com/coqui-ai/TTS>
- <https://github.com/neonbjb/tortoise-tts>
- <https://github.com/gitmylo/bark-voice-cloning-HuBERT-quantizer>

<https://github.com/Plachtaa/VITS-fast-fine-tuning>

- https://huggingface.co/spaces/Plachta/VITS-Umamusume-voice-synthesizer/blob/main/ONNXVITS_to_onnx.py
- <https://huggingface.co/spaces/zomehwh/vits-models>
- <https://github.com/elevenlabs>
- <https://github.com/Plachtaa/VALL-E-X>
- <https://github.com/lifeiteng/vall-e>
- <https://github.com/alibaba-damo-academy/FunCodec>
- <https://github.com/alibaba-damo-academy/KAN-TTS>
- <https://github.com/CjangCjengh/MoeGoe>
- <https://github.com/weirdseed/vits-ncnn-convert-tool>
- https://colab.research.google.com/github/espnet/espnet_onnx/blob/master/demo/tts_onnx_demo.ipynb
- https://github.com/espnet/espnet_onnx
- <https://huggingface.co/spaces/Plachta/VITS-Umamusume-voice-synthesizer>
- <https://github.com/suno-ai/bark>
- <https://github.com/serp-ai/bark-with-voice-clone>
- <https://github.com/snakers4/silero-models#text-to-speech>

51.8 Matcha-TTS

The model has 3 inputs:

- `x`: (batch_size, max_num_tokens)
- `x_lengths`: (batch_size,)
- `spks`: None, or of shape (batch_size,)

If `spks` is not None, then it uses an embedding layer to convert it to a tensor of shape (batch_size, spk_embedding_dim).

`x`, `x_lengths`, `spks`, are sent to the TextEncoder.

Inside the TextEncoder, `x` is sent to another embedding layer, the output is a tensor of shape (batch_size, max_num_tokens, embedding_dim), then it is transposed to (batch_size, embedding_dim, max_num_tokens)

It builds a mask from `x_lengths`. The mask is of shape (batch_size, max_num_tokens). Valid positions in the mask is True. Padded positions in the mask is False.

The mask is then reshaped to (batch_size, 1, max_num_tokens).

Then `x` is sent to a `prenet`.

`Prenet` contains 3 blocks. Each block consists of: `Conv1D`, `LayerNorm`, `ReLU`, `Dropout`. The last layer of a `prenet` is `Conv1D`, which is equivalent to a `Linear` layer. The advantage of use a `Conv1D` to replace `Linear` is that it does not need to transpose `x`. Also note that the `LayerNorm` in `Prenet` is user-implemented, which can save two transpose operations.

`Prenet` also contains a residual connection, which means the input channels and output channels have to be the same.

Note: Inside `Prenet`, only the input of the `Conv1D` is multiplied with the mask. Also, the output of `Prenet` is also multiplied with the mask.

After the `Prenet`, it concatenates `x` and `spk` embeddings, so the resulting `x` is of shape `(batch_size, embedding_dim + spk_embedding_dim, max_num_tokens)`.

At this point, we only need to process `x` and `mask`, where

- `x`: `(batch_size, embedding_dim + spk_embedding_dim, max_num_tokens)`
- `mask`: `(batch_size, 1, max_num_tokens)`

The next part is `Encoder` inside the `TextEncoder`.

`attn_mask` is of shape `(1, 1, max_num_tokens, max_num_tokens)` and is generated from `mask`.

The `Encoder` in the `TextEncoder` is a `Transformer` encoder.

After processing with the `Encoder`, it uses a `Conv1D` to convert the dim to `n_feats`.

The duration predictor outputs a tensor of shape `(batch_size, 1, max_num_tokens)`.

Note that the duration predictor predicts log durations of each token. In the end, it uses `exp` to get the duration for each token.

It then uses `ceil` to get integral durations. It can optionally use `length_scale` to scale the duration of each token.

It uses `sum` to compute the total duration of each utterance.

`fix_len_compatibility()`: Convert to the next multiple of 4 if it is not a multiple of 4

`generate_path`: It actually returns a mask of shape `(batch, max_num_tokens, max_y)`.

51.8.1 The decoder model

Its input:

- `x`: `(batch_size, num_feats, max_y)`
- `mask`: `(batch_size, 1, max_y)`
- `t`: a scalar tensor

First, an embedding is computed from `t` using `SinusoidPosEmb`. Note that it is not a standard `Sinusoid` positional encoding. The shape of the embedding is `(batch_size, in_channels)`.

The embedding is processed by `TimestepEmbedding`, which contains a linear layer, an activation layer, and a linear layer. The output shape is `(batch_size, time_embed_dim)`.

52.1 books

52.1.1 Learn Rust the Dangerous Way

<https://cliffle.com/p/dangerust/>

52.1.2 Rust By Practice

<https://practice.course.rs/>

52.1.3 100-exercises-to-learn-rust

- <https://github.com/mainmatter/100-exercises-to-learn-rust>
- <https://rust-exercises.com/100-exercises/>

52.1.4 Welcome to Comprehensive Rust

<https://google.github.io/comprehensive-rust/>

52.1.5 The Rustonomicon

<https://doc.rust-lang.org/nomicon/repr-rust.html>

52.1.6 The Rust Programming Language

Also known as the rust book

<https://doc.rust-lang.org/book/>

52.1.7 Rust by example

<https://doc.rust-lang.org/rust-by-example/>

52.1.8 The cargo book

<https://doc.rust-lang.org/cargo/>

52.1.9 The Rustonomicon

<https://doc.rust-lang.org/nomicon/>

52.1.10 A little C with your Rust

<https://docs.rust-embedded.org/book/interoperability/c-with-rust.html#a-little-c-with-your-rust>

52.1.11 The bindgen User Guide

<https://rust-lang.github.io/rust-bindgen/introduction.html>

52.1.12 Yet Another Rust Resource (YARR!)

<https://yarr.fyi/>

52.1.13 The Rust Edition Guide

<https://doc.rust-lang.org/edition-guide/rust-2021/c-string-literals.html>

52.2 Lectures

- Software Fundamentals

<https://cese.ewi.tudelft.nl/software-fundamentals/index.html>

Slides pdf: <https://cese.ewi.tudelft.nl/software-fundamentals/part-1/slides.html>

- CSCI 181G PO

<https://cs.pomona.edu/classes/cs181g/>

- CS 110L: Safety in Systems Programming

<https://reberhardt.com/cs110l/spring-2020/>

52.3 Install

```
rustc --version
```

```
rustc 1.74.0 (79e9716c9 2023-11-13)
```

```
cargo --version
```

```
cargo 1.74.0 (ecb9851af 2023-10-18)
```

```
# install the latest version
rustup update
```

after update, `rustc --version` prints `rustc 1.87.0 (17067e9ac 2025-05-09)` and `cargo --version` prints `cargo 1.87.0 (99624be96 2025-05-06)`.

52.4 hello

Listing 1: `./code/hello/hello.rs`

```
fn main() {
    println!("Hello world!");
}
```

File extension is `.rs`.

To build a statically linked executable, run:

```
rust ./hello.rs
```

It generates an executable hello:

```
(py38) fangjuns-MacBook-Pro:hello fangjun$ file hello
hello: Mach-O 64-bit executable x86_64

(py38) fangjuns-MacBook-Pro:hello fangjun$ ls -lh hello
-rwxr-xr-x  1 fangjun  staff   460K Jun 12 11:41 hello

(py38) fangjuns-MacBook-Pro:hello fangjun$ otool -L ./hello
./hello:
    /usr/lib/libSystem.B.dylib (compatibility version 1.0.0, current version 1319.
    ↪0.0)
```

52.4.1 naming convention

- my_var
- my_func

52.4.2 comments

Like C/C++, it uses `/**/` and `//`.

52.4.3 hello2 using cargo

```
cargo new hello2
```

It creates a new directory hello2 inside the current directory:

```
(py38) fangjuns-MacBook-Pro:code fangjun$ tree hello2/
hello2/
├── Cargo.toml
└── src
    └── main.rs

1 directory, 2 files
```

Listing 2: ./code/hello2/Cargo.toml

```
[package]
name = "hello2"
version = "0.1.0"
edition = "2024"

[dependencies]
```

Listing 3: ./code/hello2/src/main.rs

```
fn main() {
    println!("Hello, world!");
}
```

```
cd hello2
cargo build
tree . > a.txt
```

Listing 4: ./code/hello2/a.txt

```
.
├─ Cargo.lock
├─ Cargo.toml
├─ a.txt
├─ src
│   └─ main.rs
└─ target
    ├─ CACHEDIR.TAG
    └─ debug
        ├─ build
        ├─ deps
        │   ├─ hello2-517dbf907ca40ba4
        │   ├─ hello2-517dbf907ca40ba4.2ejis4jtcc22k78hwpfevmpn9.rcgu.o
        │   ├─ hello2-517dbf907ca40ba4.2u847h470kx4lrmip8o7wbm3j.rcgu.o
        │   ├─ hello2-517dbf907ca40ba4.5xtf1o3fzpwipxjpgh6vuq5nb.rcgu.o
        │   ├─ hello2-517dbf907ca40ba4.biyzt084nbvwnn6rzalk2kv68.rcgu.o
        │   ├─ hello2-517dbf907ca40ba4.d
        │   ├─ hello2-517dbf907ca40ba4.d4ptlawbgayqhc0mck8d1zb7y.rcgu.o
        │   └─ hello2-517dbf907ca40ba4.ebi7az3oo64vunpcvr2ytqd4k.rcgu.o
        ├─ examples
        ├─ hello2
        └─ hello2.d
```

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```

└─ incremental
  └─ hello2-21n5fdb2vj2ji
    ├── s-h889axkwsg-0ius9rz-eplmacagf6o0mvmceghnny7ip
    │   ├── 2ejis4jtcc22k78hwpfevmpn9.o
    │   ├── 2u847h470kx4lrmip8o7wbm3j.o
    │   ├── 5xtf1o3fzpwipxjpg6vuq5nb.o
    │   ├── biyzt084nbvwnn6rzalk2kv68.o
    │   ├── d4ptlawbgayqhc0mck8d1zb7y.o
    │   ├── dep-graph.bin
    │   ├── ebi7az3oo64vunpcvr2ytqd4k.o
    │   ├── query-cache.bin
    │   └─ work-products.bin
    └─ s-h889axkwsg-0ius9rz.lock

```

9 directories, 25 files

```

cd hello2
cargo build --release
tree . > b.txt

```

Listing 5: ./code/hello2/b.txt

```

.
├─ Cargo.lock
├─ Cargo.toml
├─ a.txt
├─ b.txt
├─ src
│   └─ main.rs
└─ target
    ├── CACHEDIR.TAG
    ├── debug
    │   ├── build
    │   ├── deps
    │   │   ├── hello2-517dbf907ca40ba4
    │   │   ├── hello2-517dbf907ca40ba4.2ejis4jtcc22k78hwpfevmpn9.rcgu.o
    │   │   ├── hello2-517dbf907ca40ba4.2u847h470kx4lrmip8o7wbm3j.rcgu.o
    │   │   ├── hello2-517dbf907ca40ba4.5xtf1o3fzpwipxjpg6vuq5nb.rcgu.o
    │   │   ├── hello2-517dbf907ca40ba4.biyzt084nbvwnn6rzalk2kv68.rcgu.o
    │   │   ├── hello2-517dbf907ca40ba4.d
    │   │   ├── hello2-517dbf907ca40ba4.d4ptlawbgayqhc0mck8d1zb7y.rcgu.o
    │   │   └─ hello2-517dbf907ca40ba4.ebi7az3oo64vunpcvr2ytqd4k.rcgu.o
    │   └─ examples

```

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```

|   └─ hello2
|   └─ hello2.d
|   └─ incremental
|       └─ hello2-21n5fdb2vj2ji
|           └─ s-h889axkwsg-0ius9rz-eplmacagf6o0mvmceghnny7ip
|               └─ 2ejis4jtcc22k78hwpfevmpn9.o
|               └─ 2u847h470kx4lrnip8o7wbm3j.o
|               └─ 5xtf1o3fzpwipxjph6vuq5nb.o
|               └─ biyzt084nbvwnn6rzalk2kv68.o
|               └─ d4ptlawbgayqhc0mck8d1zb7y.o
|               └─ dep-graph.bin
|               └─ ebi7az3oo64vunpcvr2ytqd4k.o
|               └─ query-cache.bin
|               └─ work-products.bin
|           └─ s-h889axkwsg-0ius9rz.lock
└─ release
    └─ build
    └─ deps
        └─ hello2-ae384e63350693e9
        └─ hello2-ae384e63350693e9.d
    └─ examples
    └─ hello2
    └─ hello2.d
    └─ incremental

```

14 directories, 30 files

```
(py38) fangjuns-MacBook-Pro:hello2 fangjun$ ls -lh target/debug/hello2 target/release/
↪hello2
```

```
-rwxr-xr-x 1 fangjun staff 463K Jun 12 21:35 target/debug/hello2
-rwxr-xr-x 1 fangjun staff 388K Jun 12 21:37 target/release/hello2
```

```
(py38) fangjuns-MacBook-Pro:hello2 fangjun$ strip target/debug/hello2 target/release/
↪hello2
```

```
(py38) fangjuns-MacBook-Pro:hello2 fangjun$ ls -lh target/debug/hello2 target/release/
↪hello2
-rwxr-xr-x 1 fangjun staff 325K Jun 12 21:39 target/debug/hello2
-rwxr-xr-x 1 fangjun staff 325K Jun 12 21:39 target/release/hello2
```

```
(py38) fangjuns-MacBook-Pro:hello2 fangjun$ cargo run
Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.00s
Running `target/debug/hello2`
```

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```
Hello, world!
```

52.4.4 hello3

```
cd code
cargo new --lib hello3
```

```
(py38) fangjuns-MacBook-Pro:code fangjun$ cargo new --lib hello3
    Creating library `hello3` package
note: see more `Cargo.toml` keys and their definitions at https://doc.rust-lang.org/
↳ cargo/reference/manifest.html

(py38) fangjuns-MacBook-Pro:code fangjun$ tree hello3
hello3
├── Cargo.toml
└── src
    └── lib.rs

1 directory, 2 files
```

Listing 6: ./code/hello3/Cargo.toml

```
[package]
name = "hello3"
version = "0.1.0"
edition = "2024"

[dependencies]
```

Listing 7: ./code/hello3/src/lib.rs

```
pub fn add(left: u64, right: u64) -> u64 {
    left + right
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn it_works() {
```

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```

    let result = add(2, 2);
    assert_eq!(result, 4);
}
}

```

Pay attention to the function signature.

```

(py38) fangjuns-MacBook-Pro:hello3 fangjun$ cargo test
  Compiling hello3 v0.1.0 (/Users/fangjun/open-source/notes/docs/source/rust/code/
↳hello3)
  Finished `test` profile [unoptimized + debuginfo] target(s) in 0.81s
  Running unittests src/lib.rs (target/debug/deps/hello3-b178d4679cda4da3)

running 1 test
test tests::it_works ... ok

test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished
↳in 0.00s

   Doc-tests hello3

running 0 tests

test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished
↳in 0.00s

```

52.4.5 use_hello3

```

cd code
cargo new use_hello3

```

```

(py38) fangjuns-MacBook-Pro:use_hello3 fangjun$ vim Cargo.toml
(py38) fangjuns-MacBook-Pro:use_hello3 fangjun$ cargo run
  Locking 1 package to latest Rust 1.87.0 compatible version
  Compiling hello3 v0.1.0 (/Users/fangjun/open-source/notes/docs/source/rust/code/
↳hello3)
  Compiling use_hello3 v0.1.0 (/Users/fangjun/open-source/notes/docs/source/rust/
↳code/use_hello3)
  Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.71s
  Running `target/debug/use_hello3`
Hello, world! 5

```

Listing 8: ./code/use_hello3/Cargo.toml

```
[package]
name = "use_hello3"
version = "0.1.0"
edition = "2024"

[dependencies]
hello3={path = "../hello3"}
```

Listing 9: ./code/use_hello3/src/main.rs

```
fn main() {
    let result = hello3::add(2, 3);
    println!("Hello, world! {}", result);
}
```

52.4.6 hello_comments

```
cd code
cargo new hello_comment
```

Listing 10: ./code/hello_comment/Cargo.toml

```
[package]
name = "hello_comment"
version = "0.1.0"
edition = "2024"

[dependencies]
```

Listing 11: ./code/hello_comment/src/main.rs

```
//! <b> hello comment </b>
//! <p> Author: my name </p>
//! <p> Apache 2.0 License</p>

/// add two numbers
pub fn add(left: u64, right: u64) -> u64 {
    left + right
}
```

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```

/// this is the main function
fn main() {
    println!("Hello, world!");
}

```

```

(py38) fangjuns-MacBook-Pro:hello_comment fangjun$ cargo doc
    Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.00s
    Generated /Users/fangjun/open-source/notes/docs/source/rust/code/hello_comment/
    ↪target/doc/hello_comment/index.html

```

52.5 array

```

fn main() {
    let data: [i8; 3] = [0; 3];
    println!("{:?}", data);
    println!("{:#?}", data);
    println!("{data:?}");
    println!("{data:#?}");
    dbg!(data);
}

```

For `println!`, we have to use `{:?}",` or `{:#?}",` for arrays.

52.6 Variables

- type cast using `as`
- pointer: `*i32`, `*const i32`
- reference: `&i32`
- a char has 4 bytes and uses utf-8 encoding

Listing 12: code/variables/src/main.rs

```

fn print_type_of<T>(_: &T) {
    println!("{}", std::any::type_name::<T>());
}

fn test_type_inference() {
    let i: i32 = 10;
}

```

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```
    let b = 2_0;
    let c: i32;
    let d;
    c = 11_000_000;
    d = i + b + c;
    print_type_of(&d); // i32
}

fn test_int() {
    let a = 43;
    println!("0x{:x}", a); // 0x2b
    println!("0x{:b}", a); // 0x101011
}

fn test_char() -> char {
    let c = 'c';
    // a char occupies 4 bytes!
    assert_eq!(std::mem::size_of::<char>(), 4);
    return c;
}

fn test_bool() {
    let t = true;
    let f = false;

    // type cast with as
    assert_eq!(t as i32, 1);
    assert_eq!(f as i32, 0);
    assert_eq!(std::mem::size_of::<bool>(), 1)
}

fn test_pointer() {
    let mut i: i32 = 10;
    let p = &mut i as *mut i32;
    unsafe {
        // rustc --explain E0133
        *p = 100;
    }
    assert_eq!(i, 100);
}

fn test_reference() {
    let i = 10;
```

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```
    let r = &i;  
    let b = r + 0; // automatically dereference for arithmetic operations  
    let c: i32 = *r;  
    assert_eq!(b, 10);  
    assert_eq!(c, 10);  
}  
  
fn main() {  
    test_type_inference();  
    test_bool();  
    test_char();  
    test_int();  
    test_pointer();  
    test_reference();  
}
```

52.7 for

```
for x in 1..5 {  
    dbg!(x);  
}  
  
// it prints: x=1, x=2, x=3, x=4  
// note that 5 is exclusive  
// also note x is not pre-defined. Outside of the for loop,  
// x is not accessible
```

```
for x in 1..=5 {  
    dbg!(x);  
}  
  
// now 5 is inclusive
```

```
for elem in [2, 4, 6] {  
    dbg!(elem);  
}  
  
// iterate an array
```

52.7.1 break with a label

```
fn main() {
    let data = [[1, 2, 3], [4, 5, 6], [7, 8, 9]];
    let target = 2;
    let mut searched = 0;
    'out_label: for i in 0..=2 {
        for k in 0..=2 {
            searched += 1;
            if data[i][k] == target {
                break 'out_label;
            }
        }
    }
    dbg!(searched);
}
```

52.8 String

```
let mut s = String::from("hello");
s.push_str(", world"); // hello, world
```

```
fn main() {
    let s = r"this is a raw string";
    let s2 = r##"this is a "second" string"##;
    let s3 = "this is a \"third\" string";
    dbg!(s);
    dbg!(s2);
    dbg!(s3);
}
```

See

- <https://doc.rust-lang.org/reference/tokens.html#characters-and-strings>
- `str`
- `CString`
- `CStr`
- `String`

Listing 13: main.rs

```

use std::ffi::{CStr, CString};

fn test_1() {
    let _a: &[u8] = b"hi";

    let b: &str = "hi"; // &str
    assert_eq!(b.len(), 2);

    let c: &str = r"hi\n"; // raw string literal
    assert_eq!(c.len(), 4);

    let d: &str = "hi\n";
    assert_eq!(d.len(), 3);

    let i: u8 = b'0';
    assert_eq!(i, 0x30);

    let k: &[u8; 2] = b"\x30\x31"; // byte string literal
    assert_eq!(k, &[48, 49]);

    let r: &str = r###"hi\n"###; // raw string literal
    assert_eq!(r, "hi\\n");

    let c1: &CStr = c"01";
    let c1_bytes_slice :&[u8] = c1.to_bytes();
    assert_eq!(c1_bytes_slice.len(), 2);
    // to_bytes() does not contain the trailing 0
    assert_eq!(c1.to_bytes(), &[0x30, 0x31]);
    println!("{}", c1.count_bytes()); // 2, like strlen()
}

fn test_2() {
    let a: &CStr = c"01";
    let b: CString = CString::from(a);
    let c: CString = CString::new("hi").unwrap();

    let d: &str = a.to_str().unwrap();
    let e: String = String::from(d);
    let f: String = c.into_string().unwrap();
    assert_eq!(&e, "01");
    assert_eq!(&f, "01");
}

```

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```
let h: CString = CString::new(f).unwrap();
assert_eq!(h.as_c_str(), a);
}

fn main() {
    test_1();
    test_2();
}
```

52.9 enum

```
pub enum Option<T> {
    None,
    Some(T),
}

pub enum Result<T, E> {
    Ok(T),
    Err(E),
}
```

52.10 mdbook

<https://rust-lang.github.io/mdBook/guide/installation.html>

52.11 Useful packages

- <https://github.com/rust-lang/regex.git>

52.12 cargo

Before publish a crate, check <https://doc.rust-lang.org/cargo/reference/publishing.html>

- license
- description
- homepage

- repository
- readme

Find some examples of existing packages/crates.

```
cargo publish --dry-run
cargo package --list # check the files to be included for publishing
```

Default CARGO_HOME is the directory ~/.cargo

We can publish a new crate in github actions, like publishing a python package.

52.13 ffi

see also <https://github.com/Picovoice/cheetah/blob/v1.1/binding/rust/src/util.rs>

52.13.1 ffi Example 1

```
cargo new ffi_ex_1
cd ffi_ex_1
cargo add cc --build
```

```
$ cargo add cc --build
  Updating crates.io index
    Adding cc v1.2.27 to build-dependencies
      Features:
        - jobserver
        - parallel
  Updating crates.io index
  Locking 2 packages to latest Rust 1.87.0 compatible versions
    Adding cc v1.2.27
    Adding shlex v1.3.0
```

After creating and editing build.rs and ./src/hello.c:

```
$ tree .
.
├── Cargo.lock
├── Cargo.toml
├── build.rs
└── src
    ├── hello.c
    └── main.rs
```

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```
1 directory, 5 files
```

Listing 14: ./Cargo.toml

```
[package]
name = "ffi_ex_1"
version = "0.1.0"
edition = "2024"

[build-dependencies]
cc = "1.2.27"
```

Listing 15: ./build.rs

```
fn main() {
    cc::Build::new().file("src/hello.c").compile("hello");
    println!("cargo:rerun-if-changed=src/hello.c");
}
```

Listing 16: ./src/hello.c

```
int my_add(int a, int b) { return a + b; }
```

Listing 17: ./src/main.rs

```
use std::os::raw::c_int;
unsafe extern "C" {
    fn my_add(a: c_int, b: c_int) -> c_int;
}

fn main() {
    let a: i32;
    unsafe {
        a = my_add(10, 20);
    }
    println!("my_add returns {a}");
}
```

```
cargo build -vv > a.txt 2>&1
```

Listing 18: a.txt

```

Compiling shlex v1.3.0
Running
`CARGO=/Users/fangjun/.rustup/toolchains/stable-x86_64-apple-darwin/bin/cargo \
CARGO_CRATE_NAME=shlex \
CARGO_MANIFEST_DIR=/Users/fangjun/.cargo/registry/src/index.crates.io-
↪1949cf8c6b5b557f/shlex-1.3.0 \
CARGO_MANIFEST_PATH=/Users/fangjun/.cargo/registry/src/index.crates.io-
↪1949cf8c6b5b557f/shlex-1.3.0/Cargo.toml \
CARGO_PKG_AUTHORS='comex <comexk@gmail.com>;Fenhl <fenhl@fenhl.net>;Adrian
↪Taylor <adetaylor@chromium.org>;Alex Touchet <alextouchet@outlook.com>;Daniel Parks
↪<dp+git@oxidized.org>;Garrett Berg <googberg@gmail.com>' \
CARGO_PKG_DESCRIPTION='Split a string into shell words, like Python\'\'s shlex.'
↪\
CARGO_PKG_HOMEPAGE='' \
CARGO_PKG_LICENSE='MIT OR Apache-2.0' \
CARGO_PKG_LICENSE_FILE='' \
CARGO_PKG_NAME=shlex \
CARGO_PKG_README=README.md \
CARGO_PKG_REPOSITORY='https://github.com/comex/rust-shlex' \
CARGO_PKG_RUST_VERSION=1.46.0 \
CARGO_PKG_VERSION=1.3.0 \
CARGO_PKG_VERSION_MAJOR=1 \
CARGO_PKG_VERSION_MINOR=3 \
CARGO_PKG_VERSION_PATCH=0 \
CARGO_PKG_VERSION_PRE='' \
DYLD_FALLBACK_LIBRARY_PATH='/Users/fangjun/open-source/notes/docs/source/rust/
↪code/ffi_ex_1/target/debug/deps:/Users/fangjun/.rustup/toolchains/stable-x86_64-
↪apple-darwin/lib:/Users/fangjun/lib:/usr/local/lib:/usr/lib' \
/Users/fangjun/.rustup/toolchains/stable-x86_64-apple-darwin/bin/rustc \
--crate-name shlex \
--edition=2015 \
/Users/fangjun/.cargo/registry/src/index.crates.io-1949cf8c6b5b557f/shlex-1.3.
↪0/src/lib.rs \
--error-format=json \
--json=diagnostic-rendered-ansi,artifacts,future-incompat \
--crate-type lib \
--emit=dep-info,metadata,link \
-C embed-bitcode=no \
--cfg 'feature="default"' \
--cfg 'feature="std"' \
--check-cfg 'cfg(docsrs,test)' \
--check-cfg 'cfg(feature, values("default", "std"))' \

```

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```

-C metadata=d38534b5664a9a84 \
-C extra-filename=-808473125270fb34 \
--out-dir /Users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↳target/debug/deps \
-L dependency=/Users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↳target/debug/deps \
--cap-lints warn`
warning: unexpected `cfg` condition name: `manual_codegen_check`

--> /Users/fangjun/.cargo/registry/src/index.crates.io-1949cf8c6b5b557f/shlex-1.3.
↳0/src/bytes.rs:353:12
|
353 | #[cfg_attr(manual_codegen_check, inline(never))]
|           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|
= help: expected names are: `docsrs`, `feature`, and `test` and 31 more
= help: consider using a Cargo feature instead
= help: or consider adding in `Cargo.toml` the `check-cfg` lint config for the
↳lint:
    [lints.rust]
    unexpected_cfgs = { level = "warn", check-cfg = ['cfg(manual_codegen_
↳check)'] }
    = help: or consider adding `println!("cargo::rustc-check-cfg=cfg(manual_codegen_
↳check)");` to the top of the `build.rs`
    = note: see <https://doc.rust-lang.org/nightly/rustc/check-cfg/cargo-specifics.
↳html> for more information about checking conditional configuration
    = note: `#[warn(unexpected_cfgs)]` on by default

compiling cc v1.2.27
running `
cargo=/users/fangjun/.rustup/toolchains/stable-x86_64-apple-darwin/bin/cargo \
cargo_crate_name=cc \
cargo_manifest_dir=/users/fangjun/.cargo/registry/src/index.crates.io-
↳1949cf8c6b5b557f/cc-1.2.27 \
cargo_manifest_path=/users/fangjun/.cargo/registry/src/index.crates.io-
↳1949cf8c6b5b557f/cc-1.2.27/cargo.toml \
cargo_pkg_authors='alex crichton <alex@alexcrichton.com>' \
cargo_pkg_description='a build-time dependency for cargo build scripts to assist
↳in invoking the native c compiler to compile native c code into a static archive to
↳be linked into rust code.' \
cargo_pkg_homepage='https://github.com/rust-lang/cc-rs' \
cargo_pkg_license='mit or apache-2.0' \
cargo_pkg_license_file='' \

```

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```

cargo_pkg_name=cc \
cargo_pkg_readme=readme.md \
cargo_pkg_repository='https://github.com/rust-lang/cc-rs' \
cargo_pkg_rust_version=1.63 \
cargo_pkg_version=1.2.27 \
cargo_pkg_version_major=1 \
cargo_pkg_version_minor=2 \
cargo_pkg_version_patch=27 \
cargo_pkg_version_pre='' \
dyld_fallback_library_path='/users/fangjun/open-source/notes/docs/source/rust/
↪code/ffi_ex_1/target/debug/deps:/users/fangjun/.rustup/toolchains/stable-x86_64-
↪apple-darwin/lib:/users/fangjun/lib:/usr/local/lib:/usr/lib' \
/users/fangjun/.rustup/toolchains/stable-x86_64-apple-darwin/bin/rustc \
--crate-name cc \
--edition=2018 \
/users/fangjun/.cargo/registry/src/index.crates.io-1949cf8c6b5b557f/cc-1.2.27/
↪src/lib.rs \
--error-format=json \
--json=diagnostic-rendered-ansi,artifacts,future-incompat \
--crate-type lib \
--emit=dep-info,metadata,link \
-c embed-bitcode=no \
--check-cfg 'cfg(docsrs,test)' \
--check-cfg 'cfg(feature, values("jobserver", "parallel"))' \
-c metadata=a7441a07b903063b \
-c extra-filename=-5dcc6a1224077a74 \
--out-dir /users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↪target/debug/deps \
-l dependency=/users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↪target/debug/deps \
--extern shlex=/users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↪target/debug/deps/libshlex-808473125270fb34.rmeta \
--cap-lints warn`
warning: `shlex` (lib) generated 1 warning

compiling ffi_ex_1 v0.1.0 (/users/fangjun/open-source/notes/docs/source/rust/code/
↪ffi_ex_1)
running `
cargo=/users/fangjun/.rustup/toolchains/stable-x86_64-apple-darwin/bin/cargo \
cargo_crate_name=build_script_build \
cargo_manifest_dir=/users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_
↪1 \
cargo_manifest_path=/users/fangjun/open-source/notes/docs/source/rust/code/ffi_

```

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```

↪ex_1/cargo.toml \
    cargo_pkg_authors='' \
    cargo_pkg_description='' \
    cargo_pkg_homepage='' \
    cargo_pkg_license='' \
    cargo_pkg_license_file='' \
    cargo_pkg_name=ffi_ex_1 \
    cargo_pkg_readme='' \
    cargo_pkg_repository='' \
    cargo_pkg_rust_version='' \
    cargo_pkg_version=0.1.0 \
    cargo_pkg_version_major=0 \
    cargo_pkg_version_minor=1 \
    cargo_pkg_version_patch=0 \
    cargo_pkg_version_pre='' \
    cargo_primary_package=1 \
    cargo_sbom_path='' \
    dyld_fallback_library_path='/users/fangjun/open-source/notes/docs/source/rust/
↪code/ffi_ex_1/target/debug/deps:/users/fangjun/.rustup/toolchains/stable-x86_64-
↪apple-darwin/lib:/users/fangjun/lib:/usr/local/lib:/usr/lib' \
    /users/fangjun/.rustup/toolchains/stable-x86_64-apple-darwin/bin/rustc \
    --crate-name build_script_build \
    --edition=2024 \
    build.rs \
    --error-format=json \
    --json=diagnostic-rendered-ansi,artifacts,future-incompat \
    --crate-type bin \
    --emit=dep-info,link \
    -c embed-bitcode=no \
    --check-cfg 'cfg(docsrs,test)' \
    --check-cfg 'cfg(feature, values())' \
    -c metadata=f2e7fe36c03b5ec1 \
    -c extra-filename=-eeb655500d051c65 \
    --out-dir /users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↪target/debug/build/ffi_ex_1-eeb655500d051c65 \
    -c incremental=/users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↪target/debug/incremental \
    -l dependency=/users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↪target/debug/deps \
    --extern cc=/users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↪target/debug/deps/libcc-5dcc6a1224077a74.rlib`

    running `

```

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```

cargo=/users/fangjun/.rustup/toolchains/stable-x86_64-apple-darwin/bin/cargo \
cargo_cfg_feature='' \
cargo_cfg_panic=unwind \
cargo_cfg_target_abi='' \
cargo_cfg_target_arch=x86_64 \
cargo_cfg_target_endian=little \
cargo_cfg_target_env='' \
cargo_cfg_target_family=unix \
cargo_cfg_target_feature=cpxchg16b,fxsr,sse,sse2,sse3,sse4.1,ssse3 \
cargo_cfg_target_has_atomic=128,16,32,64,8,ptr \
cargo_cfg_target_os=macos \
cargo_cfg_target_pointer_width=64 \
cargo_cfg_target_vendor=apple \
cargo_cfg_unix='' \
cargo_encoded_rustflags='' \
cargo_manifest_dir=/users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_
↪1 \
cargo_manifest_path=/users/fangjun/open-source/notes/docs/source/rust/code/ffi_
↪ex_1/cargo.toml \
cargo_pkg_authors='' \
cargo_pkg_description='' \
cargo_pkg_homepage='' \
cargo_pkg_license='' \
cargo_pkg_license_file='' \
cargo_pkg_name=ffi_ex_1 \
cargo_pkg_readme='' \
cargo_pkg_repository='' \
cargo_pkg_rust_version='' \
cargo_pkg_version=0.1.0 \
cargo_pkg_version_major=0 \
cargo_pkg_version_minor=1 \
cargo_pkg_version_patch=0 \
cargo_pkg_version_pre='' \
debug=true \
dyld_fallback_library_path='/users/fangjun/open-source/notes/docs/source/rust/
↪code/ffi_ex_1/target/debug/deps:/users/fangjun/open-source/notes/docs/source/rust/
↪code/ffi_ex_1/target/debug:/users/fangjun/.rustup/toolchains/stable-x86_64-apple-
↪darwin/lib/rustlib/x86_64-apple-darwin/lib:/users/fangjun/.rustup/toolchains/stable-
↪x86_64-apple-darwin/lib:/users/fangjun/lib:/usr/local/lib:/usr/lib' \
host=x86_64-apple-darwin \
num_jobs=8 \
opt_level=0 \
out_dir=/users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/target/

```

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```

↪ debug/build/ffi_ex_1-c1751820b0caf7b9/out \
    profile=debug \
    rustc=/users/fangjun/.rustup/toolchains/stable-x86_64-apple-darwin/bin/rustc \
    rustdoc=/users/fangjun/.rustup/toolchains/stable-x86_64-apple-darwin/bin/rustdoc
↪ \
    target=x86_64-apple-darwin \
    /users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/target/debug/
↪ build/ffi_ex_1-eeb655500d051c65/build-script-build`

[ffi_ex_1 0.1.0] out_dir = some(/users/fangjun/open-source/notes/docs/source/rust/
↪ code/ffi_ex_1/target/debug/build/ffi_ex_1-c1751820b0caf7b9/out)
[ffi_ex_1 0.1.0] opt_level = some(0)
[ffi_ex_1 0.1.0] target = some(x86_64-apple-darwin)
[ffi_ex_1 0.1.0] host = some(x86_64-apple-darwin)
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=cc_x86_64-apple-darwin
[ffi_ex_1 0.1.0] cc_x86_64-apple-darwin = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=cc_x86_64_apple_darwin
[ffi_ex_1 0.1.0] cc_x86_64_apple_darwin = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=host_cc
[ffi_ex_1 0.1.0] host_cc = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=cc
[ffi_ex_1 0.1.0] cc = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=cc_enable_debug_output
[ffi_ex_1 0.1.0] rustc_wrapper = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=crate_cc_no_defaults
[ffi_ex_1 0.1.0] crate_cc_no_defaults = none
[ffi_ex_1 0.1.0] debug = some(true)
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=macosx_deployment_target
[ffi_ex_1 0.1.0] macosx_deployment_target = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=cflags
[ffi_ex_1 0.1.0] cflags = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=host_cflags
[ffi_ex_1 0.1.0] host_cflags = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=cflags_x86_64_apple_darwin
[ffi_ex_1 0.1.0] cflags_x86_64_apple_darwin = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=cflags_x86_64-apple-darwin
[ffi_ex_1 0.1.0] cflags_x86_64-apple-darwin = none
[ffi_ex_1 0.1.0] cargo_encoded_rustflags = some()
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=ar_x86_64-apple-darwin
[ffi_ex_1 0.1.0] ar_x86_64-apple-darwin = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=ar_x86_64_apple_darwin
[ffi_ex_1 0.1.0] ar_x86_64_apple_darwin = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=host_ar

```

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```
[ffi_ex_1 0.1.0] host_ar = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=ar
[ffi_ex_1 0.1.0] ar = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=arflags
[ffi_ex_1 0.1.0] arflags = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=host_arflags
[ffi_ex_1 0.1.0] host_arflags = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=arflags_x86_64_apple_darwin
[ffi_ex_1 0.1.0] arflags_x86_64_apple_darwin = none
[ffi_ex_1 0.1.0] cargo:rerun-if-env-changed=arflags_x86_64-apple-darwin
[ffi_ex_1 0.1.0] arflags_x86_64-apple-darwin = none
[ffi_ex_1 0.1.0] cargo:rustc-link-lib=static=hello
[ffi_ex_1 0.1.0] cargo:rustc-link-search=native=/Users/fangjun/open-source/notes/docs/
↳source/rust/code/ffi_ex_1/target/debug/build/ffi_ex_1-c1751820b0caf7b9/out
[ffi_ex_1 0.1.0] cargo:rerun-if-changed=src/hello.c
```

```
Running `
CARGO=/Users/fangjun/.rustup/toolchains/stable-x86_64-apple-darwin/bin/cargo \
CARGO_BIN_NAME=ffi_ex_1 \
CARGO_CRATE_NAME=ffi_ex_1 \
CARGO_MANIFEST_DIR=/Users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_
↳1 \
CARGO_MANIFEST_PATH=/Users/fangjun/open-source/notes/docs/source/rust/code/ffi_
↳ex_1/Cargo.toml \
CARGO_PKG_AUTHORS='' \
CARGO_PKG_DESCRIPTION='' \
CARGO_PKG_HOMEPAGE='' \
CARGO_PKG_LICENSE='' \
CARGO_PKG_LICENSE_FILE='' \
CARGO_PKG_NAME=ffi_ex_1 \
CARGO_PKG_README='' \
CARGO_PKG_REPOSITORY='' \
CARGO_PKG_RUST_VERSION='' \
CARGO_PKG_VERSION=0.1.0 \
CARGO_PKG_VERSION_MAJOR=0 \
CARGO_PKG_VERSION_MINOR=1 \
CARGO_PKG_VERSION_PATCH=0 \
CARGO_PKG_VERSION_PRE='' \
CARGO_PRIMARY_PACKAGE=1 \
CARGO_SBOM_PATH='' \
DYLD_FALLBACK_LIBRARY_PATH='/Users/fangjun/open-source/notes/docs/source/rust/
↳code/ffi_ex_1/target/debug/deps:/Users/fangjun/.rustup/toolchains/stable-x86_64-
↳apple-darwin/lib:/Users/fangjun/lib:/usr/local/lib:/usr/lib' \
```

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```

OUT_DIR=/Users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/target/
↪ debug/build/ffi_ex_1-c1751820b0caf7b9/out \
  /Users/fangjun/.rustup/toolchains/stable-x86_64-apple-darwin/bin/rustc \
    --crate-name ffi_ex_1 \
    --edition=2024 \
    src/main.rs \
    --error-format=json \
    --json=diagnostic-rendered-ansi,artifacts,future-incompat \
    --crate-type bin \
    --emit=dep-info,link \
    -C embed-bitcode=no \
    -C debuginfo=2 \
    -C split-debuginfo=unpacked \
    --check-cfg 'cfg(docsrs,test)' \
    --check-cfg 'cfg(feature, values())' \
    -C metadata=068e65a8f5ae7900 \
    -C extra-filename=-48ca45584c8a6533 \
    --out-dir /Users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↪ target/debug/deps \
    -C incremental=/Users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↪ target/debug/incremental \
    -L dependency=/Users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↪ target/debug/deps \
    -L native=/Users/fangjun/open-source/notes/docs/source/rust/code/ffi_ex_1/
↪ target/debug/build/ffi_ex_1-c1751820b0caf7b9/out \
    -l static=hello`
Finished `dev` profile [unoptimized + debuginfo] target(s) in 4.25s

```

```
tree . > b.txt
```

Listing 19: b.txt

```

.
├─ Cargo.lock
├─ Cargo.toml
├─ a.txt
├─ b.txt
├─ build.rs
├─ src
│   └─ hello.c
│       └─ main.rs
└─ target
    └─ CACHEDIR.TAG

```

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```

└─ debug
  └─ build
    │   └─ ffi_ex_1-c1751820b0caf7b9
    │   │   └─ invoked.timestamp
    │   │   └─ out
    │   │       └─ ea708c7824d36062-hello.o
    │   │       └─ libhello.a
    │   │   └─ output
    │   │   └─ root-output
    │   │   └─ stderr
    │   └─ ffi_ex_1-eeb655500d051c65
    │       └─ build-script-build
    │       └─ build_script_build-eeb655500d051c65
    │       └─ build_script_build-eeb655500d051c65.d
  └─ deps
    │   └─ cc-5dcc6a1224077a74.d
    │   └─ ffi_ex_1-48ca45584c8a6533
    │   └─ ffi_ex_1-48ca45584c8a6533.17mg2lvz5zsf4xisp4ugqnwo7.rcgu.o
    │   └─ ffi_ex_1-48ca45584c8a6533.3l4c4w7v8kf8llhfrx4i7p5yc.rcgu.o
    │   └─ ffi_ex_1-48ca45584c8a6533.4rpejftjpoen151ivfn5qariz.rcgu.o
    │   └─ ffi_ex_1-48ca45584c8a6533.7hrnv0dz5cw72d51erex2v0bi.rcgu.o
    │   └─ ffi_ex_1-48ca45584c8a6533.8yulq6zedu5kz2pppsyts0xlz.rcgu.o
    │   └─ ffi_ex_1-48ca45584c8a6533.d
    │   └─ ffi_ex_1-48ca45584c8a6533.dr47jdmpb57wqvb5e2le5xqf3.rcgu.o
    │   └─ ffi_ex_1-48ca45584c8a6533.ehb6gqlxf57d6a0eh76iai9b0.rcgu.o
    │   └─ libcc-5dcc6a1224077a74.rlib
    │   └─ libcc-5dcc6a1224077a74.rmeta
    │   └─ libshlex-808473125270fb34.rlib
    │   └─ libshlex-808473125270fb34.rmeta
    │   └─ shlex-808473125270fb34.d
  └─ examples
  └─ ffi_ex_1
  └─ ffi_ex_1.d
  └─ incremental
    │   └─ build_script_build-35tgcqt68vk2k
    │   │   └─ s-h8i0xk63ma-001vdkg-5w1mt9vbwdwfd0geysyn3a1ou
    │   │       └─ 5c19nz8leg4fp9dhwm8o3xtui.o
    │   │       └─ 7qa59eaxgrwnvrui6xu6yolac.o
    │   │       └─ 8lgj301y0ro8zv9b7ytlreq9j.o
    │   │       └─ 8mvjxzqpmj3gh3t2p3cawrsmd.o
    │   │       └─ 9bchr3q0syk5pwys1z7vb67cl.o
    │   │       └─ b725p88zt33dujvfs20u5ushb.o
    │   │       └─ dep-graph.bin

```

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```

|   |   └─ e7jd1kyjm81jr46nyg9x9hgut.o
|   |   └─ ef35xgfwzws48flp98hjc2bvh.o
|   |   └─ query-cache.bin
|   |   └─ work-products.bin
|   └─ s-h8i0xk63ma-001vdkg.lock
└─ ffi_ex_1-22x8jcwretezy
   └─ s-h8i0xkv65h-0ah7faj-e9zhf3dbt7y4n2l8y3g56e2uy
      └─ 17mg2lvz5zsf4xisp4ugqnwo7.o
         └─ 3l4c4w7v8kf8llhfrx4i7p5yc.o
            └─ 4rpejftjpoen151ivfn5qariz.o
               └─ 7hrnv0dz5cw72d51erex2v0bi.o
                  └─ 8yulq6zedu5kz2pppsyts0xlz.o
                     └─ dep-graph.bin
                        └─ dr47jdmpb57wqvb5e2le5xqf3.o
                           └─ ehb6gqlxf57d6a0eh76iai9b0.o
                              └─ query-cache.bin
                                 └─ work-products.bin
                                    └─ s-h8i0xkv65h-0ah7faj.lock

```

14 directories, 57 files

52.13.2 ffi Example 2

```

(py38) fangjuns-MacBook-Pro:code fangjun$ cargo new ffi_ex_2
    Creating binary (application) `ffi_ex_2` package
note: see more `Cargo.toml` keys and their definitions at https://doc.rust-lang.org/
↪cargo/reference/manifest.html

(py38) fangjuns-MacBook-Pro:code fangjun$ cd ffi_ex_2/

(py38) fangjuns-MacBook-Pro:ffi_ex_2 fangjun$ cargo add cc --build
Updating crates.io index
Adding cc v1.2.27 to build-dependencies
Features:
- jobserver
- parallel
Updating crates.io index
Locking 2 packages to latest Rust 1.87.0 compatible versions

```

Now create `build.rs` and `src/hello.c` and change `src/main.rs`.

Listing 20: Cargo.toml

```
[package]
name = "ffi_ex_2"
version = "0.1.0"
edition = "2024"

[dependencies]

[build-dependencies]
cc = "1.2.27"
```

Listing 21: build.rs

```
fn main() {
    cc::Build::new().file("src/hello.c").compile("hello");
    println!("cargo:rerun-if-changed=src/hello.c");
}
```

Listing 22: src/hello.c

```
#include <stdint.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

float sum_float_arr(const float *v, int32_t n) {
    float s = 0;
    for (int32_t i = 0; i < n; ++i) {
        s += v[i];
    }
    return s;
}

void add_one_vec(int32_t *v, int32_t n) {
    for (int32_t i = 0; i < n; ++i) {
        v[i] += 1;
    }
}

void add_one_scalar(int32_t *i) { *i += 1; }

int32_t *return_i32_array(int32_t *n) {
```

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```

    int32_t *p = malloc(3 * sizeof(int32_t));
    p[0] = 100;
    p[1] = 101;
    p[2] = 102;
    printf("Return p %p\n", p);
    *n = 3;
    return p;
}

void free_i32_array(int32_t *p) {
    printf("Now p %p is: %d, %d, %d\n", p, p[0], p[1], p[2]);

    free(p);
}

int32_t my_str_len(const char *s) { return s ? strlen(s) : 0; }
void my_upper_str(char *s) {
    if (!s) {
        return;
    };
    while (s && *s) {
        if (*s >= 'a' && *s <= 'z') {
            *s -= 'a' - 'A';
        }
        s += 1;
    }
}

const char *const *my_str_array() {
    static const char *s[] = {"first", "second", NULL};
    return s;
}

typedef struct MyStruct1 {
    int32_t a;
    char b;
    float c;
    const char *s;
    double d;
} MyStruct1;

float compute_struct1_sum(const MyStruct1 *s) {
    printf("%d, %d, %.3f, %.3f, %s\n", s->a, (int32_t)s->b, s->c, s->d, s->s);
    return s->a + s->b + s->c + s->d + strlen(s->s);
}

```

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```

}

typedef struct MyStruct2 {
    const char *s1;
    MyStruct1 s2;
    int32_t i;
} MyStruct2;

float compute_struct2_sum(const MyStruct2 *s) {
    printf("%s, %d, %d, %.3f, %.3f, %d\n", s->s1, s->s2.a, (int32_t)s->s2.b,
        s->s2.c, s->s2.d, s->i);
    if (!s->s2.s) {
        printf("empty string in s->s2.s\n");
    } else {
        printf("string in s->s2.s is: %s\n", s->s2.s);
    }
    return strlen(s->s1) + s->s2.a + s->s2.b + s->s2.c + s->s2.d + s->i;
}

typedef struct MyOpaque MyOpaque;

MyOpaque *create_my_opaque() {
    MyStruct1 *s = malloc(sizeof(MyStruct1));
    s->a = 1;
    s->b = 2;
    s->c = 3.5;
    s->s = 0;
    s->d = 5.5;
    printf("return s: %p\n", s);
    return (MyOpaque *)s;
}

void free_my_opaque(MyOpaque *p) {
    MyStruct1 *s = (MyStruct1 *)p;
    printf("get s: %p\n", s);
    printf("my opaque: %d, %d, %.3f, %.3f, %s\n", s->a, (int32_t)s->b, s->c, s->d,
        s->s ? s->s : "<empty str>");
    free(s);
}

```

Listing 23: src/main.rs

```
use std::ffi::{CStr, CString};

#[repr(C)]
#[derive(Debug)]
struct MyStruct1 {
    a: i32,
    b: i8,
    c: f32,
    p: *const i8,
    d: f64,
}

#[repr(C)]
#[derive(Debug)]
struct MyStruct2 {
    s1: *const i8,
    s2: MyStruct1,
    i: i32,
}

#[repr(C)]
struct MyOpaque {
    _private: [u8; 0],
}

unsafe extern "C" {
    fn sum_float_arr(v: *const f32, n: i32) -> f32;
    fn add_one_vec(v: *mut i32, n: i32);
    fn add_one_scalar(i: *mut i32);
    fn return_i32_array(n: *mut i32) -> *mut i32;
    fn free_i32_array(v: *mut i32);
    fn my_str_len(v: *const i8) -> i32;
    fn my_upper_str(v: *mut i8);
    fn my_str_array() -> *const *const i8;
    fn compute_struct1_sum(p: *const MyStruct1) -> f32;
    fn compute_struct2_sum(p: *const MyStruct2) -> f32;
    fn create_my_opaque() -> *const MyOpaque;
    fn free_my_opaque(p: *const MyOpaque);
}

fn test_sum_float_arr() {
    println!("---test_sum_float_arr---");
}
```

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```

let v: Vec<f32> = vec![1., 2., 3.];
let sum: f32;
unsafe {
    sum = sum_float_arr((&v).as_ptr(), v.len() as i32);
}
println!("Sum of {v:?} is {sum}");
}

fn test_add_one_vec() {
    println!("---test_add_one_vec---");
    let mut v: Vec<i32> = vec![1, 2, 3];
    println!("Before add one: {v:?}");
    unsafe {
        add_one_vec((&mut v).as_mut_ptr(), v.len() as i32);
    }
    println!("After add one: {v:?}");
}

fn test_add_one_scalar() {
    println!("---test_add_one_scalar---");
    let mut i = 10;
    println!("Before add one scalar: {i}");
    unsafe {
        add_one_scalar(&mut i); //implicit conversion
        add_one_scalar((&mut i) as *mut i32); // explicit
    }
    println!("After add one scalar (two times): {i}");
}

fn test_return_i32_array() {
    println!("---test_return_i32_array---");
    unsafe {
        let mut i = 0;
        let p = return_i32_array(&mut i);
        println!("i is {i}, p is: {p:p}");

        let v = std::slice::from_raw_parts_mut(p, i as usize);
        println!("v is {v:?}");
        add_one_vec(v.as_mut_ptr(), v.len() as i32);
        println!("after adding one, v is {v:?}");
        free_i32_array(p);
    }
}

```

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```

fn test_my_str_len() {
    println!("---test_my_str_len---");
    let a = "hi";
    unsafe {
        let b = CString::new(a).unwrap();
        let n = my_str_len(b.as_ptr());
        assert_eq!(n, 2);
    }

    unsafe {
        let n = my_str_len(std::ptr::null::<i8>());
        assert_eq!(n, 0);

        let p = std::ptr::null::<i8>();
        println!("p is {p:p}"); // p is 0x0
    }
}

fn test_my_upper_str() {
    println!("---test_my_upper_str---");
    let mut a = String::from("hi");
    a.push('0');
    unsafe {
        my_upper_str(a.as_bytes_mut().as_mut_ptr() as *mut i8);
    }
    a.pop();
    assert_eq!(a, "HI");

    let mut a = CString::new("world").unwrap();
    unsafe {
        let p = a.into_raw();
        my_upper_str(p);

        a = CString::from_raw(p);
    }
    assert_eq!(a.as_c_str(), c"WORLD");
}

// convert an array of c strings to Vec<CString>
fn test_my_str_array() {
    let p: *const *const i8;
    unsafe {
        p = my_str_array();
    }
}

```

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```

        println!("{:p}", *p);
        println!("{:p}", *p.add(1));
        println!("{:p}", *p.add(2));
    }
    let mut i = 0;
    let mut v = Vec::<CString>::new();
    unsafe {
        loop {
            // if *p.add(i) == std::ptr::null() {
            if (*p.add(i)).is_null() {
                break;
            }
            let s = CStr::from_ptr(*p.add(i));
            v.push(CString::from(s));

            i += 1;
        }
    }
    assert_eq!(i, 2);
    println!("{v:?}");
}

fn test_compute_struct1_sum() {
    let c = c"hello world";
    let s = MyStruct1 {
        a: 1,
        b: 2,
        c: 3.0,
        p: c.as_ptr(),
        d: 4.0,
    };
    let sum: f32;
    unsafe {
        sum = compute_struct1_sum(&s);
    }
    assert_eq!(sum, 10.0 + c.count_bytes() as f32);
}

fn test_compute_struct2_sum() {
    let c = c"hello";
    let s = MyStruct2{
        s1: c.as_ptr().cast(),
        s2: MyStruct1 {

```

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```

        a: 1,
        b: 2,
        c: 3.0,
        p: std::ptr::null(),
        d: 4.0,
    },
    i: 10,
};
let sum: f32;
unsafe {
    sum = compute_struct2_sum(&s);
}
assert_eq!(sum, c.count_bytes() as f32 + s.s2.a as f32 + s.s2.b as f32 + s.s2.c as f32 + s.s2.d as f32 + s.i as f32);
}

fn test_my_opaque() {
    let p: *const MyOpaque;
    unsafe {
        p = create_my_opaque();
    }

    println!("p is {p:p}");
    unsafe {
        free_my_opaque(p);
    }
}

fn main() {
    test_sum_float_arr();
    test_add_one_vec();
    test_add_one_scalar();
    test_return_i32_array();
    test_my_str_len();
    test_my_upper_str();
    test_my_str_array();
    test_compute_struct1_sum();
    test_compute_struct2_sum();
    test_my_opaque();
}

```

52.14 Exercise 1 gcd

52.14.1 simple gcd

```
cargo new gcd_1
cd gcd_1/src
# edit main.rst
```

```
fn gcd(mut m: u64, mut n: u64) -> u64 {
    assert!(m != 0 && n != 0);
    while m != 0 {
        if m < n {
            let t = m;
            m = n;
            n = t;
        }
        m = m % n;
    }
    n
}

fn main() {
    println!("gcd(6, 2) is {}", gcd(6, 2));
}

#[test]
fn test_gcd() {
    assert_eq!(gcd(2, 6), 2);
    assert_eq!(gcd(3, 5), 1);
}
```

cargo run prints:

```
Compiling gcd_1 v0.1.0 (/Users/fangjun/open-source/notes/docs/source/rust/code/gcd_
→1)
Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.12s
Running `target/debug/gcd_1`
gcd(6, 2) is 2
```

cargo test prints:

```
Compiling gcd_1 v0.1.0 (/Users/fangjun/open-source/notes/docs/source/rust/code/gcd_
→1)
```

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```
Finished `test` profile [unoptimized + debuginfo] target(s) in 0.08s
Running unittests src/main.rs (target/debug/deps/gcd_1-74f037a3d4552c7d)

running 1 test
test test_gcd ... ok

test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished
↳ in 0.00s
```

52.14.2 gcd from command line

```
cargo new gcd_2
cd gcd_2
cd gcd_2/src
# edit main.rs
```

```
use std::env;

fn gcd(mut m: u64, mut n: u64) -> u64 {
    assert!(m != 0 && n != 0);
    while m != 0 {
        if m < n {
            let t = m;
            m = n;
            n = t;
        }
        m = m % n;
    }
    n
}

fn main() {
    let args: Vec<String> = env::args().collect();
    if args.len() != 3 {
        eprintln!("Usage: {} <int1> <int2>", args[0]);
        std::process::exit(1);
    }

    // args[0] is the program name
    let num1: u64 = args[1]
        .parse()
        .expect("Please provide a valid integer for the first argument");
```

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```

let num2: u64 = args[2]
    .parse()
    .expect("Please provide a valid integer for the second argument");
let n = gcd(num1, num2);

println!("GCD of {} and {} is {}", num1, num2, n);
}

```

cargo run 5 15 prints:

```

Compiling gcd_2 v0.1.0 (/Users/fangjun/open-source/notes/docs/source/rust/code/gcd_
→2)
Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.09s
Running `target/debug/gcd_2 5 15`
GCD of 5 and 15 is 5

```

52.15 TODO

- <https://github.com/sitelift/Chirp>
It uses sherpa-onnx with tauri + rust
- vosk rust: <https://github.com/Bear-03/vosk-rs/blob/main/crates/vosk/src/recognition/mod.rs>
- <https://github.com/Rust-SDL2/rust-sdl2>
- <https://github.com/rust-openssl/rust-openssl>
- <https://github.com/alexcrichton/curl-rust>

52.16 rustfmt

Create a file `rustfmt.toml` or `.rustfmt.toml` inside the project directory.

```

# Put each method in a chain on its own line
chain_width = 0

# Optional: make sure calls break vertically
fn_call_width = 60

# Optional: control general line width
max_width = 100

```

52.17 Hello Tauri

```
npm create tauri-app@latest
```

Name the project as `hello_world` and use default settings.

```
cd hello_world
npm install
npm run tauri build --release
```

For macOS, it generates both `hello_world.app` and `hello_world_0.1.0_aarch64.dmg`.

```
npm run tauri dev
npm run tauri dev -- --release
```

52.18 Course

- <https://github.com/LearningOS/rust-based-os-comp2025/blob/main/2025-spring-summary.md>
- <https://fontlos.com/post/RustAlgorithm-01>
- <https://rcore-os.cn/>

52.19 The Rust Programming Language - book notes

See <https://rust-book.cs.brown.edu/> and <https://doc.rust-lang.org/book/>

看了这本书之后，要去看 <https://doc.rust-lang.org/rust-by-example/index.html> - <https://rust-exercises.com/100-exercises/>

还要去看 <https://doc.rust-lang.org/nomicon/intro.html>

- <https://github.com/rust-lang/rustlings/>

```
cargo install rustlings
mkdir tmp
cd tmp
rustlings init
cd rustlings
rustlings
```

52.19.1 Chapter 1 Installation

```
rustc --version
```

输出:

```
rustc 1.93.1 (01f6ddf75 2026-02-11)
```

查看帮助: `rustup doc`. 会打开一个浏览器页面, 纯本地, 无需联网。

可以查看的帮助举例: 查看 `str` 的文档。可以查看代码里, 任意一个函数或者类型的帮助文档。

如果想提前下载依赖, 可以使用下面的方法 (新建一个项目, 添加依赖, 然后把依赖下载到 `cache` 里之后, 就可以删掉这个项目里。 `cache` 是全局的):

```
cargo new test-dep
cd test-dep
cargo add sherpa-onnx@1.13.6
cargo fetch
```

依赖默认保存在 `~/.cargo/` 目录下。可以通过设置环境变量 `CARGO_HOME` 来改变。默认情况下, `CARGO_HOME` 是空的, 如果用 `export CARGO_HOME=$PWD/tmp`, 那么, 下载下来的依赖就存在 ```$PWD/tmp```:

```
cd ./tmp

find . -name "*sherpa*"

./registry/cache/index.crates.io-1949cf8c6b5b557f/sherpa-onnx-sys-1.13.6.crate
./registry/cache/index.crates.io-1949cf8c6b5b557f/sherpa-onnx-1.13.6.crate
./registry/index/index.crates.io-1949cf8c6b5b557f/.cache/sh/er/sherpa-onnx
./registry/index/index.crates.io-1949cf8c6b5b557f/.cache/sh/er/sherpa-onnx-sys
./registry/src/index.crates.io-1949cf8c6b5b557f/sherpa-onnx-1.13.6
./registry/src/index.crates.io-1949cf8c6b5b557f/sherpa-onnx-sys-1.13.6
```

对于通过 `build.rs` 下载的, 需要 `cargo check` 或者 `cargo build` 等命令运行时, 才会下载。

当前项目的缓存, 存在当前目录 `./target` 下。

如果想删除缓存, 可以直接删掉 `~/.cargo/registry`。如果是删除项目当前目录下的 `target` 目录里的缓存, 可以直接用 `cargo clean`。

52.19.2 Chapter 2 Programming a Guessing Game

- 直接输出, 用 `println!("hello world");`, string interpolation 用 `println!("a is {a}");`.
- 展示了 `Result` 和 `match` 的用法
- `rust` 常规, 比较大小用 `x.cmp(y)`, 配合 `match`, 虽然, 也可以用 `if (a < b)` 这种方式
- 演示了如何用 `rand` 这个 `crate`
- 查看帮助, 用 `cargo doc --open`, 可以看项目里用到的所有的 `crate` 的帮助
- `cargo check`, `cargo build`, `cargo run` 的用法
- 循环用 `loop`, `break`, `continue`
- `if` 语句, 必须用 `{}` 括起来。可以写成 `if a < b {}`, 不需要写成 `if (a < b) {}`, 虽然不算错, 但编译时, 会报 `warnings`, 提示没必要加 `()`
- 从命令行输入 `std::io` 这个模块
- 字符串的 `new`, `trim`, `parse` (字符串转为数字)
- 创建一个项目, 用 `cargo new`, 自从生成 `Cargo.toml`, `src/main.rs`, 里面有一个 `hello word` 的 `main` 程序

52.19.3 Chapter 3 Common Programming Concepts

- 有哪些基本数据类型 - `int`: `i32`, `u32`, `i8`, `u8`, etc - `float`: `f32`, `f64` - `bool`, 一个字节 - `char`, 4 个字节
- `let` 和 `const` 的区别
- 如何定义 `mutable` 变量
- **`tuple` 与数组的定义**
 - `tuple` 可以有 `x.0`, `x.1` 等访问, 也有 `tupe destructure`
 - 数组, `runtime` 会有 `out of boundary check`
- 如何 `iterate` 一个数组
- `if`, `loop`, `while`, `for` 的用法
- `break`, `continue` 的用法
- 如何定义一个函数
- 大括号是一个 `expression`, 虽然没有 `?:` 三元操作符, 但是可以用 `if/else` 实现同样的效果, 类似 `kotlin` 里的

53.1 Basics

53.1.1 What is this GLIBCXX error?

See <https://omairmajid.com/posts/2020-07-08-what-is-glibcxx-error/>

```
app: /lib64/libc.so.6: version `GLIBC_3.1.45' not found (required by ./app)
app: /lib64/libstdc++.so.6: version `GLIBCXX_3.4.20' not found (required by ./app)
```

It shows how to build and use glibc.

```
readelf --dyn-syms onnxruntime-linux-aarch64-1.16.3/lib/libonnxruntime.so | grep ↵
↵GLIBCXX
```

```
$ readelf --dyn-syms onnxruntime-linux-aarch64-1.16.3/lib/libonnxruntime.so | grep ↵
↵GLIBCXX
   6: 0000000000000000      0 FUNC      GLOBAL DEFAULT  UND _Znam@GLIBCXX_3.4 (3)
   8: 0000000000000000      0 FUNC      GLOBAL DEFAULT  UND _ZNSo3putEc@GLIBCXX_3.4 (3)
  10: 0000000000000000      0 FUNC      GLOBAL DEFAULT  UND _ZNSt6chrono3_V212system_
↵@GLIBCXX_3.4.19 (5)
  11: 0000000000000000      0 FUNC      GLOBAL DEFAULT  UND _ZNSt8__detail15_List_
↵nod@GLIBCXX_3.4.15 (6)
  15: 0000000000000000      0 FUNC      GLOBAL DEFAULT  UND _ZNSirsERs@GLIBCXX_3.4 (3)
  16: 0000000000000000      0 FUNC      GLOBAL DEFAULT  UND _ZSt29_Rb_tree_insert_
↵and@GLIBCXX_3.4 (3)
  19: 0000000000000000      0 FUNC      GLOBAL DEFAULT  UND _ZNSdD2Ev@GLIBCXX_3.4 (3)
```

See also

- How the GNU C Library handles backward compatibility

<https://developers.redhat.com/blog/2019/08/01/how-the-gnu-c-library-handles-backward-compatibility#>

54.1 Docker

See

- <https://docs.ros.org/en/iron/Installation/Ubuntu-Install-Debians.html>
- <https://docs.ros.org/en/iron/Tutorials.html>

The QA forum is <https://robotics.stackexchange.com/>

```
docker pull osrf/ros2:nightly
docker run -it osrf/ros2:nightly
docker run --rm -it --network host --name ros2 -v /host/path:/container/path osrf/
↪ros2:nightly /bin/bash
# connect to the container
docker exec -it ros2 /bin/bash
```

After starting the docker container,

```
# rolling is the name of the distro
#
# which ros2 prints: /opt/ros/rolling/bin/ros2

source /opt/ros/rolling/setup.bash

ros2 run demo_nodes_cpp talker

# In another terminal
source /opt/ros/rolling/setup.bash
ros2 run demo_nodes_py listener
```

```
printenv | grep -i ROS
```

prints the following:

```
ROS_VERSION=2
PKG_CONFIG_PATH=/opt/ros/rolling/lib/x86_64-linux-gnu/pkgconfig:/opt/ros/rolling/lib/
↳pkgconfig
ROS_PYTHON_VERSION=3
ROSDISTRO_INDEX_URL=https://raw.githubusercontent.com/osrf/docker_images/master/ros2/
↳nightly/nightly/index-v4.yaml
AMENT_PREFIX_PATH=/opt/ros/rolling
CMAKE_PREFIX_PATH=/opt/ros/rolling
ROS_AUTOMATIC_DISCOVERY_RANGE=SUBNET
COLCON_PREFIX_PATH=/opt/ros/rolling
PYTHONPATH=/opt/ros/rolling/lib/python3.10/site-packages
LD_LIBRARY_PATH=/opt/ros/rolling/opt/rviz_ogre_vendor/lib:/opt/ros/rolling/opt/mimick_
↳vendor/lib:/opt/ros/rolling/lib
PATH=/opt/ros/rolling/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
ROS_DISTRO=rolling
```

See <https://mirrors.tuna.tsinghua.edu.cn/help/ubuntu/> to change the apt source. Also see <https://mirrors.tuna.tsinghua.edu.cn/help/ros2/>.

Install the package `ros-rolling-turtlesim`:

```
sudo apt update
sudo apt install ros-rolling-turtlesim

ros2 pkg -h

# Check that it is installed
ros2 pkg executables turtlesim
```

It prints:

Now we can run

```
# ros2 run <package_name> <executable_name>

ros2 run turtlesim turtlesim_node
ros2 node list
ros2 topic list
ros2 service list
ros2 action list
ros2 node info <node_name>
```

Source code for turtlesim: https://github.com/ros/ros_tutorials/tree/iron/turtlesim

55.1 Basics

The latest version as of 2023.03.21 is v1.8.3

<https://github.com/csukuangfj/openfst-1.8.3/blob/master/BUILD.bazel> lists the build rules. The base lib is:

```
cc_library(  
    name = "base",  
    srcs = [  
        prefix_dir + "lib/compat.cc",  
        prefix_dir + "lib/flags.cc",  
    ],  
    hdrs = [  
        prefix_dir + "include/fst/compat.h",  
        prefix_dir + "include/fst/flags.h",  
        prefix_dir + "include/fst/icu.h",  
        prefix_dir + "include/fst/lock.h",  
        prefix_dir + "include/fst/log.h",  
        prefix_dir + "include/fst/windows_defs.inc",  
    ],  
    defines = select({  
        ":has_absl": ["OPENFST_HAS_ABSL=1"],  
        "//conditions:default": [],  
    }),  
    includes = [prefix_dir + "include"],  
    deps = select({  
        ":has_absl": ["@com_google_absl//absl/synchronization"],  
        "//conditions:default": [],  
    }),  
)
```

55.1.1 compat.h

```
#if defined(__GNUC__) || defined(__clang__)
#define OPENFST_DEPRECATED(message) __attribute__((deprecated(message)))
#elif defined(_MSC_VER)
#define OPENFST_DEPRECATED(message) [[deprecated(message)]]
#else
#define OPENFST_DEPRECATED(message)
#endif
```

56.1 Hello

To setup an environment variable:

```
%env a=3
```

then we can in other cells, use \$a.

%%shell so that all commands in the current cell are executed using bash.

57.1 Install

1. Visit <https://dart.dev/get-dart>
2. We use <https://dart.dev/get-dart/archive>

```
wget https://storage.googleapis.com/dart-archive/channels/stable/release/3.4.0/sdk/  
↳ dartsdk-macos-x64-release.zip
```

```
-rw-r--r--  1 fangjun  staff   221M May 15 11:40 dartsdk-macos-x64-release.zip
```

```
cd ~/software  
unzip ~/Downloads/dartsdk-macos-x64-release.zip  
  
# It will create a directory ~/software/dart-sdk  
rm ~/Downloads/dartsdk-macos-x64-release.zip  
  
export PATH=/Users/fangjun/software/dart-sdk/bin:$PATH
```

```
(py38) fangjuns-MacBook-Pro:software fangjun$ dart --version  
Dart SDK version: 3.4.0 (stable) (Mon May 6 07:59:58 2024 -0700) on "macos_x64"  
  
(py38) fangjuns-MacBook-Pro:software fangjun$ which dart  
/Users/fangjun/software/dart-sdk/bin/dart
```

57.1.1 codefmt in vim

```
# in the terminal, run
alias dartfmt="dart format"
```

57.1.2 Install Dart plugin in IntelliJ IDEA

Name of the plugin is `Dart`. It has more than 18.8M downloads.

57.2 Package

Dart packages can be found at <https://pub.dev/>.

57.2.1 example

```
name: my_app

environment:
  sdk: ^3.4.0

dependencies:
  js: ^0.6.0
  intl: ^0.17.0
```

Save it to `pubspec.yaml`, and run:

```
dart pub get
```

It prints:

```
Resolving dependencies...
Downloading packages...
+ clock 1.1.1
+ intl 0.17.0 (0.19.0 available)
+ js 0.6.7 (0.7.1 available)
+ meta 1.15.0
+ path 1.9.0
Changed 5 dependencies!
2 packages have newer versions incompatible with dependency constraints.
Try `dart pub outdated` for more information.
```

To add a new dependency from the command line, run:

```
dart pub add vector_math
```

It prints:

```
Resolving dependencies...
Downloading packages...
  intl 0.17.0 (0.19.0 available)
  js 0.6.7 (0.7.1 available)
+ vector_math 2.1.4
Changed 1 dependency!
2 packages have newer versions incompatible with dependency constraints.
Try `dart pub outdated` for more information.
```

It changes dependencies in `pubspec.yaml` to:

```
dependencies:
  js: ^0.6.0
  intl: ^0.17.0
  vector_math: ^2.1.4
```

Note: There is a `.dart_tool` in the current directory generated automatically by `dart pub get`. Part of it is given below:

```
{
  "configVersion": 2,
  "packages": [
    {
      "name": "clock",
      "rootUri": "file:///Users/fangjun/.pub-cache/hosted/pub.dev/clock-1.1.1",
      "packageUri": "lib/",
      "languageVersion": "2.12"
    },
  ],
```

To upgrade all dependencies:

```
dart pub upgrade
```

To upgrade a specific dependency:

```
dart pub upgrade
```

There is an automatically generated file `pubspec.lock`.

See <https://dart.dev/guides/libraries/create-packages> to create a new package.

```
dart create -t package hello
```

will create a directory `hello`. There are template files in `hello`.

57.3 Basics

Listing 1: `./code/hello.dart`

```
1 // run it with
2 // dart ./hello.dart
3 // or
4 // dart run ./hello.dart
5 void main() {
6   print('hello world');
7 }
```

Listing 2: `./code/hello2.dart`

```
1 // run it with
2 // dart ./hello2.dart
3 // or
4 // dart run ./hello2.dart
5 void main(List<String> args) {
6   print('args: $args');
7 }
8 /*
9 dart hello2.dart --> args: []
10 dart hello2.dart a --> args: [a]
11
12 Note that arg[0] is not the exe name!
13
14 dart hello2.dart a bc --> args: [a, bc]
15 */
```

The entrypoint is the `main` function. It can take 0 argument or a list argument.

Note that `print` outputs a newline.

The help info for `print` can be found at <https://api.dart.dev/dart-core/print.html>. Note that its prototype is:

```
void print(Object? object)
```

It can print any type!

Listing 3: ./code/test-math.dart

```

1 // dart run ./test-math.dart
2 import "dart:math" as math;
3
4 void main() {
5   print(math.pi); // 3.141592653589793
6   print(math.sin(30 * math.pi / 180)); // 0.49999999
7   print(math.max(2, 3.5)); // 3.5
8 }

```

Naming convention `MyClass.myMethod()`, `myVariable`, `sourceUrl`, `urlDestination`.

A string uses utf-16 encoding, like JavaScript. We can use either a single quote or double quotes for string literals, e.g., `'hello'` and `"hello"` are equal.

Type of a string is `String`.

String interpolation is like javascript, e.g., `"hello ${world}"`, `"hello $world"`. We can put arbitrary expression inside `${}`.

Note that `String` is immutable in Dart, like Python.

Doc for functions <https://dart.dev/language/functions>.

Doc for packages <https://dart.dev/guides/packages>.

`num` is a super class of `int` and `double`. `int` is 64-bit.

There is also a `bool` type with `true` and `false`.

`String` and `num` are subtypes of `Object`. All non-nullable types in Dart are subtype of `Object`.

Like kotlin, it has nullable types.

Collection type examples:

- a list, `List<int>`. See help at <https://api.dart.dev/stable/3.4.0/dart-core/List-class.html>
- a set, `Set<String>`
- a map, `Map<String, int>`

Two ways to create a list:

```

var s = <String>[];
List<String> s = [];

```

```

int? a; // default initialized to null

a ??= 3; // assign 3 to a if a is not null; else do nothing.

```

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```
a = null;

int b = a ?? 10; // if a is not null, assign a to b, else assign 10 to b
```

Listing 4: ./code/test1.dart

```
1 // run it with
2 // dart --enable-asserts ./test1.dart
3
4 void testDouble() {
5     // control the number of decimal points when converting it to a string
6     double i = 2.354;
7     assert(i.toString() == '2.354');
8     assert(i.toStringAsPrecision(1) == '2');
9     assert(i.toStringAsPrecision(2) == '2.4');
10    assert(i.toStringAsPrecision(3) == '2.35');
11    assert(i.toStringAsPrecision(4) == '2.354');
12
13    i = double.parse('2.32');
14    assert(i == 2.32);
15
16    assert(i.ceil() == 3);
17    assert(i.floor() == 2);
18
19    int a = i.toInt();
20    assert(a == 2);
21 }
22
23 void testInt() {
24     int i = int.parse('23');
25     assert(i == 23);
26
27     // note that i/2 is a double
28     assert(i / 2 == 11.5);
29     assert((i / 2) is double);
30
31     assert((i / 1) is double);
32     assert((i / 1) == 23.0);
33
34     // truncation division: ~/
35     assert((i ~/ 2) is int);
36     assert((i ~/ 2) == 11);
```

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```
37
38 // see https://api.flutter.dev/flutter/dart-core/int/tryParse.html
39 // int.tryParse() returns null if the input is not an integer
40 int a = int.tryParse('30a') ?? -1;
41 assert(a == -1);
42
43 a = 10;
44
45 a += 1;
46 assert(a == 11);
47
48 a -= 1;
49 assert(a == 10);
50
51 // note that both ++a and a++ are valid
52 a++;
53 assert(a == 11);
54
55 ++a;
56 assert(a == 12);
57 }
58
59 // the type is String, not string
60 void testString() {
61     var s = 'hello';
62     var s2 = s + " world";
63     assert(s2 == "hello world");
64
65     assert("${s} world" == s2); // string interpolation
66     assert("$s world" == s2); // string interpolation
67
68     s = '''like
69 python. multi line string
70     ''';
71     print(s);
72
73     s = """another
74         multi-line string
75         """;
76     print(s);
77
78     String a = 'abc';
79     assert(a.toUpperCase() == 'ABC');
```

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```
80
81   a = 'ABC';
82
83   // we can print a string on assertion failure
84   assert(a.toLowerCase() == 'abc', "${a.toLowerCase()}");
85 }
86
87 void testBoolean() {
88   bool a = true;
89   assert(a);
90   bool c = 1 == 2;
91   assert(!c);
92 }
93
94 void testList() {
95   var a = [1, 2, 'a'];
96   assert(a[0] == 1);
97   assert(a[1] == 2);
98   assert(a[2] == 'a');
99   a.add(40.5);
100  assert(a[3] == 40.5);
101  print(a);
102
103  var b = List.filled(3, 0); // b's capacity is fixed, cannot be changed.
104  assert(b.length == 3);
105  assert(b[0] == 0);
106  assert(b[1] == 0);
107  assert(b[2] == 0);
108  // b.add(3); // compile time error. cannot change the size of b
109
110  var c = [1, 3, 5];
111  List<String> d = [];
112  c.forEach((i) {
113    d.add(i.toString());
114  });
115  assert(d.length == c.length);
116  assert(d[0] == c[0].toString());
117  assert(d[1] == c[1].toString());
118  assert(d[2] == c[2].toString());
119
120  // map returns an iterable; we use .toList() to convert it to a list
121  List<String> f = c.map((i) {
122    return i.toString();
```

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```

123     }).toList();
124     assert(f.length == c.length);
125     assert(f[0] == d[0]);
126     assert(f[1] == d[1]);
127     assert(f[2] == d[2]);
128
129     List<int> li = [0, 1, 2, 3];
130     for (int i = 0; i != li.length; ++i) {
131         assert(i == li[i]);
132     }
133
134     for (var value in li) {
135         assert(value == li[value]);
136     }
137
138     // we can also use final if we don't change value inside the loop
139     for (final value in li) {
140         assert(value == li[value]);
141     }
142 }
143
144 void testList2() {
145     // type is List<String>
146     final aListOfStrings = ['one', 'two', 'three'];
147     assert(aListOfStrings is List<String>);
148     assert(aListOfStrings.isNotEmpty);
149
150     // type is Set<String>
151     final aSetOfStrings = {'one', 'two', 'three'};
152     assert(aSetOfStrings is Set<String>);
153     assert(aSetOfStrings.length == 3);
154     assert(aSetOfStrings.contains('one'));
155     assert(aSetOfStrings.contains('two'));
156     assert(aSetOfStrings.contains('three'));
157
158     // type is Map<String, int>
159     final aMapOfStringsToInts = {
160         'one': 1,
161         'two': 2,
162         'three': 3,
163     };
164     assert(aMapOfStringsToInts is Map<String, int>);
165     assert(aMapOfStringsToInts.length == 3);

```

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```
166  assert(aMapOfStringsToInts['one'] == 1);
167  assert(aMapOfStringsToInts['two'] == 2);
168  assert(aMapOfStringsToInts['three'] == 3);
169
170  final aListOfInt = <int>[];
171  assert(aListOfInt is List<int>);
172
173  assert(aListOfInt.isEmpty);
174
175  aListOfInt.add(10);
176  aListOfInt.add(20);
177  assert(aListOfInt.length == 2);
178  assert(aListOfInt[0] == 10);
179  assert(aListOfInt[1] == 20);
180
181  final aSetOfInt = <int>{};
182  assert(aSetOfInt is Set<int>);
183
184  final aMapOfIntToDouble = <int, double>{};
185  assert(aMapOfIntToDouble is Map<int, double>);
186
187  // final aListOfBaseType = <BaseType>[SubType(), SubType()];
188 }
189
190 void testNullable() {
191   int? a;
192   assert(a == null);
193   a = 3;
194   assert(a == 3);
195   int b = a!; // kotlin uses two !, while dart uses only one
196   assert(b == 3);
197
198   a = null;
199
200   b = a ?? 10; // if a is null ,then assign 10 to b, else assign a to b
201   assert(b == 10);
202
203   a = 20;
204   b = a ?? 10;
205   assert(b == 20);
206 }
207
208 void testTypeInference() {
```

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```
209  var a = 1;
210  assert(a is int);
211
212  var b = 's';
213  assert(b is String);
214
215  var c = 3.7;
216  assert(c is double);
217
218  var d = ['1', '2', '3'];
219  assert(d is List<String>);
220
221  var f = {
222      'one': 1,
223      'two': 2,
224  };
225  assert(f is Map<String, int>);
226 }
227
228 void testDynamic() {
229     dynamic a = 1;
230     assert(a is int);
231
232     a = 2.5;
233     assert(a is double);
234
235     a = 'a';
236     assert(a is String);
237 }
238
239 void testObject() {
240     Object a = 1;
241     assert(a is int);
242     assert(a.runtimeType == int);
243
244     int i = a as int; // type casting
245     assert(i == 1);
246
247     a = 2.5;
248     assert(a is double);
249
250     a = 'a';
251     assert(a is String);
```

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```
252 }
253
254 void testCompileTimeConstant() {
255     const a = 1;
256     assert(a is int);
257
258     // specify the type explicitly
259     const String b = 'hello';
260     assert(b == 'hello');
261 }
262
263 void testRuntimeConstant() {
264     double a = 5;
265     final b = a / 2;
266     assert(b is double);
267     assert(b == 2.5);
268
269     final int c = 3;
270     assert(c == 3);
271 }
272
273 void main() {
274     testDouble();
275     testInt();
276     testString();
277     testBoolean();
278     testList();
279     testList2();
280     testNullable();
281     testTypeInference();
282     testDynamic();
283     testObject();
284     testCompileTimeConstant();
285     testRuntimeConstant();
286 }
```

Listing 5: ./code/test2.dart

```
1 // run it with
2 // dart --enable-asserts ./test2.dart
3
4 int add(int a, int b) {
5     return a + b;
6 }
```

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```

6 }
7
8 // if it contains a single return statement, it can be simplified
9 int add2(int a, int b) => a + b;
10
11 // optional positional arguments.
12 // we usually don't use positional arguments
13 int func(int a, [int? b, int? c]) {
14     return a - (b ?? 0) + (c ?? 0);
15 }
16
17 int func2(int a, {int? b, int? c}) {
18     return a - (b ?? 0) + (c ?? 0);
19 }
20
21 int myMax(int a, int b) {
22     // like ?: in C/C++
23     return (a > b) ? a : b;
24 }
25
26 // both ways are fine to define a function type
27 typedef int MyAdd(int a, int b); // argument names cannot be omitted
28
29 // argument names can be omitted
30 typedef MyAdd2 = int Function(int, int);
31 // typedef MyAdd2 = int Function(int a, int b);
32
33 void main() {
34     assert(add(2, 3) == 5);
35     assert(add2(2, 3) == 5);
36
37     // for positional arguments, we cannot ignore b if we want to pass c,
38     assert(func(10) == 10); // a is 10
39     assert(func(10, 3) == 7); // a is 10, b is 3, c is null
40     assert(func(10, null, 5) == 15); // a is 10, b is null, c is 5
41     assert(func(10, 3, 5) == 12); // a is 10, b is 3, c is 5
42
43     assert(func2(10) == 10);
44     assert(func2(10, c: 3) == 13);
45     assert(func2(10, b: 2) == 8);
46     assert(func2(10, c: 3, b: 2) == 11);
47     assert(func2(10, b: 2, c: 3) == 11);
48

```

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```
49 MyAdd myAdd = add;
50 assert (myAdd (2, 3) == 5);
51
52 MyAdd2 myAdd2 = add;
53 assert (myAdd2 (2, 3) == 5);
54
55 // define a function type
56
57 var f = (a, b) => a + b;
58 assert (f (2, 3) == 5);
59
60 assert (myMax (1, 2) == 2);
61 }
```

57.3.1 control

Listing 6: ./code/test-control.dart

```
1 // dart --enable-asserts ./test-control.dart
2
3 void TestIf() {
4   int a = 2;
5   int b = 0;
6   if (a > 1) {
7     b = 3;
8   } else {
9     b = 4;
10  }
11
12  assert (b == 3);
13
14  //like C/C++
15  b = (a > 1) ? 10 : 20;
16  assert (b == 10);
17 }
18
19 void TestSwitchInt() {
20   int a = 1;
21   int b;
22   switch (a) {
23     case 10:
24       b = 2;
```

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```
25     break;
26     case 1:
27         b = 3;
28         break;
29     default:
30         b = 4;
31         break;
32 }
33 assert(b == 3);
34 }
35
36 void TestSwitchString() {
37     String s = 'zero';
38     int b;
39     switch (s) {
40         case 'zero':
41             b = 0;
42             break;
43         case 'one':
44             b = 1;
45             break;
46         default:
47             b = -1;
48             break;
49     }
50
51     assert(b == 0);
52 }
53
54 enum MyNumber {
55     zero,
56     one,
57     two,
58 } // no trailing ;
59
60 void TestSwitchEnum() {
61     final n = MyNumber.one;
62
63     // counts from 0
64     assert(n.index == 1);
65
66     int b = -1;
67     switch (n) {
```

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```
68     case MyNumber.zero:
69         b = 0;
70         break;
71     case MyNumber.one:
72         b = 1;
73         break;
74     case MyNumber.two:
75         b = 2;
76         break;
77     // no need to use default
78 }
79
80 assert(b == 1);
81 }
82
83 void TestWhile() {
84     int i = 0;
85     int sum = 0;
86     while (i <= 3) {
87         sum += i;
88         i++;
89     }
90     assert(sum == 1 + 2 + 3);
91 }
92
93 void TestDoWhile() {
94     int i = 0;
95     int sum = 0;
96     do {
97         sum += i;
98         i++;
99     } while (i <= 3);
100
101     assert(sum == 1 + 2 + 3);
102 }
103
104 void main() {
105     TestIf();
106     TestSwitchInt();
107     TestSwitchString();
108     TestSwitchEnum();
109     TestWhile();
110     TestDoWhile();
111 }
```

57.3.2 record

Listing 7: ./code/test_record.dart

```
1 // run it with
2 // dart --enable-asserts ./test_record.dart
3
4 // records requires dart>=3.0
5 void test1() {
6   (int, int) a;
7
8   // (2, 3) is immutable
9   a = (2, 3);
10  assert(a.$1 == 2); // count from 1
11  assert(a.$2 == 3);
12
13  // for named fields in a record, we must use named assignment
14  ({int a, int b}) c = (a: 10, b: 20);
15  assert(c.a == 10);
16  assert(c.b == 20);
17
18  // mix of positional and named
19  (int, {int a, int b}) d;
20  d = (10, a: 20, b: 30);
21  assert(d.$1 == 10);
22  assert(d.a == 20);
23  assert(d.b == 30);
24
25  var f = (100, a: 200, b: 300, 400);
26  assert(f == (100, 400, a: 200, b: 300));
27
28  (int, int, {int a, int b}) h;
29
30  // named fields can appear any where
31  h = (a: 1, b: 2, 0, 3);
32
33  assert(h.$1 == 0);
34  assert(h.$2 == 3);
35  assert(h.a == 1);
36  assert(h.b == 2);
37
38  (num, Object) k = (42, 'a');
39  assert(k.$1 is int);
40  assert(k.$2 is String);
```

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```
41
42 // as kotlin, to convert types
43 int k1 = k.$1 as int;
44 String k2 = k.$2 as String;
45
46 assert(k1 == k.$1);
47 assert(k2 == k.$2);
48 }
49
50 void main() {
51     test1();
52 }
```

57.3.3 function

Listing 8: ./code/test-function.dart

```
1 // dart --enable-asserts ./test-function.dart
2 int add(int a, int b) {
3     return a + b;
4 }
5
6 // b is optional.
7 // if b is not passed, it is null by default
8 // optional arguments must be placed at the end
9 int add2(int a, [int? b]) {
10     if (b == null) {
11         return a + 2;
12     }
13     return a + b;
14 }
15
16 // multiple optional arguments
17 int add3(int a, [int? b, int? c]) {
18     return a + (b ?? 1) + (c ?? 2);
19 }
20
21 // optional argument with a default value
22 int add4(int a, [int b = 10]) {
23     return a + b;
24 }
25
```

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```
26 // b must be passed by name
27 int add5(int a, {required int b}) {
28     return a + b;
29 }
30
31 // b is an optional name parameter
32 int add6(int a, {int? b}) {
33     return a + (b ?? 3);
34 }
35
36 void main() {
37     int c = add(2, 3);
38     assert(c == 5);
39
40     c = add2(10);
41     assert(c == 12);
42
43     c = add2(10, 5);
44     assert(c == 15);
45
46     c = add3(10);
47     assert(c == 13);
48
49     c = add3(10, 2);
50     assert(c == 14);
51
52     c = add4(2);
53     assert(c == 12);
54
55     c = add5(2, b: 3);
56     assert(c == 5);
57
58     c = add6(5);
59     assert(c == 8);
60
61     c = add6(6);
62     assert(c == 9);
63 }
```

57.3.4 class

Listing 9: ./code/test-class.dart

```
1 // dart run --enable-asserts ./test-class.dart
2
3 class User {
4   int id = 0;
5   String name = '';
6
7   @override
8   String toString() {
9     return 'User(id: $id, name: $name)';
10  }
11
12   String toJson() {
13     return '{"id":$id, "name":"$name"}';
14   }
15 } // no trailing ;
16
17 void testUser() {
18   final user = User();
19   print(user); // User(id: 0, name: )
20
21   user.id = 20;
22   print(user); // User(id: 20, name: )
23   print(user.toJson()); // {"id":20, "name":""}
24
25   var user2 = User();
26   user2.name = 'hello';
27   user2.id = 10;
28   print(user2); // User(id: 10, name: hello)
29
30   // cascade notation
31   user2 = User()
32     ..name = 'hello'
33     ..id = 10;
34   print(user2); // User(id: 10, name: hello)
35
36   // assign by reference
37   var user3 = user2;
38   user3.id = 100;
39   assert(user2.id == 100);
40 }
```

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```
41
42 class Point {
43     int _x = 0;
44
45     // getter
46     int get x => _x;
47
48     // getter
49     int get x2 => _x + 2;
50
51     // computed property
52     int get x3 {
53         return _x + 3;
54     }
55
56     // setter
57     set myX(int v) => _x = v;
58
59     // define another getter
60     int get myX => _x;
61
62     // validate the value
63     set myX2(int v) {
64         if (v < 10) {
65             print('v should be >= 10. Given: $v');
66             return;
67         }
68
69         _x = v;
70     }
71 }
72
73 void testPoint() {
74     final p = Point();
75     assert(p._x == 0);
76     assert(p.x == 0);
77     assert(p.x2 == 2);
78
79     p._x = 100;
80     assert(p.x == 100);
81     assert(p.x2 == 102);
82     assert(p.x3 == 103);
83 }
```

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```
84 p.myX = 5;
85 assert(p._x == 5);
86 assert(p.x == 5);
87 assert(p.myX == 5);
88
89 p.myX2 = 12;
90 assert(p.x == 12);
91
92 p.myX2 = 9; // we have required the value should be >= 10
93 assert(p.x == 12);
94
95 p.myX2 = 13;
96 assert(p.x == 13);
97 }
98
99 class Point2 {
100   // long form constructor
101   // x and y are required positional arguments
102   Point2(int x, int y) {
103     this.x = x;
104     this.y = y;
105   }
106   int x = 0;
107   int y = 0;
108 }
109
110 void testPoint2() {
111   final p = Point2(10, 20);
112   assert(p.x == 10);
113   assert(p.y == 20);
114 }
115
116 class Point3 {
117   // short form constructor
118   Point3(this.x, this.y);
119   int x = 0;
120   int y = 0;
121 }
122
123 void testPoint3() {
124   final p = Point3(100, 200);
125   assert(p.x == 100);
126   assert(p.y == 200);
```

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```
127 }
128
129 // we can have many named constructors
130 class Point4 {
131     Point4(this.x, this.y);
132
133     // named constructor
134     Point4.myPoint() {
135         x = 10;
136         y = 20;
137     }
138
139     Point4.myPoint2(int x) {
140         this.x = x;
141         y = 200;
142     }
143
144     // forwarding constructor, this(1, 2) is Point4(1, 2)
145     Point4.myPoint3() : this(1, 2);
146
147     factory Point4.myPoint4(int x, int y) {
148         // there is no this inside the factory constructor
149         // we need to create and return an object
150
151         // we can validate, and even change the passed argument
152         return Point4(x + 2, y + 3);
153     }
154
155     int x = 0;
156     int y = 0;
157 }
158
159 void testPoint4() {
160     var p = Point4.myPoint();
161     assert(p.x == 10);
162     assert(p.y == 20);
163
164     p = Point4.myPoint2(-1);
165     assert(p.x == -1);
166     assert(p.y == 200);
167
168     p = Point4.myPoint3();
169     assert(p.x == 1);
```

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```
170  assert(p.y == 2);
171
172  p = Point4.myPoint4(11, 22);
173  assert(p.x == 11 + 2);
174  assert(p.y == 22 + 3);
175 }
176
177 void main() {
178   testUser();
179   testPoint();
180   testPoint2();
181   testPoint3();
182   testPoint4();
183 }
```

57.3.5 null

Listing 10: ./code/test-null.dart

```
1  // dart run --enable-asserts ./test-null.dart
2
3  class User {
4    int id = 1;
5  }
6
7  void test1() {
8    int? a; // default to null
9
10   assert(a == null);
11
12   a ??= 3;
13   assert(a == 3);
14
15   a ??= 5; // a is not null, so it is a no-op
16   assert(a == 3);
17
18   a = null;
19
20   int b = a ?? 10; // a is null, so 10 is assigned to b
21   assert(b == 10);
22
23   a = 1;
```

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```

24  b = a ?? 5; // a is not null, so a is assigned to b
25  assert(b == 1);
26
27  a = null;
28  // b = a; // compile time error since a is null
29
30  // b = a!; // runtime exception: Null check operator used on a null value
31
32  User? u;
33  int id = u?.id ?? 10;
34  assert(id == 10);
35
36  u = User();
37  id = u?.id ?? 10;
38  assert(id == 1);
39 }
40
41 void main() {
42   test1();
43 }

```

57.3.6 list

Listing 11: ./code/test-list.dart

```

1  // dart run --enable-asserts ./test-list.dart
2
3  void test1() {
4    var a = [1, 2, 3]; // infer types automatically
5
6    List<int> b = [1, 2, 3]; // Use List<int> explicitly
7    print(b);
8
9    var c = <int>[]; // an empty list with a specified type
10   assert(c.isEmpty == true);
11   c.add(1);
12
13   assert(c.isEmpty == false);
14
15   c.add(3);
16   c.add(2);
17   assert(c.length == b.length); // get number of elements

```

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```
18  print(c); // [1, 3, 2]
19  c.sort(); // sort in-place
20  print(c); // [1, 2, 3]
21  }
22
23  // use a for loop like c/c++
24  void test2() {
25      var a = [1, 2, 3];
26      var sum = 0;
27      for (int i = 0; i < a.length; i++) {
28          sum += a[i];
29      }
30      assert(sum == 6);
31  }
32
33  void test3() {
34      var a = [1, 2, 3];
35      var sum = 0;
36      for (final i in a) {
37          sum += i;
38      }
39      assert(sum == 6);
40  }
41
42  // the spread operator
43  void test4() {
44      var a = [1, 2];
45      var b = [3];
46      var c = [...a, ...b];
47      print(c); // 1, 2, 3
48
49      c = [10, ...b, ...a];
50      print(c); // 10, 3, 1, 2
51  }
52
53  // collection for
54  void test5() {
55      var a = [1, 2, 3];
56
57      // add all elements from a
58      var b = [10, for (final i in a) i];
59      print(b); // [10, 1, 2, 3]
60
```

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```

61 // add elements from a whose value > 1
62 b = [
63     10,
64     for (final i in a)
65         if (i > 1) i
66 ];
67 print(b); // [10, 2, 3]
68
69 // convert a list of integers to a list of strings
70 var s = [for (final i in a) i.toString()];
71 print(s); // [1, 2, 3]
72 }
73
74 void main() {
75     test1();
76     test2();
77     test3();
78     test4();
79     test5();
80 }

```

57.3.7 set

Listing 12: ./code/test-set.dart

```

1 // dart run --enable-asserts ./test-set.dart
2
3 // two ways to define an empty set
4 void test1() {
5     final Set<int> a = {};
6     assert(a.isEmpty == true);
7
8     final b = <int>{};
9     assert(b.isEmpty == true);
10 }
11
12 void test2() {
13     var a = {1, 2, 1};
14     assert(a.length == 2);
15
16     assert(a.contains(1)); // 1 is inside the set
17     assert(a.contains(10) == false); // 10 is not inside the set

```

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```
18
19 a.add(10);
20 assert(a.contains(10)); // now 10 is inside the set
21 assert(a.length == 3);
22
23 a.remove(2);
24 assert(a.length == 2);
25 assert(a.contains(2) == false); // now 2 is not inside the set
26
27 var b = [5, 1];
28 a.addAll(b);
29 print(a); // {1, 10, 5}
30
31 // iterating a set
32 var sum = 0;
33 for (final i in a) {
34     sum += i;
35 }
36 assert(sum == 1 + 10 + 5);
37 }
38
39 // convert a list to a set
40 // convert a set to a list
41 void test3() {
42     var a = [1, 2, 1];
43     var b = a.toSet();
44     print(b); // {1, 2}
45     a = b.toList(); // [1, 2]
46     print(a);
47 }
48
49 void test4() {
50     var a = <int>{1, 2};
51     var b = a; // reference assignment
52     b.add(3);
53     assert(a.contains(3));
54
55     b = a.toSet(); // real copy
56     b.add(4);
57     assert(a.contains(4) == false);
58 }
59
60 void main() {
```

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```
61 test1();
62 test2();
63 test3();
64 test4();
65 }
```

57.3.8 map

Listing 13: ./code/test-map.dart

```
1 // dart run --enable-asserts ./test-map.dart
2
3 // two ways to create an empty map
4 void test1() {
5     final Map<String, int> a = {};
6     final b = <String, int>{};
7     assert(a.isEmpty);
8     assert(b.isEmpty);
9 }
10
11 // create a map from literals
12 void test2() {
13     final a = {
14         'apple': 20,
15         'cat': 10,
16     };
17
18     assert(a['cat'] == 10);
19     assert(a['dog'] == null); // return null if the key does not exist
20
21     a['dog'] = 50;
22     assert(a['dog'] == 50);
23
24     a['dog'] = 1;
25     assert(a['dog'] == 1);
26
27     a.remove('dog'); // remove an element from a map
28     assert(a['dog'] == null);
29
30     print(a); // {apple: 20, cat: 10}
31     assert(a.length == 2);
32     assert(a.isEmpty == false);
```

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```
33  assert(a.isNotEmpty == true);
34
35  assert(a.containsKey('dog') == false);
36 }
37
38 void test3() {
39   var a = <String, int>{
40     'one': 1,
41     'two': 2,
42   };
43
44   var sum = 0;
45
46   // iterate the keys
47   //
48   // a[k] is of type int?
49   for (final k in a.keys) {
50     sum += a[k]!;
51   }
52   assert(sum == 3);
53
54   sum = 0;
55
56   // iterate the values
57   for (final v in a.values) {
58     sum += v;
59   }
60   assert(sum == 3);
61
62   sum = 0;
63
64   // iterate with key and value
65   for (final entry in a.entries) {
66     sum += entry.key.length + entry.value;
67   }
68   assert(sum == 3 + 3 + 1 + 2);
69 }
70
71 void main() {
72   test1();
73   test2();
74   test3();
75 }
```

57.3.9 string

Listing 14: ./code/test-string.dart

```
1 // dart run --enable-asserts ./test-string.dart
2
3 // there is no char type
4 // both ' and " are ok
5 void test1() {
6     var s = 'hello';
7     s = "hello";
8 }
9
10 void test2() {
11     var s = 'hello';
12     var s2 = s.toUpperCase();
13     assert(s2 == 'HELLO');
14
15     var s3 = s2.toLowerCase();
16     assert(s3 == 'hello');
17
18     // trim
19     s = ' hello ';
20     s2 = s.trim();
21     assert(s2 == 'hello');
22
23     // trim can also trim \n \r \t
24     s = ' hello\n\r\t';
25     s2 = s.trim();
26     assert(s2 == 'hello');
27
28     s = ' hello\n\r\t';
29     s2 = s.trimLeft();
30     assert(s2 == 'hello\n\r\t');
31
32     s = ' hello\n\r\t';
33     s2 = s.trimRight();
34     assert(s2 == ' hello');
35 }
36
37 void test3() {
38     var s = '1'.padLeft(2, '0');
39     assert(s == '01');
40 }
```

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```
41 // 11 is already of length 2, so the pad is a no-op
42 s = '11'.padLeft(2, '0');
43 assert(s == '11');
44 }
45
46 // several ways to convert an integer to a string
47 void test4() {
48     var s = 11.toString();
49     assert(s == '11');
50
51     // string interpolation
52     var i = 11;
53     s = '$i';
54     assert(s == '11');
55 }
56
57 // split a string
58 // join a list of string
59 void test5() {
60     var s = '1,2,3,4';
61     var a = s.split(',');
62     assert(a is List<String>);
63
64     s = a.join('-');
65     assert(s == '1-2-3-4');
66 }
67
68 void test6() {
69     var a = '1,2,3';
70     var b = a.replaceAll(',', '-');
71     assert(b == '1-2-3');
72
73     a = '1,2,3';
74     b = a.replaceFirst(',', '-');
75     assert(b == '1-2,3');
76 }
77
78 // use a string buffer to build a string
79 void test7() {
80     var s = '0,1,2,3';
81     final buffer = StringBuffer();
82     var sep = ',';
83     for (int i = 0; i < 4; i++) {
```

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```

84     buffer.write(sep);
85     buffer.write('$i');
86     sep = ', ';
87 }
88 var s2 = buffer.toString();
89 assert(s == s2);
90 }
91
92 void test8() {
93     var s = 'hello';
94     assert(s.startsWith('h') == true);
95     assert(s.startsWith('hel') == true);
96     assert(s.startsWith('hi') == false);
97
98     assert(s.endsWith('o') == true);
99     assert(s.endsWith('lo') == true);
100    assert(s.endsWith('alo') == false);
101
102    assert(s.contains('ll'));
103    assert(s.contains('lol') == false);
104 }
105
106 void main() {
107     test1();
108     test2();
109     test3();
110     test4();
111     test5();
112     test6();
113     test7();
114     test8();
115 }

```

57.3.10 events

Listing 15: ./code/test-events.dart

```

1  // dart run --enable-asserts ./test-events.dart
2
3  /*
4  There are two queues:
5  - event queue (low priority)

```

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```
6   - micro task queue (high priority)
7
8   */
9
10  /*
11  it prints
12  first
13  third
14  second
15  */
16  void test1() {
17    print('first');
18
19    // add task to the event queue
20    Future(() => print('second'));
21    print('third');
22  }
23
24  /*
25  it prints
26  first
27  fourth
28  third
29  second
30  */
31  void test2() {
32    print('first');
33
34    // add task to the event queue
35    Future(() => print('second'));
36
37    // add task to the micro task queue
38    Future.microtask(() => print('third'));
39    print('fourth');
40  }
41
42  void main() {
43    // test1();
44    test2();
45  }
```

57.3.11 isolate

Listing 16: ./code/test-isolate.dart

```
1 // dart run --enable-asserts ./test-isolate.dart
2
3 import 'dart:isolate';
4
5 void test1Isolate(SendPort sendPort) {
6   sendPort.send('hello');
7   sendPort.send('world');
8   sendPort.send(null);
9 }
10
11 Future<void> test1() async {
12   final receivePort = ReceivePort();
13   final isolate = await Isolate.spawn<SendPort>(
14     test1Isolate,
15     receivePort.sendPort,
16   );
17
18   receivePort.listen((Object? message) {
19     if (message is String) {
20       print('message: $message');
21     } else if (message == null) {
22       receivePort.close();
23       isolate.kill();
24     }
25   });
26 }
27
28 void main() {
29   test1();
30 }
```

57.4 ToDo

- https://github.com/gtbluesky/onnxruntime_flutter

Wrap onnxruntime to dart.

- <https://github.com/dart-lang/path>

A very simple but popular package.

- <https://docs.flutter.dev/platform-integration/android/c-interop>
- <https://dart.dev/tutorials/server/cmdline>
- <https://blog.flutter.wtf/dart-ffi/>

57.5 ffi

Note there are two ffi packages:

- 1. One is builtin: `dart:ffi`
- 2. One is `package:ffi/ffi.dart` <https://pub.dev/documentation/ffi/latest/ffi/ffi-library.html>

Doc for `dart:io`:

- <https://dart.dev/libraries/dart-io>
- <https://api.dart.dev/stable/3.4.0/dart-io/dart-io-library.html>

Doc for `dart:ffi`: <https://api.dart.dev/stable/3.4.0/dart-ffi/dart-ffi-library.html>

Examples for `ffi`: <https://github.com/dart-lang/samples/tree/main/ffi>

57.5.1 types

- `char` -> `Char`
- `double` -> `Double`
- `float` -> `Float`
- `int` -> `Int`
- `int8` -> `Int8`
- `int16` -> `Int16`
- `int32` -> `Int32`
- `int64` -> `Int64`
- `intptr_t` -> `IntPtr`
- `long` -> `Long`

```
// void xxx();  
typedef HelloWorldFunc = ffi.Void Function();
```

```
# create a plugin  
flutter create --template=plugin --platforms=android,ios mybatteryplugin
```


Listing 17: ./code/ffi/a.h

```
1 extern "C" int add(int a, int b);
```

Listing 18: ./code/ffi/a.cc

```
1 #include "a.h"
2 int add(int a, int b) { return a + b; }
```

Listing 19: ./code/ffi/hello.dart

```
1 import 'dart:ffi' as ffi;
2 import 'dart:io' show Platform, Directory;
3 import 'package:path/path.dart' as path;
4
5 typedef HelloWorldFunc = ffi.Int Function(ffi.Int, ffi.Int);
6
7 typedef HelloWorld = int Function(int, int);
8
9 void main() {
10   var libraryPath = path.join(Directory.current.path, 'libfoo.so');
11   if (Platform.isMacOS) {
12     libraryPath = path.join(Directory.current.path, 'libfoo.dylib');
13   }
14   final dylib = ffi.DynamicLibrary.open(libraryPath);
15
16   // Look up the C function 'hello_world'
17   final HelloWorld hello =
18     dylib.lookup<ffi.NativeFunction<HelloWorldFunc>>('add').asFunction();
19   print(hello(2, 3));
20 }
```

Listing 20: ./code/ffi/pubspec.yaml

```
1 name: hello_ffi
2
3 environment:
4   sdk: ^3.4.0
5
6 dependencies:
7   path: ^1.8.0
```

Listing 21: ./code/ffi/run.sh

```
1  #!/usr/bin/env bash
2
3  g++ -c -fPIC -o a.o a.cc
4  g++ -shared -o libfoo.dylib ./a.o
5  ls -lh
6  nm libfoo.dylib
```

57.6 Flutter

Create a project from the command line:

```
flutter create my_app
```

will create a directory `my_app` in the current directory.

57.6.1 recording

<https://github.com/Picovoice/flutter-voice-processor> supports only Android and iOS. It is used in <https://github.com/Picovoice/porcupine/tree/master/binding/flutter>.

<https://github.com/llfbandit/record/blob/master/record/pubspec.yaml> is a good candidate since it supports many platforms. It uses https://github.com/flutter/packages/blob/main/packages/plugin_platform_interface/lib/plugin_platform_interface.dart to define an interface for different platforms.

58.1 Install

To install the flutter SDK,

1. Visit <https://docs.flutter.dev/get-started/install>
2. For macOS, download `flutter_macos_3.22.0-stable.zip`

```
cd ~/software
unzip /Users/fangjun/Downloads/flutter_macos_3.22.0-stable.zip
export PATH=/Users/fangjun/software/flutter/bin:$PATH
```

3. It also installs dart. There is no need to install dart separately.
4. Open Android Studio, and install the flutter plugin
 - Android studio settings, Plugins, Marketplace, search for flutter, click install
 - Install required plugings (the flutter plugin requires Dart to be installed), install
 - Restart IDE
 - Click File, New, New Flutter Project..
4. **export ANDROID_HOME=/Users/fangjun/software/my-android**
or use `flutter config --android-sdk /Users/fangjun/software/my-android/``
5. In the terminal, run `flutter doctor -v`.

```
[✓] Flutter (Channel stable, 3.22.0, on macOS 13.1 22C65 darwin-x64,
↳ locale en-CN)
    • Flutter version 3.22.0 on channel stable at /Users/fangjun/
↳ software/flutter
    • Upstream repository https://github.com/flutter/flutter.git
    • Framework revision 5dcb86f68f (8 days ago), 2024-05-09
↳ 07:39:20 -0500
```

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```

• Engine revision f6344b75dc
• Dart version 3.4.0
• DevTools version 2.34.3

[!] Android toolchain - develop for Android devices (Android SDK
↳ version 34.0.0)
  • Android SDK at /Users/fangjun/software/my-android
  • Platform android-34, build-tools 34.0.0
  • ANDROID_HOME = /Users/fangjun/software/my-android
  • Java binary at: /Applications/Android Studio.app/Contents/jbr/
↳ Contents/Home/bin/java
  • Java version OpenJDK Runtime Environment (build 17.0.10+0-17.0.
↳ 10b1087.21-11572160)
    ! Some Android licenses not accepted. To resolve this, run:
↳ flutter doctor --android-licenses

[!] Xcode - develop for iOS and macOS (Xcode 14.2)
  • Xcode at /Applications/Xcode.app/Contents/Developer
  • Build 14C18
  ! Flutter recommends a minimum Xcode version of 15.
    Download the latest version or update via the Mac App Store.
  ☑ CocoaPods not installed.
    CocoaPods is used to retrieve the iOS and macOS platform side
↳ 's plugin code that responds to your plugin usage on the Dart side.
    Without CocoaPods, plugins will not work on iOS or macOS.
    For more info, see https://flutter.dev/platform-plugins
    To install see https://guides.cocoapods.org/using/getting-
↳ started.html#installation for instructions.

[✓] Chrome - develop for the web
  • Chrome at /Applications/Google Chrome.app/Contents/MacOS/
↳ Google Chrome

[✓] Android Studio (version 2023.3)
  • Android Studio at /Applications/Android Studio.app/Contents
  • Flutter plugin can be installed from:
    ☑ https://plugins.jetbrains.com/plugin/9212-flutter
  • Dart plugin can be installed from:
    ☑ https://plugins.jetbrains.com/plugin/6351-dart
  • Java version OpenJDK Runtime Environment (build 17.0.10+0-17.0.
↳ 10b1087.21-11572160)

[✓] IntelliJ IDEA Community Edition (version 2024.1.1)

```

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```

    • IntelliJ at /Applications/IntelliJ IDEA CE.app
    • Flutter plugin can be installed from:
      🔗 https://plugins.jetbrains.com/plugin/9212-flutter
    • Dart plugin version 241.15989.9
[✓] Connected device (2 available)
    • macOS (desktop) • macos • darwin-x64 • macOS 13.1 22C65
↳ darwin-x64
    • Chrome (web) • chrome • web-javascript • Google Chrome 125.
↳ 0.6422.60

[✓] Network resources
    • All expected network resources are available.

! Doctor found issues in 2 categories.

```

6. In the terminal, run `flutter doctor --android-licenses`.

7. for macos

```

sudo sh -c 'xcode-select -s /Applications/Xcode.app/Contents/
↳ Developer && xcodebuild -runFirstLaunch'
sudo xcodebuild -license
sudo gem install cocoapods

```


KEYWORD SPOTTING (KWS)

59.1 Papers

Please usually use 40-dim features.

Small-footprint keyword spotting using deep neural networks [CPH14] is the first paper using DNN for KWS.

Convolutional neural networks for small-footprint keyword spotting [SP15] replaces DNN with CNN to reduce computation overhead and number of parameters.

60.1 Transformers

60.1.1 2017-Attention is all you need

See [VSP+17], the first paper about transformer.

60.1.2 2020-An image is worth 16x16 words: Transformers for image recognition at scale

See [DBK+20], known as ViT.

Transformers

60.2 Programming

60.2.1 1971-Program development by stepwise refinement

See [Wir71], written the author of the Pascal programming language.

Programming

60.3 Diffusion

60.3.1 Blogs

- Generating images with DDPMs: A PyTorch Implementation

<https://medium.com/@brianpulfer/enerating-images-with-ddpms-a-pytorch-implementation-cef5a2ba8cb1>

It also provides a colab notebook and a github repository <https://github.com/BrianPulfer/PapersReimplementations/tree/main/src/cv/ddpm>

- What are Diffusion Models?

<https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>

60.3.2 2020-Denoising diffusion probabilistic models

See [HJA20]. It is short for DDPM and has been cited by more than 11358 times.

60.3.3 2022-Denoising diffusion implicit models

See [SME20]. It is short for DDIM and has been cited by more than 4216 times.

Diffusion

60.4 books

- <https://libgen.re/>
- <https://libgen.yt/>

60.5 tts

60.5.1 2024-Voicebox: Text-guided multilingual universal speech generation at scale

See [LVS+24].

60.5.2 2024-E2 TTS: Embarrassingly easy fully non-autoregressive zero-shot TTS

See [:cte:p`eskimez2024e2`](#).

Text to speech

61.1 Install

61.1.1 online pascal compiler

- https://www.onlinegdb.com/online_pascal_compiler

61.1.2 Free pascal

macos

Visit <https://www.freepascal.org/> to download a *.pkg file for macOS. After installing, we will get an executable `fpc` in the command line.

which `fpc` prints `/usr/local/bin/fpc`.

`/usr/local/share/src/fpc-3.2.2/demo` contains examples.

```
mkdir ~/t
cd ~/t
cp /usr/local/share/src/fpc-3.2.2/demo/text/hello.pp ./
fpc ./hello.pp

It throws the following error::
Free Pascal Compiler version 3.2.2 [2021/05/16] for x86_64
Copyright (c) 1993-2021 by Florian Klaempfl and others
Target OS: Darwin for x86_64
Compiling ./hello.pp
Assembling hello
Linking hello
ld: library not found for -lc
An error occurred while linking
hello.pp(19,27) Error: Error while linking
```

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```
hello.pp(19,27) Fatal: There were 1 errors compiling module, stopping
Fatal: Compilation aborted
Error: /usr/local/bin/ppcx64 returned an error exitcode
```

The fix is to run:

```
fpc -XR/Applications/Xcode.app/Contents/Developer/Platforms/MacOSX.platform/Developer/
↳ SDKs/MacOSX.sdk/ ./hello.pp
```

The logs are:

```
Free Pascal Compiler version 3.2.2 [2021/05/16] for x86_64
Copyright (c) 1993-2021 by Florian Klaempfl and others
Target OS: Darwin for x86_64
Compiling ./hello.pp
Assembling hello
Linking hello
21 lines compiled, 0.2 sec
```

The following command:

```
fpc -h | grep XR
```

prints:

```
-Xd          Do not search default library path (sometimes required for cross-compiling
↳ when not using -XR)
-XR<x>       Prepend <x> to all linker search paths (BeOS, Darwin, FreeBSD, Linux, Mac
↳ OS, Solaris)
```

Linux

Visit <https://www.freepascal.org/> to download a *.tar file for Linux.

After unzipping, we get:

```
(py311) fangjun@ubuntu23-04:/mnt/sdb/shared/fpc-3.2.2.x86_64-linux$ ls
binary.x86_64-linux.tar  demo.tar.gz  doc-pdf.tar.gz  install.sh
```

Run ./install.sh.

1. Install prefix /mnt/sdb/shared/fpc-3.2.2
2. Install documentation
3. Install demo to /mnt/sdb/shared/fpc-3.2.2/share/doc/fpc-3.2.2/examples

4. Add /mnt/sdb/shared/fpc-3.2.2/bin to PATH

```
mkdir t
cd t
cp /mnt/sdb/shared/fpc-3.2.2/share/doc/fpc-3.2.2/examples/text/hello.pp .
fpc ./hello.pp
file ./hello

hello: ELF 64-bit LSB executable, x86-64, version 1 (SYSV), statically linked, for_
↳ GNU/Linux 2.4.0, stripped

ls -lh hello
-rwxrwxr-x 1 fangjun fangjun 187K Aug  4 15:18 hello

./hello
Hello world
```

Note that it is statically linked!

61.2 Hello

61.2.1 file extensions

- `xx.pp`
- `xx.ppu`, binary file, u is unit. A `xx.ppu` must be associated with a `xx.o` file to be usable
- `xx.pas`
- `xx.o`

`fpc ./hello.pp` will generate two files: `./hello` and `./hello.o`. We can delete `./hello.o` in this case since `./hello` is an executable.

```
fpc hello
```

61.2.2 unit search path

Unit search path:

```
-Fu/some/path/to/some/dir
```

By default, the environment variable `FPCDIR` is also on the unit search path. If `FPCDIR` is not set, it is `/usr/local/lib/fpc/FPCVERSION` by default.

For instance it is `/usr/local/lib/fpc/3.2.2` on my macOS.

```
# on macOS

ls /usr/local/lib/fpc/3.2.2/units/
aarch64-darwin i386-darwin      x86_64-darwin

# on Ubuntu

fangjun@ubuntu23-04:/mnt/sdb/shared/fpc-3.2.2/lib/fpc/3.2.2$ ls units/
x86_64-linux
```

See unit search paths:

```
fpc -vu ./hello.pp
```

To specify a compiler config file:

```
fpc @./my.cfg ./hello.pp
```

where `./my.cfg` is something like below:

```
-Fu/etc/
```

61.2.3 include file search path

```
-I/some/path
-Fi/some/path
```

Both `-I` and `-Fi` can be used.

61.2.4 Config file

The default config file `fpc.cfg` is

- `$PWD/fpc.cfg`
- `$HOME/.fpc.cfg`
- `$PPC_CONFIG_PATH/fpc.cfg`, where `PPC_CONFIG_PATH` is an environment variable In the `etc` directory above the `fpc bin` directory
- `/etc/fpc.cfg`

61.3 fpc -h

The output of `fpc -help` or `fpc -h` is given below:

```
Free Pascal Compiler version 3.2.2 [2021/05/16] for x86_64
Copyright (c) 1993-2021 by Florian Klaempfl and others
fpc [options] <inputfile> [options]

Only options valid for the default or selected platform are listed.
Put + after a boolean switch option to enable it, - to disable it.
@<x>    Read compiler options from <x> in addition to the default fpc.cfg
-a      The compiler does not delete the generated assembler file
-a5     Don't generate Big Obj COFF files for GNU Binutils older than 2.25
↪(Windows, NativeNT)
-al     List sourcecode lines in assembler file
-an     List node info in assembler file (-dEXTDEBUG compiler)
-ao     Add an extra option to external assembler call (ignored for internal)
-ap     Use pipes instead of creating temporary assembler files
-ar     List register allocation/release info in assembler file
-at     List temp allocation/release info in assembler file
-A<x>   Output format:
-Adefault Use default assembler
-Aas     Assemble using GNU AS
-Agas    Assemble using GNU GAS
-Agas-darwin Assemble darwin Mach-O64 using GNU GAS
-Amasm   Win64 object file using ml64 (Microsoft)
-Apecoff PE-COFF (Win64) using internal writer
-Aelf    ELF (Linux-64bit) using internal writer
-Ayasm   Assemble using Yasm (experimental)
-Anasm   Assemble using Nasm (experimental)
-Anasmwin64 Assemble Win64 object file using Nasm (experimental)
-Anasmelf Assemble Linux-64bit object file using Nasm (experimental)
-Anasmdarwin Assemble darwin macho64 object file using Nasm (experimental)
-b      Generate browser info
-bl     Generate local symbol info
-B      Build all modules
-C<x>   Code generation options:
-C3     Turn on ieee error checking for constants
-Ca<x>  Select ABI; see fpc -i or fpc -ia for possible values
-Cb     Generate code for a big-endian variant of the target architecture
-Cc<x>  Set default calling convention to <x>
-CD     Create also dynamic library (not supported)
-Ce     Compilation with emulated floating point opcodes
-Cf<x>  Select fpu instruction set to use; see fpc -i or fpc -if for
↪possible values
```

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```

-CF<x>      Minimal floating point constant precision (default, 32, 64)
-Cg         Generate PIC code
-Ch<n>[,m]  <n> bytes min heap size (between 1023 and 67107840) and optionally
→[m] max heap size
-Ci         IO-checking
-Cn         Omit linking stage
-Co         Check overflow of integer operations
-CO         Check for possible overflow of integer operations
-Cp<x>      Select instruction set; see fpc -i or fpc -ic for possible values
-CP<x>=<y>   packing settings
            -CPPACKSET=<y>  <y> set allocation: 0, 1 or DEFAULT or NORMAL, 2, 4 and 8
            -CPPACKENUM=<y> <y> enum packing: 0, 1, 2 and 4 or DEFAULT or NORMAL
            -CPPACKRECORD=<y> <y> record packing: 0 or DEFAULT or NORMAL, 1, 2, 4, 8,
→16 and 32
-Cr         Range checking
-CR         Verify object method call validity
-Cs<n>      Set stack checking size to <n>
-Ct         Stack checking (for testing only, see manual)
-CT<x>      Target-specific code generation options
            -CTcld          Emit a CLD instruction before using the x86
→string instructions
-CX         Create also smartlinked library
-d<x>       Defines the symbol <x>
-D          Generate a DEF file
            -Dd<x>         Set description to <x>
            -Dv<x>         Set DLL version to <x>
-e<x>       Set path to executable
-E          Same as -Cn
-fPIC       Same as -Cg
-F<x>       Set file names and paths:
            -Fa<x>[,y] (for a program) load units <x> and [y] before uses is parsed
            -Fc<x>         Set input codepage to <x>
            -FC<x>         Set RC compiler binary name to <x>
            -Fd            Disable the compiler's internal directory cache
            -FD<x>         Set the directory where to search for compiler utilities
            -Fe<x>         Redirect error output to <x>
            -Ff<x>         Add <x> to framework path (Darwin only)
            -FE<x>         Set exe/unit output path to <x>
            -Fi<x>         Add <x> to include path
            -Fl<x>         Add <x> to library path
            -FL<x>         Use <x> as dynamic linker
            -Fm<x>         Load unicode conversion table from <x>.txt in the compiler dir
            -FM<x>         Set the directory where to search for unicode binary files

```

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```

-FN<x>      Add <x> to list of default unit scopes (namespaces)
-Fo<x>      Add <x> to object path
-Fr<x>      Load error message file <x>
-FR<x>      Set resource (.res) linker to <x>
-Fu<x>      Add <x> to unit path
-FU<x>      Set unit output path to <x>, overrides -FE
-FW<x>      Store generated whole-program optimization feedback in <x>
-Fw<x>      Load previously stored whole-program optimization feedback from <x>
-g          Generate debug information (default format for target)
-gc          Generate checks for pointers (experimental, only available on some
→targets, might generate false positive)
-gh          Use heaptrace unit (for memory leak/corruption debugging)
-gl          Use line info unit (show more info with backtraces)
-gm          Generate Microsoft CodeView debug information (experimental)
-go<x>       Set debug information options
    -godwarfsets  Enable DWARF 'set' type debug information (breaks gdb < 6.5)
    -gostabsabsincludes  Store absolute/full include file paths in Stabs
    -godwarfmethodclassprefix  Prefix method names in DWARF with class name
    -godwarfcpp   Simulate C++ debug information in DWARF
    -godwarfomflinum  Generate line number information in OMF LINNUM records in
→MS LINK format in addition to the DWARF debug information (Open Watcom Debugger/
→Linker compatibility)
    -gp          Preserve case in stabs symbol names
    -gs          Generate Stabs debug information
    -gt          Trash local variables (to detect uninitialized uses; multiple 't'
→changes the trashing value)
    -gv          Generates programs traceable with Valgrind
    -gw          Generate DWARFv2 debug information (same as -gw2)
    -gw2         Generate DWARFv2 debug information
    -gw3         Generate DWARFv3 debug information
    -gw4         Generate DWARFv4 debug information (experimental)
-i          Information
    -iD          Return compiler date
    -iSO         Return compiler OS
    -iSP         Return compiler host processor
    -iTO         Return target OS
    -iTP         Return target processor
    -iV          Return short compiler version
    -iW          Return full compiler version
    -ia          Return list of supported ABI targets
    -ic          Return list of supported CPU instruction sets
    -if          Return list of supported FPU instruction sets
    -ii          Return list of supported inline assembler modes

```

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```

-io          Return list of supported optimizations
-ir          Return list of recognized compiler and RTL features
-it          Return list of supported targets
-iu          Return list of supported microcontroller types
-iw          Return list of supported whole program optimizations
-I<x>       Add <x> to include path
-k<x>       Pass <x> to the linker
-l          Write logo
-M<x>       Set language mode to <x>
    -Mfpc          Free Pascal dialect (default)
    -Mobjfp          FPC mode with Object Pascal support
    -Mdelphi        Delphi 7 compatibility mode
    -Mtp            TP/BP 7.0 compatibility mode
    -Mmacpas        Macintosh Pascal dialects compatibility mode
    -Miso           ISO 7185 mode
    -Mextendedpascal ISO 10206 mode
    -Mdelphiunicode Delphi 2009 and later compatibility mode
-n          Do not read the default config files
-o<x>       Change the name of the executable produced to <x>
-O<x>       Optimizations:
    -O-          Disable optimizations
    -O1          Level 1 optimizations (quick and debugger friendly)
    -O2          Level 2 optimizations (-O1 + quick optimizations)
    -O3          Level 3 optimizations (-O2 + slow optimizations)
    -O4          Level 4 optimizations (-O3 + optimizations which might have
↳unexpected side effects)
    -Oa<x>=<y>   Set alignment
    -Oo[NO]<x>  Enable or disable optimizations; see fpc -i or fpc -io for possible
↳values
    -Op<x>       Set target cpu for optimizing; see fpc -i or fpc -ic for possible
↳values
    -OW<x>       Generate whole-program optimization feedback for optimization <x>;
↳see fpc -i or fpc -iw for possible values
    -Ow<x>       Perform whole-program optimization <x>; see fpc -i or fpc -iw for
↳possible values
    -Os          Optimize for size rather than speed
-pg          Generate profile code for gprof (defines FPC_PROFILE)
-P<x>       Target CPU / compiler related options:
    -PB          Show default compiler binary
    -PP          Show default target cpu
    -P<x>        Set target CPU (aarch64, arm, avr, i386, i8086, jvm, m68k, mips, mipsel,
↳powerpc, powerpc64, sparc, x86_64)
-R<x>       Assembler reading style:

```

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```

-Rdefault  Use default assembler for target
-Ratt      Read AT&T style assembler
-Rintel    Read Intel style assembler
-S<x>      Syntax options:
-S2        Same as -Mobjfpc
-Sc        Support operators like C (*=, +=, /= and -=)
-Sa        Turn on assertions
-Sd        Same as -Mdelphi
-Se<x>      Error options. <x> is a combination of the following:
    <n> : Compiler halts after the <n> errors (default is 1)
    w : Compiler also halts after warnings
    n : Compiler also halts after notes
    h : Compiler also halts after hints
-Sf        Enable certain features in compiler and RTL; see fpc -i or fpc -ir_
↳for possible values)
-Sg        Enable LABEL and GOTO (default in -Mtp and -Mdelphi)
-Sh        Use reference counted strings (ansistring by default) instead of_
↳shortstrings
-Si        Turn on inlining of procedures/functions declared as "inline"
-Sj        Allows typed constants to be writeable (default in all modes)
-Sk        Load fpcylx unit
-SI<x>      Set interface style to <x>
    -SIcom   COM compatible interface (default)
    -SIcorba CORBA compatible interface
-Sm        Support macros like C (global)
-So        Same as -Mtp
-Sr        Transparent file names in ISO mode
-Ss        Constructor name must be init (destructor must be done)
-Sv        Support vector processing (use CPU vector extensions if available)
-Sx        Enable exception keywords (default in Delphi/ObjFPC modes)
-Sy        @<pointer> returns a typed pointer, same as $T+
-s         Do not call assembler and linker
-sh        Generate script to link on host
-st        Generate script to link on target
-sr        Skip register allocation phase (use with -alr)
-T<x>      Target operating system:
    -Taros   AROS
    -Tdarwin Darwin/Mac OS X
    -Tdragonfly DragonFly BSD
    -Tembedded Embedded
    -Tfreebsd FreeBSD
    -Tiphonesim iPhoneSimulator
    -Tlinux   Linux

```

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```

-Tnetbsd    NetBSD
-Topenbsd   OpenBSD
-Tsolaris   Solaris
-Twin64     Win64 (64 bit Windows systems)
-u<x>       Undefines the symbol <x>
-U          Unit options:
  -Un       Do not check where the unit name matches the file name
  -Ur       Generate release unit files (never automatically recompiled)
  -Us       Compile a system unit
-v<x>       Be verbose. <x> is a combination of the following letters:
  e : Show errors (default)          0 : Show nothing (except errors)
  w : Show warnings                  u : Show unit info
  n : Show notes                     t : Show tried/used files
  h : Show hints                     c : Show conditionals
  i : Show general info               d : Show debug info
  l : Show linenumbers               r : Rhide/GCC compatibility mode
  s : Show time stamps               q : Show message numbers
  a : Show everything                 x : Show info about invoked tools
  b : Write file names messages      p : Write tree.log with parse tree
      with full path                  v : Write fpcdebug.txt with
  z : Write output to stderr          lots of debugging info
  m<x>,<y> : Do not show messages numbered <x> and <y>
-V<x>       Append '-<x>' to the used compiler binary name (e.g. for version)
-W<x>       Target-specific options (targets)
  -WA       Specify native type application (Windows)
  -Wb       Create a bundle instead of a library (Darwin)
  -WB       Create a relocatable image (Windows)
  -WB<x>    Set image base to <x> (Windows)
  -WC       Specify console type application (Windows)
  -WD       Use DEFFILE to export functions of DLL or EXE (Windows)
  -We       Use external resources (Darwin)
  -WG       Specify graphic type application (Windows)
  -Wi       Use internal resources (Darwin)
  -WI       Turn on/off the usage of import sections (Windows)
  -WM<x>    Minimum Mac OS X deployment version: 10.4, 10.5.1, ... (Darwin)
  -WN       Do not generate relocation code, needed for debugging (Windows)
  -WP<x>    Minimum iOS deployment version: 8.0, 8.0.2, ... (iphonesim)
  -WR       Generate relocation code (Windows)
  -WX       Enable executable stack (Linux)
-X          Executable options:
  -X9       Generate linkerscript for GNU Binutils ld older than version 2.19.1
↳ (Linux)
  -Xc       Pass --shared/-dynamic to the linker (BeOS, Darwin, FreeBSD, Linux)

```

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```

-Xd      Do not search default library path (sometimes required for cross-
↳compiling when not using -XR)
-Xe      Use external linker
-Xf      Substitute pthread library name for linking (BSD)
-Xg      Create debuginfo in a separate file and add a debuglink section to
↳executable
-XD      Try to link units dynamically          (defines FPC_LINK_DYNAMIC)
-Xi      Use internal linker
-XLA     Define library substitutions for linking
-XLO     Define order of library linking
-XLD     Exclude default order of standard libraries
-Xm      Generate link map
-XM<x>   Set the name of the 'main' program routine (default is 'main')
-Xn      Use target system native linker instead of GNU ld (Solaris, AIX)
-Xp<x>   First search for the compiler binary in the directory <x>
-XP<x>   Prepend the binutils names with the prefix <x>
-Xr<x>   Set the linker's rlink-path to <x> (needed for cross compile, see
↳the ld manual for more information) (BeOS, Linux)
-XR<x>   Prepend <x> to all linker search paths (BeOS, Darwin, FreeBSD, Linux,
↳Mac OS, Solaris)
-Xs      Strip all symbols from executable
-XS      Try to link units statically (default, defines FPC_LINK_STATIC)
-Xt      Link with static libraries (-static is passed to linker)
-Xv      Generate table for Virtual Entry calls
-XV      Use VLink as external linker          (default on Amiga, MorphOS)
-XX      Try to smartlink units                (defines FPC_LINK_SMART)

-?       Show this help
-h       Shows this help without waiting

```

61.4 Call C

- 7.1 Using external code and variables

<https://www.freepascal.org/docs-html/prog/progse28.html>

The official fpc documentation for calling external functions and variables.

- Using C object files in Delphi

<http://rvelthuis.de/articles/articles-cobjjs.html>

- How to use C code in Free Pascal projects

<https://downloads.freepascal.org/fpc/docs-pdf/CinFreePascal-old.pdf>

- Creating Pascal bindings for C

[https://github.com/csukuangfj/pascal-bindings-for-c/blob/master/docs/Creating%20Pascal%20bindings%20for%20C%20\(v1.0\).pdf](https://github.com/csukuangfj/pascal-bindings-for-c/blob/master/docs/Creating%20Pascal%20bindings%20for%20C%20(v1.0).pdf)

It shows how to use `xxx.o`, `libxxx.a` and `libxxx.so` in free pascal It also contains source files hosted on GitHub.

- <https://github.com/PascalGameDevelopment/SDL2-for-Pascal/blob/master/units/sdl2.pas>

Read it carefully. It has lots of examples!

- See also <https://github.com/MagicFoundation/Alcinoe/blob/master/Source/ZLibEx.pas>

- <https://github.com/gidesa/ocvWrapper24/blob/main/Delphi/OPENCVWrapper.pas>

OpenCV wrapper for delphi

- <https://github.com/Kagamma/whisper-pas/blob/master/src/whisper.pas>

Wrapper for whisper

- Manual conversion of C headers to Pascal

https://wiki.lazarus.freepascal.org/Creating_bindings_for_C_libraries#Manual_conversion_of_C_headers_to_Pascal

- See the following link for how to wrap `printf`, which accepts unconstrained number of arguments

<https://www.freepascal.org/docs-html/prog/progsu149.html>

Note that it proposes two methods.

61.4.1 ctypes

`ctypes.inc` can be found at <https://github.com/fpc/FPCSource/blob/main/rtl/unix/ctypes.inc>.

It defines commonly used integer types for C/C++.

- `cint32`, `pcint32`
- `cfloat`, `pcfloat`
- `cdouble`, `pcdouble`
- `csize_t`, `pcsize_t`

61.4.2 linklib

To link a lib in the code:

```
{ $L xxx }
// or
{ $LinkLib xxx }

// Note xxx can be on object file, e..g, use { $L foo.o }
// or it can be a library { $L foo } to link with libfoo.so, libfoo.dylib

// On windows, we have to use { $L foo.dll }
// See https://www.freepascal.org/docs-html/prog/progsu149.html
```

Or at the commandline use `-Flxxx`

To specify the object file in the commandline, use `-Fo`

61.4.3 Example 1 on macOS

See also <https://www.freepascal.org/docs-html/prog/progsu147.html>

Things to take away:

1. We can link an object file in pascal
2. The c variable name is mangled by adding a leading `_` by the compiler.
3. If we use `cvar` without a name after external, then the fpc compiler handles the name mangling by ourselves.
4. If we don't use `cvar` and instead use a name, then we must add a leading `_`.
5. If we don't use `cvar` and don't use name, then the variable must be named as the mangled c variable name.

`extvar.c` is given below:

```
/*
Declare a variable, allocate storage
*/
int extvar = 12;
```

`extdemo.pas` is given below:

```
Program ExtDemo;

{ $L extvar.o }
// { $LinkLib extvar.o } // either one is ok
// By default, it tries to find extvar.o in the current directory.
```

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```
// If it is not found, then an error is throw.
//
// To fix it, we can use -Fl/some/path
// where we extvar.o is inside /some/path

var
  { Case sensitive declaration }
  extvar : longint; cvar; external;
  _extvar : longint; external;
  I : longint; external name '_extvar';

begin
  { Extvar can be used case insensitively }
  Writeln ('Variable 'extvar' has value: ',ExtVar);
  Writeln ('Variable '_extvar' has value: ',_ExtVar);
  Writeln ('Variable 'I' has value: ',i);
  I := 13;
  Writeln ('Variable 'extvar' has value: ',ExtVar);
  Writeln ('Variable '_extvar' has value: ',_ExtVar);
  Writeln ('Variable 'I' has value: ',i);
end.
```

Build command:

```
gcc -c -o extvar.o extvar.c
fpc ./extdemo.pas
```

Output log is:

```
Free Pascal Compiler version 3.2.2 [2021/05/16] for x86_64
Copyright (c) 1993-2021 by Florian Klaempfl and others
Target OS: Darwin for x86_64
Compiling ./extdemo.pas
Assembling extdemo
Linking extdemo
ld: warning: object file (extvar.o) was built for newer macOS version (13.0) than_
↳being linked (10.8)
17 lines compiled, 0.6 sec
```

The command:

```
nm ./extvar.o
```

prints:

```
0000000000000000 D _extvar
```

And:

```
./extdemo
```

prints:

```
Variable 'extvar' has value: 12  
Variable '_extvar' has value: 12  
Variable 'I'      has value: 12  
Variable 'extvar' has value: 13  
Variable '_extvar' has value: 13  
Variable 'I'      has value: 13
```

61.5 Basics

61.5.1 system

- `Chr (b)`, where `b` is a byte, it returns an ascii character
See <https://www.freepascal.org/docs-html/rtl/system/chr.html>
- `Ord ()`, see <https://www.freepascal.org/docs-html/rtl/system/ord.html>

61.5.2 sysutils

- `IntToStr (i)` to convert an integer to a string
See <https://www.freepascal.org/docs-html/rtl/sysutils/inttostr.html>
- `StrToInt (s)` to convert a string to an integer
See <https://www.freepascal.org/docs-html/rtl/sysutils/strtoint.html>. If the conversion failed, the above link gives an example about how to catch the exception.
See also `StrToInDef`, which returns a specified default value on conversion failure.
- `Format ()`, similar to `sprintf ()` in C/C++.
See <https://www.freepascal.org/docs-html/rtl/sysutils/format.html>

61.5.3 Commonly used controls

```
Label.Caption := "hello"
```

61.6 char

A character `char` is 1 byte in pascal.

`AnsiChar` is an alias to `Char`. See <https://www.freepascal.org/docs-html/rtl/system/ansichar.html>.

'a' is a `char`. Note only single quote is used. Double quotes are not correct.

See https://wiki.freepascal.org/Character_and_string_types for different types of strings:

- `AnsiChar`
- Static array of `char`
- Dynamic array of `char`, see also <https://www.freepascal.org/docs-html/rtl/system/setlength.html> We need to free the memory by passing 0 to `SetLength`

```
var
  a: char;
begin
  a := 'a';
  a := #65; // equivalent to a := 'a'

  a := '''; // a is a single quote here.
end.
```

There is also a `string` type, which can contain at most 255 characters.

```
var
  s: string;
  s2: string[255]; // both s and s2 have 256 bytes.
begin
  writeln ('SizeOf(s) is ', SizeOf(s)); // Sizeof(s) is 256
  writeln ('SizeOf(s2) is ', SizeOf(s2)); // Sizeof(s2) is 256
  s := 'hello';
  writeln('s[0] is ', ord(s[0])); // s[0] is 5
end.
```

Note `var s: string[size]` will allocate `size + 1` bytes. `size` can be any value from 1 to 255. `s[0]` contains the length of the string. If `[size]` is not specified, then it is equivalent to `[255]`.

`pchar`:

- It is similar to the strings in C, which ends with 0. pchar is a pointer
- See <https://www.freepascal.org/docs-html/ref/refsu12.html>

```
var
  p: pchar;
begin
  p := 'hello world';
  writeln ('p is ', p); // p is hello world
end.
```

Example 2:

```
var
  s: string;
  p: pchar;
begin
  s := 'hello world'#0; // Use #0 to append a \0 at the end
  p := @s[1]; // s[0] is the length of the string, so we use @s[1] here
  writeln ('p is ', p); // p is hello world
end.
```

- Use StrLen for pchar, but use Length for string.

61.7 Pointers

See <https://www.freepascal.org/docs-html/ref/refse15.html>

null pointers in pascal is nil.

See also https://en.wikibooks.org/wiki/Pascal_Programming/Pointers

```
var
  p: ^integer;
  i: integer;
begin
  i := 3;
  p := @i; // get the address of i
  p^ += 2; // equivalent to *p += 2 in c/c++
  Writeln('i is ', i); // is is 5
end.
```

- The new function from <https://www.freepascal.org/docs-html/3.2.0/rtl/system/new.html>

It is declared as:

```
procedure New(  
  var P: Pointer  
);
```

Note that the argument type is `Pointer`, which is an untyped pointer. Also note the argument is `var`, meaning it is an input/output argument

Example:

```
var  
  p: ^integer;  
begin  
  New(p); // allocate space and save the address to p  
  p^ := 10;  
  writeln ('p^ is ', p^); // p^ is 10  
  Dispose(p); // free p  
end.
```

- `GetMem()` and `FreeMem()`

See <https://www.freepascal.org/docs-html/3.2.0/rtl/system/freemem.html>

- `FillByte()`, `FillChar()`, `FillWord()`, `FillDWord()`, are similar to `memset` in C/C++.

<https://www.freepascal.org/docs-html/3.2.0/rtl/system/fillchar.html>

61.8 class

See <https://www.freepascal.org/docs-html/ref/refch6.html#refse39.html>

Note that we have to call the constructor and destructor manually by ourselves.

61.9 Conditional compiling

61.9.1 predefined symbols

Please see <https://www.freepascal.org/docs-html/3.0.0/prog/progap7.html#x333-348000G>.

`unix` is defined for both macos and unbuntu.

`darwin` is defined only for macos.

`windows` is for both 32-bit and 64-bit windows.

61.9.2 user defined symbols

To pass from the commandline:

```
fpc -dmy_symbol
```

that is, use `-d` option. It is equivalent to:

```
{ $define my_symbol }
```

in the code.

To un-define a symbol from the commandline:

```
fpc -umy_symbol
```

That is, use `-u` option. It is equivalent to:

```
{ #undef my_symbol }
```

in the code.

61.9.3 if and else

There are multiple forms:

Example 1

```
{ $ifdef my_symbol }  
    { some code }  
{ $endif }  
  
{ $ifdef my_symbol with some comments }  
    { some code }  
    { everything after my_symbol is a comment }  
{ $endif }  
  
{ $ifdef my_symbol }  
    { some code }  
{ $else }  
    { some other code }  
{ $endif }  
  
{ $ifdef my_symbol }  
    { some code }  
{ $elseif defined(foo) }
```

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```
    {some code}
{$endif}

{$if defined(foo) or defined(bar)}
{$endif}

{$if defined(foo) and defined(bar)}
{$endif}


{$if defined(my_symbol)}
    {some code}
{$elseif defined(foo)}
    {some code}
{$endif}

{$if defined(my_symbol)}
    {some code}
{$elseif defined(foo) or defined(bar) or defined(foobar)}
    {some code}
{$endif}
```

Example2

```
{$ifndef foo}
    {code}
{$endif}

{$if not defined(foo)}
    {code}
{$endif}
```

To summarize, we can use

- `ifdef, endif`
- `ifndef`
- `else, elseif`
- `if` (we can use conditions, e.g., `if defined()` or `defined()`)
- `defined`
- `undef`

Some more examples from <https://github.com/fpc/FPCSource/blob/main/rtl/unix/dl.pp>


```

{$if defined(linux) or defined(freebsd) or defined(openbsd) or defined(dragonfly)}
  {$define ELF} // ELF symbol versioning.
{$endif}

{$if defined(linux)}
  { if libc is not linked explicitly, FPC might chose the wrong startup code, as
    libdl depends on libc on linux, this does not hurt }
  {$linklib c}
{$endif}

function dladdr(Lib: pointer; info: Pdl_info): Longint; cdecl; {$if not defined(aix) ↵
↪and not defined(android)} external; {$endif}

const
{$ifdef BSD}    // dlopen is in libc on FreeBSD.
  LibDL = 'c';
{$else}
  {$ifdef HAIKU}
    LibDL = 'root';
  {$else}
    LibDL = 'dl';
  {$endif}
{$endif}

```

Some more examples from <https://github.com/fpc/FPCSource/blob/main/rtl/inc/ctypes.pp>

```

{$ifdef FPC}
  {$inline on}
  {$ifndef DISABLE_INLINE}
    {$define CTYPES_INLINE}
  {$endif}
  {$define dummy}
{$endif}

```

61.10 Compiler options

To enable packed record:

```
{ $packrecords c }
```

61.11 dl

<https://github.com/fpc/FPCSource/blob/main/rtl/unix/dl.pp>

<https://github.com/fpc/FPCSource/blob/main/rtl/unix/dynlibs.inc>

- The inside story on shared libraries and dynamic loading

https://cseweb.ucsd.edu/~gbournou/CSE131/the_inside_story_on_shared_libraries_and_dynamic_loading.pdf

- Program Library HOWTO

<https://tldp.org/HOWTO/pdf/Program-Library-HOWTO.pdf>

- ELF: From The Programmer' s Perspective

<https://www.linux.co.cr/free-unix-os/review/acrobat/950517.pdf>

- Load-time relocation of shared libraries

<https://eli.thegreenplace.net/2011/08/25/load-time-relocation-of-shared-libraries>

- The True Story of Hello World

<https://lisha.ufsc.br/teaching/os/exercise/hello.html>

The smallest program.

- dladdr,
- dlopen
- dlsym
- dlclose

TLibHandle is an integer that can be used to hold an address.

RawByteString is an alias to AnsiString.

61.11.1 TODO

Read

- A look at dynamic linking

<https://lwn.net/Articles/961117/>

- <https://www.cs.dartmouth.edu/~sergey/cs258/dynamic-linking.txt>

61.12 Lazarus

Follow https://wiki.freepascal.org/Installing_Lazarus_on_macOS to install Lazarus.

After installation on macOS, the directory `/Applications/Lazarus/examples` contains many examples.

We need to download two files from <https://sourceforge.net/projects/lazarus/files/Lazarus%20macOS%20x86-64/>:

- `Lazarus-3.4-macosx-x86_64.pkg`
- `fpc-src-3.2.2-20210709-macosx.dmg`

`fpc-src` is installed in `/usr/local/share/src/fpc-3.2.2`. The source code of `fpc` can also be found at <https://github.com/fpc/FPCSource>

To reduce `.exe` file size:

1. Select Project, Project options..., compiler options, Debugging
2. Uncheck Generate info for the debugger (slower / increase exe size)
3. Check Strip symbols from executable
4. Note: We can create release and debug build mode instead of using the default mode.

See https://wiki.freepascal.org/Mac_Installation_FAQ to fix the following error:

```
ld: framework not found Cocoa
```

whole program optimization: https://wiki.freepascal.org/Whole_Program_Optimization#Concrete_Lazarus_IDE_example

- Multiplatform Programming Guide
https://wiki.freepascal.org/Multiplatform_Programming_Guide
- Application Bundle
https://wiki.freepascal.org/Application_Bundle
- https://wiki.lazarus.freepascal.org/Lazarus_Tutorial

61.12.1 Marcos

- For operating systems: DARWIN for macOS and iOS
- See <https://www.freepascal.org/docs-html/3.2.0/prog/progap7.html#x341-357000G> for a complete list

61.12.2 Samples and tutorials

- <https://wiki.lazarus.freepascal.org/Category:Tutorials>
- https://wiki.lazarus.freepascal.org/Lazarus_Documentation

- Manual conversion of C headers to Pascal

https://wiki.lazarus.freepascal.org/Creating_bindings_for_C_libraries#Manual_conversion_of_C_headers_to_Pascal

61.12.3 TForm

See https://wiki.lazarus.freepascal.org/Form_Tutorial

To change the caption of the form, assign a string to its `Caption` property.

To add a hint to the form, we need to first change its property `Hint` and then set property `ShowHint` to true.

1. How to show a second form from the main form?

- There are two methods.
 - (1) The form is created automatically by the application, which is the default behavior. If we want to disable auto creation, then we can either edit *.lpr or use menu -> project options -> form, disable auto creating the form.

Note that the above code assumes the form is created automatically.

```
// unit1.pas
var
  r: Integer;
begin
  r := Form2.ShowModal;

  case r of
    1: begin ShowMessage('1'); end;
    2: begin ShowMessage('2'); end;
    3: begin ShowMessage('3'); end;
  end;
end;

// unit2.pas
// to exit form2, we can either use self.Close
// or use
//
// Self.ModalResult := 10;
//
```

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```
// we can assign any integer value to Self.ModalResult
// and it will close the form immediately.
```

<https://lazarus-ccr.sourceforge.io/docs/lcl/forms/tmodalresult.html> and <https://lazarus-ccr.sourceforge.io/docs/lazutils/uitypes/tmodalresult.html> and <https://fossies.org/linux/lazarus/components/lazutils/uitypes.pas> define some constant, e.g., mrOK, mrCancel, etc.

If we use `Self.Close` or click the X to close the Form, then the returned `ModalResult` is `mrCancel`, which is 2.

- (2) If the form is not created automatically, then we can use

```
procedure TForm1.Button1Click(Sender: TObject);
begin
    Form2:=TForm2.Create(nil); //Form2 is created
    Form2.ShowModal;           //Form2 is displayed
    FreeAndNil(Form2);         //Free Form2
end;
```

2. How to create a new form dynamically?

See https://wiki.lazarus.freepascal.org/Form_Tutorial#Generate_the_form_dynamically

```
procedure TForm1.Button1Click(Sender: TObject);
var
    MyForm: TForm;
    MyButton: TButton;
begin
    MyForm:=TForm.Create(nil);
    MyForm.SetBounds(100, 100, 220, 150);
    MyForm.Caption:='My dynamic created form';

    MyButton:=TButton.create(MyForm);
    MyButton.Caption:='Close my form';
    MyButton.SetBounds(10, 10, 200, 30);
    MyButton.Parent:=MyForm;

    MyButton.OnClick:=@MyButtonClick;

    MyForm.ShowModal;

    FreeAndNil(MyForm);
end;
```

61.12.4 Tips for manual generation of controls

See https://wiki.lazarus.freepascal.org/LCL_Tips

61.12.5 TButton

See

- <https://wiki.freepascal.org/TButton>
- <https://lazarus-ccr.sourceforge.io/docs/lcl/stdctrls/tbutton.html>

Note that we can set its `tag` property. Several buttons can share the same onclick event handler, we can use its `tag` attribute to distinguish them.

Example 1:

```
// from https://wiki.lazarus.freepascal.org/Lazarus_Tutorial
procedure TForm1.Button1Click(Sender: TObject);
{ Makes use of the Tag property, setting it to either 0 or 1}
begin
    if Button1.tag = 0 then
    begin
        Button1.caption := 'Press again';
        Button1.tag := 1;
    end else
    begin
        Button1.caption := 'Press';
        Button1.tag := 0;
    end;
end;
```

Example 2:

```
// from
// https://wiki.lazarus.freepascal.org/TButton
procedure TForm1.FormCreate(Sender: TObject);
var
    i: Integer;
    aButton: TButton;
begin
    for i := 0 to 9 do begin                                // create 10 Buttons
        aButton := TButton.Create(Self);                    // create Button, Owner is Form1, where the
        ↪button is released later
        aButton.Parent := Self;                             // determine where it is to be displayed
        aButton.Width := aButton.Height;                    // Width should correspond to the height of
    end;
```

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```

↪the buttons
    aButton.Left      := i * aButton.Width; // Distance from left
    aButton.Caption := IntToStr(i);         // Captions of the buttons (0.9)
    aButton.OnClick := @aButtonClick;      // the event handler for the button -> will
↪be created yet
end;
Self.Height := aButton.Height;             // Height of the form should correspond to
↪the height of the buttons
Self.Width := aButton.Width * 10;          // Width of the form to match the width of
↪all buttons
end;

procedure TForm1.aButtonClick(Sender: TObject);
const
    Cnt: Integer = 0;
var
    i: Integer;
begin
    if (Sender is TButton) and              // called the event handler of a button
↪out?
        TryStrToInt (TButton(Sender).Caption, i) // then try to convert the label in a
↪integer
    then begin
        Cnt := Cnt + i;                     // the adding counter is incremented by
↪the number of entrechende
        Caption:='QuickAdd: '+IntToStr(Cnt); // write the result to the caption of
↪the form
    end;
end;
end;

```

61.12.6 TMemor

Support multiline texts.

It has an attributes Lines, which is of type TStringList. Lines.Count returns number of lines. If it is 0, then the text is empty. Lines.Add('a new line') to add a new line. Lines.Clear() to clear all lines.

See also <https://wiki.lazarus.freepascal.org/TMemo>.

Example 1:

```

// from https://wiki.lazarus.freepascal.org/TMemo
// assign a TStringList to a TMemo
// Note that we have to call Free to free the stringlist

```

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```
procedure TForm1.Button1Click(Sender: TObject);
var
    myStringList: TStringList;
begin
    myStringList:=TStringList.Create;           //Create my StringList
    myStringList.Add('This is the first line.');//add a line
    myStringList.Add('This is the second line.');//add a line
    myStringList.Add('This is the third line.');//add a line
    myStringList.Add('etc.');//add a line
    Memo1.Lines.Assign(myStringList);           //assign text content
    myStringList.Free;                          //free my StringList
end;
```

Example 2:

```
procedure TForm1.Button1Click(Sender: TObject);
begin
    Memo1.Lines.Clear;                          //delete all lines of Memo1
    Memo1.Lines.Add('This is the first line.');//add a line
    Memo1.Lines.Add('This is the second line.');//add a line
    Memo1.Lines.Add('This is the third line.');//add a line
    Memo1.Lines.Add('etc.');//add a line
end;
```

61.12.7 TListBox

See <https://wiki.lazarus.freepascal.org/TListBox>.

1. How to add an item
2. How to get the selected item
3. How to get multi-selected items

61.12.8 Dialogs

See https://wiki.lazarus.freepascal.org/Dialog_Examples.

1. ShowMessage
2. Application.MessageBox, we can provide several buttons, ask users to click one button, figure out which button is clicked, e.g., the yes or no button, execute some code.
3. MessageDlg, like Application.MessageBox but we can specify which buttons to show to users. We should prefer this one.

4. `QuestionDlg`: Users can customize the caption of the buttons.
5. There are also dialogues for accepts user inputs.

61.12.9 TMainMenu

See <https://wiki.lazarus.freepascal.org/TMainMenu>

Naming convention: `MenuFile`, `MenuFileOpen` (for File -> Open), `MenuHelp`.

61.12.10 macOS related

- Locating the macOS application resources directory

Info.plist

`Info.plist` is short for information property list.

We can either use `vim` to edit `Info.plist` or use `open Info.plist` to start `xcode` to edit it.

References:

- Edit property lists
 - It shows how to use `xcode` to edit `Info.plist`.
- About Info.plist Keys and Values
- About Information Property List Files

Bundle Programming Guide

- Introduction
- Application Bundles

62.1 Install

62.2 Basics

`tensor->ne[4]`: For a matrix, `ne[0]` is the number cols. `ne[1]` is the number of rows.

63.1 Basics

HARMONYOS

64.1 Install

Install <https://developer.huawei.com/consumer/cn/deveco-studio/>

64.1.1 macos

```
# How to upload files to the emulator from my macOS
# https://developer.huawei.com/consumer/cn/forum/topic/0207152795297975671

export PATH=/Users/fangjun/software/huawei/OpenHarmony/Sdk/10/toolchains:$PATH

hdc -c file send ~/open-source/sherpa-onnx/lei-jun-test.wav /storage/media/100/local/
↪files/Docs/Download/

hdc shell
cd /storage/media/100/local/files/Docs/Download/
ls -lh
```

64.1.2 todo

<https://medium.com/huawei-developers/building-a-note-app-with-arkui-and-arkts-on-harmonyos-67fd7c4c1086>

- TabsController
- aboutToAppear
- changeTab
- tagBarBuilder

64.1.3 Native 侧实现文件访问开发实践

<https://developer.huawei.com/consumer/cn/doc/best-practices-V5/bpta-file-native-side-V5#section1052945819523>

- 应用沙箱目录 <https://developer.huawei.com/consumer/cn/doc/harmonyos-guides-V5/app-sandbox-directory-V5>
- Rawfile 开发指导 <https://developer.huawei.com/consumer/cn/doc/harmonyos-guides-V5/rawfile-guidelines-V5>

64.1.4 Node-API Development Process

See <https://developer.huawei.com/consumer/en/doc/harmonyos-guides-V5/use-napi-process-V5>

It has an example about calling `Add()` from C/C++.

64.1.5 资源分类与访问

<https://developer.huawei.com/consumer/cn/doc/harmonyos-guides-V5/resource-categories-and-access-V5>

64.1.6 module.json5 配置文件

<https://developer.huawei.com/consumer/cn/doc/harmonyos-guides-V5/module-configuration-file-V5>

64.1.7 Building an NDK Project with the DevEco Studio Template

See <https://developer.huawei.com/consumer/en/doc/harmonyos-guides-V5/build-with-ndk-ide-V5>

It contains an example about adding abi filters.

64.1.8 Getting Started with the NDK

See <https://developer.huawei.com/consumer/en/doc/harmonyos-guides-V5/ndk-development-overview-V5>

64.1.9 Node-API Overview

See <https://developer.huawei.com/consumer/en/doc/harmonyos-guides-V5/napi-introduction-V5>

64.1.10 在 GitHub Action 上构建 HarmonyOS 应用

See <https://www.phodal.com/blog/github-action-for-harmonyos/>

64.1.11 OpenHarmony Linux 环境 SDK 使用说明

See https://gitee.com/han_jin_fei/lycium/blob/master/doc/ohos_use_sdk/OHOS_SDK-Usage.md

64.1.12 Command Line Tools for HarmonyOS

See <https://github.com/harmonyos-dev/hos-sdk>

64.1.13 贡献应用资源

See <https://repo.harmonyos.com/#/cn/help/contributeapp>

64.1.14 Rust 登陆【华为鸿蒙】操作系统之 Native 模块开发

See <https://rustcc.cn/article?id=568d35d6-b782-49e9-b9b1-5d870d28f927>

64.1.15 简易 Native C++ 示例

- https://developer.huawei.com/consumer/cn/codelabsPortal/carddetails/tutorials_NEXT-NativeTemplateDemo
- https://gitee.com/harmonyos_codelabs/NativeTemplate

64.1.16 编译一份适用于鸿蒙 ArkTs 的 so 动态库教学，提供给第三方导入并使用

See https://blog.csdn.net/qq_17432441/article/details/134545608?spm=1001.2014.3001.5501

64.1.17 鸿蒙 ArkTs 加载各种 so 动态链接库教程，包括适用于鸿蒙的动态库和通用 linux 动态链接库，堕胎级教程，一篇就够了，玩转 so 加载

See https://blog.csdn.net/qq_17432441/article/details/134564221

64.1.18 The first example

See <https://device.harmonyos.com/en/docs/apiref/doc-guides/start-with-ets-0000001168323542>

It shows how to switch between two pages.

64.1.19 Linear Layout

See <https://developer.huawei.com/consumer/en/doc/harmonyos-guides-V5/arkts-layout-development-linear-V5>

64.1.20 RelativeContainer

See <https://developer.huawei.com/consumer/en/doc/harmonyos-guides-V5/arkts-layout-development-relative-layout-V5>

- Vertical: top, center, bottom
- Horizontal: left, middle, right

```
RelativeContainer() {  
  
  Text(this.message)  
    .id('HelloWorld')  
    .fontSize(50)  
    .fontWeight(FontWeight.Bold)  
    .alignRules({  
      top: { anchor: '__container__', align: VerticalAlign.Top },  
      left: { anchor: '__container__', align: HorizontalAlign.Start }  
    })  
}
```

65.1 bbpe

- Neural Machine Translation with Byte-Level Subwords
 - https://github.com/facebookresearch/fairseq/blob/main/examples/byte_level_bpe/README.md
 - <https://arxiv.org/abs/1909.03341>

66.1 Install

- User manual: <https://drive.google.com/drive/u/0/folders/1kzMRI95yaXLbQuK86fUbs92NJ6QOYIGO>
- Ubuntu images: https://drive.google.com/drive/u/0/folders/1qSA5yr_UOll3LKMzdn74i_SCYygnIcJ8
- Android images: <https://drive.google.com/drive/u/0/folders/1SQcrQI8Qn1MbAcv3ggxhAPyIMTQ2-Xog?usp=sharing>
- Official tools: https://drive.google.com/drive/mobile/folders/19OaB7_ND4dqxyzOTODb4hnIXGZ1C2g0?usp=sharing

I have downloaded `Orangepi5max_1.0.0_ubuntu_jammy_server_linux6.1.43.7z`. Unzip it and we get a folder `Orangepi5max_1.0.0_ubuntu_jammy_server_linux6.1.43`.

There are two files in the folder: `Orangepi5max_1.0.0_ubuntu_jammy_server_linux6.1.43.img` and `Orangepi5max_1.0.0_ubuntu_jammy_server_linux6.1.43.img.sha`

Go to <https://etcher.balena.io/#download-etcher> and download it. Burn the `.img` file into a tf card.

After the board gets an IP, ssh to it.

Both the username and password are the same, `orangepi`.

```
ifconfig
```

It shows the wired network interface is `enP3p49s0`

Now we assign a static address to `enP3p49s0`.

```
sudo vim /etc/network/interfaces
```

The default content is:

```
source /etc/network/interfaces.d/*  
#
```

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```
auto lo
iface lo inet loopback
```

We append the following to `/etc/network/interfaces`:

```
# this is a comment line
iface enP3p49s0 inet static
    address 192.168.1.9
    netmask 255.255.255.0
```

```
sudo /etc/init.d/networking restart
```

If it does not work, use `nmcli connection show` to list current connections, which displays:

NAME	UUID	TYPE	DEVICE
Wired connection 1	xxxx	ethernet	enP3p49s0

Then run `nmcli connection mod "Wired connection 1" ipv4.addresses "192.168.1.9/24" ipv4.method "manual"`

Then run `sudo reboot`.

Power off the board and connect it to my mac.

66.1.1 wireless network

```
$ ifconfig
enP3p49s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.9 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::e08a:3fa:a528:4b1f prefixlen 64 scopeid 0x20<link>
    ether c0:74:2b:fd:cd:a1 txqueuelen 1000 (Ethernet)
    RX packets 143 bytes 15875 (15.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 111 bytes 41059 (41.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

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```
wlan0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
        ether c0:f5:35:ed:fb:25 txqueuelen 1000 (Ethernet)
        RX packets 0 bytes 0 (0.0 B)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 0 bytes 0 (0.0 B)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

so we have wlan0.

Use `sudo iwlist wlan0 scan` to scan available wireless networks.

It prints something like below:

```
wlan0      Scan completed :
          Cell 01 - Address: 8C:53:C3:52:CC:3E
                  ESSID:"Xiaomi_CC3D"
                  Mode:Managed
                  Frequency:2.412 GHz (Channel 1)
                  Quality:5/5  Signal level:-55 dBm  Noise level:-81 dBm
                  IE: IEEE 802.11i/WPA2 Version 1
                        Group Cipher : TKIP
                        Pairwise Ciphers (2) : TKIP CCMP
                        Authentication Suites (1) : PSK
                  IE: Unknown:~
          ↪DD7C0050F204104A0001101044000102103B00010310470010BC329E001DD811B286018C53C352CC3E102100067869616F
                  IE: WPA Version 1
                        Group Cipher : TKIP
                        Pairwise Ciphers (2) : TKIP CCMP
                        Authentication Suites (1) : PSK
                  Encryption key:on
                  Bit Rates:1 Mb/s; 2 Mb/s; 5.5 Mb/s; 11 Mb/s; 9 Mb/s
                              18 Mb/s; 36 Mb/s; 54 Mb/s; 6 Mb/s; 12 Mb/s
                              24 Mb/s; 48 Mb/s
          Cell 06 - Address: 1A:8C:3A:63:6D:1B
                  ESSID:"meine"
                  Mode:Managed
                  Frequency=2.437 GHz (Channel 6)
                  Quality:5/5  Signal level:-44 dBm  Noise level:-81 dBm
                  IE: IEEE 802.11i/WPA2 Version 1
                        Group Cipher : CCMP
                        Pairwise Ciphers (1) : CCMP
                        Authentication Suites (1) : PSK
                  Encryption key:on
                  Bit Rates:1 Mb/s; 2 Mb/s; 5.5 Mb/s; 11 Mb/s; 6 Mb/s
```

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```
9 Mb/s; 12 Mb/s; 18 Mb/s; 24 Mb/s; 36 Mb/s
48 Mb/s; 54 Mb/s
```

```
$ wpa_passphrase your_wifi_ssid_name your_wifi_password | sudo tee /etc/wpa_
↳supplicant.conf
$ sudo wpa_supplicant -B -c /etc/wpa_supplicant.conf -i wlan0
$ sudo dhclient wlan0
```

Now `ifconfig wlan0` should show its IP address.

66.1.2 Android

1. Download `SDDiskTool_v1.72.zip` from https://drive.google.com/drive/u/0/folders/1RFBMuHwbN8V_LsQTnW05KpRKYSgJB6MD and unzip it to get the folder `SDDiskTool_v1.72`.
2. Download `OrangePi5Max_RK3588_Android13_v1.0.0.tar.gz` from <https://drive.google.com/drive/folders/1SQcrQI8Qn1MbAcv3ggxhAPyIMTQ2-Xog> and unzip it to get `OrangePi5Max_RK3588_Android13_v1.0.0.img`.
3. Switch to the folder `SDDiskTool_v1.72` and double click `SD_Firmware_Tool.exe`, - (1) select the sd card - (2) Start from sd card and then click the button on the bottom to format the sd card - (3) Select the android image and burn it

```
adb shell
getprop ro.build.version.release
getprop ro.product.model
getprop ro.product.manufacturer
```

66.2 Basics

```
from rknn.api.rknn import RKNN
```

`RKNN.config` has a parameter `custom_string`.

```
https://github.com/rockchip-linux/rknn-toolkit2/raw/refs/heads/master/rknn-toolkit2/
↳packages/rknn_toolkit2-1.6.0+81f21f4d-cp311-cp311-linux_x86_64.whl
```

See https://github.com/rockchip-linux/rknpu2/blob/master/runtime/RK3588/Linux/librknn_api/include/rknn_api.h#L470


```
int rknn_query(rknn_context context, rknn_query_cmd cmd, void* info, uint32_t size);
```

https://github.com/rockchip-linux/rknpu2/blob/master/runtime/RK3588/Linux/librknn_api/include/rknn_api.h#L117

```
RKNN_QUERY_CUSTOM_STRING = 7, /* query the custom string */
```

https://github.com/rockchip-linux/rknpu2/blob/master/runtime/RK3588/Linux/librknn_api/include/rknn_api.h#L326

```
typedef struct _rknn_custom_string {
    char string[1024]; /* the string of custom, ↵
↵lengths max to 1024 bytes */
} rknn_custom_string;
```

So the max length of the custom string is 1023.

Examples:

- Python on PC: <https://github.com/rockchip-linux/rknn-toolkit2/blob/master/rknn-toolkit2/examples/onnx/resnet50v2/test.py#L87>
- Python on the board: https://github.com/rockchip-linux/rknn-toolkit2/blob/master/rknn_toolkit_lite2/examples/resnet18/test.py

To fix the following error:

```
I RKNN: [14:24:11.881] RKNN Runtime Information: librknnrt version: 1.4.0↵
↵(a10f100eb@2022-09-09T09:07:14)
I RKNN: [14:24:11.881] RKNN Driver Information: version: 0.9.6
E RKNN: [14:24:11.881] 6, 1
E RKNN: [14:24:11.881] Invalid RKNN model version 6
E RKNN: [14:24:11.881] rknn_init, load model failed!
E Catch exception when init runtime!
E Traceback (most recent call last):
  File "/home/orangepi/packages/rknnlite/api/rknn_lite.py", line 148, in init_runtime
    self.rknn_runtime.build_graph(self.rknn_data, self.load_model_in_npu)
  File "rknnlite/api/rknn_runtime.py", line 919, in rknnlite.api.rknn_runtime.
↵RKNNRuntime.build_graph
Exception: RKNN init failed. error code: RKNN_ERR_FAIL

Failed to init rknn runtime for ./decoder.rknn
```

See <https://github.com/rockchip-linux/rknn-toolkit2/issues/316> Download librknnrt.so from https://huggingface.co/csukuangfj/rknn-toolkit2/tree/main/rknn-toolkit2-v2.1.0-2024-08-08/rknpu2/runtime/Linux/librknn_api/aarch64

Use:

```
watch -n 1 cat /sys/kernel/debug/rknpu/load
```

to view the usage of npu cores.

To view the versions of rknn driver,

```
dmesg | grep -i rknpu
```

It prints:

```
[ 6.784814] RKNPU fdab0000.npu: Adding to iommu group 0
[ 6.784923] RKNPU fdab0000.npu: RKNPU: rknpu iommu is enabled, using iommu mode
[ 6.786126] RKNPU fdab0000.npu: can't request region for resource [mem 0xfdab0000-
↪0xfdabffff]
[ 6.786146] RKNPU fdab0000.npu: can't request region for resource [mem 0xfdac0000-
↪0xfdacffff]
[ 6.786156] RKNPU fdab0000.npu: can't request region for resource [mem 0xfdad0000-
↪0xfdadffff]
[ 6.786580] [drm] Initialized rknpu 0.9.6 20240322 for fdab0000.npu on minor 1
[ 6.791864] RKNPU fdab0000.npu: RKNPU: bin=0
[ 6.792047] RKNPU fdab0000.npu: leakage=6
[ 6.792083] debugfs: Directory 'fdab0000.npu-rknpu' with parent 'vdd_npu_s0'
↪already present!
[ 6.805376] RKNPU fdab0000.npu: pvtm=874
[ 6.809605] RKNPU fdab0000.npu: pvtm-volt-sel=3
[ 6.809648] debugfs: Directory 'fdab0000.npu-rknpu' with parent 'vdd_npu_s0'
↪already present!
[ 6.811600] RKNPU fdab0000.npu: avs=0
[ 6.811748] RKNPU fdab0000.npu: l=10000 h=85000 hyst=5000 l_limit=0 h_
↪limit=8000000000 h_table=0
```

rknpu 0.9.6 20240322 contains the driver version.

cat /sys/module/rknpu/version can also show the driver version, which is 0.9.6.

To show the version of the /usr/lib/librknnrt.so, use:

```
strings /usr/lib/librknnrt.so | grep "librknnrt"
```

It prints:

```
librknnrt.so
librknnrt version: 1.4.0 (a10f100eb@2022-09-09T09:07:14)
```

To show the version of rknn_server, run:

```
rknn_server
```

It shows:

```
start rknn server, version:1.3.0 (121b661 build: 2022-04-29 11:12:02)
I NPUTransfer: Starting NPU Transfer Server, Transfer version 2.1.0 (b5861e7@2020-11-
↪23T11:50:51)
```

We need to support RK3568、RK3566、RK3562、RK3588、RV1106, and RV1103.

66.3 lubancat

- user: cat, passwd: temppwd
- user: root, passwd: root

67.1 630c

67.1.1 ax-samples

header files: https://github.com/AXERA-TECH/ax620e_bsp_sdk/tree/main/msp/out/arm64_glibc/include

See https://github.com/AXERA-TECH/ax-samples/blob/main/.github/workflows/build_630c_glibc.yaml#L35 for how to build it.

`AX_SYS_Init()` is defined in https://github.com/AXERA-TECH/ax620e_bsp_sdk/blob/main/msp/out/arm64_glibc/include/ax_sys_api.h It returns 0 on success.

`AX_SYS_Deinit()` can be used to deinitialize the system.

ax sys api 文档, ax engine api 文档: https://dl.sipeed.com/shareURL/MaixIV/M4N-Dock/07_Software_Doc/board

- sense voice: <https://github.com/ml-inory/sensevoice.axera/tree/main>
model export: https://github.com/ml-inory/sensevoice.axera/blob/main/model_convert/README.md

67.1.2 axcl-samples

- whisper: <https://github.com/ml-inory/whisper.axcl/tree/main>

68.1 AudioSource

69.1 Tutorials

- <https://github.com/leandromoreira/ffmpeg-libav-tutorial>
- <https://leixiaohua1020.github.io/#ffmpeg-development-examples>

69.2 API

- https://ffmpeg.org/doxygen/trunk/group__lavf__core.html
- https://ffmpeg.org/doxygen/4.0/group__lavc__decoding.html
- https://ffmpeg.org/doxygen/2.0/group__lavc__decoding.html

70.1 macos

VLC is installed in /Applications/VLC.app.

- /Applications/VLC.app/Contents/MacOS/include/vlc/libvlc.h
- /Applications/VLC.app/Contents/MacOS/include/vlc/vlc.h
- /Applications/VLC.app/Contents/MacOS/lib/libvlc.dylib

API doc https://videolan.videolan.me/vlc/group__libvlc__core.html

70.2 hello

The simplest program

Listing 1: ./code/hello/hello.cc

```
1  #include "vlc/libvlc.h"
2  #include <stdio.h>
3
4  int main() {
5      const char *version = libvlc_get_version();
6      const char *compiler_version = libvlc_get_compiler();
7      const char *change_set = libvlc_get_changeset();
8      printf("libvlc version: %s\n", version);
9      printf("compiler version: %s\n", compiler_version);
10     printf("changeset: %s\n", change_set);
11     return 0;
12 }
```

Listing 2: ./code/hello/Makefile

```
1 CXX_FLAGS := -I /Applications/VLC.app/Contents/MacOS/include
2 LDFLAGS := -L /Applications/VLC.app/Contents/MacOS/lib
3 LDFLAGS += -lvlc
4 LDFLAGS += -Wl,-rpath,/Applications/VLC.app/Contents/MacOS/lib
5
6 hello: hello.cc
7     g++ -o ./hello $(CXX_FLAGS) ./hello.cc $(LDFLAGS)
```

70.3 build

```
sudo apt-get install flex gettext bison autopoint libtool
git clone https://github.com/videolan/vlc
cd vlc
./bootstrap
./configure --prefix=$PWD/my-install
make install
```

71.1 Basics

- gguf naming convention: <https://github.com/ggml-org/ggml/blob/master/docs/gguf.md>
- safetensors doc: <https://hf-mirror.com/docs/safetensors/index>
- `save_gguf`
- `save_safetensors`
- `savez`

71.1.1 mean

- torch: `keepdim`
- mlx: `keepdims`
- torch: `dim`
- mlx: `axis`

71.1.2 cat

torch uses `cat`, while mlx uses `concatenate`.

torch uses `dim` in `cat`, mlx uses `axis` in `concatenate`

71.1.3 Sequential

torch uses `this_layer[0]`, while mx uses `this_layer.layers[0]`

71.2 LSTM

- LSTM in mx does not support multi-layers
- batch is always at the 0 dim, i.e., batch first is always True
- also, does not support bidirectional lstm
- there is only a single bias, if not None

<https://github.com/ml-explore/mlx-examples/blob/main/encodec/encodec.py#L14> has a meta kernel implementation for LSTM.

Listing 1: `./code/test_lstm_1_layer.py`

```

1  #!/usr/bin/env python3
2
3  import mlx.core as mx
4  import mlx.nn as nn
5  import numpy as np
6  import torch
7  from mlx.utils import tree_flatten, tree_map, tree_unflatten
8
9
10 class TorchLstm(torch.nn.Module):
11     def __init__(self):
12         super().__init__()
13         self.lstm = torch.nn.LSTM(input_size=2, hidden_size=6, batch_first=True)
14
15     def forward(self, x, h=None, c=None):
16         if h is not None:
17             y, (h, c) = self.lstm(x, (h, c))
18         else:
19             y, (h, c) = self.lstm(x)
20         return y, h, c
21
22
23 class MlxLstm(nn.Module):
24     def __init__(self):
25         super().__init__()
26         self.lstm = nn.LSTM(input_size=2, hidden_size=6)

```

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```

27
28 def forward(self, x, h=None, c=None):
29     """
30     Args:
31         x: (N, L, H)
32         h: None or (N, H)
33         c: None or (N, H)
34     Returns:
35         y: (N, L, H)
36         h: (N, H)
37         c: (N, H)
38     """
39     if h is None:
40         h, c = self.lstm(x)
41     else:
42         h, c = self.lstm(x, h, c)
43         # now both h and c are (N, L, H)
44
45     return h, h[:, -1, :], c[:, -1, :]
46
47
48 def test_with_seq_len(torch_lstm, mx_lstm, N, T):
49     x = torch.rand(N, T, 2)
50     print("x", x.shape)
51
52     y, h, c = torch_lstm(x)
53     assert y.shape == (N, T, 6), y.shape
54     assert h.shape == (1, N, 6), h.shape # 1 is number of layers
55     assert c.shape == (1, N, 6), h.shape
56
57     ix = mx.array(x)
58     mx_y, mx_h, mx_c = mx_lstm.forward(ix)
59     assert mx_y.shape == (N, T, 6), mx_y.shape
60     assert mx_h.shape == (N, 6), mx_h.shape
61     assert mx_c.shape == (N, 6), mx_c.shape
62
63     assert torch.allclose(y, torch.from_numpy(np.array(mx_y)))
64     assert torch.allclose(h[0], torch.from_numpy(np.array(mx_h)))
65     assert torch.allclose(c[0], torch.from_numpy(np.array(mx_c)))
66
67     # now with states
68     y, h, c = torch_lstm(x, h, c)
69     assert y.shape == (N, T, 6), y.shape

```

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```

70     assert h.shape == (1, N, 6), h.shape # 1 is number of layers
71     assert c.shape == (1, N, 6), h.shape
72
73     mx_y, mx_h, mx_c = mx_lstm.forward(ix, mx_h, mx_c)
74     assert mx_y.shape == (N, T, 6), mx_y.shape
75     assert mx_h.shape == (N, 6), mx_h.shape
76     assert mx_c.shape == (N, 6), mx_c.shape
77
78     assert torch.allclose(y, torch.from_numpy(np.array(mx_y)))
79     assert torch.allclose(h[0], torch.from_numpy(np.array(mx_h)))
80     assert torch.allclose(c[0], torch.from_numpy(np.array(mx_c)))
81
82
83 def test_single_layer():
84     torch_lstm = TorchLstm()
85     mx_lstm = MlxLstm()
86
87     # ['lstm.weight_ih_l0', 'lstm.weight_hh_l0', 'lstm.bias_ih_l0', 'lstm.bias_hh_l0']
88     print(list(torch_lstm.state_dict().keys()))
89     for k, v in torch_lstm.state_dict().items():
90         print(k, v.shape)
91     """
92     lstm.weight_ih_l0 torch.Size([24, 6])
93     lstm.weight_hh_l0 torch.Size([24, 6])
94     lstm.bias_ih_l0 torch.Size([24])
95     lstm.bias_hh_l0 torch.Size([24])
96     """
97
98     # lstm
99     print(mx_lstm.parameters().keys())
100
101     # dict_keys(['Wx', 'Wh', 'bias'])
102     print(mx_lstm.parameters()["lstm"].keys())
103     for k, v in mx_lstm.parameters()["lstm"].items():
104         print(k, v.shape)
105     """
106     Wx (24, 6)
107     Wh (24, 6)
108     bias (24,)
109     """
110     # convert
111     state_dict = torch_lstm.state_dict()
112     new_state_dict = dict()

```

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```

113 for k, v in state_dict.items():
114     basename, pname = k.rsplit(".", 1)
115     if "lstm" not in basename:
116         # we convert only lstm in this for loop
117         continue
118     w_or_b, ih_or_hh, ln = pname.split("_")
119     if w_or_b == "weight":
120         new_name = "Wx" if ih_or_hh == "ih" else "Wh"
121     elif w_or_b == "bias" and ih_or_hh == "ih":
122         continue
123     else:
124         v = v + state_dict[k.replace("_hh_", "_ih_")]
125         new_name = "bias"
126         # k = basename + "." + ln[1:] + "." + new_name # for multi-layer lstm
127         k = basename + "." + new_name # for multi-layer lstm
128         new_state_dict[k] = v
129     # print("here", new_state_dict)
130     # print("here2", list(new_state_dict.items()))
131     # print("here3", tree_unflatten(list(new_state_dict.items())))
132     # print("here4", tree_flatten(tree_unflatten(list(new_state_dict.items()))))
133
134     # convert torch.tensor to mx.array
135     new_state_dict = tree_map(mx.array, new_state_dict)
136     mx_lstm.update(tree_unflatten(list(new_state_dict.items())))
137
138 for N in [1, 2, 3]:
139     for T in [1, 2, 3]:
140         test_with_seq_len(torch_lstm=torch_lstm, mx_lstm=mx_lstm, N=N, T=T)
141
142
143 @torch.no_grad()
144 def main():
145     test_single_layer()
146
147
148 if __name__ == "__main__":
149     torch.manual_seed(20250716)
150     main()

```

Listing 2: ./code/test_lstm_multi_layers

```

1 #!/usr/bin/env python3

```

```

2

```

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```

3 import mlx.core as mx
4 import mlx.nn as nn
5 import numpy as np
6 import torch
7 from mlx.utils import tree_flatten, tree_map, tree_unflatten
8
9
10 class TorchLstm(torch.nn.Module):
11     def __init__(self):
12         super().__init__()
13         self.lstm = torch.nn.LSTM(
14             input_size=2, hidden_size=5, num_layers=3, batch_first=True
15         )
16
17     def forward(self, x, h=None, c=None):
18         if h is not None:
19             y, (h, c) = self.lstm(x, (h, c))
20         else:
21             y, (h, c) = self.lstm(x)
22         return y, h, c
23
24
25 class MlxLstm(nn.Module):
26     def __init__(self):
27         super().__init__()
28         self.lstm = []
29         input_size = 2
30         hidden_size = 5
31         for i in range(3):
32             in_size = input_size if i == 0 else hidden_size
33             self.lstm.append(nn.LSTM(input_size=in_size, hidden_size=hidden_size))
34
35     def forward(self, x, states=None):
36         """
37         Args:
38             x: (N, L, C)
39             states: None or a list containing 2*num_layers tensors
40                     - states[2*i] is the h of the i-th layer, of shape (N, C)
41                     - states[2*i+1] is the c of the i-th layer, of shape (N, C)
42         Returns:
43             y: (N, L, C)
44             states: a list containing 2*num_layers tensors
45                     - states[2*i] is the h of the i-th layer, of shape (N, C)

```

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```

46         - states[2*i+1] is the c of the i-th layer, of shape (N, C)
47
48         """
49
50         if states is None:
51             states = [None] * (len(self.lstm) * 2)
52
53         new_states = []
54         for i, layer in enumerate(self.lstm):
55             h = states[2 * i]
56             c = states[2 * i + 1]
57             x, c = layer(x, h, c)
58             new_states.append(x[:, -1, :])
59             new_states.append(c[:, -1, :])
60         return x, new_states
61
62
63 def test_with_seq_len(torch_lstm, mx_lstm, N, T):
64     x = torch.rand(N, T, 2)
65     print("x", x.shape)
66
67     y, h, c = torch_lstm(x)
68     assert y.shape == (N, T, 5), y.shape
69     assert h.shape == (3, N, 5), h.shape # 3 is number of layers
70     assert c.shape == (3, N, 5), h.shape
71
72     ix = mx.array(x)
73     mx_y, states = mx_lstm.forward(ix)
74
75     assert mx_y.shape == (N, T, 5), mx_y.shape
76     assert torch.allclose(y, torch.from_numpy(np.array(mx_y)))
77
78     for i in range(3):
79         assert torch.allclose(h[i], torch.from_numpy(np.array(states[2 * i])))
80         assert torch.allclose(c[i], torch.from_numpy(np.array(states[2 * i + 1])))
81     print("mx_y", mx_y.shape, [s.shape for s in states])
82
83     # now with states
84     y, h, c = torch_lstm(x, h, c)
85     assert y.shape == (N, T, 5), y.shape
86     assert h.shape == (3, N, 5), h.shape # 3 is number of layers
87     assert c.shape == (3, N, 5), h.shape
88

```

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```

89     mx_y, states = mx_lstm.forward(ix, states)
90     assert torch.allclose(y, torch.from_numpy(np.array(mx_y)))
91
92     for i in range(3):
93         assert torch.allclose(h[i], torch.from_numpy(np.array(states[2 * i])))
94         assert torch.allclose(c[i], torch.from_numpy(np.array(states[2 * i + 1])))
95     print("mx_y", mx_y.shape, [s.shape for s in states])
96
97
98 def test_multi_layers():
99     torch_lstm = TorchLstm()
100    mx_lstm = MlxLstm()
101    # mx.eval(mx_lstm.parameters())
102
103    # ['lstm.weight_ih_10', 'lstm.weight_hh_10', 'lstm.bias_ih_10',
104    # 'lstm.bias_hh_10', 'lstm.weight_ih_11', 'lstm.weight_hh_11',
105    # 'lstm.bias_ih_11', 'lstm.bias_hh_11', 'lstm.weight_ih_12',
106    # 'lstm.weight_hh_12', 'lstm.bias_ih_12', 'lstm.bias_hh_12']
107    print(list(torch_lstm.state_dict().keys()))
108    for k, v in torch_lstm.state_dict().items():
109        print(k, v.shape)
110
111    """
112    lstm.weight_ih_10 torch.Size([12, 2])
113    lstm.weight_hh_10 torch.Size([12, 3])
114    lstm.bias_ih_10 torch.Size([12])
115    lstm.bias_hh_10 torch.Size([12])
116    lstm.weight_ih_11 torch.Size([12, 3])
117    lstm.weight_hh_11 torch.Size([12, 3])
118    lstm.bias_ih_11 torch.Size([12])
119    lstm.bias_hh_11 torch.Size([12])
120    lstm.weight_ih_12 torch.Size([12, 3])
121    lstm.weight_hh_12 torch.Size([12, 3])
122    lstm.bias_ih_12 torch.Size([12])
123    lstm.bias_hh_12 torch.Size([12])
124    """
125
126    # lstm
127    print(mx_lstm.parameters().keys())
128    assert isinstance(mx_lstm.parameters()["lstm"], list)
129    assert len(mx_lstm.parameters()["lstm"]) == 3, len(mx_lstm.parameters())
130
131    # dict_keys(['Wx', 'Wh', 'bias'])
132    print(mx_lstm.parameters()["lstm"][0].keys())

```

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```

132     for k, v in mx_lstm.parameters()["lstm"][0].items():
133         print(k, v.shape)
134
135     """
136     Wx (12, 2)
137     Wh (12, 3)
138     bias (12,)
139     """
140     print(mx_lstm.parameters()["lstm"][1].keys())
141     for k, v in mx_lstm.parameters()["lstm"][1].items():
142         print(k, v.shape)
143     """
144     Wx (12, 3)
145     Wh (12, 3)
146     bias (12,)
147     """
148     # a list of tuple
149     # [('lstm.0.Wx', array(...)),
150     #  ('lstm.1.Wh', array(...)) ]
151     # print(tree_flatten(mx_lstm.parameters()))
152
153     # convert
154     state_dict = torch_lstm.state_dict()
155     new_state_dict = dict()
156     for k, v in state_dict.items():
157         basename, pname = k.rsplit(".", 1)
158         if "lstm" not in basename:
159             # we convert only lstm in this for loop
160             continue
161         w_or_b, ih_or_hh, ln = pname.split("_")
162         if w_or_b == "weight":
163             new_name = "Wx" if ih_or_hh == "ih" else "Wh"
164         elif w_or_b == "bias" and ih_or_hh == "ih":
165             continue
166         else:
167             v = v + state_dict[k.replace("_hh_", "_ih_")]
168             new_name = "bias"
169         k = basename + "." + ln[1:] + "." + new_name
170         new_state_dict[k] = v
171
172     # convert torch.tensor to mx.array
173     new_state_dict = tree_map(mx.array, new_state_dict)
174     mx_lstm.update(tree_unflatten(list(new_state_dict.items())))

```

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```
175     print("updated")
176
177     for N in [1, 2, 3]:
178         for T in [1, 2, 3]:
179             test_with_seq_len(torch_lstm=torch_lstm, mx_lstm=mx_lstm, N=N, T=T)
180
181
182 @torch.no_grad()
183 def main():
184     test_multi_layers()
185
186
187 if __name__ == "__main__":
188     torch.manual_seed(20250716)
189     main()
```

71.3 conv1d

There is no `nn.functional.conv1d` in `mx`, use `mx.core.conv1d`.

The weight matrix in `pytorch` is `(out_channels, in_channels, kernel_size)`, The weight matrix in `mx` is `(out_channels, kernel_size, in_channels)`,

The input in `pytorch` is `(N, C, L)`.

The input in `mx` is `(N, L, C)`

71.4 Conv2d

Input of `pytorch` `NCHW`.

Input of `mx` `NHCW`.

Weight shape in `PyTorch` `(out_channels, in_channels/groups, kernel_size[0], kernel_size[1])`

Weight shape in `mx` `(out_channels, kernel_size[0], kernel_size[1], in_channels/groups)`

71.5 todo

- Learn other mlx-based project, e.g.,
 - <https://lmstudio.ai/blog/lmstudio-v0.3.4>
 - <https://github.com/lmstudio-ai/mlx-engine>

71.6 BatchNorm

71.6.1 2d

Input shape for PyTorch NCHW.

Input shape for mlx NHWC

72.1 dmd

72.1.1 Linear regression

scatter plot

1. Select the two columns of the data (left column x, right y), including the headers
2. Insert, scatter plot
3. Click any of the point on the plot, Click the "+" symbol, click the "trend line", select more options

P 值, 0.05, 是一个分界线。P 越小越好, 代表对应这个系数是 0 的概率越小。

问豆包的问题:

- 方差分析里的 MS 和 SS 代表什么, 如何计算的
- excel 里的线性回归里的 SS 和 MS
- p 值如何算

看懂 t 统计量, p value

看懂 F 建议和显著性. F 针对整个模型, t 只针对具体某个参数.

ESS 是 explained sum of squared, 也是 SSR, 即 regression sum of squares, 是回归平方和。

SSE 是 residual sum of squared errors 或者 residual sum of squares, 是残差平方和。

ANOVA 是方差分析的意思, analysis of variance

自由度: 回归平方和里的自由度, 是参数总数减去 1。残差平方和里的自由度是 $n - 1$ - 自变量个数。

回归均方是回归平方和除以回归平方的自由度。用 MSR 表示. 这个自由度也是自变量的个数。

残差均方是残差平方和除以残差平方的自由度。用 MSE 表示。

F 统计量为 MSR/MSE. 分子是 MSR, 它的自由度是自变量个数, 即参数个数减去 1。分母是 $n - k - 1$ (k 是自变量个数)

得到 F 的值后，可以根据分子和分母的自由度，去查 F 分布的表。表里表示大于 F 的面积，这个面积就是 F significance，也是原假设的概率。原假设是所有参数都为 0。所以 F significance 越小，就代表原假设的概率越小。一般的，用 0.05 做阈值。如果 F significance 小于 0.05，即原假设的对立面的概率大于 0.95，即至少有一个参数不为 0。

t 统计量里的 P 值，也是原假设成立的概率。越小越好。0.05 是一个阈值。

73.1 GPT

- <https://github.com/karpathy/minGPT>
- <https://github.com/openai/gpt-2>
- Improving language understanding by generative pre-training
<https://www.mikecaptain.com/resources/pdf/GPT-1.pdf>
GPT-1, cited 15k, published in 2018
- Language Models are Unsupervised Multitask Learners
https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf
Cited 18k, published in 2019
It is GPT-2. Code: <https://github.com/openai/gpt-2>

73.2 bpe

- Japanese and Korean voice search
<https://static.googleusercontent.com/media/research.google.com/en//pubs/archive/37842.pdf>
Proposes wordpieces, 2012
- Neural Machine Translation of Rare Words with Subword Units
<https://arxiv.org/abs/1508.07909>
2016
- Subword Regularization: Improving Neural Network Translation Models with Multiple Subword Candidates
<https://arxiv.org/pdf/1804.10959>, 2018

- Sentencepiece: A Simple and Language Independent Subword Tokenizer and Detokenizer for Neural Text Processing

<https://arxiv.org/abs/1808.06226>, 2018

73.3 TODO

- <https://github.com/zhaochenyang20>
- <https://github.com/zhaochenyang20/Awesome-ML-SYS-Tutorial>

73.4 The Big LLM Architecture Comparison

<https://magazine.sebastianraschka.com/p/the-big-llm-architecture-comparison>

- positional encoding: absolute, RoPE
- attention: multihead attention, Grouped-Query Attention
- activation: GELU, SwiGLU

73.5 Papers

- 2017 Attention is all you need

<https://proceedings.neurips.cc/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf>

transformer

- 2018 Improving Language Understanding by Generative Pre-Training

GPT-1

- 2019.06 Language models are unsupervised multitask learners

<https://storage.prod.researchhub.com/uploads/papers/2020/06/01/language-models.pdf>

GPT-2, 1.5B parameters

- 2020 Language models are few-shot learners

<https://proceedings.nips.cc/paper/2020/file/1457c0d6bfc4967418bfb8ac142f64a-Paper.pdf>

GPT-3

- 2023.02 LLaMA: Open and Efficient Foundation Language Models

<https://arxiv.org/pdf/2302.13971>

- 2023.12 GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints
<https://arxiv.org/pdf/2305.13245>
- 2025.5 Qwen3 Technical Report
<https://arxiv.org/pdf/2505.09388>
- 2023.11 A Survey of Large Language Models
https://www.researchgate.net/profile/Tang-Tianyi-3/publication/369740832_A_Survey_of_Large_Language_Models/links/665fd2e3637e4448a37dd281/A-Survey-of-Large-Language-Models.pdf
- 2025.5 Large language models: A survey
<https://arxiv.org/pdf/2402.06196>

73.6 ai-coding

```
npm install -g @anthropic-ai/claude-code
```

After installation, run:

```
claude
```

If it throws:

```
Unable to connect to Anthropic services

Failed to connect to api.anthropic.com: ECONNREFUSED

Please check your internet connection and network settings.

Note: Claude Code might not be available in your country. Check supported
countries at https://anthropic.com/supported-countries
```

Edit `~/.claude.json` and add a line to the end of the file:

```
"hasCompletedOnboarding": true
```

Then change `~/.claude/settings.json` to setup the keys.

After that, run:

```
claude

alias claude='claude --dangerously-skip-permissions'
```


74.1 Basics

74.1.1 npu-smi

```
npu-smi info

# It's similar to nvida-smi
```

```
yyy@xxx:~$ npu-smi info

+-----+
| npu-smi 24.1.0                               Version: 24.1.0 |
+-----+
+-----+-----+-----+-----+-----+
| NPU   Name           | Health       | Power (W)    | Temp (C)     | Hugepages-  |
| Usage (page) |
| Chip           | Bus-Id       | AICore (%)   | Memory-Usage (MB) | HBM-        |
| Usage (MB)      |
+=====+=====+=====+=====+=====+
| 0      910B          | OK           | xx.x         | xxx          | 0 / 0 |
|
| 0                  | 1234:d0:12.1 | 0            | xxxx / 1xxxx | 0 /    |
|xxxxxx
+=====+=====+=====+=====+=====+
| NPU   Chip           | Process id   | Process name  | Process      |
| memory (MB)          |
+=====+=====+=====+=====+=====+
| No running processes found in NPU 0
```

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```

compiler_installed_version=[7.6.0.1.220:8.0.0]
hccl_installed_version=[7.6.0.1.220:8.0.0]
opp_installed_version=[7.6.0.1.220:8.0.0]
toolkit_installed_version=[7.6.0.1.220:8.0.0]
aoe_installed_version=[7.6.0.1.220:8.0.0]
ncs_installed_version=[7.6.0.1.220:8.0.0]

```

```

echo $PATH | grep ascend-toolkit

/usr/local/Ascend/ascend-toolkit/latest/bin:/usr/local/Ascend/ascend-toolkit/latest/
↪ compiler/cccec_compiler/bin:/usr/local/Ascend/ascend-toolkit/latest/tools/cccec_
↪ compiler/bin

```

```

ls /usr/local/Ascend/ascend-toolkit/

8.0  8.0.0  latest  set_env.sh

```

74.1.3 atc --help

ascend tool chain

```

ATC start working now, please wait for a moment.
usage: atc <args>
generate offline model example:
atc --model=./alexnet.prototxt --weight=./alexnet.caffemodel --framework=0 --output=./
↪ domi --soc_version=<soc_version>
generate offline model for single op example:
atc --singleop=./op_list.json --output=./op_model --soc_version=<soc_version>

===== Basic Functionality =====
[General]
  --h/help          Show this help message
  --mode            Run mode.
                    0: default, generate offline model;
                    1: convert model to JSON format;
                    3: only pre-check;
                    5: convert ge dump txt file to JSON format;
                    6: display model info;
                    30: convert original graph to execute-om for nano(offline_
↪ model)

[Input]

```

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```

--distributed_cluster_build      build model for distribute mode, 1: enable
↪ distribute; 0(default): disable distribute
--model                          Model file
--weight                        Weight file. Required when framework is Caffe
--om                             The model file to be converted to json
--framework                     Framework type. 0:Caffe; 1:MindSpore; 3:Tensorflow; 5:Onnx
--input_format                  Format of input data. E.g.: "NCHW"
--input_shape                   Shape of static input data or shape range of dynamic input.
↪ Separate multiple nodes with semicolons (;). Use double quotation marks (") to
↪ enclose each argument.
                                E.g.: "input_name1:n1,c1,h1,w1;input_name2:n2,c2,h2,w2"
                                "input_name1:n1~n2,c1,h1,w1;input_name2:n3~n4,c2,h2,w2"
--input_shape_range             This option is deprecated and will be removed in future version,
↪ please use input_shape instead. Shape range of input data. Separate multiple nodes
↪ with semicolons (;).
                                Use double quotation marks (") to enclose each argument.
                                E.g.: "input_name1:[n1~n2,c1,h1,w1];input_name2:[n2,c2~c3,h2,w2]"
↪ "
--dynamic_batch_size            Set dynamic batch size. E.g.: "batchsize1,batchsize2,batchsize3"
↪ "
--dynamic_image_size            Set dynamic image size. Separate multiple nodes with
↪ semicolons (;). Use double quotation marks (") to enclose each argument.
                                E.g.: "imagesize1_height,imagesize1_width;imagesize2_height,
↪ imagesize2_width"
--dynamic_dims                  Set dynamic dims. Separate multiple nodes with semicolons (;).
↪ Use double quotation marks (") to enclose each argument.
                                E.g.: "dims1_n1,dims1_n2;dims2_n1,dims2_n2"
--singleop                      Single op definition file. atc will generate offline.model(s)
↪ for single op if --singleop is set.
--shard_model_dir               directory of split models
--model_relation_config         relation of split models
--enable_graph_parallel         whether to enable parallel
--graph_parallel_option_path    parallel strategy configuration file
--cluster_config                logic cluster configuration of target environment

[Output]
--output                        Output file path&name(needn't suffix, will add .om/.exeom
↪ automatically).
                                If --mode is set to 30, an additional dbg file will be
↪ generated.
                                If --singleop is set, this arg specifies the directory to which
↪ the single op offline model will be generated.
--output_type                   Set net output type. Support FP32, FP16, UINT8, INT8. E.g.:

```

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```

↪FP16, indicates that all out nodes are set to FP16.
        "node1:0:FP16;node2:1:FP32", indicates setting the datatype of
↪multiple out nodes.
    --check_report    The pre-checking report file. Default value is: "check_result.
↪json"
    --json            The output json file path&name which is converted from a model
    --host_env_os      OS type of the target execution environment.
                        The parameters that support setting are the OS types of
↪the opp package
                        Supported host env os as list:
                        minios linux
                        default: linux
    --host_env_cpu     CPU type of the target execution environment.
                        The parameters that support setting are the CPU types of
↪the opp package
                        Supported host env cpu as list:
                        support cpu: aarch64 , respond to os: minios
                        support cpu: aarch64 x86_64 , respond to os: linux
                        default: aarch64

[Target]
    --soc_version      The soc version.
    --virtual_type     Set whether offline model can run on the virtual devices under
↪compute capability allocation.
                        0 (default) : Disable virtualization; 1 : Enable virtualization.
    --core_type        Set core type AiCore or VectorCore. VectorCore: use vector core.
↪ Default value is: AiCore
    --aicore_num       Set aicore num
===== Advanced Functionality =====
[Feature]
    --out_nodes        Output nodes designated by users. Separate multiple nodes with
↪semicolons (;).Use double quotation marks (") to enclose each argument.
                        E.g.: "node_name1:0;node_name1:1;node_name2:0"
    --input_fp16_nodes Input node datatype is fp16. Separate multiple nodes with
↪semicolons (;). Use double quotation marks (") to enclose each argument. E.g.:
↪"node_name1;node_name2"
    --insert_op_conf    Config file to insert new op
    --op_name_map       Custom op name mapping file
                        Note: A semicolon(;) cannot be included in each path, otherwise
↪the resolved path will not match the expected one.
    --is_input_adjust_hw_layout Input node datatype is fp16 and format is NC1HWC0,
↪used with input_fp16_nodes. true: enable; false(default): disable. E.g.: "true,true,
↪false,true"

```

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```

--is_output_adjust_hw_layout    Net output node datatype is fp16 and format is_
↪NC1HWC0, used with out_nodes. true: enable; false(default): disable. E.g.: "true,
↪true,false,true"
--external_weight              Convert const to file constant, and save weight in file.
                                0 (default): save weight in om. 1: save weight in file.

[Model Tuning]
--disable_reuse_memory          The switch of reuse memory. Default value is : 0. 0 means_
↪reuse memory, 1 means do not reuse memory.
--fusion_switch_file           File for fusion rule(graph fusion and UB fusion).
                                Enter as the configuration file path, disable specified_
↪fusion rules
--enable_scope_fusion_passes    validate the non-general scope fusion passes,_
↪multiple names can be set and separated by ','. E.g.: ScopePass1,ScopePass2,...
--enable_single_stream          Enable single stream. true: enable; false(default):_
↪disable
--ac_parallel_enable            Enable engines such as Aicpu to parallel with other_
↪engines in dynamic shape graphs. 1: enable; 0(default): disable
--tiling_schedule_optimize      Enable tiling schedule optimize. 1: enable; 0(default):_
↪disable
--quant_dumpable                Ensure that the input and output of quant nodes can be_
↪dumped. 1: enable; 0(default): disable.
--enable_small_channel          Set enable small channel. 0(default): disable; 1: enable
--enable_compress_weight        Enable compress weight. true: enable; false(default):_
↪disable
--compress_weight_conf          Config file to compress weight
--compression_optimize_conf     Config file to compress optimize
--sparsity                      Optional; enable structured sparse. 0(default): disable;_
↪1: enable
--buffer_optimize               Set buffer optimize. Support "l2_optimize" (default), "l1_
↪optimize", "off_optimize"
--mdl_bank_path                 Set the path of the custom repository generated after_
↪model tuning.
--oo_level                       The graph optimization level. Support "O1", "O3"(default).
--topo_sorting_mode             The option of graph topological sort, 0: BFS; 1:_
↪DFS(default); 2: RDFS; 3: StableRDFS(stable topo).
--oo_constant_folding           The switch of constant folding, false: disable;_
↪true(default): enable.
--oo_dead_code_elimination       The switch of dead code elimination, false:_
↪disable; true(default): enable.

[Operator Tuning]
--op_precision_mode             Set the path of operator precision mode configuration file_

```

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```

→ (.ini)
  --allow_hf32          enable hf32. false: disable; true: enable. (not support,
→ reserved)
  --precision_mode      precision mode, support force_fp16(default), force_fp32,
→ cube_fp16in_fp32out, allow_mix_precision, allow_fp32_to_fp16, must_keep_origin_
→ dtype, allow_mix_precision_fp16, allow_mix_precision_bf16, allow_fp32_to_bf16.
  --precision_mode_v2    precision mode v2, support fp16(default), origin, cube_
→ fp16in_fp32out, mixed_float16, mixed_bfloat16, cube_hif8, mixed_hif8.
  --modify_mixlist       Set the path of operator mixed precision configuration file.
  --keep_dtype           Retains the precision of certain operators in inference.
→ scenarios by using a configuration file.
  --customize_dtypes      Set the path of custom dtypes configuration file.
  --is_weight_clip       Ensure weight is finite by clipped when its datatype is
→ floating-point data, 0: disable; 1(default): enable.
  --op_bank_path          Set the path of the custom repository generated after
→ operator tuning with Auto Tune.
  --op_select_implmode    Set op select implmode. Support high_precision, high_
→ performance, high_precision_for_all, high_performance_for_all. default: high_
→ performance
  --optypelist_for_implmode  Appoint which op to select implmode, cooperated with
→ op_select_implmode.

                                Separate multiple nodes with commas (,). Use double
→ quotation marks (") to enclose each argument. E.g.: "node_name1,node_name2"

[Debug]
  --op_debug_level        Debug enable for TBE operator building.
                                0 (default): Disable debug; 1: Enable TBE pipe_all, and
→ generate the operator CCE file and Python-CCE mapping file (.json);
                                2: Enable TBE pipe_all, generate the operator CCE file and
→ Python-CCE mapping file (.json), and enable the CCE compiler -O0-g.
                                3: Disable debug, and keep generating kernel file (.o and .
→ json)
                                4: Disable debug, keep generation kernel file (.o and .
→ json) and generate the operator CCE file (.cce) and the UB fusion computing
→ description file (.json)
  --save_original_model    Control whether to output original model. E.g.: true:
→ output original model
  --log                   Generate log with level. Support debug, info, warning,
→ error, null(default)
  --dump_mode              The switch of dump json with shape, to be used with mode 1.
→ 0(default): disable; 1: enable.
  --debug_dir              Set the save path of operator compilation intermediate
→ files.

```

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```

                                Default value: ./kernel_meta
--status_check                  switch for op status check such as overflow.
                                0(default): disable; 1: enable.
--op_compiler_cache_dir         Set the save path of operator compilation cache files.
                                Default value: $HOME/atc_data
--op_compiler_cache_mode        Set the operator compilation cache mode. Options are
↪disable(default), enable and force(force to refresh the cache)
--display_model_info            enable for display model info; 0(default): close display,
↪1: open display.
--shape_generalized_build_mode   For selecting the mode of shape generalization
↪when build graph.
                                shape_generalized: Shape will be generalized
↪during graph build
                                shape_precise(default): Shape will not be
↪generalized, use precise shape
--op_debug_config               Debug enable for Operator memory detection, enter as the
↪configuration file path.
                                If option is default, debug for Operator memory detection
↪is disable.
--atomic_clean_policy           For selecting the atomic op clean memory policy.
                                0 (default): centralized clean. 1: separate clean.
--deterministic                 For deterministic calculation.
                                0 (default): deterministic off. 1: deterministic on.
--export_compile_stat           The option of configuring statistics of the graph
↪compiler, 0: Not Generate; 1: Generated when the program exits(default); 2:
↪Generated when graph compilation complete.

```

74.1.4 atc

To show debug messages:

```
export ASCEND_SLOG_PRINT_TO_STDOUT=1
```

74.1.5 fix errors

```

[INFO] acl init success [INFO] open device 0 success [INFO] create new context
[ACL ERROR] E19999: Inner Error! E19999:
[PID: 3370899] 2025-10-17-18:40:43.617.804 Invalid opp version [8.3.T14.0.B101] or
compiler_version [],Please check if it is within the required
range[FUNC:CheckOsCpuInfoAndOppVersion] [FILE:model_helper.cc] [LINE:973]

```

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```

TraceBack (most recent call last): Assert ((error_code) == ge::SUCCESS)
failed[FUNC:LoadExecutorFromModelData][FILE:api.cc][LINE:111]
[Model][FromData]call gert::LoadExecutorFromModelDataWithMem load model from data_
↳ failed,
ge result[4294967295][FUNC:ReportCallError][FILE:log_inner.cpp][LINE:161]
[ERROR] load model from file failed, model file is ./encoder.om
[WARN] Check failed:processModel->LoadModelFromFile(modelPath), ret:1

```

I am using ascendai/cann:latest from <https://github.com/Ascend/cann-container-image> in GitHub actions It shows:

```

source /usr/local/Ascend/ascend-toolkit/set_env.sh

/usr/local/Ascend/ascend-toolkit/8.3.RC1.alpha003/fwkacllib/lib64/libascend_protobuf.
↳ so
/usr/local/Ascend/ascend-toolkit/8.3.RC1.alpha003/fwkacllib/lib64/libascend_dump.so

```

And on my device, I have:

```

cat /usr/local/Ascend/driver/version.info

Version=24.1.0
ascendhal_version=7.35.23
aicpu_version=1.0
tdt_version=1.0
log_version=1.0
prof_version=2.0
dvppkernels_version=1.1
tsfw_version=1.0
Innerversion=V100R001C19SPC002B226
compatible_version=[V100R001C13],[V100R001C15],[V100R001C17],[V100R001C18],
↳ [V100R001C19]
compatible_version_fw=[7.0.0,7.6.99]
package_version=24.1.0

```

24.1.0 requires CANN 8.0 / 8.1 LTS, but i was using 8.3.RC1.alpha003, which is too new.

Switch to 8.1.rc1-910b-ubuntu22.04-py3.10. See <https://github.com/Ascend/cann-container-image/tree/main/cann/8.1.rc1-910b-ubuntu22.04-py3.10>

- CANN 7.0.x supports 23.0.x, for 910/310
- CANN 7.1.x supports 23.1.x and 23.2x, for A310B, 910B
- CANN 8.0.x/8.1.x supports 24.0.x/24.1.x
- CANN 8.3.x supports 25.0.x/ 25.1.x

74.2 Paraformer to ascend npu

74.2.1 V1: Static shape

The following code is from ChatGPT:

```
#!/usr/bin/env python3
# -*- coding: utf-8 -*-

import acl
import numpy as np
import librosa
import kaldi_native_fbank as knf

# -----
# 特征提取
# -----

def compute_feat(filename):
    sample_rate = 16000
    samples, _ = librosa.load(filename, sr=sample_rate)
    opts = knf.FbankOptions()
    opts.frame_opts.dither = 0
    opts.frame_opts.snip_edges = False
    opts.frame_opts.samp_freq = sample_rate
    opts.mel_opts.num_bins = 80

    online_fbank = knf.OnlineFbank(opts)
    online_fbank.accept_waveform(sample_rate, (samples * 32768).tolist())
    online_fbank.input_finished()

    features = np.stack([online_fbank.get_frame(i) for i in range(online_fbank.num_
→frames_ready)])
    assert features.data.contiguous is True
    assert features.dtype == np.float32

    window_size = 7
    window_shift = 6
    T = (features.shape[0] - window_size) // window_shift + 1
    features = np.lib.stride_tricks.as_strided(
        features,
        shape=(T, features.shape[1] * window_size),
        strides=((window_shift * features.shape[1]) * 4, 4),
    )
    return np.copy(features)
```

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```

# -----
# Acoustic embedding
# -----
def get_acoustic_embedding(alpha: np.array, hidden: np.array):
    alpha = alpha.tolist()
    acc = 0
    embeddings = []
    cur_embedding = np.zeros((hidden.shape[1],), dtype=np.float32)

    for i, w in enumerate(alpha):
        acc += w
        if acc >= 1:
            overflow = acc - 1
            remain = w - overflow
            cur_embedding += remain * hidden[i]
            embeddings.append(cur_embedding)
            cur_embedding = overflow * hidden[i]
            acc = overflow
        else:
            cur_embedding += w * hidden[i]

    if len(embeddings) == 0:
        raise ValueError("No speech in the audio file")
    return np.array(embeddings)

# -----
# Tokens
# -----
def load_tokens():
    ans = dict()
    i = 0
    with open("tokens.txt", encoding="utf-8") as f:
        for line in f:
            ans[i] = line.strip().split()[0]
            i += 1
    return ans

# -----
# ACL 模型加载
# -----
ACL_MEM_MALLOCC_NORMAL_ONLY = 0
ACL_MEMCPY_HOST_TO_DEVICE = 0

```

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```

ACL_MEMCPY_DEVICE_TO_HOST = 1

def load_om_model(path):
    model_id, ret = acl.mdl.load_from_file(path)
    if ret != 0:
        raise RuntimeError(f"Load model failed: {path}, ret={ret}")
    model_desc = acl.mdl.create_desc()
    acl.mdl.get_desc(model_desc, model_id)
    return model_id, model_desc

# -----
# 单输入模型推理
# -----

def run_model_single_input(model_id, model_desc, input_array):
    input_size = input_array.nbytes
    input_buf, _ = acl.rt.malloc(input_size, ACL_MEM_MALLOC_NORMAL_ONLY)
    acl.rt.memcpy(input_buf, input_size, input_array.ctypes.data, input_size, ACL_
↪MEMCPY_HOST_TO_DEVICE)

    input_dataset = acl.mdl.create_dataset()
    input_db = acl.create_data_buffer(input_buf, input_size)
    acl.mdl.add_dataset_buffer(input_dataset, input_db)

    output_size = acl.mdl.get_output_size_by_index(model_desc, 0)
    output_buf, _ = acl.rt.malloc(output_size, ACL_MEM_MALLOC_NORMAL_ONLY)
    output_dataset = acl.mdl.create_dataset()
    output_db = acl.create_data_buffer(output_buf, output_size)
    acl.mdl.add_dataset_buffer(output_dataset, output_db)

    ret = acl.mdl.execute(model_id, input_dataset, output_dataset)
    if ret != 0:
        raise RuntimeError(f"Execute failed, ret={ret}")

    output_host, _ = acl.rt.malloc_host(output_size)
    acl.rt.memcpy(output_host, output_size, output_buf, output_size, ACL_MEMCPY_
↪DEVICE_TO_HOST)
    output_data = np.frombuffer(acl.util.ptr_to_bytes(output_host, output_size),
↪dtype=np.float32)

    acl.rt.free(input_buf)
    acl.rt.free(output_buf)
    acl.rt.free_host(output_host)
    acl.mdl.destroy_dataset(input_dataset)

```

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```

acl.mdl.destroy_dataset(output_dataset)

    return output_data

# -----
# 多输入 decoder 推理
# -----
def run_model_decoder(model_id, model_desc, encoder_out, acoustic_embedding, mask):
    # 准备输入 Dataset
    input_dataset = acl.mdl.create_dataset()

    # encoder_out
    buf1, _ = acl.rt.malloc(encoder_out.nbytes, ACL_MEM_MALLOC_NORMAL_ONLY)
    acl.rt.memcpy(buf1, encoder_out.nbytes, encoder_out.ctypes.data, encoder_out.
↪nbytes, ACL_MEMCPY_HOST_TO_DEVICE)
    db1 = acl.create_data_buffer(buf1, encoder_out.nbytes)
    acl.mdl.add_dataset_buffer(input_dataset, db1)

    # acoustic_embedding
    buf2, _ = acl.rt.malloc(acoustic_embedding.nbytes, ACL_MEM_MALLOC_NORMAL_ONLY)
    acl.rt.memcpy(buf2, acoustic_embedding.nbytes, acoustic_embedding.ctypes.data, ↪
↪acoustic_embedding.nbytes, ACL_MEMCPY_HOST_TO_DEVICE)
    db2 = acl.create_data_buffer(buf2, acoustic_embedding.nbytes)
    acl.mdl.add_dataset_buffer(input_dataset, db2)

    # mask
    buf3, _ = acl.rt.malloc(mask.nbytes, ACL_MEM_MALLOC_NORMAL_ONLY)
    acl.rt.memcpy(buf3, mask.nbytes, mask.ctypes.data, mask.nbytes, ACL_MEMCPY_HOST_
↪TO_DEVICE)
    db3 = acl.create_data_buffer(buf3, mask.nbytes)
    acl.mdl.add_dataset_buffer(input_dataset, db3)

    # 输出
    output_size = acl.mdl.get_output_size_by_index(model_desc, 0)
    output_buf, _ = acl.rt.malloc(output_size, ACL_MEM_MALLOC_NORMAL_ONLY)
    output_dataset = acl.mdl.create_dataset()
    output_db = acl.create_data_buffer(output_buf, output_size)
    acl.mdl.add_dataset_buffer(output_dataset, output_db)

    # 执行
    ret = acl.mdl.execute(model_id, input_dataset, output_dataset)
    if ret != 0:
        raise RuntimeError(f"Decoder execute failed, ret={ret}")

```

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```

# 拷贝回 host
output_host, _ = acl.rt.malloc_host(output_size)
acl.rt.memcpy(output_host, output_size, output_buf, output_size, ACL_MEMCPY_
↪DEVICE_TO_HOST)
output_data = np.frombuffer(acl.util.ptr_to_bytes(output_host, output_size), ↪
↪dtype=np.float32)

# 释放
acl.rt.free(buf1)
acl.rt.free(buf2)
acl.rt.free(buf3)
acl.rt.free(output_buf)
acl.rt.free_host(output_host)
acl.mdl.destroy_dataset(input_dataset)
acl.mdl.destroy_dataset(output_dataset)

return output_data

# -----
# 主函数
# -----
def main():
    acl.init()
    device_id = 0
    acl.rt.set_device(device_id)
    context, _ = acl.rt.create_context(device_id)

    # 特征
    features = compute_feat("./1.wav")
    if features.shape[0] >= 83:
        features = features[:83]
    else:
        padding = features[-(83 - features.shape[0]):]
        features = np.concatenate([features, padding])
    features = features[None, ...] # batch=1

    # 加载 OM 模型
    enc_id, enc_desc = load_om_model("encoder.om")
    pred_id, pred_desc = load_om_model("predictor.om")
    dec_id, dec_desc = load_om_model("decoder.om")

    # encoder

```

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```

encoder_out = run_model_single_input(enc_id, enc_desc, features)
encoder_out = encoder_out.reshape(-1, 512)
print("encoder_out", encoder_out.shape, encoder_out.sum(), encoder_out.mean())

# predictor
alpha = run_model_single_input(pred_id, pred_desc, encoder_out)
print("alpha", alpha.shape, alpha.sum(), alpha.mean())

# acoustic embedding
acoustic_embedding = get_acoustic_embedding(alpha, encoder_out)
print('acoustic_embedding', acoustic_embedding.shape)
num_tokens = acoustic_embedding.shape[0]
print('num_tokens', num_tokens)
padding = np.zeros((83 - num_tokens, 512), dtype=np.float32)
acoustic_embedding = np.concatenate([acoustic_embedding, padding], axis=0)
mask = np.zeros((83,), dtype=np.float32)
mask[:num_tokens] = 1
print('mask', mask.sum(), mask.shape)

decoder_input = acoustic_embedding[None, ...]

# decoder
decoder_out = run_model_decoder(dec_id, dec_desc, encoder_out, decoder_input, ↵
↵mask)
decoder_out = decoder_out.reshape(83, -1)

print("decoder_out", decoder_out.shape, decoder_out.sum(), decoder_out.mean())

# 输出文字
yseq = decoder_out[:num_tokens].argmax(axis=-1).tolist()
print('yseq', yseq)
tokens = load_tokens()
words = [tokens[i] for i in yseq if i not in (1,2)]
text = "".join(words)
print("Recognized text:", text)

# 卸载模型 & 清理
for mid, mdesc in [(enc_id, enc_desc), (pred_id, pred_desc), (dec_id, dec_desc)]:
    acl.mdl.unload(mid)
    acl.mdl.destroy_desc(mdesc)

acl.rt.destroy_context(context)
acl.rt.reset_device(device_id)

```

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```

acl.finalize()

if __name__ == "__main__":
    main()

```

74.2.2 V2: Static shape + simplified API

```

#!/usr/bin/env python3
# Copyright (c) 2025 Xiaomi Corporation

import kaldi_native_fbank as knf
import librosa
import torch
import numpy as np
from ais_bench.infer.interface import InferSession

class SinusoidalPositionEncoder(torch.nn.Module):
    def encode(
        self,
        positions: torch.Tensor = None,
        depth: int = None,
        dtype: torch.dtype = torch.float32,
    ):
        """
        Args:
            positions: (batch_size, )
        """
        batch_size = positions.size(0)
        positions = positions.type(dtype)
        device = positions.device
        log_timescale_increment = torch.log(
            torch.tensor([10000], dtype=dtype, device=device)
        ) / (depth / 2 - 1)
        inv_timescales = torch.exp(
            torch.arange(depth / 2, device=device).type(dtype)
            * (-log_timescale_increment)
        )
        inv_timescales = torch.reshape(inv_timescales, [batch_size, -1])
        scaled_time = torch.reshape(positions, [1, -1, 1]) * torch.reshape(
            inv_timescales, [1, 1, -1]
        )

```

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```

        encoding = torch.cat([torch.sin(scaled_time), torch.cos(scaled_time)], dim=2)
        return encoding.type(dtype)

    def forward(self, batch_size, timesteps, input_dim):
        positions = torch.arange(1, timesteps + 1)[None, :]
        position_encoding = self.encode(positions, input_dim, torch.float32)

        return position_encoding

def compute_feat(filename):
    sample_rate = 16000
    samples, _ = librosa.load(filename, sr=sample_rate)
    opts = knf.FbankOptions()
    opts.frame_opts.dither = 0
    opts.frame_opts.snip_edges = False
    opts.frame_opts.samp_freq = sample_rate
    opts.mel_opts.num_bins = 80

    online_fbank = knf.OnlineFbank(opts)
    online_fbank.accept_waveform(sample_rate, (samples * 32768).tolist())
    online_fbank.input_finished()

    features = np.stack(
        [online_fbank.get_frame(i) for i in range(online_fbank.num_frames_ready)]
    )
    assert features.data.contiguous is True
    assert features.dtype == np.float32, features.dtype
    print("features sum", features.sum(), features.shape)

    window_size = 7 # lfr_m
    window_shift = 6 # lfr_n

    T = (features.shape[0] - window_size) // window_shift + 1
    features = np.lib.stride_tricks.as_strided(
        features,
        shape=(T, features.shape[1] * window_size),
        strides=((window_shift * features.shape[1]) * 4, 4),
    )
    return np.copy(features)

def load_tokens():

```

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```

ans = dict()
i = 0
with open("tokens.txt", encoding="utf-8") as f:
    for line in f:
        ans[i] = line.strip().split()[0]
        i += 1
return ans

class OmModel:
    def __init__(self):
        print("init encoder")
        self.encoder = InferSession(device_id=0, model_path='./encoder.om',
↪ debug=False)
        self.decoder = InferSession(device_id=0, model_path='./decoder.om',
↪ debug=False)
        self.predictor = InferSession(device_id=0, model_path='./predictor.om',
↪ debug=False)

        print("---encoder---")
        for i in self.encoder.get_inputs():
            print(i.name, i.datatype, i.shape)

        print("-----")

        for i in self.encoder.get_outputs():
            print(i.name, i.datatype, i.shape)

        print("---decoder---")
        for i in self.decoder.get_inputs():
            print(i.name, i.datatype, i.shape)

        print("-----")

        for i in self.decoder.get_outputs():
            print(i.name, i.datatype, i.shape)

        print("---predictor---")
        for i in self.predictor.get_inputs():
            print(i.name, i.datatype, i.shape)

        print("-----")

```

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```

        for i in self.predictor.get_outputs():
            print(i.name, i.datatype, i.shape)

# def run_encoder(self, features, pos_emb):
def run_encoder(self, features):
    encoder_out = self.encoder.infer([features], mode='static')[0]
    return encoder_out

def run_predictor(self, encoder_out):
    alphas = self.predictor.infer([encoder_out], mode='static')[0]
    return alphas

# def run_decoder(self, encoder_out, acoustic_embedding, mask):
def run_decoder(self, encoder_out, acoustic_embedding, mask):
    decoder_out = self.decoder.infer([encoder_out, acoustic_embedding, mask],
↪mode='static')[0]
    return decoder_out

def get_acoustic_embedding(alpha: np.array, hidden: np.array):
    """
    Args:
        alpha: (T,)
        hidden: (T, C)
    Returns:
        acoustic_embeds: (num_tokens, C)
    """
    alpha = alpha.tolist()
    acc = 0
    num_tokens = 0

    embeddings = []
    cur_embedding = np.zeros((hidden.shape[1],), dtype=np.float32)

    for i, w in enumerate(alpha):
        acc += w
        if acc >= 1:
            overflow = acc - 1
            remain = w - overflow
            cur_embedding += remain * hidden[i]
            embeddings.append(cur_embedding)

            cur_embedding = overflow * hidden[i]

```

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```

        acc = overflow
    else:
        cur_embedding += w * hidden[i]

if len(embeddings) == 0:
    raise ValueError("No speech in the audio file")

embeddings = np.array(embeddings)
return embeddings

def main():
    features = compute_feat("./1.wav")
    print("here", features.shape, features.shape[0] > 83)
    if features.shape[0] >= 83:
        features = features[:83]
    else:
        padding = features[-(83 - features.shape[0]) :]
        print("padding", features.shape, padding.shape)
        features = np.concatenate([features, padding])

    pos_emb = (
        SinusoidalPositionEncoder()(1, features.shape[0], features.shape[1])
        .squeeze(0)
        .numpy()
    )

    print("features.shape", features.shape, pos_emb.shape)

    print("sum", features.sum(), features.mean(), pos_emb.sum(), pos_emb.mean())

    model = OmModel()

    # encoder_out = model.run_encoder(features[None], pos_emb[None])
    encoder_out = model.run_encoder(features[None])
    print("encoder_out.shape", encoder_out.shape)
    print("encoder_out.sum", encoder_out.sum(), encoder_out.mean())

    alpha = model.run_predictor(encoder_out)
    print("alpha.shape", alpha.shape)
    print("alpha.sum()", alpha.sum(), alpha.mean())

    acoustic_embedding = get_acoustic_embedding(alpha[0], encoder_out[0])

```

(continues on next page)

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```

print("acoustic_embedding.shape", acoustic_embedding.shape)
num_tokens = acoustic_embedding.shape[0]

padding = np.zeros((83 - acoustic_embedding.shape[0], 512), dtype=np.float32)
print("padding.shape", padding.shape, acoustic_embedding.shape)

acoustic_embedding = np.concatenate([acoustic_embedding, padding], axis=0)
print("acoustic_embedding.shape", acoustic_embedding.shape)
print("acoustic_embedding.sum", acoustic_embedding.sum(), acoustic_embedding.
↪mean())

mask = np.zeros((83,), dtype=np.float32)
mask[:num_tokens] = 1
print(mask)

decoder_out = model.run_decoder(encoder_out, acoustic_embedding[None], mask)
# decoder_out = model.run_decoder(encoder_out, acoustic_embedding[None])
print("decoder_out", decoder_out.shape)
print("decoder_out.sum", decoder_out.sum(), decoder_out.mean())
yseq = decoder_out[0, :num_tokens].argmax(axis=-1).tolist()
print(yseq, "-->", len(yseq))

tokens = load_tokens()
words = [tokens[i] for i in yseq if i not in (1, 2)]
print(words)
text = "".join(words)
print(text)

if __name__ == "__main__":
    main()

```

74.3 acl

- 动态 Shape 输入（设置 Shape 范围）

https://www.hiascend.com/document/detail/zh/canncommercial/80RC2/developmentguide/appdevg/aclcppdevg/aclcppdevg_000044.html

- 头文件和库文件说明

https://www.hiascend.com/document/detail/zh/CANNCommunityEdition/83RC1alpha003/API/appdevgapi/aclcppdevg_03_1952.html

https://www.hiascend.com/document/detail/zh/canncommercial/82RC1/appdevg/acldevg/aclcppdevg_000000.html

```
ls /usr/local/Ascend/ascend-toolkit/latest
aarch64-linux  arm64-linux  bin          conf          hccl          lib64          opp  ↵
↪platform.ini  python          runtime       toolkit       var
acllib         atc             compiler      fwkacllib     include       mindstudio-toolkit ops  ↵
↪pyACL         remove_latest_softlink.sh test-ops      tools         version.cfg
```

```
ls /usr/local/Ascend/ascend-toolkit/latest/include

acl          aclnn          aclnnop  amct  ascende  experiment  flow_func  ge  ↵
↪hccl        llm_datadist  mstx     proto  securec.h  tiling     utils
acldevppop  aclnn_kernels  aicpu    aoe    exe_graph  external   flow_graph  graph ↵
↪kernel_tiling mspti        parser   register  securectype.h toolchain  version
```

```
ls /usr/local/Ascend/ascend-toolkit/latest/include/acl

acl_base.h  acl_dump.h  acl.h  acl_md1.h  acl_op_compiler.h  acl_op.h  acl_prof.h  ↵
↪acl_rt_allocator.h  acl_rt.h  acl_tdt.h  acl_tdt_queue.h  dvpp  error_codes  media  ↵
↪ops
```

```
ls /usr/local/Ascend/ascend-toolkit/latest/include/acl/ops

acl_cblas.h  acl_dvpp.h  acl_fv.h
```

```
ls /usr/local/Ascend/ascend-toolkit/latest/lib64/

auto_deploy_utils-0.1.0-py3-none-any.whl  libaclnn_ops_infer.so  libbuilt_
↪in_flowfunc.so  libhccl_alg_research.so  libregister.so
auto_tune-0.1.0-py3-none-any.whl          libaclnn_ops_train.so  lib_caffe_
↪parser.so          libhccl_alg.so          libs.so
c                                          libaclnn_rand.so       libcann_kb.
↪so                  libhccl_heterog.so     librt2_registry.a
d2d_bandwidth_test_310b.o                 libacl_op_compiler.so  libcce_
↪aicore.so          libhccl_plf.so         librunttime_kb.so
d2d_bandwidth_test.o                     libacl_retr.so         libcce_
↪aicpudev_online.so  libhccl.so             librunttime.so
d2d_bandwidth_write_test.o               libacl_tde_mpi.so      libcce.so  ↵
↪                  libhcom_executor.so    libslicec.so
dataflow-0.0.1-py3-none-any.whl          libacl_tdt_channel.so  libcce_
↪tools.so           libhcom_gradient_split_tune.so  libte_fusion.so
hbm_power.o                               libacl_tdt_queue.so    ↵
```

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```

↪libcompress.so          libhcom_graph_adaptor.so      libtiling_api.a
hbm_test_0.o              libacl_vi_mpi.so             ↪
↪libcompressweight.so    libhdds_base.so              libtiling.so
hbm_test_1.o              libacl_vo_mpi.so             libc_sec.a↪
↪                          libhost_aicpu_scheduler.so    libtptrt.so
hbm_test_2.o              libacl_vpss_mpi.so           libc_sec.
↪so                      libhybrid_executor.so        libtsdclient.so
hbm_test_3.o              libadcore.a                  libdata_
↪flow_base.so            libindextransform.so         libvm_runtime.so
hbm_test_4.o              libadump_server.a           ↪
↪libdatatransfer.so      libllm_engine.so             libvm.so
hbm_test_5.o              libaicore_utils.so           libdavinci_
↪executor.so             libmmpa.a                    libudf_dump.so
hccl-0.1.0-py3-none-any.whl  libaicpu_engine_common.so    libdgw_
↪client.so               libmmpa.so                   libudf_profiling.so
hccl_aiv_op_ascend910B.o    libamctacl.so                libdgw_
↪client_so.so            libmodel_deployer.so         llm_datadist-0.0.1-py3-none-any.
↪whl
hccl_reduce_op_Ascend310P3.o  libaoe_compiler.so           libdummy_
↪tls.so                  libmsprofiler.so             op_compile_tool-0.1.0-py3-none-
↪any.whl
hccl_reduce_op_ascend910B.o    libaoe_executor.so           libdvpp_
↪cmdlist.so              libmspti.so                   opc_tool-0.1.0-py3-none-any.whl
hccl_reduce_op_ascend910.o    libaoe_opat.so                libdvpp_op_
↪base.so                  libms_tools_ext.so            plugin
infer_310b_model_int8.o      libaoe_plugin.so              liberror_
↪manager.a               libnca.so                     schedule_search-0.1.0-py3-none-
↪any.whl
infer_310b_model.o           libaoe_runner.so              liberror_
↪manager.so              libnnopbase.so                stub
infer_mmad_model_int8.o      libaoe_tuning.so              libexe_
↪graph.so                 libopapi.so                   switch_by_index.o
infer_mmad_model.o           libascendalog.so              libflow_
↪func.so                  libopat.so                     te-0.4.0-py3-none-any.whl
infer_mmad_v200_model_int8.o  libascend_aoe_runner_offline.so  libflow_
↪graph.so                 libop_common.so                train_matmul_model_fp32.o
infer_mmad_v200_model.o      libascendcl.a                 libfmk_
↪onnx_parser.so          libopskernel.so               train_matmul_model_hf32.o
libacl_audio_mpi.so          libascendcl.so                libfmk_
↪parser.so               libopt_feature.so             train_matmul_model_int8.o
libacl_cblas.so             libascendc_runtime.a          libge_
↪common_base.so          libparser_common.so           train_matmul_model.o
libacl_dvpp_mpi.so          libascend_dump.a              libge_

```

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```

↪common.so          libplatform.so          train_mmad_model_int8.o
libacl_dvpp_op.so    libascend_dump.so          libge_
↪compiler.so        libprofapi.so              train_mmad_model.o
libacl_dvpp.so       libascend_ml.so            libge_
↪executor.so        libprofimpl.so            train_mmad_vec_model_int8.o
libacl_hdmi_mpi.so   libascendprofhal.so        libgert.so↪
↪                  libqos_manager.so          train_mmad_vec_model.o
libacl_isp_ae_mpi.so libascend_protobuf.so      libge_
↪runner.so          libra_hdc.so               vector_multiplication_fp32.o
libacl_isp_awb_mpi.so libascend_protobuf.so.3.13.0.0 libgraph_
↪base.so            libra_peer.so              vector_multiplication.o
libacl_isp_mpi.so    libascend_trace.so         libgraph.
↪so                 libra.so
libacl_mfusion_mpi.so libascend_watchdog.so      ↪
↪libgraphtuner_executor.so libreader_writer.so
libaclnn_math.so     libauto_tiling.so          libhccd.so↪
↪                  libregister.a

```

74.4 cann

- 快速安装 CANN

https://www.hiascend.com/document/detail/zh/CANNCommunityEdition/83RC1alpha003/softwareinst/instg/instg_quick.html?Mode=PmIns&OS=Debian&Software=cannToolKit

First, use `npu-smi info` to check the driver version, and then select a version of CANN toolkit that is compatible with the driver version.

```

source ~/py38/bin/activate
./Ascend-cann-toolkit_8.5.0.alpha001_linux-aarch64.run --full

# To uninstall
./Ascend-cann-toolkit_8.5.0.alpha001_linux-aarch64.run --uninstall
cd /usr/local/Ascend
rm -rf ascend-toolkit/

```

Part of the installation logs:

```

=====
= Summary =
=====

Driver:   Installed in /usr/local/Ascend/driver.

```

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```

Toolkit: Ascend-cann-toolkit_8.5.0.alpha001_linux-aarch64 install success,
↳ installed in /usr/local/Ascend.

Please make sure that the environment variables have been configured.
- To take effect for all users, you can add "source /usr/local/Ascend/ascend-
↳ toolkit/set_env.sh" to /etc/profile.
- To take effect for current user, you can exec command below: source /usr/local/
↳ Ascend/ascend-toolkit/set_env.sh or add "source /usr/local/Ascend/ascend-toolkit/
↳ set_env.sh" to ~/.bashrc.

```

To fix the following error:

```

+ atc --model=./model.onnx --framework=5 --output=model --input_format=ND --input_
↳ shape=x:1,-1 --soc_version=Ascend910B
ATC start working now, please wait for a moment.
...
ATC run failed, Please check the detail log, Try 'atc --help' for more information
E40001: Value [/usr] for environment variable [ld_library_path or ldconfig] is
↳ invalid when load python dynamic library.
    Solution: Reset the environment variable by referring to the installation
↳ guide.
    TraceBack (most recent call last):
    [GraphOpt][InitializeInner][InitTbeFunc] Failed to init tbe.
↳ [FUNC:InitializeInner][FILE:tbe_op_store_adapter.cc][LINE:1623]
    [SubGraphOpt][PreCompileOp][InitAdapter] InitializeAdapter adapter [tbe_op_
↳ adapter] failed! Ret [4294967295][FUNC:InitializeAdapter][FILE:op_store_adapter_
↳ manager.cc][LI
NE:85]
    [SubGraphOpt][PreCompileOp][Init] Initialize op store adapter failed,
↳ OpsStoreName[tbe-custom].[FUNC:Initialize][FILE:op_store_adapter_manager.
↳ cc][LINE:126]
    [FusionMgr][Init] Op store adapter manager init failed.
↳ [FUNC:Initialize][FILE:fusion_manager.cc][LINE:124]
    PluginManager InvokeAll failed.[FUNC:Initialize][FILE:ops_kernel_manager.
↳ cc][LINE:96]
    OpsManager initialize failed.[FUNC:InnerInitialize][FILE:gelib.cc][LINE:237]
    GELib::InnerInitialize failed.[FUNC:Initialize][FILE:gelib.cc][LINE:165]

```

Run:

```

# python3 -c "import ctypes; ctypes.CDLL('libpython3.10.so')"

# find /root/miniconda3/ -name "libpython3.10.so"
/root/miniconda3/pkgs/python-3.10.0-hc137634_5/lib/libpython3.10.so

```

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```
/root/miniconda3/envs/py310/lib/libpython3.10.so

# export LD_LIBRARY_PATH=/root/miniconda3/envs/py310/lib:$LD_LIBRARY_PATH
```

74.5 orange pi ai pro

- <https://www.hiascend.com/developer/devboard>

74.6 samples

- <https://gitee.com/ascend/samples/tree/master/inference/ACLHelloWorld>

74.7 ais_bench

Use Python >= 3.9 to install ais_bench!

See https://gitee.com/ascend/tools/tree/7526b3c5ae6e8217e15a61b4a15851de32630e1d/ais-bench_workload/tool/ais_bench

See https://gitee.com/ascend/tools/tree/master/ais-bench_workload/tool/ais_bench

```
source py310/bin/activate
source /usr/local/Ascend/ascend-toolkit/set_env.sh
git clone https://gitee.com/ascend/tools

cd tools/ais-bench_workload/tool/ais_bench

pip3 wheel ./backend/ -v
pip3 wheel ./ -v
pip3 install ./aclruntime-0.0.2-cp310-cp310-linux_aarch64.whl
pip3 install ./ais_bench-0.0.2-py3-none-any.whl
```


74.8 HiDevLab

- 连接环境 https://www.hikunpeng.com/document/detail/zh/techcert/certtest/envapp/kunpeng_hidevlabeg_0002.html

install UniVPN

74.8.1 Install CANN

See <https://www.hiascend.com/cann/download>

```
groupadd HwHiAiUser
useradd -g HwHiAiUser -d /home/HwHiAiUser -m HwHiAiUser -s /bin/bash

sudo apt-get update
sudo apt-get install -y gcc python3 python3-pip linux-headers-$(uname -r)

wget https://ascend-repo.obs.cn-east-2.myhuaweicloud.com/CANN/CANN%208.5.T63/Ascend-
↪cann_8.5.0_linux-aarch64.run

bash ./Ascend-cann_8.5.0_linux-aarch64.run --install
```

```
=====
= Summary =
=====

Driver: Not installed.
CANN: Ascend-cann_8.5.0_linux-aarch64 install success, installed in /usr/local/
↪Ascend.

Please make sure that the environment variables have been configured.
- To take effect for all users, you can add "source /usr/local/Ascend/cann-8.5.0/set_
↪env.sh" to /etc/profile.
- To take effect for current user, you can exec command below: source /usr/local/
↪Ascend/cann-8.5.0/set_env.sh or add "source /usr/local/Ascend/cann-8.5.0/set_env.sh
↪" to ~/.bashrc.
```

```
wget https://ascend-repo.obs.cn-east-2.myhuaweicloud.com/CANN/CANN%208.5.T63/Ascend-
↪cann-310p-ops_8.5.0_linux-aarch64.run
bash ./Ascend-cann-310p-ops_8.5.0_linux-aarch64.run --install
```

```
=====
= Summary =
```

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```
=====
```

```
Driver:    Not installed.
```

```
ops_310p:  Ascend-cann-310p-ops_8.5.0_linux-aarch64 install success, installed in /  
↳usr/local/Ascend.
```

```
Please make sure that the environment variables have been configured.
```

- To take effect **for** all users, you can add "source /usr/local/Ascend/cann-8.5.0/set_
↳env.sh" to /etc/profile.
- To take effect **for** current user, you can exec command below: source /usr/local/
↳Ascend/cann-8.5.0/set_env.sh **or** add "source /usr/local/Ascend/cann-8.5.0/set_env.sh
↳" to ~/.bashrc.

```
source /usr/local/Ascend/ascend-toolkit/set_env.sh
```

75.1 basics

gmail, B1!

- https://scalable.uni-jena.de/opt/eml/chapters/assignment_mobile_inference.html

- QNN Execution Provider

<https://onnxruntime.ai/docs/execution-providers/QNN-ExecutionProvider.html>

It mentions Qualcomm AI Engine Direct SDK (QNN SDK)

- PR: Refine ggml-qnn backend(QNN, Qualcomm Neural Network, aka Qualcomm AI Engine Direct) for latest ggml, whisper.cpp, llama.cpp #246

<https://github.com/kantv-ai/kantv/issues/246>

- PoC: Add Qualcomm mobile SoC native backend for GGML #121

<https://github.com/kantv-ai/kantv/issues/121>

- Qualcomm AI Engine Direct SDK

<https://www.qualcomm.com/developer/software/qualcomm-ai-engine-direct-sdk>

```
wget https://apigwx-aws.qualcomm.com/qsc/public/v1/api/download/software/sdks/  
↪Qualcomm_AI_Runtime_Community/All/2.39.0.250926/v2.39.0.250926.zip
```

- qnn doc

https://docs.qualcomm.com/bundle/publicresource/topics/80-63442-10/general_introduction.html?product=1601111740009302

- Linux Setup

https://docs.qualcomm.com/bundle/publicresource/topics/80-63442-10/linux_setup.html?product=1601111740009302

- Tutorial: Converting and executing a CNN model with QNN

https://docs.qualcomm.com/bundle/publicresource/topics/80-63442-10/tutorial_convert_execute_cnn_model.html?product=1601111740009302

- Install AI Engine Direct

<https://docs.qualcomm.com/bundle/publicresource/topics/80-70015-15B/qnn-download.html>

75.2 Install qnn

Go to https://softwarecenter.qualcomm.com/catalog/item/Qualcomm_AI_Runtime_Community

```
wget https://apigwx-aws.qualcomm.com/qsc/public/v1/api/download/software/sdks/  
↳Qualcomm_AI_Runtime_Community/All/2.32.0.250228/v2.32.0.250228.zip  
  
wget https://softwarecenter.qualcomm.com/api/download/software/qualcomm_neural_  
↳processing_sdk/v2.26.2.240911.zip  
  
wget https://apigwx-aws.qualcomm.com/qsc/public/v1/api/download/software/sdks/  
↳Qualcomm_AI_Runtime_Community/All/2.39.0.250926/v2.39.0.250926.zip  
unzip v2.39.0.250926.zip
```

```
Archive:  ./v2.39.0.250926.zip  
  creating: qairt/  
  creating: qairt/2.39.0.250926/  
  creating: qairt/2.39.0.250926/benchmarks/  
  creating: qairt/2.39.0.250926/benchmarks/QNN/
```

```
cd qairt/2.39.0.250926/  
cd bin  
source ./envsetup.sh
```

```
[INFO] QAIRT_SDK_ROOT=/star-fj/fangjun/download/qnn/qairt/2.39.0.250926  
[WARN] QNN_SDK_ROOT/SNPE_ROOT set to QAIRT_SDK_ROOT for backwards compatibility and  
↳will be deprecated in a future release.  
[INFO] QAIRT SDK environment setup complete
```

```
$ echo $QAIRT_SDK_ROOT  
/star-fj/fangjun/download/qnn/qairt/2.39.0.250926  
  
$ echo $QNN_SDK_ROOT  
/star-fj/fangjun/download/qnn/qairt/2.39.0.250926
```

```
sudo ./check-linux-dependency.sh
```

For android:

```
sdkmanager --list | grep ndk
sdkmanager "ndk;26.2.11394342"
```

```
[=====] 26% Downloading android-ndk-r26c-linux
```

```
libc++.so.1: cannot open shared object file: No such file or directory

sudo apt install libc++-dev libc++abi-dev
# Installed inside /usr

ImportError: libunwind.so.1: cannot open shared object file: No such file or directory

sudo apt install libunwind8 libunwind-dev
```

75.3 SenseVoice

```
qnn-onnx-converter \
  --input_network ./model.onnx \
  --output_path ./qnn_model_dynamic \
  --out_node logits \
  --input_dtype x float32 \
  --input_dtype prompt int32 \
  --input_dim x 1,300,560 \
  --input_dim prompt 4 \
  --float_bitwidth 16 \
  --use_per_channel_quantization

qnn-onnx-converter \
  --input_network ./model.onnx \
  --output_path ./qnn_model_dynamic \
  --out_node logits \
  --input_dtype x float32 \
  --input_dtype prompt int32 \
  --input_layout x NTF \
  --define_symbol N 1 \
  --define_symbol T 93 \
  --float_bitwidth 16 \
```

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```
--float_fallback \
--use_per_row_quantization \
--input_dim x 1,93,560 \
--input_dim prompt 4
```

```
2025-10-29 05:45:28,229 - 240 - WARNING - Symbolic shape inference Failed. Exception:
↳ Incomplete symbolic shape inference. Running normal shape inference.
2025-10-29 05:45:44,439 - 240 - WARNING - Shape not found for tensor: /ctc_lo/MatMul_
↳ output_0
WARNING: The argument `input_shapes` is deprecated. Please use
`overwrite_input_shapes` and/or `test_input_shapes` instead. An error will be
raised in the future.
WARNING: the simplification stopped because of timeout. Please set environment
↳ variable `ONNXSIM_FIXED_POINT_ITERS` to a number higher than 50 if you want further
↳ simplification.
2025-10-29 05:56:30,769 - 235 - INFO - Simplified model validation is successful
2025-10-29 05:58:56,511 - 235 - INFO - Skipping quantization, no input_list provided
2025-10-29 05:58:56,511 - 235 - INFO - Saving QNN Model...
2025-10-29 05:59:02,856 - 235 - INFO - Model CPP saved at: ./qnn_model_dynamic
2025-10-29 05:59:02,857 - 235 - INFO - Model BIN saved at: /content/qnn_model_dynamic.
↳ bin
2025-10-29 05:59:02,978 - 235 - INFO - Conversion complete!
```

```
file qnn_model_dynamic

qnn_model_dynamic: C source, ASCII text, with very long lines (2378)

file qnn_model_dynamic.bin

qnn_model_dynamic.bin: POSIX tar archive
```

```
ls -lh qnn_model_dynamic*
```

```
-rw-r--r-- 1 root root 9.2M Oct 29 05:59 qnn_model_dynamic
-rw-r--r-- 1 root root 444M Oct 29 05:59 qnn_model_dynamic.bin
-rw-r--r-- 1 root root 4.4M Oct 29 05:59 qnn_model_dynamic_net.json
```

```
mv qnn_model_dynamic qnn_model_dynamic.cpp
```

```
export QNN_TARGET_ARCH="x86_64-linux-clang"

python3 "${QNN_SDK_ROOT}/bin/x86_64-linux-clang/qnn-model-lib-generator" \
```

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```
-c "qnn_model_dynamic.cpp" \
-b "qnn_model_dynamic.bin" \
-o model_libs \
-t ${QNN_TARGET_ARCH}
```

```
ls -lh model_libs/x86_64-linux-clang/

-rwxr-xr-x 1 root root 448M Oct 29 06:18 libqnn_model_dynamic.so
```

If we don't specify the `-t`, it will show:

```
2025-10-29 10:41:45,757 - INFO - qnn-model-lib-generator: Target: x86_64-linux-
↳clang      Library: /content/model_libs/x86_64-linux-clang/libqnn_model_dynamic.so
2025-10-29 10:41:45,757 - INFO - qnn-model-lib-generator: Target: aarch64-android
↳Library: /content/model_libs/aarch64-android/libqnn_model_dynamic.so
```

75.3.1 First working version

```
wget https://hf-mirror.com/csukuangfj/qnn-toolkit/resolve/main/tmp/model-op17-static-
↳93.onnx

if [ ! -f input0-en.raw ]; then
wget https://modelscope.cn/models/csukuangfj/2025-10-14/resolve/master/2025-10-29-qnn/
↳input0-en.raw
wget https://modelscope.cn/models/csukuangfj/2025-10-14/resolve/master/2025-10-29-qnn/
↳input1-en.raw
fi

if [ ! -f input0.raw ]; then
wget https://modelscope.cn/models/csukuangfj/2025-10-14/resolve/master/2025-10-29-qnn/
↳input0.raw
wget https://modelscope.cn/models/csukuangfj/2025-10-14/resolve/master/2025-10-29-qnn/
↳input1.raw
fi

echo "/content/input0-en.raw /content/input1-en.raw" > input_list.txt

echo "./input0-en.raw ./input1-en.raw" > input_list.txt

#echo "/content/input0.raw /content/input1.raw" >> input_list.txt

cat input_list.txt
```

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```

source py310/bin/activate

pushd qairt/2.33.0.250327/bin
source envsetup.sh
popd

qnn-onnx-converter \
  --input_network ./model-op17-static-93.onnx \
  --output_path ./model-op17-static-93-quantized \
  --out_node logits \
  --input_list ./input_list.txt \
  --use_native_input_files \
  --input_dtype x float32 \
  --input_dtype prompt int32 \
  --act_bitwidth 16 \
  --bias_bitwidth 32 \
  --input_layout x NTF

mv model-op17-static-93-quantized model-op17-static-93-quantized.cpp

python3 "${QNN_SDK_ROOT}/bin/x86_64-linux-clang/qnn-model-lib-generator" \
  -c "model-op17-static-93-quantized.cpp" \
  -b "model-op17-static-93-quantized.bin" \
  -o model_libs_quantized \
  -t aarch64-android

```

```

-rwxr-xr-x 1 root root 228M Nov  4 04:29 model_libs_quantized/aarch64-android/
↳ libmodel-op17-static-93-quantized.so

```

In `model-op17-static-93-quantized_net.json`, we have:

```

"x": {
  "id": 1,
  "type": 0,
  "dataFormat": 0,
  "data_type": 1046,
  "unquantized_data_type": 562,
  "permute_order_to_src": [
    0,
    1,
    2
  ],

```

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```

    "quant_params": {
        "definition": 1,
        "encoding": 0,
        "is_overridden": false,
        "scale_offset": {
            "bitwidth": 16,
            "minimum": -1.078358769416809,
            "maximum": 24.36089515686035,
            "scale": 0.00038817813037894666,
            "offset": -2778,
            "is_symmetric": false,
            "is_fixed_point": true
        }
    },
    "dims": [
        1,
        93,
        560
    ],
    "is_dynamic_dims": [],
    "is_quantizable": true,
    "is_updateable": false
}

"prompt": {
    "id": 2,
    "type": 0,
    "dataFormat": 0,
    "data_type": 50,
    "unquantized_data_type": 50,
    "permute_order_to_src": [
        0,
        1
    ],
    "quant_params": {
        "definition": 2147483647,
        "encoding": 2147483647,
        "is_overridden": false,
        "scale_offset": {
            "bitwidth": 0,
            "minimum": 0.0,
            "maximum": 0.0,
            "scale": 0.0,

```

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```

        "offset": 0,
        "is_symmetric": false,
        "is_fixed_point": true
    },
    },
    "dims": [
        1,
        4
    ],
    "is_dynamic_dims": [],
    "is_quantizable": false,
    "is_updateable": false
},
"logits": {
    "id": 4700,
    "type": 1,
    "dataFormat": 0,
    "data_type": 1046,
    "unquantized_data_type": 562,
    "permute_order_to_src": [
        0,
        1,
        2
    ],
    "quant_params": {
        "definition": 1,
        "encoding": 0,
        "is_overridden": false,
        "scale_offset": {
            "bitwidth": 16,
            "minimum": -33.50352096557617,
            "maximum": 38.97942352294922,
            "scale": 0.0011060187825933099,
            "offset": -30292,
            "is_symmetric": false,
            "is_fixed_point": true
        }
    },
    "dims": [
        1,
        97,
        25055
    ]
}

```

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```

    ],
    "is_dynamic_dims": [],
    "is_quantizable": true,
    "is_updateable": false
  },

```

Output from my qnn-demo:

```

input tensor 0, version 2
  id: 1
  name: x
  type: QNN_TENSOR_TYPE_APP_WRITE
  data format: 0
  data type: QNN_DATATYPE_UFIXED_POINT_16
  quantize info:
    encodingDefinition: 0x1
    quantizationEncoding: QNN_QUANTIZATION_ENCODING_SCALE_OFFSET
    scale: 0.000388178
    offset: -2778
  rank: 3
  dimensions: 1, 93, 560,
input tensor 1, version 2
  id: 2
  name: prompt
  type: QNN_TENSOR_TYPE_APP_WRITE
  data format: 0
  data type: QNN_DATATYPE_INT_32
  quantize info:
    encodingDefinition: 0x7fffffff
    quantizationEncoding: QNN_QUANTIZATION_ENCODING_UNDEFINED
  rank: 2
  dimensions: 1, 4,
output tensor 2940, version 2
  id: 4840
  name: logits
  type: QNN_TENSOR_TYPE_APP_READ
  data format: 0
  data type: QNN_DATATYPE_UFIXED_POINT_16
  quantize info:
    encodingDefinition: 0x1
    quantizationEncoding: QNN_QUANTIZATION_ENCODING_SCALE_OFFSET
    scale: 0.00110602
    offset: -30292
  rank: 3

```

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```
dimensions: 1, 97, 25055,
```

To generate test data for x:

```
features = compute_feat(
    samples=samples,
    sample_rate=sample_rate,
)
print("features.shape", features.shape)
if features.shape[0] > args.num_frames:
    features = features[: args.num_frames]
elif features.shape[0] < args.num_frames:
    pad_width = ((0, args.num_frames - features.shape[0]), (0, 0))
    padding = np.zeros(())
    features = np.pad(features, pad_width, mode="constant", constant_values=0)

scale = 0.000388178
offset = -2778
features = np.round(features / scale - offset).clip(0, 65535).astype(np.uint16)

features.tofile("input0.raw")
```

To generate test data for prompt:

```
language_auto = 0
language_zh = 3
language_en = 4
language_yue = 7
language_ya = 11
language_ko = 12
language_nospeech = 13

language = language_auto

with_itn = 14
without_itn = 15

text_norm = with_itn

prompt = np.array([language, 1, 2, text_norm], dtype=np.int32)
prompt.tofile("input1.raw")
```

Test with qnn-net-run:

```
./qnn-net-run --log_level verbose --model ./libmodel-op17-static-93-quantized.so --
↪backend ./libQnnHtp.so --input_list ./input_list.txt --output_dir ./out --use_
↪native_input_files=true
```

```
pandora:/data/local/tmp $ ls -l out/Result_0/logits.raw
-rw-rw-rw- 1 shell shell 9721340 2025-11-04 12:52 out/Result_0/logits.raw
pandora:/data/local/tmp $ bc
>>> 9721340/25055/4
97
```

```
import numpy as np

def load_tokens(filename):
    ans = dict()
    i = 0
    with open(filename, encoding="utf-8") as f:
        for line in f:
            ans[i] = line.strip().split()[0]
            i += 1
    return ans

logits = np.fromfile("./logits.raw", dtype=np.float32).reshape((-1, 25055))

idx = logits.argmax(axis=-1)
print("idx", idx)
print(len(idx))
prev = -1
ids = []
for i in idx:
    if i != prev:
        ids.append(i)
    prev = i
ids = [i for i in ids if i != 0]
print(ids)

tokens = load_tokens("./tokens.txt")
text = "".join([tokens[i] for i in ids])

text = text.replace("_", " ")
print(text)
```

75.4 qnn-onnx-converter

```
qnn-onnx-converter --help
```

```
usage: qnn-onnx-converter [--out_node OUT_NAMES] [--input_type INPUT_NAME INPUT_TYPE]
                           [--input_dtype INPUT_NAME INPUT_DTYPE] [--input_encoding ..
↳.]]
                           [--input_layout INPUT_NAME INPUT_LAYOUT] [--custom_io_
↳CUSTOM_IO]
                           [--preserve_io [PRESERVE_IO ...]]
                           [--dump_qairt_io_config_yaml DUMP_QAIRT_IO_CONFIG_YAML]
                           [--enable_framework_trace] [--dry_run [DRY_RUN]] [-d INPUT_
↳NAME INPUT_DIM]
                           [-n] [-b BATCH] [-s SYMBOL_NAME VALUE]
                           [--dump_custom_io_config_template DUMP_CUSTOM_IO_CONFIG_
↳TEMPLATE]
                           [--quantization_overrides QUANTIZATION_OVERRIDES] [--keep_
↳quant_nodes]
                           [--disable_batchnorm_folding] [--keep_disconnected_nodes]
                           [--apply_masked_softmax {compressed,uncompressed}]
                           [--packed_masked_softmax_inputs PACKED_MASKED_SOFTMAX_
↳INPUTS [PACKED_MASKED_SOFTMAX_INPUTS ...]]
                           [--packed_max_seq PACKED_MAX_SEQ] [--input_list INPUT_LIST]
                           [--param_quantizer PARAM_QUANTIZER] [--act_quantizer ACT_
↳QUANTIZER]
                           [--algorithms ALGORITHMS [ALGORITHMS ...]] [--bias_bitwidth_
↳BIAS_BITWIDTH]
                           [--bias_bw BIAS_BW] [--act_bitwidth ACT_BITWIDTH] [--act_bw_
↳ACT_BW]
                           [--weights_bitwidth WEIGHTS_BITWIDTH] [--weight_bw WEIGHT_
↳BW]
                           [--ignore_encodings] [--use_per_channel_quantization]
                           [--use_per_row_quantization] [--float_fallback] [--use_
↳native_input_files]
                           [--use_native_dtype] [--use_native_output_files]
                           [--disable_relu_squashing]
                           [--restrict_quantization_steps ENCODING_MIN, ENCODING_MAX]
                           [--pack_4_bit_weights]
                           [--act_quantizer_calibration ACT_QUANTIZER_CALIBRATION]
                           [--param_quantizer_calibration PARAM_QUANTIZER_CALIBRATION]
                           [--act_quantizer_schema ACT_QUANTIZER_SCHEMA]
                           [--param_quantizer_schema PARAM_QUANTIZER_SCHEMA]
                           [--percentile_calibration_value PERCENTILE_CALIBRATION_
```

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```

↪VALUE]
                                [--dump_qairt_quantizer_command DUMP_QAIRT_QUANTIZER_
↪COMMAND]
                                --input_network INPUT_NETWORK [--debug [DEBUG]] [-o OUTPUT_
↪PATH]
                                [--copyright_file COPYRIGHT_FILE] [--float_bitwidth FLOAT_
↪BITWIDTH]
                                [--float_bw FLOAT_BW] [--float_bias_bitwidth FLOAT_BIAS_
↪BITWIDTH]
                                [--float_bias_bw FLOAT_BIAS_BW] [--overwrite_model_prefix]
                                [--exclude_named_tensors] [--model_version MODEL_VERSION]
                                [--op_package_lib OP_PACKAGE_LIB]
                                [--converter_op_package_lib CONVERTER_OP_PACKAGE_LIB]
                                [-p PACKAGE_NAME | --op_package_config CUSTOM_OP_CONFIG_
↪PATHS [CUSTOM_OP_CONFIG_PATHS ...]]
                                [-h] [--arch_checker] [--validate_models]

```

Script to convert ONNX model into QNN

required arguments:

```

--input_network INPUT_NETWORK, -i INPUT_NETWORK
                                Path to the source framework model.

```

optional arguments:

```

--out_node OUT_NAMES, --out_name OUT_NAMES
                                Name of the graph's output Tensor Names. Multiple output_
↪names should be
                                provided separately like:
                                --out_name out_1 --out_name out_2
--input_type INPUT_NAME INPUT_TYPE, -t INPUT_NAME INPUT_TYPE
                                Type of data expected by each input op/layer. Type for each_
↪input is
                                |default| if not specified. For example: "data" image.Note_
↪that the quotes
                                should always be included in order to handle special_
↪characters, spaces, etc.
                                For multiple inputs specify multiple --input_type on the_
↪command line.
                                Eg:
                                --input_type "data1" image --input_type "data2" opaque
                                These options get used by DSP runtime and following_
↪descriptions state how
                                input will be handled for each option.

```

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```

Image:
Input is float between 0-255 and the input's mean is 0.0f and
↳the input's
max is 255.0f. We will cast the float to uint8ts and pass the
↳uint8ts to the
DSP.
Default:
Pass the input as floats to the dsp directly and the DSP will
↳quantize it.
Opaque:
Assumes input is float because the consumer layer(i.e next
↳layer) requires
it as float, therefore it won't be quantized.
Choices supported:
    image
    default
    opaque
--input_dtype INPUT_NAME INPUT_DTYPE
The names and datatype of the network input layers specified
↳in the format
[input_name datatype], for example:
    'data' 'float32'
Default is float32 if not specified
Note that the quotes should always be included in order to
↳handlespecial
characters, spaces, etc.
For multiple inputs specify multiple --input_dtype on the
↳command line like:
    --input_dtype 'data1' 'float32' --input_dtype 'data2'
↳'float32'
--input_encoding ..., -e ...]
Usage:    --input_encoding "INPUT_NAME" INPUT_ENCODING_IN
[INPUT_ENCODING_OUT]
Input encoding of the network inputs. Default is bgr.
e.g.
    --input_encoding "data" rgba
Quotes must wrap the input node name to handle special
↳characters,
spaces, etc. To specify encodings for multiple inputs, invoke
--input_encoding for each one.
e.g.
    --input_encoding "data1" rgba --input_encoding "data2"
↳other

```

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Optionally, an output encoding may be specified **for** an input_

↳node by

providing a second encoding. The default output encoding **is**_

↳bgr.

e.g.

```
--input_encoding "data3" rgba rgb
```

Input encoding types:

- image color encodings: bgr,rgb, nv21, nv12, ...
- time_series: **for** inputs of rnn models;
- other: **not** available above **or is** unknown.

Supported encodings:

- bgr
- rgb
- rgba
- argb32
- nv21
- nv12
- time_series
- other

```
--input_layout INPUT_NAME INPUT_LAYOUT, -l INPUT_NAME INPUT_LAYOUT
```

↳the default

Layout of each input tensor. If **not** specified, it will use_

↳encoding.

based on the Source Framework, shape of input **and** input_

Accepted values are-

```
NCDHW, NDHWC, NCHW, NHWC, NFC, NCF, NTF, TNF, NF, NC, F,_
```

↳NONTRIVIAL

N = Batch, C = Channels, D = Depth, H = Height, W = Width, F_

↳= Feature, T =

Time

NDHWC/NCDHW used **for** 5d inputs

NHWC/NCHW used **for** 4d image-like inputs

NFC/NCF used **for** inputs to Conv1D **or** other 1D ops

NTF/TNF used **for** inputs **with** time steps like the ones used_

↳**for** LSTM op

NF used **for** 2D inputs, like the inputs to Dense/

↳FullyConnected layers

NC used **for** 2D inputs **with 1 for** batch **and** other **for** Channels_

↳(rarely used)

F used **for** 1D inputs, e.g. Bias tensor

NONTRIVIAL **for** everything elseFor multiple inputs specify_

↳multiple

--input_layout on the command line.

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```

Eg:
--input_layout "data1" NCHW --input_layout "data2" NCHW
--custom_io CUSTOM_IO
    Use this option to specify a yaml file for custom IO.
--preserve_io [PRESERVE_IO ...]
    Use this option to preserve IO layout and datatype. The
↳different ways of
    using this option are as follows:
        --preserve_io layout <space separated list of names of
↳inputs and
        outputs of the graph>
        --preserve_io datatype <space separated list of names of
↳inputs and
        outputs of the graph>
    In this case, user should also specify the string - layout or
↳datatype in
        the command to indicate that converter needs to
        preserve the layout or datatype. e.g.
        --preserve_io layout input1 input2 output1
        --preserve_io datatype input1 input2 output1
    Optionally, the user may choose to preserve the layout and/or
↳datatype for
        all the inputs and outputs of the graph.
    This can be done in the following two ways:
        --preserve_io layout
        --preserve_io datatype
    Additionally, the user may choose to preserve both layout and
↳datatypes for
        all IO tensors by just passing the option as follows:
        --preserve_io
    Note: Only one of the above usages are allowed at a time.
    Note: --custom_io gets higher precedence than --preserve_io.
--dump_qairt_io_config_yaml DUMP_QAIRT_IO_CONFIG_YAML
    Use this option to dump a yaml file which contains the
↳equivalent I/O
        configurations of QAIRT Converter along with the QAIRT
↳Converter Command and
        can be passed to QAIRT Converter using the option --io_config.
--enable_framework_trace
    Use this option to enable converter to trace the op/tensor
↳change
        information.
--dry_run [DRY_RUN]    Evaluates the model without actually converting any ops, and

```

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↳ returns unsupported ops/attributes **as** well **as** unused inputs **and/or**

↳ outputs **if** any. Leave empty **or** specify "info" to see dry run **as** a table, **or**

↳ specify "debug" to show more detailed messages only"

-d INPUT_NAME INPUT_DIM, --input_dim INPUT_NAME INPUT_DIM
The name **and** dimension of **all** the **input** buffers to the

↳ network specified **in** the **format** [input_name comma-separated-dimensions],
for example: 'data' 1,224,224,3.
Note that the quotes should always be included **in** order to

↳ handle special characters, spaces, etc.
NOTE: This feature works only **with** Onnx 1.6.0 **and** above

-n, --no_simplification
Do **not** attempt to simplify the model automatically. This may

↳ prevent some models **from properly** converting
when sequences of unsupported static operations are present.

-b BATCH, --batch BATCH
The batch dimension override. This will take the first

↳ dimension of **all** inputs **and** treat it **as** a batch dim, overriding it **with** the

↳ value provided here. For example:
--batch 6
will result **in** a shape change **from** [1,3,224,224] to [6,3,224,

↳ 224].
If there are inputs without batch dim this should **not** be used

↳ **and** each input should be overridden independently using -d option **for** input

↳ dimension overrides.

-s SYMBOL_NAME VALUE, --define_symbol SYMBOL_NAME VALUE
This option allows overriding specific **input** dimension

↳ symbols. For instance you might see **input** shapes specified **with** variables such **as** :
data: [1,3,height,width]
To override these simply **pass** the option **as**:
--define_symbol height 224 --define_symbol width 448
which results **in** dimensions that look like:
data: [1,3,224,448]

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```

--dump_custom_io_config_template DUMP_CUSTOM_IO_CONFIG_TEMPLATE
    Dumps the yaml template for Custom I/O configuration. This
↪file canbe edited
    as per the custom requirements and passed using the option --
↪custom_ioUse
    this option to specify a yaml file to which the custom IO
↪config template is
    dumped.
--disable_batchnorm_folding
--keep_disconnected_nodes
    Disable Optimization that removes Ops not connected to the
↪main graph.
    This optimization uses output names provided over commandline
↪OR
    inputs/outputs extracted from the Source model to determine
↪the main graph
--debug [DEBUG]    Run the converter in debug mode.
-o OUTPUT_PATH, --output_path OUTPUT_PATH
    Path where the converted Output model should be saved.If not
↪specified, the
    converter model will be written to a file with same name as
↪the input model
--copyright_file COPYRIGHT_FILE
    Path to copyright file. If provided, the content of the file
↪will be added
    to the output model.
--float_bitwidth FLOAT_BITWIDTH
    Use the --float_bitwidth option to convert the graph to the
↪specified float
    bitwidth, either 32 (default) or 16.
--float_bw FLOAT_BW    Note: --float_bw is deprecated, use --float_bitwidth.
--float_bias_bitwidth FLOAT_BIAS_BITWIDTH
    Use the --float_bias_bitwidth option to select the bitwidth
↪to use for float
    bias tensor
--float_bias_bw FLOAT_BIAS_BW
    Note: --float_bias_bw is deprecated, use --float_bias_
↪bitwidth.
--overwrite_model_prefix
    If option passed, model generator will use the output path
↪name to use as
    model prefix to name functions in <qnn_model_name>.cpp
↪(Useful for running

```

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```

multiple models at once) eg: modelName_composeGraphs. Default.
↳ is to use
        generic "QnnModel_".
--exclude_named_tensors
        Remove using source framework tensorNames; instead use a
↳ counter for naming
        tensors. Note: This can potentially help to reduce the final
↳ model library
        that will be generated(Recommended for deploying model).
↳ Default is False.
--model_version MODEL_VERSION
        User-defined ASCII string to identify the model, only first
↳ 64 bytes will be
        stored
-h, --help
        show this help message and exit
--validate_models
        Validate the original onnx model against optimized onnx model.
        Constant inputs with all value 1s will be generated and will
↳ be used
        by both models and their outputs are checked against each
↳ other.
        The {'option_strings': ['--validate_models'], 'dest':
↳ 'validate_models',
        'nargs': 0, 'const': True, 'default': False, 'type': None,
↳ 'choices': None,
        'required': False, 'help': 'Validate the original onnx model
↳ against
        optimized onnx model.\nConstant inputs with all value 1s will
↳ be generated
        and will be used \nby both models and their outputs are
↳ checked against each
        other.\nThe % average error and 90th percentile of output
↳ differences will
        be calculated for this.\nNote: Usage of this flag will incur
↳ extra time due
        to inference of the models.', 'metavar': None, 'container':
        <argparse._ArgumentGroup object at 0x135d9d9f6500>, 'prog':
↳ 'qnn-onnx-
        converter'}verage error and 90th percentile of output
↳ differences will be
        calculated for this.
        Note: Usage of this flag will incur extra time due to
↳ inference of the
        models.

```

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Custom Op Package Options:

```
--op_package_lib OP_PACKAGE_LIB, -opl OP_PACKAGE_LIB
```

Use this argument to **pass** an op package library **for** quantization. Must be **in** the form <op_package_lib_path:interfaceProviderName> **and** be separated by a comma **for** multiple package libs

```
--converter_op_package_lib CONVERTER_OP_PACKAGE_LIB, -cpl CONVERTER_OP_PACKAGE_LIB
```

Absolute path to converter op package library compiled by the OpPackage generator. Must be separated by a comma **for** multiple package libraries.

Note: Order of converter op package libraries must follow the order of xmls.

Ex1: --converter_op_package_lib absolute_path_to/libExample.so
Ex2: -cpl absolute_path_to/libExample1.so,absolute_path_to/libExample2.so

```
-p PACKAGE_NAME, --package_name PACKAGE_NAME
```

A **global** package name to be used **for** each node **in** the Model. **cpp** file.

Defaults to Qnn header defined package name

```
--op_package_config CUSTOM_OP_CONFIG_PATHS [CUSTOM_OP_CONFIG_PATHS ...], -opc CUSTOM_OP_CONFIG_PATHS [CUSTOM_OP_CONFIG_PATHS ...]
```

Path to a Qnn Op Package XML configuration file that contains user defined custom operations.

Quantizer Options:

```
--quantization_overrides QUANTIZATION_OVERRIDES
```

Use this option to specify a json file **with** parameters to use **for** quantization. These will override **any** quantization data **carried from** **conversion** (eg TF fake quantization) **or** calculated during the **normal** quantization process. Format defined **as** per AIMET **specification**.

```
--keep_quant_nodes
```

Use this option to keep activation quantization nodes **in** the **graph** rather than stripping them.

```
--input_list INPUT_LIST
```

Path to a file specifying the **input** data. This file should be

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```

↪ a plain text
↪ Each path is
↪ the "raw" format,
↪ preprocessing.
↪ inputs to the
↪ specified for
↪ when this is
    --param_quantizer PARAM_QUANTIZER
↪ use. Must be
↪ bitwidth (default).
↪ with long tails
↪ values about zero.
↪ always 0. Note:
    --act_quantizer ACT_QUANTIZER
↪ use. Must be
↪ bitwidth (default).
↪ with long tails
↪ values about zero.

```

file, containing one **or** more absolute file paths per line.

expected to point to a binary file containing one **input in**

ready to be consumed by the quantizer without **any** further

Multiple files per line separated by spaces indicate multiple

network. See documentation **for** more details. Must be

quantization. All subsequent quantization options are ignored

not provided.

Optional parameter to indicate the weight/bias quantizer to

followed by one of the following options:

"tf": Uses the real **min/max** of the data **and** specified

"enhanced": Uses an algorithm useful **for** quantizing models

present **in** the weight distribution.

"adjusted": Note: "adjusted" mode **is** deprecated.

"symmetric": Ensures **min and max** have the same absolute

Data will be stored **as int#_t data such that the offset is**

Legacy option --param_quantizer will be deprecated, use --param_quantizer_calibration instead

Optional parameter to indicate the activation quantizer to

followed by one of the following options:

"tf": Uses the real **min/max** of the data **and** specified

"enhanced": Uses an algorithm useful **for** quantizing models

present **in** the weight distribution.

"adjusted": Note: "adjusted" mode **is** deprecated.

"symmetric": Ensures **min and max** have the same absolute

Data will be stored **as int#_t data such that the offset is**

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```

↪always 0.Note:
    Legacy option --act_quantizer will be deprecated, use
    --act_quantizer_calibration instead
    --algorithms ALGORITHMS [ALGORITHMS ...]
        Use this option to enable new optimization algorithms. Usage
↪is:
    --algorithms <algo_name1> ... The available optimization
↪algorithms are:
    "cle" - Cross layer equalization includes a number of methods
↪for equalizing
    weights and biases across layers in order to rectify
↪imbalances that cause
    quantization errors.
    --bias_bitwidth BIAS_BITWIDTH
        Use the --bias_bitwidth option to select the bitwidth to use
↪when quantizing
    the biases, either 8 (default) or 32.
    --bias_bw BIAS_BW    Note: --bias_bw is deprecated, use --bias_bitwidth.
    --act_bitwidth ACT_BITWIDTH
        Use the --act_bitwidth option to select the bitwidth to use
↪when quantizing
    the activations, either 8 (default) or 16.
    --act_bw ACT_BW    Note: --act_bw is deprecated, use --act_bitwidth.
    --weights_bitwidth WEIGHTS_BITWIDTH
        Use the --weights_bitwidth option to select the bitwidth to
↪use when
    quantizing the weights, either 4 or 8 (default).
    --weight_bw WEIGHT_BW
        Note: --weight_bw is deprecated, use --weights_bitwidth.
    --ignore_encodings
        Use only quantizer generated encodings, ignoring any user or
↪model provided
    encodings.
        Note: Cannot use --ignore_encodings with --quantization_
↪overrides
    --use_per_channel_quantization
        Use this option to enable per-channel quantization for
↪convolution-based op
    weights.
        Note: This will replace built-in model QAT encodings when
↪used for a given
    weight.
    --use_per_row_quantization
        Use this option to enable rowwise quantization of Matmul and

```

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```

↪FullyConnected
ops.
--float_fallback Use this option to enable fallback to floating point (FP)
↪instead of fixed point.
This option can be paired with --float_bitwidth to indicate
↪the bitwidth for FP (by default 32).
If this option is enabled, then input list must not be
↪provided and --ignore_encodings must not be provided.
The external quantization encodings (encoding file/FakeQuant
↪encodings) might be missing quantization parameters for some interim
↪tensors. First it will try to fill the gaps by propagating across math-
↪invariant functions. If the quantization params are still missing,
then it will apply fallback to nodes to floating point.
--use_native_input_files Boolean flag to indicate how to read input files:
1. float (default): reads inputs as floats and quantizes if
↪necessary based on quantization parameters in the model.
2. native: reads inputs assuming the data type to be
↪native to the model. For ex., uint8_t.
Note: This option is deprecated, use --use_native_input_files
↪option in future.
Boolean flag to indicate how to read input files:
1. float (default): reads inputs as floats and quantizes if
↪necessary based on quantization parameters in the model.
2. native: reads inputs assuming the data type to be
↪native to the model. For ex., uint8_t.
--use_native_output_files Use this option to indicate the data type of the output files
1. float (default): output the file as floats.
2. native: outputs the file that is native to the
↪model. For ex., uint8_t.

```

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```

--disable_relu_squashing
    Disables squashing of Relu against Convolution based ops for
↳ quantized
        models
--restrict_quantization_steps ENCODING_MIN, ENCODING_MAX
    Specifies the number of steps to use for computing
↳ quantization encodings
    such that scale = (max - min) / number of quantization steps.
    The option should be passed as a space separated pair of
↳ hexadecimal string
    minimum and maximum values i.e. --restrict_quantization_steps
↳ "MIN MAX".
    Please note that this is a hexadecimal string literal and
↳ not a signed
    integer, to supply a negative value an explicit minus sign is
↳ required.
    E.g. --restrict_quantization_steps "-0x80 0x7F" indicates an
↳ example 8 bit
    range,
    --restrict_quantization_steps "-0x8000 0x7F7F" indicates
↳ an example 16
    bit range.
    This argument is required for 16-bit Matmul operations.
--pack_4_bit_weights Store 4-bit quantized weights in packed format in a single
↳ byte i.e. two
    4-bit quantized tensors can be stored in one byte
--act_quantizer_calibration ACT_QUANTIZER_CALIBRATION
    Specify which quantization calibration method to use for
↳ activations
    supported values: min-max (default), sqnr, entropy, mse,
↳ percentile
    This option can be paired with --act_quantizer_schema to
↳ override the
    quantization
    schema to use for activations otherwise default
↳ schema(asymmetric) will be
    used
--param_quantizer_calibration PARAM_QUANTIZER_CALIBRATION
    Specify which quantization calibration method to use for
↳ parameters
    supported values: min-max (default), sqnr, entropy, mse,
↳ percentile
    This option can be paired with --param_quantizer_schema to

```

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```

↪override the
    quantization
    schema to use for parameters otherwise default.
↪schema(asymmetric) will be
    used
    --act_quantizer_schema ACT_QUANTIZER_SCHEMA
    Specify which quantization schema to use for activations
    supported values: asymmetric (default), symmetric,
↪unsignedsymmetric
    This option cannot be used with legacy quantizer option --act_
↪quantizer
    --param_quantizer_schema PARAM_QUANTIZER_SCHEMA
    Specify which quantization schema to use for parameters
    supported values: asymmetric (default), symmetric,
↪unsignedsymmetric
    This option cannot be used with legacy quantizer option --
↪param_quantizer
    --percentile_calibration_value PERCENTILE_CALIBRATION_VALUE
    Specify the percentile value to be used with Percentile.
↪calibration method
    The specified float value must lie within 90 and 100,
↪default: 99.99
    --dump_qairt_quantizer_command DUMP_QAIRT_QUANTIZER_COMMAND
    Use this option to dump a file which contains the equivalent.
↪Commandline
    input for QAIRT Quantizer

```

Masked Softmax Optimization Options:

```

    --apply_masked_softmax {compressed,uncompressed}
    This flag enables the pass that creates a MaskedSoftmax Op and
    rewrites the graph to include this Op. MaskedSoftmax Op may
↪not
    be supported by all the QNN backends. Please check the
    supplemental backend XML for the targeted backend.
    This argument takes a string parameter input that selects
    the mode of MaskedSoftmax Op.
    'compressed' value rewrites the graph with the compressed.
↪version of
    MaskedSoftmax Op.
    'uncompressed' value rewrites the graph with the uncompressed.
↪version of
    MaskedSoftmax Op.
    --packed_masked_softmax_inputs PACKED_MASKED_SOFTMAX_INPUTS [PACKED_MASKED_SOFTMAX_

```

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```

↪INPUTS ...]
    Mention the input ids tensor name which will be packed in the
↪single
    inference.
    This is applicable only for Compressed MaskedSoftmax Op.
    This will create a new input to the graph named 'position_ids'
    with same shape as the provided input name in this flag.
    During runtime, this input shall be provided with the token
    locations for individual sequences so that the same will be
    internally passed to positional embedding layer.
    E.g. If 2 sequences of length 20 and 30 are packed together
    in single batch of 64 tokens then this new input 'position_ids'
↪' should have
    value [0, 1, ..., 19, 0, 1, ..., 29, 0, 0, 0, ..., 0]
    Usage: --packed_masked_softmax input_ids
    Packed model will enable the user to pack multiple sequences
↪into
    single batch of inference.
    --packed_max_seq PACKED_MAX_SEQ
    Number of sequences packed in the single input ids and
    single attention mask inputs. Applicable only for
    Compressed MaskedSoftmax Op.

Architecture Checker Options(Experimental):
    --arch_checker      Pass this option to enable architecture checker tool.
    This is an experimental option for models that are intended
↪to run on HTP
    backend.

Note: Only one of: {'op_package_config', 'package_name'} can be specified Note: Only
↪one of:
{'op_package_config', 'package_name'} can be specified

```

75.5 qnn-model-lib-generator

```
qnn-model-lib-generator --help
```

```
usage: qnn-model-lib-generator
    [-h]
    -c
```

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```

CPP
[-b BIN]
[-t LIB_TARGETS [LIB_TARGETS ...]]
[-l LIB_NAME]
[-o OUTPUT_DIR]
[-d]

```

options:

```

-h, --help
    show this help message and exit

```

required arguments:

```

-c CPP, --cpp CPP
    Filepath for the qnn model .cpp file

```

optional arguments:

```

-b BIN, --bin BIN
    Filepath for the qnn model .bin file. (Note: if not passed, runtime will fail if .
    ↪cpp
    needs any items from a .bin file.)
-t LIB_TARGETS [LIB_TARGETS ...], --lib_targets LIB_TARGETS [LIB_TARGETS ...]
    Specifies the targets to build the models for. Supported
    targets: x86_64-linux-clang, aarch64-oe-linux-gcc8.2,
    aarch64-oe-linux-gcc11.2, aarch64-oe-linux-gcc9.3, aarch64-ubuntu-gcc9.4,
    aarch64-android.
    Default: [x86_64-linux-clang, aarch64-android]
-l LIB_NAME, --lib_name LIB_NAME
    Specifies the name to use for libraries.
    Default:
    uses name in <model>.bin if provided, else generic qnn_model
-o OUTPUT_DIR, --output_dir OUTPUT_DIR
    Location for saving output libraries. Default: /content/lib
-d, --debug
    Set log level as debug mode

```

75.6 qnn-net-run

75.6.1 help

```
qnn-net-run --help
```

75.6.2 On Android

1. Copy `/star-fj/fangjun/download/qnn/qairt/2.32.0.250228/bin/aarch64-android/qnn-net-run` to android

```
adb push ./qnn-net-run /data/local/tmp/
```

Then `adb shell` and switch to `/data/local/tmp` and run `chmod +x ./qnn-net-run`

2. Copy the following files from `/star-fj/fangjun/download/qnn/qairt/2.32.0.250228/lib/aarch64-android/` to android

- `libQnnGpu.so`
- `libQnnHtp.so`

75.7 qairt-converter

```
echo "Maintained by k2-fsa" > copyright.txt
```

```
qairt-converter \
  --set_model_version "2025-10-31" \
  --copyright_file ./copyright.txt \
  --input_network ./model-opset-17.onnx \
  --source_model_input_shape 'x' 1,93,560 \
  --source_model_input_shape 'prompt' 1,4 \
  --source_model_input_datatype 'x' float32 \
  --source_model_input_datatype 'prompt' int32
```

```
usage: qairt-converter [--source_model_input_shape INPUT_NAME INPUT_DIM]
                      [--out_tensor_node OUT_NAMES]
                      [--source_model_input_datatype INPUT_NAME INPUT_DTYPE]
                      [--source_model_input_layout INPUT_NAME INPUT_LAYOUT]
                      [--desired_input_layout INPUT_NAME DESIRED_INPUT_LAYOUT]
                      [--source_model_output_layout OUTPUT_NAME OUTPUT_LAYOUT]
                      [--desired_output_layout OUTPUT_NAME DESIRED_OUTPUT_LAYOUT]
                      [--desired_input_color_encoding ...]
                      [--preserve_io_datatype [PRESERVE_IO_DATATYPE ...]]
                      [--dump_config_template DUMP_IO_CONFIG_TEMPLATE] [--config IO_
↪CONFIG]
                      [--dry_run [DRY_RUN]] [--enable_framework_trace] [--gguf_
↪config GGUF_CONFIG]
                      [--quantization_overrides QUANTIZATION_OVERRIDES]
                      [--lora_weight_list LORA_WEIGHT_LIST] [--onnx_skip_
```

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```

↪simplification]
                                [--onnx_override_batch BATCH] [--onnx_define_symbol SYMBOL_
↪NAME VALUE]
                                [--onnx_validate_models] [--onnx_summary]
                                [--onnx_perform_sequence_construct_optimizer] [--tf_summary]
                                [--tf_disable_optimization] [--tf_show_unconsumed_nodes]
                                [--tf_saved_model_tag SAVED_MODEL_TAG]
                                [--tf_saved_model_signature_key SAVED_MODEL_SIGNATURE_KEY]
                                [--tf_validate_models] [--tflite_signature_name SIGNATURE_NAME]
                                --input_network INPUT_NETWORK [--debug [DEBUG]] [--output_path_
↪OUTPUT_PATH]
                                [--copyright_file COPYRIGHT_FILE] [--float_bitwidth FLOAT_
↪BITWIDTH]
                                [--float_bias_bitwidth FLOAT_BIAS_BITWIDTH]
                                [--set_model_version MODEL_VERSION]
                                [--converter_op_package_lib CONVERTER_OP_PACKAGE_LIB]
                                [--package_name PACKAGE_NAME | --op_package_config CUSTOM_OP_
↪CONFIG_PATHS [CUSTOM_OP_CONFIG_PATHS ...]]
                                [-h] [--target_backend BACKEND] [--target_soc_model SOC_MODEL]

```

required arguments:

```

--input_network INPUT_NETWORK, -i INPUT_NETWORK
    Path to the source framework model.

```

optional arguments:

```

--source_model_input_shape INPUT_NAME INPUT_DIM, -s INPUT_NAME INPUT_DIM
    The name and dimension of all the input buffers to the
↪network specified in
    the format [input_name comma-separated-dimensions],
    for example: --source_model_input_shape 'data' 1,224,224,3.
    Note that the quotes should always be included in order to
↪handle special
    characters, spaces, etc.
    NOTE: Required for TensorFlow and PyTorch. Optional for Onnx
↪and Tflite
    In case of Onnx, this feature works only with Onnx 1.6.0 and
↪above
    --out_tensor_node OUT_NAMES, --out_tensor_name OUT_NAMES
    Name of the graph's output Tensor Names. Multiple output
↪names should be
    provided separately like:
        --out_tensor_name out_1 --out_tensor_name out_2
    NOTE: Required for TensorFlow. Optional for Onnx, Tflite and

```

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```

↪PyTorch
  --source_model_input_datatype INPUT_NAME INPUT_DTYPE
                                The names and datatype of the network input layers specified.
↪in the format
                                [input_name datatype], for example:
                                'data' 'float32'
                                Default is float32 if not specified
                                Note that the quotes should always be included in order to
↪handlespecial
                                characters, spaces, etc.
                                For multiple inputs specify multiple --source_model_input_
↪datatype on the
                                command line like:
                                --source_model_input_datatype 'data1' 'float32'
                                --source_model_input_datatype 'data2' 'float32'
  --source_model_input_layout INPUT_NAME INPUT_LAYOUT
                                Layout of each input tensor. If not specified, it will use
↪the default based
                                on the Source Framework, shape of input and input encoding.
                                Accepted values are-
                                NCDHW, NDHWC, NCHW, NHWC, HWIO, OIHW, NFC, NCF, NTF, TNF,
↪NF, NC, F
                                N = Batch, C = Channels, D = Depth, H = Height, W = Width, F
↪= Feature,
                                T = Time, I = Input, O = Output
                                NDHWC/NCDHW used for 5d inputs
                                NHWC/NCHW used for 4d image-like inputs
                                HWIO/IOHW used for Weights of Conv Ops
                                NFC/NCF used for inputs to Conv1D or other 1D ops
                                NTF/TNF used for inputs with time steps like the ones used
↪for LSTM op
                                NF used for 2D inputs, like the inputs to Dense/
↪FullyConnected layers
                                NC used for 2D inputs with 1 for batch and other for Channels
↪(rarely used)
                                F used for 1D inputs, e.g. Bias tensor
                                For multiple inputs specify multiple --source_model_input_
↪layout on the
                                command line.
                                Eg:
                                --source_model_input_layout "data1" NCHW --source_model_
↪input_layout
                                "data2" NCHW

```

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```

--desired_input_layout INPUT_NAME DESIRED_INPUT_LAYOUT
    Desired Layout of each input tensor. If not specified, it
will use the
    default based on the Source Framework, shape of input and
input encoding.
    Accepted values are-
        NCDHW, NDHWC, NCHW, NHWC, HWIO, OIHW, NFC, NCF, NTF, TNF,
NF, NC, F
    N = Batch, C = Channels, D = Depth, H = Height, W = Width, F
= Feature,
    T = Time, I = Input, O = Output
    NDHWC/NCDHW used for 5d inputs
    NHWC/NCHW used for 4d image-like inputs
    HWIO/OIHW used for Weights of Conv Ops
    NFC/NCF used for inputs to Conv1D or other 1D ops
    NTF/TNF used for inputs with time steps like the ones used
for LSTM op
NF used for 2D inputs, like the inputs to Dense/
FullyConnected layers
NC used for 2D inputs with 1 for batch and other for Channels
(rarely used)
F used for 1D inputs, e.g. Bias tensor
For multiple inputs specify multiple --desired_input_layout
on the command
    line.
    Eg:
        --desired_input_layout "data1" NCHW --desired_input_
layout "data2" NCHW
--source_model_output_layout OUTPUT_NAME OUTPUT_LAYOUT
    Layout of each output tensor. If not specified, it will use
the default
    based on the Source Framework, shape of input and input
encoding.
    Accepted values are-
        NCDHW, NDHWC, NCHW, NHWC, HWIO, OIHW, NFC, NCF, NTF, TNF,
NF, NC, F
    N = Batch, C = Channels, D = Depth, H = Height, W = Width, F
= Feature, T =
    Time
    NDHWC/NCDHW used for 5d inputs
    NHWC/NCHW used for 4d image-like inputs
    NFC/NCF used for inputs to Conv1D or other 1D ops
    NTF/TNF used for inputs with time steps like the ones used

```

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```

→for LSTM op
    NF used for 2D inputs, like the inputs to Dense/
→FullyConnected layers
    NC used for 2D inputs with 1 for batch and other for Channels.
→(rarely used)
    F used for 1D inputs, e.g. Bias tensor
    For multiple inputs specify multiple --source_model_output_
→layout on the
    command line.
    Eg:
        --source_model_output_layout "data1" NCHW --source_model_
→output_layout
        "data2" NCHW
    --desired_output_layout OUTPUT_NAME DESIRED_OUTPUT_LAYOUT
        Desired Layout of each output tensor. If not specified, it
→will use the
        default based on the Source Framework.
        Accepted values are-
            NCDHW, NDHWC, NCHW, NHWC, HWIO, OIHW, NFC, NCF, NTF, TNF,
→NF, NC, F
            N = Batch, C = Channels, D = Depth, H = Height, W = Width, F =
→= Feature, T =
            Time
            NDHWC/NCDHW used for 5d outputs
            NHWC/NCHW used for 4d image-like outputs
            NFC/NCF used for outputs to Conv1D or other 1D ops
            NTF/TNF used for outputs with time steps like the ones used
→for LSTM op
    NF used for 2D outputs, like the outputs to Dense/
→FullyConnected layers
    NC used for 2D outputs with 1 for batch and other for
→Channels (rarely used)
    F used for 1D outputs, e.g. Bias tensor
    For multiple outputs specify multiple --desired_output_layout
→on the command
    line.
    Eg:
        --desired_output_layout "data1" NCHW --desired_output_
→layout "data2"
        NCHW
    --desired_input_color_encoding [...], -e ...]
        Usage: --input_color_encoding "INPUT_NAME" INPUT_ENCODING_
→IN

```

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```
[INPUT_ENCODING_OUT]
Input encoding of the network inputs. Default is bgr.
e.g.
    --input_color_encoding "data" rgba
Quotes must wrap the input node name to handle special_
↳characters,
spaces, etc. To specify encodings for multiple inputs, invoke
--input_color_encoding for each one.
e.g.
    --input_color_encoding "data1" rgba --input_color_
↳encoding "data2" other
Optionally, an output encoding may be specified for an input_
↳node by
providing a second encoding. The default output encoding is_
↳bgr.
e.g.
    --input_color_encoding "data3" rgba rgb
Input encoding types:
    image color encodings: bgr,rgb, nv21, nv12, ...
    time_series: for inputs of rnn models;
    other: not available above or is unknown.
Supported encodings:
    bgr
    rgb
    rgba
    argb32
    nv21
    nv12
--preserve_io_datatype [PRESERVE_IO_DATATYPE ...]
Use this option to preserve IO datatype. The different ways_
↳of using this
option are as follows:
    --preserve_io_datatype <space separated list of names of_
↳inputs and
outputs of the graph>
e.g.
    --preserve_io_datatype input1 input2 output1
The user may choose to preserve the datatype for all the_
↳inputs and outputs
of the graph.
    --preserve_io_datatype
Note: --config gets higher precedence than --preserve_io_
↳datatype.
```

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```

--dump_config_template DUMP_IO_CONFIG_TEMPLATE
    Dumps the yaml template for I/O configuration. This file can
↳ be edited as
    per the custom requirements and passed using the option --
↳ configUse this
    option to specify a yaml file to which the IO config template
↳ is dumped.
--config IO_CONFIG
    Use this option to specify a yaml file for input and output
↳ options.
--dry_run [DRY_RUN]
    Evaluates the model without actually converting any ops, and
↳ returns
    unsupported ops/attributes as well as unused inputs and/or
↳ outputs if any.
--enable_framework_trace
    Use this option to enable converter to trace the op/tensor
↳ change
    information.
    Currently framework op trace is supported only for ONNX
↳ converter.
--gguf_config GGUF_CONFIG
    This is an optional argument that can be used when input
↳ network is a GGUF
    File.It specifies the path to the config file for building
↳ GenAI model.(the
    config.json file generated when saving the huggingface model)
--debug [DEBUG]
    Run the converter in debug mode.
--output_path OUTPUT_PATH, -o OUTPUT_PATH
    Path where the converted Output model should be saved.If not
↳ specified, the
    converter model will be written to a file with same name as
↳ the input model
--copyright_file COPYRIGHT_FILE
    Path to copyright file. If provided, the content of the file
↳ will be added
    to the output model.
--float_bitwidth FLOAT_BITWIDTH
    Use the --float_bitwidth option to convert the graph to the
↳ specified float
    bitwidth, either 32 (default) or 16.
--float_bias_bitwidth FLOAT_BIAS_BITWIDTH
    Use the --float_bias_bitwidth option to select the bitwidth
↳ to use for float
    bias tensor, either 32 or 16 (default '0' if not provided).

```

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```

--set_model_version MODEL_VERSION
                                User-defined ASCII string to identify the model, only first
↪64 bytes will be
                                stored
-h, --help                      show this help message and exit

```

Custom Op Package Options:

```

--converter_op_package_lib CONVERTER_OP_PACKAGE_LIB, -cpl CONVERTER_OP_PACKAGE_LIB
                                Absolute path to converter op package library compiled by the
↪OpPackage
                                generator. Must be separated by a comma for multiple package
↪libraries.
                                Note: Order of converter op package libraries must follow the
↪order of xmls.
                                Ex1: --converter_op_package_lib absolute_path_to/libExample.so
                                Ex2: -cpl absolute_path_to/libExample1.so,absolute_path_to/
↪libExample2.so
--package_name PACKAGE_NAME, -p PACKAGE_NAME
                                A global package name to be used for each node in the Model.
↪cpp file.
                                Defaults to Qnn header defined package name
--op_package_config CUSTOM_OP_CONFIG_PATHS [CUSTOM_OP_CONFIG_PATHS ...], -opc
↪CUSTOM_OP_CONFIG_PATHS [CUSTOM_OP_CONFIG_PATHS ...]
                                Path to a Qnn Op Package XML configuration file that contains
↪user defined
                                custom operations.

```

Quantizer Options:

```

--quantization_overrides QUANTIZATION_OVERRIDES, -q QUANTIZATION_OVERRIDES
                                Use this option to specify a json file with parameters to use
↪for
                                quantization. These will override any quantization data
↪carried from
                                conversion (eg TF fake quantization) or calculated during the
↪normal
                                quantization process. Format defined as per AIMET
↪specification.

```

LoRA Converter Options:

```

--lora_weight_list LORA_WEIGHT_LIST
                                Path to a file specifying a list of tensor names that should
↪be updateable.

```

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Onnx Converter Options:

```

--onnx_skip_simplification, -oss
    Do not attempt to simplify the model automatically. This may
    prevent some
    models from
    properly converting when sequences of unsupported static
    operations are
    present.

--onnx_override_batch BATCH
    The batch dimension override. This will take the first
    dimension of all
    inputs and treat it as a batch dim, overriding it with the
    value provided
    here. For example:
    --onnx_override_batch 6
    will result in a shape change from [1,3,224,224] to [6,3,224,
    224].

    If there are inputs without batch dim this should not be used
    and each input
    should be overridden independently using -d option for input
    dimension
    overrides.

--onnx_define_symbol SYMBOL_NAME VALUE
    This option allows overriding specific input dimension
    symbols. For instance
    you might see input shapes specified with variables such as :
    data: [1,3,height,width]
    To override these simply pass the option as:
    --onnx_define_symbol height 224 --onnx_define_symbol width 448
    which results in dimensions that look like:
    data: [1,3,224,448]

--onnx_validate_models
    Validate the original ONNX model against optimized ONNX model.
    Constant inputs with all value 1s will be generated and will
    be used
    by both models and their outputs are checked against each
    other.

    The % average error and 90th percentile of output differences
    will be
    calculated for this.
    Note: Usage of this flag will incur extra time due to
    inference of the
    models.

```

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```

--onnx_summary          Summarize the original onnx model and optimized onnx model.
                        Summary will print the model information such as number of
↳ parameters,
                        number of operators and their count, input-output tensor name,
↳ shape and
                        dtypes.
--onnx_perform_sequence_construct_optimizer
                        This option allows optimization on SequenceConstruct Op.
                        When SequenceConstruct op is one of the outputs of the graph,
↳ it removes
                        SequenceConstruct op and makes its inputs as graph outputs to
↳ replace the
                        original output of SequenceConstruct.
--tf_summary            Summarize the original TF model and optimized TF model.
                        Summary will print the model information such as number of
↳ parameters,
                        number of operators and their count, input-output tensor name,
↳ shape and
                        dtypes.

```

TensorFlow Converter Options:

```

--tf_disable_optimization
                        Do not attempt to optimize the model automatically.
--tf_show_unconsumed_nodes
                        Displays a list of unconsumed nodes, if there any are found.
↳ Nodes which are
                        unconsumed do not violate the structural fidelity of
↳ the generated graph.
--tf_saved_model_tag SAVED_MODEL_TAG
                        Specify the tag to select a MetaGraph from savedmodel. ex:
                        --saved_model_tag serve. Default value will be 'serve' when
↳ it is not
                        assigned.
--tf_saved_model_signature_key SAVED_MODEL_SIGNATURE_KEY
                        Specify signature key to select input and output of the model.
↳ ex:
                        --tf_saved_model_signature_key serving_default. Default value
↳ will be
                        'serving_default' when it is not assigned
--tf_validate_models    Validate the original TF model against optimized TF model.
                        Constant inputs with all value 1s will be generated and will
↳ be used
                        by both models and their outputs are checked against each

```

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```

↳ other.
    The % average error and 90th percentile of output differences
↳ will be
    calculated for this.
    Note: Usage of this flag will incur extra time due to
↳ inference of the
    models.

Tflite Converter Options:
  --tflite_signature_name SIGNATURE_NAME
    Use this option to specify a specific Subgraph signature to
↳ convert

Backend Options:
  --target_backend BACKEND
    Use this option to specify the backend on which the model
↳ needs to run.
    Providing this option will generate a graph optimized for the
↳ given backend
    and this graph may not run on other backends.
    Supported backends are CPU,GPU,DSP,HTP,HTA,LPAI.

  --target_soc_model SOC_MODEL
    Use this option to specify the SOC on which the model needs
↳ to run.
    This can be found from SOC info of the device and it starts
↳ with strings
    such as SDM, SM, QCS, IPQ, SA, QC, SC, SXR, SSG, STP, QRB, or
↳ AIC.
    NOTE: --target_backend option must be provided to use --
↳ target_soc_model
    option.

Note: Only one of: {'op_package_config', 'package_name'} can be specified Note: Only
↳ one of:
{'op_package_config', 'package_name'} can be specified

```